A-infinity Chiral Algebras and 3D Holomorphic–Topological QFT

Volume II of the programme: Hamiltonian jets, shifted cotangent fields, universal branes, and the open–closed map

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Abstract

Bulk reconstruction requires coherent actions on boundaries and on the operators between them. This book constructs finite Hamiltonian models, supported collision operations and marked reconstruction maps, while retaining their geometric and physical comparison problems.

Finite Hamiltonian jets give shifted-cotangent models satisfying the quantum master equation by vanishing adjoint traces. For a finite-dimensional complex Lie algebra, relative Hochschild cochains of its Rees envelope correspond to adjoint Chevalley cochains with differential $\hbar d_{\mathrm{CE}}$, including finite coefficient quotients and the formal limit. Quadratic Moyal Hamiltonians give the even Weyl algebra through the oscillator Casimir quotient. Separate input and output jets determine the continuous cochain carrier; its operadic refinement remains additional.

Polynomial-root current actions retain a nonzero Massey coset, integral comparison paths and derived specialization. Finite interface centres have brackets and higher compositions beyond their cup products. Tree expansions construct action profiles and multiplicative recipients for finite nilpotent families. Explicit pointed and ordinary multiplicative actions obstruct finality in their specified categories.

Wick fields and mixed forest homotopies give meromorphic coefficients with point-source contacts. Outer principal-part classes survive in every arity at least three. Supported Heisenberg pair maps on regular singleton inputs admit a complete triple homotopy; the common four-input comparison remains open. Finite Gaussian elimination retains cumulants, determinant, conditional differential and sewing. Infinite-mode evolution has its numerical Fock states, formal cubic sewing and a positive quartic stabilization with a sharp continuity threshold.

Hamiltonians act in all arities on the resolved plane and derived matrix spaces. Rank-two relative quantum restriction is an integral quasi-isomorphism; absolute reduction retains its gauge class and higher operations. Cyclic traces retain the central rank, while the full Hamiltonian/cyclic comparison remains open.

Gaussian half-space fields give supported boundary actions and an incoming cap on a specified finite-jet domain. Derived normal support reconstructs on the normal Weyl carrier. Marked Morita reconstruction holds on generated sectors. The Ramond junction has nonzero polynomial right orthogonals, a local residue cone without an original-quartet lift, and a polynomial field prescription that fails derived invariance. Shuffle lattices distinguish integral multiplication from rational cooperation. Tensor reconstruction and gravitational examples specify the additional maps required to preserve local fields, nonlinear equations and physical sewing.

Operations and reconstruction

A boundary algebra records compositions. Its derived centre records coherent actions on its module category; the derived end of one selected module instead detects the sector that module generates. Reconstructing bulk fields requires an independently defined bulk, its boundary action and a comparison preserving positions, supports and collisions.

Quantization must deform the map as well as its endpoints. Over $k[\varepsilon]/(\varepsilon^2)$, the relation $x^2 = \varepsilon$ defines a free rank-two deformation of $k[x]/(x^2)$. Its augmentation cannot lift to the undeformed coefficient algebra: any lift sends x to εc, whose square is zero. An algebra deformation alone therefore does not provide the required bulk–boundary deformation.

Chapters 1–7 construct the algebraic and supported ingredients of this problem. Morphism homotopies lead to coefficient modules on ordered configurations, finite Hamiltonian jets and continuous cochains. Polynomial-root interfaces retain Massey products, comparison fibres and gauge stabilizers. Action profiles give explicit recipients, while two distinct nonfinality arguments constrain universal pointed and multiplicative actions. Supported pair operations admit a triple comparison; four inputs require a common recipient and the pentagon primitive for the prescribed lower homotopies.

Chapters 8–13 retain quantum and state data. Gaussian elimination transports the differential, odd inclusion, determinant and all cumulants. The cubic matrix trace defect forces higher Hamiltonian components. Integral relative rank-two restriction, absolute gauge cohomology and cyclic trace calculations preserve data lost by degree-zero invariants. Infinite-mode states require a specified completion; a positive quartic stabilization changes the cubic theory and has its own continuity threshold.

Chapters 14–15 begin with native fields and an independent junction. The Gaussian boundary action, corrected cutoff system and incoming cap have explicit coefficient domains. Normal Weyl reconstruction uses derived adic support, with a separate comparison to physical fusion. The Ramond junction's reverse mode algebra is computed, but the junction misses nonzero polynomial boundary objects. Its local residue also produces a cone that cannot carry the original quartet. An acyclic localization shows why unchanged polynomial local fields cannot be the derived mapping objects on every pair.

Chapter 16 compares the remaining algebraic and geometric structures. Ordinary bars, integral shuffle lattices and rational cooperation retain different operations. Matrix coefficients reconstruct a specified tensor category. Gravitational examples distinguish tangent complexes from nonlinear equations and connection coefficients from holonomy. The full reconstruction problem must preserve local mapping objects, both tensor actions, units, preparations and coherent sewing.

Notation and normalization

The ground field has characteristic zero. Ambient stable categories used for colimit constructions are presentable and idempotent-complete, and their tensor products preserve colimits separately. Perfect sectors retain their stated finite-generation conditions. Each completion specifies its coefficient ring, filtration and transition maps; ordinary and derived inverse limits are distinguished. The Igusa normalization is

$$\Delta_5^{\theta} = 64 D_5, \qquad \chi_{10} = D_5{}^2,$$

with D_5 monic. Physical charge is additive; Gram degree is carried through its polarization cocycle. Bar, factorization, and Hopf coproducts retain distinct notation and distinct geometric meanings.

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Chapter 1

Boundary actions and their comparison homotopies

A bulk observable acts on boundary conditions and on the operators between them. Its action may vanish on each boundary while remaining nonzero on the category of boundaries. Two cubic boundaries exhibit this phenomenon. They lead to a complex that retains the actions, the homotopies comparing them, and the equations imposed by composition. The resulting ordinary derived centre gives an algebraic model of coherent actions. In a mixed holomorphic–topological field theory, which of these actions come from local bulk observables?

1.1 A relation detected between two boundaries

Let k be a field of characteristic zero and put $R = k[x]$. A *matrix factorization* of $W \in R$ is a finite free parity-graded R-module $E = E^{\bar{0}} \oplus E^{\bar{1}}$ with an odd endomorphism Q_E satisfying $Q_E^2 = W \,\mathrm{id}_E$.

For two factorizations of the same potential, set

$$\partial f = Q_F f - (-1)^{|f|} f Q_E, \qquad f \in \mathrm{Hom}_R(E, F). \tag{1.1}$$

Here $|f| \in \mathbb{Z}/2$ is its parity. Since W is central, $\partial^2 f = Wf - fW = 0$. Composition satisfies the graded Leibniz rule. Thus the factorizations form a differential category with parity grading.

For the cubic potential, take the factorizations on $R^{1|1}$

$$Q_X = \begin{pmatrix} 0 & x \\ x^2 & 0 \end{pmatrix}, \qquad Q_Y = \begin{pmatrix} 0 & x^2 \\ x & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix} : X \longrightarrow Y. \tag{1.2}$$

The map f is even and closed. Multiplication by x has an odd primitive on each object: there are h_X, h_Y with $\partial h_X = x \,\mathrm{id}_X$ and $\partial h_Y = x \,\mathrm{id}_Y$. Their compatibility along f is measured by

$$c(f) = h_Y f - f h_X.$$

It is closed, because $\partial c(f) = xf - fx = 0$.

Proposition 1.1. *For the objects and map in* (1.2)*, every choice of h_X, h_Y gives the same nonzero class*

$$[c(f)] = [J] \in H^{\bar{1}} \mathrm{Hom}_R(X, Y), \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

In particular, the objectwise primitives cannot be extended to a nullhomotopy compatible with this morphism.

Proof. The most general primitives are

$$h_X = \begin{pmatrix} 0 & b \\ 1 - xb & 0 \end{pmatrix}, \qquad h_Y = \begin{pmatrix} 0 & 1 - xc \\ c & 0 \end{pmatrix}, \qquad b, c \in R.$$

Indeed, an odd matrix with upper entry u and lower entry v is a primitive on X precisely when $xv + x^2u = x$, so $v = 1 - xu$. The equation on Y is analogous.

Multiplication gives

$$c(f) = (1 - x(b + c))J. \tag{1.3}$$

For an even morphism $a = \operatorname{diag}(a_0, a_1)$ and an odd morphism with entries b_0, c_0, the two differentials are

$$\partial a = x(xa_1 - a_0)J, \qquad \partial \begin{pmatrix} 0 & b_0 \\ c_0 & 0 \end{pmatrix} = x(b_0 + c_0)f.$$

Consequently the even cycles and boundaries are Rf and xRf, and the odd cycles and boundaries are RJ and xRJ. The coefficient of J in (1.3) is one modulo x, independently of b, c. □

For three objects with such primitives and closed even maps $f : X \to Y$, $g : Y \to Z$, the same definition gives

$$c(gf) = c(g)f + gc(f).$$

Thus even a primitive for each $c(f)$ would leave a compatibility equation for composition. The complex of coherent actions must contain components on morphisms and on longer composable strings.

1.2 The ordinary derived centre

Let $\mathcal{E}$ be a small unital differential graded category over k. Complexes have cohomological grading, $V[1]^j = V^{j+1}$. The same constructions apply with degrees reduced modulo two to the preceding category of matrix factorizations. All tensors in this section are over k, and total Hom complexes use products. In particular, no boundedness in the cohomological degrees is implicit. Write $D(k)$ for the derived category of coefficient complexes.

The *diagonal bimodule* is $\Delta_{\mathcal{E}}(a, b) = \mathcal{E}(a, b)$, with the two actions given by composition. Write $\mathcal{E}^e = \mathcal{E}^{\mathrm{op}} \otimes_k^{\mathbb{L}} \mathcal{E}$ for the derived enveloping category. The *ordinary derived centre* and its cohomology are

$$Z_k(\mathcal{E}) = \mathrm{RHom}_{\mathcal{E}^e}(\Delta_{\mathcal{E}}, \Delta_{\mathcal{E}}), \qquad \mathrm{HH}_k^*(\mathcal{E}) = H^* Z_k(\mathcal{E}). \tag{1.4}$$

This definition fixes the coefficient tensor. Relative centres over a larger ring will use that ring's tensor and derived Hom instead.

To compute (1.4), resolve the diagonal by its two-sided bar. After bimodule Hom, its cochains have factors

$$\prod_{n \geq 0} \prod_{w_0, \ldots, w_n} \mathrm{Hom}_k(\mathcal{E}(w_{n-1}, w_n)[1] \otimes \cdots \otimes \mathcal{E}(w_0, w_1)[1], \mathcal{E}(w_0, w_n)). \tag{1.5}$$

At $n = 0$ the input is k. Every intermediate object is an index of the product. The normalized cochains vanish when an input is an identity. Denote their product totalization by $\mathrm{CC}_k^\bullet(\mathcal{E})$.

The differential has an internal part and a part that composes adjacent inputs or acts at an endpoint. An unsuspended convention makes their signs explicit. An arity-n cochain F of internal degree r has total degree $r + n$. Put $\partial F = [\delta, F]$, the graded Hom differential.

For homogeneous composable inputs, ordered so that the rightmost acts first, set

$$\begin{aligned}(b_H F)(a_1, \ldots, a_{n+1}) &= (-1)^{r|a_1|} a_1 F(a_2, \ldots, a_{n+1}) \\ &\quad + \sum_{j=1}^{n} (-1)^j F(a_1, \ldots, a_j a_{j+1}, \ldots, a_{n+1}) \\ &\quad + (-1)^{n+1} F(a_1, \ldots, a_n) a_{n+1}, \end{aligned} \tag{1.6}$$

$$(\mathcal{D}F)_n = (-1)^n \partial F_n + b_H F_{n-1}, \qquad F_{-1} = 0. \tag{1.7}$$

Associativity and the Leibniz rule give $b_H^2 = \partial^2 = 0$ and $\partial b_H = b_H \partial$. Hence $\mathcal{D}^2 = 0$. Each component involves only two arities, so the differential is defined on the product totalization.

Proposition 1.2 (Bar representation and evaluation)**.** *The normalized complex* $\mathrm{CC}_k^\bullet(\mathcal{E})$ *represents* $Z_k(\mathcal{E})$*. For every coefficient complex* $P \in D(k)$*, evaluation induces a natural equivalence*

$$\mathrm{RHom}_k(P, Z_k(\mathcal{E})) \simeq \mathrm{RHom}_{\mathcal{E}^e}(P \otimes_k^{\mathbb{L}} \Delta_{\mathcal{E}}, \Delta_{\mathcal{E}}). \tag{1.8}$$

Proof. Use cofibrant models for the coefficient complexes. The augmented two-sided bar contracts by insertion of a unit at an outside end.

Its filtration by word length has induced bimodules as successive quotients. Over the field k, their coefficient complexes are homotopically projective: Hom from them into an acyclic complex is acyclic. A finite length stage therefore has the same property.

Hom out of the full direct-sum bar is the inverse limit of the Hom complexes of these stages. The transition maps are surjective in every degree, since the length filtration splits on graded modules. The kernels are acyclic by the same coefficient argument, so primitives can be lifted successively through the tower. Its inverse limit is therefore acyclic. Thus the bar is a homotopically projective resolution.

The degeneracy contraction splits off the cochains containing an identity, leaving the normalized complex. Applying bimodule Hom to the remaining resolution gives (1.5). For a projective model of P, the tensor–Hom adjunction on each bar term identifies the two sides of (1.8). Products commute with this adjunction, and the identifications commute with the differential. □

The identity action is the nullary family id_w. The cup product composes outputs after cutting a composable word. The brace operations insert the output of one cochain into an input of another, with the Koszul signs of the suspended inputs. These are operations on the full cochain complex. Retaining only the cohomology of the endomorphisms of each object removes the positive-arity terms where Proposition 1.1 found its obstruction.

1.3 The scalar action on all cubic boundaries

The cubic calculation extends to every finite free boundary. It distinguishes a relation in the category from a relation on each of its objects. Put $R = k[[x]]$. Let C be the parity differential category of all matrix factorizations of x^3 on standard finite free parity-graded R-modules. Fixing standard modules makes the objects a set. Its morphisms are R-linear, with differential (1.1).

Here the centre is *absolute over* k: its cochains are k-multilinear, with no continuity condition on those maps. They are normalized and product-totalized as above. For $g \in R$, scalar multiplication defines a cocycle $\rho(g)$ whose nullary value is $g\,\mathrm{id}_E$ and whose higher values vanish.

Theorem 1.3. *The kernel of the scalar action*

$$R \longrightarrow \mathrm{HH}_k^{\bar{0}}(C), \qquad g \longmapsto [\rho(g)],$$

is (x^2)*. Nevertheless, multiplication by* x *is nullhomotopic on every object of* C*.*

Proof. For a cochain primitive with nullary component h_E and unary component u, equations (1.6) and (1.7) begin with

$$\delta h_E = g\,\mathrm{id}_E, \tag{1.9}$$

$$\delta u(f) - u(\partial f) = (-1)^{|f|} f h_E - h_F f. \tag{1.10}$$

The map h_E is odd and u preserves parity. If all later components vanish, the binary equation is $u(vf) = u(v)f + vu(f)$.

Differentiate the matrices in their fixed bases and take

$$h_E = \tfrac{1}{3}Q_E', \qquad u(f) = \tfrac{1}{3}f', \qquad F_n = 0 \quad (n \geq 2).$$

Differentiating $Q_E^2 = x^3\,\mathrm{id}_E$ proves (1.9) for $g = x^2$. For a homogeneous morphism,

$$(\delta f)' = \delta(f') + Q_F' f - (-1)^{|f|} f Q_E',$$

which proves (1.10). The derivative kills identities and satisfies the Leibniz rule, so every remaining component of $\mathcal{D}F$ vanishes. Multiplying the components by any $a \in R$ gives a primitive for ax^2, since both cochain differentials commute with the central scalar action on the output.

Conversely, reduce g modulo (x^2) to $\alpha + \beta x$. On the object X of (1.2), every odd matrix has the form

$$h = \begin{pmatrix} 0 & b \\ d & 0 \end{pmatrix}, \qquad \delta h = x(d + xb)\,\mathrm{id}_X\,.$$

The nullary equation forces $\alpha = 0$ and $d + xb = \beta$.

The odd endomorphism

$$\psi = \begin{pmatrix} 0 & 1 \\ -x & 0 \end{pmatrix}$$

is closed. The unary equation on ψ would give

$$\delta u(\psi) = -(\psi h + h\psi) = -(\beta - 2xb)\,\mathrm{id}_X\,.$$

Every matrix boundary on X is divisible by x. Evaluation at $x = 0$ forces $\beta = 0$. Higher arities do not enter these two equations, so they cannot remove the obstruction.

It remains to prove the objectwise assertion for arbitrary rank. Over the fraction field of R, the two odd blocks of Q_E are inverse up to x^3. The even and odd ranks therefore agree. Smith normal form over the discrete valuation ring R diagonalizes one block. The equation $Q_E^2 = x^3\,\mathrm{id}_E$ forces the other block to be diagonal in the complementary bases. The factorization is consequently a direct sum of blocks (x^a, x^{3-a}), where $0 \leq a \leq 3$.

For $a = 0, 3$ one block is invertible, so the object is contractible. For $a = 1$ the matrix

$$H = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \qquad \text{satisfies} \qquad \delta H = x\,\mathrm{id}\,.$$

Exchanging the two parity summands gives a primitive for $a = 2$. Direct sums and changes of basis preserve these homotopies. □

The scalar image in this absolute centre is therefore $k[[x]]/(x^2)$. This computes the scalar image, without asserting that it is the entire centre. The coefficient convention enters the proof at the unary map:

$$u(xf) = \tfrac{1}{3}f + xu(f).$$

It is k-linear and is not R-linear. A relative centre over R asks a different question, since it excludes this primitive.

1.4 Two scalar endpoints joined by a complex

The cubic example required an infinite bar of a category. There is also a finite model in which the entire relative centre, including its multiplication in a family, can be computed.

Let R be a commutative ring and M a bounded complex of finite projective R-modules. Use integer cohomological degrees and set

$$T(M) = Re_0 \oplus Re_1 \oplus M, \qquad e_1 M e_0 = M, \qquad M^2 = 0. \tag{1.11}$$

The idempotents are orthogonal, closed, and sum to the unit. All other idempotent actions on M vanish. The differential on M is its given differential. Thus M is the complex of arrows from object zero to object one, and the endpoint endomorphisms are scalar. All tensors in this section are over R unless another ring is displayed.

Put $T = T(M)$ and $T^e = T \otimes_R T^{\mathrm{op}}$, with graded opposite multiplication. The stated projectivity makes this a model of the derived enveloping algebra. Define

$$Z_R(T) = \mathrm{RHom}_{T^e}(T, T), \qquad E = \mathrm{End}_R(M), \qquad d_E F = d_M F - (-1)^{|F|} F d_M.$$

The difference of the two scalar actions is the chain map

$$\chi : R^2 \longrightarrow E, \qquad \chi(a, b) = (a - b)\,\mathrm{id}_M,$$

where a acts at the source and b at the target. Write $D(R)$ for the derived category of R-module complexes.

Theorem 1.4. *The relative centre is represented by the homotopy fibre*

$$Z_R(T(M)) \simeq \mathrm{Cone}(\chi)[-1]. \tag{1.12}$$

With the fibre sign convention, its terms and differential are

$$Z^n = (R^2)^n \oplus E^{n-1}, \qquad d_Z(a, b, F) = (da, db, \chi(a, b) - d_E F).$$

For every $P \in D(R)$, evaluation gives

$$\mathrm{RHom}_R(P, Z_R(T)) \simeq \mathrm{RHom}_{T^e}(P \otimes^{\mathbb{L}}_R T, T). \tag{1.13}$$

Proof. Let $S = Re_0 \oplus Re_1$. The normalized relative bimodule bar of T over S stops at length one, since $M \otimes_S M = 0$:

$$T \otimes_S M \otimes_S T \longrightarrow T \otimes_S T \longrightarrow T. \tag{1.14}$$

The first term has the extra degree -1 of the bar suspension. Choose the boundary on an arrow to be $e_1 \otimes m - m \otimes e_0$, with the corresponding suspended internal differential. Multiplication adds the two copies of M in the middle term. Their difference is its kernel. On the idempotents, multiplication is the identity. This proves exactness of the augmented sequence of complexes.

The terms are induced from the split idempotent bimodules of S, with bounded projective coefficients R or M. They are therefore homotopically projective as T-bimodules. Applying bimodule Hom into T yields R^2 at length zero and $\mathrm{End}_R(M)$ at length one. The two endpoint actions give $(a - b)\,\mathrm{id}_M$, and suspension gives $-d_E$. This is (1.12). Tensoring the same bar with a projective model of P and using tensor–Hom adjunction proves (1.13). □

Example 1.5 (Parallel arrows)**.** For $R = k$ and $M = k^m$ in degree zero, with $m > 0$, the centre complex is

$$k^2 \longrightarrow \mathrm{Mat}_m(k), \qquad (a, b) \longmapsto (a - b)\,\mathrm{id},$$

in degrees zero and one. Hence

$$\mathrm{HH}^0_k(T) = k, \qquad \mathrm{HH}^1_k(T) = \mathrm{Mat}_m(k)/k\,\mathrm{id}, \qquad \mathrm{HH}^j_k(T) = 0 \quad (j \neq 0, 1).$$

Insertion of one unary cochain into another is composition on M, so the degree-one Gerstenhaber bracket is the matrix commutator modulo scalars. The product of two positive-degree classes vanishes: a cup product would require two composable arrows, and there is no such word. Thus a single arrow has no positive centre cohomology, whereas three parallel arrows give eight independent degree-one classes.

1.5 A differential varying with a parameter

The joining complex can change while both endpoint algebras remain fixed. Put $R = k[t]$ and take

$$M_t = [Ru \xrightarrow{t} Rv], \qquad |u| = -1, \qquad |v| = 0.$$

Its endomorphism complex is

$$E^{-1} = R \xrightarrow{s \mapsto (ts, ts)} E^0 = R^2 \xrightarrow{(p,q) \mapsto t(p-q)} E^1 = R.$$

Here s sends v to su, the pair (p, q) acts on (u, v), and the last scalar sends u to a multiple of v. The centre in Theorem 1.4 is therefore

$$R^3 \xrightarrow{D_0} R^2 \xrightarrow{D_1} R, \qquad \begin{aligned} D_0(a, b, s) &= (a - b - ts, a - b - ts), \\ D_1(p, q) &= -t(p - q), \end{aligned} \tag{1.15}$$

in degrees zero, one, and two.

For a complex of free R-modules, its *Bockstein* on cohomology modulo t sends $[\bar{c}]$ to $[t^{-1}dc \bmod t]$, where c lifts the cocycle $\bar{c}$. This is the connecting map for the first-order thickening $k[t]/(t^2)$, with the submodule (t) identified with k.

Theorem 1.6. *The full relative Hochschild cohomology algebra is*

$$\mathrm{HH}^*_R(T(M_t)) \cong \frac{R[z, \omega]}{(z^2 - tz,\ t\omega,\ z\omega,\ \omega^2)}, \qquad |z| = 0, \qquad |\omega| = 2. \tag{1.16}$$

After inverting t, it is $R[t^{-1}] \times R[t^{-1}]$ in degree zero. The derived fibre $Z_R(T(M_t)) \otimes^{\mathbb{L}}_R k$ has degree-zero algebra $k[z]/(z^2)$, together with one class ϵ in degree one and one class ω in degree two. In that fibre,

$$z\epsilon = z\omega = \epsilon^2 = \epsilon\omega = \omega^2 = 0.$$

Its Bockstein is

$$\beta(\epsilon) = -\omega, \qquad \beta(z) = 0. \tag{1.17}$$

Proof. A degree-zero cycle obeys $a - b = ts$, so its free coordinates are (b, s). There are no incoming boundaries. The two cuts of a one-arrow word give the cup product

$$(a, b, s)(a', b', s') = (aa', bb', bs' + sa'). \tag{1.18}$$

The unit is $(1, 1, 0)$, and $z = (t, 0, 1)$ satisfies $z^2 = tz$. Every degree-zero cycle is uniquely $b\,1 + s\,z$.

Since t is a non-zero-divisor, $\ker D_1$ is the diagonal copy of R, which equals $\operatorname{im} D_0$. Thus $H^1 Z = 0$. The last cokernel is $R/(t)$, generated by the one-arrow cochain $\omega : u \mapsto v$. Products of two one-arrow cochains vanish. The left product $z\omega$ uses the target scalar zero, while the right product uses the source scalar t and vanishes in $R/(t)$. These are all relations in (1.16), by the computed cohomology groups.

Every term of (1.15) is free over R, so reduction modulo t computes the derived fibre. The reduced differential is

$$D_0(a, b, s) = (a - b, a - b), \qquad D_1 = 0.$$

The cohomology dimensions are $2, 1, 1$ in degrees $0, 1, 2$. The degree-one class is represented by $(1, 0) \in Z^1$. At $t = 0$, the representative of z has zero endpoint components, so its products with ϵ and ω contain two one-arrow cochains and vanish. Formula (1.18) also gives $z^2 = 0$.

The lift $(1, 0)$ of ϵ has differential $-t$. The lift $(t, 0, 1)$ of z is already closed. This proves (1.17). After inverting t, the roots $0, t$ of $z(z - t)$ separate and the torsion class disappears. □

The map $b\,1 + s\,z \mapsto (b + ts, b)$ identifies the degree-zero centre with

$$R \times_{R/(t)} R.$$

Its spectrum consists of two affine lines joined at the origin. The degree-two torsion has a further effect: derived specialization produces ϵ, which tensoring the cohomology groups alone with k would omit. At the special fibre, z itself is a nonzero central class with zero endpoint components. It is carried entirely by the homotopy on the joining complex.

1.6 Gluing endpoints with their own operations

For differential graded endpoint algebras, the derived centres act on the joining bimodule, and their difference gives the same compatibility problem. Work over the field k. Let A_0, A_1 be unital differential graded algebras and let M be an A_1–A_0 bimodule. Form the two-object category $\mathcal{E}_M$ with these endomorphism algebras, $\mathcal{E}_M(0, 1) = M$, and $\mathcal{E}_M(1, 0) = 0$. Use derived coefficient models throughout.

Tensoring an endomorphism of the diagonal A_0-bimodule with M gives its right characteristic action on M. The left action is defined similarly using A_1. Write χ for right action minus left action.

Proposition 1.7. *There is a fibre triangle of complexes*

$$\mathrm{CC}^\bullet_k(\mathcal{E}_M) \longrightarrow \mathrm{CC}^\bullet_k(A_0) \oplus \mathrm{CC}^\bullet_k(A_1) \xrightarrow{\chi} \mathrm{RHom}_{A_1 \otimes^{\mathbb{L}}_k A_0^{\mathrm{op}}}(M, M). \tag{1.19}$$

The cochains use product totalization over all bar lengths.

Proof. Normalize the two-sided bar relative to the object idempotents. A composable word stays in one corner or crosses once from zero to one. Put $\overline{A}_i = A_i/k1$.

The crossing cochains have factors

$$\mathrm{Hom}_k\Big((\overline{A}_1[1])^{\otimes p} \otimes M[1] \otimes (\overline{A}_0[1])^{\otimes q}, M\Big), \qquad p, q \geq 0.$$

The double bar resolving M as an A_1–A_0 bimodule is the direct sum of

$$A_1 \otimes (\overline{A}_1[1])^{\otimes p} \otimes M \otimes (\overline{A}_0[1])^{\otimes q} \otimes A_0.$$

Its differential combines the internal differentials, adjacent products, and the two actions on M. Unit insertion contracts the augmented bar, first on one side and then on the other. Its length filtration has induced bimodules with homotopically projective coefficient complexes as successive quotients. The surjective inverse-tower argument of Proposition 1.2 proves that the whole double bar is homotopically projective, including when the coefficient complexes are unbounded.

Hom from this resolution into M computes the right-hand term of (1.19). The crossing factor has one additional suspension, so the kernel of restriction to the corners is this complex shifted by -1. The remaining endpoint mergers give the right and left characteristic actions with opposite signs. The resulting short exact sequence splits on graded cochains and gives the stated fibre triangle. □

This is a triangle of complexes. The cup products and braces remain on the Hochschild cochains and their composable strings. The last term can be replaced by an ordinary endomorphism complex only when the chosen bimodule model computes derived Hom. Likewise, selecting fewer boundary objects computes the centre of that selected category. Comparison with all boundaries is a further assertion.

1.7 From coefficient actions to local bulk observables

The bar evaluation gives a precise universal property over the declared coefficient ring. In a mixed holomorphic–topological theory, operators vary on a curve and compose in an ordered topological direction. Their products may acquire terms supported on collision divisors. A comparison must preserve these operations and their coefficient modules as well as the ordinary composition of morphisms.

The physical source must be defined independently. Its fields, action, boundary conditions, and quantization determine a complex of bulk observables and their action on boundary insertions. Identifying that complex with a geometric centre requires a geometric category of boundary probes and its coefficient tensor. The evaluation map must be compatible with the supported operations. The comparison must also retain homotopies on morphisms and their compatibility with composition, as the cubic example requires.

The ordinary centres above supply complete algebraic models of that compatibility, with evaluation maps and explicit dependence on coefficients. Their identification with a native chiral centre remains a separate construction. A geometric issue is already visible before quantization. Coefficients with poles along a divisor and coefficients supported on it do not form one ordinary algebra of functions. Their respective modules and the operations between them must be specified.

Chapter 2

Local coefficients and ordered collisions

When insertion points collide, an operation can acquire a pole or a term supported on the collision. Both depend on the normal coordinate, but they obey different multiplication rules. A pole permits division by that coordinate. A supported term can be annihilated by it. Their relation is an attachment between coefficient modules. To place the comparison homotopies of Chapter 1 on ordered configurations, we first construct these modules and their derived maps.

2.1 Principal parts and their residue currents

Work over $\mathbb{C}$, with one algebraic normal coordinate z. Put

$$A = \mathbb{C}[z], \qquad L = \mathbb{C}[z, z^{-1}], \qquad E = L/A.$$

The quotient E is the module of principal parts at the origin. It has basis $t_n = z^{-n} \bmod A$, $n \geq 1$, over $\mathbb{C}$. Every element is a finite linear combination of these basis vectors.

Multiplication and differentiation give

$$z t_1 = 0, \qquad z t_n = t_{n-1} \quad (n > 1), \qquad \partial_z t_n = -n t_{n+1}. \tag{2.1}$$

Thus E retains derivatives of every normal order, even though each of its elements is killed by a power of z.

Let $D = A\langle \partial_z \rangle$ be the algebra with relation $\partial_z z - z \partial_z = 1$. A left D-module is an A-module with an operator ∇ satisfying $\nabla(am) = a'm + a\nabla m$. Such an operator is its connection in the coordinate z. The usual derivative makes

$$0 \longrightarrow A \longrightarrow L \longrightarrow E \longrightarrow 0 \tag{2.2}$$

an exact sequence of left D-modules.

Proposition 2.1. *The map $D/Dz \to E$ sending the class of 1 to t_1 is an isomorphism of left D-modules. There is no unital associative algebra extending the multiplication on L and containing a nonzero element T with $zT = 0$.*

Proof. Using $z\partial_z = \partial_z z - 1$, every element of D/Dz has a representative that is a polynomial in ∂_z. Its monomials act on t_1 by $\partial_z^r t_1 = (-1)^r r! t_{r+1}$. These images form a basis of E, so the map is both injective and surjective. In the proposed algebra, associativity and the inverse of z give $T = z^{-1}(zT) = 0$, a contradiction. □

The relation with an analytic contact term fixes the normalization. Write $z = x + iy$, orient the plane by $dx \wedge dy$, and set $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$.

A distribution acts on compactly supported smooth test functions. Its derivatives are defined by integration by parts. Let δ_0 be the distribution $\langle \delta_0, \phi \rangle = \phi(0)$. The locally integrable function $1/z$ defines a distribution P_1. For $n \geq 1$, define its higher principal values by

$$P_n = \frac{(-1)^{n-1}}{(n-1)!} \partial_z^{n-1} P_1.$$

They agree with z^{-n} off the origin.

Proposition 2.2. *With these conventions,*

$$\bar\partial P_n = \frac{\pi(-1)^{n-1}}{(n-1)!}\,\partial_z^{n-1}\delta_0\, d\bar z. \tag{2.3}$$

The assignment of the right-hand side to t_n identifies E with the left D-module spanned by these residue currents. The coefficient module is placed in cochain degree zero. Its currents have Dolbeault degree one when used in a Dolbeault complex.

Proof. For a test function ϕ, Stokes' formula on the complement of the disc of radius ε gives

$$-\lim_{\varepsilon\to 0}\int_{|z|>\varepsilon}\frac{\partial_{\bar z}\phi}{z}\,dx\,dy = \lim_{\varepsilon\to 0}\frac{1}{2i}\int_{|z|=\varepsilon}\frac{\phi(z)}{z}\,dz = \pi\phi(0).$$

The circle on the middle expression is counterclockwise. The outer boundary contributes zero because ϕ is compactly supported. This proves $\bar\partial P_1 = \pi\delta_0 d\bar z$. Distributional derivatives commute, giving (2.3) for every n.

The identity $z\partial_z^r\delta_0 = -r\partial_z^{r-1}\delta_0$ follows by differentiating $z\delta_0 = 0$. It gives precisely the multiplication rule in (2.1). Differentiation gives its connection rule. The derivatives of δ_0 are linearly independent: test them against z^r times a smooth function equal to one near the origin. Thus the assignment is an isomorphism onto the displayed span. □

The residue is therefore a map between specified coefficient objects. It supplies no multiplication of a principal value by an arbitrary supported distribution. To compare operations using these objects, we need resolutions that retain their connections.

2.2 Resolutions and derived coefficient maps

Complexes have cohomological grading, with $V[1]^j = V^{j+1}$. A complex of differential modules has a connection commuting with its differential.

Define a free A-complex Q_L by

$$Q_L^{-1} = \bigoplus_{n\ge 0} A\alpha_n, \qquad Q_L^0 = \bigoplus_{n\ge 0} Ae_n, \qquad d\alpha_n = e_n - ze_{n+1}. \tag{2.4}$$

Its augmentation sends e_n to z^{-n}. Give it the connection

$$\nabla e_n = -ne_{n+1}, \qquad \nabla\alpha_n = -(n+1)\alpha_{n+1}, \tag{2.5}$$

extended by the Leibniz rule. Put $Q_A = A$ in degree zero, and let $r : Q_A \to Q_L$ send 1 to e_0. Set $Q_E = \mathrm{Cone}(r)$, with the convention

$$\mathrm{Cone}(r)^j = Q_L^j \oplus Q_A^{j+1}, \qquad d(\zeta,\xi) = (d_{Q_L}\zeta + r\xi, -d_{Q_A}\xi). \tag{2.6}$$

Lemma 2.3. *The augmentations $Q_L \to L$ and $Q_E \to E$ are quasi-isomorphisms of differential modules. All terms of Q_A, Q_L, Q_E are free over A.*

Proof. If a finite sum $\sum a_n(e_n - ze_{n+1})$ vanishes, its coefficient of e_0 gives $a_0 = 0$. The successive coefficients give $a_n = 0$ for every n. Thus the differential in (2.4) is injective. Its cokernel imposes $e_n = ze_{n+1}$ and is the direct limit of $A \xrightarrow{z} A \xrightarrow{z} \cdots$, identified with L by $e_n \mapsto z^{-n}$.

The connection commutes with the differential because

$$\nabla(e_n - ze_{n+1}) = -(n+1)e_{n+1} + (n+1)ze_{n+2} = d\big(-(n+1)\alpha_{n+1}\big).$$

It also commutes with both augmentations and with r. The cone of the injection $A \to L$ has cohomology E in degree zero and no other cohomology. The cone in (2.6) consequently resolves E. Its terms are direct sums of the indicated free modules. □

These resolutions are free over A. To obtain derived maps of differential modules, we also resolve the derivative action. For any one of the three complexes Q, use the two Spencer columns

$$D \otimes_A Q \xrightarrow{s} D \otimes_A Q, \qquad s(P \otimes q) = P\partial_z \otimes q - P \otimes \nabla q, \tag{2.7}$$

in degrees -1, 0. In column degree p, the total differential is $s + (-1)^p d_Q$. Denote this total complex by P_Q. The Leibniz rule makes s well defined under A-balancing.

Proposition 2.4. *The complexes P_{Q_A}, P_{Q_L}, P_{Q_E} are bounded free D-resolutions of A, L, E. Their complexes of D-linear maps, with ordinary graded composition, form a differential graded category computing all derived maps among these three objects.*

For any complex N of left D-modules, put $C = \mathrm{Hom}_A(Q, N)$, using product totalization, and define $\mathcal{D}f = \nabla_N f - f\nabla_Q$. Then $\mathrm{RHom}_D(Q, N)$ is represented by the complex with degree-j term $C^j \oplus C^{j-1}$ and differential

$$d(a, b) = (d_C a, \mathcal{D}a - d_C b), \qquad d_C f = d_N f - (-1)^{|f|} f d_Q. \tag{2.8}$$

Proof. For a differential module M, augment the second column of (2.7) by $P \otimes m \mapsto Pm$. Write D in its right A-basis $1, \partial_z, \partial_z^2, \ldots$. The highest derivative of a nonzero finite sum acquires a nonzero term of one higher order under s, so s is injective. Its relations successively replace $\partial_z^r \otimes m$ by $1 \otimes \nabla^r m$. Every class in its cokernel therefore has a unique representative $1 \otimes m$, proving exactness.

Apply this resolution in each degree of Q. The double complex is bounded in both degrees, so the total augmentation $P_Q \to Q$ is a quasi-isomorphism. Its terms are free over D because the terms of Q are free over A. A bounded complex of projective modules computes derived Hom, including when its free terms have countably many generators.

The adjunction $\mathrm{Hom}_D(D \otimes_A Q, N) = \mathrm{Hom}_A(Q, N)$ identifies the map induced by s with $\mathcal{D}$. With the sign of the second summand chosen as in (2.8), the total Hom differential is the one displayed there. For every A-linear f, the map $\mathcal{D}f$ is again A-linear. The operator $\mathcal{D}$ is $\mathbb{C}$-linear, obeys $\mathcal{D}(af) = a'f + a\mathcal{D}f$ for $a \in A$, and commutes with d_C. Hence the displayed differential squares to zero.

The direct sums in Q give products in its Hom complexes. Using the free D-resolutions on both sides supplies associative composition and the graded Leibniz identity. □

The extra column in (2.8) records compatibility with normal differentiation. The underlying A-linear Hom complex alone cannot express that compatibility. A second distinction appears when two supported coefficients meet: their tensor product must also be derived.

2.3 Supported tensors and redundant equations

Write $D(A)$ for the derived category of A-modules. The symbol $\otimes^{\mathbb{L}}_A$ means the tensor computed after a flat resolution. All tensors in the next proposition are over A, rather than over D. Let $k_0 = A/(z)$ be the coherent closed fibre. It has no left D-action extending its A-action: since z acts by zero, the commutator $[\partial_z, z]$ could not act by one. The differential module supported at the origin is instead E, as in Proposition 2.1.

Proposition 2.5. *In $D(A)$ one has*

$$E \otimes_A E = 0, \qquad E \otimes^{\mathbb{L}}_A E \simeq E[1], \qquad k_0 \otimes^{\mathbb{L}}_A k_0 \simeq k_0 \otimes \Lambda(\epsilon), \quad |\epsilon| = -1. \tag{2.9}$$

The last expression has zero differential.

Proof. Every element of E is z-power torsion, whereas multiplication by z on E is surjective. For a simple tensor $e \otimes e'$, choose m with $z^m e = 0$ and write $e' = z^m f$. Then $e \otimes e' = z^m e \otimes f = 0$.

The flat resolution $[A \to L]$ of E is in degrees -1, 0. Since $L \otimes_A E = 0$, its tensor with E is $[E \to 0]$, which is $E[1]$.

For the closed fibre, tensor the free resolution $[A \xrightarrow{z} A]$ with k_0. Its differential becomes zero. The Koszul multiplication gives the exterior generator ϵ. □

Several collision equations introduce a further choice. For a commutative ring B and an element $f \in B$, let $B_f = B[f^{-1}]$ and define

$$C_f = [B \longrightarrow B_f]$$

in degrees 0, 1. For a finite list $\mathbf{f} = (f_1, \dots, f_s)$, its *stable support complex* is

$$C_{\mathbf{f}} = C_{f_1} \otimes_B \cdots \otimes_B C_{f_s}, \tag{2.10}$$

with the usual tensor differential. Its terms are flat. The projection $C_f \to B$ is the identity in degree zero and zero in degree one.

Lemma 2.6. *If g lies in the radical of the ideal $(f_1, \dots, f_s)$, then the projection*

$$C_{\mathbf{f}} \otimes_B C_g \longrightarrow C_{\mathbf{f}}$$

is a quasi-isomorphism.

Proof. The complex C_f is the filtered direct limit of $[B \xrightarrow{f^n} B]$, $n \geq 1$. The transition is the identity in degree zero and multiplication by f in degree one. Consequently $C_{\mathbf{f}}$ is the direct limit of $\bigotimes_i [B \xrightarrow{f_i^n} B]$. On the ith two-term factor, the identity from degree one to degree zero is a homotopy for multiplication by f_i^n. Its signed tensor with the identities on the other factors gives the same homotopy on the full tensor product. Filtered direct limits of modules are exact, so each cohomology class of $C_{\mathbf{f}}$ is annihilated by some common powers $f_1^n, \dots, f_s^n$.

Choose N with $g^N \in (f_1, \dots, f_s)$. The ideal $(f_1, \dots, f_s)^{s(n-1)+1}$ is contained in $(f_1^n, \dots, f_s^n)$, since each generating monomial contains one f_i at least n times. A power of g therefore annihilates the given class. It follows that $C_{\mathbf{f}} \otimes_B B_g$ is acyclic, since localization is exact. Tensoring the triangle $C_g \to B \to B_g$ with the flat complex $C_{\mathbf{f}}$ proves the assertion. □

The finite Koszul complex retains different information. Take $B = \mathbb{C}[u, v]$ and a third equation $w = u + v$. The finite complex $K(u, v, w)$ is the exterior algebra over B on degree--1 generators e_u, e_v, e_w, with differential

$$de_u = u, \qquad de_v = v, \qquad de_w = u + v.$$

The change of basis $\eta = e_w - e_u - e_v$ gives

$$K(u, v, u + v) = K(u, v) \otimes \Lambda(\eta), \qquad d\eta = 0, \tag{2.11}$$

and hence $H^* K(u, v, u + v) = \mathbb{C} \otimes \Lambda(\eta)$. Indeed, u, v is a regular sequence: multiplication by u is injective in B, and multiplication by v is injective in $B/(u)$. The two successive two-term resolutions prove that $K(u, v)$ resolves $B/(u, v) = \mathbb{C}$.

By contrast, $C_{u,v,u+v} \simeq C_{u,v}$ by Lemma 2.6. The latter has only degree-two cohomology,

$$H^2 C_{u,v} = E_2 = \frac{\mathbb{C}[u^{\pm 1}, v^{\pm 1}]}{\mathbb{C}[u^{\pm 1}, v] + \mathbb{C}[u, v^{\pm 1}]}, \tag{2.12}$$

with basis $u^{-m} v^{-n}$, $m, n \geq 1$. To see the concentration, write $C_{u,v}$ as the tensor over $\mathbb{C}$ of the one-variable complexes for u and v. Each has its principal-parts cohomology in degree one.

The vanishing of $(E_2)_{u+v}$ also has a direct calculation:

$$\frac{1}{u^m v^n} = \frac{1}{u+v}\Big(\frac{1}{u^{m-1} v^n} + \frac{1}{u^m v^{n-1}}\Big).$$

Repeated application decreases $m + n$ until every term is a power of $(u+v)^{-1}$ times a fraction regular in one normal coordinate. That fraction represents zero in E_2, so the term vanishes in $(E_2)_{u+v}$. The odd class η in (2.11) belongs to the finite closed-intersection complex. It does not supply an extra class in the stable support complex.

2.4 Differential modules on configuration spaces

The coordinate calculations extend over a smooth separated complex algebraic curve X. For a smooth scheme Y, let $\mathcal{D}_Y$ be its sheaf of differential operators. Let $\mathrm{DMod}(Y)$ denote the stable enhancement of the unbounded derived category of right $\mathcal{D}_Y$-modules with quasi-coherent underlying $\mathcal{O}_Y$-modules. The enhancement retains mapping complexes and coherent homotopies. We use the enhanced differential-module formalism of Francis–Gaitsgory [38], §1.4, including extraordinary inverse image, direct image and external product, with their composition coherences.

Right modules give a coordinate-independent form of our local left modules. If M is a left differential module, the associated right module is $\omega_Y \otimes_{\mathcal{O}_Y} M$, where ω_Y is the line of top forms. In one coordinate its action satisfies

$$(dz \otimes m)\,\partial_z = -dz \otimes \nabla m.$$

For a nonconstant form, the derivative also acts on its coefficient. This is the transposition that fixes functions, negates coordinate derivatives and reverses products. It preserves the normal derivatives in E.

Here are the functors and shifts used below. For a map $f : Z \to Y$ of smooth schemes, with dimensions d_Z, d_Y, the transfer modules are

$$\mathcal{D}_{Z \to Y} = \mathcal{O}_Z \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{D}_Y, \qquad \mathcal{D}_{Y \leftarrow Z} = \omega_Z \otimes_{\mathcal{O}_Z} \mathcal{D}_{Z \to Y} \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\omega_Y^{-1}.$$

The first has a left $\mathcal{D}_Z$-action and a right $f^{-1}\mathcal{D}_Y$-action. Transposition gives the second a left $f^{-1}\mathcal{D}_Y$-action and a right $\mathcal{D}_Z$-action.

For right-module complexes M on Z and N on Y, set

$$f_+ M = Rf_*^{\mathcal{O}}(M \otimes^{\mathbb{L}}_{\mathcal{D}_Z} \mathcal{D}_{Z \to Y}), \tag{2.13}$$

$$f^! N = (f^{-1} N \otimes^{\mathbb{L}}_{f^{-1}\mathcal{D}_Y} \mathcal{D}_{Y \leftarrow Z})[d_Z - d_Y]. \tag{2.14}$$

Here f^{-1} is inverse image of sheaves, and $Rf_*^{\mathcal{O}}$ is derived sheaf direct image. External product, denoted $\boxtimes$, places two differential modules on independent factors.

For the structure map $p_Y : Y \to \operatorname{Spec} \mathbb{C}$, our normalization is

$$J_Y = p_Y^! \mathbb{C} = \omega_Y[d_Y], \qquad J_Y \boxtimes J_Z = J_{Y \times Z}. \tag{2.15}$$

For an open immersion j, write j_* for j_+. Its functor $j^!$ is restriction. For a closed immersion i, write $i_* = i_+$. The functor i_* retains all normal derivatives along the closed subspace.

Differential modules on a possibly singular reduced closed subspace are defined by the corresponding supported modules in smooth embeddings, using Kashiwara's equivalence. Thus the open–closed recollement identities apply to unions of diagonals:

$$i_* i^! N \longrightarrow N \longrightarrow j_* j^! N, \qquad i^! j_* = 0, \qquad j^! j_* = \mathrm{id}\,. \tag{2.16}$$

These identities, and the coherent composition laws of the functors, are part of the specified coefficient formalism.

The cut product will first restrict to a disjoint locus and then apply j_*. Its compatibility with a collision rests on the following consequence of recollement.

Lemma 2.7. *For a Cartesian square with j an open immersion,*

$$\begin{array}{ccc} U' & \xrightarrow{f_U} & U \\ {\scriptstyle j'}\downarrow & & \downarrow{\scriptstyle j} \\ Y' & \xrightarrow[f]{} & Y, \end{array} \qquad f^! j_* M \simeq j'_* f_U^! M.$$

These equivalences are coherent under composition of f.

Proof. Let $i : Z \hookrightarrow Y$ and $i' : Z' \hookrightarrow Y'$ be the reduced closed complements. The map $f i'$ factors as ih. Transitivity and (2.16) give

$$i'^! f^! j_* M = h^! i^! j_* M = 0.$$

The recollement triangle on Y' therefore identifies $f^! j_* M$ with $j'_* j'^! f^! j_* M$. On the open set U', transitivity identifies its restriction with $f_U^! M$.

For a composite f, all such comparisons lie in the fully faithful image of j'_*. Their restrictions are the given coherent composition maps for extraordinary inverse image. Full faithfulness lifts those maps and all their higher homotopies uniquely. □

2.5 Ordered words and consecutive cuts

An ordered insertion retains its place even when its point coincides with another point. The word (x, y, x), with $x \neq y$, is an elementary example. Its two consecutive cuts are $(x) \mid (y, x)$ and $(x, y) \mid (x)$. Neither separates disjoint sets of points.

The intervening operation cannot be deleted. For the matrix units E_{ij} on $\mathbb{C}^2$, which send the jth basis vector to the ith,

$$E_{12} E_{21} E_{12} = E_{12}, \qquad E_{12}^2 = 0.$$

We accordingly retain words and allow adjacent positions to collapse without identifying a word with its unordered set of values.

Let $[n] = \{1, \dots, n\}$ with its usual order, and let $\mathrm{OrdSurj}_+$ be the category of these nonempty ordered sets and monotone surjections. A surjection $\alpha : [n] \twoheadrightarrow [m]$ defines

$$d_\alpha : X^m \longrightarrow X^n, \qquad d_\alpha(x_1, \dots, x_m)_i = x_{\alpha(i)}.$$

Write $D(\mathbb{C})$ for complexes of complex vector spaces up to quasi-isomorphism, with their stable enhancement. The category of *word coefficients* is

$$\mathcal{W}_X = D(\mathbb{C}) \times \varprojlim_{\mathrm{OrdSurj}_+} \mathrm{DMod}(X^n), \tag{2.17}$$

where the transition from n to m is $d_\alpha^!$. An object F consists of an empty coefficient F_0, coefficients F_n on X^n, equivalences $d_\alpha^! F_n \simeq F_m$, and their coherent compatibility under composition. The empty coefficient has no collision arrow to a nonempty one.

For $0 < p < n$, define the cut open and its inclusion by

$$U_{p,n} = \{(x_1, \dots, x_n) : x_i \neq x_j \text{ whenever } i \leq p < j\}, \qquad j_{p,n} : U_{p,n} \hookrightarrow X^n. \tag{2.18}$$

For $p = 0, n$, put $U_{p,n} = X^n$ and let $j_{p,n}$ be the identity. Define

$$(F \star_{\mathrm{cut}} G)_n = \bigoplus_{p=0}^{n} j_{p,n,*} j_{p,n}^! (F_p \boxtimes G_{n-p}). \tag{2.19}$$

An empty factor means scalar tensoring. In particular, $(F \star_{\mathrm{cut}} G)_0 = F_0 \otimes_{\mathbb{C}} G_0$.

Theorem 2.8. *Formula* (2.19) *defines an associative monoidal product on $\mathcal{W}_X$. Its unit K has $K_0 = \mathbb{C}$ and $K_n = 0$ for $n > 0$. Its associativity maps are the identifications of the iterated products with the sums over consecutive partitions whose blocks have pairwise disjoint values in X.*

Proof. First pull a summand of (2.19) back along d_α. If its cut lies inside a fibre of α, two positions on opposite sides of the cut are forced to have the same value. Its inverse-image open is empty, and Lemma 2.7 makes the summand zero. Otherwise the cut lies between fibres, including either end, hence determines a unique cut q of $[m]$. Its inverse-image open is $U_{q,m}$. External-product compatibility, the word equivalences of F, G, and the same lemma identify the pulled-back summand with the qth summand of $(F \star_{\mathrm{cut}} G)_m$.

The coherence in the lemma supplies the coherence for composite surjections, so the product is a word coefficient system.

For three factors, either parenthesization is a direct sum over $0 \leq p \leq q \leq n$. Each summand starts with the external product on the three consecutive blocks. Restrict it to the locus where values in different blocks are distinct, then apply open direct image. Successive cut conditions impose precisely this common open. Base change and composition of open direct images identify the two expressions. Empty blocks impose no condition.

The same description holds for any number of factors. All associativity diagrams, including the pentagon, restrict to the identity of this common expression. The functorial coherences therefore make them commute. With K, only the empty-block summand survives on the appropriate side, giving the unit maps and their compatibility. □

No permutation of the blocks has been introduced. In particular, this construction supplies an associative product without a chosen symmetry. The empty unit is distinct from a local unit field. A local vacuum in a boundary algebra is extra data on nonempty configurations, whereas $K_1 = 0$.

Example 2.9. At three positions,

$$U_{1,3} = X^3 \setminus (\Delta_{12} \cup \Delta_{13}), \qquad U_{2,3} = X^3 \setminus (\Delta_{13} \cup \Delta_{23}),$$

where Δ_{ij} is the locus $x_i = x_j$. Their union has complement exactly Δ_{13}. If $x_1 \neq x_3$ and the first cut fails, then $x_1 = x_2$, so the second cut succeeds. Conversely, $x_1 = x_3$ defeats both cuts. Thus every nonadjacent equality crossing a cut is required already in the first three-position calculation.

On an affine coordinate chart, $u = z_1 - z_2$ and $v = z_2 - z_3$ give the outer equation $u + v = 0$. The stable and finite complexes of Section 2.3 compute different coefficient constructions on this same arrangement.

Call F *reduced* if $F_0 = 0$. For reduced objects $F^{(1)}, \dots, F^{(r)}$, every nonzero summand of their iterated cut product at n has r nonempty consecutive blocks. Hence

$$(F^{(1)} \star_{\mathrm{cut}} \cdots \star_{\mathrm{cut}} F^{(r)})_n = 0 \qquad (r > n). \tag{2.20}$$

This bound concerns word length. It makes tensor-length constructions finite at a fixed configuration size. It does not bound the order of a pole or the normal derivatives in a supported coefficient.

2.6 Comparing attachments

The localization sequence (2.2) ties regular, meromorphic and supported coefficients together. A comparison must respect this attachment in addition to the separate objects. Let $r : U \to Z$ and $r' : U' \to Z'$ be cochain maps of complexes of left D-modules, and take D-linear Hom complexes. For derived maps among A, L, E, use the bounded free resolutions of Proposition 2.4.

Give each cone the two-step filtration $0 \subset Z \subset \mathrm{Cone}(r)$, with the convention of (2.6). A degree-zero map preserving these filtrations has the form

$$(\zeta, \xi) \longmapsto (g\zeta + H\xi, f\xi), \qquad |f| = |g| = 0, \quad |H| = -1. \tag{2.21}$$

Here $f : U \to U'$, $g : Z \to Z'$ and $H : U \to Z'$.

Lemma 2.10. *The map* (2.21) *is closed if and only if* f, g *are cochain maps and*

$$d_{\mathrm{Hom}} H = gr - r'f. \tag{2.22}$$

The complex of filtration-preserving cone maps is isomorphic, with the shifted-coordinate signs, to the homotopy fibre of

$$\mathrm{Hom}(U, U') \oplus \mathrm{Hom}(Z, Z') \longrightarrow \mathrm{Hom}(U, Z'), \qquad (f, g) \longmapsto gr - r'f.$$

For closed degree-zero maps, composition is

$$(f_2, g_2, H_2)(f_1, g_1, H_1) = (f_2 f_1, g_2 g_1, g_2 H_1 + H_2 f_1). \tag{2.23}$$

Proof. Insert (2.21) into the cone differential $(\zeta, \xi) \mapsto (d_Z \zeta + r\xi, -d_U \xi)$. The two diagonal entries in the chain-map equation say $df = dg = 0$. The off-diagonal entry says $d_{Z'} H + H d_U = gr - r'f$, which is (2.22).

In general degree j, put a factor $(-1)^j$ on the $U[1] \to U'[1]$ entry and a factor $(-1)^j$ on the $U[1] \to Z'$ entry. In these coordinates the differential is

$$(f, g, H) \longmapsto (df, dg, gr - r'f - dH).$$

This is the homotopy-fibre differential, since $|H| = j - 1$. Finally, multiplication of the degree-zero upper triangular block matrices gives (2.23) and its associativity. □

Equations (2.8) and (2.22) retain two kinds of compatibility: with differentiation and with the residue attachment. The cut product adds the geometry of ordered insertion. Together they specify coefficient objects and comparison complexes for local operations.

An action on them must still give operations in every arity and homotopies compatible with substitution and collision restriction. The construction of a field theory must, independently, supply its fields, differential, interaction and quantum observables. It must also give a map from those observables to the geometric actions. The next calculation begins this theory-dependent part with finite Hamiltonian jets.

Chapter 3

Finite Hamiltonian jets and relative quantization

A Hamiltonian determines an infinitesimal motion of a symplectic space. To study a boundary fixed at a chosen point, we keep the Hamiltonians whose vector fields vanish there. Their finite jets give Lie algebras with an explicit quantization. The resulting associative algebras have relative Hochschild complexes, so they also give finite models of the coherent actions considered in Chapter 1.

The lowest nonzero jet is already instructive. Quadratic Hamiltonians form $\mathfrak{sl}_2$; cubic Hamiltonians supply its four-dimensional representation; the bracket of two cubics first survives at quartic order. We shall construct these algebras, their Poisson cochains and their finite Batalin–Vilkovisky actions before comparing the quantized algebra with its Hochschild cochains.

3.1 Hamiltonians fixing a point

Work over $\mathbb{C}$ on the formal plane with coordinates z_1, z_2 and symplectic form $\omega = dz_1 \wedge dz_2$. Set $\mathfrak{m} = (z_1, z_2) \subset \mathbb{C}[\![z_1, z_2]\!]$. Our Poisson convention is

$$\{f, g\} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g. \tag{3.1}$$

Thus the vector field $X_f(g) = \{f, g\}$ vanishes at the origin exactly when the linear part of f vanishes. Constants give the zero vector field. Choosing the representative with zero constant term identifies the pointed Hamiltonian Lie algebra with $\mathfrak{h} = \mathfrak{m}^2$.

For the quadratic polynomials

$$e = \tfrac{1}{2} z_1^2, \qquad f = -\tfrac{1}{2} z_2^2, \qquad h = -z_1 z_2,$$

direct differentiation gives

$$\{h, e\} = 2e, \qquad \{h, f\} = -2f, \qquad \{e, f\} = h. \tag{3.2}$$

They are the Chevalley generators of $\mathfrak{sl}_2$.

Write R_r for the homogeneous polynomials of degree r. Each derivative lowers degree by one, so $\{R_r, R_s\} \subset R_{r+s-2}$. The filtration

$$F^p\mathfrak{h} = \mathfrak{m}^{p+2} \quad (p \geq 0)$$

is therefore additive: $\{F^p\mathfrak{h}, F^q\mathfrak{h}\} \subset F^{p+q}\mathfrak{h}$. It is separated and complete in the formal power-series topology. For $d \geq 2$, define

$$\mathfrak{h}_{\leq d} = \mathfrak{h}/\mathfrak{m}^{d+1} = \bigoplus_{r=2}^{d} R_r. \tag{3.3}$$

The equality on the right specifies the polynomial representatives; their bracket is projected to degrees at most d.

Proposition 3.1 (Finite Hamiltonian jets)**.** *The bracket makes $\mathfrak{h}_{\le d}$ a Lie algebra of dimension*

$$\dim \mathfrak{h}_{\le d} = \frac{d(d+3)}{2} - 2.$$

Its quadratic subalgebra is $\mathfrak{sl}_2$, and R_r is its irreducible representation of highest weight r.

Proof. The ideal $\mathfrak{m}^{d+1}$ is stable under bracketing with $\mathfrak{m}^2$. Hence Jacobi descends from the Poisson bracket. In coordinates, this bracket is induced by the constant bivector $\partial_{z_1} \wedge \partial_{z_2}$; commuting partial derivatives cancel the Jacobiator. Its monomial formula is

$$\{z_1^a z_2^b, z_1^c z_2^s\} = (as - bc) z_1^{a+c-1} z_2^{b+s-1}, \tag{3.4}$$

where a term with zero coefficient is zero, including when a displayed exponent would be negative. For three exponent vectors α, β, γ and $\epsilon = (1, 1)$, the Jacobi coefficient is

$$\det(\alpha, \beta) \det(\alpha + \beta - \epsilon, \gamma) + \text{cyclic permutations} = 0.$$

The terms quadratic in the pairwise determinants cancel. The remaining terms vanish by the two-dimensional identity

$$\det(\alpha, \beta)\gamma + \det(\beta, \gamma)\alpha + \det(\gamma, \alpha)\beta = 0.$$

The dimension is $\sum_{r=2}^{d}(r+1) = d(d+3)/2 - 2$.

Put $u_j = z_1^{r-j} z_2^j$ for $0 \le j \le r$. The quadratic action is

$$\{h, u_j\} = (r - 2j)u_j, \qquad \{e, u_j\} = j u_{j-1}, \qquad \{f, u_j\} = (r - j)u_{j+1}. \tag{3.5}$$

Terms outside the index range are zero. A nonzero invariant subspace contains a weight vector because the eigenvalues of h are distinct. Repeated application of e reaches u_0, and repeated application of f then reaches every u_j. This proves irreducibility. □

The pointed condition is needed for these quotients. The affine algebra

$$\mathfrak{h}^{\mathrm{aff}} = \mathbb{C}[\![z_1, z_2]\!]/\mathbb{C}$$

also contains linear Hamiltonians, which generate translations. In it,

$$\{z_1, z_2^{d+1}\} = (d+1) z_2^d.$$

Consequently $\mathfrak{m}^{d+1}$ is no longer an ideal. A moving boundary or a centre-of-mass coordinate requires an affine construction in addition to the pointed jet quotient.

3.2 The first nonlinear brackets

For $d = 2$ the quotient is $\mathfrak{sl}_2$ itself. For $d = 3$, set $v_j = z_1^{3-j} z_2^j$ for $0 \le j \le 3$. Equation (3.5) gives the quadratic action on their span.

Proposition 3.2 (The cubic jet)**.** *The degree-three quotient is the semidirect product*

$$\mathfrak{h}_{\le 3} = \mathfrak{sl}_2 \ltimes R_3, \qquad [R_3, R_3] = 0.$$

Proof. A bracket of two cubics has degree four and vanishes in this quotient. The brackets with quadratics are exactly the action in (3.5). □

At $d = 4$, put $w_k = z_1^{4-k} z_2^k$ for $0 \le k \le 4$. The cubic bracket becomes

$$\{v_i, v_j\} = 3(j - i) w_{i+j-1}. \tag{3.6}$$

When the index is outside $0, \dots, 4$, the coefficient is zero. In particular,

$$\{v_0, v_3\} = 9w_2, \qquad \{v_1, v_2\} = 3w_2.$$

The first equality differentiates z_1^3 and z_2^3. The second is $4z_1^2 z_2^2 - z_1^2 z_2^2 = 3z_1^2 z_2^2$. The two cubic pairs reach the same quartic monomial with coefficients in the ratio $3 : 1$.

Proposition 3.3 (The quartic jet)**.** *On the vector space $R_2 \oplus R_3 \oplus R_4$, all brackets are determined by* (3.2)*,* (3.5) *for $r = 3, 4$,* (3.6)*, and*

$$[R_3, R_4] = 0, \qquad [R_4, R_4] = 0.$$

This Lie algebra is $\mathfrak{h}_{\le 4}$. Its quadratic and cubic elements generate it; its ideal $R_3 \oplus R_4$ is two-step nilpotent with commutator subalgebra R_4.

Proof. The omitted brackets have degrees five and six, so they vanish in the quotient. Every quartic is a cubic bracket: use the pairs $(i, j) = (0, 1), (0, 2), (0, 3), (1, 3), (2, 3)$ to obtain w_0, w_1, w_2, w_3, w_4, respectively, with nonzero coefficients. These formulas exhaust the brackets of a basis. Jacobi follows from Proposition 3.1. They also give the stated commutator and its centrality inside the ideal. □

The adjoint trace will measure the divergence of the finite cotangent action. The homogeneous decomposition already determines this trace.

Proposition 3.4 (Unimodularity)**.** *For every $d \ge 2$ and $x \in \mathfrak{h}_{\le d}$, $\operatorname{Tr}(\operatorname{ad}_x) = 0$.*

Proof. A homogeneous Hamiltonian of degree greater than two strictly raises polynomial degree. Its adjoint matrix is strictly triangular in the decomposition (3.3). For a quadratic Hamiltonian, use the basis e, f, h. On each R_r, the matrices of e and f have zero diagonal, while the trace of h is $\sum_{j=0}^{r} (r - 2j) = 0$. Linearity proves the assertion. □

3.3 Chevalley cochains of the linear Poisson structure

Fix a finite-dimensional complex Lie algebra $\mathfrak{l}$, with basis e_a and bracket $[e_a, e_b] = f^c{}_{ab} e_c$; repeated indices are summed. We shall apply the constructions to $\mathfrak{l} = \mathfrak{h}_{\le d}$. Use cohomological grading, with $(W[k])^n = W^{n+k}$. In particular, a vector of degree zero has degree one in $W[-1]$.

Let $\mathfrak{l}^\vee$ carry the coadjoint action $(\operatorname{ad}_x^* \lambda)(y) = -\lambda([x, y])$. Define the graded Lie algebra

$$\mathfrak{q}(\mathfrak{l}) = \mathfrak{l} \ltimes \mathfrak{l}^\vee[-1]$$

with zero internal differential and abelian second summand. Its elements in the second summand have degree one. For a finite graded Lie algebra $\mathfrak{q}$, we use polynomial Chevalley cochains $C^*_{\mathrm{CE}}(\mathfrak{q}) = \operatorname{Sym}(\mathfrak{q}^\vee[-1])$ with differential dual to the bracket. Here Sym means the graded symmetric algebra, without completion in its even generators.

Proposition 3.5 (Chevalley–Poisson complex)**.** *There is a natural isomorphism of commutative differential graded algebras*

$$C^*_{\mathrm{CE}}(\mathfrak{q}(\mathfrak{l})) \cong C^*(\mathfrak{l}, \operatorname{Sym} \mathfrak{l}) = \operatorname{Sym} \mathfrak{l} \otimes \Lambda^* \mathfrak{l}^\vee.$$

On the right the differential is the Chevalley differential for the adjoint action. It is also the linear Poisson differential on polynomial polyvector fields on $\mathfrak{l}^*$.

Proof. Dualizing and shifting the degree-one summand gives $(\mathfrak{l}^\vee[-1])^\vee[-1] = \mathfrak{l}$ in degree zero. The degree-zero summand gives exterior generators c^a of degree one. Let $x_a \in \operatorname{Sym}\mathfrak{l}$ denote the linear function $x_a(\lambda) = \lambda(e_a)$. Dualizing the semidirect bracket gives

$$d_{\mathrm{CE}} c^c = -\tfrac{1}{2} f^c{}_{ab} c^a c^b, \qquad d_{\mathrm{CE}} x_a = f^c{}_{ba} c^b x_c. \tag{3.7}$$

These are precisely the bracket and action terms of the Chevalley differential. On a polynomial F the second formula reads

$$d_{\mathrm{CE}} F = c^a \{x_a, F\}_{\mathrm{lin}}, \qquad \{x_a, x_b\}_{\mathrm{lin}} = f^c{}_{ab} x_c.$$

Identify c^a with the constant vector field ∂_{x_a}. Use the Schouten bracket extending $[X, F] = X(F)$ by graded skew-symmetry and the graded derivation rule. For the bivector $\pi = \tfrac{1}{2} f^c{}_{ab} x_c \partial_{x_a} \wedge \partial_{x_b}$, the Schouten differential $[\pi, -]$ gives both formulas in (3.7). Since both operators are derivations of the exterior product, they agree on all polyvectors. Jacobi is $[\pi, \pi] = 0$, and hence $d_{\mathrm{CE}}^2 = 0$. □

3.4 The finite BV action

The preceding complex has degree-zero polynomial coordinates x_a. The BV construction uses the different graded space $T^*[-1](\mathfrak{l}[1])$. Its polynomial functions are

$$\mathcal{B}(\mathfrak{l}) = \Lambda(c^1, \dots, c^m) \otimes \mathbb{C}[b_1, \dots, b_m], \qquad |c^a| = 1, \quad |b_a| = -2,$$

where $m = \dim \mathfrak{l}$. Thus the cotangent coordinates b_a are antifields of degree -2. Fix left derivatives in the odd variables and set

$$\Delta = \sum_a \partial_{b_a} \partial_{c^a}, \qquad \{F, G\}_{\mathrm{BV}} = (-1)^{|F|} \big(\Delta(FG) - \Delta(F) G - (-1)^{|F|} F \Delta(G)\big). \tag{3.8}$$

The bracket has degree one, with $\{b_a, c^b\}_{\mathrm{BV}} = \delta_a^b$ and $\{c^b, b_a\}_{\mathrm{BV}} = -\delta_a^b$. Commutation of the even derivatives and anticommutation of the odd ones give $\Delta^2 = 0$. These Darboux coordinates specify a constant Berezinian up to a nonzero scalar; its normalization does not change Δ. The quantum master equation for this measure is proved in Theorem 3.6.

With these conventions the degree-zero cotangent action is

$$S = -\tfrac{1}{2} b_c f^c{}_{ab} c^a c^b. \tag{3.9}$$

In particular, $\{S, c^a\}_{\mathrm{BV}} = d_{\mathrm{CE}} c^a$.

Theorem 3.6 (Finite quantum master equation)**.** *For every finite-dimensional Lie algebra* $\mathfrak{l}$*, the action* (3.9) *satisfies*

$$\tfrac{1}{2}\{S, S\}_{\mathrm{BV}} = 0, \qquad \Delta S = \sum_b \operatorname{Tr}(\operatorname{ad}_{e_b}) c^b.$$

Consequently, for $\mathfrak{l} = \mathfrak{h}_{\le d}$ *and a formal parameter* $\hbar$ *of degree zero,*

$$\tfrac{1}{2}\{S, S\}_{\mathrm{BV}} + \hbar \Delta S = 0.$$

This identity holds in $\mathcal{B}(\mathfrak{h}_{\le d})[\![\hbar]\!]$.

Proof. Put $Q^a = -\tfrac{1}{2} f^a{}_{bc} c^b c^c$, so that $S = b_a Q^a$. The coordinate bracket gives

$$\tfrac{1}{2}\{S, S\}_{\mathrm{BV}} = \sum_{a,k} b_k Q^a \partial_{c^a} Q^k.$$

The coefficient of b_k is $d_{\mathrm{CE}}^2 c^k$, which vanishes by Jacobi. For the second identity, left differentiation gives

$$\Delta S = -\sum_{a,b} f^a{}_{ab} c^b = \sum_{a,b} f^a{}_{ba} c^b.$$

The last coefficient is the trace of ad_{e_b}. Proposition 3.4 makes it zero for every finite Hamiltonian jet. □

The trace condition is substantive. In the two-dimensional Lie algebra with basis u, v and $[u, v] = v$, one has

$$S = -b_v c^u c^v, \qquad \Delta S = c^u.$$

The classical master equation still holds, but this constant-measure action fails the quantum master equation. For Hamiltonian jets, unimodularity removes this finite-dimensional divergence. A local field theory has further analytic and renormalization data; the finite identity alone does not construct its quantum observables.

3.5 Rees multiplication and interval operations

Quantization replaces commuting linear functions on $\mathfrak{l}^*$ by generators whose commutator is $\hbar$ times the Lie bracket. For $N \geq 0$, let

$$R_N = \mathbb{C}[\hbar]/(\hbar^{N+1}), \qquad \mathfrak{l}_N = R_N \otimes_{\mathbb{C}} \mathfrak{l}, \qquad [x, y]_N = \hbar[x, y].$$

Here N is the coefficient order, independent of the jet order d or a Chan–Paton rank. Define the augmented R_N-algebra

$$A_N = U_{R_N}(\mathfrak{l}_N) = T_{R_N}(\mathfrak{l}_N)/(xy - yx - \hbar[x, y]),$$

whose augmentation sends every $x \in \mathfrak{l}$ to zero. The Poincaré–Birkhoff–Witt (PBW) theorem has the following form over this coefficient ring.

Proposition 3.7 (PBW over the finite coefficient ring)**.** *Choose an ordered basis of* $\mathfrak{l}$*. Its ordered monomials form an* R_N*-basis of* A_N*. If* $S_0 = \mathrm{Sym}_{\mathbb{C}} \mathfrak{l}$*, symmetrization is a filtered* R_N*-module isomorphism*

$$\mathrm{sym}_N : S_0 \otimes R_N \longrightarrow A_N, \qquad x_1 \cdots x_r \longmapsto \frac{1}{r!} \sum_{\sigma \in S_r} x_{\sigma(1)} \cdots x_{\sigma(r)}.$$

It intertwines the adjoint action through

$$[x, \mathrm{sym}_N(F)]_{A_N} = \hbar\, \mathrm{sym}_N(\mathrm{ad}_x F). \tag{3.10}$$

Proof. Replace every inverted adjacent pair yx by $xy + \hbar[y, x]$. In the first term the number of inversions decreases; the second has shorter length. Induction on length and then inversions proves termination. Disjoint replacements commute. For an overlapping triple $z > y > x$, the difference between the two reductions, after the shorter words have been ordered, is $\hbar^2$ times

$$[[z, y], x] + [[y, x], z] + [[x, z], y] = 0.$$

Induction therefore makes the ordered result independent of every choice of replacement. This defines a normal-form map killing the relations and equal to the identity on ordered monomials. Those monomials are consequently a basis, even though R_N has nilpotents.

Give generators PBW filtration degree one. Symmetrization induces the identity on the associated graded polynomial algebra. Subtracting the highest-degree symbol and descending in degree constructs its inverse. This uses only finite degrees and the invertibility of factorials in R_N. Finally, commuting x past each factor of a symmetrized word gives one term $\hbar[x, x_i]$ in each position. Their sum is (3.10). □

Transporting multiplication by sym_N gives the associative PBW product on $S_0 \otimes R_N$. Its reduction modulo $\hbar$ is the ordinary product. For $N \geq 1$, the first commutator is the linear Poisson bracket. Indeed, the coefficient of $\hbar$ in a commutator is a biderivation of the commutative reduction; its value on $x, y \in \mathfrak{l}$ is $[x, y]$. These values determine the biderivation on the polynomial algebra.

For disjoint intervals $I_1 < \cdots < I_r$ contained in an interval J, assign the operation

$$A_N^{\otimes_{R_N} r} \longrightarrow A_N, \qquad a_1 \otimes \cdots \otimes a_r \longmapsto a_1 \cdots a_r.$$

Use the unit for an empty collection and the identity for an inclusion of one interval. Associativity makes these operations compatible with nested collections, and their independence of interval lengths makes them locally constant. They supply the algebraic ordered interval operations expected of a boundary quantization. For a cotangent field theory on a half-plane, fix the cotangent field at the boundary and leave the Lie algebra field free. Identifying the resulting boundary observables with these interval operations requires a map from that theory's quantum observables, compatible with interval inclusions and fusion, and a proof that the map is a quasi-isomorphism. PBW constructs the target algebra, while this realization remains a separate comparison.

3.6 The relative Hochschild comparison

The coefficient ring affects the action complex. For the quadratic jet, $[h, e] = 2e$, whereas $[h, e]_{A_N} = 2\hbar e$. Modulo $\hbar$, the Hochschild differential of the zero-cochain e vanishes because the algebra is commutative. The ordinary Chevalley differential has $(d_{\mathrm{CE}} e)(h) = 2e$. Thus it is the scaled differential $\hbar d_{\mathrm{CE}}$ that can compare with relative Hochschild cochains.

Write $\bar{A}_N$ for the augmentation ideal. Use normalized relative Hochschild cochains

$$C^n_{\mathrm{HH}}(A_N/R_N, A_N) = \mathrm{Hom}_{R_N}(\bar{A}_N^{\otimes_{R_N} n}, A_N)$$

with differential

$$\begin{aligned}(bF)(a_1, \ldots, a_{n+1}) = {} & a_1 F(a_2, \ldots, a_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i F(a_1, \ldots, a_i a_{i+1}, \ldots, a_{n+1}) \\ & + (-1)^{n+1} F(a_1, \ldots, a_n) a_{n+1}.\end{aligned}$$

Let d_{CE} act on $C^n(\mathfrak{l}, S_0) = \mathrm{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{l}, S_0)$ by the adjoint Chevalley differential.

Theorem 3.8 (Relative Rees comparison)**.** *Antisymmetrization defines a quasi-isomorphism of R_N-complexes*

$$\mathrm{Alt}_N : C^\bullet_{\mathrm{HH}}(A_N/R_N, A_N) \longrightarrow \big(C^\bullet(\mathfrak{l}, S_0) \otimes R_N, \hbar d_{\mathrm{CE}}\big),$$

given by

$$(\mathrm{Alt}_N F)(x_1, \ldots, x_n) = \mathrm{sym}_N^{-1} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) F(x_{\sigma(1)}, \ldots, x_{\sigma(n)}).$$

These maps commute with coefficient reduction in N.

Proof. Write $A = A_N$, $R = R_N$, and $A^e = A \otimes_R A^{\mathrm{op}}$. In homological degree n, set

$$K_n = A \otimes_R \Lambda^n_R \mathfrak{l}_N \otimes_R A.$$

Its augmentation is multiplication. For $X = x_1 \wedge \cdots \wedge x_n$, let $X_{\widehat{i}}$ and $X_{\widehat{ij}}$ omit the indicated entries without reordering. Define

$$\partial(a \otimes X \otimes b) = \sum_i (-1)^{i+1}\big(ax_i \otimes X_{\widehat{i}} \otimes b - a \otimes X_{\widehat{i}} \otimes x_i b\big) + \hbar \sum_{i<j} (-1)^{i+j} a \otimes [x_i, x_j] \wedge X_{\widehat{ij}} \otimes b.$$

The normalized relative bimodule bar has terms $\mathcal{B}_n = A \otimes_R \bar{A}^{\otimes_R n} \otimes_R A$. Define

$$\iota_n(a \otimes X \otimes b) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a[x_{\sigma(1)}| \cdots |x_{\sigma(n)}]b.$$

The outer bar faces give the first sum in ∂. Pair adjacent transpositions in the internal faces and use $xy - yx = \hbar[x, y]$ to obtain the second sum. For two inputs this reads

$$b\big(1[x|y]1 - 1[y|x]1\big) = x[y]1 - y[x]1 + 1[x]y - 1[y]x - \hbar\, 1[[x, y]]1.$$

Thus $b\iota = \iota\partial$. PBW makes $\mathfrak{l}_N$ a free summand of $\bar{A}$. Since every factorial is invertible in R, ι is injective. The identity $b^2 = 0$ then gives $\partial^2 = 0$.

Give K_n filtration degree equal to its outer PBW degrees plus n. The associated graded is the diagonal Koszul complex of $\operatorname{Sym}_R(\mathfrak{l}_N)$. Its differential uses differences of left and right polynomial generators. A linear change of variables makes these independent polynomial variables. They form a regular sequence even when R has nilpotents. For an augmented cycle, lift a primitive of its highest-degree symbol and subtract its differential. The filtration degree decreases, so this procedure terminates. Hence $K_\bullet \to A$ is a free bimodule resolution.

The relative bar is also free over A^e, with its usual unit-insertion contraction. The map ι lifts the identity of A. To construct a reverse chain map, first lift the augmentation in degree zero. At each higher degree its required boundary is a cycle; exactness of $K_\bullet$ and projectivity of the bar term give a lift. Each composite and the identity induce the same augmentation. Their difference is therefore a boundary in degree zero, and the same lifting argument, degree by degree, constructs a chain homotopy. Applying $\operatorname{Hom}_{A^e}(-, A)$ gives a cochain homotopy equivalence.

The induced differential on $\operatorname{Hom}_R(\Lambda^n_R \mathfrak{l}_N, A)$ is

$$(\delta_N \phi)(x_1, \ldots, x_{n+1}) = \sum_i (-1)^{i+1}[x_i, \phi(x_1, \ldots, \widehat{x_i}, \ldots, x_{n+1})]_A + \hbar \sum_{i<j} (-1)^{i+j} \phi([x_i, x_j], x_1, \ldots, \widehat{x_i}, \ldots, \widehat{x_j}, \ldots, x_{n+1}).$$

By (3.10), the commutator term also carries one factor of $\hbar$. Both sums therefore become $\hbar$ times the ordinary Chevalley differential. Precomposition with ι gives Alt_N. All formulas commute with coefficient reduction. □

3.7 The formal coefficient limit

Put $R = \mathbb{C}[\![\hbar]\!]$ and $A = \varprojlim_N A_N$. PBW identifies its underlying module with $S_0[\![\hbar]\!]$. Each coefficient of $\hbar$ is a polynomial. We also write $U_\hbar(\mathfrak{l}) = A$ and obtain the continuous module isomorphism

$$\mathrm{sym}_\hbar = \varprojlim_N \mathrm{sym}_N : S_0[\![\hbar]\!] \xrightarrow{\sim} U_\hbar(\mathfrak{l}).$$

Define

$$A^{\widehat{\otimes}_R n} = \varprojlim_N A_N^{\otimes_{R_N} n}, \qquad C^\bullet_{\mathrm{HH,cont}}(A/R, A) = \varprojlim_N C^\bullet_{\mathrm{HH}}(A_N/R_N, A_N).$$

Equivalently, these are continuous R-multilinear normalized cochains on the completed relative tensors.

Corollary 3.9 (Formal relative Rees comparison)**.** *The compatible antisymmetrizations induce a quasi-isomorphism*

$$C^{\bullet}_{\mathrm{HH},\mathrm{cont}}(A/R, A) \longrightarrow \big(C^{\bullet}(\mathfrak{l}, S_{\mathrm{o}})[\![\hbar]\!], \hbar d_{\mathrm{CE}}\big).$$

Proof. Coefficient reduction is surjective in each cochain degree. For Hochschild cochains, lift outputs on the PBW input basis. The Chevalley exterior input has finite rank, so its outputs also lift. The comparison cones are acyclic and their transition maps are degreewise surjective. The long exact sequence shows that every transition kernel is acyclic. Choose a primitive of a compatible cycle at level zero. Lift it one level and correct its differential by a primitive in the transition kernel. Iteration gives compatible primitives at every order. The inverse-limit cone is therefore acyclic. □

The coefficient ring is part of this result. Absolute $\mathbb{C}$-linear cochains also detect variations of the coefficient parameter. For $\mathfrak{l} = 0$, the continuous absolute derivation $\partial_\hbar$ on R has no relative counterpart. For nonabelian $\mathfrak{l}$, the associated graded for the $\hbar$ filtration has zero differential, whereas d_{CE} need not vanish. The theorem asserts a quasi-isomorphism of complexes. A comparison of cup products, braces and their higher coherences requires additional maps and identities. The inverse limit above changes only the coefficient order N, at a fixed finite-dimensional Lie algebra. Passing to all Hamiltonian jets also requires compatible transition maps for the chosen cochain complexes and a proof that their limit computes the intended continuous carrier.

Chapter 4

Continuous Hamiltonian cochains

A formal Hamiltonian is determined by its finite jets. An operation on the quantized algebra need not return as many coefficients as it inspects. For example, extraction of a cubic coefficient can have a scalar output. A limit of self-valued cochains at equal input and output jet orders would miss this operation.

We therefore keep the two orders separate. The resulting continuous Hochschild complex has braces and a relative bar resolution. Its Chevalley comparison is compatible with both jet orders and with the quantization parameter. This constructs a continuous algebraic centre. Relating it to geometric collision operations or to bulk observables requires maps preserving those additional operations.

4.1 Residue duality and opposite directions of variance

Let k be a field of characteristic zero. The polynomial formulas of Chapter 3 apply over k. Put

$$\mathfrak{m} = (z_1, z_2), \qquad \mathfrak{h} = \mathfrak{m}^2 \subset k[\![z_1, z_2]\!], \qquad \mathfrak{h}_{\le d} = \mathfrak{h}/\mathfrak{m}^{d+1} \quad (d \ge 2).$$

The bracket is (3.1). Give $\mathfrak{h}$ its order topology and k the discrete topology. A continuous linear functional on $\mathfrak{h}$ vanishes on $\mathfrak{m}^{d+1}$ for some d. Consequently

$$\mathfrak{h}^\vee_{\mathrm{cont}} = \varinjlim_d \mathfrak{h}^\vee_{\le d} = \bigoplus_{a+b\ge 2} k\rho_{ab}, \qquad \rho_{ab} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2. \tag{4.1}$$

The notation on the right represents the iterated residue functional, normalized by

$$\mathrm{Res}_0(z_1^i z_2^j \rho_{ab}) = \delta_{ia}\delta_{jb}.$$

On the whole power-series ring, the same formula includes all $a, b \ge 0$. The two degree-one modes pair with the translations in $\mathfrak{h}^{\mathrm{aff}}$, whereas the pointed algebra starts in degree two. The continuous dual has the discrete topology.

Proposition 4.1 (Coadjoint currents)**.** *For $f = z_1^i z_2^j \in \mathfrak{h}$,*

$$\mathrm{ad}_f^*\, \rho_{ab} = \big(j(a+1) - i(b+1)\big)\rho_{a-i+1,b-j+1}. \tag{4.2}$$

A mode with a negative index or with total degree less than two is zero as a functional on $\mathfrak{h}$. The formula extends to formal f, and the coadjoint action is continuous.

Proof. By definition, $(\mathrm{ad}_f^*\, \rho_{ab})(g) = -\rho_{ab}(\{f, g\})$. For $g = z_1^p z_2^q$, the monomial bracket has coefficient $iq - jp$ and exponent $(i + p - 1, j + q - 1)$. The residue can be nonzero only at $(p, q) = (a - i + 1, b - j + 1)$, where its negative coefficient is $j(a+1) - i(b+1)$. This proves the formula. If $a + b \le d$, a Hamiltonian of order greater than d acts by zero: the possible output mode would have degree at most one. Thus the action on any finite sum of residue modes uses only finitely many Taylor coefficients of f. Its output is again a finite sum. This also proves continuity for the stated topologies. □

The graded semidirect Lie algebra

$$\mathfrak{q}_{\mathrm{cont}} = \mathfrak{h} \ltimes \mathfrak{h}_{\mathrm{cont}}^{\vee}[-1]$$

has degree-zero Hamiltonians and degree-one covectors. The second summand is abelian. The order topology on the first summand and the discrete topology on the second make it separated and complete. Evaluation gives the invariant pairing between the two summands: its invariance is the defining negative-transpose identity above.

The quotient $\mathfrak{h} \to \mathfrak{h}_{\le d}$ induces an inclusion of duals. Its image is a coadjoint submodule, and there is a surjective Lie map

$$\mathfrak{h} \ltimes \mathfrak{h}_{\le d}^{\vee}[-1] \longrightarrow \mathfrak{q}(\mathfrak{h}_{\le d}) = \mathfrak{h}_{\le d} \ltimes \mathfrak{h}_{\le d}^{\vee}[-1], \qquad (x, \alpha) \longmapsto (x \bmod \mathfrak{m}^{d+1}, \alpha). \tag{4.3}$$

Its kernel is $\mathfrak{m}^{d+1}$ in the first summand. Indeed, that ideal acts trivially on the displayed finite dual, and the coadjoint formula descends to the finite quotient. Thus the finite current algebra is a quotient of a subalgebra of $\mathfrak{q}_{\mathrm{cont}}$.

Simultaneous coordinate truncation on both full summands is not a Lie map. At $d = 2$, it kills both z_1^3 and ρ_{21}, but

$$\mathrm{ad}^*_{z_1^3}\, \rho_{21} = -6\rho_{02}$$

has a nonzero quadratic component. The quotient of Hamiltonians and the inclusion of covectors must retain their opposite directions.

4.2 The completed algebra and its continuous operations

Set $R = k[\![\hbar]\!]$ and $R_N = R/(\hbar^{N+1})$. For $d \ge 2$ and $N \ge 0$, define

$$A_{d,N} = U_{R_N}(\mathfrak{h}_{\le d} \otimes_k R_N, \hbar\{-,-\}), \qquad A = \widehat{U}_\hbar(\mathfrak{h}) = \varprojlim_{d,N} A_{d,N}. \tag{4.4}$$

The defining relation is $xy - yx = \hbar\{x, y\}$. Choose one ordering of the Hamiltonian monomials, compatible with the inclusions of their finite bases. The PBW theorem proved in Proposition 3.7 gives free R_N-modules with the corresponding ordered-monomial bases. Every quotient map in (4.4) is onto.

Give A the inverse-limit topology with discrete quotients $A_{d,N}$. At each fixed (d, N), PBW expressions are finite polynomials. In particular, this construction does not complete the polynomial degree in the enveloping generators. For $n \ge 1$, put

$$A^{\widehat{\otimes}_R n} = \varprojlim_{d,N} A_{d,N}^{\otimes_{R_N} n}, \qquad A^{\widehat{\otimes}_R 0} = R. \tag{4.5}$$

Different cutoffs in the tensor factors give the same limit because a common larger cutoff is cofinal. All continuity statements below refer to this quotient tensor.

Here is an operation requiring different jet orders. Let X_{30} be the enveloping generator represented by the Hamiltonian z_1^3. Projection to $A_{3,N}$ followed by extraction of the PBW coefficient of the single monomial X_{30} is an R_N-linear scalar-valued map. The compatible maps in N, followed by the scalar inclusion into A, define a continuous R-linear map $\lambda_{30} : A \to A$. Its output in $A_{2,N}$ is nonzero on X_{30}, although the input projection to $A_{2,N}$ kills X_{30}. That output therefore cannot be computed from the quadratic input jet.

Definition 4.2. The continuous Hochschild cochains for the quotient tensor are

$$C^n_{\mathrm{uc}}(A/R, A) = \varprojlim_{d,N} \varinjlim_{e \geq d} \mathrm{Hom}_{R_N}(A_{e,N}^{\otimes_{R_N} n}, A_{d,N}). \tag{4.6}$$

The direct maps precompose with input quotients. The inverse maps reduce the output and the parameter precision. Give each direct limit the discrete topology and each C^n_{uc} its inverse-limit topology. The subscript specifies uniform continuity for (4.5).

Lemma 4.3 (An input cutoff for each output)**.** *The quotient projections in* (4.5) *are onto, and there is a natural identification*

$$C^n_{\mathrm{uc}}(A/R, A) = \mathrm{Hom}_{R,\mathrm{cont}}(A^{\widehat{\otimes}_R n}, A). \tag{4.7}$$

Both the tensor source and the cochain space are separated and complete.

Proof. A finite PBW tensor lifts by the same ordered monomials to all larger cutoffs; hence every projection is onto. On underlying modules, symmetrization identifies the tensor source with

$$\left(\varprojlim_{d} (\mathrm{Sym}_k \mathfrak{h}_{\leq d})^{\otimes_k n}\right)[\![\hbar]\!].$$

Each parameter coefficient is a compatible system of finite polynomials. Coefficientwise division by $\hbar^{N+1}$ therefore identifies its quotient by that parameter power with $\varprojlim_d A_{d,N}^{\otimes n}$. The inverse-limit description also gives separatedness and completeness.

Let f be a continuous map on the right of (4.7). Its composite with the output (d, N) kills the kernel of some input projection (e, M). Increase that projection so that $e \geq d$ and $M \geq N$. Surjectivity makes the induced map R-linear. Since the target is killed by $\hbar^{N+1}$, it factors through $A_{e,N}^{\otimes n}$. Two choices give the same map precisely after a common input refinement. The output compatibilities give (4.6). Conversely, its compatible families define a map into A, and their finite input factorizations at each output make it continuous. Finally, an inverse limit of discrete modules is a closed submodule of their product and is complete, proving the assertion for cochains. □

4.3 Composition and the relative derived centre

All enveloping generators have cohomological degree zero, and an n-cochain has cohomological degree n. Its Hochschild differential is

$$\begin{aligned}(bf)(a_1, \dots, a_{n+1}) = {} & a_1 f(a_2, \dots, a_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(a_1, \dots, a_i a_{i+1}, \dots, a_{n+1}) \\ & + (-1)^{n+1} f(a_1, \dots, a_n) a_{n+1}.\end{aligned}$$

For brace signs, use the reduced degree $|f| = n - 1$. If g_j has arity q_j, define

$$f\{g_1, \dots, g_r\} = \sum_{\text{ordered disjoint insertions}} (-1)^{\sum_j (q_j - 1) i_j} f(\dots, g_1(\dots), \dots, g_r(\dots), \dots), \tag{4.8}$$

where i_j is the number of arguments preceding the block of g_j in the resulting argument list. Equivalently, these are the Koszul insertion signs on the suspended tensor coalgebra. For f of arity p and g of arity q, the cup product is

$$(f \smile g)(a_1, \dots, a_{p+q}) = f(a_1, \dots, a_p) g(a_{p+1}, \dots, a_{p+q}).$$

Theorem 4.4 (Continuous braces)**.** *The differential, cup product and every finite brace preserve $C^\bullet_{\mathrm{uc}}(A/R, A)$. They are continuous for the stated cochain topologies and obey the Hochschild brace identities.*

Proof. Fix an output quotient (d, N) for the outer cochain f. It uses some input cutoff e. Each inner cochain g_j is then needed only at output (e, N), and that output uses some finite input cutoff e_j. Choose E at least as large as $e, e_1, \dots, e_r$. Every summand is now a map from a tensor power of $A_{E,N}$ to $A_{d,N}$. There are finitely many summands at fixed arities, and further refinement leaves their class unchanged. This constructs the brace. Multiplication is defined at each finite quotient, so the same argument constructs b and the cup product.

To prove continuity at a specified tuple of cochains, fix its outer component at (d, N) and the finitely many inner components just used. These equalities describe open neighborhoods, and all outputs at (d, N) agree throughout them. The argument applies to every output quotient. For a brace identity, choose one common quotient for all the intermediate outputs in its two sides. Both sides enumerate the same nested or disjoint insertions there. The Koszul sign is the sign of the same permutation of suspended inputs, so the terms agree. Thus the identity holds at every output and in the limit. Associativity of multiplication gives $b^2 = 0$. □

An individual rectangular complex with coefficients in $A_{d,N}$ and inputs in $A_{e,N}$ is not a self-valued brace algebra when $e \neq d$. The total cochain space supplies the intermediate output needed by the next insertion.

We now specify the derived category for this construction. Work with complete separated linearly topologized R-modules and quotient completed tensors. A map out of a tensor is continuous when each discrete output factors through discrete quotients of all its inputs. Use complete A-bimodules whose action is continuous for this tensor, and write $A^e = A\widehat{\otimes}_R A^{\mathrm{op}}$. A short exact sequence is *relatively admissible* when it has a continuous R-linear splitting, which need not be A-bilinear. Relative resolutions and derived Hom refer to this exact structure.

For a complete coefficient module E and a complete A-bimodule M, restriction to $1 \otimes E \otimes 1$ gives the induction adjunction

$$\mathrm{Hom}_{A^e,\mathrm{cont}}(A\widehat{\otimes}_R E\widehat{\otimes}_R A, M) = \mathrm{Hom}_{R,\mathrm{cont}}(E, M). \tag{4.9}$$

The inverse sends f to $a \otimes e \otimes b \mapsto af(e)b$. First refining the output for the action and then the input for f proves its continuity. If $M' \to M$ has a continuous R-linear section, a map on the right lifts by composition with that section. The adjunction then lifts the bimodule map. Thus induced modules are relatively projective.

Theorem 4.5 (The completed relative bar)**.** *The augmented complex*

$$B_n(A) = A\widehat{\otimes}_R A^{\widehat{\otimes}_R n}\widehat{\otimes}_R A \quad (n \geq 0), \qquad B_0(A) \longrightarrow A,$$

with alternating multiplication boundary and B_n in degree $-n$, is a relative projective resolution. Hence

$$\mathrm{RHom}^{\mathrm{rel,cont}}_{A^e}(A, A) \simeq C^\bullet_{\mathrm{uc}}(A/R, A). \tag{4.10}$$

Proof. Each term is induced, hence relatively projective by (4.9). On the augmented bar, put

$$s_n(a_0 \otimes \cdots \otimes a_{n+1}) = 1 \otimes a_0 \otimes \cdots \otimes a_{n+1}, \qquad s_{-1}(a) = 1 \otimes a.$$

These maps are continuous and R-linear. In $\partial s + s\partial$, every multiplication except the newly inserted first one occurs twice with opposite signs. The remaining term is the identity. On cycles, s is therefore a continuous R-linear section of the boundary onto those cycles. This proves admissible exactness. Continuous bimodule Hom identifies the term in degree n with (4.7). Its induced differential is exactly b, proving (4.10). □

This is the derived centre in the stated relative continuous theory. An absolute derived category of topological modules or an analytic tensor product requires a comparison with this resolution.

4.4 The Chevalley comparison and the inverse limit

For $e \geq d$, let $A_{d,N}$ be an $A_{e,N}$-bimodule through the quotient. Define

$$P^n_{e,d,N} = \mathrm{Hom}_k\big(\Lambda^n \mathfrak{h}_{\leq e}, \mathrm{Sym}_k\, \mathfrak{h}_{\leq d}\big) \otimes_k R_N. \tag{4.11}$$

The coefficient action of $\mathfrak{h}_{\leq e}$ factors through $\mathfrak{h}_{\leq d}$ and extends its adjoint action as a derivation of the symmetric algebra. Let d_{CE} be the Chevalley differential with this action and with the Poisson Lie bracket. For example,

$$(d_{\mathrm{CE}} f)(x, y) = x \cdot f(y) - y \cdot f(x) - f(\{x, y\})$$

for a one-cochain. In general the action term indexed by i has sign $(-1)^i$ on inputs $x_0, \ldots, x_n$, and the bracket term indexed by $i < j$ has sign $(-1)^{i+j}$, with $\{x_i, x_j\}$ put first and the remaining inputs in their original order. The differential on $P^\bullet_{e,d,N}$ is $\hbar d_{\mathrm{CE}}$.

Theorem 4.6 (Chevalley comparison with two jet orders)**.** *Output PBW symmetrization and input antisymmetrization define natural quasi-isomorphisms*

$$C^\bullet_{\mathrm{HH}}(A_{e,N}/R_N, A_{d,N}) \longrightarrow (P^\bullet_{e,d,N}, \hbar d_{\mathrm{CE}}), \tag{4.12}$$
$$f \longmapsto \left[(x_1, \ldots, x_n) \longmapsto \mathrm{PBW}^{-1} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) f(x_{\sigma(1)}, \ldots, x_{\sigma(n)})\right].$$

They commute with input refinement, output quotient and parameter reduction. Consequently they induce a quasi-isomorphism

$$C^\bullet_{\mathrm{uc}}(A/R, A) \longrightarrow \mathcal{P}_{\mathrm{uc}} := \varprojlim_{d,N} \varinjlim_{e \geq d} (P^\bullet_{e,d,N}, \hbar d_{\mathrm{CE}}). \tag{4.13}$$

For both systems in this formula, the ordinary inverse limit computes the derived inverse limit.

Proof. Use the finite Chevalley bimodule resolution from the proof of Theorem 3.8, now for $A_{e,N}$ and with coefficient bimodule $A_{d,N}$. Its terms are

$$A_{e,N} \otimes_{R_N} \Lambda^n(\mathfrak{h}_{\leq e} \otimes_k R_N) \otimes_{R_N} A_{e,N}.$$

They are free bimodules. Filtering by PBW degree plus exterior degree gives the Koszul resolution of the diagonal of a polynomial algebra over R_N. The diagonal differences form a regular sequence even over this nilpotent coefficient ring. A cycle has finite filtration degree, and lifting an associated-graded primitive lowers that degree. This proves exactness without a convergence assumption.

Antisymmetrization of the exterior factors is a map into the bar resolution: internal bar products combine in pairs to give the scaled Lie commutator, and the two endpoint terms give the left and right actions. It is therefore a chain homotopy equivalence between projective resolutions. Applying bimodule Hom into $A_{d,N}$ gives (4.12). Under PBW, the commutator action of x on the coefficient algebra is $\hbar$ times its adjoint Poisson action; the bracket term also carries $\hbar$. This proves the stated differential. All formulas commute with the three quotient maps, proving naturality.

Filtered direct limits of modules are exact. For the output inverse limit, use the countable cofinal sequence $(d, N) = (r + 2, r)$. Every cochain at one output factors through a finite input order e. For the next output, increase e if necessary and precompose with the input quotient. Lift the value of each PBW tensor basis element to the next output and parameter precision, then extend linearly over the new coefficient ring. Each value is a finite polynomial, so this defines an allowed cochain. The exterior basis gives the same construction on $P^n_{e,d,N}$. These are degreewise lifts, not chain sections; no R_N-linear section of $R_{N+1} \to R_N$ is used.

Thus both cochain towers are degreewise surjective. For a countable tower of complexes with transitions $p_r : K_{r+1} \to K_r$, the derived inverse limit is the homotopy fibre of

$$1 - \mathrm{shift} : \prod_r K_r \longrightarrow \prod_r K_r.$$

Degreewise surjectivity of the transition maps makes this product map degreewise onto: choose the next coordinate recursively to solve $x_r - p_r x_{r+1} = y_r$. Its kernel is $\varprojlim_r K_r$. The homotopy fibre is consequently quasi-isomorphic to that kernel. Applying the derived inverse limit to the levelwise comparisons therefore gives (4.13). □

At $\hbar = 0$, the Chevalley differential vanishes, as required by the commutative polynomial special fibre. At positive parameter precision the two Chevalley terms remain scaled together. The theorem identifies complexes. It does not identify a chosen polyvector wedge with the Hochschild cup product or make antisymmetrization a strict brace map.

Finite comparison homotopies can also be computed. At parameter zero, write $t_i = x_i - y_i$ for the diagonal differences in the Koszul resolution and ϵ_i for its exterior generators. On positive total t, ϵ degree, the operators

$$d_0 = \sum_i t_i \partial_{\epsilon_i}, \qquad h_0 = \frac{\sum_i \epsilon_i \partial_{t_i}}{\deg_t + \deg_\epsilon}$$

satisfy $d_0 h_0 + h_0 d_0 = 1$; on t, ϵ-constant terms the homotopy is zero. Let i_0 include this constant summand and let p_0 project to it. The anticommutator before division is the total Euler operator, so

$$d_0 h_0 + h_0 d_0 = 1 - i_0 p_0, \qquad h_0^2 = h_0 i_0 = p_0 h_0 = 0.$$

After PBW identification the quantum differential is $d_K = d_0 + \delta$, where δ has order $\hbar$. Modulo $\hbar^{N+1}$, put

$$T = \sum_{j=0}^{N} (-\delta h_0)^j, \qquad h_K = h_0 T, \qquad p_K = p_0 T, \qquad i_K = \sum_{j=0}^{N} (-h_0 \delta)^j i_0.$$

These sums are finite. Multiplying them and using $d_K^2 = 0$ and the preceding identities gives

$$d_K h_K + h_K d_K = 1 - i_K p_K, \qquad p_K i_K = 1.$$

The differential transferred to the constant summand is zero by degree. That summand represents the diagonal quotient $A_{e,N}$, so p_K is its augmentation in PBW coordinates. This supplies the contraction on the augmented resolution.

Starting with the augmentation lift, define the bar-to-Chevalley comparison on a free middle generator by

$$\psi_n(1[a_1|\cdots|a_n]1) = h_K \psi_{n-1} \partial (1[a_1|\cdots|a_n]1),$$

and extend bimodule-linearly. Its argument is a cycle by the previous stage, so the contraction identity proves the chain-map equation. If α is the antisymmetrization map into the bar, define on its free middle generators

$$H_n = s_n\big((1 - \alpha\psi)_n - H_{n-1}\partial\big), \qquad H_{-1} = 0,$$

and again extend bimodule-linearly. The expression in parentheses is a cycle by induction, so $\partial H + H\partial = 1 - \alpha\psi$. These are finite algorithms at every fixed cutoff. Transporting higher operations through a simultaneous continuous choice requires the compatibility of those homotopies as well.

4.5 A localization test for limits and adjoints

The surjectivity used above is a substantive hypothesis. A scalar example shows both what can fail without it and why changing the completion changes the problem.

Let $S = k[t]$, $L = k[t, t^{-1}]$, and write $D(S)$ and $D(L)$ for the derived categories of modules. Restriction of scalars $i : D(L) \to D(S)$ is fully faithful, since localization is flat and $L \otimes_S L = L$. Its left adjoint is $L \otimes^{\mathbb{L}}_S (-)$; its right adjoint is $\mathrm{RHom}_S(L, -)$. The right adjoint represents maps from localized objects into a given object, whereas the left adjoint represents maps out of that object into localized ones.

Proposition 4.7 (The two localization adjoints)**.** *Their values at S are*

$$L \otimes^{\mathbb{L}}_S S = L, \qquad \mathrm{RHom}_S(L, S) \simeq (k[\![t]\!]/k[t])[-1].$$

The latter equivalence is one of S-complexes. Multiplication by t is invertible on its displayed cohomology and gives the L-module structure. For $\widehat{S} = k[\![t]\!]$ and $\widehat{L} = k((t))$, the right adjoint takes $\widehat{S}$ to zero, while the left adjoint takes it to $\widehat{L}$.

Proof. The direct system $S \xrightarrow{t} S \xrightarrow{t} \cdots$ has colimit L, where the nth copy maps by multiplication by t^{-n}. Its free telescope resolution has terms $\bigoplus_{n\geq 0} S$ and differential $e_n \mapsto e_n - te_{n+1}$. Injectivity follows by inspecting the least nonzero coordinate of a finitely supported vector. Its cokernel imposes exactly the relations of that colimit. Applying $\mathrm{Hom}_S(-, S)$ gives the complex in degrees zero and one

$$\prod_{n\geq 0} S \xrightarrow{D} \prod_{n\geq 0} S, \qquad D(a)_n = a_n - ta_{n+1}. \tag{4.14}$$

If $D(a) = 0$, each a_n is divisible by every power of t, so $a_n = 0$. Define

$$\Theta(b) = \left[\sum_{n\geq 0} t^n b_n\right] \in k[\![t]\!]/k[t].$$

The series converges coefficientwise. For $b = D(a)$ it telescopes to the polynomial a_0, so Θ annihilates the image of D. It is onto, by taking the b_n to be scalar coefficients of a series. If its value is zero, the series equals a polynomial c. Put

$$a_n = t^{-n}\left(c - \sum_{j<n} t^j b_j\right).$$

The expression in parentheses is a polynomial divisible by t^n, so $a_n \in S$, and $D(a) = b$. This proves the asserted cokernel. On that cokernel, the inverse of multiplication by t sends $[f]$ to $[(f - f(0))/t]$.

Over $\widehat{S}$, the same telescope differential has the inverse

$$a_n = \sum_{j\geq 0} t^j b_{n+j}.$$

Each series converges t-adically, and the homogeneous equation still has only the zero solution. Thus the derived right-adjoint value is zero. Flat localization computes both left-adjoint values. □

For $b_n = 1$, the class $\Theta(b) = [(1 - t)^{-1}]$ is nonzero. Every finite subsystem nevertheless has a polynomial solution: with $a_{M+1} = 0$, take

$$a_n = 1 + t + \cdots + t^{M-n} \qquad (0 \leq n \leq M).$$

These degrees grow with M and give no polynomial solution of the infinite system. Completion supplies such a solution by admitting a different class of coefficients. The continuous Hamiltonian comparison avoids this inference by proving surjectivity of its actual cochain transitions.

4.6 Compatible models and geometric actions

The algebra A of (4.4) is the formal Hamiltonian boundary algebra, and (4.10) constructs its relative continuous derived centre. The explicit model $\mathcal{P}_{\mathrm{uc}}$ records the corresponding Hamiltonian cochains with their two variances. A different polyvector realization must be compared with this complex in its own topology.

To compare another realization obtained from finite models, one must also specify its transitions. At a fixed jet order let

$$A_d = \varprojlim_N A_{d,N}, \qquad \mathcal{P}_d = \big(C^\bullet(\mathfrak{h}_{\le d}, \operatorname{Sym}\mathfrak{h}_{\le d})[\![\hbar]\!], \hbar d_{\mathrm{CE}}\big).$$

A *compatible jet comparison* consists of complete R-linear cochain models P_d, H_d, continuous transition maps, levelwise comparisons $P_d \simeq \mathcal{P}_d$ and $H_d \simeq C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$, and compatible quasi-isomorphisms $P_d \to H_d$. All comparison maps and limit identifications are continuous. Require degreewise surjective transitions or use derived inverse limits. Specify the identifications of their chosen limits with a proposed complex $\mathcal{P}_\infty$ and with $C^\bullet_{\mathrm{uc}}(A/R, A)$. An ordinary limit may be used only when its comparison with the derived limit is supplied. All identifications use the quotient topology fixed above.

Theorem 4.8 (Continuous comparison from compatible jets)**.** *A compatible jet comparison induces a continuous quasi-isomorphism*

$$\Phi_\infty : \mathcal{P}_\infty \xrightarrow{\sim} C^\bullet_{\mathrm{uc}}(A/R, A).$$

Compatible homotopy-E_2 *and Swiss-cheese refinements, whose operations converge in the chosen topology, give the corresponding complete open–closed operations.*

Proof. The derived inverse limit preserves levelwise equivalences. For degreewise surjective towers it agrees with the ordinary limit by the product calculation in the proof of Theorem 4.6. Apply the specified limit identifications. For the final assertion, the supplied operadic comparisons identify each operation and its coherence homotopies; convergence defines them on the limits. □

The two-index construction already gives the algebraic cochain comparison (4.13). It supplies neither a chosen geometric polyvector product nor its equivalence with the brace operations. In particular, an invariant continuous Duflo map requires a multiplicative comparison and a specified prescription for its wheel coefficients.

Finite matrix branes are representation-theoretic images of the formal Hamiltonian boundary algebra whenever their representations are continuous for this completion. Its relative derived centre is then an algebraic target for coherent bulk actions. A field-theory identification starts instead from that theory's observables and constructs their action map. It must preserve the insertion supports, collisions, vacuum and higher substitutions. The continuous algebraic resolution provides the target of this comparison; it does not construct its geometric or physical source.

Chapter 5

Current actions and derived interface centres

A boundary action has to commute with fields inserted near the boundary. When this commutation holds up to homotopy, two fields require a compatibility between the two homotopies and their collision. The compatibility is additional data. We first recover its cochains from supported maps and construct them for a resolved current action.

The Heisenberg algebra makes the resulting distinctions explicit. At a polynomial root, forgetting the current action retains many more operators than the current cohomology sees. Their derived composition has a nonzero Massey product. Over the level ring, the path in the forgetful fibre turns a torsion current class into a free class. Finite categories of interfaces then give centres in which cup products, insertion brackets and parameter fibres can all be computed. These calculations specify operations that a geometric bulk–boundary comparison must preserve.

Throughout the chapter, k is a field of characteristic zero. Complexes are cohomological, with $K[1]^n = K^{n+1}$. For a homogeneous map f the Hom differential is $d_{\mathrm{Hom}} f = df - (-1)^{|f|} f d$. Polynomial states have finite support in their variables. A completion will always be stated separately.

5.1 Current homotopies from a supported map

Let $O = k[y]$ and $D_y = O\langle \partial_y \rangle$, with $[\partial_y, y] = 1$. The Heisenberg vacuum at level $\kappa \in k^\times$ is

$$V_\kappa = k[x_1, x_2, \dots], \qquad J_{-n} = L_{x_n}, \quad J_n = \kappa n \partial_{x_n} \quad (n > 0), \qquad J_0 = 0. \tag{5.1}$$

Here L_f denotes multiplication by f. Its vacuum is 1, its current state is $J = x_1$, and its translation is

$$T = \sum_{n \geq 1} n x_{n+1} \partial_{x_n}, \qquad \mathcal{V} = O \otimes_k V_\kappa, \qquad \nabla = \partial_y + T.$$

For a formal variable u, the field $Y(J,u) = \sum_{m \in \mathbb{Z}} J_m u^{-m-1}$ determines the other fields by normal products and translation. Direct calculation gives $[J_m, J_n] = \kappa m \delta_{m+n,0}$ and $[\nabla, J_m] = -m J_{m-1}$. Each field is lower truncated on each state.

Take a second coordinate z and put $u = z - y$. A map supported on $u = 0$ takes values in principal parts, with all finite normal derivatives retained as in Section 2.1. The current generates a free D_y-submodule: $T^n J = n! x_{n+1}$. Denote it by $L_J = D_y J$.

A semi-free complex has an increasing filtration by subcomplexes, split on underlying graded modules, whose successive quotients are direct sums of shifts of D_y.

Lemma 5.1 (Laurent coefficients of supported maps)**.** *Let Q be a semi-free complex of left D_y-modules. Let j include the complement of the diagonal and let Δ include the diagonal. Then*

$$R\,\mathrm{Hom}_{D_{z,y}}\big(j_* j^*(L_J \boxtimes Q), \Delta_+ \mathcal{V}\big)$$

is computed by ordinary differential-module maps. Taking the residue identifies this complex with families of O-linear maps $H_m : Q \to \mathcal{V}$, $m \in \mathbb{Z}$, satisfying

$$[\nabla, H_m] + mH_{m-1} = 0. \tag{5.2}$$

For each input, only finitely many H_m with $m \geq 0$ are nonzero. There is no bound required uniformly over all inputs.

Proof. For one free generator the localized source has the telescope presentation

$$D_{z,y}[u^{-1}] = \varinjlim_{k} D_{z,y}u^{-k},$$

whose maps are right multiplication by u. The Ore localization exists because $\operatorname{ad} u$ is locally nilpotent. Applying Hom gives the inverse system of principal parts under multiplication by u. These maps are onto: u^{-r} has preimage u^{-r-1}. Thus its ordinary limit computes its derived limit, by the product model in the proof of Theorem 4.6. Direct sums of free generators become products of these exact calculations.

Use the semi-free filtration of Q, split on underlying graded modules. At each finite stage the assertion follows by extensions. The restriction maps between the ordinary Hom complexes are degreewise onto, so the assertion passes to the entire filtration. The resulting totalization uses products.

The map with residue family (H_m) is reconstructed by

$$\varphi(u^m J \boxtimes q) = \sum_{r \geq 0} H_{m+r}(q)u^{-r-1}. \tag{5.3}$$

The stated finiteness makes this a principal part for each m, q. Multiplication by u shifts the formula. Linearity in ∂_z then determines the values on every $T^n J$. Finally $\partial_y|_z = \partial_y|_u - \partial_u$ and $\operatorname{Res}_u \partial_u = 0$ give

$$\nabla H_m(q) = -mH_{m-1}(q) + H_m(\partial_y q),$$

which is precisely the remaining differential-module relation. These constructions are inverse and commute with the Hom differential. □

To extract these maps from a collision, retain its attaching map. Write a one-current Cousin complex as $A[1] \oplus B$, where A is the open term, B is the supported term, and $\mu : A \to B$ is the unshifted residue map. For the suspension $s : A \to A[1]$ of degree -1, its differential is

$$d(sa, b) = (-s\, d_A a, \mu a + d_B b).$$

A map to a second such complex has open and closed components f_U, f_Δ and a degree -1 cross component h. Its equation is $d_{\mathrm{Hom}} h = f_\Delta \mu - \mu' f_U$.

Suppose that the prescribed unary operation is $a : Q \to \mathcal{V}$, and that the open and closed restrictions are identified with $1 \boxtimes a$ and a by specified homotopies γ, δ. Orient them by $d\gamma = f_U - 1 \boxtimes a$ and $d\delta = f_\Delta - a$. Writing ∂_P, ∂_C for the source and target residue maps, set

$$\widetilde{h} = h + \partial_C \gamma - \delta\, \partial_P, \qquad d\widetilde{h} = a\, \partial_P - \partial_C (1 \boxtimes a). \tag{5.4}$$

For a semi-free P, take $Q = P \otimes_O \mathcal{V}$ with its prescribed boundary current action. On a free D_y-summand, the change from $\partial_y \otimes 1$ to the diagonal connection $\partial_y \otimes 1 + 1 \otimes \nabla$ is triangular in PBW degree. It identifies the tensor with the module induced from the O-free module $\mathcal{V}$, so the tensor is D_y-free. Applying this to the filtration quotients makes Q semi-free. The residue map ∂_P acts on the factor $\mathcal{V}$. The identity

$$\operatorname{Res}_u \operatorname{pp}_u(u^m Y(J, u)w) = J_m w$$

therefore gives

$$H_m = -\operatorname{Res}_u \widetilde{h}(u^m J \boxtimes -), \qquad d_{\mathrm{Hom}} H_m = J_m a - a J_m. \tag{5.5}$$

The source action is the rightmost J_m in this formula. In particular, the zero and negative modes are part of the comparison. Equality of the restrictions in cohomology would not specify γ, δ or the corrected supported coordinate.

5.1.1 *The second equation and all higher equations*

Let $\mathfrak{a}$ be an ordinary Lie algebra and let M, V be dg $\mathfrak{a}$-modules. Put $E = \mathrm{Hom}_k(M, V)$ and $T_x f = \rho_V(x) f - f \rho_M(x)$. For an alternating p-cochain set

$$\begin{aligned}(\delta f)(x_0, \dots, x_p) = {} & \sum_i (-1)^i T_{x_i} f(x_0, \dots, \widehat{x_i}, \dots, x_p) \\ & + \sum_{i<j} (-1)^{i+j} f([x_i, x_j], x_0, \dots, \widehat{x_i}, \dots, \widehat{x_j}, \dots, x_p).\end{aligned} \tag{5.6}$$

The bracket occupies the first slot. The representation identity and Jacobi identity give $\delta^2 = 0$; moreover $\delta d_E = d_E \delta$. Use the product totalization with differential $\delta + (-1)^p d_E$ on Chevalley degree p.

A coherent current-linear map is a family $H_p : \Lambda^p \mathfrak{a} \to E^{-p}$ satisfying

$$H_0 = a, \qquad d_E a = 0, \qquad d_E H_p = \delta H_{p-1} \quad (p \geq 1). \tag{5.7}$$

Indeed, $\alpha_p = (-1)^{p(p-1)/2} H_p$ is then a total degree-zero cocycle. At two inputs the equation is

$$d_E H_2(x, y) = T_x H_1(y) - T_y H_1(x) - H_1([x, y]). \tag{5.8}$$

Its right side is closed once the first equation holds. Its class in $\mathrm{Hom}(\Lambda^2 \mathfrak{a}, H^{-1} E)$ must vanish before this chosen first homotopy extends. The three terms keep the two orders of current–root collision and the current–current collision. The last is the third channel.

Theorem 5.2 (A resolved current action). *Suppose I is a dg $\mathfrak{a}$-module and $i : V \to I$ is equivariant. Let (p, i, h) be a contraction of complexes with*

$$pi = 1, \qquad d_I h + h d_I = 1 - ip, \qquad hi = ph = h^2 = 0.$$

For every equivariant cochain map $j : M \to I$ there is a coherent current-linear map with unary part pj. It is given in all arities by a series finite in each output arity. Neither p nor h need be equivariant.

Proof. Postcomposition gives the same contraction on the Hom complexes. On Chevalley degree p its homotopy is $h_C = (-1)^p h$. Treat δ as a perturbation of $(-1)^p d_{\mathrm{Hom}}$ and define

$$p' = p(1 + \delta h_C)^{-1} = \sum_{r \geq 0} (-1)^r p(\delta h_C)^r. \tag{5.9}$$

Each perturbation raises Chevalley degree, so this defines a map on the product totalization. Since i is equivariant and $hi = 0$, $h_C \delta i = 0$. The transferred differential is consequently the ordinary total Chevalley differential on E. Multiplying the geometric series and using the contraction identity shows that p' intertwines the total differentials. The map j is a total cocycle in arity zero. Set $H_p = (-1)^{p(p-1)/2} (p' j)_p$ to obtain (5.7).

For the first two coefficients the formulas reduce to

$$H_1(x) = -p \rho_I(x) h j, \tag{5.10}$$

$$H_2(x, y) = p\big(\rho_I(x) h \rho_I(y) h - \rho_I(y) h \rho_I(x) h\big) j. \tag{5.11}$$

For example, applying d_E to the first gives $-p\rho_I(x)(1-ip)j = \rho_V(x)pj - pj\rho_M(x)$. In the second, differentiation of the two copies of h leaves the two action terms and their commutator, giving (5.8). The terms containing ph, hi or h^2 vanish by the side conditions. □

Example 5.3. The second homotopy may be nonzero for commuting currents. Let I have basis z, c in degree -2, d, a in degree -1, and b in degree zero, with $d_I c = d$ and $d_I a = b$. Put $V = kz$, let p be projection, and take $h(d) = c$, $h(b) = a$. Define commuting cochain operators by $\rho_x(c) = z$, $\rho_y(a) = d$, with all other values zero. For $M = I$ and $j = 1$ the formulas give

$$H_1(x)(d) = -z, \qquad H_1(y) = 0, \qquad H_2(x, y)(b) = z.$$

All other values vanish. Thus $d_E H_2(x, y)(a) = -z$, which is the nonzero right side of (5.8) on a.

For an infinite current algebra the same construction requires either pointwise finiteness of the operations or a specified completed Chevalley domain. The theorem begins with an equivariant resolved presentation. Applying it to a geometric boundary action still requires a map from that action's supported collision complex to this presentation, preserving its unary path and all three channels.

5.2 Several roots and upward extensions

Return to the Heisenberg state space $V = V_\kappa$ of (5.1). The acting current adds a root to the source of a module cochain. This explains why a bound on the roots of the original module is not a bound on every output of its derived endomorphisms. We make the coefficient category explicit before proving the bound that does hold.

For a finite set I, take a chart X_σ^I for each total order σ of I, where $X = \mathbb{A}^1_k$. Glue two charts along the open set obtained by removing $x_i = x_j$ whenever their orders of i, j disagree. Denote the result by $\mathscr{R}_I$. The triple overlap condition holds: disagreement of two orders on a pair forces one of them to disagree with any third order on that pair. Thus these opens define a smooth, generally nonseparated scheme. At a tuple with equality blocks B, its fibre over X^I has $\prod_B |B|!$ points, one order for each block.

A surjection $\alpha : I \twoheadrightarrow J$ with an order on each fibre defines $\delta_\alpha : \mathscr{R}_J \to \mathscr{R}_I$ by $x_i = y_{\alpha(i)}$: expand every ordered label of J into its ordered fibre. Composition concatenates these orders. The maps glue because the preimage of a chart overlap is the corresponding overlap on J. Let OSurj_+ be the category of nonempty finite sets and these ordered-fibre surjections. Define

$$C_+ = \varprojlim_{\mathrm{OSurj}_+} D\,\mathrm{Mod}(\mathscr{R}_I), \qquad C = D(k) \times C_+, \tag{5.12}$$

using extraordinary inverse images $\delta_\alpha^!$ on right differential modules. Differential modules on the atlas are defined by descent from its charts. An object retains the equivalences $\delta_\alpha^! F_I \simeq F_J$ and all their coherences. The factor $D(k)$ is the empty coefficient; a positive object has zero empty coefficient.

For a decomposition $I = A \sqcup B$ let $U_{A,B}$ be the locus on which every cross pair is distinct, and let $j_{A,B}$ be its open inclusion. Restriction of orders identifies this open with the disjoint locus in $\mathscr{R}_A \times \mathscr{R}_B$. Set

$$(F \star G)_I = \bigoplus_{I = A \sqcup B} j_{A,B,*}(F_A \boxtimes G_B)|_{U_{A,B}}. \tag{5.13}$$

An empty subset contributes scalar tensor product with its coefficient in $D(k)$; at $I = \varnothing$ the value is $F_\varnothing \otimes_k G_\varnothing$. Both subsets may be empty. This is the all-subset convolution; the cut product of Section 2.5 used only consecutive decompositions of a fixed word.

Proposition 5.4. *The product (5.13) is symmetric monoidal. Its unit K is k at the empty coefficient and zero elsewhere. Every parenthesized product of r positive objects has the common formula*

$$(F_1 \star \cdots \star F_r)_I = \bigoplus_{I=I_1 \sqcup \cdots \sqcup I_r} j_{P,*}(F_{1,I_1} \boxtimes \cdots \boxtimes F_{r,I_r})|_{U_P}, \tag{5.14}$$

where $P = (I_1, \dots, I_r)$ and U_P removes all cross-block diagonals. In particular this coefficient is zero for $r > |I|$.

Proof. Under an ordered collision, a summand whose blocks meet the same collapsed fibre has zero extraordinary inverse image: that fibre misses its open locus. Every remaining block is the inverse image of a unique block downstairs. Open base change (Lemma 2.7) identifies these remaining summands with the convolution downstairs. This also proves compatibility with composed collisions.

Nested open conditions impose exactly the cross-block inequalities of U_P. Localization commutes with external products, so all parenthesizations have (5.14) as their common expression. The coherent base-change identifications give the associativity diagrams. Permutation of the blocks gives the symmetry with the usual Koszul signs. Only an empty block can contribute the object K, proving the unit assertion. Positivity forces every I_j to be nonempty, proving the last assertion. □

For a right differential module $\mathcal{M}$ on X, its main-diagonal family $i\mathcal{M}$ is $\Delta_{I,*}\mathcal{M}$ on each ordered chart. The branch maps are the extraordinary restrictions of this diagonal direct image. Let $V^r = \omega_X \otimes_O \mathcal{V}$. The residue pairings of the Heisenberg fields give the degree-zero Lie object

$$\mathfrak{g} = L_{V_\kappa} = iV^r$$

for $\star$. Explicitly, on a two-block summand its bracket is the principal part of $Y(v, z-y)w$, with its right transfer-module normal derivatives. The field skew symmetry and Jacobi identity are the antisymmetry and Jacobi identities of this pairing, on every ordered branch. The local vacuum is the state $1 \in V$; it is distinct from the empty convolution unit K.

Write

$$N_a = \mathrm{Sym}^a_\star \mathfrak{g}, \qquad A = U_\star \mathfrak{g}.$$

Symmetric powers use coinvariants, identified with invariants by averaging. We use the full derived category of A-modules and the adjoint action on N_a and A, without imposing preservation of conformal weight. PBW symmetrization gives $A = \bigoplus_{a \geq 0} N_a$ as adjoint modules. The usual multilinear PBW identities apply here: convolution preserves colimits, and finite symmetric-group coinvariants are exact in characteristic zero. Since $N_a(I) = 0$ for $a > |I|$, this sum is also the product in compatible coefficient families. On each chart, localization and diagonal direct image are exact, so each N_a has differential-module coefficients in degree zero.

Theorem 5.5 (The degree-dependent support bound)**.** *For $w, a \geq 1$,*

$$\mathrm{Ext}^q_{\mathfrak{g}}(N_w, N_a) = 0 \qquad (q < a - w). \tag{5.15}$$

Consequently, for $w \geq 1$ and $q \geq 0$,

$$\tau_{\leq q} R\,\mathrm{Hom}_{\mathfrak{g}}(N_w, A) \simeq \tau_{\leq q} \prod_{0 \leq a \leq w+q} R\,\mathrm{Hom}_{\mathfrak{g}}(N_w, N_a). \tag{5.16}$$

The truncation is applied to the full complexes. In a connective coefficient resolution, degree-q cycle equations retain Chevalley columns through $q+1$.

Proof. The coefficient family N_a is a sum of open direct images from exactly a distinct roots, with diagonal support within each block. If a coefficient family B is supported on at most $a-1$ roots, ordinary open adjunction gives $R\operatorname{Hom}_C(B, N_a) = 0$. This is an adjunction before imposing module equivariance.

The module Chevalley resolution has columns

$$R\operatorname{Hom}_C(\Lambda^n_\star \mathfrak{g} \star N_w, N_a)[-n].$$

Their sources have at most $n+w$ roots, so a column vanishes when $n+w<a$. Otherwise it is n-connective: its coefficient objects are differential modules in degree zero and their derived Homs have no negative cohomology. The compatible-family limits preserve this bound. In each total degree only finitely many columns contribute. This proves (5.15).

Derived Hom into A is the product of derived Homs into the N_a. Every omitted factor in (5.16) starts above degree q; products preserve that lower bound. Finally column $q+1$ can receive the outgoing differential of a degree-q cochain, whereas later columns cannot enter its cycle equation. □

The boundary of this vanishing range is attained. Let $\mathsf{D} = \partial_{x_1}$ on polynomial states. It satisfies $\mathsf{D}1 = 0$, $\mathsf{D}J = 1$ and $[\mathsf{D}, T] = 0$. Moreover $[\mathsf{D}, Y(J,u)] = 1 = Y(\mathsf{D}J, u)$. Normal products and translation extend this derivation identity to every state. Thus $\phi_\tau = e^{\tau\mathsf{D}}$ is a vertex-algebra automorphism over $k[\tau]$, finite on each state, and induces a Lie derivation of $\mathfrak{g}$. This is the current translation subgroup described by Matsuo–Nagatomo [76, §3.3, Theorem 3.6].

Proposition 5.6. *Let $m_w : \mathfrak{g} \star N_w \to N_{w+1}$ be symmetric multiplication. For every $w \geq 1$,*

$$\gamma_w(x; m) = m_w(\mathsf{D}x, m)$$

defines a nonsplit extension of N_w by N_{w+1} as $\mathfrak{g}$-modules, preserving the vacuum action.

Proof. The Chevalley differential of γ_w is

$$m_w([x, \mathsf{D}y] + [\mathsf{D}x, y] - \mathsf{D}[x, y], m) = 0.$$

Hence the action on $N_{w+1} \oplus N_w$ is $x(n,m) = (xn + \gamma_w(x;m), xm)$. Its off-diagonal vacuum action vanishes because $\mathsf{D}1 = 0$.

Ordinary coefficient adjunction gives $\operatorname{Hom}_C(N_w, N_{w+1}) = 0$, so the underlying split extension has exactly one coefficient splitting. This splitting is not equivariant: at $w+1$ distinct roots, γ_w sends J at the acting root and vacua at the other roots to a nonzero external vacuum. Degree-zero maps between these degree-zero modules are ordinary equivariant maps by the nonnegative Chevalley resolution. The extension therefore remains nonzero in the derived category. □

An unrestricted derived open-support adjunction for these modules would force this extension to vanish. Such an adjunction cannot be inferred from the corresponding adjunction for their underlying coefficient families. For instance, input weight two has a genuine degree-one output in N_3.

5.2.1 The extension integrates to an interface

For any degree-zero Lie object with a natural derivation D, the formula

$$\rho_{0,\tau}(x)m = \operatorname{ad}_x(m) + \tau(\mathsf{D}x)m \tag{5.17}$$

defines a module structure on $\operatorname{Sym}_\star \mathfrak{g}[\tau]$. Indeed, if M_z is multiplication by z, then

$$[\operatorname{ad}_x, M_{\mathsf{D}y}] = M_{[x,\mathsf{D}y]}, \qquad [M_{\mathsf{D}x}, \operatorname{ad}_y] = M_{[\mathsf{D}x,y]}, \qquad [M_{\mathsf{D}x}, M_{\mathsf{D}y}] = 0.$$

Their sum is the representation identity. The added term raises symmetric degree by one. For $w \geq 1$ and $r \geq 0$, quotienting the degrees at least w by those above $w + r$ therefore gives a flat family of modules on $\bigoplus_{a=w}^{w+r} N_a$. For $r = 1$ its extension coordinate is $\tau\gamma_w$.

Suppose D is locally nilpotent and the Rees envelope $A_\nu = U_{\nu,\star}\mathfrak{g}$, with $xy - yx = \nu[x, y]$, is PBW flat. Write $\phi_t = e^{t\mathsf{D}}$. PBW gives a unique integral operator $\mathrm{ad}_\nu(x)$ with $\nu\,\mathrm{ad}_\nu(x)m = xm - mx$. Then

$$\rho_{\nu,\tau}(x)m = \mathrm{ad}_\nu(x)m + \sum_{j\geq 1} \frac{\nu^{j-1}\tau^j}{j!}(\mathsf{D}^j x)m \tag{5.18}$$

is integral over $k[\nu, \tau]$. After inverting ν it equals $\nu^{-1}(\phi_{\nu\tau}(x)m - mx)$. Commuting left and right multiplications gives

$$[L_{\phi x} - R_x, L_{\phi y} - R_y] = \nu(L_{\phi[x,y]} - R_{[x,y]}).$$

Torsion-freeness then proves the representation identity before inversion. At $\nu = 0$ only the $j = 1$ shifted term survives, giving (5.17).

For an associative algebra A and its automorphisms ϕ_τ, let $B_\tau = {}_{\phi_\tau}A$ have ordinary right action and left action $x \cdot b = \phi_\tau(x)b$. These twisted regular bimodules compose by

$$B_\sigma \otimes_A^L B_\tau \xrightarrow{\sim} B_{\sigma+\tau}, \qquad a \otimes b \longmapsto \phi_\tau(a)b. \tag{5.19}$$

They are free on either side, so ordinary tensor computes the derived tensor. The formula is balanced and left equivariant; its inverse is $c \mapsto 1 \otimes c$. Both associations of three factors give $\phi_{\tau+\upsilon}(a)\phi_\upsilon(b)c$. Thus $B_0 = A$ is the unit and $B_{-\tau}$ is inverse to B_τ. Equation (5.18) is the normalized left-minus-right action on $B_{\nu\tau}$. This constructs the algebraic interface underlying the upward extension. A field realization must also supply its collision and interchange maps.

Choose semi-free resolutions $P_w \to N_w$ and form the dg category $\mathcal{E}(w, a) = \mathrm{Hom}_{\mathfrak{g}}(P_w, P_a)$. Composition is composition of resolution maps. The dg Yoneda functor identifies the subcategory generated by the P_w under finite cones, shifts and retracts with $\mathrm{Perf}(\mathcal{E})$: it is fully faithful on the generators, this property extends by those operations in each variable, and represented modules generate the perfect modules. This is the precise generated-sector conclusion of dg Morita theory [60, Theorem 3.8(b)]. It supplies no generation statement for all physical boundaries.

5.3 Operators at a polynomial root

Forgetting the current action leaves an endomorphism of the moving differential module. The allowed endomorphisms must be specified before this forgetful map can be computed. Put

$$N = \mathrm{End}_O(\mathcal{V}), \qquad \nabla_N A = \nabla A - A\nabla, \qquad G(N) = [N \xrightarrow{\nabla_N} N], \tag{5.20}$$

where $G(N)$ is a complex of k-vector spaces in degrees zero and one. The identity $\nabla_N(fA) = f'A + f\nabla_N A$ for $f \in O$ makes N a differential module. In a monomial state basis, N consists of column-finite matrices over O: each input has a finite polynomial output, with no common bound on all outputs. For example, $v_j \mapsto y^j v_j$ lies in N but not in $O \otimes_k \mathrm{End}_k(V)$.

Lemma 5.7. *For an O-free left D_y-module M in degree zero, the internal derived endomorphism for $\otimes_O^L$ is $\mathrm{End}_O(M)$ in degree zero, with its commutator connection. This holds without a finite-rank hypothesis.*

Proof. Evaluation gives

$$\mathrm{Hom}_{D_y}(P, \mathrm{Hom}_O(M, Q)) = \mathrm{Hom}_{D_y}(P \otimes_O M, Q).$$

The connection on the Hom is forced by this equality. Tensoring with M is exact. Its right adjoint is exact as well: after choosing an O-basis, its underlying module is a product of copies of Q, and products of modules are exact. Both functors preserve quasi-isomorphisms, so their adjunction descends to derived categories. Taking $Q = M$ proves the assertion. □

This affine internal Hom need not commute with localization of an infinite-rank source. The supported calculation below uses the restriction of a specified coefficient object, not a new internal Hom formed on the open set.

5.3.1 The supported degrees

Use right densities dy on X and $dy \wedge du$ on $Y = X^2$. The monoidal left-to-right normalization for the diagonal tensor is $M \mapsto M^r[1]$, with $M^r = \omega_X \otimes_O M$; its units are $J_X = \omega_X[1]$ and $J_Y = \omega_Y[2]$. Subscripts $\mathcal{D}$ on Hom denote right differential modules. For any left D_y-module L with connection ∇_L, write $G(L) = [L \xrightarrow{\nabla_L} L]$ in degrees zero and one. Its normal direct image has the model

$$\Delta_+ L = L[\eta], \qquad \partial_u = \eta, \qquad u = -\partial_\eta.$$

Its normal Spencer complex is $[L[\eta] \xrightarrow{\eta} L[\eta]]$ in degrees zero and one. Project the degree-one term to its constant coefficient and include that coefficient back. The homotopy sends $\eta^m \ell$ in degree one to $\eta^{m-1}\ell$ for $m \geq 1$ and to zero for $m = 0$. It satisfies $dh + hd = 1 - ip$ on every monomial. In tangential form degree a, give both the normal differential and this homotopy the sign $(-1)^a$. The tangential terms then cancel, proving

$$R\,\mathrm{Hom}_{\mathcal{D}_Y}(J_Y, \Delta_*(L^r[1])) \simeq G(L)[-2]. \tag{5.21}$$

One shift comes from the normal Spencer complex, and the other is the density shift $[1-2]$.

In particular, suppose a coefficient object E_2 has $\Delta^! E_2 = N^r[1]$. Recollement and open adjunction give

$$\mathrm{Fib}\Big(R\,\mathrm{Hom}_{\mathcal{D}_Y}(J_Y, E_2) \longrightarrow R\,\mathrm{Hom}_{\mathcal{D}_U}(J_U, j^! E_2)\Big) \simeq G(N)[-2]. \tag{5.22}$$

The fibre has cohomology only in degrees two and three. Restriction therefore induces an equivalence on the unshifted mapping spaces. For a source $J_Y[-d]$, the fibre instead is $G(N)[d-2]$. Thus shifted sources detect supported operators invisible to the unshifted comparison.

For example, let P_0 be projection to the constant state. Both P_0 and $\mathtt{D} = \partial_{x_1}$ commute with T and are horizontal endomorphisms. Their commutator is nonzero, since $(P_0\mathtt{D} - \mathtt{D}P_0)J = 1$. They give classes in the degree-zero supported fibre for $d = 2$. The inclusion of that fibre need not be injective on cohomology; realizing the operators by full boundary actions is an additional question.

5.3.2 Translation towers

Give x_n energy n and let V_+ be the positive-energy states. If x_N is the largest variable occurring in a polynomial f, the coefficient of x_{N+1} in Tf is $N\partial_{x_N} f$. Descent on N proves $\ker T = k1$. Choose a graded complement U of TV_+ in V_+, containing J and J^2. Induction on energy gives

$$V = k1 \oplus k[T] \otimes_k U, \qquad \mathcal{V} \cong O \oplus P, \qquad P = D_y \otimes_k U. \tag{5.23}$$

The second map sends $f(y)\partial_y^n \otimes u$ to $f(y)T^n u$; PBW and the connection identify the two sides. If $p(E)$ is the partition number, $\dim U_1 = 1$ and $\dim U_E = p(E) - p(E-1)$ for $E \geq 2$.

On one tower write t for T, so its connection is $\partial_y + t$ on $k[y,t]$. Define

$$\operatorname{red}\left(\sum_n f_n(y)t^n\right) = \sum_n (-1)^n f_n^{(n)}(y), \tag{5.24}$$

$$H_P(f(y)t^n) = \sum_{j=0}^{n-1} (-1)^j f^{(j)}(y)t^{n-1-j}, \qquad H_P(f) = 0. \tag{5.25}$$

If j includes the constant tower coefficient, cancellation gives

$$\operatorname{red}\nabla = 0, \qquad \nabla H_P = 1 - j\operatorname{red}, \qquad H_P\nabla = 1.$$

On the vacuum use evaluation at $y = 0$ and $H_O(f) = \int_0^y f(s)\,ds$. Together these maps define a contraction

$$G(\mathcal{V}) \rightleftarrows k \oplus W[-1], \qquad W = k[y] \otimes_k U. \tag{5.26}$$

Write $H_{\mathcal{V}}$ for its homotopy and red_+ for reduction of its positive-state summand.

Evaluation at the vacuum splits N as differential modules:

$$N = \sigma(\mathcal{V}) \oplus N_0, \qquad \sigma(v) = vP_0, \qquad N_0 = \{A : A_1 = 0\}.$$

For $A \in N_0$ define an O-linear operator $R(A)$ by

$$R(A)1 = R(A)u = 0, \qquad R(A)(T^{n+1}u) = \nabla R(A)(T^n u) - A(T^n u). \tag{5.27}$$

Every column requires finitely many operations, so $R(A)$ is an allowed endomorphism. If $F \in N_0$, let $\Phi(F)$ be the horizontal operator with $\Phi(F)(T^n u) = \nabla^n F(u)$ and $\Phi(F)1 = 0$. Recursion then gives

$$\nabla_N R(A) = A, \qquad R(\nabla_N F) = F - \Phi(F).$$

Theorem 5.8. *The operator complex has a contraction*

$$G(N) \rightleftarrows B_0 \oplus W[-1], \qquad B_0 = \operatorname{End}_{D_y}(\mathcal{V}).$$

Its degree-one projection is $A \mapsto \operatorname{red}_+(A_1)$*, and its homotopy is*

$$H_N(A) = \sigma(H_{\mathcal{V}}(A_1)) + R(A - \sigma(A_1)). \tag{5.28}$$

For a basis $\mathcal{U}$ *of* U*, in the order* (O, P) *one has*

$$B_0 = \begin{pmatrix} k & \prod_{u\in\mathcal{U}} k[y] \\ 0 & \operatorname{Mat}_{\mathrm{col},\mathcal{U}}(D_y^{\mathrm{op}}) \end{pmatrix}. \tag{5.29}$$

The complex computes $R\operatorname{Hom}_{D_y}(\mathcal{V},\mathcal{V})$.

Proof. The preceding contractions apply to the two summands of N. The degree-zero projection is the horizontal operator whose value on the vacuum is the constant-state coefficient of A_1 at $y = 0$ and whose value on each primitive u is $A(u)$. The degree-one section is $w \mapsto \sigma(jw)$. The displayed identities prove $dH_N + H_N d = 1 - ip$, $pi = 1$, $H_N i = pH_N = H_N^2 = 0$.

The free module P has arbitrary images on its basis U. The connection $\partial_y + t$ is injective on a polynomial tower, so $\mathrm{Hom}_{D_y}(O, P) = 0$. A map $O \to O$ is a scalar; a map $P \to O$ chooses a polynomial for each generator. Endomorphisms of the direct sum of copies of D_y are the column-finite matrices over D_y^{op}. This proves (5.29).

Finally P is projective and $[D_y \xrightarrow{(-)\partial_y} D_y]$ resolves O in degrees $-1, 0$. Equivalently, the one-dimensional relative Spencer resolution for the O-free module $\mathcal{V}$ gives the commutator complex (5.20). Thus it is a model of the stated derived Hom. □

Its cohomology product has $W^2 = 0$. A matrix $b = \left(\begin{smallmatrix} a & v \\ 0 & C \end{smallmatrix}\right)$ acts on the left of W through C and on the right by a. On $D_y/\partial_y D_y = k[y]$, right multiplication by ∂_y acts as $f \mapsto -f'$. This identifies the cohomology algebra. The contraction was a contraction of complexes; it has made no assertion about higher compositions.

5.3.3 *The charge line and the comparison homotopy*

Define the polynomial current cochain algebra

$$Q_\kappa = k[y, x_1, x_2, \dots] \otimes \Lambda(\xi_0, \xi_1, \dots), \qquad dx_n = \kappa n \xi_n, \qquad d\xi_n = 0.$$

The ghosts ξ_n have degree one. Translation extends by $T\xi_n = (n+1)\xi_{n+1}$. Take its root de Rham complex $C_\kappa = Q_\kappa \oplus dy\, Q_\kappa$, with total differential $d + dy(\partial_y + T)$ and forms written on the left; in particular, $d(dy\, q) = -dy\, dq$. Ghost deletion and multiplication define the cochain map

$$\rho_\kappa : C_\kappa \longrightarrow G(N), \qquad f \longmapsto L_f, \tag{5.30}$$

with every positive ghost monomial sent to zero. This is the coefficient map that forgets the regular-current equivariance. It keeps the differential-module connection.

Proposition 5.9. *The current complex has cohomology k in degree zero and $k\gamma$ in degree one, where*

$$\gamma = \xi_0 + dy\, J/\kappa.$$

The maps induced by ρ_κ are $k \hookrightarrow B_0$, $a \mapsto a\,\mathrm{id}$, and $k\gamma \hookrightarrow W$, $\gamma \mapsto J/\kappa$.

Proof. Contract every Koszul pair (x_n, ξ_n), $n > 0$, by the Euler homotopy: on positive total polynomial-plus-ghost degree divide $\sum_{n>0} x_n(\kappa n)^{-1}\partial_{\xi_n}$ by that degree. The remaining complex is $k[y] \otimes \Lambda(\xi_0)$. Perturb by root translation. The perturbation series terminates after one root form, since $dy^2 = 0$, and the induced translation on the zero ghost vanishes because $T\xi_0 = \xi_1$. The transferred differential is $dy\partial_y$, while the corrected ghost representative is γ. Directly, its two differentials are $dy\xi_1$ and $-dy\xi_1$. Polynomial integration in y leaves the two stated classes. Finally $\rho_\kappa \gamma = dy\, L_J/\kappa$, whose normal form is J/κ by Theorem 5.8. □

Corollary 5.10. *For $A \in N$, the class of $dy\, A$ lifts through ρ_κ exactly when* $\mathrm{red}_+(A_1) \in kJ$. *If this normal form is cJ/κ, a lift is $c\gamma$, with comparison homotopy $H_N(A - cL_J/\kappa)$. The fibre satisfies*

$$H^1 \mathrm{Fib}(\rho_\kappa) = B_0/k\,\mathrm{id}, \qquad H^2 \mathrm{Fib}(\rho_\kappa) = W/kJ,$$

and has no other cohomology.

Proof. The image criterion follows from Proposition 5.9. When it holds, the degree-one projection of $A - cL_J/\kappa$ vanishes, so the contraction identity gives $\nabla_N H_N(A - cL_J/\kappa) = A - cL_J/\kappa$. The two injections in that proposition give the fibre groups. □

For example, L_{J^2} and yL_J do not lift: their normal forms are J^2 and yJ. A scalar charge q does integrate, with $J_0 = q\,\mathrm{id}$ and $\nabla_q = \partial_y + T + (q/\kappa)L_J$. Only the $n = 1$ moving-mode equation changes, by $[(q/\kappa)L_J, \kappa\partial_{x_1}] = -q$. When $q \neq 0$ the vector 1 is no longer horizontal.

There are two distinct completions. For a new parameter g, suppose $A(g) \in gN[[g]]$ kills 1. The map

$$U_g 1 = 1, \qquad U_g(T^n u) = (\partial_y + T + A(g))^n u$$

extends $k[y][[g]]$-linearly and intertwines the two connections. It is the identity modulo g, so it has a g-adic inverse. Every coefficient of every column is a polynomial. This trivializes the connection with a chosen horizontal vacuum, but leaves its horizontal automorphisms as stabilizers.

Completing the root instead gives $\widehat{V} = V[[y]]$. The gauge e^{-yT} turns its connection into ∂_y; on continuous endomorphisms it acts by conjugation. Formal integration is onto, hence

$$H^0G(\widehat{N}) = \mathrm{End}_k(V), \qquad H^1G(\widehat{N}) = 0, \qquad \widehat{N} = \mathrm{End}_{k[[y]],\mathrm{cont}}(V[[y]]). \tag{5.31}$$

In this category the charge has the primitive

$$g_\kappa(y) = \frac{1}{\kappa}\sum_{n\geq 0}\frac{(-1)^n y^{n+1}}{(n+1)!}T^n J, \qquad (\partial_y + T)g_\kappa = J/\kappa.$$

Its infinitely many nonzero state coefficients prevent it from being a polynomial-root primitive. Completion has changed the coefficient category and the cohomology.

5.4 A NONFORMAL DERIVED OPERATOR ALGEBRA

One free translation tower and the vacuum already detect the composition missing from the linear contraction. Write $D = D_y$ and choose the tower generated by J. Let $P = D$ and $O = D/D\partial$. Resolve O by

$$Q = [D \xrightarrow{R_\partial} D], \qquad Q^{-1} = D, \quad Q^0 = D, \quad R_\partial(a) = a\partial.$$

The Weyl PBW basis proves injectivity of R_∂ and identifies its cokernel with O. Put

$$B = D^{\mathrm{op}} = k\langle y, p\rangle/(py - yp + 1), \qquad \mathscr{E} = \mathrm{End}_D(P \oplus Q).$$

Thus p represents right multiplication by ∂, with $[p, y] = -1$. Products in $\mathscr{E}$ are matrix compositions, the rightmost map acting first. The free cells P, Q^0, Q^{-1} have degrees $(0, 0, -1)$; the only nonzero entry of their differential is $(Q^0, Q^{-1}) = p$. Hence

$$\partial X = dX - (-1)^{|X|}Xd$$

makes $\mathscr{E}$ a dg algebra of ranks 2, 5, 2 over B in degrees $-1, 0, 1$. It models the marked derived endomorphisms of $P \oplus O$. The splitting (5.23) embeds it as a corner in a projective model of the full Heisenberg module.

5.4.1 Polynomial division gives a contraction

Use normal monomials $y^a p^b$. Define

$$r_L(y^a p^b) = \partial_y^b(y^a), \qquad \ell(y^a p^b) = \sum_{j=0}^{\min(a,b-1)} \frac{a!}{(a-j)!} y^{a-j} p^{b-1-j}, \tag{5.32}$$

$$r_R\left(\sum_b f_b(y) p^b\right) = f_0(y), \qquad r\left(\sum_b f_b(y) p^b\right) = \sum_{b\geq 1} f_b(y) p^{b-1}.$$

Empty sums are zero. The relation $yp = py + 1$ gives

$$F = p\ell(F) + r_L(F) = r(F)p + r_R(F). \tag{5.33}$$

In particular $pB = \ker r_L$ and $Bp = \ker r_R$, and neither contains 1.

The block $\mathrm{Hom}(P, Q)$ is $[B \xrightarrow{p(-)} B]$ in degrees -1, 0. Its projection is r_L, its section includes polynomials, and its homotopy is ℓ. The block $\mathrm{Hom}(Q, P)$ is $[B \xrightarrow{-(-)p} B]$ in degrees 0, 1; its maps are r_R, polynomial inclusion and $-r$.

On $\mathrm{End}(Q)$ write a degree-zero map as (a, c), acting on Q^0, Q^{-1}, and a degree-one map as $z : Q^{-1} \to Q^0$. Then

$$\delta h = (ph, hp), \qquad \delta(a, c) = pc - ap. \tag{5.34}$$

With $I(f) = \int_0^y f(t)dt$, its projection to k is $r_L(a)(0)$, the section is $s \mapsto (s, s)$, and its homotopy is

$$H(a, c) = \ell(a), \qquad H(z) = (-I(r_L z), \ell(z) - I(r_L z)). \tag{5.35}$$

It is zero on degree -1.

Proposition 5.11. *These block maps, and the identity on* $\mathrm{End}(P) = B$*, form a contraction of* $\mathscr{E}$ *onto its cohomology with* $\pi i = 1$*,* $H^2 = Hi = \pi H = 0$*. In the order* (P, O) *the result is*

$$H^0\mathscr{E} = \begin{pmatrix} B & 0 \\ B/pB & k \end{pmatrix}, \qquad H^1\mathscr{E} = \begin{pmatrix} 0 & B/Bp \\ 0 & 0 \end{pmatrix}, \qquad H^{\neq 0,1}\mathscr{E} = 0.$$

Every map in this contraction is polynomial on each input.

Proof. The off-diagonal identities are the two divisions in (5.33). For the vacuum block, the maps $(a, c) \mapsto r_L(a)$ and $z \mapsto r_L(z)$ map the complex to $[k[y] \xrightarrow{-\partial_y} k[y]]$. A section sends f to (f, f) and g to g, since $pf - fp = -f'$. In the kernel, write $a = pa_1$, $z = pz_1$. The differential becomes

$$B \longrightarrow B \oplus B \longrightarrow B, \qquad h \longmapsto (h, hp), \quad (a_1, c) \longmapsto c - a_1 p.$$

The homotopy sends (a_1, c) to a_1 and z_1 to $(0, z_1)$. Contract the remaining derivative by $-I$. Composition gives (5.35) and $\delta H + H\delta = 1 - i\pi$. The identity $\ell(ap) = \ell(a)p + r_L(a)$ verifies the middle degree directly. Integration vanishes at zero and ℓ kills polynomials, giving the side conditions. All maps preserve the two object idempotents. □

5.4.2 *The full Massey coset*

Consider the closed maps

$$a : P \to Q, \quad a^{\circ} = 1; \qquad b : Q \to P[1], \quad b^{-1} = 1; \qquad c : P \to P, \quad c = p.$$

Their degrees are 0, 1, 0, and $ba = 0$, while $cb = \delta h$ for $h : Q^{\circ} \to P$, $h = -1$. For primitives $\delta h = cb$, $\delta k = ba$, use the defining Massey representative $ha - ck$. Taking $k = 0$ gives -1_P.

Theorem 5.12. *The exact composable Massey coset is*

$$\langle c, b, a\rangle = -1 + pB \ \subset H^{\circ}\operatorname{End}(P) = B. \tag{5.36}$$

It excludes zero. Both $\mathscr{E}$ and the full polynomial-root derived endomorphism algebra of $\mathcal{V}$ are nonformal.

Proof. A second primitive of cb differs by a degree-zero cocycle in $\operatorname{Hom}(Q, P)$. There are no such nonzero cocycles, since its differential is right multiplication by $-p$. A primitive of $ba = 0$ can differ by any element of B. The indeterminacy is therefore exactly pB. The remainder map r_L annihilates this ideal and sends 1 to 1, so zero is excluded.

In a dg algebra with zero differential, a defined triple Massey product contains zero, by choosing zero primitives. A dg quasi-isomorphism preserves the Massey coset: it sends defining systems to defining systems, and surjectivity on cohomology recovers all changes of primitives, whose images are the two indeterminacy subspaces. Applying this in both directions along a quasi-isomorphism zigzag proves nonformality; equivalently one may use the minimal-model criterion of [59, §§3.3–3.4].

For the full module, sandwich a defining system between the idempotents of the chosen $P \oplus O$ summand. Its $P \to P$ projection is a defining system above. Additional matrix blocks do not change the projected indeterminacy: the degree-zero $O \to P$ cohomology is still zero, and the $P \to P$ corner is B. A zero total Massey class would thus give zero in (5.36), which is impossible. □

Replacing a by the representative $a_f = f(y)$ gives $\langle c, b, a_f\rangle = -f(y) + pB$, nonzero for every nonzero polynomial f. The obstruction concerns the whole coset, not a particular transferred ternary coefficient. At level κ, a is the restriction of $J_1/\kappa = \partial_{x_1}$ to the J-tower, b is its charge class $[J]$, and c is its translation. Thus this composition uses three of the operators already present in the current comparison.

5.4.3 *Higher compositions and the ordinary centre*

The polynomial contraction gives all higher operations explicitly. Let $\mathcal{H} = H^*\mathscr{E}$, and let $s : \mathscr{E} \to \mathscr{E}[1]$ have degree -1. On the tensor coalgebra of suspended matrices use

$$b_1(sx) = -s\partial x, \qquad b_2(sx, sy) = (-1)^{|x|}s(xy), \qquad h_s = -sHs^{-1}.$$

Extend b_1, b_2 as coderivations, and write i_s, π_s for the suspended inclusion and projection. The tensor contraction is

$$\mathbf{H}|_{\mathscr{E}[1]^{\otimes n}} = \sum_{j=1}^{n} (i_s\pi_s)^{\otimes(j-1)} \otimes h_s \otimes 1^{\otimes(n-j)},$$

with graded tensor signs. Let i_T, π_T be the tensor extensions of i_s, π_s. Homological perturbation gives

$$\begin{aligned} b_{\mathcal{H}} &= \pi_T b_2(1 + \mathbf{H}b_2)^{-1} i_T, \\ \mathbf{I} &= (1 + \mathbf{H}b_2)^{-1} i_T, \qquad \mathbf{P} = \pi_T(1 + b_2\mathbf{H})^{-1}, \qquad \mathbf{H}' = \mathbf{H}(1 + b_2\mathbf{H})^{-1}. \end{aligned} \tag{5.37}$$

Each binary merger lowers length, so at input length n these inverse series have at most $n - 1$ merger stages. Expanding the contraction identity yields $b_{\mathcal{H}}^2 = 0$, $(b_1 + b_2)\mathbf{I} = \mathbf{I}b_{\mathcal{H}}$ and $\mathbf{PI} = 1$, with comparison

homotopy $\mathbf{H}'$. Deconcatenation and the tensor homotopy give the coalgebra identities. Tensoring over the two scalar idempotents retains composable strings; the side conditions give strict units. This constructs a minimal A_∞ category in every arity. Its deciding value is

$$(b_{\mathcal{H}})_3(sc, sb, sa) = -s1_P :$$

the merger ba vanishes, while the other merger has homotopy $h = -1$. The Massey coset prevents removal of all higher operations by a different minimal model.

Its ordinary bimodule centre has a different answer. The free module $P = D$ generates $\mathrm{Perf}(D)$ and Q is a two-term cone of free modules. Thus $P \oplus Q$ is another generator. The functors

$$R\,\mathrm{Hom}_D(P \oplus Q, -), \qquad - \otimes^L_{\mathscr{E}} (P \oplus Q)$$

are inverse on the generated perfect categories: evaluation is an equivalence on the generator and extends by cones, shifts and retracts. Derived Morita invariance of Hochschild cochains [60, §5.4] gives

$$\mathrm{HH}^*(\mathscr{E}) \cong \mathrm{HH}^*(D^{\mathrm{op}}) \cong \mathrm{HH}^*(D) = k. \tag{5.38}$$

For the last equality, resolve the Weyl algebra as a bimodule by its two-generator PBW Koszul complex. Its cochains have differential given by $[\partial, -] = \partial_y$ and $[y, -] = -\partial_\partial$, hence are the polynomial de Rham complex after a change of one exterior generator. Exactness of the bimodule resolution follows from its associated graded polynomial Koszul resolution and the bounded-below PBW filtration. Euler contraction, divided by positive polynomial-plus-form degree, leaves only the constant class. This proves (5.38) without a root completion.

Nonformality of a selected derived operator algebra is compatible with a scalar ordinary centre of its perfect module sector. The centre of the full geometric boundary theory requires its own category and its own action comparison.

5.5 The level ring and the forgetful fibre

The nonzero-level calculation divided by κ. To retain the singular level, work over $R = k[\lambda]$ and use a separate symbol for the Koszul differential:

$$Q = k[x_1, x_2, \ldots] \otimes \Lambda(\xi_0, \xi_1, \ldots), \qquad d_0 x_n = n\xi_n, \quad d_0 \xi_n = 0.$$

Translation is $T x_n = n x_{n+1}$ and $T\xi_n = (n+1)\xi_{n+1}$. It commutes with d_0 on generators and hence everywhere. With forms on the left define

$$\begin{aligned} C_\lambda &= Q_R[y] \oplus dy\, Q_R[y], \qquad Q_R = R \otimes_k Q, \\ \mathsf{D}_\lambda(q + dy\, r) &= \lambda d_0 q + dy\big((\partial_y + T)q - \lambda d_0 r\big). \end{aligned} \tag{5.39}$$

This is the polynomial-root current complex at the universal level. There are only two root-form degrees, so its totalization needs no additional convention. The parameter λ has degree zero.

5.5.1 *Translation coinvariants before specialization*

Let Q_+ be the augmentation ideal. Translation is injective on each Q_+^q. To see this, choose the largest occupied index of an even or an odd variable. Its translate introduces a new variable of that type, with coefficient its corresponding partial derivative. No other term can cancel that new variable. Descending

on the occupied indices removes every variable, including ξ_0; only the unit in the full algebra can be translation-invariant.

Put $K^q = k[y] \otimes Q_+^q$. The operator $\nabla = \partial_y + T$ is also injective on K^q: the highest power of y in a nonzero kernel element would have a nonzero coefficient killed by T. Define the vector spaces

$$\mathsf{W}^q = K^q/\nabla K^q, \qquad \overline{d}[q] = [d_0 q], \qquad \mathsf{B}^q = \operatorname{im}(\overline{d} : \mathsf{W}^{q-1} \to \mathsf{W}^q) \quad (q \geq 1). \tag{5.40}$$

They are k-vector spaces. The quotient is not an O-module quotient, since $[\nabla, y] = 1$.

For computation, choose translation primitives in each state energy and exterior degree. These pieces are finite dimensional, including the energy-zero ghost. The same tower division as (5.24) gives the finite normal form

$$[f(y)T^n u] = (-1)^n f^{(n)}(y)[u].$$

Apply d_0, decompose its result into towers, and reduce to compute $\overline{d}$. In degree zero $\mathsf{W}^0 = W$ of (5.26), as vector spaces.

Lemma 5.13. *The cohomology of* $(\mathsf{W}, \overline{d})$ *is the one-dimensional span* $k \cdot [J]$ *in degree zero and zero in positive degrees.*

Proof. The Euler homotopy for the positive Koszul pairs is $\sum_{n>0}(x_n/n)\partial_{\xi_n}$ divided by positive total degree. It leaves only ξ_0 in the augmentation ideal, so $H^*(K, d_0) = k[y]\xi_0$ in degree one. Translation acts trivially there, because $T\xi_0 = \xi_1 = d_0 J$. Thus ∇ acts on this cohomology as ∂_y, which is onto with kernel $k\xi_0$. The exact sequence of complexes

$$0 \longrightarrow K \xrightarrow{\nabla} K \longrightarrow \mathsf{W} \longrightarrow 0$$

now proves the assertion. The connecting preimage of ξ_0 is $[J]$, since $d_0 J = \nabla \xi_0$. □

Theorem 5.14. *There is an integral quasi-isomorphism of complexes*

$$C_\lambda \simeq R \oplus (\mathsf{W} \otimes_k R, \lambda\overline{d})[-1]. \tag{5.41}$$

Consequently

$$\begin{aligned} H^0 C_\lambda &= R, & H^1 C_\lambda &= R\Gamma_\lambda, & \Gamma_\lambda &= \lambda\xi_0 + dy\, J, \\ H^{q+1} C_\lambda &= \mathsf{B}^q \otimes_k R/(\lambda) & & (q \geq 1). \end{aligned} \tag{5.42}$$

Proof. The constant-state root complex contracts to R. On the positive part the map $q + dy\, r \mapsto [r]$ has target differential $-\lambda\overline{d}$, the sign of the shift. In its kernel $r = \nabla s$ has a unique polynomial solution s. The homotopy sends $dy\nabla s$ to s and the root-degree-zero part to zero. The terms $\lambda d_0 s$ cancel in $\mathsf{D}_\lambda H + H\mathsf{D}_\lambda$, leaving the identity. This proves the quasi-isomorphism without inverting the level.

Multiplication by λ is injective on each extended vector space. Thus the cycles of $\lambda\overline{d}$ are the cycles of $\overline{d}$, and its boundaries are $\lambda\mathsf{B}^q$. Lemma 5.13 gives the groups. Finally Γ_λ is closed, since its two nonzero differentials are $\lambda dy\xi_1$ and $-\lambda dy\xi_1$, and its image in the quotient is $[J]$. □

This is a linear quotient calculation. Multiplication by y does not descend to W, and no multiplicative quotient of C_λ has been asserted.

5.5.2 An equivalence of arrows

Over the level ring use the actual full operator modules

$$\mathcal{N}_R = \mathrm{End}_{R[y]}(R[y] \otimes_k V), \qquad \mathcal{B}_R = \mathrm{End}_{D_y \otimes R}(R[y] \otimes_k V).$$

These permit different parameter degrees in different columns; they are not replaced by tensoring N or B_0 with R. The tower and column recurrences work directly over R, giving

$$G(\mathcal{N}_R) \simeq \mathcal{B}_R \oplus (\mathsf{W}^0 \otimes_k R)[-1].$$

Let $\rho : C_\lambda \to G(\mathcal{N}_R)$ be ghost deletion followed by multiplication. Define its fibre by

$$\mathcal{F}_\lambda^n = C_\lambda^n \oplus G(\mathcal{N}_R)^{n-1}, \qquad d(q, A) = (\mathsf{D}_\lambda q, \rho q - d_G A). \tag{5.43}$$

Its second coordinate is the retained comparison path.

The contractions of source and target alone would not identify this fibre. We also need their comparison on the arrow ρ. Vacuum evaluation splits $\mathcal{N}_R = \sigma(\mathcal{V}_R) \oplus \mathcal{N}_{0,R}$, with $\sigma(v) = vP_0$ and $\mathcal{N}_{0,R}\,1 = 0$. On the first summand, evaluation of ρ is exactly ghost deletion, so the translation quotients commute.

On the second summand, column recursion contracts positive root degree. Its remaining horizontal component is homotopic to $a \mapsto a(1 - P_0)$ on constants and zero on the positive quotient. Here is the homotopy. On $d\,y\,v$ with no ghosts, take the horizontal operator whose vacuum value is zero and whose primitive-column value on u is $H_{\mathcal{V}}(v)u$; extend by $T^n u \mapsto \nabla^n(H_{\mathcal{V}}(v)u)$. On positive ghost inputs it is zero. The identity

$$H_{\mathcal{V}} \nabla f = f - \epsilon(f)(0)$$

shows that its composite with D_λ is exactly the horizontal component of multiplication by f minus $\epsilon(f)(0)(1 - P_0)$. Here ϵ extracts the constant state. The first summand contributes aP_0. Thus the arrow is equivalent, with its homotopy supplied, to

$$R \oplus (\mathsf{W}_R, \lambda\overline{d})[-1] \longrightarrow \mathcal{B}_R \oplus \mathsf{W}_R^0[-1], \qquad (a, w) \longmapsto (a\,\mathrm{id}, \pi_0 w), \tag{5.44}$$

where $\mathsf{W}_R = \mathsf{W} \otimes_k R$ and π_0 is the brutal degree-zero quotient.

Theorem 5.15. *There is a constructed equivalence of complexes*

$$\mathcal{F}_\lambda \simeq (\mathcal{B}_R / R\,\mathrm{id})[-1] \oplus (\mathsf{W}_R^{\geq 1}, \lambda\overline{d})[-1], \tag{5.45}$$

where $\mathsf{W}^{\geq 1}$ *is the brutal subcomplex. Its cohomology is*

$$\begin{aligned} H^1\mathcal{F}_\lambda &= \mathcal{B}_R / R\,\mathrm{id}, \\ H^2\mathcal{F}_\lambda &= \mathsf{B}^1 \otimes_k R, \\ H^n\mathcal{F}_\lambda &= \mathsf{B}^{n-1} \otimes_k R/(\lambda) \quad (n \geq 3), \end{aligned} \tag{5.46}$$

and all other groups vanish.

Proof. In (5.44), the scalar inclusion splits by vacuum constant evaluation. The second map is degreewise onto, so its fibre is its kernel $\mathsf{W}_R^{\geq 1}[-1]$. Explicitly, the redundant two terms in its cone are the graph $w \mapsto (-\lambda\overline{d}w, w)$. Projection $(v, u) \mapsto v + \lambda\overline{d}u$ and homotopy $(v, u) \mapsto u$ cancel that graph without dividing by λ. This proves the equivalence. In degree two the incoming $\lambda\mathsf{B}^1$ disappeared with the cancelled term, leaving the free module B^1_R. In higher degrees the boundaries remain $\lambda\mathsf{B}^{n-1}$, giving the formula. □

The exact sequence in degree two is therefore

$$0 \longrightarrow (\mathsf{W}^0/(k\cdot[J]))\otimes_k R \xrightarrow{\lambda\overline{d}} \mathsf{B}^1\otimes_k R \longrightarrow \mathsf{B}^1\otimes_k R/(\lambda) \longrightarrow 0. \tag{5.47}$$

Under the isomorphism $\overline{d}:\mathsf{W}^0/(k\cdot[J])\to\mathsf{B}^1$, the first map is multiplication by λ. A concrete instance uses $F=J^2/2$, for which $d_0F=x_1\xi_1$. The fibre differential gives

$$d(-dy\,F,0)=(\lambda dy\,x_1\xi_1,-dy\,L_F).$$

Hence

$$\boxed{\lambda[(dy\,x_1\xi_1,0)]=[(0,dy\,L_{J^2/2})].} \tag{5.48}$$

After its path is forgotten, the current class on the left is λ-torsion. In the fibre its lift is a nonzero free class. Indeed, J^2 is a translation primitive independent of J, and the kernel of $\overline{d}$ in degree zero is only $k\cdot[J]$.

The integral charge-lifting test follows from the same arrow:

$$[dy\,A]\text{ lifts through }\rho \quad\Longleftrightarrow\quad \mathrm{red}_+(A_1)\in RJ. \tag{5.49}$$

If the normal form is $a(\lambda)J$, the lift is $a\Gamma_\lambda$ and the comparison homotopy is $H_N(A-aL_J)$. No inverse level is used. Once a base lift for a horizontal source shifted by s has been chosen, its lift space has homotopy groups $H^{s-j}\mathcal{F}_\lambda$ in degrees $j\geq 0$. Without a base lift the fibre may be empty.

5.5.3 *The singular level*

The terms of W_R are free over R. The quotient $\mathcal{B}_R/R\,\mathrm{id}$ is a direct summand of the torsion-free module $\mathcal{B}_R$ and hence is flat over the PID R. Derived specialization is therefore computed by reducing the displayed fibre complex. At $\lambda=0$ its positive differential vanishes, giving

$$H^{q+1}(\mathcal{F}_\lambda\otimes^L_R k)=\mathsf{W}^q \qquad (q\geq 1). \tag{5.50}$$

In degree two the new classes occur in

$$0\longrightarrow\mathsf{B}^1\longrightarrow\mathsf{W}^1\xrightarrow{\overline{d}}\mathsf{B}^2\longrightarrow 0.$$

They are the Tor contribution of integral H^3.

For example, $[\xi_0\xi_1]\neq 0$ in W^2. A polynomial preimage under ∇ would have root degree zero, by injectivity of T on $\mathcal{Q}_+$. It would then give a two-ghost translation preimage in energy zero, where there is no such monomial. Since $d_0(x_1\xi_0)=-\xi_0\xi_1$, this is a nonzero element of B^2. Thus $dy\,x_1\xi_0$ becomes a degree-two cocycle at zero level without being an integral cocycle. The same torsion prevents an R-linear deformation retraction of this fibre complex onto its cohomology: its nonzero λ-torsion could not have a cochain section into the torsion-free terms. This excludes that linear retraction, not every possible derived replacement.

5.5.4 *The strict mode-preserving gauge groupoid*

Complete only a new deformation parameter g. On $\mathcal{V}_R[[g]]$ fix

$$J_{-n}=L_{x_n},\qquad J_n=\lambda n\partial_{x_n}\quad(n>0),\qquad J_0=q(g)\,\mathrm{id},\quad q(g)\in gR[[g]].$$

Seek connections $\nabla+A(g)$ with $A\in g\mathcal{N}_R[[g]]$ satisfying $[\nabla+A,J_n]=-nJ_{n-1}$ for every integer n. A mode-preserving gauge is an element of $1+g\mathcal{N}_R[[g]]$ commuting with every nonzero mode.

Theorem 5.16. *Such a connection exists exactly when $q(g) = \lambda a(g)$ for $a(g) \in gR[[g]]$. Every one is*

$$\partial_y + T + a(g)L_J + c(y,g)\,\mathrm{id}, \qquad c(y,g) \in gR[y][[g]]. \tag{5.51}$$

It is mode-preserving gauge equivalent to $\partial_y + T + a(g)L_J$. The stabilizer of that normal form is $1 + gR[[g]]$. Thus the strict formal groupoid is the disjoint union, indexed by $a(g)$, of the classifying groupoids of these scalar units.

Proof. The negative-mode equations force A to commute with all multiplications by x_n. These generate the state module from 1, so $A = L_f$. The equations for $n \geq 2$ give $\lambda\partial_{x_n} f = 0$. Torsion-freeness makes f independent of those variables. The $n = 1$ equation is $\lambda\partial_{x_1} f = q$. Polynomial coefficient comparison forces $q = \lambda a$ and $f = aJ + c$. Conversely this formula satisfies every mode relation.

A gauge commuting with the negative modes is multiplication by a formal unit. Commutation with the positive modes and torsion-freeness make it independent of the states. The unit

$$U = \exp\!\left(-\int_0^y c(t,g)\,dt\right)$$

has polynomial coefficients in every g-degree and satisfies $(\nabla + aL_J + c)U = U(\nabla + aL_J)$. Between two normal forms a state-independent gauge cannot change a, and its derivative must be zero. These are exactly the stated scalar stabilizers. □

This groupoid fixes all nonzero modes. It is not the full derived deformation groupoid, nor the quotient by gauges that change those modes. At zero level the positive modes vanish; then arbitrary multiplication corrections are permitted when $q = 0$. Most are not specializations of (5.51).

Root completion also changes this problem. The series $g_0 = \sum_{n\geq 0}(-1)^n y^{n+1}T^n J/(n+1)!$ satisfies $\nabla g_0 = J$, so the integral charge image $dy\,L_J$ becomes exact. In the completed J-tower the section $1 \mapsto e^{-yT}J = J - Tg_0$ splits the quotient by state translation. This is the completion of the polynomial extension $0 \to P \xrightarrow{R_\partial} P \to O \to 0$. The polynomial projective model need not remain projective in the continuous category. Its nonzero Massey coset and its integral path relation therefore cannot be transported by an undeclared equivalence across this change of coefficients.

5.6 Finite categories of interfaces

An invertible bimodule composes by derived tensor product, as in (5.19). A collection of interfaces with specified junction maps has more data: the centre must act compatibly with every composition of those maps. For two objects, Proposition 1.7 expresses this as the fibre of the difference between the two endpoint characteristic actions on the joining bimodule. With more objects, the longer composable strings contribute further terms.

Let R be a commutative k-algebra and let $\mathcal{A}$ be a dg category with ordered objects $0, \dots, n-1$, scalar endomorphisms $\mathcal{A}(i,i) = R$, and $\mathcal{A}(i,j) = 0$ for $i > j$. Suppose each forward complex M_{ij} is bounded and finite projective over R, and all dg compositions are specified and associative. Let A be the sum algebra, $S = \bigoplus_i Re_i$, and $\overline{A} = \ker(A \to S)$. The normalized relative bimodule bar is

$$P_\ell = A \otimes_S (\overline{A}[1])^{\otimes_S \ell} \otimes_S A, \qquad 0 \leq \ell \leq n-1. \tag{5.52}$$

For an unsuspended tensor of internal degree q and bar length ℓ, the total degree is $q - \ell$. One sign convention is $d_{\mathrm{int}} + (-1)^q \sum_{j=0}^{\ell}(-1)^j d_j$, where d_j multiplies adjacent factors, including the two endpoint actions. Suspension translates this to (5.52).

Theorem 5.17. *The total complex P is a finite homotopically projective A-bimodule resolution of A. Thus the relative derived centre is*

$$Z_R(A) = R\operatorname{Hom}_{A\otimes_R A^{\mathrm{op}}}(A, A) \simeq \operatorname{Hom}_{A^e}(P, A).$$

Its evaluation gives, for every complex F of R-modules,

$$R\operatorname{Hom}_R(F, Z_R(A)) \simeq R\operatorname{Hom}_{A^e}(F \otimes_R^L A, A). \tag{5.53}$$

For every derived commutative scalar extension $R \to R'$,

$$Z_R(A) \otimes_R^L R' \simeq Z_{R'}(A \otimes_R^L R'), \tag{5.54}$$

compatibly with evaluation, cup products and Hochschild insertions.

Proof. A nonzero string of ℓ forward arrows visits $\ell + 1$ strictly increasing objects, so $\ell \le n - 1$. The unit-insertion contraction of the augmented unnormalized bar proves its exactness after forgetting an outer action. The degenerate subcomplex splits off, leaving the normalized resolution.

Each remaining term is induced from a tensor of bounded finite projective coefficient complexes and two representable idempotent modules. Hom into an acyclic bimodule reduces to Hom from that coefficient complex into an acyclic idempotent component, hence is acyclic. The finite totalization is therefore homotopically projective. Tensor–Hom adjunction applied to $F \otimes_R P$ proves (5.53).

Every cochain coefficient is a Hom between bounded finite projective R-complexes. These Homs and the finite sums commute with derived scalar extension. The bar differential, cuts and insertions use the same composition maps after extension. This proves (5.54) with its operations. □

Strictly increasing object indices made the bar finite. With arbitrary dg endpoint algebras the repeated corner inputs of Proposition 1.7 instead require product totalization. We now use the finite case to see what the operations and the parameter fibres retain.

5.7 Koszul parameters and contraction brackets

Let $r \ge 1$, $R = k[h_1, \dots, h_r]$, $I = (h_1, \dots, h_r)$, and $S_0 = R/I$. Let $U_0 = k^r$ have degree-one basis $u_1, \dots, u_r$, and write L_i for exterior multiplication by u_i. Take the arrow complex

$$\begin{aligned} M &= R \otimes_k \Lambda U_0, \qquad d_M = \sum_i h_i L_i, \\ T(M) &= Re_0 \oplus Re_1 \oplus M, \qquad e_1 M e_0 = M, \quad M^2 = 0. \end{aligned} \tag{5.55}$$

The differential is a differential of the arrow module; it is not a derivation of the auxiliary exterior multiplication. Anticommutation gives $d_M^2 = 0$, and the Koszul complex has cohomology S_0 in degree r.

Put $E = \operatorname{End}_R(M)$ with $\delta f = d_M f - (-1)^{|f|} f d_M$. Reverse the one-arrow coordinate in Theorem 1.4. The centre then has the finite model

$$Z^n = (R^2)^n \oplus E^{n-1}, \qquad d(a, b; f) = (0, 0; (b - a)\,\mathrm{id} - \delta f). \tag{5.56}$$

The endpoint terms occur only in degree zero. Use the source-first opposite-cup convention

$$(a, b; f)(a', b'; f') = (aa', bb'; fb' + af'). \tag{5.57}$$

In this convention, for $\sigma f = (0, 0; f)$ of degree $|f| + 1$, the insertion bracket is

$$[\sigma f, \sigma g] = \sigma\big(f g - (-1)^{|f||g|} g f\big). \tag{5.58}$$

Endpoint scalars have zero bracket with the mixed terms. The bracket has degree -1.

These operations follow directly from the one-arrow bar. Its two cuts give (5.57), while insertion into the only arrow input is composition in E. Two mixed cup factors would require two composable forward arrows, which do not exist. The differential identity on endpoints is $bb' - aa' = (b - a)b' + a(b' - a')$; the other identities reduce to R-linearity of δ. On $Z[1]$ the mixed differential is δ, so the bracket differential and Jacobi identities are the graded-commutator identities. This fixes the signs of the comparison as well as its underlying complex.

5.7.1 The integral ring and its bracket

Let ι_i contract by the dual of u_i. The relations are

$$\{L_i, L_j\} = 0, \qquad \{\iota_i, \iota_j\} = 0, \qquad \{\iota_i, L_j\} = \delta_{ij}\,\mathrm{id}\,. \tag{5.59}$$

The operators $L_A \iota_B$, with increasing subsets A, B, form a basis of E: for one generator 1, L, ι, $L\iota$ span all matrices, and graded tensor products give the general case. Moreover $\delta L_i = 0$ and $\delta \iota_i = h_i\,\mathrm{id}$. At fixed creation monomial L_A, the contraction variables are the Koszul resolution of (h_i). Therefore

$$H^q(E) = S_0 \otimes_k \Lambda^q U_0 \quad (0 \le q \le r), \qquad H^q(E) = 0 \text{ otherwise.}$$

Theorem 5.18. *The relative centre of* (5.55) *has*

$$\begin{aligned} \mathrm{HH}^0_R(T) &= R \times_{S_0} R, & \mathrm{HH}^1_R(T) &= 0, \\ \mathrm{HH}^{q+1}_R(T) &= S_0 \otimes_k \Lambda^q U_0 & &(1 \le q \le r), \end{aligned}$$

and no other cohomology. Its positive-degree cup ideal is square-zero; a pair in degree zero acts through its common residue in S_0. For $t_i = [(0, h_i; \iota_i)]$,

$$\mathrm{HH}^0_R(T) = R[t_1, \dots, t_r]/(h_i t_j - h_j t_i,\ t_i t_j - h_i t_j)_{i,j}. \tag{5.60}$$

Proof. The map $R^2 \to H^0(E) = S_0$ is $(a, b) \mapsto b - a$. It is onto with kernel $R \times_{S_0} R$, so the fibre exact sequence gives the groups. The mixed cup product is zero; the endpoint residue gives its scalar action.

The representative of t_i is closed because $\delta \iota_i = h_i$. It maps to $(0, h_i)$ in the fibre product. Every pair is a diagonal scalar plus $(0, c)$ with $c \in I$, so these classes generate. The quadratic relations reduce each polynomial to $a + \sum_i b_i t_i$. It maps to zero exactly when $a = 0$ and $\sum_i b_i h_i = 0$. The first Koszul syzygies $h_i t_j - h_j t_i$ generate this kernel, proving the presentation. □

For $A = \{a_1 < \cdots < a_q\}$ set $\alpha_A = [\sigma L_{a_1} \cdots L_{a_q}]$, of degree $q + 1$. Then for $q \ge 2$,

$$[t_i, \alpha_A] = \begin{cases} (-1)^{j-1} \alpha_{A \setminus \{i\}}, & i = a_j, \\ 0, & i \notin A. \end{cases} \tag{5.61}$$

Indeed, the graded commutator of ι_i with the creation product is precisely this contraction. When $q = 1$ the result is the scalar identity, which is the boundary of an endpoint difference in (5.56), so the bracket is zero. All brackets between two positive classes or two degree-zero classes vanish; the generators supercommute in their respective creation or contraction factors.

For $r = 2$, with $\omega = \alpha_{\{1,2\}}$, this gives

$$[t_1, \omega] = \alpha_{\{2\}}, \qquad [t_2, \omega] = -\alpha_{\{1\}}.$$

The same t_i annihilate every positive class under cup product. Thus the bracket detects a composition invisible to that product. The proof of the integral calculation also works for any regular sequence (h_i) in a commutative $\mathbb{Q}$-algebra: regularity is used in the Koszul cohomology and in its first syzygies.

5.7.2 *The full singular fibre*

The finite bar commutes with derived scalar extension by Theorem 5.17. If some h_i is invertible, $h_i^{-1}\iota_i$ contracts the arrow M and the centre becomes the two scalar endpoint sectors. At the common zero, put $V_0 = \Lambda U_0$. The arrow differential vanishes, and the only centre differential sends the endpoint difference to $\mathrm{id} \in E^0$, in centre degree one. Hence

$$\mathrm{HH}^*_k(T \otimes_R k) = k \ltimes (\mathrm{End}_k(V_0)/k\,\mathrm{id})[-1]. \tag{5.62}$$

Its displayed ideal has square-zero cup product and shifted graded-commutator bracket. The unit has zero bracket. Since

$$\dim_k \mathrm{End}^p(\Lambda U_0) = \sum_j \binom{r}{j}\binom{r}{j+p} = \binom{2r}{r+p},$$

the dimensions are

$$\dim_k \mathrm{HH}^n_k(T \otimes_R k) = \begin{cases} 1 + \binom{2r}{r-1}, & n = 0, \\ \binom{2r}{r} - 1, & n = 1, \\ \binom{2r}{r+n-1}, & n \neq 0, 1. \end{cases} \tag{5.63}$$

Out-of-range binomial coefficients are zero. The total dimension is 4^r. For $r = 2$, degrees $-1, 0, 1, 2, 3$ have dimensions $1, 5, 5, 4, 1$. In contrast, ordinary specialization of integral degree-zero cohomology has basis $1, t_1, t_2$ and dimension three. Specializing the complex retains the missing extension data.

There is no R-linear deformation retraction of (5.56) onto its integral cohomology. The nonzero $\mathrm{HH}^2_R(T)$ is annihilated by h_1, whereas multiplication by h_1 is injective on each free cochain term. A cochain section of this torsion group must be zero and cannot have a left inverse. This argument concerns a linear retraction over R; it is not a classification of all formality zigzags.

5.7.3 *A second-order obstruction*

At $r = 2$ and zero parameters, use the basis $v_0 = 1, v_1 = u_1, v_2 = u_2, v_3 = u_1u_2$ of V_0. Define the degree-one endomorphism f by

$$f(v_0) = v_1, \qquad f(v_1) = v_3, \qquad f(v_2) = f(v_3) = 0.$$

Over $k[\epsilon]/(\epsilon^2)$, the arrow differential ϵf defines a first-order deformation of T_0. But $f^2(v_0) = v_3$, and

$$\tfrac{1}{2}[\sigma f, \sigma f] = \sigma f^2 \neq 0 \quad \text{in } \mathrm{HH}^3_k(T_0). \tag{5.64}$$

It is nonzero in (5.62) because an internal degree-two map is not a scalar identity. Thus the second-order Hochschild equation $d_{\mathrm{Hoch}}g + \frac{1}{2}[\sigma f, \sigma f] = 0$ has no solution. This excludes even curvature-free A_∞ corrections on the same graded module with the same first-order class, although the cup square of σf is zero. By contrast, the original Koszul directions $xL_1 + yL_2$ square to zero and integrate over $k[x, y]$.

5.8 Four composable objects

The preceding category had one arrow direction and no composable pair of nonidentity arrows. Four ordered objects provide the first three-arrow string. Its two composition orders give an associativity equation, and the centre retains that equation in its own differential.

Use chronological products in this section: $a_{ij}a_{jk}$ means first $i \to j$, then $j \to k$, so $e_i a_{ij} = a_{ij} = a_{ij}e_j$. Take objects $0 < 1 < 2 < 3$, scalar identities, and a degree-zero basis arrow a_{ij} for each $i < j$. Set

$$a_{01}a_{12} = pa_{02}, \quad a_{12}a_{23} = qa_{13}, \quad a_{02}a_{23} = ra_{03}, \quad a_{01}a_{13} = sa_{03}. \tag{5.65}$$

All other products are forced by identities or are zero. The only associator of nonidentity arrows is

$$(a_{01}a_{12})a_{23} - a_{01}(a_{12}a_{23}) = Fa_{03}, \qquad F = pr - qs.$$

Thus the associative family is the rank-ten algebra A_B over $B = k[p, q, r, s]/(pr - qs)$.

5.8.1 The centre differential and its operations

Order the edges as 01, 02, 03, 12, 13, 23 and the triangles as 012, 123, 023, 013. A normalized cochain assigns a scalar to each increasing string, multiplying its endpoint arrow. The finite bar gives

$$Z_B(A_B) \simeq [B^4 \xrightarrow{d_0} B^6 \xrightarrow{d_1} B^4 \xrightarrow{d_2} B] \tag{5.66}$$

in degrees zero through three, with

$$\begin{aligned} (d_0\lambda)_{ij} &= \lambda_j - \lambda_i, \\ d_1 f &= \big(p(f_{01} + f_{12} - f_{02}), q(f_{12} + f_{23} - f_{13}), \\ &\qquad\qquad r(f_{02} + f_{23} - f_{03}), s(f_{01} + f_{13} - f_{03})\big), \\ d_2(\alpha, \beta, \gamma, \delta) &= -r\alpha + s\beta - p\gamma + q\delta. \end{aligned} \tag{5.67}$$

Indeed, in arity zero the two Hochschild endpoint terms give d_0. On a triangle the three faces have the same composition scalar, giving d_1. On 0123 the four terms give d_2. Before imposing the relation, direct substitution gives

$$d_1 d_0 = 0, \qquad d_2 d_1 f = -F(f_{01} + f_{12} + f_{23} - f_{03}). \tag{5.68}$$

The square-zero condition remembers the original associator.

Write $m_{012} = p, m_{123} = q, m_{023} = r, m_{013} = s$. For positive arities n, m, the cup product is

$$(f \smile g)_{i_0 \cdots i_{n+m}} = m_{i_0 i_n i_{n+m}} f_{i_0 \cdots i_n} g_{i_n \cdots i_{n+m}}. \tag{5.69}$$

An arity-zero factor acts by the appropriate first or last endpoint scalar. A product with no increasing string is zero. The bar cut proves this formula; two successive cuts prove associativity using $pr = qs$, and the faces give the Leibniz identity. Its unit is (1, 1, 1, 1).

For insertion, use $f \circ g = \sum_{j=0}^{n-1} (-1)^{j(m-1)} f \circ_j g$ and $[f, g] = f \circ g - (-1)^{(n-1)(m-1)} g \circ f$. For unary f and binary g this gives

$$[f, g]_{ijk} = (f_{ik} - f_{ij} - f_{jk}) g_{ijk}.$$

For binary cochains it gives

$$[f, g]_{0123} = f_{023}g_{012} - f_{013}g_{123} + g_{023}f_{012} - g_{013}f_{123}. \tag{5.70}$$

These are the two insertion slots on the unique three-arrow string. With the Hochschild signs of (5.67), a positive arity-n cochain has $df = (-1)^{n-1}[\mu, f]$, where μ is the binary product. Rephasing arities gives the dg Lie convention $[\mu, -]$. In particular $[\mu, \mu]/2 = Fa_{03}$.

5.8.2 The parameter fibres and the integral module

Let $B \to K$ be a map to a field, and let c count the nonzero parameters among p, q, r, s. Theorem 5.17 makes the specialized complex the full centre over K. Its cohomology dimensions are

c	HH^0	HH^1	HH^2	HH^3
0	1	3	4	1
1	1	2	2	0
2	1	1	1	0
4	1	0	0	0

(5.71)

The proof is a rank calculation. The incidence matrix d_0 has rank three. The four unweighted triangle rows of d_1 have one relation with all four coefficients nonzero, and any three are independent. Thus rank d_1 is $\min(c, 3)$. The last row has rank one off the origin and zero at the origin. Subtracting adjacent ranks gives the table. The equation $pr = qs$ excludes $c = 3$ and excludes the supports $\{p, r\}$ and $\{q, s\}$ when $c = 2$.

Integrally B is a domain: $pr - qs$ is primitive and irreducible as a polynomial in r over $k[p, q, s]$. The answer retains a module supported at the origin:

$$\begin{aligned} \mathrm{HH}^0_B(A_B) &= B, \qquad \mathrm{HH}^1_B(A_B) = 0, \\ \mathrm{HH}^3_B(A_B) &= B/(p, q, r, s), \\ \mathrm{HH}^2_B(A_B) &= \ker(-r, s, -p, q)/\operatorname{im} L. \end{aligned} \tag{5.72}$$

Here $L(x, y, z) = (-px, -qy, r(x - z), s(y - z))$. For the degree-one assertion, nonzero parameters in the domain force all four triangle additivity equations. Setting $\lambda_0 = 0$ and $\lambda_i = f_{0i}$ gives $f_{ij} = \lambda_j - \lambda_i$. More generally, modulo a vertex coboundary an edge cochain has a unique representative with $f_{01} = f_{12} = f_{23} = 0$. Its other coordinates $(f_{02}, f_{13}, f_{03}) = (x, y, z)$ give the displayed image L. The cokernel of d_2 gives degree three.

There is no B-linear deformation retraction onto these cohomology groups. Its nonzero degree-three group is killed by p, and therefore cannot have a section into the free degree-three cochain term. Scalar extension must act on (5.66), not merely on (5.72).

At the origin, let the binary cochain f have $f_{012} = f_{023} = 1$ and $f_{123} = f_{013} = 0$. The first-order product ϵf is associative modulo ϵ^2, but

$$\tfrac{1}{2}[f, f]_{0123} = 1 \in \mathrm{HH}^3 = k.$$

The positive-arity differential is zero there, so no second-order correction removes this obstruction. Equivalently, $p = r = \epsilon$, $q = s = 0$ gives $F = \epsilon^2$. A ternary operation cannot help on the unchanged degree-zero arrow module over an ordinary ring: it would have degree -1 and no target. This is the deformation problem with the objects, arrow bases and units fixed.

5.8.3 A primitive and a finite coherent resolution

Introduce a coefficient of degree -1 with the required differential:

$$R_{\mathrm{coh}} = k[p, q, r, s] \otimes \Lambda(\chi), \qquad d\chi = pr - qs. \tag{5.73}$$

Its cohomology is B in degree zero, since F is a non-zero-divisor. A map to a derived test algebra nevertheless retains the image of χ as a specified primitive. On the rank-ten arrow module, the products (5.65) and

$$m_3(a_{01}, a_{12}, a_{23}) = -\chi a_{03} \tag{5.74}$$

define a strictly unital A_∞ category, with convention $dm_3 = -\operatorname{Assoc}(m_2)$ on closed degree-zero arrows. There is no increasing string of four nonidentity inputs, so all higher normalized identities vanish.

The binary homotopies and their compatibility can also be retained in a strict dg algebra. Let $\boldsymbol{Q}$ be the free directed path algebra over R_{coh} on

$$X_{ij} : i \to j \ (|X_{ij}| = 0), \qquad H_{ijk} : i \to k \ (|H_{ijk}| = -1), \qquad K_{03} : 0 \to 3 \ (|K_{03}| = -2),$$

for increasing indices. Extend the coefficient differential by the graded Leibniz rule and

$$\begin{aligned} dX_{ij} &= 0, \\ dH_{ijk} &= X_{ij}X_{jk} - m_{ijk}X_{ik}, \\ dK_{03} &= H_{012}X_{23} - X_{01}H_{123} + pH_{023} - qH_{013} + \chi X_{03}. \end{aligned} \tag{5.75}$$

Odd coefficients are moved to the left with their Koszul sign. In d^2K_{03} the first four terms cancel all paths except $-FX_{03}$; the last gives FX_{03}. Thus $d^2 = 0$. The free basis has fifteen degree-zero paths (four identities, six arrows, four length-two paths and one length-three path), six degree--1 paths (the four H and their two composable extensions), and the single K_{03}. Its rank is 22.

Let V_{ar} be the submodule spanned by the identities and the six X_{ij}, with its coefficient differential. Let i include it, and let π be the identity there and zero on the H_{ijk}, K_{03}. Specify its remaining values by

$$\begin{aligned} \pi(X_{ij}X_{jk}) &= m_{ijk}X_{ik}, \\ \pi(X_{01}X_{12}X_{23}) &= prX_{03}, \qquad \pi(X_{01}H_{123}) = \chi X_{03}, \qquad \pi(H_{012}X_{23}) = 0. \end{aligned} \tag{5.76}$$

The degree--1 homotopy is

$$\begin{aligned} h(X_{ij}X_{jk}) &= H_{ijk}, \\ h(X_{01}X_{12}X_{23}) &= H_{012}X_{23} + pH_{023}, \qquad h(X_{01}H_{123}) = -K_{03}, \end{aligned} \tag{5.77}$$

and is zero on all other basis paths. Extend π linearly and h graded-linearly, so $h(\chi a) = -\chi h(a)$.

Theorem 5.19. *Over all of R_{coh} these maps satisfy*

$$\pi i = 1, \qquad dh + hd = 1 - i\pi, \qquad h^2 = hi = \pi h = 0, \qquad d\pi = \pi d.$$

The transferred product on V_{ar} is (5.65)*, with the ternary value* (5.74)*; every normalized higher operation vanishes. No parameter is inverted.*

Proof. The intervals $0 \to 2$ and $1 \to 3$ contain the pairs $H_{012} \mapsto X_{01}X_{12} - pX_{02}$ and $H_{123} \mapsto X_{12}X_{23} - qX_{13}$. On $0 \to 3$ put

$$\begin{aligned} v_1 &= X_{02}X_{23} - rX_{03}, & h_1 &= H_{023}, \\ v_2 &= X_{01}X_{13} - sX_{03}, & h_2 &= H_{013}, \\ v_3 &= X_{01}X_{12}X_{23} - pX_{02}X_{23}, & h_3 &= H_{012}X_{23}, \\ \ell &= H_{012}X_{23} - X_{01}H_{123} + pH_{023} - qH_{013} + \chi X_{03}. \end{aligned}$$

Together with K_{03}, X_{03} these are another free basis, obtained by a triangular change with constant invertible pivots. Its differential has $dh_j = v_j$, $dK_{03} = \ell$, $d\ell = 0$, leaving X_{03}. Contracting these four pairs and returning to the

original basis gives (5.76)–(5.77). In particular $\pi(\ell) = 0$ forces the nonzero value $\pi(X_{01}H_{123}) = \chi X_{03}$. Graded linearity of h cancels the coefficient-differential terms.

The transferred binary operation is $\pi\mu(i \otimes i)$. Its two ternary trees give

$$\pi\big(h(X_{01}X_{12})X_{23} - X_{01}h(X_{12}X_{23})\big) = -\chi X_{03}.$$

There are no longer composable nonidentity strings, and the side conditions kill higher trees with an identity input. This proves the strictly unital structure. □

After an ordinary specialization $F = \chi = 0$, the augmentation $X_{ij} \mapsto a_{ij}$, $H_{ijk}, K_{03} \mapsto 0$ is a strict dg algebra quasi-isomorphism, by this contraction. Over R_{coh} the projection is a cochain map to the A_∞ category, not a strict algebra map. The primitive has not removed the ordinary obstruction: a dg map to $k[\epsilon]/(\epsilon^3)$ with $p = r = \epsilon$ and $q = s = 0$ would need a degree--1 primitive of ϵ^2, which that ring does not contain.

The finite categories in this chapter specify algebraic compositions of interfaces. Their object labels alone supply no field-theoretic realization. Such a realization must identify the arrows, junctions, collision maps and local vacuum operations. Likewise, a native centre comparison must carry the full current hierarchy, the polynomial Massey coset and the integral path relation, together with derived specialization. These are tests on an actual evaluation map; they do not construct a common recipient for arbitrary geometric boundary actions.

Chapter 6

Action profiles and universal recipients

A universal boundary action must receive actions together with their comparison homotopies. A multiplication on the underlying coefficient algebra does not suffice. For an action whose first nonzero term is quadratic in its parameter, doubling that parameter multiplies the term by four, whereas composing two copies adds their first terms and multiplies it by two. The standard additive multiplication is therefore incompatible with such an action whenever its first class is nonzero. Nevertheless, a nonlinear change of actor can place the action inside a multiplicative one.

The higher terms of these actions have a simple scalar example. Over a characteristic-zero field, the formal solution of

$$p = q + \frac{z^2}{2}p^2, \qquad p \equiv q \pmod{z},$$

is

$$p = \sum_{n\geq 0} \frac{1}{n+1}\binom{2n}{n}\left(\frac{z^2}{2}\right)^n q^{n+1}.$$

Writing $p = qC(z^2q/2)$ reduces the equation to $C = 1 + uC^2$, whose solution with constant term one is $(1-\sqrt{1-4u})/(2u)$. For noncommuting coefficients the same implicit construction uses ordered trees. Its terms define actual actions on the collision coefficients. The universal question then concerns all maps between these actions, including their higher paths.

6.1 The coefficients of an action

Fix a field k of characteristic zero and $X = \mathbb{A}^1_k$. At a collision, the order of the coinciding labels is retained. For a nonempty finite set I, take a copy X^I_σ of X^I for each total order σ on I. Glue two copies by the identity on the open

$$V_{\sigma\rho} = X^I \setminus \bigcup_{\substack{\{i,j\}\subset I \\ \sigma,\rho \text{ order } i,j \text{ oppositely}}} \{z_i = z_j\}.$$

The resulting scheme, denoted by $\mathscr{R}_I$, is the ordered configuration scheme of [2, §4.1]. The gluing condition holds because, whenever two orders disagree on a pair, at least one disagrees with any third order on that pair. Thus $V_{\sigma\rho}\cap V_{\sigma\eta} = V_{\sigma\rho}\cap V_{\rho\eta}$. The scheme is smooth, with a finite affine-chart cover and quasi-compact intersections. It can be nonseparated. Over a tuple with equality blocks B, the map $\mathscr{R}_I \to X^I$ has $\prod_B |B|!$ points, one for each order inside each block.

A surjection $\alpha : I \twoheadrightarrow J$ equipped with an order on each fibre defines

$$\delta_\alpha : \mathscr{R}_J \longrightarrow \mathscr{R}_I, \qquad z_i = z_{\alpha(i)}.$$

On charts it expands each label into its ordered fibre. Fibre orders compose by concatenation. The chart maps glue, since the inverse image of an overlap is the overlap of the corresponding orders on J.

Their compositions are the maps for the composite ordered surjections. In particular, a nonconsecutive collision uses its actual map into $\mathscr{R}_I$; it is not replaced by the small diagonal of one chart.

Let $\mathscr{C}$ be the enhanced category of compatible complexes of right differential modules on these schemes:

$$\mathscr{C} = D(k) \times \varprojlim_{I,\alpha} D\mathrm{Mod}(\mathscr{R}_I), \qquad \delta_\alpha^! F_I \simeq F_J. \tag{6.1}$$

Differential modules on $\mathscr{R}_I$ are defined by descent on its smooth chart cover. The displayed equivalences include all composition homotopies. The first factor is the isolated empty coefficient $F_\varnothing$. An object is *positive* if this coefficient is zero. Complexes are cohomological, suspension $s = [1]$ has degree -1, and every permutation uses the Koszul sign. Extraordinary inverse images and the diagonal tensor retain their canonical density lines and shifts. Passing to left modules for a coordinate computation uses the canonical-bundle equivalence.

For an ordered decomposition $I = A \sqcup B$, let $U_{A,B}$ be the open where every point in A is distinct from every point in B. Collisions inside either part remain. Restriction of orders identifies this open with the disjointness open in $\mathscr{R}_A \times \mathscr{R}_B$. With $j_{A,B}$ its inclusion, set

$$(F \star G)_I = \bigoplus_{I=A\sqcup B} j_{A,B,*}\big((F_A \boxtimes G_B)|_{U_{A,B}}\big). \tag{6.2}$$

An empty part contributes scalar tensor product. The unit K has $K_\varnothing = k$ and zero nonempty coefficients. Iterating (6.2) sums over ordered partitions into the required number of parts. All parenthesizations impose the same cross-part inequalities. This proves associativity and its coherences; permuting the parts gives symmetry. Under a collision, a summand vanishes if one collision fibre meets two parts. Every surviving partition descends uniquely to the quotient labels. Localization and closed-support base change therefore give the collision coherences. These operations preserve colimits separately.

The second tensor product uses coefficients at the same configuration:

$$(F \otimes^! G)_I = \Delta^!_{\mathscr{R}_I}(F_I \boxtimes G_I).$$

Its unit J is the family of dualizing objects $J_I = (\mathscr{R}_I \to \operatorname{Spec} k)^! k$, with empty coefficient k. The objects J and K have different nonempty coefficients. The forward interchange

$$(F \star G) \otimes^! (H \star L) \longrightarrow (F \otimes^! H) \star (G \otimes^! L) \tag{6.3}$$

keeps summands with the same ordered partition and applies the localization comparison there. In a rectangular iterated product, either order of interchange keeps exactly the common partitions. This proves its coherence and gives a diagonal tensor on coassociative $\star$-coalgebras.

Write $\mathscr{M}$ for connected coaugmented counital coherent coassociative coalgebras in $\mathscr{C}$. Connected means that the empty coefficient and its unit maps are k. A coalgebra F is *factorizing* if every ordered partition $P = (I_1|\cdots|I_r)$ into nonempty parts has an equivalence

$$\delta_{P,F} : j_P^! F_I \xrightarrow{\sim} (F_{I_1} \boxtimes \cdots \boxtimes F_{I_r})|_{U_P}. \tag{6.4}$$

The coproduct supplies this map and its refinement homotopies. Let $\mathscr{F} \subset \mathscr{M}$ be the full subcategory of such coalgebras. The diagonal tensor preserves it by (6.3).

Fix a boundary $C \in \mathscr{F}$. A *pointed action* consists of $F \in \mathscr{F}$, a point $u : J \to F$, an action $\alpha : F \otimes^! C \to C$ in $\mathscr{M}$, and a specified path $\alpha(u \otimes^! \mathrm{id}_C) \simeq \mathrm{id}_C$. A morphism $(F, \alpha) \to (G, \beta)$ includes

a pointed coalgebra map $f : F \to G$, a path $\alpha \simeq \beta(f \otimes^{!} \mathrm{id}_C)$, and its compatibility on J. Their higher homotopies are retained. Denote this category by $\mathscr{A}_C$. The functor it classifies is

$$\Phi_C(F, u) = \mathrm{Map}^{\mathscr{M}}_{C/}(F \otimes^{!} C, C), \tag{6.5}$$

the homotopy fibre of restriction over the identity of C. Representability of Φ_C is equivalent to a final object of $\mathscr{A}_C$.

The ordered product of two actions is their composition, with actor $F \otimes^{!} G$. Denote it by $a \circledast b$; the first action is outermost. The unit action on J is initial. A *multiplicative action* has an associative unital monoid actor for $\otimes^{!}$ and a coherent module structure on C. Its binary path compares acting after multiplying the two actors with their composed action. All associativity and unit homotopies are part of the structure. Write $\mathscr{B}_C$ for the category of multiplicative actions, with monoid maps and coherent action paths. A local centre, in this categorical sense, is a final object of $\mathscr{B}_C$. A final pointed action has a unique coherent multiplicative structure. The converse requires a comparison with pointed actions whose actors have no specified multiplication.

6.2 Trees integrating a derivation

Let $\mathfrak{g}$ be a positive Lie object in $\mathscr{C}$ whose coefficient modules lie in degree zero, and put $A = U_\star\mathfrak{g}$. Fix a degree-zero derivation D of the full coefficient family. Its extension to A satisfies Leibniz and commutes with all collision and normal-transfer maps. Products below are ordered products in A. PBW symmetrization uses averaging in characteristic zero.

Give an odd symbol e degree one and zero differential, and form $\mathfrak{h} = \mathfrak{g} \ltimes e\mathfrak{g}$, with

$$[x, em] = e[x, m], \qquad [em, en] = 0.$$

The Chevalley coalgebra is $\mathrm{CE}_\star(\mathfrak{h}) = K \oplus \bigoplus_{r \geq 1} \mathrm{Sym}^r_\star(s\mathfrak{h})$, with unshuffle coproduct. Its coderivation has $Q_1(sx) = -s(dx)$ and $Q_2(sx\,sy) = (-1)^{|x|} s[x, y]$. The *actor weight* counts the factors $s(em)$, which have degree zero. It is independent of PBW output number.

A degree-one twisting cochain to A can be nonzero only on one ordinary suspended input. Write its value as $\theta_w(x; m_1, \ldots, m_w)$ and put $\theta_0(x) = x$. For $m_S = \circ_{i \in S} m_i$ in $P_{|S|} = \mathrm{Sym}^{|S|}_\star \mathfrak{g}$, the twisting equation is

$$\begin{aligned} 0 = {} & [x, \theta_w(y; m)] - \theta_w(y; [x, m]) - [y, \theta_w(x; m)] + \theta_w(x; [y, m]) - \theta_w([x, y]; m) \\ & + \sum_{\substack{S \sqcup T = [w] \\ S, T \neq \varnothing}} \big(\theta_{|S|}(x; m_S)\theta_{|T|}(y; m_T) - \theta_{|S|}(y; m_S)\theta_{|T|}(x; m_T) \big). \end{aligned} \tag{6.6}$$

Here $[x, m]$ is the sum of brackets with each module input, and the splitting (S, T) is ordered. To verify the signs, use convolution

$$(f * g)(c) = \sum_{(c)} (-1)^{|g||c'|} f(c')\, g(c'').$$

On two ordinary suspended inputs, the tensor sign of two degree-one cochains is minus. The terms with no actor input on one side give the four adjoint terms above; the Chevalley bracket gives the fifth. All other components vanish by degree. Thus (6.6) is both necessary and sufficient.

For a nonempty set B of actor labels, define

$$u_B = \frac{1}{|B|!} \sum_{\sigma \in S_B} m_{\sigma(1)} \cdots m_{\sigma(|B|)}.$$

Let $\mathcal{T}_r$ consist of nonnegative tuples $(a, b_1, \dots, b_r)$ such that

$$a + \sum_i b_i = r, \qquad a + \sum_{i=1}^{j} b_i \geq j + 1 \quad (0 \leq j < r).$$

These are the child counts of plane rooted trees in preorder, with the root first. For envelope inputs $v_1, \dots, v_r$, set

$$T_r^D(x; v_1, \dots, v_r) = \sum_{(a,b)\in\mathcal{T}_r} \frac{D^a x}{a!} \sum_{\sigma\in S_r} \frac{D^{b_1} v_{\sigma(1)} \cdots D^{b_r} v_{\sigma(r)}}{b_1! \cdots b_r!}, \qquad T_0^D(x) = x. \tag{6.7}$$

The slots are symmetric; each displayed product remains ordered. For a scalar sequence $\mathbf{t} = (t_1, t_2, \dots)$, define

$$\theta_w^{\mathbf{t}}(x; m_1, \dots, m_w) = \sum_{\pi\in\mathrm{Part}([w])} \left(\prod_{B\in\pi} t_{|B|}\right) T_{|\pi|}^D(x; u_B : B \in \pi). \tag{6.8}$$

The partitions are unordered, with nonempty blocks. For example,

$$\begin{aligned} \theta_1^{\mathbf{t}}(x; m) &= t_1 Dx\, m, \\ \theta_2^{\mathbf{t}}(x; m, n) &= t_2 Dx\, \frac{mn + nm}{2} + t_1^2 \Big(D^2 x\, \frac{mn + nm}{2} + Dx(Dm\, n + Dn\, m) \Big). \end{aligned}$$

In weight three the term proportional to $t_1 t_2$ is the sum of $T_2^D(x; m_i, u_B)$ over $\{i\} \sqcup B = [3]$.

Theorem 6.1 (Simultaneous profiles)**.** *The maps* (6.8) *satisfy* (6.6)*. They define a map of coassociative coalgebras*

$$\mathrm{CE}_\star(\mathfrak{g} \ltimes e\mathfrak{g}) \longrightarrow \mathrm{CE}_\star(\mathfrak{g})$$

under $\mathrm{CE}_\star(\mathfrak{g})$*, in the derived category of coalgebras. Every nonconstant term on a configuration I has actor weight $w < |I|$. Thus each scalar sequence defines coefficient maps without completing states or normal derivatives.*

Proof. Use a central formal variable z and define right substitution by

$$f\{R\} = \sum_{j\geq 0} \frac{D^j f}{j!} R^j, \qquad U = \sum_{r\geq 1} \frac{t_r z^r}{r!} m^r, \qquad R = U\{R\}, \qquad G_x = x\{R\}.$$

Every coefficient of $R \in zA[[z]]$ is determined by the preceding ones. Let ∂_n vary m in direction n, including its D-derivatives. Adjoint equivariance gives $\partial_{[x,m]} U = [x, U]$. Put $U_j = D^j U / j!$ and $X_i = D^i x / i!$. Differentiating the fixed-point equation gives a linear filtered equation for $\partial_{[x,m]} R$. The same equation is satisfied by $[G_x, R]$: the difference of its expanded sides is

$$\sum_{i,j} X_i [R^i, U_j] R^j = \sum_i X_i \sum_{p+q=i-1} R^p \Big(\sum_j [R, U_j] R^j \Big) R^q = 0,$$

because the inner sum is $[R, R]$. The linear term of the equation raises the z-filtration, so uniqueness gives $\partial_{[x,m]} R = [G_x, R]$.

Leibniz for D now implies

$$[G_x, G_y] - G_{[x,y]} - \partial_{[x,m]} G_y + \partial_{[y,m]} G_x = 0. \tag{6.9}$$

Indeed, with $Y_j = D^j y/j!$, the first two terms differ by $\sum_{i,j} X_i[R^i, Y_j]R^j - \sum_{i,j} Y_j[R^j, X_i]R^i$. The two derivative terms cancel precisely these sums. Expanding the root and its ordered subtrees gives (6.7); allocating the original inputs to the subtree slots gives (6.8). Its normalization is $G_x = x + \sum_{w\geq 1} z^w \theta^{\mathbf{t}}_w(x; m, \dots, m)/w!$. Coefficient extraction in (6.9), followed by polarization, proves the twisting equation.

A twisting cochain defines a map into $\mathrm{B}_\star A$. The Chevalley–bar comparison identifies $\mathrm{B}_\star A$ with $\mathrm{CE}_\star(\mathfrak{g})$. Explicitly, this comparison sends a symmetric word in suspended Lie inputs to the sum of its signed bar words, without a factorial divisor. It preserves the unshuffle and deconcatenation coproducts. PBW reduces its associated graded to the Koszul comparison for a symmetric algebra. The usual Koszul homotopy, divided by positive polynomial weight, contracts its augmentation complex. At I, both filtrations stop at $|I|$, so the comparison is an equivalence. Its inverse in the pointed derived category gives the asserted map.

Each nonconstant term consumes one positive ordinary input and w positive actor inputs. Hence $w + 1 \leq |I|$. All operations in each surviving term are full coefficient maps, so the finite formulas retain collision and normal-transfer compatibility. □

For $D = \partial_q$, $x = m = q$ and $t_2 = 1$ with the other parameters zero, right substitution becomes $f(q + R)$. Thus $p = q + R$ satisfies the scalar equation at the chapter's opening. The noncommutative proof explains why the order of the powers of R is fixed.

The same construction supplies mixed homotopies. In the relative convolution algebra with differential d_0 at the scalar action, write the positive cochain as

$$\theta_{\mathbf{t}} = \sum_r t_r u_r + \sum_{p\leq q} t_p t_q \eta_{pq} + \cdots .$$

Its equation gives $d_0\eta_{pq} = -[u_p, u_q]$ for $p < q$ and $d_0\eta_{pp} = -u_p^2$. These are compatible primitives because they are coefficients of one Maurer–Cartan element. More explicitly, give a free graded Lie generator e_α degree one for every nonzero finite multi-index α, with weight $\sum r\alpha_r$, and set

$$de_\alpha = -\frac{1}{2} \sum_{\substack{\beta+\gamma=\alpha \\ \beta,\gamma\neq 0}} [e_\beta, e_\gamma]. \tag{6.10}$$

This is a differential. In the tensor envelope it is $de_\alpha = -\sum_{\beta+\gamma=\alpha} e_\beta e_\gamma$; applying d twice gives opposite sums over ordered three-part splittings. Coefficient extraction from the profile equation shows that $e_\alpha \mapsto [\mathbf{t}^\alpha]\theta_{\mathbf{t}}$ is a dg Lie map. Only the positive-weight completion is used.

6.3 A change of actor and an actual bar composite

For a vertex algebra V, let L_V be its main-diagonal Lie family on (6.1), with singleton the right differential module of states and connection induced by translation. In each ordered chart its higher coefficient is the full normal direct image of this module on the small diagonal. The chiral bracket uses principal parts. Put $C_V = \mathrm{CE}_\star(L_V)$. For the vacuum vertex algebra k, this is the dualizing coalgebra J. For $E = \Lambda(e)$, $|e| = 1$, with zero differential and translation, write C_E for its factorizing coalgebra.

The following tensor comparison specifies how the algebraic profiles act on the coefficient family.

Lemma 6.2. *If C is a cocommutative factorizing coalgebra with empty coefficient k, then $F \mapsto F \otimes^! C$ is a colimit-preserving strong symmetric monoidal functor for $\star$. For $C = C_V$ it sends L_k to L_V and L_E to $L_V \ltimes eL_V$, with their Lie structures. Consequently*

$$C_E \otimes^! C_V \simeq \mathrm{CE}_\star(L_V \ltimes eL_V). \tag{6.11}$$

The comparison preserves the vacuum, actor grading, finite scalar extensions, and all collision maps and normal derivatives.

Proof. For each ordered partition, tensor its convolution summand with C_I and use (6.4) for C. The projection formula for localization identifies the result with the corresponding summand of $(F \otimes^! C) \star (G \otimes^! C)$. At the empty coefficient the map is the scalar identity. All iterated comparisons use the same ordered partition and the same iterated coproduct of C. This gives associativity and units; cocommutativity gives symmetry. Collision pullback keeps precisely the saturated partitions on both sides, with the same base-change maps. Thus the functor is strong monoidal on the enhanced diagram and commutes with its Chevalley complexes, envelopes and bars.

On the main diagonal, the projection formula identifies $L_k \otimes^! C_V$ with L_V. The principal-part boundary of the vacuum line, tensored with the factorization identification of C_V on the punctured two-point locus, is the chiral bracket of V. This checks the Lie identity on both ordered branches with the full normal direct image. The even vacuum and odd e summands give the stated semidirect product. These coefficient formulas preserve actor scaling and tensor powers. Finite free scalar extensions preserve their equivalences, which proves the remaining assertions. □

For $\mathfrak{g} = L_V$ and this comparison, write $\alpha_{\mathbf{t}} : C_E \otimes^! C_V \to C_V$ for the profile action. The sequence $t_1 = 1, t_r = 0$ for $r > 1$ gives α_D. The actor maps receiving the other profiles are nonlinear even though a map between the exterior coefficient algebras is linear on its degree-one generator.

Let $\mathfrak{l} = L_k$. In $U_\star(\mathfrak{l} \ltimes e\mathfrak{l})$ define

$$\rho_r(m_1, \dots, m_r) = \frac{1}{r!} \sum_{\sigma \in S_r} m_{\sigma(1)} \cdots m_{\sigma(r-1)}(em_{\sigma(r)}). \tag{6.12}$$

This is the averaged PBW product with one marked odd factor. In particular its diagonal value is $m^{r-1}em$. The cochain

$$\tau_{\mathbf{t}}(sx) = x, \qquad \tau_{\mathbf{t}}(s(em_1) \cdots s(em_r)) = t_r \rho_r(m_1, \dots, m_r)$$

has all other reduced components zero.

Proposition 6.3. *The cochain $\tau_{\mathbf{t}}$ is twisting and defines a map $\Sigma_{\mathbf{t}} : C_E \to C_E$ under and over J. Its formal coordinate series is $\sum_{r \geq 1} t_r p^r / r!$. There is a pointed action path*

$$\alpha_D(\Sigma_{\mathbf{t}} \otimes^! \mathrm{id}_{C_V}) \simeq \alpha_{\mathbf{t}}. \tag{6.13}$$

It preserves every ordered cut, collision, normal transfer and unit homotopy. In the associative bar its composite corestriction is exactly (6.8).

Proof. The maps ρ_r are adjoint-equivariant, giving the twisting equation with one ordinary input. With two ordinary inputs it is the equation for the canonical cochain. On pure actor inputs, polarization reduces the equation to the square of

$$\sum_{r \geq 1} \frac{t_r}{r!} m^{r-1} em.$$

It is zero because $[m, em] = 0$ and $(em)^2 = 0$. The induced map to the associative bar is the finite sum of tensor powers of $s\tau_{\mathbf{t}}$ applied to the iterated reduced coproduct. The inverse Chevalley–bar comparison gives $\Sigma_{\mathbf{t}}$. Its restriction to J is the identity and its augmentation is fixed. Removing the marked odd factor in (6.12) gives the scalar coordinate coefficient $t_r / r!$, with exponential polarization.

To calculate the composite, put $B = U_\star(\mathfrak{g} \ltimes e\mathfrak{g})$ and let $\pi : B \to A$ kill the odd ideal. There is a degree--1 projection derivation

$$q(x) = 0, \quad q(em) = m, \quad q(1) = 0, \qquad q(uv) = q(u)\pi(v) + (-1)^{|u|}\pi(u)q(v). \tag{6.14}$$

It respects the envelope relations: the mixed relation reduces to the same bracket identity, while a relation with two odd inputs maps to zero. In $\mathrm{Hom}(\mathrm{B}_\star B, A)$ let a be the canonical length-one twisting cochain induced by π and p the length-one cochain induced by q. They have degrees one and zero and satisfy

$$da + a^2 = 0, \qquad dp + [a, p] = 0.$$

Apply D on the output and solve

$$R = \sum_{j\ge 0} \frac{D^j p}{j!} R^j, \qquad A_R = \sum_{j\ge 0} \frac{D^j a}{j!} R^j$$

in positive bar length. These equations give a twisting cochain. For completeness, put $a_j = D^j a/j!$, $p_j = D^j p/j!$. The differentiated identities are $da_j = -\sum_{u+v=j} a_u a_v$ and $dp_j = -\sum_{u+v=j} [a_u, p_v]$. Moreover,

$$\sum_{r,s} [a_r, p_s] R^{r+s} = [A_R, R] - \sum_s p_s [A_R, R^s],$$

since their difference is $\sum_r a_r [R^r, \sum_s p_s R^s] = 0$. The differentiated fixed-point equation therefore has the unique solution $dR = -[A_R, R]$. Using that each a_j is odd gives

$$dA_R = -\sum_{r,s} a_r a_s R^{r+s} + \sum_r a_r [A_R, R^r], \qquad A_R^2 - \sum_{r,s} a_r a_s R^{r+s} = \sum_r a_r [R^r, A_R].$$

Thus $dA_R + A_R^2 = 0$.

The associated bar map $\mathcal{F}_D : \mathrm{B}_\star B \to \mathrm{B}_\star A$ has length-$(w+1)$ corestriction on $[v_0|\cdots|v_w]$

$$\sum_{(a,b)\in \mathcal{T}_w} \frac{D^a \pi(v_0)}{a!} \frac{D^{b_1} q(v_1)\cdots D^{b_w} q(v_w)}{b_1!\cdots b_w!}. \tag{6.15}$$

It vanishes unless the first input has exterior degree zero and the others have exterior degree one. Restriction to the Chevalley coalgebra is the tree action α_D.

Lemma 6.2 carries the actor map from $\mathfrak{l}$ to $\mathfrak{g}$. Its bar representative partitions the actor inputs into marked blocks and places the one ordinary input in a singleton. Formula (6.15) forces this singleton first. Equation (6.14) gives $q\rho_r = u_{[r]}$. For each unordered partition, the sum over orders of its blocks is therefore $T^D_{|\pi|}$, with coefficient $\prod_{B\in\pi} t_{|B|}$. This proves equality of the actual corestrictions. The pointed Chevalley–bar inverses and the strong tensor comparison transport that equality to (6.13), including all unit paths. Each calculation is finite on every configuration. □

The quadratic map with $t_2 = 1$ has coordinate series $p^2/2$. The map with coordinate series p^2 has $t_2 = 2$. This normalization is independent of the unnormalized permutation sum in the Chevalley–bar comparison.

6.4 Natural homotopies and multiplication

The profile comparison gives a map to α_D. To receive compositions, that target must carry its own coherent multiplication. The obstruction groups can be computed before evaluating on a particular boundary.

Let W be finite-dimensional in degree zero and set $E_W = \Lambda(W[-1])$, whose generators have degree one. Let $\mathfrak{d}$ be a Lie algebra of degree-zero derivations of an ordinary Lie algebra $\mathfrak{g}$. A *natural operation* uses each Lie input once, applies elements of $U(\mathfrak{d})$ to these inputs, and multiplies the results in a specified order in $U(\mathfrak{g})$. Brackets are commutators. Operations on $\mathrm{CE}(\mathfrak{g}\otimes E_W)$ include signed symmetrization of suspended inputs. Two formulas are identified if they agree for every Lie algebra with such a derivation action. This restriction defines a universal algebra of operations; it does not define all maps of geometric coefficient families.

Put $P = W^*$. Denote the natural convolution algebra by $\mathcal{N}(P;\mathfrak{d})$, with differential twisted by the scalar cochain τ. Thus

$$\delta f = -(-1)^{|f|} f Q + \tau * f - (-1)^{|f|} f * \tau.$$

Complete in actor weight and allow only finite sums of differential operations at each weight and cochain degree. The ideal of positive actor weight is $\mathcal{N}_+$. Natural actions and homotopies mean its Maurer–Cartan space, using polynomial differential forms on simplices as parameters.

For commuting derivations $D_1, \dots, D_r$, the unary cochain $x_{S,a}$ extracts an exterior coefficient e_S and applies $D^a/a!$. It has degree $1 - |S|$. Write $t = x_{\varnothing,0}$. If $\epsilon(B, C)$ is the exterior shuffle sign, the unary differential is

$$\begin{aligned} \delta x_{S,a} = & - \sum_{\substack{B \sqcup C = S \\ b+c=a}} (-1)^{|B|(|C|+1)} \epsilon(B, C) x_{B,b} x_{C,c} \\ & + t x_{S,a} - (-1)^{1-|S|} x_{S,a} t. \end{aligned} \tag{6.16}$$

It follows from $Q_2(s(e_B x)s(e_C y)) = (-1)^{|B|} s(e_B e_C [x, y])$ and divided Leibniz. The two terms containing t cancel when $x_{S,a} \neq t$, while $\delta t = t^2$. For one actor and one derivation,

$$\delta x_{\{1\},1} = x_{\{1\},0} x_{\varnothing,1} - x_{\varnothing,1} x_{\{1\},0}, \qquad \delta x_{\varnothing,2} = -x_{\varnothing,1}^2.$$

These identities already show the polynomial actor coordinate and the exterior derivation class in cohomology.

Theorem 6.4. *There is a surjective filtered dg algebra map*

$$\pi : \mathcal{N}(P; \mathfrak{d}) \longrightarrow \prod_{q \geq 0} \operatorname{Sym}^q(P) \otimes \Lambda(\mathfrak{d}[-1]) \tag{6.17}$$

with acyclic kernel in each actor weight. The target differential is zero; its actor coordinates have degree zero and its derivation generators degree one. On positive actor weights it induces an equivalence of Maurer–Cartan spaces. Their components are

$$\prod_{q \geq 1} \operatorname{Sym}^q(W^*) \otimes \mathfrak{d}. \tag{6.18}$$

At each point the fundamental group is the additive group $\mathfrak{m}_W = \prod_{q \geq 1} \operatorname{Sym}^q(W^*)$, and all higher homotopy groups vanish.

Proof. Unary operations freely generate the convolution algebra. To prove this, evaluate on the free Lie algebra on named copies of the free $\mathfrak{d}$-module $U(\mathfrak{d})$. In its tensor envelope, an ordered monomial records the input, its exterior label and the operator applied to it. Distinct such words are independent. Polarization recovers repeated exterior types with the suspended signs. A cochain of actor weight q and degree p uses $q + p$ inputs.

The Hopf coproduct of $U(\mathfrak{d})$ expresses its Leibniz rule. Rescaling the exterior-label generator by $(-1)^{\binom{|S|}{2}}$ identifies the differential with the cobar differential of $\Lambda(P[1]) \otimes U(\mathfrak{d})$, apart from the scalar generator t. Thus the natural algebra is

$$(k\langle t \rangle, \delta t = t^2) * \Omega\big(\Lambda(P[1]) \otimes U(\mathfrak{d})\big). \tag{6.19}$$

Here $*$ is free product, with the stated weight completion. The augmentation ideal of the first factor contracts by $h(t^{2j}) = t^{2j-1}$ and $h(t^{2j-1}) = 0$. In the free product, contract the first such factor of each alternating word, with the sign of its prefix. This removes t without changing cohomology.

PBW symmetrization identifies $U(\mathfrak{d})$ with the symmetric coalgebra on $\mathfrak{d}$. This is a coalgebra identification, not an identification of products. The cobar projection sends primitive actor and derivation generators to their namesakes and all other generators to zero. The cobar differential maps to graded commutators or odd squares, so the projection is a dg algebra map.

Its cohomology is computed in finite auxiliary weights. The dual algebra of $\Lambda(P[1]) \otimes \mathrm{Sym}^c(\mathfrak{d})$ is exterior on degree-one variables α_i and polynomial on degree-zero variables z_j. A free resolution of its scalar module is

$$\Lambda(\alpha_i) \otimes k[z_j] \otimes k[v_i] \otimes \Lambda(\eta_j), \qquad |v_i| = 0, \quad |\eta_j| = -1, \quad dv_i = \alpha_i, \quad d\eta_j = z_j.$$

For an odd pair, $d(v^{a+1}) = (a+1)\alpha v^a$; for an even pair this is the two-term Koszul resolution. Each pair has scalar cohomology, so their tensor product resolves k. Applying Hom into k gives zero differential and the algebra on the right of (6.17). The bar resolution computes the same Ext. The primitive projection is surjective on cohomology and both sides have the same finite dimension in each auxiliary weight, proving the quasi-isomorphism. For an infinite-dimensional derivation algebra take the filtered union of finite-dimensional symmetric-coalgebra computations.

To lift a Maurer–Cartan element, solve successively in actor weight. At the first unsolved weight, its curvature is a closed degree-two element of the kernel of π. A degree-one primitive removes it. The same argument with forms relative to a simplex boundary lifts every compatible simplex. Tensoring a weightwise acyclic complex with these forms preserves acyclicity over k. The resulting fibres are contractible. In the target, the commutator bracket and differential vanish. Its degree-one elements give (6.18), its degree-zero elements give the stated loops, and its negative degrees vanish. □

For one derivation, retain also the derivative order. The sector where actor weight and derivative order both equal q has cohomology

$$\mathrm{Sym}^q(P) \otimes \Lambda^q(kD) \quad \text{in degree } q.$$

It is therefore acyclic for $q \geq 2$. For several commuting derivations, the same formula has their span in place of kD; its degree-two term need not vanish.

Lemma 6.5. *Restriction along* $\mathrm{CE}(\mathfrak{h}) \to \mathrm{B}\,U(\mathfrak{h})$ *is a surjective quasi-isomorphism on natural convolution complexes twisted by the scalar action. Its kernel contracts naturally, preserving actor weights and derivative orders. It has the same property relative to any union of actor unit faces.*

Proof. For abelian $\mathfrak{h}$, multilinear PBW weight d of the bar is the finite complex of ordered partitions of d named inputs, with differential merging adjacent blocks. The Koszul resolution used above identifies its homology with suspended symmetrization. Choose rational complements to cycles and boundaries and average over S_d. Inverting the differential on the boundary complement gives a permutation-equivariant contraction, normalized to

$$pj = 1, \quad hj = ph = h^2 = 0, \qquad dh + hd = 1 - jp.$$

Evaluation on tensor powers gives a natural contraction for every graded $\mathfrak{h}$, compatible with its internal differential.

For nonabelian $\mathfrak{h}$, PBW identifies the change in the bar differential with a perturbation b that lowers primitive degree. The formulas $p' = p(1+bh)^{-1}$ and $h' = h(1+bh)^{-1}$ are finite on each input. The inclusion stays j, since $bj = jQ_2$ and $hj = 0$. The transferred differential is the Chevalley differential. Precomposition by p' splits restriction, and $f \mapsto (-1)^{|f|} f h'$ contracts its untwisted kernel.

The scalar twist preserves that kernel because j is a coalgebra map. Its commutator raises the least primitive input degree on which a cochain is nonzero. Perturbing the kernel contraction again therefore gives finite sums on each input. All these maps use permutations, PBW operations, products and brackets. They commute with exterior face projections and preserve derivative orders. Restriction to cochains vanishing on the chosen faces proves the relative assertion. □

Natural identities evaluate on the coefficient category because they are identities of these multilinear operations. On a configuration of m positions with n exterior generators, actor weight is at most mn. Their formal weight products introduce no completion of a state or of a normal derivative.

Theorem 6.6. *For a vertex derivation D commuting with translation, the action α_D admits a coherent unital multiplicative structure for the ordinary exterior product of C_E. Within natural operations of equal actor and derivative weight, the higher fillings are contractible after fixing the unary action and a compatible binary bar path. Every profile $\alpha_{\mathbf{t}}$ has a map of pointed actions to this multiplicative action. In particular its two unit insertions have the common recipient*

$$\mu(\Sigma_{\mathbf{t}} \otimes^{!} \Sigma_{\mathbf{t}}) : (C_E, \alpha_{\mathbf{t}})^{\circledast 2} \longrightarrow (C_E, \alpha_D). \tag{6.20}$$

Proof. Use the natural bar complex for n exterior actors. Restriction to their proper unit faces is surjective: the coordinate projections commute, so inclusion–exclusion extends compatible face data. Its kernel is the fully mixed part. For $n \geq 2$ that part has actor weight at least two and is acyclic in the equal derivative-weight sector, by Theorem 6.4 and Lemma 6.5.

Consequently compatible Maurer–Cartan data on the actor faces and the boundary of a parameter cube extend across the cube. Lift the data linearly, then kill the first remaining curvature in each actor weight by a primitive in the fully mixed complex tensored with the forms relative to the cube boundary. The Bianchi identity makes the curvature closed. Further parameter simplices give the same lifting, proving contractibility of the relative choices.

The component for n actors uses the cube $[0,1]^{n-1}$. At $t_i = 0$ multiply the adjacent actors; at $t_i = 1$ compose the two shorter bar maps. An end unit deletes its adjacent parameter. An interior unit combines the two parameters by

$$a \vee b = a + b - ab.$$

This polynomial operation is associative, with identity zero and absorbing element one. It makes all repeated unit restrictions agree with the endpoint faces. The preceding lifting constructs the binary path and then every higher component, or extends a specified compatible binary path.

The endpoint compositions are the geometric ones. Indeed, $\mathrm{id} + hD$ over $k[h]/(h^2)$ is a vertex automorphism, and Lemma 6.2 carries D to $D \otimes 1$ on exterior coefficient extensions. Its naturality identifies every bar operation and contraction with the corresponding coefficient map. Thus the constructed families retain collisions, normal transfers and the vacuum.

Proposition 6.3 supplies each incoming map. Composing its tensor square with μ gives (6.20). On either unit axis it is $\Sigma_{\mathbf{t}}$ with its action path. The unital coherence identifies these restrictions on their common vacuum, supplying the required relative two-simplex. The same higher product coherences receive any finite ordered product of profile maps. □

The contractibility is in the specified natural-operation complex. The space of all geometric multiplicative structures can have additional components or homotopy groups.

6.5 First-weight classes and actor size

Fix $\kappa \in k^{\times}$ and the Heisenberg vertex algebra

$$V_\kappa = k[x_1, x_2, \ldots], \qquad \mathrm{j} = x_1, \qquad x_n = \mathrm{j}_{-n}\mathbf{1}.$$

Its states are polynomials. Translation and the positive current modes are

$$T x_n = n x_{n+1}, \qquad \mathrm{j}_n = \kappa n \partial_{x_n} \quad (n > 0).$$

Let $L_0 x_n = n x_n$, extended as a derivation of the polynomial algebra. Put $\mathfrak{g} = L_{V_\kappa}$, $A = U_\star \mathfrak{g}$, $P_w = \mathrm{Sym}^w_\star \mathfrak{g}$ and $C = C_{V_\kappa}$. Lie-module actions on P_w and A are adjoint actions. A coefficient derivation is a degree-zero map of the full family $\mathfrak{g}$ which obeys Leibniz for its Lie bracket. Their Lie algebra is $\mathfrak{d}_V = \mathrm{Der}_{\mathscr{C}}(\mathfrak{g})$.

In left-module coordinates, the singleton coefficient has connection $\partial_y + T$. A coefficient derivation is therefore represented there by a $k[y]$-linear map $B(y)$ with

$$\partial_y B + [T, B] = 0, \tag{6.21}$$

together with compatibility with the chiral bracket. Each input state has a finite polynomial image. A common polynomial-degree bound on all inputs is not imposed. For example, $B(y) = L_0 + yT$ is horizontal because $[T, L_0] = -T$. The conformal identity for $Y(a, z)$ gives

$$B(q)Y(a, p-q)b = Y(B(p)a, p-q)b + Y(a, p-q)B(q)b.$$

Taking arbitrary Laurent principal parts proves its coefficient derivation identity, including the normal derivatives.

The current shift $D = \partial_{x_1}$ is another such derivation: it is the derivative of $\mathrm{j} \mapsto \mathrm{j} + t\mathbf{1}$. Every state zero mode $a_{(0)}$ is a vertex derivation commuting with T, by the mode commutator identity. Put $Z_r = (\mathrm{j}^r)_{(0)}$ for $r \geq 2$. Normal ordering gives

$$Z_2 = 2\kappa T, \qquad Z_r\mathrm{j} = r(r-1)\kappa x_1^{r-2}x_2, \qquad (\mathrm{j}^2)_{(1)} = 2\kappa L_0. \tag{6.22}$$

Indeed the contraction of $\mathrm{j}^r(z)$ with one current is $r\kappa z^{-2}{:}\mathrm{j}(z)^{r-1}{:}$, whose residue is $r\kappa T(\mathrm{j}^{r-1})$. There is no multiple contraction with the single current on the right. The quadratic zero and first modes give the displayed formulas for T and L_0 on every polynomial state. The images of j show that $D, Z_2, Z_3, \dots$ are linearly independent.

6.5.1 An additional output root

The family P_w is the open direct image of its coefficient on w distinct roots, supported on configurations with at most w distinct positions. On the complement of the locus with fewer roots, each ordered partial diagonal is a copy of $\mathrm{Conf}_w(X)$. Its coefficient is the tensor of w state modules, equivariant under S_w. Reconstruction is exact: use closed direct image on each partial diagonal and localization along its cross-root divisors. This also retains every normal derivative.

Lemma 6.7. *For $a, w \geq 1$,*

$$\mathrm{Ext}^q_{\mathfrak{g}}(P_w, P_a) = 0 \quad \text{if } q < a - w.$$

For every coefficient derivation B, the formula

$$\gamma_B^{(w)}(x; v_1 \cdots v_w) = m_{w+1}(Bx, v_1, \dots, v_w) \tag{6.23}$$

defines a class in $\mathrm{Ext}^1_{\mathfrak{g}}(P_w, P_{w+1})$, where m_{w+1} is symmetric multiplication. The map from $\mathfrak{d}_V$ to this extension group is injective.

Proof. A coefficient family supported on at most $a-1$ roots has zero derived Hom to P_a, by open adjunction. The nth Chevalley term computing the desired module maps is

$$\mathrm{RHom}_{\mathscr{C}}(\Lambda^n_\star\mathfrak{g} \star P_w, P_a)[-n].$$

It vanishes for $w + n < a$. Before the shift it has no negative cohomology, since its inputs are coefficient modules in degree zero. The first-quadrant totalization proves the bound.

Give $P_{w+1} \oplus P_w$ the triangular action with off-diagonal term (6.23). Its module defect is symmetric multiplication by $[x, By] + [Bx, y] - B[x, y]$, hence is zero on the full coefficient diagram. Open adjunction also

gives $\mathrm{Hom}_{\mathscr{C}}(P_w, P_{w+1}) = 0$. The underlying split extension has only its canonical coefficient splitting. If $B \neq 0$, this splitting is not equivariant: on $w+1$ distinct roots put a state on which B is nonzero at the acting root and the vacuum at all other roots. Its image under (6.23) is nonzero. The same argument for differences proves injectivity. □

To compare homotopies, resolve the pointed arrow before taking relative convolution cochains. Bar–cobar duality gives maps under A from $U_\star(\mathfrak{g} \ltimes e\mathfrak{g})$ to A. At the scalar action, the weight-w complex is

$$T_w = \mathrm{RHom}_{\mathfrak{g}}(P_w, A). \tag{6.24}$$

Indeed the underlying relative Chevalley comodule is $\mathrm{CE}_\star(\mathfrak{g}) \star \mathrm{Sym}^c_\star(e\mathfrak{g}[1])$, and $e\mathfrak{g}[1] = \mathfrak{g}$. The coefficient resolutions are taken in the full enhanced diagram. This computes the homotopy fibre specifying the pointing, rather than the kernel of an unresolved restriction.

Scaling e by a parameter over $k[t]/(t^2)$ defines

$$\nu : \pi_0 \Phi_C(C_E) \longrightarrow H^1(T_1).$$

The first coefficient is a cocycle; a pointed homotopy changes it by a boundary. More generally a relative homotopy has a normalized convolution gauge $G = 1 + g_1 + g_2 + \cdots$ satisfying $d_0 G = G\theta - \theta' G$. This follows by integrating the polynomial interval equation successively in positive weight. Every inverse and exponential is finite on a fixed configuration. If both actions vanish in weights below r, the equations give

$$d_0 g_j = 0 \quad (j < r), \qquad d_0 g_r = \theta_r - \theta'_r. \tag{6.25}$$

The lower gauge coefficients need not be zero.

Theorem 6.8. *There are vector-space identifications*

$$\mathrm{Ext}^1_{\mathfrak{g}}(P_1, P_2) \cong \mathfrak{d}_V, \qquad H^1(T_1) = k\xi \oplus \mathfrak{d}_V. \tag{6.26}$$

The first identification is $B \mapsto [\gamma_B^{(1)}]$. The class ξ changes the zero-mode charge at one root. For every $B \in \mathfrak{d}_V$, the tree action α_B has first class $[\gamma_B^{(1)}]$.

Proof. In the Chevalley extension spectral sequence

$$E_1^{n,q} = \mathrm{Ext}^q_{\mathscr{C}}(\Lambda^n_\star \mathfrak{g} \star \mathfrak{g}, P_2), \qquad n, q \geq 0,$$

open adjunction kills the $n = 0$ column. Only $(n, q) = (1, 0)$ can contribute in total degree one. There are no incoming boundaries, and every higher differential has a negative index. Thus the extension group consists precisely of module cocycles $\Gamma : \mathfrak{g} \star \mathfrak{g} \to P_2$.

On the two-root locus write $R = k[p, q, (p-q)^{-1}]$. Such a map is an R-linear endomorphism of $R \otimes (V_\kappa \otimes V_\kappa)$ with

$$\partial_p \Gamma + [T \otimes 1, \Gamma] = 0, \qquad \partial_q \Gamma + [1 \otimes T, \Gamma] = 0.$$

Its individual input images are finite; no uniform pole bound is assumed. In the module cocycle equation give the two acting inputs positions r, p and the module input position q. Away from the triple diagonal the target has three disjoint pieces $r = q$, $p = q$, $r = p$, whose open direct images detect equality.

On $r = q$, comparison of the zeroth normal coefficients after testing $(r-q)^n$ gives

$$\Gamma(y \otimes x_{(n)} v) = (1 \otimes x_{(n)})\Gamma(y \otimes v) \quad (n \geq 0).$$

The full normal transfer is retained; this projection supplies necessary commutation identities. Take $x = \mathrm{j}$, $n > 0$, and $x = \mathrm{j}^2/(2\kappa)$, $n = 1$. Then Γ commutes with all partial derivatives and L_0 in the second state factor. Its restriction to the weight-zero line of that factor defines B. By induction on second-state weight, $\Gamma(a \otimes v) - Ba \otimes v$ is killed by all second-factor partial derivatives. It is therefore constant in that factor, while having positive weight, and is zero. Hence $\Gamma = B(p, q) \otimes 1$.

The second horizontality equation gives $\partial_q B = 0$. A q-invariant element of R lies in $k[p]$: it depends only on p in the fraction field, and substitution $q = p - 1$ makes it polynomial. This applies to each matrix coefficient, so $B = B(p)$ and it obeys (6.21). On $r = p$ the cocycle equation is the derivation identity for B, tensored with the unchanged spectator at q. The normal transfer differentiates its coordinate dependence; $B(r)$ is not replaced by $B(p)$ before that transfer. Tensoring with a spectator vacuum and inverting $q - p$ is faithful, so this identity holds in the full coefficient family. This proves surjectivity; Lemma 6.7 proves injectivity.

Equivariant PBW decomposes A by output number. It is both a sum and a product, because only finitely many output numbers occur at any configuration. Lemma 6.7 leaves only one and two outputs in $H^1(T_1)$. At one root, extensions are extensions of the unital moving Heisenberg vacuum. The positive modes give the polynomial Koszul contraction for $\kappa n \partial_{x_n}$. Its nonconstant monomials contract by the Euler homotopy, leaving the zero-mode class. The de Rham complex of the affine root coordinate has only its scalar class, so this contribution is $k\xi$. Concretely, over $k[\epsilon]/(\epsilon^2)$ take $\mathrm{j}_0 = \epsilon$ and connection $\partial_y + T + \epsilon x_1/\kappa$. Its commutator with j_1 is $-\epsilon$, and the other current translation relations remain unchanged. This is the charge extension; its zero-mode action prevents a split. These calculations identify the one-output summand.

Finally the first tree term is $Bx\,v$. PBW writes it as $m_2(Bx, v) + [Bx, v]/2$. For the zero-cochain B, $(dB)(x; v) = [x, Bv] - B[x, v] = -[Bx, v]$. The output-one term is exact and the first class is the asserted output-two extension. □

Horizontality in the proof does not mean that an arbitrary two-root map has bounded poles on the entire state space. For $z = p - q$, the operator

$$\mathcal{K}_z = \kappa \sum_{m,n \geq 1} \frac{(-1)^{m-1}(m + n - 1)!}{(m - 1)!(n - 1)!} z^{-m-n} \partial_{x_m} \otimes \partial_{x_n} \tag{6.27}$$

is horizontal and finite on each polynomial input. For its scalar coefficients $c_{m,n}$, $mc_{m+1,n} = -(m + n)c_{m,n}$ and $nc_{m,n+1} = (m + n)c_{m,n}$; together with $[T, \partial_{x_m}] = -(m - 1)\partial_{x_{m-1}}$ these prove the two connection equations. Its pole order on $x_m \otimes x_n$ grows with $m + n$. The commutant condition, not a uniform pole assumption, is what removes the poles in Theorem 6.8.

6.5.2 *Bounds on an actor*

Corollary 6.9. *For every $N \geq 1$ there is a nontrivial pointed Heisenberg action which is the trivial action on every configuration with at most N positions, including all its collisions and normal derivatives.*

Proof. Take $t_N = 1$ and the other profile parameters zero. Its first nonzero cochain is $Dx\, u_{[N]}$. Its PBW output-$N + 1$ component is the nonzero extension $[\gamma_D^{(N)}]$ of Lemma 6.7. Equation (6.25) prevents a homotopy to the trivial action. Every nonconstant term uses at least $N + 1$ positive supports, so vanishes on the stated smaller configurations. □

Proposition 6.10. *Suppose a pointed action of C_E has zero positive weights below $r > 1$ and nonzero first class $[\theta_r]$. It cannot be multiplicative for the standard exterior-algebra product.*

Proof. Pull the two candidate binary actions back along the primitive diagonal induced by $e \mapsto e_1 + e_2$. Multiplication followed by the action has first class $2^r[\theta_r]$. Composition has first class $2[\theta_r]$. To see the latter with its paths, scale the two actors independently by s, t and truncate at total degree r. The first nonzero jet of composition is the sum of the pure jets, since a product of two perturbations has degree at least $2r$. Unit homotopies identify

the pure terms. A pointed binary path would therefore force $(2^r - 2)[\theta_r] = 0$ by (6.25), contrary to characteristic zero. □

The profile actions satisfy this obstruction when their first parameter has index greater than one. They still have the multiplicative recipient (6.20).

For any pointed factorizing actor F, let B_F be a cobar model under $U_\star L_k$, with its induced adjoint L_k-action. At the augmented map $C_E \to J \to F$, the first-weight mapping complex is

$$T_1^{E,F} = \mathrm{RHom}_{L_k}(L_k, B_F).$$

The relative Chevalley argument of (6.24) gives this identification, without a degree bound on B_F. A fixed action β of F induces a linear map

$$b_\beta : H^1(T_1^{E,F}) \longrightarrow H^1(T_1), \qquad \nu\big(\beta(f \otimes^! \mathrm{id}_C)\big) = b_\beta(\nu_F(f)). \tag{6.28}$$

Indeed, terms with one perturbation in a compatible bar composite give a chain map; terms with two perturbations vanish over $k[t]/(t^2)$. Resolving the pointed arrow retains its prescribed homotopy. This finite free scalar extension commutes with derived coefficient maps and diagram limits.

Lemma 6.11. *For W finite-dimensional in degree zero and $F = C_{E_W}$, one has $H^1(T_1^{E,F}) = W$. Its vectors are represented by the linear exterior maps $e \mapsto w$.*

Proof. In PBW for $U_\star L_{E_W}$, let a count output roots and b their total exterior degree. The support bound puts Chevalley degree at least $a - 1$. Total degree one allows only $(a, b) = (1, 0), (1, 1), (2, 0)$. The first is $H^1_{\mathrm{dR}}(\mathbb{A}^1; k) = 0$. The second is $\mathrm{Hom}_{L_k}(L_k, L_k) \otimes W = W$. For $(2, 0)$, open adjunction leaves only the scalar multiple m of symmetric multiplication in Chevalley degree one. Its differential is $(dm)(x, y; v) = m([x, y], v)$, nonzero on a simple pole between the acting points with the module root distinct. The vacuum principal-part bracket has residue one there. Thus this last summand supplies no class. □

Theorem 6.12. *If a pointed action (F, β) represents Φ_C, then $H^1(T_1^{E,F})$ is infinite-dimensional. The same conclusion holds if (F, β) is a local centre in $\mathscr{B}_C$. Consequently no coalgebra equivalent as a pointed object to C_{E_W}, with W finite-dimensional and $d = T = 0$ on its exterior generators, can underlie either object.*

Proof. Representability supplies an incoming map from every α_{Z_r}. Equation (6.28) puts every $[\gamma^{(1)}_{Z_r}]$ in the image of b_β. These classes are linearly independent by (6.22) and Theorem 6.8. If only finality in $\mathscr{B}_C$ is assumed, the same argument applies because each α_{Z_r} is multiplicative by Theorem 6.6. Lemma 6.11 excludes the stated finite exterior actors in both cases. □

A finite exterior actor can still receive a specified finite family. The theorem excludes its universal property for all actions. It does not exclude a larger factorizing centre or identify the whole formal deformation functor of a recipient.

6.6 Recipients for nilpotent families

Several derivations need not commute. Their action classes compose by the Baker–Campbell–Hausdorff law. An exterior actor can carry that law, but its multiplication must retain which input supplies each output generator. We first describe these marked coefficients.

6.6.1 Natural actor maps

In this subsection all actor maps are under and over J, with both comparison paths. Let W, V be finite-dimensional vector spaces in degree zero, and write $C_W = C_{E_W}$ and $C_V = C_{E_V}$. Natural operations between C_W and C_V use products, brackets, permutations and linear maps of exterior coefficients. Each ordinary Lie input is used once. No additional derivation acts on it. The relative convolution cochains vanish on purely ordinary inputs and take values in the augmentation ideal of $U_\star L_{E_V} \to U_\star L_k$.

Theorem 6.13. *The space of natural maps $C_W \to C_V$ is equivalent to the discrete space of formal matrices*

$$A(t) \in \mathrm{Hom}(W, V)[[W^*]].$$

The coefficient of polynomial degree $r-1$ has source actor weight r. Its coordinate contraction is $\Phi_A(t) = A(t)t$. Composition and diagonal tensor are

$$(B \circ A)(t) = B(A(t)t)A(t), \qquad (A \oplus B)(t,s) = \begin{pmatrix} A(t) & 0 \\ 0 & B(s) \end{pmatrix}. \tag{6.29}$$

The identity is the constant identity matrix.

Proof. Let $K_W = (E_W^+)^\vee$ and let $\mathfrak{q}_W$ be its Lie cobar. For a basis of W, a generator y_I indexed by a nonempty exterior subset has degree $1 - |I|$ and differential

$$dy_I = \frac{1}{2} \sum_{\substack{B \sqcup D = I \\ B, D \neq \varnothing}} (-1)^{|B|-1} \epsilon(B, D)[y_B, y_D].$$

The sign is obtained by the exterior-label rescaling used in (6.19). In particular $dy_{\{i,j\}} = [y_i, y_j]$ for $i < j$. Adjoin a degree-one Lie generator z with $dz = [z,z]/2$ by Lie coproduct, writing $\mathfrak{h}_W = \mathrm{Lie}(z) \amalg \mathfrak{q}_W$.

The natural convolution algebra identifies with the summand of $U(\widehat{\mathfrak{h}_W \otimes E_V})$ having positive source weight and positive target exterior degree. To check this assertion, polarize the inputs and evaluate on the free Lie algebra on their names. A Lie word specifies an output root, and a PBW product specifies a forest of roots, each with its exterior label. Distinct forests give independent multilinear operations. Evaluation on the group-like exponential of the suspended inputs turns convolution into multiplication. Transposing exterior multiplication gives the displayed Lie cobar differential. The scalar twist cancels the couplings between z and the nonempty labels by equivariance; its remaining equation is $dz = z^2$.

Killing z is a quasi-isomorphism in each source weight. Before tensoring with E_V, the associative envelope is

$$(k\langle z \rangle, dz = z^2) * U(\mathfrak{q}_W).$$

Contract the first positive z factor in every alternating word by $h(z^{2j}) = z^{2j-1}$, $h(z^{2j-1}) = 0$, with the prefix sign. PBW identifies the Lie algebra with a direct summand of its envelope, so the Lie projection is also a quasi-isomorphism. Tensoring with E_V and taking envelopes preserve it: tensor powers and symmetric coinvariants are exact over k.

The projection $\mathfrak{q}_W \to W^*$ sends y_i to its namesake in an abelian Lie algebra and kills longer exterior labels. The resolution $E_W \otimes k[v_i]$, $dv_i = e_i$, contracts in positive combined weight N by $N^{-1} \sum_i v_i \partial_{e_i}$. Dualizing the bar comparison in each finite weight proves $H(U\mathfrak{q}_W) = \mathrm{Sym}(W^*)$, with this projection. The PBW length-one summand again gives the Lie quasi-isomorphism. Consequently the relative natural algebra has a surjective dg algebra projection, with weightwise acyclic kernel, to

$$\mathcal{B}(W, V) = \left[\widehat{\mathrm{Sym}}(W^* \otimes E_V)\right]_{E_V^+}.$$

The subscript denotes the ideal generated by the positive target exterior labels. Its differential is zero. It has no terms in degrees at most zero and

$$\mathcal{B}^1(W, V) = \widehat{\mathrm{Sym}}(W^*) \otimes W^* \otimes V.$$

Relative Maurer–Cartan lifting, as in Theorem 6.4, has contractible fibres. In the target a degree-one simplex is a polynomial function with zero differential, hence constant. This proves the claimed discreteness.

The surviving $W^* \otimes V$ factor marks the input supplying the target exterior generator. The other input factors are unmarked. Exponential polarization supplies their factorials. In a bar composite, unmarked coefficients substitute $t \mapsto A(t)t$, while the marked factor is transported by $A(t)$ and then by $B(A(t)t)$. All joined-root terms vanish under the projection. This proves the first formula in (6.29); disjoint actor groups give the block-diagonal formula. The relative bar contraction transports these calculations to actual maps and paths. At m positions the source actor weight is at most $m \dim W$, so every formula is finite there. □

The theorem classifies the specified natural maps. Full geometric mapping spaces can contain coefficient extensions. For example, with left differential modules and $D_x = k\langle x, \partial_x\rangle$, put $M = k[x, x^{-1}]$ and $\Delta = D_x/D_x x$. The presentation $M = D_x/D_x(x\partial_x + 1)$ gives a two-term free resolution. On $\partial_x^n \delta \in \Delta$, the resulting Hom differential $x\partial_x + 1$ is multiplication by $-n$. Thus $\mathrm{Hom}_{D_x}(M, \Delta) = k$ and $\mathrm{Ext}^1_{D_x}(M, \Delta) = k$. This normal-chart extension alone does not construct a compatible operation on the whole collision diagram.

6.6.2 *Multiplication with a chosen frame*

An associative multiplication on C_W has a matrix $M(x, y) = (L(x, y) \mid R(x, y))$ and coordinate contraction $F(x, y) = L(x, y)x + R(x, y)y$. Its unit conditions are $L(x, 0) = I$ and $R(0, y) = I$. By (6.29), associativity is precisely

$$\begin{aligned} L(F(x, y), z)L(x, y) &= L(x, F(y, z)), \\ L(F(x, y), z)R(x, y) &= R(x, F(y, z))L(y, z), \\ R(F(x, y), z) &= R(x, F(y, z))R(y, z). \end{aligned} \tag{6.30}$$

It gives the fibrewise-linear formal group law

$$(x, u)(y, v) = (F(x, y), L(x, y)u + R(x, y)v).$$

Both its zero section and its diagonal are homomorphisms. Conversely, such a law gives the displayed matrix equations. Discreteness of the natural mapping spaces supplies a contractible space of higher monoid coherences.

Let $\mathfrak{a}$ be the tangent Lie algebra of F. A compatible left-symmetric product on $\mathfrak{a}$ is $x \cdot y = \ell_x y$ such that

$$\ell_x y - \ell_y x = [x, y], \qquad [\ell_x, \ell_y] = \ell_{[x,y]}.$$

These products describe the fibre laws up to changes of frame which fix the diagonal and are the identity on the fibre at zero. Here is the construction, including the frame itself.

The affine Lie map

$$x \longmapsto \begin{pmatrix} \ell_x & x \\ 0 & 0 \end{pmatrix}$$

integrates formally to a representation ρ_0 and a cocycle b, with $db_0 = I$ and $b(F(x, y)) = b(x) + \rho_0(x)b(y)$. In exponential coordinates these are $\rho_0(x) = e^{\ell_x}$ and $b(x) = \int_0^1 e^{t\ell_x} x\, dt$. The formal

matrix exponential and its Baker–Campbell–Hausdorff identity prove both equations. Choose an invertible frame $M(x)$ with $M(0) = I$ and $M(x)x = b(x)$. One always exists: if $b = \sum_{r\geq 1} b_r$ is its homogeneous expansion, take $M = \sum_{r\geq 1} Db_r/r$ and use Euler's identity. Then

$$L(x, y) = M(F(x, y))^{-1} M(x), \qquad R(x, y) = M(F(x, y))^{-1} \rho_0(x) M(y) \tag{6.31}$$

satisfies (6.30) and has coordinate contraction F.

For the converse, the zero section splits the fibre group. Multiplication $(0, w)(x, 0) = (x, L(0, x)w)$ trivializes it. The frame $M(x) = L(0, x)^{-1}$ transforms the law to a semidirect product, and the diagonal becomes a cocycle $b(x) = M(x)x$. Differentiating its identities gives exactly the two identities for ℓ. Another frame M' changes the original law by $S(x) = M'(x)^{-1} M(x)$, with $S(x)x = x$ and $S(0) = I$. Conversely such an isomorphism leaves the infinitesimal product unchanged. This proves the assertion about changes of frame, while retaining the distinct multiplication matrices on a fixed actor.

For instance, on $W = k^2$ with additive F, put $N(t) = t(-t_2, t_1)$ and $S(t) = I + N(t)$. Then $N^2 = 0$, $S^{-1} = I - N$ and $S(t)t = t$. The frame gives $L(x, y) = S(x + y)^{-1} S(x)$ and $R(x, y) = S(x + y)^{-1} S(y)$. Although its fibre conjugation is trivial, its multiplication is not the constant additive matrix: $L((1, 0)^T, (0, 1)^T) = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$.

For a finite-dimensional positively graded Lie algebra $\mathfrak{a} = \bigoplus_{j>0} \mathfrak{a}_j$, let $G|_{\mathfrak{a}_j} = j$. The product

$$\ell_x y = G^{-1}[x, Gy] \tag{6.32}$$

is compatible: the derivation identity for G proves its antisymmetrization, and $\ell_x = G^{-1} \operatorname{ad}_x G$ proves its representation identity. In Baker–Campbell–Hausdorff coordinates choose $M(x) = \int_0^1 e^{t\ell_x}\, dt$. Every ℓ_x raises positive grading. The exponentials, M^{-1} and F are therefore polynomials. Formula (6.31) constructs a natural coherent monoid on $C_{\mathfrak{a}}$.

6.6.3 *Composition of action classes*

For a natural action with full twisting cochain τ, adjoin independent central degree-zero test elements z_i annihilated by the derivations and put $\zeta = \sum_i e_i z_i$. The operator

$$T_\tau(z)x = \tau\big(sx \exp(s\zeta)\big)$$

has constant term x. It defines a series in $\prod_{q\geq 0} \operatorname{Sym}^q(W^*) \otimes U(\mathfrak{d})$. The twisting equation on two ordinary inputs gives $T_\tau[x, y] = [T_\tau x, T_\tau y]$. Evaluate on the free Lie algebra on the named free modules $U(\mathfrak{d})x \oplus U(\mathfrak{d})y$. The map $a \otimes b \mapsto [ax, by]$ is injective, by projection to the corresponding ordered word in its tensor envelope. The preceding identity therefore says $\Delta T_\tau = T_\tau \otimes T_\tau$.

Its logarithm is primitive, hence belongs to the completed $\mathfrak{d}$-valued series. PBW projection to degree one identifies this logarithm with the action class ϕ of (6.18): powers of the one element ϕ are already symmetric powers. Thus $T_\tau = e^{\phi}$. The actual tensor extension of an inner action carries D to $D \otimes 1$, by Lemma 6.2. It fixes the outer central test parameters. Composition with an outer action of class ϕ and an inner action of class ψ therefore has operator $e^{\phi} e^{\psi}$ and class

$$\mathrm{BCH}(\phi, \psi) = \log(e^{\phi} e^{\psi}). \tag{6.33}$$

The bracket is $[fD, gB] = fg[D, B]$ and raises actor weight. For example $[D, Z_3] = 6\kappa T$ and T is central, so

$$\mathrm{BCH}(pD, qZ_3) = pD + qZ_3 + 3\kappa pqT.$$

Reversing the action order reverses the last sign.

Pullback along a natural actor map of matrix $A(t)$ sends an action class ψ and a scalar loop σ to

$$\psi(A(t)t), \qquad \sigma(A(t)t). \tag{6.34}$$

Indeed removing the marked generator gives $q_j \mapsto \sum_i A_{ji}(t)t_i$. Precomposition on the natural bar preserves products; its scalar derivation generators are unchanged. This proves the formula on the whole cohomology algebra. Postcomposition by a fixed outer action preserves the scalar loop of the inner action: the outer operator fixes its central, derivation-inert test coefficients.

For a finite-dimensional nilpotent Lie algebra $\mathfrak{a}$, let $F(x, y) = \mathrm{BCH}(x, y)$. Its normalized formal q-cochains with trivial scalar coefficients are formal series $c(x_1, \dots, x_q)$ which vanish whenever any argument is zero. The completion is in total coordinate degree. Their differential is

$$\begin{aligned}(\delta c)(x_1, \dots, x_{q+1}) = {} & c(x_2, \dots, x_{q+1}) \\ & + \sum_{i=1}^{q} (-1)^i c(x_1, \dots, F(x_i, x_{i+1}), \dots, x_{q+1}) \\ & + (-1)^{q+1} c(x_1, \dots, x_q).\end{aligned} \tag{6.35}$$

Associativity and the unit give $\delta^2 = 0$. Degree zero is k with zero differential. Denote the cohomology by $H^\bullet_{\mathrm{for}}(F; k)$. These are functions near the formal identity, rather than cochains on the discrete group of k-points.

Theorem 6.14. *Let $\mathfrak{a}$ be finite-dimensional and positively graded, and let $\rho : \mathfrak{a} \to \mathfrak{d}_V$ be a Lie map. There is a natural action of $C_{\mathfrak{a}}$ on the Heisenberg boundary with class $x \mapsto \rho(x)$, compatible with the monoid (6.31). Compatibility includes every unit and associativity homotopy. Every natural action in this class admits such a structure.*

Proof. First work universally with derivation algebra $\mathfrak{a}$ and class $\theta(x) = x$. The natural convolution projection constructs an action in this class. Its two binary composites have the same class, since

$$\theta(F(x, y)) = \mathrm{BCH}(\theta(x), \theta(y)).$$

Choose a comparison path $h(x, y)$ relative to both unit axes. The fully mixed summand is a direct summand of the acyclic kernel of that projection, so relative Maurer–Cartan lifting retains these unit restrictions.

Let K_n be the associahedron of bracketings of n ordered inputs; its dimension is $n - 2$. For a module the last input is C and the others are actors. The path h occupies K_3. The boundary of K_4 is now a scalar loop $o(x, y, z)$. It vanishes on each unit face. Replacing h by a normalized scalar loop $\sigma(x, y)$ changes it by

$$(\delta\sigma)(x, y, z) = \sigma(y, z) - \sigma(F(x, y), z) + \sigma(x, F(y, z)) - \sigma(x, y).$$

These transports are (6.34) and the unchanged inner scalar loop. The boundary in the next associahedron gives $\delta o = 0$: each transported edge occurs twice with opposite orientation, the square faces express functoriality, and the pure actor face is supplied by the monoid coherence. Hence o defines a class in the normalized formal-group cohomology $H^3_{\mathrm{for}}(F; k)$, with trivial scalar coefficients.

This cohomology equals $H^3(\mathfrak{a}; k)$ equivariantly for the grading torus. To see this, finite-order distributions at the identity form $U(\mathfrak{a})$, with PBW associated graded. Formal functions are its full linear dual. The normalized group cochain complex is the dual bar resolution of the scalar module. The Chevalley resolution computes the same derived Hom; its exactness follows from the polynomial Koszul resolution under PBW. Antisymmetrization into the bar dualizes to differentiation at the identity. This gives the asserted comparison in every degree; see also [67, Theorem 4.2 and Corollary 4.3].

The obstruction class is independent of the chosen lift and binary path. The grading torus preserves θ, (6.32), its frame and the natural bar comparisons, so this class is invariant. But every weight of $\Lambda^q \mathfrak{a}^*$ is strictly negative for $q > 0$. Thus its invariant cohomology vanishes. Choose a normalized σ with $\delta\sigma = -o$ and correct the binary path. The resulting boundary fills.

All later attaching spheres have dimension at least two. Their action spaces have zero homotopy groups in these dimensions by Theorem 6.4. Successive module associahedra therefore fill, with their specified unit faces. Positive grading has finite pieces, so the formal corrections respect the completion. Evaluation is finite at every configuration. Finally applying $U(\rho)$ to the universal operations preserves all their equations, paths and units. A natural homotopy to another action in the same class transports the constructed module structure. □

The binary path in this proof is chosen and corrected. The theorem does not require it to equal an independently prescribed path. Nor does the invariant vanishing assert that all third cohomology vanishes. For $[a, b] = c$ with weights 1, 1, 2, the Chevalley differential is $dc^* = -a^* \wedge b^*$. All degree-two cochains are closed, so $[a^* \wedge b^* \wedge c^*]$ is a nonzero top class, of grading weight -4.

Corollary 6.15. *A finite family of natural exterior-actor actions whose classes take values in one finite-dimensional nilpotent Lie subalgebra of $\mathfrak{d}_V$ has a common coherent multiplicative recipient on a finite-dimensional exterior actor. Their classes may be formal series.*

Proof. If the subalgebra is zero, use J. Otherwise choose its generators and nilpotence class c. The free nilpotent Lie algebra $\mathfrak{a}$ of class c on these generators is finite-dimensional, positively graded by bracket length, and surjects onto the subalgebra. A linear section lifts each formal action class to a series $\Phi(t) \in \mathfrak{a}[[W^*]]$ with zero constant term. If $\Phi = \sum_{r \geq 1} \Phi_r$, the matrix

$$A_\Phi(t) = \sum_{r \geq 1} \frac{1}{r} D\Phi_r(t)$$

satisfies $A_\Phi(t)t = \Phi(t)$. Theorem 6.13 realizes it. Pullback of the action in Theorem 6.14 has the given class by (6.34), so equality of natural classes supplies the required pointed action path. □

The incoming maps in this corollary need not preserve multiplication. For additive one-dimensional actors, the map $\Phi(t) = t^2$ has matrix $A(t) = t$, but $\Phi(s + t) - \Phi(s) - \Phi(t) = 2st$.

There is also a precise limitation on paths within this natural augmented category. Fix an actor map $f : C_W \to C_V$ and an action path p over it. For a nonzero scalar loop $\sigma \in \mathfrak{m}_W$, the paths p and $p \cdot \sigma$ define morphisms in different components. Their difference persists after postcomposition through any natural exterior actor. Indeed the path fibre has components forming a torsor under $\mathfrak{m}_W$, and the actor mapping spaces are discrete, so no actor loop identifies these components. Postcomposition concatenates both paths with the same pulled-back path. This obstruction concerns the specified natural maps preserving augmentation. A geometric recipient can have additional actor loops, whose images must be included in the full question.

6.7 The full density object

Fix an arbitrary factorizing boundary $C \in \mathscr{F}$. Consider all pointed factorizing actions on C, with their geometric mapping spaces. Their ambient colimit is an action, but factorization requires comparing one common actor index with independent indices in the separated blocks. This comparison has a section. Its kernel is the remaining obstruction to pointed representability.

6.7.1 Ambient endomorphisms and a small index

The enhanced coefficient category $\mathscr{C}$ is presentable. Indeed its differential-module categories are presentable, its extraordinary pullbacks preserve colimits, and the small limit defining (6.1) is a limit of

presentable categories and colimit-preserving functors. Colimits are computed on the coefficients. The same assertions hold for its positive subcategory $\mathscr{C}_+$.

Lemma 6.16. *The category $\mathscr{M}$ is presentable. Its colimits are computed componentwise on reduced coefficients, and $\otimes^!$ preserves them separately. Thus $- \otimes^! C$ has a right adjoint and*

$$\mathcal{E}_C = [C, C]_{\otimes^!}$$

is an ambient associative endomorphism monoid, with unit $\eta : J \to \mathcal{E}_C$ and evaluation $\mathcal{E}_C \otimes^! C \to C$. Maps of monoids to $\mathcal{E}_C$ classify coherent actions on C.

Proof. For positive coefficient families, the nth convolution power vanishes on configurations with fewer than n positions. Bar and cobar therefore use finite filtrations at each configuration. Normalized bar terms are $a^{\star n}[n]$ for the positive augmentation ideal a, and normalized cobar terms are $c^{\star n}[-n]$ for the positive reduced coalgebra c. Simplicial realization and cosimplicial totalization give their coherent structures. Collision pullbacks preserve these finite normalized constructions.

In ΩBA, flatten a term to its n original a-factors and filter increasingly by n. Each of the $n-1$ gaps is either inside a bar block or between cobar factors. On the associated graded, multiplication of original inputs disappears and the differential changes one inside gap to an outside gap. After the suspension signs, this is the tensor product of $n-1$ two-term identity complexes, tensored with the coefficient word. Opposite signs around each square give the usual tensor differential. For $n > 1$ a contraction in its first gap contracts the product. For $n = 1$ the surviving coefficient is a; the empty term is K. The comparison is the identity on these surviving terms.

For $B\Omega C$, use the decreasing filtration by original c-factors. Coproduct splitting increases that filtration. Its associated graded has the same gap complexes with the direction of the cuts reversed, again leaving only K and c. Both filtrations end at $|I|$ on configuration I, so their comparisons are equivalences. Higher coherent products lower the first filtration and higher coproducts raise the second; the same associated graded proof applies. The bar and cobar adjunction retains the triangle coherences. Thus connected coalgebras and positive augmented algebras are equivalent as enhanced categories.

Such algebras are nonunital associative algebras in $\mathscr{C}_+$. The tensor-word monad is accessible and preserves sifted colimits, giving a presentable algebra category [75, Corollary 3.2.3.5]. This proves presentability of $\mathscr{M}$. On the coalgebra side, the reduced objects are nonunital algebras in $\mathscr{C}_+^{\mathrm{op}}$. Limits of algebras are computed on their underlying objects: the operations and their coherences are obtained by composing with each projection. Reversing arrows proves the assertion about coalgebra colimits.

The reduced part of a diagonal tensor is the diagonal tensor of the reduced parts. Its pullback and external product preserve colimits. The adjoint functor theorem [74, Corollary 5.5.2.9] gives the asserted right adjoint. Composing two evaluations and adjointing defines multiplication on $\mathcal{E}_C$. Associativity and units are those of composition, including their higher coherences. Adjunction transports precisely the monoid-map identities to the coherent action identities. □

This lemma constructs an ambient object; it imposes no factorization on $\mathcal{E}_C$. Write

$$\mathscr{V}_C = (\mathscr{M}_{/\mathcal{E}_C})_{(J,\eta)/}.$$

Then $\mathscr{A}_C$ is its full subcategory with factorizing underlying object. An action a will be written (F_a, u_a, β_a), where $\beta_a : F_a \to \mathcal{E}_C$ includes the path $\beta_a u_a \simeq \eta$.

The category $\mathscr{A}_C$ is accessible. For each finite ordered partition, factorization is the condition that a natural arrow between accessible coefficient functors be an equivalence. The category of equivalences in an arrow category is accessible. Pulling it back for the small set of partitions, and then adding the pointing and slice, gives the assertion. Choose one sufficiently large regular cardinal λ for these functors. The inclusion $\mathscr{A}_C \to \mathscr{V}_C$ preserves λ-filtered presentations: on a filtered index the diagonal to any finite power is cofinal, so finite external products of coefficients preserve the factorization comparisons.

Choose a small skeleton $\mathscr{A}_0$ of the λ-compact generators, and let $\mathscr{T}_C$ be the small full monoidal subcategory generated by it and the unit action. Fullness retains all mapping spaces between the selected objects. Define

$$L_C = \operatorname*{colim}_{a \in \mathscr{T}_C} F_a \quad \text{in } \mathscr{V}_C, \qquad \lambda_a : F_a \to L_C. \tag{6.36}$$

The unit action is initial, so this index has contractible nerve. Its underlying colimit is consequently the colimit in $\mathscr{M}$, with the specified point and map $\beta_L : L_C \to \mathcal{E}_C$.

Proposition 6.17. *The density object is a unital ambient monoid over $\mathcal{E}_C$. Its cocone extends naturally to all of $\mathscr{A}_C$, retaining every mapping-space coherence, and is multiplicative for the ordered action product $\circledast$.*

Proof. Separate preservation of colimits defines multiplication by

$$m_L(\lambda_a \otimes^! \lambda_b) = \lambda_{a \circledast b}.$$

The iterated products of the index supply associativity; its initial unit supplies both unit laws. The maps β_a make β_L multiplicative.

For any action a, the filtered slice $\mathscr{A}_0/a$ presents F_a. Thus the inclusion into $\mathscr{V}_C$ is the left Kan extension of its restriction to $\mathscr{A}_0$. The same holds after restricting it to $\mathscr{T}_C$, because that subcategory is full and has the same slices at its objects. The universal property of left Kan extension therefore extends the cocone, including its higher naturality, to all actions. Passing to compact filtered presentations in each variable extends the multiplication identity and its coherences as well. □

6.7.2 The section and its kernel

For an ordered partition $P = (I_1 | \cdots | I_r)$, set

$$X_P(F) = j_P^! F_I, \qquad Y_P(F) = \Big(\boxtimes_{j=1}^{r} F_{I_j} \Big)|_{U_P}.$$

These objects lie in the stable differential-module category on U_P. Internal block collisions remain in that open. The coproduct is $\delta_{P,F} : X_P(F) \to Y_P(F)$. For factorizing F it is an equivalence; its inverse is taken naturally in the functor category.

Proposition 6.18. *The map δ_{P,L_C} has a section s_P. If*

$$K_{C,P} = \operatorname{fib}(\delta_{P,L_C}), \qquad p_P = 1 - s_P \delta_{P,L_C}, \tag{6.37}$$

then

$$X_P(L_C) \simeq Y_P(L_C) \oplus K_{C,P}, \qquad \operatorname{cofib}(\delta_{P,L_C}) \simeq K_{C,P}[1].$$

For $h = a^{\circledast r}$ and its unit insertions $v_j : a \to h$, put

$$r_{a,P} = X_P(v_1) - \delta_{P,F_h}^{-1} \Big(\boxtimes_j (v_j)_{I_j} \Big) \delta_{P,F_a}. \tag{6.38}$$

Then $p_P X_P(\lambda_a) = X_P(\lambda_h) r_{a,P}$, with its natural homotopy.

Proof. For independently indexed actions $a_1, \dots, a_r$, let $h = a_1 \circledast \cdots \circledast a_r$ and let $v_j : a_j \to h$ insert the other units. On this tuple define

$$s_P\big|_{(a_j)} = X_P(\lambda_h) \delta_{P,F_h}^{-1} \Big(\boxtimes_j (v_j)_{I_j} \Big). \tag{6.39}$$

This is a natural cocone on $\mathscr{T}_C^r$. The external product preserves its separate colimits, giving a map $Y_P(L_C) \to X_P(L_C)$. Naturality of δ makes its composite with δ_{P,L_C} equal to the target cocone, since $\lambda_h v_j \simeq \lambda_{a_j}$. Hence $\delta s \simeq 1$. The split fibre sequence gives the two decompositions. Using the tuple $(a, \dots, a)$ in (6.39) gives (6.38). Compact presentations extend the formula to every action. Internal collisions and refinements use the same iterated coproduct coherences. □

Theorem 6.19. *The following are equivalent: the pointed-action functor Φ_C is representable; L_C factorizes; and all $K_{C,P}$ vanish. It suffices to test every ordered binary block partition, with its full block coefficients. When these conditions hold, L_C is also the final factorizing monoid action.*

Proof. Binary factorization iterated inside blocks gives all partitions. The split fibre sequence identifies factorization with kernel vanishing.

If L_C factorizes, it belongs to $\mathscr{A}_C$. Naturality of the extended cocone gives $\lambda_{L_C}\lambda_a \simeq \lambda_a$ on the small index, hence $\lambda_{L_C} \simeq 1$ by its colimit property. Naturality on the whole space $\mathrm{Map}_{\mathscr{A}_C}(a, L_C)$ now homotopes its identity map to the constant map with value λ_a. This space is contractible, so L_C is final and represents Φ_C.

Conversely a final pointed action Z gives a cocone and thus a map $\rho : L_C \to Z$. The extended cocone gives $\lambda_Z : Z \to L_C$. Naturality on the full small index gives $\lambda_Z \rho \simeq 1$. The reverse composite is the identity by finality of Z. Thus $L_C \simeq Z$ and factorizes. Finally all operation spaces from finite products to a final object are contractible. They give its unique coherent monoid structure and a contractible space of monoid-action maps into it from every factorizing monoid actor. □

This proof uses finality among all pointed actions. Finality among multiplicative actions alone does not supply the cocone from arbitrary pointed actors.

6.7.3 Specified monoid extensions

A monoid extension is (a, M, f) with $M \in \mathscr{B}_C$ and a pointed-action morphism $f : a \to \mathsf{U}M$, where U forgets multiplication. Its morphisms include an action map on a, a monoid-action map on M, and a specified homotopy making their square commute. Denote this category, with its higher coherences, by $\mathscr{E}_C = \mathscr{A}_C^{\Delta^1} \times_{\mathscr{A}_C} \mathscr{B}_C$, using target evaluation on the arrow category and U on the monoid category.

Theorem 6.20. *On $\mathscr{E}_C$ there is a natural nullhomotopy $p_P X_P(\lambda_a) \simeq 0$. It retains the vacuum paths, ordered coproducts and all internal collision comparisons.*

Proof. For $h = a^{\circledast r}$ set $q = \mu_M^{(r)} f^{\otimes^1 r} : h \to \mathsf{U}M$. The monoid-action identity makes this a pointed-action map in the prescribed order. Its unit paths give $qv_j \simeq f$ for every j. Naturality of factorization then gives

$$X_P(q)\delta_{P,F_h}^{-1}\Big(\boxtimes_j (v_j)_{I_j}\Big)\delta_{P,F_a} \simeq \delta_{P,M}^{-1} Y_P(f)\delta_{P,F_a} \simeq X_P(f).$$

The other term of $X_P(q)r_{a,P}$ is also $X_P(f)$. Thus $X_P(q)r_{a,P} \simeq 0$. The density cocone has $\lambda_M q \simeq \lambda_h$, proving the assertion.

These are equations of natural transformations on $\mathscr{E}_C$: the unit paths come from its coherent monoid structures, and the factorization inverses are natural inverses there. Their composite is therefore a homotopy on every simplex, not a collection of unrelated objectwise homotopies. Restriction to either ordered collision branch and to internal block collisions preserves the calculation. The projector's domain remains U_P; no extension through cross-block collisions is asserted. □

The full subcategory of pointed actions which happen to admit multiplicative structures also has maps that do not preserve those structures. It is not the extension category just defined. This distinction matters when passing the nullhomotopy to a colimit.

For a small diagram $\mathscr{E} \to \mathscr{E}_C$, define

$$A_P^{\mathscr{E}} = \operatorname*{colim}_{e \in \mathscr{E}} X_P(F_{a_e}), \qquad \chi_P : A_P^{\mathscr{E}} \longrightarrow X_P(L_C).$$

The map is the extended density cocone. Theorem 6.20 gives a nullhomotopy of $p_P \chi_P$, so

$$\mathrm{cofib}(\chi_P) \simeq \mathrm{cofib}(\delta_{P,L_C}\chi_P) \oplus K_{C,P}. \tag{6.40}$$

Indeed, put $A = A_P^{\mathscr{E}}$, $X = X_P(L_C)$, $Y = Y_P(L_C)$ and $K = K_{C,P}$. In the splitting $X = Y \oplus K$, write $\chi = (\chi_Y, \chi_K)$ and represent the specified homotopy by $\chi_K = d_K h + h d_A$. Our cone convention is

$$\operatorname{Cone}(\chi)^n = Y^n \oplus K^n \oplus A^{n+1},$$
$$d(y,k,a) = (d_Y y + \chi_Y a, d_K k + \chi_K a, -d_A a).$$

The cochain isomorphism is the shear

$$(y,k,a) \longmapsto ((y,a), k + h(a)). \tag{6.41}$$

Its last differential is $d_K k + \chi_K a - h d_A a = d_K(k + ha)$, and its inverse subtracts ha. In particular the retraction to the kernel includes the chosen homotopy.

For example take $A = (k \xrightarrow{5} k)$, $Y = (k \xrightarrow{2} k)$ and $K' = (k \xrightarrow{(3,0)} k^2)$ in degrees -1, 0. Let $\chi^{-1} = (10, 35)$ and $\chi^0 = (4, 21, 0)$. Its kernel component has homotopy $h^0 = 7$. Projection to k alone is not a chain map on the cone, because of the term $21a$; the corrected projection is $k + 7a$. The nonzero class in $H^0(K')$ remains a retract.

Corollary 6.21. *Pointed representability holds if and only if a small diagram of specified monoid extensions makes χ_P an equivalence for every ordered binary partition. Equivalently, one may use a different such diagram for each partition.*

Proof. If the maps are equivalences, their cofibres vanish. Equation (6.40) makes every $K_{C,P}$ a retract of zero, and Theorem 6.19 applies. Conversely a final pointed action Z has its coherent monoid structure. Its final cocone supplies a functor $\mathscr{T}_C \to \mathscr{E}_C$ with constant target Z. The resulting χ_P is the original density colimit on every block-disjoint open, hence an equivalence. □

Nonzero relative cofibres alone do not obstruct this criterion: they may come from the first summand in (6.40). Nor is a defect before the density colimit a class of its kernel. Take an exterior actor with a multiplicative action, such as (C_E, α_D). At two separated positions let $c_{L,i}, c_{R,i}$ be the degree-one generators in the exterior state factors of two copies, ordered first by copy and then by position. They give the nonzero expression

$$c_{L,1}c_{L,2} - c_{L,1}c_{R,2}.$$

Multiplication sends both monomials to $c_1 c_2$ and annihilates their difference. When this multiplication is a pointed-action map, the full density cocone annihilates the same difference. Its action paths are part of this conclusion.

6.8 A pair surviving every pointed factorizing action

A common recipient of objects need not equalize their maps. For the polynomial Heisenberg boundary, two prescribed maps remain distinct after every receiving pointed factorizing action. The vacuum imposes the restriction responsible for this obstruction: a degree-zero operation coming from one root of a factorizing actor must commute with the boundary-module action.

Retain $C = C_{V_\kappa}$ with $\kappa \neq 0$, and put $\mathfrak{l} = L_k$, $\mathfrak{g} = L_{V_\kappa}$ and $A = U_\star \mathfrak{g}$. Write $\ell = \mathfrak{l}_{\{1\}} = \omega_X$ and $G = \mathfrak{g}_{\{1\}}$ for the right differential modules on X. Thus $J_{\{1\}} = \ell[1]$ and $C_{\{1\}} = G[1]$. All state spaces and tensors at a fixed configuration are algebraic.

For a coalgebra Z, let $\overline{Z} = \operatorname{cofib}(K \to Z)$. With $t = [-1]$ of degree one, the normalized cobar is

$$\Omega Z = K \oplus \bigoplus_{r \geq 1} (t\overline{Z})^{\star r}, \qquad d(tc) = -t(d_Z c) + \sum_{\overline{\Delta} c = \sum c' \otimes c''} (-1)^{|c'|+1} (tc')(tc'').$$

Its product is concatenation. The sums are finite on every configuration. These are the cobar and suspension conventions used in the bar–cobar comparison above.

6.8.1 An antisymmetric actor map

Put $F_1 = C_k$ and $F_2 = C_{k^2}$ in the notation $C_W = C_{\Lambda(W[-1])}$ of Section 6.6. Denote their exterior generators by ε and ε_1, ε_2, respectively. The subscript k here denotes the one-dimensional actor space; the vacuum coalgebra remains $J = C_0$.

Lemma 6.22. *There is a natural actor map $f : F_2 \to F_1$, under and over J, whose marked matrix and coordinate contraction are*

$$A_f(t_1, t_2) = \tfrac{1}{2}(t_2, -t_1), \qquad \Phi_f = 0.$$

Its pure linear components vanish. In a cobar representative its pure mixed binary component is

$$\tfrac{1}{2}(a_m M_n - a_n M_m), \qquad a_m = t(s(\varepsilon m)), \quad M_n = t(sn), \quad m, n \in \mathbb{I}. \tag{6.42}$$

Every positive-weight natural action pulled back along f is naturally homotopic to the trivial action.

Proof. The matrix classification in Theorem 6.13 already constructs an actual map with this matrix in every weight. An explicit composite fixes its binary representative. Take two auxiliary odd generators z_1, z_2, and let D_z be the even derivation with $D_z z_1 = z_2$ and $D_z z_2 = 0$. The tree action extends to these graded coefficients by inserting the Koszul sign in each permutation of the actor coefficients.

Here is a check of that extension. A nonzero term in the twisting equation has exactly two ordinary inputs: a source bracket joins them or joins one to an actor input, and a product uses one in each factor. For each homogeneous coefficient input of nonnegative degree d, multiply it by a product of d distinct auxiliary odd variables of degree -1. Use disjoint variables for different inputs. The modified inputs have degree zero. Extend D_z trivially over these variables and apply the fixed-point proof of Theorem 6.1. Extracting their ordered product gives the graded twisting identity. Permuting actor inputs gives their Koszul sign; moving the ordinary suspension uses its degree one lower. Thus the graded action is an actual coefficient map, with the same finite-configuration bound.

On one z input and w pure auxiliary actor inputs its value is $(D_z^w z)n \operatorname{pbw}(m_1 \cdots m_w)$. Act on z_1, z_2 by the auxiliary generator, then project $z_1 \mapsto 0$, $z_2 \mapsto \varepsilon$. The resulting map $b : C_{k^3} \to F_1$ has marked matrix

$$B(y_1, y_2, y_3) = (y_2, 0, 1).$$

Choose the inner map $a : F_2 \to C_{k^3}$ with matrix

$$A_a(t) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -t_2/2 & -t_1/2 \end{pmatrix}.$$

Its coordinate contraction is $(t_1, t_2, -t_1 t_2)$. The composition law (6.29) gives $A_{ba} = (t_2, -t_1)/2$ and $\Phi_{ba} = 0$.

For the binary calculation, relabel the intermediate generators as z_1, η, z_2, with η the auxiliary actor. Write $a_z(m) = t(s(zm))$. A lift of the mixed inner component is

$$L_{12}(m, n) = t\big(s(z_1 m)s(\eta n)\big) - \tfrac{1}{2}\big(a_{z_2}(m)M_n + a_{z_2}(n)M_m\big).$$

The source input $s(\varepsilon_1 \varepsilon_2 p)$ lifts to $-t(s(\eta z_1 p))$. The cobar differential gives

$$dL_{12} = -t\big(s(\eta z_1[m, n])\big) - a_{z_1}(m)a_\eta(n) - a_\eta(n)a_{z_1}(m).$$

This is the image of the source differential. The primitive product correction is closed. The relative bar contraction extends these compatible components and keeps their binary values. Applying b sends the first term of L_{12} to

$(\varepsilon m)n$ and its correction to $-\big((\varepsilon m)n + (\varepsilon n)m\big)/2$. This proves (6.42). Both pure diagonal pairs and both pure linear components vanish.

Finally, (6.34) identifies pullback of a natural action class ψ with $\psi(\Phi_f)$. Its constant term is zero, so this is zero in every weight. The Maurer–Cartan equivalence of Theorem 6.4 supplies the homotopy. Evaluation and the relative bar comparison preserve its vacuum path and all collision coefficients. □

Let $f_0 : F_2 \to J \to F_1$ be the augmented map. For $D = \partial_{x_1}$ the action α_D has natural class pD. Choose the all-weight homotopy from the preceding lemma, with the constant homotopy for f_0. They give two actual morphisms

$$(f, p_f), (f_0, p_0) : (F_2, \mathrm{triv}) \rightrightarrows (F_1, \alpha_D) \quad \text{in } \mathscr{A}_C. \tag{6.43}$$

Equality of their first action classes alone would not have supplied these paths.

6.8.2 *The operation at one root*

For a right differential module N on X, let $\iota_{\mathrm{diag}} N$ be its positive coefficient family, with the direct image of N on every small-diagonal branch. For any positive family H there is an equivalence

$$\mathrm{RHom}_{\mathscr{C}}(\iota_{\mathrm{diag}} N, H) \simeq \mathrm{RHom}_X(N, H_{\{1\}}). \tag{6.44}$$

Indeed the union of small-diagonal branches is closed on each $\mathscr{R}_I$. Local cohomology there is the sum $\bigoplus_\sigma \delta_{\sigma,*}\delta_\sigma^! H_I$. Collision pullback preserves these unions and their local cohomology. The resulting coefficient family is $\iota_{\mathrm{diag}} H_{\{1\}}$, by the collision identifications. Its counit induces an equivalence on maps from a family supported on the branches. Restriction to one position is inverse to ι_{diag} on such supported families, proving the formula.

Fix a pointed factorizing action (Z, u, γ) and set

$$B = \Omega Z, \qquad \mathcal{T}_C = (-) \otimes^! C, \qquad \Phi : \mathcal{T}_C(B) \longrightarrow A.$$

Here Φ is the cobar of γ, followed by the Chevalley–enveloping comparison. Lemma 6.2 identifies $\mathcal{T}_C(\Omega Z)$ with $\Omega(Z \otimes^! C)$. The pointing gives $\iota : \mathbb{1} \to B$, and its action path identifies $\Phi\mathcal{T}_C(\iota)$ with $\mathfrak{g} \to A$.

For an affine open $Y \subset X$ and a class $a \in H^\circ \mathrm{RHom}_Y(\ell, B_{\{1\}})$, use (6.44) to regard a as a one-root coefficient map. Define

$$\Upsilon(a) = \Phi\mathcal{T}_C(a).$$

Since $A_{\{1\}} = G$, the same adjunction identifies $\Upsilon(a)$ with an endomorphism of the full family $\mathfrak{g}$ over Y. No cohomological bound on B is imposed.

Lemma 6.23. *On the open U of two distinct roots, both ordered products*

$$\mu_{12}, \mu_{21} : (B_{\{1\}} \boxtimes B_{\{2\}})|_U \longrightarrow B_{\{1,2\}}|_U$$

are equivalences, with the source order understood in each case. For the external symmetry σ, put $R_B = \mu_{12}^{-1}\mu_{21}\sigma$. Every algebra map $F : B \to B'$ between such algebras satisfies

$$(F_1 \boxtimes F_1) R_B = R_{B'}(F_1 \boxtimes F_1).$$

For $A = U_\star\mathfrak{g}$, the exchange is the identity.

Proof. On two roots the reduced cobar has one generator $t\overline{Z}_2$ and two product summands $t\overline{Z}_1 \boxtimes t\overline{Z}_1$. Its two differential components are the coproducts of Z, with the displayed suspension signs. Both are equivalences on U. Quotienting by either product summand leaves the cone of the other coproduct equivalence. Thus both product inclusions are equivalences, even for unbounded complexes. The exchange identity follows after composition with μ'_{12} from preservation of the two products. In A, the difference of the two generator products is the chiral bracket, supported on the diagonal. Hence its open exchange is the identity. □

Write $\mu_{\mathfrak{g}} : \mathfrak{g} \star \mathfrak{g} \to \mathfrak{g}$ for the coefficient bracket. An endomorphism M of the module $\mathfrak{g}$ obeys $M\mu_{\mathfrak{g}} = \mu_{\mathfrak{g}}(\mathrm{id} \star M)$.

Lemma 6.24. *Every one-root operation $M = \Upsilon(a)$ satisfies*

$$M\mu_{\mathfrak{g}} = \mu_{\mathfrak{g}}(\mathrm{id} \star M) = \mu_{\mathfrak{g}}(M \star \mathrm{id}) \tag{6.45}$$

on both ordered branches. If it is also a coefficient derivation, then $M = 0$.

Proof. All maps are derived coefficient maps over Y. On the two-root scheme write $j : U \hookrightarrow \mathscr{R}_2$ for the disjoint open and i for the union of the two closed diagonal branches. Put $R = \ell \boxtimes \ell$, pulled back from Y^2. Both convolution summands have source $j_* j^* R$. The difference $\kappa_a = [\iota, a]$ of their ordered products obeys

$$\Phi\mathcal{T}_C(\kappa_a) = \mu_{\mathfrak{g}}(\mathrm{id} \star M)$$

by the envelope relation and the chosen unit path.

The commutator may have an open part. Define on U

$$w = (R_B - \mathrm{id})(\iota \boxtimes a), \qquad P(w) = \mu_{12} j_*(w) : j_* j^* R \longrightarrow B_2.$$

The second formula uses the global product from the first convolution summand. Thus $\kappa_a + P(w)$ restricts to zero on U. The exchange identity for Φ, and the strong symmetric monoidal comparison for $\mathcal{T}_C$, give

$$\Phi\mathcal{T}_C(P(w)) = 0.$$

Indeed the image of w is killed by $R_A - \mathrm{id}$. The open projection formula carries this zero map and its nullhomotopy through j_* and multiplication. No primitive is extended from the open.

Put $Q = \mathrm{cofib}(j_! j^* R \to j_* j^* R)$. Open adjunction lifts $\kappa_a + P(w)$ through Q. The source R is regular holonomic, so $j_!$ is defined for this source regardless of the unbounded target. The two localization triangles give

$$i_* i^* R \longrightarrow Q \longrightarrow i_* i^! R[1].$$

On each branch $i^! R = \ell[-1]$ and $i^* R = \ell[1]$. The exact sequence of principal parts $\omega \to \omega(*0) \to \delta_0$ fixes the first shift; duality fixes the second. These are the right-module conventions of [9, §§2.1.1–2.1.3]. The map $j_* j^* R \to Q \to i_* \ell$ is the vacuum principal-part bracket.

The closed projection formula and $i^! C_2 = C_1$ give

$$\mathcal{T}_C(i_* i^* R) = i_* G[1], \qquad \mathcal{T}_C(i_* i^! R[1]) = i_* G.$$

The finite PBW filtration puts A_2 in the right-module heart. Its quotients are open and closed direct images of degree-zero external products; these images are exact on the affine collision charts. Also $i^! A_2 = G$. Consequently

$$\mathrm{Hom}(i_* G[1], A_2) = \mathrm{Ext}^{-1}(G, G) = 0.$$

After applying $\mathcal{T}_C$ and Φ, the lifted map from Q therefore factors on each branch through an endomorphism N of G. It follows that

$$\mu_{\mathfrak{g}}(\mathrm{id} \star M) = N\mu_{\mathfrak{g}}.$$

Insert the vacuum in the first slot. The bracket becomes can : $j_* j^*(\ell \boxtimes G) \to i_* G$, and the last equality becomes $(N - M)\mathrm{can} = 0$. Precomposition by can is injective on degree-zero maps to A_2: its kernel is an image of $\mathrm{Hom}((\ell \boxtimes G)[1], A_2) = 0$, again by the right-module heart. Thus $N = M$. Skew symmetry gives the identity with M in the first slot, proving (6.45).

If M is a derivation, subtract its second-slot module identity from the derivation identity. The result is $\mu_{\mathfrak{g}}(M \star \mathrm{id}) = 0$. Inserting the vacuum in the second slot and using the same principal-part injectivity gives $M = 0$. □

Corollary 6.25. *Every $\Upsilon(a)$ over a nonempty affine open is a constant scalar multiple of the identity.*

Proof. The module identity implies commutation with every state mode. Normal transfer can be written with the normal derivative in the first coordinate; the matrix of M on the second coordinate then acquires no derivatives. Put $c = M\mathbf{1}$. The regular coefficient of $J(z)$ multiplied by $(z - w)^{-n}$ detects the creation mode J_{-n} in the principal-part module identity. Commutation with these modes gives $M(f) = fc$ for every polynomial state. Commutation with $J_n = \kappa n \partial_{x_n}$ for $n > 0$ gives $\partial_{x_n} c = 0$. Thus c is a scalar function of the root. Horizontality makes it constant. □

Both kinds of modes are necessary. The derivation $D = \partial_{x_1}$ commutes with all annihilation modes, but

$$(DJ_{-1} - J_{-1}D)\mathbf{1} = \mathbf{1}.$$

It is not a module map.

6.8.3 A singleton class that no recipient kills

Normalize the fibre of a right differential module at $q \in X(k)$ by

$$N_q = i_q^! N[1].$$

Then $\ell_q = k$ and $G_q = V_\kappa$ lie in degree zero. This is a derived restriction, so its induced maps on coefficient complexes preserve boundaries.

Lemma 6.26. *For a nonempty affine open $Y \subset X$, a complex N of differential modules on Y and a complex W of k-vector spaces,*

$$\mathrm{RHom}_Y(\ell, N \otimes_k W) \simeq \mathrm{RHom}_Y(\ell, N) \otimes_k W.$$

No bound on either complex is required. Every cohomology class has finite support in the algebraic Künneth direct sum.

Proof. The unshifted left–right conversion takes ℓ to $\mathcal{O}_Y$. For the affine coordinate z, the Spencer resolution is

$$0 \longrightarrow D_Y \xrightarrow{P \mapsto P\partial_z} D_Y \xrightarrow{P \mapsto P(1)} \mathcal{O}_Y \longrightarrow 0.$$

Affineness makes it a two-term resolution by finite free modules for the derived Hom in question. Its Hom complex commutes with tensoring by any complex W. Over the field k, the cohomology of an algebraic tensor product is the direct sum of the tensor products of cohomology groups. Every element has finite support. □

Let $D_0 \in \mathfrak{d}_V$ be a coefficient derivation, and let α be any action of F_1 with first-weight class $[\gamma^{(1)}_{D_0}]$ in Theorem 6.8. This includes the natural action α_{D_0}. For a receiving morphism

$$h : (F_1, \alpha) \longrightarrow (Z, \gamma),$$

let h_0 be its twisting component on the pure singleton $s(\varepsilon m)$, $m \in \mathbf{I}$. It defines a class

$$[h_0] \in H^1 \operatorname{RHom}_X(\ell, B_{\{1\}}).$$

This class forgets equivariance by $\mathbf{I}$. Nonvanishing of an equivariant extension would not by itself prove its nonvanishing here.

Theorem 6.27. *If $D_0 \neq 0$, then $[h_0]|_Y \neq 0$ on every nonempty affine open $Y \subset X$, for every receiving pointed factorizing action.*

Proof. Suppose $[h_0]|_Y = 0$. Resolve the pointed arrow before taking its relative convolution, retaining the full coefficient differential. A primitive of h_0 defines a positive-actor-weight gauge. Its weight-one equation replaces h_0 by zero exactly; higher weights change by the usual gauge series, finite at each configuration. This gives $h' \simeq h$ under J with $h'_0 = 0$. Composition transports the specified action path.

The first mixed component of h' has one ordinary input and one actor input. It is a closed degree-zero cochain c_1. It need not be supported on collisions. The normalized bar formula for $\mathcal{T}_C(h')$ on these inputs has the term $\mathcal{T}_C(c_1)$ and products involving $\mathcal{T}_C(h'_0)$. Those products vanish. Its composite with the action therefore has mixed component $\Phi\mathcal{T}_C(c_1)$.

Choose $q \in Y(k)$ and put $Y' = Y \setminus \{q\}$. Such a choice exists because k is infinite and Y is a nonempty open of $\mathbb{A}^1_k$. Keep the actor root at q and let the ordinary root vary over Y'. The first ordered product and normalized fibre identify the target on $Y' \times \{q\}$ with $B_{\{1\}}|_{Y'} \otimes_k B_q$. Lemma 6.26 writes the class of c_1 as a finite sum with bidegrees $(r, -r)$. After the action, its second factors lie in

$$H^{-r} \operatorname{RHom}_k(V_\kappa, V_\kappa) = 0 \qquad (r \neq 0).$$

Thus the resulting operation has the form

$$\sum_{i=1}^{m} M_i \otimes N_i, \qquad M_i = \Upsilon(a_i), \quad N_i = \Upsilon_q(b_i),$$

where a_i and b_i have degree zero and Υ_q is the normalized point action. The module maps M_i act on the moving ordinary-root factor.

The action path identifies the first-weight class of this composite with $[\gamma^{(1)}_{D_0}]$. The two-output part of the proof of Theorem 6.8 identifies such classes with actual module cocycles, without boundaries. Indeed open adjunction kills the zero Chevalley column in every degree, and total degree one leaves only its first column in coefficient degree zero. This argument also holds on Y. Restricting the resulting equality of maps gives

$$D_0|_{Y'} \otimes \mathrm{id}_{V_\kappa} = \sum_{i=1}^{m} M_i \otimes N_i. \tag{6.46}$$

Algebraic tensors of endomorphisms inject into external endomorphisms. For the finite list of second factors, finitely many vector evaluations and linear functionals separate every linearly independent sublist; applying them recovers each first factor. Choose a functional λ on $\operatorname{span}_k\{\mathrm{id}, N_1, \dots, N_m\}$ with $\lambda(\mathrm{id}) = 1$. Equation (6.46) then gives

$$D_0|_{Y'} = \sum_i \lambda(N_i) M_i.$$

Lemma 6.24 makes the right side a module map. Since D_0 is a derivation, that lemma forces $D_0|_{Y'} = 0$. The coefficient module $\mathcal{O}_X \otimes_k V_\kappa$ is torsion-free, and every state has an algebraic image. Localization is faithful on each such image, so $D_0 = 0$ on X, a contradiction. □

6.8.4 Homotopy invariance on two roots

The obstruction must survive homotopies of coalgebra maps, including those with nonzero intermediate singleton components. Use compatible differential graded models of the full coefficient mapping complexes, with the pointed arrow resolved, and write ∂ for their differential. The two pure inputs $s(\varepsilon_1 m)$ and $s(\varepsilon_2 n)$ have degree zero. Their bidegrees record the numbers of ε_1 and ε_2. Let j denote restriction to the two-root disjoint open and put $P_{1,1} = \mathrm{I} \star \mathrm{I}$, keeping the labels.

Lemma 6.28. *Suppose maps $\phi_0, \phi_1 : F_2 \to Z$ under J have twisting representatives with zero pure singleton actor components. If they are homotopic under J, their restricted pure binary components give the same class in*

$$H^1 \operatorname{RHom}(j^* P_{1,1}, j^* B).$$

Proof. Represent the relative homotopy by $\tau(u) + c(u)\,du$, where τ and c have degrees one and zero. Polynomial interval forms suffice in the positive filtration, which is finite on each configuration. On a pure singleton input the source differential and reduced coproduct vanish. Hence

$$\tau_{1,0} = \partial E, \quad \tau_{0,1} = \partial F, \qquad E(u) = \int_0^u c_{1,0}(v)\,dv, \quad F(u) = \int_0^u c_{0,1}(v)\,dv.$$

We have $E(0) = F(0) = 0$ and $\partial E(1) = \partial F(1) = 0$. On the disjoint open the source bracket vanishes. The mixed interval equation, with a prime denoting u-derivative, is

$$\tau_{1,1}' = \partial c_{1,1} + [\partial E, F'] + [\partial F, E'].$$

The brackets use both labelled external products followed by the target multiplication. Since E and F have degree zero, the graded Leibniz rule gives

$$[\partial E, F'] + [\partial F, E'] = \partial [E, F'] - \frac{d}{du}[E, \partial F]. \tag{6.47}$$

Therefore

$$\tau_{1,1}(1) - \tau_{1,1}(0) = \partial \int_0^1 \big(c_{1,1} + [E, F']\big)\,du - [E, \partial F]\big|_0^1.$$

The endpoint term is zero. At both endpoints the Maurer–Cartan equation makes the restricted binary term closed, because the singleton terms vanish. Their difference is thus a boundary in the stated full coefficient complex. □

The commutator correction need not be zero as a cochain. For $E = u(1-u)E_0$ and $F = u^2(1-u)F_0$, with E_0, F_0 of degree zero in a differential graded algebra,

$$\int_0^1 [E, F']\,du = \frac{1}{60}[E_0, F_0].$$

Its differential can be nonzero; the lemma uses its exactness.

Theorem 6.29. *For every morphism $h : (F_1, \alpha_D) \to (Z, \gamma)$ in $\mathscr{A}_C$, the composites hf and hf_0 from (6.43) are not homotopic under J. Thus $\mathscr{A}_C$ is not filtered, and Φ_C is not representable on pointed factorizing coalgebras for the polynomial Heisenberg boundary at nonzero level.*

Proof. Both composites have zero pure singleton actor components. On two distinct roots the source cobar differential gives

$$d\,t\big(s(\varepsilon n)s(m)\big) = -t\big(s(\varepsilon n)\big)t\big(s(m)\big) + t\big(s(m)\big)t\big(s(\varepsilon n)\big).$$

Applying Ωh reorders $h_0(n)\iota(m)$ into $\iota(m)h_0(n)$ modulo a boundary. The boundary comes from the cocommutative source; no centrality of the vacuum in Z is assumed.

Fix $q \in X(k)$ and put $Y = X \setminus \{q\}$. Restrict to $Y \times \{q\}$ by the normalized derived fibre. In the first product order, the binary class of hf is

$$\tfrac{1}{2}\big([h_0]|_Y \otimes [\iota_q] - [\iota]|_Y \otimes [h_0]_q\big). \tag{6.48}$$

Lemma 6.26 puts the two terms in the distinct Künneth summands of bidegrees $(1,0)$ and $(0,1)$. The first is nonzero by Theorem 6.27 and $\Psi_q(\iota_q) = \mathrm{id}_{V_\kappa}$. The second cannot cancel it and need not itself be nonzero. The composite hf_0 has zero binary class. Lemma 6.28 excludes a homotopy under J, and hence a homotopy of pointed actions with its action two-cell.

A filtered infinity-category equalizes every parallel pair after some receiving morphism. The pair (6.43) violates that condition. A representing object for Φ_C would be a final object of $\mathscr{A}_C$; its mapping space from (F_2, triv) would be contractible. That too would identify the two composites, which is impossible. □

Corollary 6.30. *For every such h, the two classes represented on the disjoint open by $h_0(m)\iota(n)$ and $h_0(n)\iota(m)$ are linearly independent in $H^1\,\mathrm{RHom}(j^*P_{1,1}, j^*B)$.*

Proof. Write the classes as c_1, c_2, with the root labels fixed. Fix $q \in X(k)$ and put $Y = X \setminus \{q\}$. Suppose $ac_1 + bc_2 = 0$. On $Y \times \{q\}$ the $(1,0)$ Künneth component is $a[h_0]|_Y \otimes [\iota_q]$, which is nonzero unless $a = 0$. Now restrict to $\{q\} \times Y$ and take the moving root first in the product order. The same component is $b[h_0]|_Y \otimes [\iota_q]$, so $b = 0$. The source cobar boundary used in the theorem permits both reorderings. Thus neither centrality of the recipient vacuum nor nonvanishing of $[h_0]_q$ is required. □

Multiplying the marked matrix of f by 2λ, for $\lambda \in k^\times$, replaces $1/2$ in (6.48) by λ. The coordinate contraction remains zero, so the same construction and obstruction apply.

The theorem concerns pointed actions with all pointed maps. For the additive exterior multiplications, f is not a monoid map. Its first marked form is $(t_2\,dt_1 - t_1\,dt_2)/2$. Pulling it back by addition and subtracting its values on the two factors leaves

$$\tfrac{1}{2}(s_2\,dt_1 - s_1\,dt_2 + t_2\,ds_1 - t_1\,ds_2),$$

which is nonzero in the first marked-matrix invariant. Thus the displayed pair does not decide finality in $\mathscr{B}_C$, where comparison maps must preserve multiplication.

6.8.5 How the ambient endomorphism object kills the class

The object $\mathcal{E}_C$ of Lemma 6.16 receives every action, without a factorization requirement. Its failure to factorize can be seen directly in its cochains. Put $c = \overline{C}$ and $a = \overline{\Omega C}$, and let $\underline{\mathrm{RHom}}^!$ be the internal Hom for $\otimes^!$ on positive coefficient families. There is an equivalence

$$\Omega\mathcal{E}_C \simeq K \oplus \underline{\mathrm{RHom}}^!(c, a). \tag{6.49}$$

The differential on the right is the internal mapping differential. Its cup product uses the coproduct of C, the interchange map, evaluation, and multiplication in ΩC.

To prove the underlying equivalence, test against the free augmented algebra $T_\star(V)$ on a positive coefficient family V. Its bar contracts to $K \oplus sV$, with zero reduced coproduct: the same cut contraction as in Lemma 6.16 leaves only the terms of tensor length one. Tensoring with C leaves the primitive reduced coefficient $sV \otimes^! c$. The bar–cobar and internal-Hom adjunctions identify maps successively with

$$\mathrm{Map}_{\mathscr{C}}(V, \overline{\Omega\mathcal{E}_C}) \simeq \mathrm{Map}_{\mathscr{M}}\big((K \oplus sV) \otimes^! C, C\big) \simeq \mathrm{Map}_{\mathscr{C}}\big(V, \underline{\mathrm{RHom}}^!(c, a)\big).$$

Yoneda gives the equivalence and its differential. For the product, take strict representatives and put $H = K \oplus \underline{\mathrm{RHom}}^!(c,a)$ with the stated cup product. Let $\pi : \overline{\mathrm{B}_\star H} \to \underline{\mathrm{RHom}}^!(c,a)$ be the bar twisting cochain. Evaluation gives

$$\tau = \mathrm{ev}(\pi \otimes^! \mathrm{I}) : \overline{\mathrm{B}_\star H} \otimes^! c \longrightarrow a.$$

Naturality of interchange identifies the quadratic term in the twisting equation for τ with the evaluation of the cup-product term for π. Evaluation is a chain map, so the linear terms agree as well. Thus τ defines $\mathrm{B}_\star H \otimes^! C \to C$. Adjunction and cobar give $H \to \Omega\mathcal{E}_C$. On a map $V \to \underline{\mathrm{RHom}}^!(c,a)$, its adjoint is exactly evaluation of that map. The underlying ideal map is therefore the identity under the equivalence just proved. This proves the algebra equivalence in (6.49).

Proposition 6.31. *For every coefficient derivation $D_{\mathrm{o}} \in \mathfrak{d}_V$, the classifying map $\beta_{\alpha_{D_{\mathrm{o}}}} : F_{\mathrm{I}} \to \mathcal{E}_C$ has zero singleton actor component. For every receiving morphism $h : (F_{\mathrm{I}}, \alpha_{D_{\mathrm{o}}}) \to (Z, \gamma)$, the classifying map $\beta_\gamma : Z \to \mathcal{E}_C$ sends $[h_{\mathrm{o}}]$ to zero. The ambient endomorphism object $\mathcal{E}_C$ is not factorizing.*

Proof. The singleton classifying map is adjoint to the singleton action. The twisting cochain for $\alpha_{D_{\mathrm{o}}}$ needs one ordinary input; a pure singleton actor input has none. Its image is therefore zero. The specified action path gives $\beta_\gamma h \simeq \beta_{\alpha_{D_{\mathrm{o}}}}$ under J. Restriction to one position proves the assertion about $[h_{\mathrm{o}}]$.

Equation (6.49) supplies a degree-zero internal cochain $a_D : \mathrm{I} \to \Omega\mathcal{E}_C$ whose one-root action is D. At a singleton, its adjunction is precisely that between $\ell \otimes^! C_{\mathrm{I}} = G$ and $D : G \to G$; (6.44) extends it to the full coefficient family. If $\mathcal{E}_C$ factorized, Lemma 6.24 would make D a module map, contradicting $(DJ_{-\mathrm{I}} - J_{-\mathrm{I}}D)\mathbf{1} = \mathbf{1}$. □

For any nonzero coefficient derivation D_{o}, the same adjunction gives $a_{D_{\mathrm{o}}} : \mathrm{I} \to \Omega\mathcal{E}_C$. Its vacuum commutator $[\iota, a_{D_{\mathrm{o}}}]$ is nonzero on the two-root disjoint open. If it vanished there, the commutator itself would lift through Q in the proof of Lemma 6.24. The supported part of that proof would give $\mu_{\mathfrak{g}}(\mathrm{id} \star D_{\mathrm{o}}) = N\mu_{\mathfrak{g}}$. The first vacuum slot forces $N = D_{\mathrm{o}}$; the derivation identity and the second vacuum slot then force $D_{\mathrm{o}} = 0$. Once open vanishing is assumed, this argument uses no factorization of the target. The nonzero open commutator becomes zero after the action, whose bracket image is supported on the diagonal. This explains how $\mathcal{E}_C$ can carry the mixed derivation while killing the singleton class.

Finally, Theorems 6.19 and 6.29 imply that L_C is not factorizing: at least one of its two-block kernel coefficients $K_{C,A|B}$ is nonzero. This consequence does not yet exhibit a representative of such a coefficient in the full density bar. That bar also retains maps capable of killing defects of individual objects.

6.9 Multiplication and a surviving commutator

The pointed sphere did not use a multiplication on its source actor. To obstruct a final multiplicative action, we need two monoid-action maps. A trivially acting submonoid supplies them: its inclusion and the map through the unit have the same action. The multiplication of a larger actor can nevertheless distinguish these maps after every receiving action.

Work over $\mathbb{C}$. Let Q be an algebraic unital vertex algebra in cohomological degree zero, with translation T, and let $C_Q = \mathrm{CE}_\star(L_Q)$ on the full ordered collision carrier. A vertex derivation is a linear map $D : Q \to Q$ satisfying

$$D\mathbf{1} = 0, \qquad DT = TD, \qquad D(x_{(n)}y) = (Dx)_{(n)}y + x_{(n)}Dy \quad (n \in \mathbb{Z}).$$

We retain the category of multiplicative actions defined in Section 6.1. Receiving actors may be unbounded. They need not have an augmentation to J.

6.9.1 An exterior monoid with a trivial direction

Let $A_0 = \mathbb{C}a \oplus \mathbb{C}b$ have multiplication

$$a^2 = a, \qquad ab = ba = b, \qquad b^2 = 0.$$

The unit a belongs to this auxiliary algebra; its exterior generator is distinct from the vacuum. Set

$$\mathsf{F} = C_{\Lambda(A_0[-1])}, \qquad \mathsf{G} = C_{\Lambda(\mathbb{C}a[-1])}, \qquad \mathsf{H} = C_{\Lambda(\mathbb{C}b[-1])}.$$

The exterior generators have degree one, zero differential and zero translation. For $x \in A_0$ let L_x be multiplication by x and put

$$R(x) = e^{L_x}, \qquad M(x) = \sum_{n \geq 0} \frac{L_x^n}{(n+1)!}.$$

The constant term of M is the identity. Define the marked multiplication matrix

$$(\mathcal{L}(x,y) \mid \mathcal{R}(x,y)) = \big(M(x+y)^{-1}M(x) \mid M(x+y)^{-1}R(x)M(y)\big). \tag{6.50}$$

Lemma 6.32. *This matrix gives a coherent unital monoid on* F *under and over* J*, with coordinate contraction* $x + y$*. The two inclusions* $i : \mathsf{G} \to \mathsf{F}$ *and* $j : \mathsf{H} \to \mathsf{F}$ *are monoid maps. The induced multiplication on* H *is additive. All these maps preserve the auxiliary grading* $\deg a = 0$, $\deg b = 1$.

Proof. The law

$$(x, U)(y, V) = (x + y, U + R(x)V)$$

is associative because $R(x+y) = R(x)R(y)$. Changing fibre coordinates by $U = M(x)u$ gives (6.50). Its three associativity identities are (6.30), and its unit identities follow from $M(0) = I$. In A_0,

$$M(x)x = e^x - a, \qquad e^{x+y} - a = (e^x - a) + e^x(e^y - a).$$

Thus the diagonal $u = x$ is a homomorphism, proving the coordinate contraction. The a- and b-lines are preserved; on the b-line, $L_{sb}b = 0$, so both multiplication blocks are the identity. The algebra product preserves auxiliary degree, hence so do R, M and M^{-1}.

Theorem 6.13 realizes these matrices and their identities, with a contractible space of coherent choices. The multigraded contractions in its proof preserve the auxiliary grading. At m positions, exterior degree bounds the source actor weight by $2m$. The formal series therefore evaluate as finite sums on the original coefficient diagrams. □

The zero auxiliary degree of a prevents the use of positive grading alone to remove action loops. Prescribing the full G-action removes those loops instead.

Lemma 6.33. *For every vertex derivation* D*, there is a coherent unital* F*-action* α *on* C_Q *whose restriction to* G *is trivial and whose restriction to* H *is the single-derivation action* α_D*. With these comparisons,* i *and* j *are monoid-action maps.*

Proof. First use one formal derivation. For r actor copies let t_ν, s_ν be coordinates dual to a, b. Give them auxiliary weights $0, -1$ and give D weight 1. Take natural cochains of weight zero which vanish when all b-inputs are absent. The scalar-bar and Koszul projection of Theorem 6.4 has target

$$[(s_1, \dots, s_r)\mathbb{C}[[t_1, s_1, \dots, t_r, s_r]] \otimes \Lambda(\mathbb{C}D[-1])]^{\mathbb{G}_m}. \tag{6.51}$$

Completion is in total actor weight. Restriction to the a-subspace has a splitting, and taking auxiliary weight zero is exact in each finite total weight and derivative order. The projection is consequently surjective with weightwise acyclic kernel on this relative complex.

The target has no nonpositive degrees: a scalar monomial in the displayed ideal has negative auxiliary weight. Its degree-one part consists of $\sum_\nu s_\nu f_\nu(t_1,\dots,t_r)D$. Relative Maurer–Cartan lifting therefore makes each corresponding component contractible. The same assertion holds with prescribed unit faces and restrictions to the b-lines. Indeed the deletion projections for actor copies and the two exterior directions commute and split. Inclusion–exclusion extends compatible data on their union; the residual kernel is a sum of the same acyclic multigraded kernels. Polynomial forms relative to parameter boundaries then supply the required fillings.

Choose the class sD on one copy. Pullback by multiplication has class $(s_1+s_2)D$, since the coordinate contraction is addition. The composite action has the same class by (6.33), or directly by $e^{s_1D}e^{s_2D}=e^{(s_1+s_2)D}$. Contractibility supplies its multiplication path, relative to the prescribed line actions and units. Each subsequent module associahedron has boundary in the component with class $(s_1+\cdots+s_r)D$. Relative contractibility fills it and all parameterized comparison boundaries.

Finally evaluate on Q. The strong symmetric monoidal functor $\mathcal{T}_Q=(-)\otimes^!C_Q$ of Lemma 6.2 sends the vacuum Lie object to L_Q, with its bracket. It therefore identifies $\mathsf{F}\otimes^! C_Q$ with $\mathrm{CE}_\star(L_Q\otimes\Lambda(A_0[-1]))$ and commutes with envelopes and bars. The infinitesimal vertex automorphism $1+uD$ over $\mathbb{C}[u]/(u^2)$ shows that this comparison carries D to the coefficient derivation, retaining collisions and normal transfers. It carries the universal polynomials and their parameter homotopies as well. At each configuration, actor weight is bounded and derivative order equals the b-count. Every evaluated operation is finite. □

6.9.2 *A commutator detected on two roots*

Let $\mathsf{S}=C_{\Lambda(\varepsilon_1,\varepsilon_2)}$. Map it to $\mathsf{F}^{\otimes^!4}$ by

$$\varepsilon_1\longmapsto a^{(1)}-a^{(3)},\qquad \varepsilon_2\longmapsto b^{(2)}-b^{(4)},$$

and multiply the four factors in that order. Call the resulting pointed map $w:\mathsf{S}\to\mathsf{F}$. Define the formal series $\chi(t)=\sum_{n\ge0}t^n/(n+1)!$.

Lemma 6.34. *There is a natural pointed map $f:\mathsf{S}\to\mathsf{H}$ with*

$$w\simeq jf,\qquad A_f(t,s)=(-s\chi(t)\mid t\chi(t)). \tag{6.52}$$

Its coordinate contraction and singleton actor components vanish. Its first marked form is $t\,ds-s\,dt$.

Proof. In the affine coordinates used above, an a-line element with coordinate t and fibre ra has linear part e^t and translation $\chi(t)ra$. A b-line element with coordinate s and fibre qb has linear part multiplication by $a+sb$ and translation qb. Negating coordinate and fibre gives inversion, since $M(-x)=R(-x)M(x)$. The linear parts commute. Their affine commutator has linear part one and translation

$$-s\chi(t)rb+(e^t-1)qb=\chi(t)(tq-sr)b.$$

Its coordinate is zero, where the frame is the identity. The marked-map classification now gives (6.52) and its pointed homotopy. Finally $A_f(t,s)(t,s)^T=0$, $A_f(0,0)=0$ and $\chi(0)=1$ prove the remaining assertions. □

The singleton detector extends from the Heisenberg algebra to Q without a scalar-endomorphism classification. Let $h:(\mathsf{H},\alpha_D)\to(Z,\beta)$ be any receiving pointed factorizing action and put $B=\Omega Z$. Write h_0 for its pure singleton twisting component and $\iota:L_{\mathbb{C}}\to B$ for its pointing.

Lemma 6.35. *If $D\neq0$, then $[h_0]|_Y\neq0$ in $H^1\,\mathrm{RHom}_Y(\ell,B_{\{1\}})$ for every nonempty affine open $Y\subset X$.*

Proof. The proof of Lemma 6.24 applies with $G = \omega_X \otimes Q$ and $\mathfrak{g} = L_Q$. Here are the hypotheses it uses. Factorization makes both ordered cobar products invertible on distinct roots. The open commutator correction has zero action by the exchange identity and open projection. Its remaining supported part factors through the vacuum localization cofiber, whose terms on each branch are $\ell[1]$ and ℓ. Tensoring with C_Q replaces them by $G[1]$ and G. The PBW envelope of L_Q lies in the right-module heart on each affine collision chart, with extraordinary diagonal restriction G. Negative Ext therefore kills the first term and makes the principal-part vacuum map injective on degree-zero maps. Inserting the vacuum in both slots gives

$$M\,\mu_{L_Q} = \mu_{L_Q}(1 \star M) = \mu_{L_Q}(M \star 1)$$

for every plain degree-zero coefficient operation M. Such an operation that is also a derivation is zero.

If $[h_0]|_Y = 0$, the relative gauge used in Theorem 6.27 removes h_0. Separate the moving ordinary root on $Y' = Y \setminus \{q\}$ from the actor root at $q \in Y$. The finite Spencer comparison of Lemma 6.26 applies to arbitrary receiving complexes. Since Q is in degree zero, only the bidegree $(0, 0)$ survives after its action. The action path gives

$$D|_{Y'} \otimes 1_Q = \sum_{\nu=1}^{m} M_\nu \otimes N_\nu.$$

This is equality of two-output cocycles. Resolve the coefficient maps in nonnegative degrees before taking the Chevalley complex. Open adjunction annihilates its zero column; total degree one then has only column one in coefficient degree zero, with no incoming boundary. Each M_ν satisfies the module identities just proved. Finite vector evaluations separate the span of $1_Q, N_1, \dots, N_m$. Applying a functional taking 1_Q to 1 makes $D|_{Y'}$ a linear combination of these module maps. It is a derivation as well, so it vanishes. Localization is faithful on the free algebraic state module $\mathcal{O}_X \otimes Q$. Hence $D = 0$, a contradiction. □

Theorem 6.36. *If Q has a nonzero vertex derivation, its category of coherent unital multiplicative actions on ordinary factorizing coalgebras has no final object. There are parallel monoid-action maps which no receiving monoid action equalizes up to homotopy.*

Proof. Let $i_0 : \mathsf{G} \to J \to \mathsf{F}$ be augmentation followed by the unit. Lemma 6.33 makes i and i_0 morphisms from the trivial G-action to (F, α).

For every receiving pointed map h from (H, α_D), the binary class of hf on $(X \setminus \{q\}) \times \{q\}$ is

$$[\iota]|_{X\setminus\{q\}} \otimes [h_0]_q - [h_0]|_{X\setminus\{q\}} \otimes [\iota]_q. \tag{6.53}$$

The two terms lie in distinct Künneth bidegrees. The second is nonzero by Lemma 6.35 and the unit identity: $[\iota]_q$ acts as 1_Q. Changing the ordered product changes the representative by the cobar boundary used in Theorem 6.29. Lemma 6.28 makes this class invariant under every pointed geometric homotopy. Thus hf cannot be homotopic to the map through J.

Now suppose a receiving monoid-action map $g : (\mathsf{F}, \alpha) \to (Z, \beta)$ equalized i and i_0. Forgetting the action retains a homotopy under J. In the four-factor word gw, multiplicativity of g and that homotopy replace the first and third factors by units. The second and fourth factors cancel, since j is a monoid map and the additive H has inverse $b \mapsto -b$. Tensoring the comparison homotopies and applying the monoid unit coherences therefore gives $gw \simeq \mathsf{S} \to J \to Z$. But $w \simeq jf$, and $g\,j$ receives α_D, contradicting (6.53).

A final action would receive (F, α) and identify the two composites from the trivial G-action, because its mapping space from that action would be contractible. The contradiction excludes finality. □

For the polynomial Heisenberg algebra at nonzero level, $D = \partial_{x_1}$ is a nonzero vertex derivation by (6.22). The theorem therefore excludes a final multiplicative action on this boundary over $\mathbb{C}$, independently of the pointed-sphere argument. For $Q = \mathbb{C}$, by contrast, the action category is the slice of transverse monoids over J: an action $F \otimes^! J \to J$ is a monoid map $F \to J$. Its final object is 1_J, with evaluation $J \otimes^! J \simeq J$. The nonzero-derivation hypothesis thus distinguishes an actual positive case. The ambient endomorphism object remains defined in both cases, with its different universal property.

6.10 Recipient bars and the role of maps

The density comparison depends on the maps of its index, not just on the available actors. A weighted bar records which two actors meet in a common recipient, and how different choices of their incoming maps are related.

Let $\mathcal{T}$ be a small infinity-category, let $Q : \mathcal{T}^{2} \to D(k)$ be a coefficient diagram, and write $\Delta : \mathcal{T} \to \mathcal{T}^{2}$ for the diagonal. For $d = (a, b)$ the category $d \downarrow \Delta$ consists of recipients t with maps $a \to t$ and $b \to t$, including their mapping spaces and coherences. Put

$$W(d) = C_{*}(|d \downarrow \Delta|; k), \qquad \widetilde{W}(d) = \operatorname{fib}(W(d) \to k).$$

Chains have nonpositive cohomological degrees and the map is augmentation. An empty recipient category gives the same formula with $W(d) = 0$.

Proposition 6.37. *For the full coefficient diagram, there is an equivalence*

$$\operatorname{fib}\left(\operatorname*{hocolim}_{a} Q(a, a) \longrightarrow \operatorname*{hocolim}_{a,b} Q(a, b)\right) \simeq \int_{\mathbb{L}}^{d \in \mathcal{T}^{2}} Q(d) \otimes_{k} \widetilde{W}(d). \tag{6.54}$$

No splitting of a vacuum coefficient is required.

Proof. The weight $W(d)$ is the homotopy colimit over t of the chains on $\operatorname{Map}_{\mathcal{T}^{2}}(d, \Delta t)$. Derived co-Yoneda identifies its coend with $\operatorname{hocolim}_{t} Q(\Delta t)$. The constant weight k gives $\operatorname{hocolim}_{d} Q(d)$, and its augmentation is the comparison displayed above. Tensor product and coend preserve colimits. In a stable category they therefore preserve finite fibres, proving the formula. □

The same proof works for coefficients in the stable differential-module category on U_{P}, with its tensoring over $D(k)$. For the binary density comparison use

$$Q(a, b) = (F_{a,I_{1}} \boxtimes F_{b,I_{2}})|_{U_{P}}.$$

Factorization identifies its diagonal value with $X_{P}(F_{a})$. Thus (6.54) computes $K_{C,P}$ on its full index, before any coefficient reduction.

For an ordinary category $\mathcal{D}$ and strict resolved diagrams, the normalized two-sided bar has length-n term

$$\bigoplus_{d_{0} \to \cdots \to d_{n}} \big(Q(d_{0}) \otimes_{k} \widetilde{W}(d_{n})\big)[n]. \tag{6.55}$$

Degenerate strings are removed. The first face transports Q, the interior faces compose arrows, and the last pulls the weight back. The differential is $\partial + (-1)^{n} d_{\mathrm{int}}$, with $\partial = \sum_{i=0}^{n} (-1)^{i} d_{i}$ for $n > 0$ and no face term at length zero. Total degree is internal degree minus n, and realization uses direct sums. For an infinity-category the bar uses chains on mapping spaces, not just sets of arrows.

If $\mathcal{D}$ has a terminal object t, the bar of a covariant diagram V contracts to $V(t)$. Let ϵ transport length-zero coefficients to t and let i include that summand. On a length-n string, append the unique arrow to t with sign $(-1)^{n+1}$, and set the result to zero if that arrow is an identity. The resulting degree-minus-one map satisfies

$$dh + hd = 1 - i\epsilon. \tag{6.56}$$

All faces except the last cancel against the corresponding faces of the original string. The last gives the identity; at length zero the other face gives $-i\epsilon$. The length change reverses the internal differential sign, so the internal terms cancel too.

6.10.1 *Equality patterns*

Let FinInj be a skeleton of finite sets and injections, with $V(S) = k[S]$. Its monoidal product is disjoint union and its initial unit is the empty set. For $r \geq 1$, consider

$$\operatorname*{hocolim}_S V(S)^{\otimes r} \longrightarrow \left(\operatorname*{hocolim}_S V(S)\right)^{\otimes r}.$$

If B_r is the number of partitions of $[r]$, this is the summation map $k^{B_r} \to k$ in degree zero. Its fibre is $k^{B_r - 1}$.

Indeed the source bar is the linearized nerve of the category of pairs (S, t) with $t \in S^r$. An injection preserves the equality partition of t. Each such component has an initial object: the set of blocks, with its canonical tuple. Its unique injection to any tuple of that pattern determines every element. Thus each partition contributes one contractible component. The case $r = 1$ computes the target as k, and every pattern maps to its generator. The discrete partition gives a section. If all finite-set maps are allowed, the singleton is terminal, so both sides contract to k and the comparison is an equivalence.

The first fibre dimensions are 0, 1, 4, 14, 51, 202, 876, 4139. For two singleton inputs, injections retain separate same-image and distinct-image recipient components. A quotient map joins them. Common upper bounds and an initial unit therefore do not imply diagonal cofinality.

A pointed variant gives the split map explicitly. Let I be finite pointed sets and point-preserving injections, and put

$$V = \operatorname*{hocolim}_{\mathrm{I}} k[S], \qquad W = \operatorname*{hocolim}_{\mathrm{I}} k[S^2].$$

The basepoint is a nonzero basis vector. There are two one-entry patterns, and five two-entry patterns

$$(0,0), \quad (0,x), \quad (x,0), \quad (x,x), \quad (x,y), \qquad 0, x, y \text{ distinct}.$$

Their minimal pointed sets are initial in their components. Hence $V = k^2$ and $W = k^5$, in degree zero. The comparison $d : W \to V \otimes V$ and its section are

$$d = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}, \qquad s = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{6.57}$$

The section comes from Cartesian products of pointed sets: (x, y) is sent to $((x, 0), (0, y))$. When both entries are nonzero this selects distinct images. Thus $ds = 1$, but the kernel is generated by $(0, 0, 0, 1, -1)^T$, on which $1 - sd$ is the identity. The cofiber is $k[1]$. This is a computation on the stated finite-set category, not on $\mathscr{A}_C$.

6.10.2 *Parallel maps*

Let $\mathcal{U}$ have two objects a, b and two distinct nonidentity arrows $f, g : a \to b$. For the constant diagram $Q = k$ on $\mathcal{U}^2$, the fibre of its diagonal comparison has

$$H^{-1} = k, \qquad H^0 = k,$$

and no other cohomology. The nerve of $\mathcal{U}$ is a circle, and that of $\mathcal{U}^2$ is a torus. The diagonal is an isomorphism on H_0 and the diagonal inclusion $k \to k^2$ on H_1. Its fibre therefore receives the target H_2

in cohomological degree -1 and the quotient of H_1 in degree zero. Adjoining $h : b \to c$ with $hf = hg$ makes c terminal. The comparison then contracts to the identity of k.

Even the recipient components do not suffice by themselves: for (a, a) there are three components. The choices (f, f) and (g, g) connect through $(a, \mathrm{id}, \mathrm{id})$, whereas (f, g) and (g, f) are isolated. The coend also retains transport in the source variables and every bar face. A functional detecting a density kernel class must be a cocycle on this whole bar. In degree zero its components must obey $\ell_b V(f) = \ell_a$ for every arrow; its higher components obey the dual bar equations.

6.ii Evaluation and the remaining homotopies

The density kernel acts trivially on the boundary. Its own vanishing is a stronger assertion. We give the action nullhomotopy first, then write the equations for a nullhomotopy of the projector itself.

Fix a binary partition P. On U_P abbreviate

$$Z = X_P(L_C), \quad W = Y_P(L_C), \quad A = X_P(C), \quad A' = Y_P(C), \quad \delta = \delta_{P,L_C}, \quad t = \delta_{P,C}.$$

The map t is an equivalence. Let s be the density section and $p = 1 - s\delta$. The action supplies $\alpha : Z \otimes^! A \to A$ and $\alpha^Y : W \otimes^! A' \to A'$. Choose resolved models of their full diagram. For homogeneous maps use $\partial u = du - (-1)^{|u|} ud$. The prescribed paths give degree-minus-one maps e, n, h and a chain map $g : A' \to A$ with

$$\partial e = \delta s - 1, \qquad \partial n = t\alpha - \alpha^Y(\delta \otimes^! t), \qquad \partial h = gt - 1.$$

Put

$$\sigma = -e\delta, \qquad N = \alpha^Y(\sigma \otimes^! t) + n(p \otimes^! 1), \qquad H_{\mathrm{ev}} = gN - h\alpha(p \otimes^! 1). \tag{6.58}$$

Then $\partial\sigma = \delta p$, $\partial N = t\alpha(p \otimes^! 1)$, and hence

$$\partial H_{\mathrm{ev}} = \alpha(p \otimes^! 1).$$

These identities prove the asserted nullhomotopy. All its paths come from the same enhanced diagram, so it restricts coherently along the density cocone and every internal collision. Evaluation on a closed boundary state, followed by a closed functional, therefore cannot detect this kernel on cohomology.

6.ii.1 A map retaining the fibre homotopy

Set $Z_E = X_P(\mathcal{E}_C)$, $W_E = Y_P(\mathcal{E}_C)$ and $v = \delta_{P,\mathcal{E}_C}$. Define its factorization defect $\mathrm{Def}_P(\mathcal{E}_C) = \mathrm{Fib}(v)$, using the cochain convention

$$\mathrm{Fib}(u)^n = U^n \oplus V^{n-1}, \qquad d(x, y) = (dx, ux - dy), \qquad u : U \to V. \tag{6.59}$$

Write $\beta_0 = X_P(\beta_L)$ and $\beta_1 = Y_P(\beta_L)$. Their coproduct square gives $\partial\eta = v\beta_0 - \beta_1\delta$. It induces

$$T_\eta : K_{C,P} \to \mathrm{Def}_P(\mathcal{E}_C), \qquad T_\eta(x, y) = (\beta_0 x, \beta_1 y + \eta x).$$

The map

$$q_P = T_\eta\pi : Z \to \mathrm{Def}_P(\mathcal{E}_C), \qquad \pi = (p, -e\delta), \qquad q_P = (\beta_0 p, \eta p - \beta_1 e\delta) \tag{6.60}$$

is a cochain map. Indeed $\partial(-e\delta) = \delta p$ is exactly the fibre equation. If $i : K_{C,P} \to Z$ projects to the first component, $i\pi = p$, while

$$R(x, y) = (sy, ey), \qquad dR + Rd = 1 - \pi i.$$

Thus $\pi i \simeq 1$, and q_P is nullhomotopic if and only if T_η is nullhomotopic.

The two components in (6.60) record an insertion discrepancy. More explicitly, let $\ell = \beta_0 s$ and $b = \eta s + \beta_1 e$. Then $\partial b = v\ell - \beta_1$, and

$$(\beta_0 - \ell\delta, \eta - b\delta) = q_P.$$

For an action a, restrict this formula along its density map and inverse factorization, defining

$$\Gamma_a = [q_P X_P(\lambda_a)\delta_{P,F_a}^{-1}]. \tag{6.61}$$

Its lift $\ell Y_P(\lambda_a)$ is precisely the ordered insertion lift: on the tuple (a, a), (6.39) uses the two unit insertions into $a \circledast a$, followed by composition in $\mathcal{E}_C$. The path b is their coproduct and unit path. Composition of homotopy squares retains the term $\eta'\beta_0 + \beta_1'\eta$, so this identification includes the second fibre component.

A nonzero Γ_a would force $K_{C,P} \neq 0$. Projection to Z_E alone can lose such a class. Evaluation loses more: its coproduct square sends $\mathrm{Def}_P(\mathcal{E}_C) \otimes^! A$ to the fibre of t, which is zero. Conversely, vanishing of q_P would leave any part of the density kernel annihilated by T_η undetermined.

6.11.2 Equations on the full action bar

Choose a simplicial-category presentation of $\mathscr{T}_C$ and a resolved enriched diagram $X_a = X_P(F_a)$. Let $Q(a, b) = C_*(\mathrm{Map}_{\mathscr{T}_C}(a, b); k)$, in nonpositive cohomological degrees. The constant right module k uses its augmentation. The two-sided bar has terms

$$B_n = \bigoplus_{a_0,\dots,a_n} Q(a_{n-1}, a_n) \otimes \cdots \otimes Q(a_0, a_1) \otimes X_{a_0}. \tag{6.62}$$

Normalize in the simplicial direction, denoting the result by NB_n, and put

$$B = \bigoplus_{n\geq 0} NB_n[n], \qquad D|_{NB_n[n]} = (-1)^n d_{\mathrm{int}} + \sum_{j=0}^{n} (-1)^j d_j.$$

The face sum is absent for $n = 0$. Simplicial identities and compatibility with the internal differential give $D^2 = 0$. This bar computes Z: resolve the constant weight by represented right modules, whose augmented bar has the extra identity degeneracy, then tensor with the coefficient diagram. All tensors are derived.

Let $p_B : B \to B$ represent p, and let $\iota_n : NB_n \to B$ be the inclusion, of degree $-n$. A degree-minus-one map H with $DH + HD = p_B$ is equivalent to maps $h_n = H\iota_n$, of degree $-n - 1$, such that

$$\partial h_n = p_B \iota_n - \sum_{j=0}^{n} (-1)^j h_{n-1} d_j, \qquad \partial h_n = Dh_n + (-1)^n h_n d_{\mathrm{int}}. \tag{6.63}$$

The graded Leibniz rule proves the forward implication; the same equations on every summand define H on the direct sum and prove the converse.

For an object a and arrow $f : a \to b$, the first equations are

$$\partial h_a = p_B \iota_a, \qquad \partial h_f = p_B \iota_f - h_b X(f) + h_a. \tag{6.64}$$

The cocone term $p_B \iota_f$ remains after the object equations are solved. A mapping chain γ of degree -1, with $d\gamma = g - f$, gives

$$\partial \iota_\gamma = \iota_b X(\gamma) - \iota_g + \iota_f, \qquad \partial h_\gamma = p_B \iota_\gamma - h_b X(\gamma) + h_g - h_f.$$

Thus paths between action maps impose their own degree-minus-three primitives.

Unit insertion supplies part of these equations. Let $D_2 a = a \circledast a$ and put

$$w_a = \delta^{-1}_{P,F_{D_2 a}}((v_1)_{I_1} \boxtimes (v_2)_{I_2})\delta_{P,F_a}, \qquad r_a = X_P(v_1) - w_a.$$

The map w_a is a coefficient map; it has not thereby become an action morphism $a \to D_2 a$. Let V and W_0 be the induced maps on the bar from D_2 with $X(v_1)$ and w, retaining their coherences. The section formula gives $W_0 \simeq s\delta$; choose $p_B = 1 - W_0$. The natural transformation $v_1 : 1 \to D_2$ has a prism P with $DP + PD = V - 1$. On an ordinary string it inserts $a_j \to D_2 a_j$ in slot j with sign $(-1)^j$ and applies D_2 thereafter. Interior faces cancel in pairs and the end faces give $V - 1$. For mapping-space chains the naturality simplices supply the same cancellations. Consequently

$$p_B = \partial(-P) + R_0, \qquad R_0 = V - W_0, \qquad R_0 \iota_a = \iota_{D_2 a} r_a. \tag{6.65}$$

The projector is nullhomotopic exactly when R_0 is.

The residual equations therefore begin with

$$\partial g_a = \iota_{D_2 a} r_a, \qquad \partial g_f = R_0 \iota_f - g_b X(f) + g_a, \tag{6.66}$$

and continue by (6.63). A specified monoid extension supplies an object primitive: let $q = \mu_M(f \circledast f)$ and let its unit and action paths give $\partial\tau = X_P(q) r_a$. Using the extended density cocone if necessary,

$$g_a = \iota_M \tau - \iota_q r_a$$

has the required differential, since $\partial \iota_q = \iota_M X_P(q) - \iota_{D_2 a}$. The extension category carries its higher coherences. Extending them to all action maps is an additional requirement.

6.11.3 Objectwise and infinite compatibility

Consider a category with initial object j and two parallel arrows $f, g : a \to b$, whose composites from j agree. Take $V_j = V_b = 0$ and $V_a = k$. Its normalized bar is

$$T = [k\{f, g\} \xrightarrow{(-1,-1)} k\{a\}],$$

in degrees $-1, 0$. Every object inclusion is nullhomotopic, since the one at a sends 1 to $-f$ under a primitive. Yet $H^{-1}(T) = k\{f - g\}$, so 1_T is not nullhomotopic. Adding a constant diagram k gives a vacuum summand. The projector onto T annihilates that vacuum and every object inclusion up to homotopy, while preserving the parallel-arrow class. This finite example isolates the missing arrow equations.

There is also an inverse-limit condition. Let $F_m B$ be the subcomplex of bar length at most m. Let $M_m = \mathrm{RHom}(F_m B, B)$ be the derived k-linear mapping complex in the differential-module category on U_P. Since $B = \operatorname{colim}_m F_m B$,

$$\mathrm{RHom}(B, B) \simeq \operatorname*{Rlim}_m M_m.$$

The fibre model of $1 -$ shift on $\prod_m M_m$, and exactness of products of vector spaces, give

$$0 \longrightarrow \lim_m {}^1 H^{-1}(M_m) \longrightarrow H^0 \,\mathrm{RHom}(B, B) \longrightarrow \lim_m H^0(M_m) \longrightarrow 0. \tag{6.67}$$

No vanishing of the first term has been proved here. Nullhomotopies on finite skeleta must be compatible before they define one on the whole bar.

For the split projector itself there is a stronger criterion. The finite skeleta in this criterion retain the full coefficient complexes and every mapping-space chain; only bar length is bounded.

Proposition 6.38. *Let $B = \operatorname{colim}_m B_m$ be an exhaustive filtered presentation in the derived category of differential modules on a block-disjoint open. Suppose $B \simeq Y \oplus K$ and $p \simeq \operatorname{diag}(0, 1_K)$. Then $K = 0$ if and only if $H^q(pi_m) = 0$ for every m, q, where $i_m : B_m \to B$ is the colimit map. In particular, $pi_m \simeq 0$ for every m implies $p \simeq 0$.*

Proof. On each affine ordered chart, filtered colimits of differential modules are exact, so

$$H^q(B) = \operatorname*{colim}_m H^q(B_m).$$

The stated vanishing makes $H^q(p)$ zero on the image of every stage, hence zero everywhere. But its restriction to $H^q(K)$ is the identity. Thus K is acyclic on every affine chart. Those restrictions are jointly conservative, so $K = 0$. The converse and the last assertion follow from the splitting. □

Apply the proposition to $p = 1 - s\partial$ and $B_m = F_m B$. A nonzero density kernel is therefore detected on some full finite bar skeleton. No bound on that skeleton follows, and no finite truncation of states, normal derivatives or mapping chains is asserted. The proposition also does not make the whole $\lim^1$ group in (6.67) vanish.

Prescribed paths retain a different obstruction. If $r : M \to N$ restricts the complex of natural transformations to a specified extension diagram, its fibre has differential $d(x, y) = (dx, rx - dy)$. A prescribed cancellation $dh = rp$ gives the cycle (p, h) in this fibre. It need not vanish when the absolute class $[p]$ vanishes: if $p = dk$, the remaining class is represented by the closed element $h - rk$ of N^{-1}, modulo the image of $H^{-1}(M)$. This relative cycle is not a split projector on a coefficient object. The proposition does not supply its required path.

Problem 6.39. For the Heisenberg boundary C_{V_κ} at $\kappa \neq 0$, exhibit a nonzero class of a binary kernel $K_{C,A|B}$ in the full action bar, with its object, arrow and higher components. Determine whether such a class can be an insertion class (6.61). Over $\mathbb{C}$, determine whether there are two objects of $\mathscr{B}_C$ with no common receiving multiplicative factorizing action. The parallel maps of Theorem 6.36 have no receiving equalizer; this does not decide the two-object question.

The tree and nilpotent constructions give actual families of incoming actions. The density section, evaluation nullhomotopy and unit prism describe their images and the remaining equations on the full bar. For this Heisenberg boundary, Theorem 6.29 proves that these equations cannot all have a compatible solution: the pointed-action functor is not representable, and some density kernel is nonzero. The kernel criterion and primitive equations still apply to other boundaries. Over $\mathbb{C}$, Theorem 6.36 also excludes finality among multiplicative actions, using the multiplication itself. Common recipients for two specified action objects remain a distinct comparison problem.

Chapter 7

Supported compositions and outer collision classes

Three quadratic current fields have a connected Gaussian correlation with a pole on each pair diagonal. Its triangle of contractions already lies beyond a forest calculation. A second operator product must act on these poles and on their residue attachments, as well as on the states left by the first product. The coefficient modules of Chapter 2 provide the domains for those maps.

We begin with the finite Wick formula and its triangle current. Mixed Green kernels give a second construction, in which each edge has a point source and a homotopy must retain contact terms. Ordered collision complexes then show which classes disappear when only consecutive cuts are kept. The final comparison concerns compositions on supported field coefficients, with the lower homotopies fixed.

Unless another base is specified, k is a field of characteristic zero. Polynomial states contain finitely many jets; infinite Taylor series occur only in the indicated formal variables. Analytic currents are taken over $\mathbb{C}$. Complexes are cohomological, with $K[1]^n = K^{n+1}$, and the differential of a homogeneous map is $\delta f = d_{\mathrm{out}} f - (-1)^{|f|} f d_{\mathrm{in}}$.

7.1 Finite fields and the Gaussian triangle

Allow the Heisenberg level to be the indeterminate λ. The polynomial representation of (5.1) is then defined over $k[\lambda]$, including at $\lambda = 0$:

$$V = k[\lambda][x_1, x_2, \dots], \qquad J_{-n} = L_{x_n}, \quad J_n = \lambda n \partial_{x_n} \quad (n > 0), \qquad J_0 = 0.$$

Set $J = x_1$, $T = \sum_{n \geq 1} n x_{n+1} \partial_{x_n}$ and $j_r = T^r J = r! x_{r+1}$. These are raw, undivided jets. Write $J(z) = \sum_{m \in \mathbb{Z}} J_m z^{-m-1}$. For a polynomial $F(j_0, j_1, \dots)$, define its field $Y(F, z)$ by replacing j_r with $\partial_z^r J(z)$ and putting every creation operator J_{-n} to the left of every annihilation operator J_n. The vacuum field is $Y(1, z) = 1$.

The commutator of an annihilation part with a creation part is the expansion of $\lambda/(z-w)^2$ in nonnegative powers of w/z. Consequently the contraction between raw jets is

$$K_{rs}(z, w) = \partial_z^r \partial_w^s \frac{\lambda}{(z-w)^2} = \frac{\lambda (-1)^r (r+s+1)!}{(z-w)^{r+s+2}}. \tag{7.1}$$

It is a meromorphic coefficient. A chosen ordering of the formal variables specifies its Laurent expansion.

For polynomial inputs F_i at distinct labels $i = 1, \dots, n$, let $j_r^{(i)}$ be independent copies of the jet variables and put

$$\Gamma_{ij} = \sum_{r,s \geq 0} K_{rs}(z_i, z_j) \partial_{j_r^{(i)}} \partial_{j_s^{(j)}} \qquad (i < j).$$

After these operators act, colons mean that the surviving variables are replaced by their fields at the stated positions and all resulting creation operators are moved to the left, without further contractions.

Proposition 7.1 (Polynomial Wick evaluator). *On each state, use the formal recipient $V((z_1))\cdots((z_n))$ prescribed by $|z_1| > \cdots > |z_n|$. Then*

$$Y(F_1, z_1)\cdots Y(F_n, z_n) = :\exp\left(\sum_{i<j}\Gamma_{ij}\right)\prod_i F_i(j^{(i)}):. \tag{7.2}$$

The exponential is finite on each list of polynomial inputs. Before Laurent expansion, every coefficient is meromorphic with a finite pole order on each pair diagonal. The formula commutes with differentiation in the positions and with insertion of the vacuum. Grouping successive inputs gives the same operator product in a common iterated Laurent recipient.

Proof. Move each annihilation operator past the creation operators to its right. Their commutators are scalars, so a term is specified by a set of disjoint pairs of field slots belonging to different inputs. Each pair contributes (7.1); the unpaired slots remain normally ordered. A product of q contraction operators counts the same unordered set of pairs $q!$ times, cancelled by the exponential coefficient. Repeated derivatives count the choices among identical slots. This proves the formula on monomials and hence on polynomials.

Each contraction removes two slots. There are only finitely many slots, so the exponential terminates, and each scalar coefficient is a finite sum of products of the displayed rational functions. Differentiating the contraction identity differentiates the corresponding slot; the product rule accounts for all choices. A vacuum slot has no jet variables, so it has no contractions and contributes the identity. Finally, the operators Γ_{ij} commute. Grouping the factors partitions the same set of pairs into internal and cross pairs, each used once. Their exponentials multiply to the single exponential in (7.2). Expanding every rational coefficient in one common order proves the last assertion. □

The meromorphic expression and its Laurent expansions have different roles. At a collision with separation u, the differential-module operation applied to a Laurent test $f(u)$ is $\mathrm{pp}_u(f(u)Y(F,u)G)$. A pole in the test can detect a regular field term. For example, $\mathrm{pp}_u(u^{-1}Y(1,u)G) = u^{-1}G$. Thus a vacuum insertion remains visible even though its field has no singular part before multiplication by the test.

Proposition 7.2 (All current modes). *For every polynomial state F and integer m,*

$$[J_m, Y(F,w)] = \lambda\sum_{r\geq 0}(m)_{r+1}w^{m-r-1}Y(\partial_{j_r}F, w), \qquad (m)_{r+1} = m(m-1)\cdots(m-r). \tag{7.3}$$

The sum is finite, also for negative m.

Proof. A single current can contract with only one slot of F. The contraction with $j_r(w)$ is $\lambda(r+1)!(z-w)^{-r-2}$. The coefficient of $(z-w)^{r+1}$ in the formal expansion of z^m about w is $\binom{m}{r+1}w^{m-r-1}$, for every integer m. Taking the residue in z therefore gives the displayed falling power. Only the finitely many jet variables occurring in F contribute. □

Write $\langle - \rangle$ for the Gaussian vacuum functional, defined on a product of fields by applying the product to $1 \in V$ and then taking its constant polynomial coefficient. In (7.2) this retains exactly the complete pairings. For fields $A_1, \ldots, A_n$, their connected vacuum coefficient is

$$\kappa_n^{\mathrm{vac}}(A_1,\ldots,A_n) = \sum_{\pi\in\mathrm{Part}([n])}(-1)^{|\pi|-1}(|\pi|-1)!\prod_{B\in\pi}\left\langle\prod_{i\in B}A_i\right\rangle.$$

Products within a block use the inherited order. This definition specifies an algebraic vacuum functional; a numerical state or a field-theoretic preparation requires its own construction.

Put $Q(z) = :J(z)^2:$ and $K_{ij} = \lambda/(z_i - z_j)^2$.

Proposition 7.3 (Connected quadratic fields)**.** *For $n \geq 3$,*

$$\kappa_n^{\mathrm{vac}}(Q(z_1), \dots, Q(z_n)) = 2^{n-1} \sum_{\sigma \in S_{\{2,\dots,n\}}} K_{1,\sigma(2)} K_{\sigma(2),\sigma(3)} \cdots K_{\sigma(n),1}. \tag{7.4}$$

For $n = 2$ it is $2K_{12}^2$. In particular,

$$\kappa_3^{\mathrm{vac}}(Q(z_1), Q(z_2), Q(z_3)) = \frac{8\lambda^3}{z_{12}^2 z_{23}^2 z_{13}^2}, \qquad z_{ij} = z_i - z_j. \tag{7.5}$$

These identities hold as meromorphic coefficients.

Proof. The partition formula cancels every disconnected pairing. Indeed, for a fixed graph with c connected components, its coefficient is the sum of $(-1)^{|\pi|-1}(|\pi|-1)!$ over partitions of those components. This sum is 1 for $c = 1$ and 0 for $c > 1$, as follows by comparing coefficients in $\log(\exp t) = t$.

Normal ordering forbids a contraction within one input. A connected pairing of two slots at each of at least three vertices is therefore a cycle. Each unoriented cycle has 2^n labelled matchings, since each vertex can assign its two slots to its neighbours in two ways. Fixing vertex 1 and permuting the other vertices counts that cycle twice, once in each orientation. This gives the factor 2^{n-1}. At two vertices there are two bijections between the slots. At three vertices there is one unoriented cycle and eight labelled matchings, giving (7.5). □

This triangle has no interaction vertex. It is already a correlation of composite fields in a free theory. Its nonzero coefficient must be retained by any comparison covering all polynomial boundary fields. A calculation restricted to forests does not contain it.

7.1.1 Changing the covariance

The normal-order convention can itself be changed algebraically. Let B be a commutative $\mathbb{Q}$-algebra and $A = B[q_1, \dots, q_N]$. For symmetric matrices K, R over B, independent of the variables q_i, set

$$\Gamma_K = \sum_{i,j} K_{ij}\, \partial_{q_i} \otimes \partial_{q_j}, \qquad m_K = m \exp(\Gamma_K), \qquad \Delta_R = \sum_{i,j} R_{ij}\, \partial_{q_i} \partial_{q_j}, \qquad \alpha_R = \exp(\Delta_R/2),$$

where m is ordinary multiplication. All exponentials terminate on polynomials. The variables may include separate copies at several positions; the output operator then includes the cross derivatives between those copies.

Proposition 7.4. *For another symmetric constant matrix S,*

$$\alpha_R m_K = m_{K+R}(\alpha_R \otimes \alpha_R), \qquad \alpha_R \alpha_S = \alpha_{R+S}.$$

Proof. The second-order product rule is

$$\tfrac{1}{2}\Delta_R m = m\big(\tfrac{1}{2}\Delta_R \otimes 1 + 1 \otimes \tfrac{1}{2}\Delta_R + \Gamma_R\big).$$

All operators on the right commute. Exponentiating and composing with $\exp(\Gamma_K)$ proves the first formula. The second follows from $\Delta_R + \Delta_S = \Delta_{R+S}$. □

An intertwiner of these products does not specify a change of the vacuum functional. Comparing states requires the functional as well as the multiplication.

7.2 Residue attachments and closed coincidences

The triangle has three dependent differences. Nevertheless its meromorphic coefficient admits an explicit current extension using only products in independent normal coordinates. Work over $\mathbb{C}$, put $u = z_{12}$ and $v = z_{23}$, and use the distributions P_j of Proposition 2.2. Write $R_j = \bar\partial P_j$. Thus

$$P_j = \frac{(-1)^{j-1}}{(j-1)!}\partial_u^{j-1} P_1, \qquad R_j = \frac{\pi(-1)^{j-1}}{(j-1)!}\partial_u^{j-1}\delta_0\, d\bar u.$$

The rational identity

$$\frac{1}{u^2v^2(u+v)^2} = \frac{1}{v^4u^2} + \frac{1}{v^4(u+v)^2} - \frac{2}{v^5u} + \frac{2}{v^5(u+v)} \tag{7.6}$$

follows by multiplying by the common denominator. It defines the following extension of the triangle coefficient:

$$\mathcal{P}_\triangle = P_4(v)\big(P_2(u) + P_2(u+v)\big) - 2P_5(v)\big(P_1(u) - P_1(u+v)\big). \tag{7.7}$$

Each summand is an external product in either (u, v) or $(u+v, v)$. Both pairs are linear coordinates, so every product in this formula is defined.

Every meromorphic Wick coefficient also has a uniform principal-value realization. For $F = G/Q$ on a smooth complex chart, it is $[F] = |Q|^{2s}G/Q\big|_{s=0}$, defined by analytic continuation. This assignment is linear, independent of the holomorphic presentation, and compatible with holomorphic derivatives; see Andersson–Wulcan (2018) [3, §2, Example 4.6, Corollary 4.26]. It applies to the finite rational coefficients of Proposition 7.1, including cycles and repeated edges. In independent coordinates it agrees with the external products above, so $\mathcal{P}_\triangle$ is the principal value of (7.6). Restriction off the poles makes the assignment injective. On its image, $[F] \diamond [G] = [FG]$ transports associative meromorphic multiplication. This specified product does not extend to arbitrary supported currents.

Proposition 7.5. *The current $8\lambda^3\mathcal{P}_\triangle$ extends (7.5). Its complete Dolbeault attachment is*

$$\begin{aligned}\bar\partial\mathcal{P}_\triangle = {}& R_4(v)\big(P_2(u) + P_2(u+v)\big) + P_4(v)\big(R_2(u) + R_2(u+v)\big)\\ & - 2R_5(v)\big(P_1(u) - P_1(u+v)\big) - 2P_5(v)\big(R_1(u) - R_1(u+v)\big).\end{aligned} \tag{7.8}$$

Holomorphic differentiation gives the attachments of its jets.

Proof. Away from the three divisors, the principal values are their ordinary rational functions, so (7.6) proves the extension assertion. The distributional differential satisfies the product rule on external products. Apply it separately in the two coordinate systems to obtain all eight terms in (7.8). Holomorphic derivatives commute with $\bar\partial$ on distributions. □

The displayed extension fixes its contact terms. The rational identity by itself would not exclude adding a current supported on the collision. Nor does this construction define the ordinary product of arbitrary singular distributions. The localization and contact objects remain separate, as required by Section 2.3.

7.2.1 *The finite Koszul complex of a graph*

A closed coincidence gives a different complex. Let Γ be a finite oriented multigraph with vertex set I and edge set E; loops and repeated edges are allowed. Put

$$S = k[z_i : i \in I], \qquad K_\Gamma = S \otimes \Lambda(e_a : a \in E), \qquad |e_a| = -1, \quad de_{(i,j)} = z_i - z_j. \tag{7.9}$$

No difference is inverted. A loop has zero differential.

Choose a spanning forest T and one root in each component. For each nonforest edge a, express its difference as a signed sum along the unique forest path and set

$$c_a = e_a - \sum_{t \in \mathrm{path}(a)} \epsilon_t e_t.$$

For a loop the sum is empty. The vectors c_a form a basis of the cycle space of the graph and satisfy $dc_a = 0$. Root coordinates and the forest differences w_t form a linear coordinate system: removing a leaf successively reconstructs every vertex coordinate. With $B = k[z_{\mathrm{roots}}]$, the complex is therefore

$$K_\Gamma = B[w_t : t \in T] \otimes \Lambda(e_t : t \in T) \otimes \Lambda(c_a : a \notin T), \qquad d = \sum_t w_t \partial_{e_t}.$$

Let p set all w_t, e_t to zero, and let i include the resulting root-and-cycle algebra.

Proposition 7.6 (Forest contraction)**.** *Let N count total polynomial degree in the w_t plus exterior degree in the e_t, excluding the c_a. Define*

$$H = \frac{1}{N} \sum_t e_t \partial_{w_t} \quad (N > 0), \qquad H = 0 \quad (N = 0).$$

Then

$$dH + Hd = 1 - ip, \qquad H^2 = Hi = pH = 0, \qquad H^*(K_\Gamma) = B \otimes \Lambda(c_a : a \notin T).$$

The homotopy and the chosen section are B-linear. The augmentation p is S-linear for the quotient action on its target.

Proof. The graded commutator of $\sum_t w_t \partial_{e_t}$ and $\sum_t e_t \partial_{w_t}$ is the Euler operator in exactly the w_t, e_t variables. Both preserve N. Dividing by N proves the contraction identity on positive degree; on degree zero ip is the identity. The numerator of H squares to zero because the exterior multiplications anticommute and the ordinary derivatives commute. Evaluation gives $pH = 0$, and $Hi = 0$ follows from the absence of forest variables in the image of i. The target has zero differential, proving the cohomology formula.

The formulas leave root coordinates unchanged. In contrast, $H(w_t) = e_t$ whereas $w_t H(1) = 0$, so H is generally not S-linear. The augmentation respects S because setting all forest differences to zero is precisely its quotient action. □

For the triangle the surviving generator is $e_{13} - e_{12} - e_{23}$. Identifying the endpoints of the path $12, 23$ instead gives the closed generator $e_{12} + e_{23}$. Keeping the original edge generators makes vertex identifications maps of Koszul complexes. The choice of spanning forest calculates their cohomology without removing those cycles.

The stable support complex of Lemma 2.6 is unchanged by a redundant support equation. The finite Koszul complex retains its excess cycle. Neither construction identifies that cycle with the meromorphic triangle current. A graph loop, for example, gives a closed exterior generator in (7.9), but no rational function $1/(z_i - z_i)$.

7.3 Mixed kernels and their forest homotopies

To compare a mixed propagator with a boundary coefficient, its point source must remain in the differential. Work on $\mathbb{R}_t \times \mathbb{C}_z$, where $z = x + iy$, with orientation $dt \wedge dx \wedge dy$. Partial forms are generated by dt, $d\bar{z}$, and

$$D = dt\,\partial_t + d\bar{z}\,\partial_{\bar{z}}, \qquad \partial_{\bar{z}} = \tfrac{1}{2}(\partial_x + i\partial_y).$$

Distribution coefficients are measured against $dt\,dx\,dy$. Put $r = (t^2 + |z|^2)^{1/2}$ and define

$$P = \frac{t\,d\bar{z} - 2\bar{z}\,dt}{4\pi r^3}, \qquad A = -\frac{\delta(t)\,dt}{\pi z}, \qquad \Delta = \delta^{(3)}(t, z)\,dt \wedge d\bar{z}. \tag{7.10}$$

The coefficients of P are locally integrable. The coefficient of A is the external product of the one-dimensional delta and the locally integrable Cauchy kernel.

Lemma 7.7. *One has $DP = DA = \Delta$ as currents.*

Proof. The coefficient of $dt \wedge d\bar{z}$ in DP is

$$\frac{1}{4\pi}\left(\partial_t \frac{t}{r^3} + 2\partial_{\bar{z}} \frac{\bar{z}}{r^3}\right).$$

The imaginary angular terms cancel. The remainder is $1/(4\pi)$ times the divergence of $(t, x, y)/r^3$, whose flux through a sphere is 4π. It is therefore $\delta^{(3)}$ distributionally. For A, use $\partial_{\bar{z}}(1/(\pi z)) = \delta^{(2)}(z)$. The minus sign in A cancels the sign from moving $d\bar{z}$ past dt. □

Let $H(t)$ be the Heaviside distribution. Its value at zero is irrelevant to the current. Define

$$S(t, z) = \frac{H(t) - \frac{1}{2} - \frac{t}{2r}}{\pi z}. \tag{7.11}$$

Proposition 7.8. *The function S is locally integrable and satisfies*

$$DS = P - A, \qquad \int_{\mathbb{R}} P = \int_{\mathbb{R}} A = -\frac{1}{\pi z}.$$

At fixed z, its two tails are $O(|\bar{z}|/t^2)$.

Proof. On $t > 0$ and $t < 0$ respectively,

$$S_+ = \frac{1 - t/r}{2\pi z}, \qquad S_- = \frac{-1 - t/r}{2\pi z}.$$

Both have derivatives $\partial_t S_\pm = -\bar{z}/(2\pi r^3)$ and $\partial_{\bar{z}} S_\pm = t/(4\pi r^3)$. Their jump across $t = 0$ is $1/(\pi z)$, supplying the $\delta(t)/(\pi z)$ term in $P - A$. For fixed $t \neq 0$ the numerator vanishes at $z = 0$, so there is no further axial delta. The function is homogeneous of degree -1 with bounded angular factor near the origin. Its first derivatives have degree -2. A distribution supported only at the origin is a finite sum of delta derivatives, whose degrees are at most -3. Thus no additional point-supported term occurs, and the identity holds on the whole space.

Integration of the dt coefficient gives

$$-\frac{\bar{z}}{2\pi} \int_{-\infty}^{\infty} (t^2 + |z|^2)^{-3/2}\,dt = -\frac{1}{\pi z}.$$

Its absolute integral is locally integrable in z, which justifies the distributional pushforward. The axial integral has the same value. Expanding t/r at either end gives the tail estimate. □

Multiplication by $-\pi$ converts this pushforward to the Cauchy coefficient $1/z$, whose contour residue is one under $(2\pi i)^{-1} \oint dz$. A physical coupling or pairing must include that normalization. Every fixed finite holomorphic or normal derivative commutes with the current identity.

7.3.1 Cutoffs and boundary conditions

For a smooth cutoff χ equal to one near the origin,

$$D(\chi P) = \Delta + D\chi \wedge P, \qquad D(\chi S) = \chi(P - A) + D\chi\, S. \tag{7.12}$$

The annular terms belong to the supported comparison. If a composition needs a proper projection, choose cutoffs giving that property and retain these terms in its localization complex.

A smooth regulator follows from $D^* = -\partial_t \iota_{dt} - 4\partial_z \iota_{d\bar z}$. For $0 < \varepsilon < L$, set

$$K_\ell = (4\pi\ell)^{-3/2} e^{-r^2/(4\ell)}, \qquad P_{\varepsilon,L} = D^*\left(\int_\varepsilon^L K_\ell \, d\ell \; dt \wedge d\bar z\right).$$

Since $DD^* + D^*D$ is the negative Euclidean Laplacian, the heat equation gives

$$DP_{\varepsilon,L} = (K_\varepsilon - K_L)\, dt \wedge d\bar z.$$

Both heat endpoints occur in its differential.

On a half-line the relative de Rham complex has Dirichlet zero-forms and Neumann one-forms. Writing $G_\ell(a) = (4\pi\ell)^{-1/2} e^{-a^2/(4\ell)}$, their kernels are

$$K^0_\ell(t,s) = G_\ell(t-s) - G_\ell(t+s), \qquad K^1_\ell(t,s) = G_\ell(t-s) + G_\ell(t+s).$$

Differentiation gives $\partial_t K^0_\ell = -\partial_s K^1_\ell$. Integration by parts, using the relative boundary value, proves cochain compatibility. Tensoring with the tangential Dolbeault complex preserves the identity. The polarization of a particular field theory determines whether its normal complex is relative or absolute.

7.3.2 Forest cubes and contact terms

Let F be an oriented forest on a finite labelled set. For $e = (i, j)$ put $t_e = t_i - t_j$ and $z_e = z_i - z_j$. Choose one root per connected component. The root coordinates and edge differences are a linear coordinate system with determinant ± 1, by successive leaf deletion. Hence the pullbacks P_e, A_e, S_e, Δ_e for different forest edges have independent normal variables. Their ordered products, including any fixed finite normal derivatives, are external products of distributions in these coordinates.

Theorem 7.9. *On every labelled forest, the mixed and axial decorations are joined by homotopy cubes compatible with contact contraction, spectators, and products or graftings that remain forests. This includes the edge* 13 *with spectator* 2 *and the crossed pair of edges* 13, 24.

Proof. Introduce $s_e \in [0,1]$ for each edge, with parameter forms written before spatial forms, and set

$$\mathbb{P}_e = A_e + s_e(P_e - A_e) + ds_e\, S_e.$$

Then $(D + d_s)\mathbb{P}_e = \Delta_e$: the term $ds_e(P_e - A_e)$ cancels $-ds_e DS_e$. Independence of the edge variables defines every forest product. Its differential is the signed sum replacing one edge by its contact current. At $s_e = 0$ or 1 the edge becomes A_e or P_e. Restrictions to two faces commute, giving the higher face identities. The diagonal interval varies all edges together.

Contact contraction identifies the two endpoints of an edge; the remaining differences are coordinates of the contracted forest. The distributional equations persist under finite normal differentiation and retain its contact jets. Adding an isolated vertex adds a spectator, and the empty product is the coefficient unit. Edge reorderings have the usual form signs. □

For two edges define $H_{e,f} = S_e P_f - A_e S_f$. Since $|P| = |A| = 1$ and $|S| = 0$, the product rule gives

$$P_e P_f - A_e A_f = D H_{e,f} + \Delta_e S_f - S_e \Delta_f. \tag{7.13}$$

The last two terms express the common point sources of the kernels. They remain even though $P - A$ is exact.

At three positions put $z_{12} = u$, $z_{23} = v$, $z_{13} = u + v$. The axial products have the same time delta and oriented time two-form. Their scalar coefficients satisfy

$$\frac{1}{uv} - \frac{1}{u(u+v)} - \frac{1}{v(u+v)} = 0.$$

Each term is locally integrable on $\mathbb{C}^2$ in its independent linear coordinates. Thus the rational identity is also a distributional identity. Define $\mathcal{K} = H_{12,23} - H_{12,13} - H_{13,23}$. Applying (7.13) to the three pairs gives

$$\begin{aligned} P_{12} P_{23} - P_{12} P_{13} - P_{13} P_{23} &= D\mathcal{K} + C, \\ C &= \Delta_{12} S_{23} - S_{12} \Delta_{23} - \Delta_{12} S_{13} + S_{12} \Delta_{13} \\ &\quad - \Delta_{13} S_{23} + S_{13} \Delta_{23}. \end{aligned} \tag{7.14}$$

Every product here has independent normal variables.

These are scalar coefficient currents on the mixed space. Decorating them by fields requires a map preserving their contact terms, translations, partial vacua and supported attachments. A cycle has dependent edge differences, so its mixed current product needs an additional extension. The meromorphic triangle of (7.7) supplies a specified boundary coefficient, without identifying it with such a mixed product.

7.4 Classes discarded by consecutive cuts

Consecutive cuts preserve an order on the labels. A collision can join labels that are separated in that order. To measure this difference, we compare two coefficient complexes before introducing a map from coefficients to fields.

Let $R_n = k[z_1, \dots, z_n]$ and $L_n = R_n[(\prod_{i<j} z_{ij})^{-1}]$. An ordered partition $P = (B_1 | \cdots | B_r)$ has coefficient module

$$M_P = R_n[z_{ij}^{-1} : i, j \text{ lie in different blocks of } P] \subset L_n. \tag{7.15}$$

Equal subrings attached to different ordered partitions remain different summands. Tensor every term with the common right density $\omega_n = \omega_{\mathbb{A}^n}$; for a constant density ν, $(f\nu)\partial_i = -(\partial_i f)\nu$. We suppress this common factor in the coordinate formulas.

Write $e_P = \xi_{B_1} \cdots \xi_{B_r}$, where $|\xi_B| = 1 - |B|$. The full complex C_n^{all} has term $M_P e_P$ in degree $r - n$. Its differential splits a block $B_i = A \sqcup C$ into two nonempty ordered parts, with coefficient the localization inclusion and sign

$$(-1)^{\sum_{j<i}(1-|B_j|)+|A|+1} \epsilon(A, C). \tag{7.16}$$

Here $\epsilon(A, C)$ is the sign of the concatenated increasing lists of A, C relative to the increasing list of B_i. A partition is *consecutive* when its blocks are intervals in increasing order. Projection to those summands gives the consecutive-cut complex C_n^{cut}. Its kernel is

$$F_n^{r-n} = \bigoplus_{\substack{P \text{ has } r \text{ blocks} \\ P \text{ is not consecutive}}} M_P e_P. \tag{7.17}$$

There is no one-block term in F_n.

Lemma 7.10. *The discarded summands form a subcomplex and $d^2 = 0$. After extending every coefficient to L_n, the full complex is the oriented cellular chain complex of the permutahedron of dimension $n-1$, with chain degree negated. For $n \geq 3$, the map from discarded two-block terms to three-block terms is injective, both after localization and on the actual modules.*

Proof. If a refinement is consecutive, merging its two adjacent new blocks recovers a consecutive partition. Thus a nonconsecutive partition has only nonconsecutive refinements. Two splits in different blocks cancel by the odd Leibniz sign. Two successive splits of the same block have the same density permutation and opposite cobar signs. Hence $d^2 = 0$.

The permutahedron is the convex hull of the permutations of $(1, \dots, n)$. A face is specified by an ordered partition; its dimension is $n - r$, and splitting one block gives a boundary face. Orient the block factor by the increasing labels, then orient the product in block order. Its boundary orientation is exactly (7.16). This identifies the fully localized incidence complex.

The permutahedron is a ball. In its cellular complex the kernel on facets is the image of the unique top cell, spanned by $d\xi_{[n]}$. Every facet has coefficient 1 or -1 in that boundary. The discarded facet space omits all $n-1$ consecutive facets, so it contains no nonzero multiple of the top boundary. This proves injectivity over k, hence over L_n. Each actual coefficient module embeds in L_n, giving the last assertion. □

7.4.1 *The outer principal-part module*

Put $u = z_{1n}$. Retain in $d\xi_{[n]}$ only those facets which keep $1, n$ together:

$$T_n = \sum_{\substack{A \sqcup C = [n],\ A, C \neq \varnothing \\ 1, n \text{ lie in the same part}}} (-1)^{|A|+1} \epsilon(A, C) \xi_A \xi_C, \qquad b_n = dT_n. \tag{7.18}$$

These facets are all discarded. The degrees of T_n and b_n are $2 - n$ and $3 - n$, respectively.

Theorem 7.11 (Outer classes in every arity)**.** *For every $n \geq 3$, the formula*

$$\omega_n \otimes_{R_n} (R_n[u^{-1}]/R_n) \hookrightarrow H^{3-n}(F_n), \qquad [f] \longmapsto [f\, b_n] \tag{7.19}$$

defines an injective map of right differential modules. Thus no arity-independent lower bound exists for the degrees of nonzero cohomology of the discarded complexes.

Proof. Consider a three-block partition which still keeps $1, n$ together. Its two facet coarsenings both occur in T_n, so their contributions to dT_n cancel. Every nonzero coordinate of b_n therefore separates $1, n$ and has a coefficient module in which u is invertible. Consequently $f\, b_n$ is an allowed cochain for every $f \in R_n[u^{-1}]$, and it is closed. If $f \in R_n$, the primitive $f\, T_n$ is allowed, so the formula factors through principal parts.

Conversely, suppose $f\, b_n = dv$ for an actual discarded facet cochain v. Over L_n, the injectivity in Lemma 7.10 forces $v = f\, T_n$. Choose a facet P occurring in T_n. Its module does not invert u, and unique factorization gives $M_P \cap R_n[u^{-1}] = R_n$. The corresponding coordinate of v is $\pm f$, so f is regular. Such a facet exists for $n \geq 3$, and the same injectivity shows $b_n \neq 0$.

The coefficients of b_n are constants. Differentiating $f\, b_n$ therefore differentiates only f. Restoring the common right density proves differential-module linearity. Every normal derivative of the simple principal part is retained. □

For four positions the numerator is

$$\begin{aligned} T_4 = {} & -\xi_{124}\xi_3 + \xi_{134}\xi_2 - \xi_{14}\xi_{23} \\ & - \xi_2\xi_{134} - \xi_{23}\xi_{14} + \xi_3\xi_{124}. \end{aligned} \tag{7.20}$$

Thus $z_{14}^{-1} dT_4$ is a nonzero class in $H^{-1}(F_4)$. Its fully localized primitive $z_{14}^{-1} T_4$ has a forbidden pole in each displayed facet coefficient. The obstruction is precisely that failure to lie in the specified module.

The relation with the cut class also fixes a sign and a degree. The degreewise exact sequence is

$$0 \longrightarrow F_n \longrightarrow C_n^{\mathrm{all}} \longrightarrow C_n^{\mathrm{cut}} \longrightarrow 0.$$

Its cut top boundary is

$$U_n = d_{\mathrm{cut}} \xi_{[n]} = \sum_{p=1}^{n-1} (-1)^{p+1} \xi_{[1,p]} \xi_{[p+1,n]}.$$

Every term separates $1, n$, so $f U_n$ is an allowed cut cocycle for every $f \in R_n[u^{-1}]$.

Proposition 7.12. *The connecting homomorphism sends*

$$H^{2-n}(C_n^{\mathrm{cut}}) \longrightarrow H^{3-n}(F_n), \qquad [f U_n] \longmapsto -[f b_n].$$

Proof. Write $d\xi_{[n]} = U_n + T_n + V_n$, where V_n consists of discarded facets separating $1, n$. Lift $f U_n$ to the same cochain in the full complex. Its differential is $d(f U_n) = -f b_n - d(f V_n)$. The latter term is a legal boundary in F_n, since every facet in V_n inverts u. □

This comparison concerns restriction of all-subset cobar coefficients to a chosen order and cobar after restriction to cuts. Its kernel does not refute bar–cobar duality on either of its stated categories. A map from these coefficient classes to a field-action complex remains additional data.

7.4.2 Global cohomology at four positions

The full four-position permutahedron has 14 facets, 36 edges and 24 vertices. Removing consecutive partitions removes three facets, three edges and the ascending vertex. Hence

$$F_4 : \qquad \bigoplus_{11} M_P \xrightarrow{A} \bigoplus_{33} M_Q \xrightarrow{B} L^{23}, \qquad L = L_4, \tag{7.21}$$

in degrees $-2, -1, 0$. The fully localized ranks are 11, 22. For the second rank, the edges connect all remaining vertices: adjacent swaps increasing inversion number connect every nonascending permutation to the reverse permutation without passing through the removed vertex. Thus over L, $\ker B = \operatorname{im} A$ and $\ker A = 0$.

For an unordered pair h, write

$$L_h = R_4[z_{ab}^{-1} : \{a, b\} \neq h], \qquad W_h = L / L_h.$$

A three-block term has a unique unsplit pair h and coefficient L_h. Let I_h be the discarded facets keeping h together and J_h the three-block terms keeping it together. Denote the corresponding submatrix of A by K_h.

Lemma 7.13. *The sizes and ranks are*

h	12	13	14	23	24	34
$\lvert I_h \rvert$	4	5	6	4	5	4
$\lvert J_h \rvert$	5	6	6	5	6	5
$\operatorname{rank} K_h$	4	5	5	4	5	4.

Only K_{14} has a kernel. In the order of the six terms in (7.20)*, that kernel is generated by* $(-1, 1, -1, -1, -1, 1)$.

Proof. Use (7.16). In the displayed order,

$$K_{14} = \begin{pmatrix} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{pmatrix}.$$

Its equations determine every coordinate from the first. For the other pairs, order the partitions lexicographically by their increasing block lists. The rows for 12, 23, 34 give respectively

$$\begin{array}{c|ccccc} 12 & -v_1 & -v_2 & v_2+v_3 & v_1-v_4 & -v_3+v_4 \\ 23 & v_1 & -v_2 & v_2-v_3 & -v_1+v_4 & v_3-v_4 \\ 34 & v_1 & v_2 & -v_2+v_3 & -v_1-v_4 & -v_3+v_4. \end{array}$$

Each zero system forces all four coordinates to vanish. For 13 the rows are $-v_1, v_1 - v_2, -v_3, v_3 + v_4, v_2 + v_5, -v_4 - v_5$; for 24 they are $v_1, v_2, -v_1 - v_4, v_3 + v_4, -v_2 + v_5, -v_3 - v_5$. These force all five coordinates to vanish. Every elimination uses only 1, −1, so the same conclusions hold with coefficients in W_h. □

Theorem 7.14. *Globally on the standard affine chart,*

$$H^{-2}(F_4) = 0, \qquad H^{-1}(F_4) = \omega_4 \otimes_{R_4} (R_4[z_{14}^{-1}]/R_4).$$

The second identification sends $[f]$ *to* $[f\,dT_4]$.

Proof. The first assertion follows from facet injectivity. Suppress the density for the second. A degree-minus-one cocycle w has a unique fully localized primitive $v \in L^{11}$, with $Av = w$. Requiring the J_h coordinates of w to lie in L_h gives

$$K_h(v_i \bmod L_h)_{i \in I_h} = 0.$$

For $h \neq 14$, Lemma 7.13 forces each indicated coordinate of v to be regular along h. Unique factorization gives, inside L,

$$M_P = \bigcap_{h \text{ internal to } P} L_h.$$

Putting $u = z_{14}$, we obtain $v_P \in M_P[u^{-1}]$. If P separates 1, 4, this is already M_P, and its coordinate is a legal boundary. The six remaining principal parts are proportional to the coefficients of T_4. Thus H^{-1} is the intersection, inside W_{14}, of the images of $M_P[u^{-1}]/M_P$ for these six facets.

To compute that intersection at the other collision divisors as well, use

$$u = z_{14}, \quad a = z_{12}, \quad b = z_{23}, \quad c = a + b = z_{13}, \quad y = z_4, \qquad D = k[y, a, b].$$

The remaining differences are $z_{24} = u - a$, $z_{34} = u - c$. Finite negative Laurent expansion at $u = 0$ identifies

$$W_{14} = \bigoplus_{j \geq 1} D[(abc)^{-1}]u^{-j}.$$

Indeed, expand $(u - a)^{-1}$, $(u - c)^{-1}$ only to the finite order required by the pole in u. A zero negative part means that the numerator is divisible by that power of u, hence the original fraction belongs to L_{14}.

The three distinct facet images, from 124|3, 134|2 and 14|23, are respectively

$$\bigoplus_{j \geq 1} D[(bc)^{-1}]u^{-j}, \quad \bigoplus_{j \geq 1} D[(ab)^{-1}]u^{-j}, \quad \bigoplus_{j \geq 1} D[(ac)^{-1}]u^{-j}.$$

The reversed facets repeat them. For example, $M_{124|3}$ inverts $b, c, u - c$, whose negative Laurent coefficients invert only b, c; all such coefficients occur. The irreducibles $a, b, c = a + b$ are pairwise nonassociated in D, so the intersection of the three coefficient rings is D. The principal-part intersection is therefore $\bigoplus_{j \geq 1} Du^{-j} = R_4[u^{-1}]/R_4$. The identification is $[f] \mapsto [f\,dT_4]$, already proved differential-linear in Theorem 7.11. □

Degree zero has a finite global presentation. Let G be the permutahedron graph on $S_4 \setminus \{1234\}$ and choose $\sigma_0 = 4321$. Label each edge by its swapped pair h and define

$$\mathcal{G}_{R_4} = \frac{\bigoplus_{\sigma \neq \sigma_0} Le_\sigma}{\sum_{\{\sigma,\tau\} \in E(G)} L_{h(\sigma,\tau)}(e_\sigma - e_\tau)}, \qquad e_{\sigma_0} = 0.$$

Then

$$H^0(F_4) = \omega_4 \otimes_{R_4} (L \oplus \mathcal{G}_{R_4}). \tag{7.22}$$

To see this, orient the singleton density basis so that B is the graph incidence map. Its image lies in the kernel of the sum of vertex coordinates. Inclusion at σ_0 splits that sum. In its kernel, the basis $e_\sigma - e_{\sigma_0}$ gives exactly the displayed relations. All basis changes are constant and preserve the differential module structure. This presents the global module by 22 generator modules and 33 relation modules; it does not decompose its extensions across all diagonal intersections.

7.4.3 The crossed four-position chart

On the open set separating $\{1, 3\}$ from $\{2, 4\}$, put

$$u = z_{13}, \quad v = z_{24}, \quad A = R_4[(z_{12}z_{14}z_{23}z_{34})^{-1}],$$
$$A_u = A[u^{-1}], \quad A_v = A[v^{-1}], \quad L = A[(uv)^{-1}].$$

The two remaining collision coordinates u, v are independent. The coefficient types in (7.21) become

$$A^2 \oplus A_u^3 \oplus A_v^3 \oplus L^3 \longrightarrow A_u^6 \oplus A_v^6 \oplus L^{21} \longrightarrow L^{23}.$$

A nonzero scalar entry a between two identical coefficient modules can be cancelled. Isolating that entry by triangular row and column operations gives a contractible two-term summand with homotopy a^{-1}. The remaining differential is the Schur complement $D'_{rc} = D_{rc} - D_{rj}a^{-1}D_{ic}$. Every entry still composes allowed localization inclusions. These changes commute with differential operators.

Order the partitions lexicographically by increasing block lists. Cancel the first available scalar between identical modules, using lower degree first, then row, then column. The nine cancellations from degree -2 have types

$$A_u, L, A_v, A_v, L, A_u, A_u, L, A_v$$

and pivots $1, 1, -1, -1, 1, 1, 1, 1, 1$. The next eighteen cancellations, from degree -1, have type L and pivot 1. The remaining complex is

$$A^2 \xrightarrow{M_0} A_u^3 \oplus A_v^3 \xrightarrow{M_1} L^5.$$

In its retained middle order $(A_u, A_v, A_v, A_u, A_u, A_v)$,

$$M_0 = \begin{pmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & -1 \\ -1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad M_1 = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{pmatrix}.$$

Use the basis matrices

$$S = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \end{pmatrix}, \qquad W = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 1 & 0 & -1 & 0 \\ -1 & 1 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \end{pmatrix}.$$

Both are invertible. The first three columns of S have coefficient A_u, the last three A_v, so this basis change respects the modules. Matrix multiplication gives

$$S^{-1}M_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{pmatrix}, \qquad W^{-1}M_1S = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{7.23}$$

Theorem 7.15 (Crossed local form)**.** *On this open set, F_4 is chain-homotopy equivalent by right differential-module maps to*

$$L[0] \oplus [A_u \to L] \oplus [A_v \to L] \oplus [A \to A_u \oplus A_v \to L]^{\oplus 2}. \tag{7.24}$$

The two-term complexes have degrees -1, 0; the three-term complexes have degrees -2, -1, 0. In particular,

$$H^{<0}(F_4) = 0, \qquad H^0(F_4) = \omega_4 \otimes_A \Big(L \oplus L/A_u \oplus L/A_v \oplus \big(L/(A_u + A_v)\big)^{\oplus 2} \Big).$$

Proof. The first two columns and their fourth and fifth partners in (7.23) give two complexes with maps $a \mapsto (a, -a)$ and $(f, g) \mapsto f + g$. Columns three and six give the two remaining two-term complexes. The last target row gives $L[0]$. All preceding cancellations were chain homotopy equivalences.

The maps $A_u \to L$ and $A_v \to L$ are injective. In a three-term summand the middle kernel is $(f, -f)$ with $f \in A_u \cap A_v = A$. The last equality follows by unique factorization in the localization of R_4: the independent prime factors u, v remain nonassociated and neither has been inverted in A. Only the displayed cokernels remain. □

The codimension-two module $L/(A_u + A_v)$ is nonzero. Near $u = v = 0$, its class $1/(uv)$ has a negative power in both independent normals, which cannot occur in a sum of terms with poles in only one of them. The crossed chart has retained these two copies of the collision module. It has inverted z_{14}, so its vanishing negative cohomology agrees with the global outer class, whose support is $z_{14} = 0$.

At a generic point of that outer divisor, where all other differences are invertible, set $u = z_{14}$ and $A = R_4[(z_{12}z_{13}z_{23}z_{24}z_{34})^{-1}]$. The same cancellations give

$$[A \to A[u^{-1}]]_{[-2,-1]} \oplus [A \xrightarrow{(1,-1)} A[u^{-1}]^2]_{[-1,0]}.$$

Thus $H^{-1} = A[u^{-1}]/A$ and $H^0 = A[u^{-1}] \oplus A[u^{-1}]/A$ there. These are full principal-part modules, including every normal derivative.

7.5 Independent normals and the full diagonal

The crossed calculation retains proper collision strata. The full diagonal has a simpler normal calculation, whose shift depends on the number of independent equations. We compute it before imposing any identification with a field coefficient.

Let $X = \mathbb{A}^b_k$ have coordinates $y_1, \dots, y_b$, let $Y = X \times \mathbb{A}^c_k$ have normal coordinates $u_1, \dots, u_c$, and let $i : X \hookrightarrow Y$ be the zero section. For a left D_X-module L, its normal direct image is

$$i_+L = L[\eta_1, \dots, \eta_c], \qquad \partial_{u_j} = \eta_j, \qquad u_j = -\partial_{\eta_j}.$$

Tangential operators act on L. All polynomial normal jets are retained. Write $\mathrm{DR}_X(L)$ for the unshifted algebraic de Rham complex in degrees $0, \dots, b$.

Proposition 7.16. *There is an explicit contraction*

$$R\,\mathrm{Hom}_{D_Y}(\mathcal{O}_Y, i_+L) \simeq \mathrm{DR}_X(L)[-c].$$

For right differential modules, put $L^r = \omega_X \otimes_{\mathcal{O}_X} L$ and $\mathbb{J}_Y = \omega_Y[b+c]$. Then

$$R\,\mathrm{Hom}_{D_Y}\big(\mathbb{J}_Y, i_*(L^r[b])\big) \simeq \mathrm{DR}_X(L)[-2c], \tag{7.25}$$

where i_* denotes right differential-module direct image.

Proof. Apply Hom from the Spencer resolution of $\mathcal{O}_Y$. One normal direction contributes the cochain complex

$$K_j = [k[\eta_j] \xrightarrow{\eta_j} k[\eta_j]]$$

in degrees zero and one. Let p_j extract the constant coefficient in degree one and let i_j insert it. Let h_j vanish on degree zero and send η_j^a in degree one to η_j^{a-1} for $a > 0$, and 1 to zero. Then

$$dh_j + h_j d = 1 - i_j p_j, \qquad p_j i_j = 1, \qquad h_j^2 = h_j i_j = p_j h_j = 0.$$

On the tensor product of the normal complexes set

$$H = \sum_{j=1}^{c} (i_1 p_1) \otimes \cdots \otimes (i_{j-1} p_{j-1}) \otimes h_j \otimes 1 \otimes \cdots \otimes 1,$$

using the Koszul signs for tensor products of graded maps. The identity follows by telescoping $1 - q_1 \cdots q_c = \sum_j q_1 \cdots q_{j-1}(1 - q_j)$, with $q_j = i_j p_j$. The cross terms in H^2 contain $h_j i_j$ or $p_j h_j$ and vanish. Tensoring with the tangential de Rham complex contracts the whole Spencer complex onto that complex in normal degree c. Every polynomial jet has finite degree; the argument also applies to an infinite-rank module L.

Left-to-right conversion identifies the source and target of (7.25) with $\mathcal{O}_Y[b+c]$ and $i_+L[b]$. Their relative shift is $-c$. Combining it with the normal Spencer degree gives $-2c$. □

For an ordered n-point affine chart, the first coordinate and its differences with the other points give one tangential coordinate and $n-1$ independent normals. Let $N = \mathrm{End}_{k[y]}(\mathcal{V})$, with its commutator connection, be the polynomial-root endomorphism module of Theorem 5.8. Its de Rham complex is $G(N)$, with cohomology B_0 in degree zero and W in degree one.

Suppose a specified collision comparison identifies the actual full-diagonal coefficient of an ambient object with $N^r[1]$. The supported linear mapping complex from $\mathbb{J}_{\mathbb{A}^n}$ is then

$$G(N)[-2(n-1)], \qquad H^{2n-2} = B_0, \quad H^{2n-1} = W, \tag{7.26}$$

and all other cohomology vanishes. This follows from Proposition 7.16 with $b = 1, c = n - 1$ and the stated endomorphism contraction. Replacing the source by $\mathbb{J}_{\mathbb{A}^n}[-d]$ changes the complex to $G(N)[d - 2(n-1)]$.

The collision identification is a hypothesis about an actual coefficient object and its maps. Matching a rank does not supply it. For $n \geq 3$, the complement of the full diagonal still contains proper collision strata; their simultaneous support requires their attachment maps. The localization triangle also controls the map from supported to ambient cohomology, which need not be injective. Finally, dependent graph equations have codimension equal to their rank, and this calculation concerns linear Hom. Coalgebra and substitution compatibility require their respective operation maps.

7.6 Heisenberg compositions on the pair support

We now construct an outer operation on a complete pair-support complex. Its source includes poles in the remaining separation and the triple-diagonal term. A correction to the binary path will make the operation closed on this larger domain.

Fix $\kappa \in k^{\times}$ and $X = \mathbb{A}^1_k$. The dg vertex algebra is $\mathcal{Z} = V_\kappa \otimes \Lambda[c]$, with $|J| = 0$, $|c| = 1$, $dJ = \kappa T c$ and $dc = 0$. Its state complex is

$$C = k[u_0, u_1, \ldots] \otimes \Lambda(c_0, c_1, \ldots), \quad u_n = T^n J, \quad c_n = T^n c, \quad du_n = \kappa c_{n+1}, \quad dc_n = 0. \tag{7.27}$$

All states and state tensor products are algebraic. The current bracket is $[J_\nu J] = \kappa \nu \mathbf{1}$; brackets with c vanish. The Fourier variable ν records normal derivatives: the contact of v^{-n-1} is the nth normal derivative divided by $n!$. On regular state inputs this is the usual bracket $[a_\nu b] = \operatorname{Res}_v e^{\nu v} Y(a, v) b \, dv$.

7.6.1 A common resolution and its state contraction

Let E be the free quartet vertex algebra on fields of degrees $(0, 0, 1, -1)$. Form $\mathcal{W} = V_\kappa \otimes E^{\otimes 3}$, with quartet generators $(t_s, p_s, \eta_s, \iota_s)$ and

$$Q t_s = \eta_s, \qquad Q \iota_s = p_s, \qquad [p_{s\,\nu} t_l] = [\iota_{s\,\nu} \eta_l] = \delta_{sl}.$$

The other generating brackets vanish, apart from those forced by graded skew symmetry and translation. In this ambient algebra take $QJ = 0$; its current commutes with the quartets. Put

$$K_3 = J + p_1 + p_2 + p_3, \qquad \mathcal{B} = \operatorname{Com}(K_3, \mathcal{W}), \qquad U_s = J + \sum_{l<s} p_l - \kappa T t_s.$$

The commutant means the common kernel of the nonnegative modes of K_3. There are vertex chain maps

$$j_s : \mathcal{Z} \longrightarrow \mathcal{B}, \qquad j_s(J) = U_s, \quad j_s(c) = -\eta_s. \tag{7.28}$$

Distinct images $j_s(\mathcal{Z})$ commute. Indeed $QU_s = -\kappa T \eta_s$ and $[U_{r\,\nu} U_s] = \delta_{rs} \kappa \nu$. For $r < s$, the mixed contribution $[-\kappa T t_{r\,\nu} p_r] = -\kappa \nu$ cancels the current bracket. All current–ghost and ghost–ghost brackets are zero, and these images commute with K_3. The assignments therefore extend to the stated vertex maps.

Here is a direct description of the state space of $\mathcal{B}$. Write $\tau_n^{(s)} = T^n t_s / n!$ for the divided quartet jets, with $n \geq 0$, and use the current modes $x_n = J_{-n}\mathbf{1}$. In these polynomial coordinates, the nonnegative modes of K_3 are

$$\sum_s \partial_{\tau_0^{(s)}}, \qquad \kappa n \partial_{x_n} + \sum_s \partial_{\tau_n^{(s)}} \quad (n > 0).$$

Changing coordinates to $x_n - \kappa n \tau_n^{(1)}$ and $\tau_n^{(s)} - \tau_n^{(1)}$ for $s = 2, 3$ leaves these as derivatives in the remaining $\tau_n^{(1)}$ coordinates. Their common kernel is therefore freely spanned by the PBW monomials in the translates of

$$U_1, \eta_1; \qquad a_2, a_3, e_2, e_3; \qquad \iota_1, \iota_2, \iota_3, p_1, p_2, p_3, \quad a_s = t_s - t_1, \quad e_s = \eta_s - \eta_1.$$

Their differentials are $QU_1 = -\kappa T \eta_1$, $Qa_s = e_s$, $Q\iota_s = p_s$, with the remaining images zero. The fields after the first semicolon are auxiliary.

Let $r : \mathcal{B} \to C$ set every auxiliary coordinate to zero, send U_1 to J and η_1 to $-c$. It is a map of state complexes. Let N count total auxiliary number and let s be the odd polynomial derivation with

$$s(T^m e_j) = T^m a_j \quad (j = 2, 3), \qquad s(T^m p_j) = T^m \iota_j \quad (j = 1, 2, 3),$$

zero on the other generators. For q tensor factors use the graded sum s_q and total number N_q, and put $h_q^{\mathrm{aux}} = s_q / N_q$ on positive auxiliary number, zero on number zero.

Lemma 7.17. *With $i = j_1$, these maps satisfy*

$$d h_q^{\mathrm{aux}} + h_q^{\mathrm{aux}} d = 1 - i^{\otimes q} r^{\otimes q}, \qquad r^{\otimes q} i^{\otimes q} = 1,$$

and $(h_q^{\mathrm{aux}})^2 = r^{\otimes q} h_q^{\mathrm{aux}} = h_q^{\mathrm{aux}} i^{\otimes q} = 0$. They commute with translation in each factor and with graded permutations. Moreover $r\, j_s = 1$ for every slot.

Proof. On generators, $Qs + sQ = N$ and $s^2 = 0$. The graded Leibniz rule extends both identities to all PBW polynomials. Their graded sums give the same identities on tensor powers, since cross terms cancel. Translation preserves each auxiliary pair and its number. Division by the positive integer N_q gives the contraction; on number zero, $i^{\otimes q} r^{\otimes q}$ is the identity. A nonzero value of h_q^{aux} has positive auxiliary number, proving the side conditions. The sums are invariant under graded permutations. Finally all U_s project to J and all $-\eta_s$ to c. □

Write $h = h_1^{\mathrm{aux}}$. The maps r, h are polynomial state maps, not vertex morphisms. For example, auxiliary fields killed by r can have a singular product equal to the vacuum. The collision-coefficient correction will account for that product.

7.6.2 Adjacent paths and all six pair lifts

Let $A_r \subset \mathcal{B}$, for $r = 1, 2$, be generated by

$$U_r, \quad D_r = t_{r+1} - t_r, \quad p_r, p_{r+1}, \eta_r, \eta_{r+1}, \iota_r, \iota_{r+1}.$$

Put $A = \mathrm{Com}(J + p_1 + p_2, V_\kappa \otimes E^{\otimes 2})$. Each A_r is a copy of this two-quartet commutant. The first commutes with $j_3(\mathcal{Z})$: it commutes with $J + p_1 + p_2$ and with the third quartet. The second is obtained by sending J to $J + p_1$ and shifting the two quartet indices; it commutes with $j_1(\mathcal{Z})$. The substitutions preserve the generating brackets, the differential and the PBW basis. Thus multiplication gives vertex maps $A_1 \otimes \mathcal{Z} \to \mathcal{B}$ and $\mathcal{Z} \otimes A_2 \to \mathcal{B}$.

Put polynomial forms $k[s, ds]$ before fields. In A_r define two paths on the generating states by

$$\begin{aligned} F_r(J) &= U_r - s\kappa T D_r, & F_r(c) &= -\eta_r - s(\eta_{r+1} - \eta_r) - ds\, D_r, \\ G_r(J) &= U_r - \kappa T D_r + s p_r + ds\, \iota_r, & G_r(c) &= -\eta_{r+1}. \end{aligned} \tag{7.29}$$

Extend them by the vertex operations. Their images have current self-bracket $\kappa\nu$, zero current–ghost bracket, and closed ghost. The tensor differential gives $dF_r(J) = \kappa T F_r(c)$ and $dG_r(J) = \kappa T G_r(c)$;

the parameter derivatives cancel the differentials of the ds terms. Hence they are vertex chain maps. They commute with the fixed spectator throughout, by the commutant calculation above. Their concatenation runs from j_r to j_{r+1}.

Let h_r be the sum of their integrated ds coefficients. Each polynomial input has a finite polynomial integrand, so integration is defined over k. Stokes' formula gives

$$Qh_r + h_r d = j_{r+1} - j_r, \qquad h_r(J) = \iota_r, \quad h_r(c) = -D_r, \quad h_r(1) = 0. \tag{7.30}$$

The maps commute with translation and satisfy $rh_1 = rh_2 = 0$. Indeed every form coefficient contains an auxiliary factor which survives the polynomial state projection until it is set to zero.

Let $q_I : \mathscr{R}_I \to X^3$ be the ordered atlas of Section 6.1. Put $N_C = \omega_X \otimes C$ with $(f\,a\,dz)\partial_z = -(f'a + fTa)\,dz$, and let $P = q_I^!(N_C \boxtimes N_C \boxtimes N_C)$ be the regular singleton source. For a partition π, let $\mathcal{A}_\pi$ be the external product of $\mathcal{B}^r = \omega_X \otimes \mathcal{B}$ over its block roots, localized where distinct roots are separated and pushed onto the corresponding diagonal. This direct image retains every normal derivative. Order the blocks by their largest labels and transport their right densities with that order. The three-position coefficient is

$$\begin{aligned} \mathcal{D}_3 &= \mathcal{A}_{1|2|3} \oplus \bigoplus_{ab} \mathcal{A}_{ab|c}[-1] \oplus \mathcal{A}_{123}[-2], \\ D(x,y,z) &= (dx, b_1 x - dy, b_2 y + dz). \end{aligned} \tag{7.31}$$

Here b_1, b_2 are full vertex contact maps, including normal transfers and the Koszul signs. On regular states, $b_1 = \varrho_{12} + \varrho_{13} - \varrho_{23}$, where

$$\begin{aligned} \varrho_{12}(X,Y,W) &= [X_\lambda Y] \boxtimes W, \\ \varrho_{13}(X,Y,W) &= (-1)^{|X||Y|} Y \boxtimes [X_\lambda W], \\ \varrho_{23}(X,Y,W) &= X \boxtimes [Y_\mu W]. \end{aligned}$$

The second contacts are $[M_{\lambda+\mu}W]$ on 12|3, $[Y_\mu M]$ on 13|2, and $[X_\lambda M]$ on 23|1. The vertex Jacobi identity gives

$$b_2 b_1(X,Y,W) = [[X_\lambda Y]_{\lambda+\mu} W] + (-1)^{|X||Y|}[Y_\mu [X_\lambda W]] - [X_\lambda [Y_\mu W]] = 0.$$

These contact formulas extend by the differential-module relations and retain their Laurent tests.

For $D_{ab} = q_I^{-1}\Delta_{ab}$ and $D_{123} = q_I^{-1}\Delta_{123}$, including every ordered branch, the support complexes are

$$M_{ab} = R\Gamma_{D_{ab}}\mathcal{D}_3 = \mathcal{A}_{ab|c}[-1] \oplus \mathcal{A}_{123}[-2], \qquad T_3 = R\Gamma_{D_{123}}\mathcal{D}_3 = \mathcal{A}_{123}[-2]. \tag{7.32}$$

To verify this description, apply the localization triangle to the finite filtration by number of blocks. The separated term is a direct image from the complement of D_{ab} and has zero extraordinary restriction there. The other pair terms meet D_{ab} only at the triple diagonal, away from which they are localized. Only the two displayed terms remain. Their inclusion is the support counit; localization agrees on chart overlaps, so the description holds on the atlas.

An order σ assigns its labels to slots 1, 2, 3 and gives a singleton insertion $E_\sigma : P \to \mathcal{D}_3$ through j_1, j_2, j_3. The commuting slot algebras make its pair and triple components zero. For a binary pair set

$$H_r(a,b) = -h_r(a) \boxtimes j_{r+1}(b) + (-1)^{|a|} j_{r+1}(a) \boxtimes h_r(b).$$

Its Hom differential is $j_r \boxtimes j_{r+1} - j_{r+1} \boxtimes j_r$. Use $H_1 \boxtimes j_3$ for slots 1, 2 and $j_1 \boxtimes H_2$ for slots 2, 3, with the graded tensor sign. Relabel source, target, coordinates and densities together; call the resulting map $H_{\sigma,r}$. The six edges are

edge	pair	r	(spectator, slot)
$123 \longrightarrow 213$	12	1	$(3,3)$
$213 \longrightarrow 231$	13	2	$(2,1)$
$231 \longrightarrow 321$	23	1	$(1,3)$
$321 \longrightarrow 312$	12	2	$(3,1)$
$312 \longrightarrow 132$	13	1	$(2,3)$
$132 \longrightarrow 123$	23	2	$(1,1)$.

(7.33)

Both occurrences of 13 exchange adjacent slots; neither moves directly through a fixed middle slot.

Proposition 7.18. *For an edge $e = (\sigma, r)$ exchanging labels a, b, put*

$$\beta_e = -(b_1 H_{\sigma,r})_{ab|c}, \qquad \widehat{\beta}_e = (\beta_e, 0) : P \longrightarrow M_{ab}.$$

This is a cochain map. Its composite with the support counit is homotopic to $E_\sigma - E_{\sigma s_r}$ by $(H_{\sigma,r}, 0, 0)$. It retains the spectator and every normal derivative. It vanishes on a vacuum in a paired input and recovers the binary lift on a vacuum spectator.

Proof. The path equation gives the separated component of the endpoint difference. The image of the paired homotopy lies in A_r, which commutes with the spectator. Thus the two unwanted first contacts vanish. The remaining contact is

$$\beta_r(a,b) = [h_r(a)_v\, j_{r+1}(b)] - (-1)^{|a|}[\, j_{r+1}(a)_v h_r(b)], \tag{7.34}$$

tensored with the spectator and the prescribed sign. Its image lies in A_r, so its second contact with that spectator also vanishes. In the total complex this says

$$(E_\sigma - E_{\sigma s_r}) - \delta(H_{\sigma,r}, 0, 0) = (0, \beta_e, 0).$$

The right side is closed and factors through M_{ab}.

All maps commute with translation and use full contacts, so their differential-module extensions retain the normal derivatives. Vertex locality bounds the normal order on each polynomial input. Relabelling transports the normal coordinate and density once; the same formulas therefore agree on chart overlaps and separation restrictions. Finally $h_r(1) = 0$ and the regular singular bracket with 1 is zero, giving the vacuum assertions. □

7.6.3 The two-colour coefficient and its endpoints

For free dg vertex algebras D, E, put

$$V(D,E) = j_* j^!(D^r \boxtimes E^r), \qquad W(D,E) = \Delta_*(D \otimes E)^r,$$

where j is the complement of the diagonal in the two-root space and $D^r = \omega_X \otimes D$. If v is the first root minus the second, the tensor-product contact is

$$c_{DE}\big(f(v)(a \boxtimes b)\big) = \mathrm{pp}_v\big(f(v)(e^{vT_D}a) \otimes b\big). \tag{7.35}$$

Only finitely many Taylor terms meet any fixed principal part. The diagonal term retains the whole tensor $D \otimes E$. This gives the two-colour complex

$$Q(D,E) = V(D,E) \oplus W(D,E)[-1], \qquad D(x,z) = (dx, c_{DE}x - dz).$$

Translation makes the contact differential-linear, and the internal differential commutes with it.

For the resolving algebra itself let $W(\mathcal{B}) = \Delta_* \mathcal{B}^r$ and put

$$\mathcal{F}_{\mathcal{B}} = V(\mathcal{B}, \mathcal{B}) \oplus W(\mathcal{B})[-1], \qquad D(x, z) = (dx, b_{\mathcal{B}} x - dz),$$

with its full vertex contact. Denote the positive diagonal inclusion by ι.

There are two orientations: R for $(A, \mathcal{Z})$ and L for $(\mathcal{Z}, A)$. Write $\alpha : A \to A_1$ and $\gamma : A \to A_2$ for the two embeddings. Their placements in the common resolution are

	$u_+\|_V$	$u_-\|_V$
R	$\alpha \boxtimes j_3$	$\gamma \boxtimes j_1$
L	$j_1 \boxtimes \gamma$	$j_3 \boxtimes \alpha$.

(7.36)

On the diagonal use the corresponding normally ordered products. The images commute, so these are vertex morphisms and hence coefficient chain maps to $\mathcal{F}_{\mathcal{B}}$.

To use them at a pair ab, apply right differential-module direct image along $i_{ab} : X^{\{ab,c\}} \hookrightarrow X^3$. It includes the entire inner normal transfer and its density. The subsequent diagonal direct image is $i_{123,+}$. After the pair-support suspension, the placements have target M_{ab}, and ι becomes the counit $T_3 \to M_{ab}$.

Let $r_A : A \to C$ be the two-quartet polynomial retraction. Projection in the A factor gives maps $P_R = r_A \otimes 1$ and $P_L = 1 \otimes r_A$, with components P_V, P_W. Since r_A commutes with translation,

$$P_W c_\eta = c_{\mathcal{Z}\mathcal{Z}} P_V, \qquad \eta = R, L. \tag{7.37}$$

This uses the tensor-product contact, not multiplicativity of r_A as a vertex map. The first-slot inclusion and the binary auxiliary homotopy give a contraction I, P, H_A of these coloured complexes. The homotopy on the diagonal has the negative of its state sign; equation (7.35) makes the off-diagonal terms cancel, so $\partial H_A = 1 - IP$.

On $V(\mathcal{Z}, \mathcal{Z})$ define

$$H_1 = -h_1 \boxtimes j_2 + j_2 \boxtimes h_1, \qquad K_R = H_1 + j_1 \boxtimes h_2, \qquad K_L = H_1 - h_2 \boxtimes j_1.$$

These tensor formulas include the graded signs. Extend them over every finite meromorphic coefficient and make them zero on the diagonal source. Their internal differentials are $j_1 \boxtimes j_3 - j_2 \boxtimes j_1$ and $j_1 \boxtimes j_2 - j_3 \boxtimes j_1$, respectively. With $f_{st}(a, b) = :j_s a\, j_t b:$, put

$$F_R = f_{13} - f_{21}, \qquad F_L = f_{12} - f_{31}, \qquad b_\eta = -b_{\mathcal{B}} K_\eta. \tag{7.38}$$

Here b_η has internal degree -1, while F_η has degree zero on the diagonal coefficient.

Lemma 7.19. *The full supported map is*

$$\mathbb{B}^\circ_\eta(x, z) = b_\eta P_V x + F_\eta P_W z : Q_\eta \longrightarrow W(\mathcal{B})[-1].$$

It is closed. If $E_A = u_+ - u_-$, then

$$E_A - \iota \mathbb{B}^\circ_\eta = \partial \mathbb{K}_A, \qquad \mathbb{K}_A = K_\eta P + E_A H_A. \tag{7.39}$$

Proof. The path equation and the chain equations of the placements give, in unshifted state degrees,

$$\delta b_\eta = -F_\eta c_{ZZ}, \qquad \delta F_\eta = 0. \tag{7.40}$$

Together with (7.37), these are the two components of closure into $W(\mathcal{B})[-1]$. On the two-singleton source, subtracting δK_η from the endpoint difference leaves exactly its supported components $b_\eta x + F_\eta z$. On arbitrary auxiliary states, use $E_A I$ for that singleton difference and $1 - IP = \delta H_A$. This gives (7.39), including the term $E_A H_A$. □

The state projection has two useful identities:

$$r F_R = r F_L = 0, \qquad (r \boxtimes r) u_+|_V = (r \boxtimes r) u_-|_V = P_V. \tag{7.41}$$

Indeed each placement is linear on free generators, and the normally ordered product of its two commuting images is their concatenated PBW polynomial. Both endpoints have the same polynomial projection. This does not identify that polynomial product with the vertex (-1)-product in C. The second equality follows from $r\alpha = r\gamma = r_A$ and $r j_s = 1$. Translation extends both to the complete coefficient modules.

Define

$$s_\eta = h F_\eta, \qquad N_\eta = b_\eta + s_\eta c_{ZZ}. \tag{7.42}$$

The auxiliary contraction and (7.41) give $\delta s_\eta = F_\eta$; thus (7.40) gives $\delta N_\eta = 0$. On regular singleton inputs the tensor-product contact vanishes. The additional h_2 term in K_η has zero regular contact with j_1, by the commutant property. Hence $N_R = N_L = \beta_1$ there. On pole coefficients those terms and the correction remain part of the operation.

7.6.4 The outer map on the actual support

Use the suspended coloured complex

$$\widetilde{Q}_\eta = V_\eta[-1] \oplus W_\eta[-2], \qquad D(y, z) = (-dy, c_\eta y + dz). \tag{7.43}$$

It is $Q_\eta[-1]$ with its diagonal coordinate multiplied by -1. Under the positive support counit, the coloured lift becomes

$$\mathbb{B}_\eta(y, z) = -b_\eta P_V y + F_\eta P_W z.$$

Theorem 7.20 (Outer extension)**.** *On the complete pair support define*

$$C_\eta(y, z) = -N_\eta(r \boxtimes r) y, \qquad L_\eta(y, z) = -s_\eta P_W z. \tag{7.44}$$

The first is a degree-zero map $M_{ab} \to T_3$ and the second has total degree -1 on the coloured source. Both placements satisfy

$$\delta C_\eta = 0, \qquad C_\eta u_\pm - \mathbb{B}_\eta = \delta L_\eta. \tag{7.45}$$

Use R for $12|3$ and L for $23|1, 13|2$, with the stated root orders and normal direct images. If $\mathbb{K}_A^{\mathrm{supp}}$ is the suspended homotopy of (7.39), then

$$E_A - \iota C_\eta u_\pm = \delta(\mathbb{K}_A^{\mathrm{supp}} - \iota L_\eta).$$

Thus the extension retains the prescribed endpoint products.

Proof. The internal degree of N_η is -1; the difference between the pair and triple support shifts makes C_η degree zero. It kills the diagonal component of M_{ab}, so its chain equation is precisely $\partial N_\eta = 0$. This proves closure on arbitrary polynomial auxiliary states and scalar poles.

On either placement, equations (7.37) and (7.41) give the difference

$$-s_\eta P_W c_\eta y - F_\eta P_W z.$$

The signs $(-,+)$ in (7.43) and the odd degree of L_η give

$$\partial L_\eta(y,z) = -s_\eta P_W c_\eta y - (Q s_\eta + s_\eta d) P_W z,$$

which is that difference. The final identity follows by subtracting $\iota \partial L_\eta$ from the suspended endpoint identity. Every equality is on the complete coloured source, including its diagonal term. □

Neither placement needs to be inverted. In fact it is not an equivalence. On diagonal state cohomology its map is

$$\Lambda(\alpha,\beta) \longrightarrow \Lambda(\varepsilon), \qquad \alpha,\beta \longmapsto \varepsilon.$$

To compute these groups, contract the pairs $(u_n, \kappa c_{n+1})$ in (7.27); only c_0 remains. The auxiliary contraction gives the same cohomology for each resolving algebra. The displayed diagonal map has kernel containing $\alpha - \beta$ and $\alpha\beta$. Its separated part is a quasi-isomorphism, so the triangle of the two-term coefficients identifies its cone with the shifted cone of this diagonal map. Thus an inverse-placement method would fail, while (7.44) supplies the outer extension directly.

The pole correction has a small explicit diagnostic. In the R orientation put

$$a = \iota_1 + \iota_2, \quad p = p_1 + p_2, \quad t = T(t_3 - t_1), \quad e = T(\eta_3 - \eta_1).$$

For $x_J = v^{-1}(1 \boxtimes J)$ and $x_{Tc} = v^{-1}(1 \boxtimes Tc)$, let z_J, z_{Tc} denote the corresponding diagonal transfers. The defining formulas give

$$Dx_J = \kappa x_{Tc} + z_J, \quad Dx_{Tc} = z_{Tc}, \quad Dz_J = -\kappa z_{Tc},$$
$$b_R x_J = -a, \quad b_R x_{Tc} = t, \quad F_R z_J = p - \kappa t, \quad F_R z_{Tc} = -e.$$

Since $h(p - \kappa t) = a$ and $h(-e) = -t$, the corrected values are $N_R x_J = -a + a = 0$ and $N_R x_{Tc} = t - t = 0$. Omitting the correction leaves chain defect $-(p - \kappa t)$ on x_J; reversing its sign leaves $-2(p - \kappa t)$. These are nonzero states. They explain the correction on poles without asserting an obstruction to the corrected map.

7.7 The triple homotopy and its coefficient transfer

The six pair lifts have two occurrences of each labelled pair. A cube filling compares these occurrences. Its triple contact currents sum to zero, whereas the actual iterated outer maps need a further comparison. We construct both maps before identifying their difference as a boundary.

7.7.1 The nine-square filling

Let I be the cellular chain complex of the interval $[1,3]$, with vertices v_1, v_2, v_3 in degree zero, edges e_1, e_2 in degree one and $\partial e_r = v_{r+1} - v_r$. A cell of $I^{\otimes 3}$ assigns a height or a moving edge to each label, in label order. Define

$$\Gamma(c_1 \otimes c_2 \otimes c_3) = \phi(c_1) \boxtimes \phi(c_2) \boxtimes \phi(c_3), \qquad \phi(v_s) = j_s, \quad \phi(e_r) = h_r.$$

Its target is the separated term of (7.31); a cell of dimension d gives an operation of degree $-d$. The graded tensor formula and (7.30) give $\delta\Gamma(c) = \Gamma(\partial c)$ for the internal differential.

For an edge $\sigma \to \sigma s_r$, let x, y be its two exchanged labels and z the spectator at height q. The corresponding chain is

$$c_{\sigma,r} = -(x : e_r,\, y : v_{r+1},\, z : v_q) + (x : v_{r+1},\, y : e_r,\, z : v_q),$$

with factors put in label order. Its boundary is the first permutation vertex minus the second, and $\Gamma(c_{\sigma,r}) = H_{\sigma,r}$. The sum around the hexagon is the boundary of

$$\Sigma = \sum_{i=1}^{3} (-1)^i \big((e_1, e_2)_i + (e_2, e_1)_i + (e_2, e_2)_i \big), \tag{7.46}$$

where $(c, c')_i$ fixes label i at v_3 and assigns c, c' to the other two labels in increasing order. To verify this, the hexagon chain has twelve edges: it passes through the six permutation vertices and through

$$(2,2,3),\ (3,1,3),\ (3,2,2),\ (3,3,1),\ (2,3,2),\ (1,3,3).$$

On each of the three faces through $(3,3,3)$, it removes the square at the opposite $(1,1)$ corner. The remaining three squares are those in (7.46). Their twelve interior edges cancel: two within each face and two along each of the three shared edges. The exterior twelve edges have precisely the hexagon orientation.

Put $\Phi = \Gamma(\Sigma)$ and define its contact maps

$$K_{ab} = (b_1\Phi)_{ab|c}, \qquad \gamma_{ab} = -b_2 K_{ab}. \tag{7.47}$$

Their internal state degree is -2. Thus $(0, K_{ab}, 0)$ has total degree -1 and $(0, 0, \gamma_{ab})$ has total degree zero in (7.31).

Proposition 7.21. *If e_+, e_- are the two hexagon occurrences of the pair ab, then*

$$\widehat{\beta}_{e_+} + \widehat{\beta}_{e_-} = \delta(0, K_{ab}, 0) + (0, 0, \gamma_{ab}), \qquad \delta\gamma_{ab} = 0, \tag{7.48}$$

and $\gamma_{12} + \gamma_{23} + \gamma_{13} = 0$. The whole supported hexagon bounds:

$$\sum_e (H_e, 0, 0) + \sum_{ab} (0, K_{ab}, 0) = \delta(\Phi, 0, 0).$$

Proof. The cube boundary gives $\delta(\Phi, 0, 0) = (\sum_e H_e, b_1\Phi, 0)$. Differentiate it. The endpoint differences telescope, and Proposition 7.18 identifies the remaining terms as the six supported lifts. Comparing each pair component gives $\beta_{e_+} + \beta_{e_-} = -\delta K_{ab}$ internally. Its triple component gives the first identity with $\gamma_{ab} = -b_2 K_{ab}$. Differentiating that identity shows γ_{ab} is closed. Finally $\sum \gamma_{ab} = -b_2 b_1 \Phi = 0$ by vertex Jacobi. □

For the comparison with iterated brackets, set $g = h_1 + h_2$ and define the merged-pair transport

$$k(a,b) = (-1)^{|a|}\big([h_1 a_\nu h_2 b] + [h_2 a_\nu h_1 b] + [h_2 a_\nu h_2 b] \big).$$

Its internal differential is $\delta k = \beta_1 - \beta_2$. This follows by differentiating the two homotopies; the contacts of h_1 with j_3 and of h_2 with j_1 vanish by the commutant property. Taking the contacts of the nine squares gives

$$\begin{aligned}
K_{12}(a,b,c) &= -k(a,b) \boxtimes j_3 c + (-1)^{|a|+|b|} \beta_2(a,b) \boxtimes g c, \\
K_{23}(a,b,c) &= (-1)^{|a|} g a \boxtimes \beta_2(b,c) + j_3 a \boxtimes k(b,c), \\
K_{13}(a,b,c) &= (-1)^{|b|(|a|+1)} g b \boxtimes \beta_2(a,c) + (-1)^{|a||b|} j_3 b \boxtimes k(a,c).
\end{aligned} \tag{7.49}$$

For the first formula, the face fixing the spectator at v_3 contributes the three terms of $-k$. On the other two faces the spectator moves along g; the h_1 contacts with j_3 vanish and the rest give β_2. The same calculation gives the second formula. Moving label 2 in front of label 1 gives the third formula and its Koszul signs.

7.7.2 The three actual compositions

Define the bracket on states by

$$B_\nu(a,b) = (-1)^{|a|} r\beta_1(a,b), \qquad x(a,b) = r\beta_1(a,b).$$

It has degree -1 and is the operation fixed by the two polynomial paths and the PBW retraction. Its three channels are

$$\begin{aligned} L(a,b,c) &= B_{\lambda+\mu}(B_\lambda(a,b),c), \\ R_B(a,b,c) &= B_\lambda(a,B_\mu(b,c)), \\ N_B(a,b,c) &= B_\mu(b,B_\lambda(a,c)), \qquad \epsilon = (-1)^{(|a|+1)(|b|+1)}. \end{aligned} \tag{7.50}$$

The Fourier variables belong to positions 1, 2 relative to position 3. Sesquilinearity sends their translations to $T_1 = -\lambda$, $T_2 = -\mu$, $T_3 = T + \lambda + \mu$.

Choose the edges $e_{12} : 123 \to 213$, $e_{23} : 231 \to 321$ and $e_{13} : 213 \to 231$. Their placements are respectively $u_{R,+}$, $u_{L,-}$ and $u_{L,+}$. Define literal compositions through the pair supports by

$$G_{12} = C_R\widehat{\beta}_{e_{12}}, \qquad G_{23} = -C_L\widehat{\beta}_{e_{23}}, \qquad G_{13} = C_L\widehat{\beta}_{e_{13}}. \tag{7.51}$$

Their source remains P. To compare with the coloured map, let $a_e : P \to \widetilde{Q}_\eta$ have the indicated binary lift in A tensored with the spectator, and zero diagonal component. Its outer scalar coefficient is regular, so $c_\eta a_e = 0$, including after any inner normal transfer. It is therefore closed, $u_e a_e = \widehat{\beta}_e$ and $L_\eta a_e = 0$. Equation (7.45) becomes

$$C_\eta\widehat{\beta}_e = \mathbb{B}_\eta a_e.$$

This is equality of operations on the whole regular singleton source.

Projection of the preferred pair components gives

pair	$(r \boxtimes r)\beta_e(a,b,c)$	outer variable
12	$x(a,b) \boxtimes c$	$\lambda + \mu$
23	$-(-1)^{\lvert a\rvert} a \boxtimes x(b,c)$	λ
13	$(-1)^{\lvert a\rvert\lvert b\rvert+\lvert b\rvert} b \boxtimes x(a,c)$	μ.

The minus sign in the second row is its first-contact sign; the third row includes moving the spectator to the first root. Since $N_R = N_L = \beta_1$ on a regular outer coefficient,

$$\begin{aligned} G_{12}(a,b,c) &= -\beta_1(x(a,b),c), \\ G_{23}(a,b,c) &= -(-1)^{|a|}\beta_1(a,x(b,c)), \\ G_{13}(a,b,c) &= -(-1)^{|a||b|+|b|}\beta_1(b,x(a,c)). \end{aligned} \tag{7.52}$$

The degree of $x(a,b)$ is $|a| + |b| - 1$, so

$$rG_{12} = (-1)^{|b|}L, \qquad rG_{23} = -(-1)^{|b|}R_B, \qquad rG_{13} = (-1)^{|b|}\epsilon N_B. \tag{7.53}$$

For example, the first sign is $-(-1)^{|a|+|b|-1}(-1)^{|a|} = (-1)^{|b|}$; the third uses $-(-1)^{|a||b|+|a|} = (-1)^{|b|}\epsilon$.

7.7.3 Comparing cube contacts with the bracket

The comparison uses a polynomial projection π retaining U_1, η_1, p_1, ι_1 and killing every other auxiliary coordinate and its translates.

Lemma 7.22. *For $Z \in \mathcal{B}$ and $a, b, y \in C$,*

$$r[Z_\nu j_1 y] = r[\pi Z_\nu j_1 y], \qquad r[j_1 y_\nu Z] = r[j_1 y_\nu \pi Z],$$
$$\pi h = h\pi, \qquad \pi Y = \pi j_2(rY) \quad (Y \in A_2), \qquad \pi h j_2 = \pi h_1,$$
$$r[h_1 a_\nu h_1 b] = 0,$$
$$r\beta_1(a,b) = r[h_1 a_\nu j_1 b] - (-1)^{|a|} r[j_1 a_\nu h_1 b].$$

Proof. Only U_1, p_1 contract with U_1, and only ι_1 contracts with η_1. A coordinate killed by π thus survives every contraction with $j_1 y$ and is killed by r. This proves the first two equalities. The auxiliary reverse derivation carries the killed coordinates among themselves, and their absence leaves auxiliary number unchanged, proving $\pi h = h\pi$. On the generators of A_2, both π and $\pi j_2 r$ send U_2 to $U_1 + p_1$, send the ghosts to η_1, and kill the remaining auxiliary fields. Hence they agree on all PBW polynomials.

For $\pi h j_2 = \pi h_1$, the first path contributes a factor of a_2, which π kills. In the second path, a monomial with m current factors choosing p_1 has form coefficient obtained by replacing one of them by ι_1 and weighting the other $m-1$ by s^{m-1}. Integration gives $1/m$, exactly the auxiliary contraction of the same monomial in j_2. The graded sign of the odd replacement is the same.

To prove $r[h_1 a_\nu h_1 b] = 0$, split $h_1 = h^F + h^G$ by the two paths. Every monomial of h^F contains one translate of $D = t_2 - t_1$; every monomial of h^G contains one translate of ι_1. Two F terms leave both D factors uncontracted, and two G terms leave both ι_1 factors. Their projections vanish. For a mixed term, use the generating polynomials represented coefficientwise by

$$a = \exp\Big(\sum_n \alpha_n u_n + \sum_n \theta_n c_n\Big), \quad b = \exp\Big(\sum_n \alpha'_n u_n + \sum_n \theta'_n c_n\Big),$$

where α, α' are even and θ, θ' odd. Only finitely many parameters are needed for any chosen coefficient. If A_s, B_t are the zero-form exponents of the two paths, their form coefficients are

$$h^F a = \int_0^1 \Big(\sum_n \theta_n T^n D\Big) e^{A_s}\, ds, \quad h^G b = \int_0^1 \Big(\sum_m \alpha'_m T^m \iota_1\Big) e^{B_t}\, dt.$$

For the projection to survive, D must contract with a $t\,p_1$ factor of B_t and ι_1 with an $-(1-s)\eta_1$ factor of A_s. Since $[D_\nu p_1] = [\eta_{1\nu} \iota_1] = 1$, both contractions have jet kernel

$$P_{nm}(z-w) = (-1)^n (n+m)!\,(z-w)^{-n-m-1}.$$

Their product is $-t(1-s)$ times the square of the odd element $\sum_{n,m} \theta_n \alpha'_m P_{nm}$, hence zero. All other contractions multiply this zero. Reversing the two paths gives the same argument with θ'. Taking polynomial coefficients proves the identity on all states.

Finally, the contact of $h_1 \otimes h_1$ followed by r is zero. Its differential, using $\delta h_1 = j_2 - j_1$, is the difference between $r\beta_1$ and the last displayed bracket expression. That difference is therefore zero, proving the final formula. □

Put

$$T_{ab} = -r b_2(h \text{ on the merged factor}) K_{ab}, \qquad \widehat{H} = -(T_{12} + T_{23} + T_{13}). \tag{7.54}$$

These maps have internal degree -3. The operation on the merged factor uses its graded tensor sign.

Proposition 7.23. *On all polynomial states,*

$$r\gamma_{12} = (-1)^{|b|}L + \delta T_{12}, \qquad r\gamma_{23} = -(-1)^{|b|}R_B + \delta T_{23}, \qquad r\gamma_{13} = (-1)^{|b|}\epsilon N_B + \delta T_{13}.$$

Thus $\delta\widehat{H} = (-1)^{|b|}(L - R_B + \epsilon N_B)$.

Proof. For 12, use $\delta(h \otimes 1) = (1 - j_1 r) \otimes 1$ and the fact that rb_2 commutes with the internal differential. Equation (7.48) gives

$$r\gamma_{12} - \delta T_{12} = -rb_2(j_1 r \otimes 1)K_{12} + rb_2(h \otimes 1)(\beta_{e_+} + \beta_{e_-}).$$

Here $\beta_{e_+} + \beta_{e_-} = \beta_1 \otimes j_3 - \beta_2 \otimes j_1$, and $r\beta_1 = r\beta_2 = x$, since the slot shift has the same polynomial projection in both binary copies. In the first term, the part with j_3 has zero contact with j_1; in the part with g, only h_1 remains. In the second term, $h\beta_1 \in A_1$ commutes with j_3. Lemma 7.22 replaces the remaining contact of $h\beta_2$ with j_1 by that of $h_1 x$. The result is

$$-rb_2(j_1 x \otimes h_1 + h_1 x \otimes j_1) = -r\beta_1 \circ (x \otimes 1) = (-1)^{|b|}L.$$

For 23, the tensor formulas are $K_{23} = g \otimes \beta_2 + j_3 \otimes k$ and $\beta_{e_+} + \beta_{e_-} = j_1 \otimes \beta_2 - j_3 \otimes \beta_1$. Apply $1 \otimes h$ to the merged factor. The same commutant vanishings and reduction give

$$r\gamma_{23} - \delta T_{23} = rb_2(j_1 \otimes h_1 x - h_1 \otimes j_1 x) = -r\beta_1 \circ (1 \otimes x) = -(-1)^{|b|}R_B.$$

The tensor $1 \otimes x$ contributes $(-1)^{|a|}$. For 13, transpose a, b and λ, μ with input sign $(-1)^{|a||b|}$. This is a chain map. The result is $-(-1)^{|a||b|+|a|}N_B = (-1)^{|b|}\epsilon N_B$. Summing the three identities and using $\sum \gamma_{ab} = 0$ proves the assertion about $\widehat{H}$. □

Theorem 7.24 (Supported triple homotopy)**.** *The maps*

$$U_{ab} = j_1 T_{ab} + h(\gamma_{ab} - G_{ab}) \tag{7.55}$$

satisfy $\delta U_{ab} = \gamma_{ab} - G_{ab}$. *Consequently*

$$\mathcal{H} = -\sum_{ab} U_{ab} = j_1\widehat{H} + h(G_{12} + G_{23} + G_{13}) \tag{7.56}$$

has total degree -1 *and*

$$\delta\mathcal{H} = G_{12} + G_{23} + G_{13}, \qquad r\mathcal{H} = \widehat{H}.$$

These are maps into the full triple-supported coefficient on every ordered branch, with all normal derivatives retained.

Proof. Both γ_{ab} and G_{ab} are closed. Their projection difference is δT_{ab} by (7.53) and Proposition 7.23. The auxiliary contraction gives

$$\delta\big(h(\gamma_{ab} - G_{ab})\big) = \gamma_{ab} - G_{ab} - j_1\delta T_{ab}.$$

Adding $\delta(j_1 T_{ab})$ proves the first equation. The sum of the cube contacts is zero, so summation gives (7.56) and its differential. The side condition $rh = 0$ gives its projection. Every constituent commutes with translation and extends over the full normal transfer; all are polynomial on each input. Thus these are identities of supported differential-module operations, rather than only identities of state values. □

Changing a preferred occurrence of a pair changes its channel by a specified boundary. Applying the closed outer map to (7.48), whose triple component it kills, gives

$$C_{ab}\widehat{\beta}_{e_+} + C_{ab}\widehat{\beta}_{e_-} = \delta\big(C_{ab}(0, K_{ab}, 0)\big).$$

Change $\mathcal{H}$ by the same signed map. Changes for different pairs add and therefore commute. This retains all eight choices of one occurrence per pair, including both 13|2 edges.

7.7.4 *Normal transfers, substitutions and vacua*

The diagram just constructed has objects P, the coloured complexes $\widetilde{Q}_\eta$, the three pair supports M_{ab}, the triple support T_3 and $\mathcal{D}_3$. Its arrows include the first lifts a_e, both placements, the outer maps and the support counits. Equation (7.45) supplies the placement homotopy, and Theorem 7.24 supplies its three-channel comparison.

If a specified collision substitution gives a chain map $v : D' \to \widetilde{Q}_\eta$, precomposition gives

$$C_\eta u_\pm v - \mathbb{B}_\eta v = \delta(L_\eta v).$$

For a further substitution w, the homotopies agree literally: $(L_\eta v)w = L_\eta(vw)$. At three positions the maps in this diagram are the vertex placements and the first lifts already constructed. This statement does not assume an unspecified system of higher substitution maps.

Normal differentiation acts on coefficients as well as states. With ν the first-root Fourier variable and T output translation, the corrected operation satisfies, for every integer pole order m,

$$\begin{aligned} N_\eta(Ta, b; v^{-m}) &= -\nu N_\eta(a, b; v^{-m}) + m N_\eta(a, b; v^{-m-1}), \\ N_\eta(a, Tb; v^{-m}) &= (T + \nu) N_\eta(a, b; v^{-m}) - m N_\eta(a, b; v^{-m-1}). \end{aligned} \tag{7.57}$$

Differentiate $v^{-m} Y(a, v) b$ and then take principal parts to obtain these identities for the contact part. The same calculation with (7.35) gives them for the correction. The initial closed embedding retains the inner collision, the two-term coloured differential retains the outer one, and the triple comparison retains their intersection. Open restrictions localize these identities. Relabelling transports states, normals and densities together. Different ordered branches of the triple diagonal have no overlap, so the formulas glue branchwise.

On a pole coefficient with either colour equal to the vacuum, $r b_\eta = 0$: the contact is a principal part of a translate of $h_s(a)$, and $r h_s = 0$. Also $r s_\eta = 0$. Thus N_η is closed with zero projection on that face, and $h N_\eta$ is its explicit null-homotopy. The canonical outer map has the negative of this homotopy. On the common all-vacuum face both are zero. The endpoint-relative homotopy remains the one of Theorem 7.20.

For the ternary maps, a vacuum in the paired inputs makes the first lift zero. A vacuum spectator makes the regular outer bracket zero. Hence every G_{ab} vanishes on a vacuum input. The formulas for K_{ab} show that each γ_{ab} and T_{ab} also vanishes there, so U_{ab} and $\mathcal{H}$ do as well. A vacuum spectator still leaves $K_{12}(a, b, 1) = -k(a, b) \boxtimes 1$; its outer contact is zero, while its lower binary transport is retained.

7.7.5 *Removing the auxiliary fields on coefficients*

The state retraction must be corrected before it acts on collision coefficients. For a finite set I of n positions, write $\mathcal{D}_{A,I} = C_{A,I}[-n]$, for $A = \mathcal{B}$ or $\mathcal{Z}$. A partition with q blocks has term $\mathcal{A}_\pi[q - n]$. Split its differential as

$$D_A = D_{0,A} + b_A, \qquad D_{0,A}|_\pi = (-1)^{n-q} d_q,$$

where b_A merges blocks by full contacts. Extend $i^{\otimes q}$, $r^{\otimes q}$, h_q^{aux} over the scalar coefficients and normal direct images. On that partition put

$$I_\pi = i^{\otimes q}, \qquad P_\pi = r^{\otimes q}, \qquad H_\pi = (-1)^{n-q} h_q^{\mathrm{aux}}.$$

Then $D_{0,\mathcal{B}} H + H D_{0,\mathcal{B}} = 1 - I P$. The inclusion is a vertex map, so $b_\mathcal{B} I = I b_\mathcal{Z}$. Define finite operators

$$R_I = \sum_{j=0}^{n-1} (-b_\mathcal{B} H)^j, \qquad \Pi_I = P R_I, \qquad \mathcal{H}_I = H R_I. \tag{7.58}$$

Every contact lowers the block number. Thus $(b_{\mathcal{B}}H)^n = 0$ and $R_I = (1 + b_{\mathcal{B}}H)^{-1}$.

Proposition 7.25. *The maps in* (7.58) *give a contraction of the actual coefficient complexes:*

$$\Pi_I D_{\mathcal{B}} = D_{\mathcal{Z}}\Pi_I, \qquad \Pi_I I = 1, \qquad D_{\mathcal{B}}\mathcal{H}_I + \mathcal{H}_I D_{\mathcal{B}} = 1 - I\Pi_I,$$

with $\mathcal{H}_I^2 = \Pi_I\mathcal{H}_I = \mathcal{H}_I I = 0$. *They preserve every fixed collision support and its counit, commute with relabelling and open restriction, and preserve the scalar vacuum maps. Their values have finite normal order on each polynomial input.*

Proof. Split the internal complex as $I\mathcal{D}_{\mathcal{Z},I} \oplus K$, where $K = \ker P$. On K let the internal differential be d_K and the homotopy be h_K. Then $d_K h_K + h_K d_K = 1$, $h_K^2 = 0$. Since I preserves contacts, the full differential has the block form

$$D_{\mathcal{B}} = \begin{pmatrix} D_{\mathcal{Z}} & a \\ 0 & D_K \end{pmatrix}, \qquad D_K = d_K + b_K, \qquad D_{\mathcal{Z}}a + aD_K = 0.$$

The contact identities give $b_K^2 = 0$ and $d_K b_K + b_K d_K = 0$. Put $A_K = 1 + b_K h_K$. Direct multiplication gives $D_K A_K = A_K d_K$, since their difference is $b_K(1 - d_K h_K - h_K d_K) = 0$. Therefore $h'_K = h_K A_K^{-1}$ satisfies $D_K h'_K + h'_K D_K = 1$. Also $A_K^{-1} h_K = h_K$, so $(h'_K)^2 = 0$.

The displayed finite formulas are exactly

$$\Pi_I(z,k) = z - a h'_K k, \qquad \mathcal{H}_I(z,k) = (0, h'_K k).$$

The asserted equations follow by matrix multiplication, using $D_{\mathcal{Z}}a + aD_K = 0$. In particular the target differential is the original one with its vertex contacts.

The state maps commute with translation in each block and are symmetric under graded permutations. The contacts have the same equivariance with their Chevalley signs. Thus every term agrees on ordered chart overlaps and open restrictions. A partition lies in a fixed collision support when it is coarser than its defining partition. State maps preserve partitions and contacts only coarsen them, proving support compatibility. The localization description of (7.32) identifies its inclusion with the counit. The vacuum map factors through I, on which H vanishes. Finally there are at most $n - 1$ contacts, each of finite normal order on a polynomial input. □

At three positions, $\Pi_3 = P - PbH + PbHbH$. For $w = \iota_1 \boxtimes \eta_1 \boxtimes 1$ and $x = p_1 \boxtimes \eta_1 \boxtimes 1$, the differential is $D_{\mathcal{B}}w = x + v_{\mathcal{B}}$, where $v_{\mathcal{B}}$ is the simple pair-contact vacuum. One has $Hx = w$, $Hw = Hv_{\mathcal{B}} = 0$, and $Pw = Px = 0$, $Pv_{\mathcal{B}} = v_{\mathcal{Z}}$. Consequently $\Pi_3(x) = -v_{\mathcal{Z}}$ and $\Pi_3(v_{\mathcal{B}}) = v_{\mathcal{Z}}$. The uncorrected projection has defect $v_{\mathcal{Z}} \neq 0$ on w.

On a pair support the formula becomes

$$\Pi_M(y,z) = \big((r \boxtimes r)\,y,\, rz + rb_2 h_2^{\mathrm{aux}} y\big). \tag{7.59}$$

There is at most one contact, and the pair homotopy has sign $-h_2^{\mathrm{aux}}$, giving the plus sign. On the triple support, $\Pi_T = r$ and $I_T = j_1$. Let I_M be its pair inclusion and put

$$\beta_e^C = \Pi_M \widehat{\beta}_e, \qquad C_{ab}^C = \Pi_T C_{ab} I_M.$$

Then

$$\Pi_T C_{ab}\widehat{\beta}_e = C_{ab}^C \beta_e^C, \qquad \Pi_T\mathcal{H} = \widehat{H}. \tag{7.60}$$

Indeed C_{ab} kills the triple component of the pair homotopy, and $(r \boxtimes r)h_2^{\mathrm{aux}} = 0$. Hence $C_{ab}\mathcal{H}_M = 0$. Apply the closed C_{ab} to $1 - I_M\Pi_M = \partial\mathcal{H}_M$ to obtain $C_{ab} = C_{ab}I_M\Pi_M$. This proves the strict composition identity. The pair-to-triple term of (7.59) still occurs in each transferred first lift and its endpoint homotopy.

7.7.6 *A nonzero state Jacobiator*

The homotopy is necessary even though the cube-contact sum vanishes. Write $u = u_0$, $c = c_0$. The bracket fixed above satisfies

$$\begin{aligned} B_\nu(u^2, cTc) &= -2uTc + 2\nu uc + 2(Tu)c, \\ B_\nu(u^2, uc) &= -2u^2 - \tfrac{\kappa}{2}\nu^2, \\ B_\nu(u^2, uTc) &= -2\nu u^2 - 2uTu - \tfrac{\kappa}{6}\nu^3, \\ B_\nu(u^2, (Tu)c) &= -2uTu - \tfrac{\kappa}{3}\nu^3, \qquad B_\nu(u^2, u^2) = 0. \end{aligned} \tag{7.61}$$

For the scalar terms, the parts of $h_1(uc)$, $h_1(uTc)$ and $h_1((Tu)c)$ without ι_1 are respectively

$$-{:}U_1D_1{:} + \tfrac{\kappa}{2}{:}(TD_1)D_1{:}, \quad -{:}U_1TD_1{:} + \tfrac{\kappa}{2}{:}(TD_1)^2{:}, \quad -{:}(TU_1)D_1{:} + \tfrac{\kappa}{2}{:}(T^2D_1)D_1{:}.$$

The factor $1/2$ is the integral of the first path parameter. Every omitted term contains ι_1, which cannot contract with the second embedding and is killed by r. The contractions of U_2 with D_1, TD_1, T^2D_1 are $-v^{-1}, -v^{-2}, -2v^{-3}$. Their double contractions with U_2^2, before the bracket's minus sign, are $\kappa v^{-3}, \kappa v^{-4}, 2\kappa v^{-4}$. Converting v^{-m-1} to $\nu^m/m!$ gives the three scalar coefficients. Single contractions and the Taylor terms give the displayed state terms. The two ordered ghost contractions give the first line, and every term in the last bracket retains an auxiliary field.

Substitution in the three channels gives

$$L(u^2, u^2, cTc) - R_B(u^2, u^2, cTc) + \epsilon N_B(u^2, u^2, cTc) = \frac{\kappa}{3}(\lambda + \mu)^3 \mathbf{1}.$$

The state terms cancel and $\epsilon = -1$. Set $H(a, b, c) = (-1)^{|b|}\widehat{H}(a, b, c)$. The ordinary Hom equation becomes

$$\begin{aligned} L - R_B + \epsilon N_B = {} & dH(a, b, c) + H(da, b, c) \\ & - (-1)^{|a|}H(a, db, c) + (-1)^{|a|+|b|}H(a, b, dc). \end{aligned} \tag{7.62}$$

Since $d(u^2) = 2\kappa uTc$ and $d(cTc) = 0$, degree and the displayed value force

$$H(uTc, u^2, cTc) - H(u^2, uTc, cTc) = \frac{(\lambda + \mu)^3}{6}.$$

The scalar vacuum need not be exact in C: the input differential contributes the boundary in this operation complex. The noncocycle example therefore does not define an obstruction class from three cohomology classes.

The construction supplies outer operations on the complete pair-support complexes and their triple homotopy for the regular singleton source. Extending these fixed maps to higher merged collision diagrams remains a further equation. It must preserve the endpoint homotopies, proper collision strata and vacuum faces already specified here.

7.8 Directed composition beyond the associator

A finite directed category isolates the next coherence equation. Four arrows have five parenthesizations, so their associativity homotopies must satisfy a pentagon relation. This calculation fixes a carrier of rank

one on every forward arrow. Its solution will not identify that carrier with a supported field-operation complex.

Fix objects $0, \dots, 4$, degree-zero arrows a_{ij} for $i < j$, scalar identities and no backward arrows. Products are written chronologically. Introduce ten degree-zero coefficients and their associativity equations:

$$a_{ij}a_{jk} = m_{ijk}a_{ik}, \qquad F_{ijkl} = m_{ijk}m_{ikl} - m_{jkl}m_{ijl}.$$

The dg coefficient algebra

$$K = k[m_{ijk}] \otimes \Lambda(\chi_{ijkl}), \qquad |\chi_{ijkl}| = -1, \quad d\chi_{ijkl} = F_{ijkl}, \quad dm_{ijk} = 0$$

retains a chosen homotopy for each binary associativity equation. All index sequences are increasing.

Proposition 7.26 (The relative pentagon class)**.** *The element*

$$\begin{aligned} S = {} & m_{034}\chi_{0123} + m_{123}\chi_{0134} + m_{014}\chi_{1234} \\ & - m_{234}\chi_{0124} - m_{012}\chi_{0234} \end{aligned} \tag{7.63}$$

is a nonzero class in $H^{-1}(K)$. Adjoining an even generator ζ of degree -2 with $d\zeta = S$ therefore fills a relation not already filled in K.

Proof. The five parenthesizations of the four-arrow path have coefficients

$$\begin{aligned} &P_1 = m_{012}m_{023}m_{034}, \quad P_2 = m_{123}m_{013}m_{034}, \quad P_3 = m_{123}m_{134}m_{014}, \\ &P_4 = m_{234}m_{124}m_{014}, \quad P_5 = m_{012}m_{234}m_{024}. \end{aligned}$$

The differentials of the five displayed terms in S are $P_1 - P_2$, $P_2 - P_3$, $P_3 - P_4$, $P_4 - P_5$, $P_5 - P_1$, so $dS = 0$. Every element of K^{-2} is a polynomial combination of products $\chi_I\chi_J$. Its differential has coefficients of m-degree at least two. The nonzero coefficients of S have m-degree one, so S cannot be a boundary. Finally $d^2\zeta = dS = 0$; its even degree requires the polynomial algebra $k[\zeta]$. □

The obstruction fixes the lower homotopies. In $A = \Lambda(\varepsilon)$ with $|\varepsilon| = -1$ and $d = 0$, set every $m_{ijk} = 1$, set $\chi_{0123} = \varepsilon$ and all other χ to zero. All F_{ijkl} vanish, giving a map $K \to A$. It cannot extend to $K[\zeta]$, since S maps to ε and $A^{-2} = 0$. Setting every χ to zero would extend. Thus the example obstructs prescribed homotopies, not all coherent structures with those binary products.

7.8.1 The coefficient algebra in every arity

For N ordered objects $0, \dots, N-1$, introduce a variable β_I of degree $2-n$ for every increasing sequence $I = (i_0 < \cdots < i_n)$, $n \geq 2$. Let R_N be their free graded-commutative algebra, with

$$d\beta_I = -\sum_{s=2}^{n-1}\sum_{r=0}^{n-s}(-1)^{r+(r+1)(2-s)}\beta_{(i_r,\dots,i_{r+s})}\beta_{(i_0,\dots,i_r,i_{r+s},\dots,i_n)}. \tag{7.64}$$

The inner coefficient is written first, fixing the sign convention. The sum is empty on binary coefficients. There are $2^N - 1 - N - \binom{N}{2}$ generators.

Theorem 7.27. *Equation (7.64) defines a differential. For every commutative dg algebra A, maps $R_N \to A$ specify strictly unital directed A_∞ structures on the fixed labelled rank-one carrier, with unary differential that of A. For $N = 5$, the substitutions*

$$\beta_{ijk} = m_{ijk}, \qquad \beta_{ijkl} = -\chi_{ijkl}, \qquad \beta_{01234} = -\zeta$$

identify it with the preceding pentagon algebra.

Proof. Suspend each forward arrow to degree -1, denoting it by $\bar{a}_{ij}$. Define the degree-one operations

$$b_n(\bar{a}_{i_0 i_1}|\cdots|\bar{a}_{i_{n-1} i_n}) = \beta_I \bar{a}_{i_0 i_n}.$$

Keep coefficients on the left and extend them as a coderivation of the tensor coalgebra over the object idempotents. Its unary part differentiates coefficients. Inserting an s-ary operation after r inputs contributes $(-1)^r$. Moving its coefficient of degree $2-s$ past those inputs and then through the outer degree-one operation contributes $(-1)^{(r+1)(2-s)}$. Thus the component of b^2 with output length one is $d\beta_I$ plus the insertion sum in (7.64).

Apply the prescribed derivation once more. Each resulting term is a planar substitution with three operation vertices. Its two internal edges can be contracted in exactly two orders. For nested substitutions these are the two associative orders of composing graded maps; for disjoint substitutions they interchange two odd insertions. The graded Leibniz rule gives opposite signs and the same coefficient permutation in each pair. Hence every term cancels and $d^2\beta_I = 0$.

A map $R_N \to A$ therefore gives exactly the coefficients with $b^2 = 0$. No nonidentity composable string has length greater than $N-1$. Normalize away strings with an identity input and impose the ordinary unit products; these give the strictly unital structure. Conversely, its coefficients give the map. Substitution at arities three and four gives $d\chi = F$ and $d\zeta = S$ with the displayed signs. □

Polynomial differential forms on simplices can be used as test algebras to retain paths and higher simplices of these structures. The carrier and its bases stay fixed. Taking a quotient by changes of basis or by general A_∞ equivalences is a separate moduli problem. Nor does the presentation assert that R_N resolves the ordinary associative-parameter ring at every singular point.

7.8.2 An algebraic contraction of the invertible chart

Assume $N \geq 3$ and invert every binary coefficient:

$$R_N^\times = R_N[m_{ijk}^{-1} : i<j<k], \qquad q_{ij} = m_{0ij}, \qquad L_N = k[q_{ij}^{\pm 1} : 1 \leq i < j \leq N-1].$$

These are algebraic localizations; no completion is used. The first object supplies a contracting partner for every coefficient on the other objects.

Theorem 7.28. *There are dg algebra maps $L_N \xrightarrow{\iota} R_N^\times \xrightarrow{p} L_N$, with $p\iota = 1$, and an L_N-linear degree-minus-one homotopy satisfying $dH + Hd = 1 - \iota p$. The inclusion sends q_{ij} to m_{0ij}, and*

$$p(m_{ijk}) = \frac{q_{ij}q_{jk}}{q_{ik}} \quad (0<i<j<k), \qquad p(\beta_I) = 0 \quad (|I| \geq 4). \tag{7.65}$$

Thus $H^0(R_N^\times) = L_N$ and all other cohomology vanishes, for every finite N.

Proof. For each increasing $I = (i_1, \dots, i_\ell)$ with $i_1 > 0$ and $\ell \geq 3$, set

$$\eta_I = q_{i_1 i_\ell}^{-1}\beta_{0I}, \qquad z_I = q_{i_1 i_\ell}^{-1} d\beta_{0I}. \tag{7.66}$$

Then $d\eta_I = z_I$ and $dz_I = 0$. In the insertion formula for $d\beta_{0I}$, the term $r=1, s=\ell-1$ is $+\beta_I q_{i_1 i_\ell}$: the exponent $1+2(3-\ell)$ is odd and the formula has an initial minus sign. Every other coefficient with no zero index has fewer than ℓ indices. Coefficients containing zero are already q's or unit multiples of η's. Thus $z_I = \beta_I +$ terms of smaller nonzero index size. Increasing induction on ℓ inverts this triangular polynomial change of generators.

For $I = (i,j,k)$, $z_I = m_{ijk} - c_I$, with $c_I = q_{ij}q_{jk}/q_{ik}$ a unit of L_N. Consequently $R_N^\times$ is the tensor product over L_N of the factors

$$A_I = \begin{cases} L_N[z_I, (z_I + c_I)^{-1}] \otimes \Lambda(\eta_I), & |I| = 3, \\ \mathrm{Sym}_{L_N}(L_N z_I \oplus L_N \eta_I), & |I| \geq 4, \end{cases}$$

with $|z_I| = 3 - |I|$, $|\eta_I| = 2 - |I|$. Evaluation at both variables zero is defined in every factor, since each localized c_I is a unit.

For $|I| = 3$, a contraction of evaluation is

$$h(f(z)) = \frac{f(z) - f(0)}{z}\eta, \qquad h(f(z)\eta) = 0.$$

The quotient belongs to the same localized ring because the evaluation kernel is the ideal (z). For odd $|I| \geq 5$, use $h(z^a) = z^{a-1}\eta$ for $a \geq 1$, $h(1) = 0$ and $h(z^a\eta) = 0$. For even $|I| \geq 4$, use

$$h(\eta^a z) = \frac{\eta^{a+1}}{a+1}, \qquad h(\eta^a) = 0 \quad (a \geq 0).$$

The graded Leibniz rule gives $dh + hd = 1 - \iota p$ in each case, with $h\iota = ph = h^2 = 0$. The denominators are invertible in characteristic zero.

Order the finite set of factors and take the signed tensor homotopy $H = \sum_j (\iota_1 p_1) \otimes \cdots \otimes (\iota_{j-1} p_{j-1}) \otimes h_j \otimes 1 \otimes \cdots$. The contraction equation telescopes, as in Proposition 7.16. After the binary substitutions, induction through the triangular coordinates sends every higher coefficient to zero. Evaluation is therefore exactly (7.65). □

The inclusion and projection are algebra maps; the displayed homotopy is a homotopy of complexes. It also determines the tangent complex before changes of arrow basis. At the point with every $m_{ijk} = 1$ and every higher coefficient zero, let $T^\bullet$ be the dual of the coefficient cotangent complex. Then

$$\dim T^j = \binom{N}{j+3}, \qquad \operatorname{rank}(d : T^j \to T^{j+1}) = \binom{N-1}{j+3} \quad (j \geq 0), \tag{7.67}$$

with out-of-range binomial coefficients zero. Indeed the cotangent complex is obtained by differentiating the free localized presentation. Each pair (7.66) gives an identity from tangent degree $|I| - 3$ to $|I| - 2$, and the $\binom{N-1}{2}$ variables q_{ij} remain unpaired. Thus its cohomology is $k^{\binom{N-1}{2}}$ in degree zero and zero otherwise. For $N = 5$, the dimensions are $(10, 5, 1)$ and ranks $(4, 1)$; for $N = 6$, they are $(20, 15, 6, 1)$ and $(10, 5, 1)$.

7.8.3 Ordinary centres on two loci

The tangent complex of the coefficient space differs from the ordinary Hochschild complex of a category at a chosen point. Let C be the directed category on these N objects, with scalar identities and a single degree-zero basis arrow a_{ij} for each $i < j$.

Proposition 7.29. *At an associative point with every $m_{ijk} \neq 0$, $HH^0(C) = k$ and $HH^{>0}(C) = 0$. At the zero-product point,*

$$\dim HH^0(C) = 1, \qquad \dim HH^1(C) = \binom{N-1}{2},$$
$$\dim HH^j(C) = \binom{N}{j+1} \quad (2 \leq j \leq N-1),$$

and the higher groups vanish.

Proof. Associativity on $(0, i, j, k)$ gives $m_{ijk} = q_{ij}q_{jk}/q_{ik}$. Replace a_{ij} by $q_{ij}^{-1}a_{ij}$ when $i > 0$ and leave a_{0i} unchanged. Every composable product then has coefficient one. The relative bar of Theorem 5.17 identifies its

normalized Hochschild cochains with the simplicial cochains of the ordered $(N-1)$-simplex. For a positive-degree cochain put

$$(hf)(i_0,\dots,i_{n-1}) = \begin{cases} f(0,i_0,\dots,i_{n-1}), & i_0 > 0,\\ 0, & i_0 = 0. \end{cases}$$

In $dh + hd$, terms deleting any vertex other than the inserted zero cancel. The remaining term is f in positive degree and $f - f(0)$ in degree zero. This contracts onto the constants.

At zero products, every differential from degree one upward is zero: each merger and endpoint action multiplies two forward arrows. The degree-zero map is the difference of vertex values on every forward edge. Its kernel is k and its rank $N-1$. Degree j has one coordinate for each increasing $(j+1)$-tuple. Subtracting that rank in degree one gives the stated dimensions. □

At the zero-product point, the coefficient differential has no linear part, so its tangent complex retains every generator. Neither this tangent complex nor its ordinary centre is obtained by substituting zero into the localized contraction.

7.9 The four-input comparison

The preceding constructions solve different composition problems. The discarded coefficient complex retains outer and crossed collision classes. The Heisenberg construction gives actual pair-support outer maps and a triple homotopy on regular singleton inputs. The directed coefficient algebra retains all arities on its fixed rank-one carrier and contracts on its invertible chart. A comparison with field operations must supply the maps relating these domains.

At four inputs, first construct a common operation complex which retains the local vacuum and every nonconsecutive collision channel. Transport the five parenthesizations and their chosen associativity homotopies into it, including the operations composing a homotopy with the remaining inputs. Their pentagon then gives a specified cycle. The unresolved equation is to construct a degree-minus-two primitive of that cycle, or to prove that its class is nonzero, with those lower homotopies fixed.

The directed algebra includes its higher generators and their prescribed differentials. Theorem 7.28 computes its localized cohomology; it does not assert that every choice of lower homotopies in another coefficient algebra extends. The common supported operation complex and the comparison into it remain to be constructed. Likewise the three-channel scalar current identity (7.14) retains the contact terms needed by a field comparison, but its residue relation alone is not that operation homotopy. These missing maps preserve the distinction between a coefficient calculation and a coherent action on fields.

Chapter 8

Gaussian elimination and higher state operations

Let p be expectation for a centred real Gaussian variable of variance $\sigma > 0$. Then $p(x) = 0$ and $p(x^2) = \sigma$. Eliminating x therefore leaves information in products that its mean does not contain. The connected contractions of Chapter 7 exhibit the same effect for fields. A reduction that preserves their operations must retain these fluctuations together with the differential and the normalization of the state.

In a finite polynomial model, an explicit homotopy identifies each observable with its expectation. The failure to preserve products becomes part of the comparison map. Formal interactions change both the expectation and the differential of the variables that remain. Successive integrations must preserve this changed differential and the state on every product. We construct these maps and compare them under changes of coordinates and of the integration data.

The normal direction of a boundary comparison has an additional datum: its endpoint condition. We first isolate that elementary support question. The finite quantum construction that follows does not require a choice of continuum propagator.

8.1 A supported normal contraction

Put $I = [0, \infty)$ and work over $\mathbb{C}$. The relative compactly supported de Rham complex

$$C = \Omega_c^\bullet(I, \{0\})$$

has $C^0 = \{f \in C_c^\infty(I) : f(0) = 0\}$ and $C^1 = C_c^\infty(I)\, dt$, with $df = f'\, dt$. Thus the relative condition constrains functions at the endpoint; it imposes no restriction on the coefficient of a one-form there. Choose $\rho \in C_c^\infty(I)$ with $\int_0^\infty \rho(t)\, dt = 1$.

Define maps to and from $\mathbb{C}[-1]$, placed in cohomological degree one, by

$$\begin{aligned} p(\beta\, dt) &= \int_0^\infty \beta(t)\, dt, & p(f) &= 0, \\ i(a) &= a\rho(t)\, dt, & H(f) &= 0, \\ H(\beta\, dt)(t) &= \int_0^t \Big(\beta(s) - \rho(s) \int_0^\infty \beta(r)\, dr\Big) ds. \end{aligned} \tag{8.1}$$

The subtraction in the primitive removes its constant tail. It is required to keep the homotopy in the stated support class.

Proposition 8.1. *The maps in* (8.1) *are continuous and satisfy*

$$pi = 1, \qquad dH + Hd = 1 - ip, \qquad H^2 = Hi = pH = 0.$$

The normal support of $H\beta$ lies in the compact interval spanned by 0, $\operatorname{supp}\beta$ *and* $\operatorname{supp}\rho$. *Tensoring this contraction with the full Dolbeault complex on an open set $U \subset \mathbb{C}^d$ retains every tangential form degree.*

Proof. The integrand defining $H\beta$ has total integral zero. Its primitive vanishes at zero and again beyond the stated interval. Differentiation gives $dH(\beta\,dt) = \beta\,dt - i\,p(\beta\,dt)$. For $f \in C^0$, compact support and $f(0) = 0$ give

$$p(df) = f(\infty) - f(0) = 0, \qquad H(df) = f.$$

These are the homotopy identities in the two degrees. The equation $pi = 1$ is the normalization of ρ; substituting $i(a)$ in H gives $Hi = 0$. The remaining side identities follow from degrees.

On functions or forms supported in a fixed compact interval, integration is bounded by a finite smooth seminorm. The same estimate for the primitive and its derivatives proves continuity into the enlarged compact interval. These estimates give continuity for the usual inductive-limit topology of compact supports. On $C \widehat{\otimes}_\pi \Omega^{0,\bullet}(U)$, the completed projective tensor product, the maps extend by continuity. The tensor differential is

$$d(\alpha \otimes \gamma) = d\alpha \otimes \gamma + (-1)^{|\alpha|}\alpha \otimes \bar\partial\gamma.$$

Its cross terms with $H \otimes 1$ cancel. The resulting target is $\Omega^{0,\bullet}(U)[-1]$, with differential $-\bar\partial$, as required by the shift. □

Changing the endpoint condition changes the answer. In the absolute complex $\Omega_c^\bullet(I)$, the primitive

$$H_{\mathrm{abs}}(\beta\,dt)(t) = -\int_t^\infty \beta(s)\,ds$$

is allowed even when it is nonzero at $t = 0$. It is compactly supported and satisfies $dH_{\mathrm{abs}} = 1$ on one-forms and $H_{\mathrm{abs}}d = 1$ on functions. The absolute complex is acyclic. In a field theory, the polarization must determine which normal complex occurs before either contraction is used.

8.2 The finite Gaussian Ward complex

Let R_0 be a nonzero commutative $\mathbb{Q}$-algebra and put $R = R_0[\hbar]$, where $\hbar$ has degree zero. Fix a symmetric invertible matrix $A \in \mathrm{Mat}_m(R_0)$ and write $C = A^{-1}$. Introduce even coordinates $q_1, \dots, q_m$ of degree zero and odd coordinates $\eta_1, \dots, \eta_m$ of degree -1. Our observable algebra is

$$\mathcal{G}_A = R[q_1, \dots, q_m] \otimes \Lambda(\eta_1, \dots, \eta_m).$$

All odd derivatives act on the left. Tensor products and reorderings use the cohomological Koszul sign.

The quadratic potential $S_0(q) = \frac{1}{2}q^{\mathrm{t}}Aq$ defines the *Ward differential*

$$D_A = \sum_i \big((Aq)_i - \hbar\partial_{q_i}\big)\partial_{\eta_i}. \tag{8.2}$$

This is the finite-dimensional homological form of Gaussian integration by parts; see Gwilliam–Johnson-Freyd [50, §§1–3]. The even operators $(Aq)_i - \hbar\partial_{q_i}$ commute, because A is symmetric. The odd derivatives anticommute, so $D_A^2 = 0$. In one variable, with $A = (a)$,

$$D_A\eta = ax, \qquad D_A(\eta x) = ax^2 - \hbar. \tag{8.3}$$

The second equality already records the fluctuation that survives elimination.

We will construct a *contraction* of this complex onto R: cochain maps $i : R \to \mathcal{G}_A$, $p : \mathcal{G}_A \to R$ and a degree-minus-one map H satisfying

$$pi = 1, \qquad D_AH + HD_A = 1 - ip, \qquad H^2 = Hi = pH = 0.$$

The last three equations are the side conditions of a strong deformation retract. The projection p, normalized by $p(1) = 1$, will be the formal Gaussian state.

8.2.1 Heat conjugation and the Euler homotopy

Let

$$d_0 = \sum_i (Aq)_i \partial_{\eta_i}, \qquad T_A = \exp\left(\frac{\hbar}{2} \sum_{i,j} C_{ij} \partial_{q_i} \partial_{q_j}\right).$$

The heat series and its inverse terminate on every polynomial. The commutator of its exponent with q_i is $\hbar \sum_j C_{ij} \partial_{q_j}$, and the next commutator is zero. Consequently

$$T_A^{-1} q_i T_A = q_i - \hbar \sum_j C_{ij} \partial_{q_j}, \qquad D_A = T_A^{-1} d_0 T_A. \tag{8.4}$$

Let N count total polynomial degree in the q_i plus exterior degree in the η_i, and put

$$R_A = \sum_{i,j} \eta_i C_{ij} \partial_{q_j}, \qquad h_0 = N^{-1} R_A \quad (N > 0), \qquad h_0(1) = 0.$$

The notation N^{-1} divides a homogeneous monomial by its positive integer degree. These divisions are valid over R_0. Let ϵ set every q_i, η_i equal to zero, and let i include constants.

Lemma 8.2. *The maps (ϵ, i, h_0) contract $(\mathcal{G}_A, d_0)$ onto R.*

Proof. The anticommutator of the odd derivations d_0 and R_A is the Euler operator:

$$d_0 R_A + R_A d_0 = \sum_i q_i \partial_{q_i} + \sum_i \eta_i \partial_{\eta_i} = N.$$

Indeed, its value on each q_i and η_i is that generator; an even derivation is determined by these values. Both odd operators preserve N. Division by N proves $d_0 h_0 + h_0 d_0 = 1$ in positive total degree, while both sides of $1 - i\epsilon$ vanish on constants. The constant-coefficient even derivatives in R_A commute and its odd coefficients anticommute, giving $R_A^2 = 0$. Hence $h_0^2 = 0$. The equations $h_0 i = \epsilon h_0 = 0$ follow from the same formulas. □

Theorem 8.3. *The maps*

$$p_A = \epsilon T_A, \qquad i(a) = a, \qquad H_A = T_A^{-1} h_0 T_A \tag{8.5}$$

give a contraction of $(\mathcal{G}_A, D_A)$ onto R. On even polynomials, p_A is the Wick functional of covariance $\hbar A^{-1}$; it kills every term of negative cohomological degree. All maps are defined over $R_0[\hbar]$.

Proof. Conjugate every identity of Lemma 8.2 by T_A. Equation (8.4) gives the new differential, and $T_A i = i$ keeps the inclusion fixed. Thus the homotopy and all side identities follow on every polynomial, without a convergence argument.

To identify the state, introduce independent degree-zero formal variables $J_1, \dots, J_m$. Applying ϵT_A coefficientwise gives

$$p_A(e^{J^t q}) = \exp\left(\frac{\hbar}{2} J^t C J\right). \tag{8.6}$$

Differentiation in the J_i pairs all input coordinates, with $\hbar C_{ij}$ for each pair. This proves Wick's formula and determines p_A on every polynomial. Heat operators preserve exterior degree, so evaluation kills negative cohomological degrees. □

For $A = (a)$, the retract gives the concrete primitive

$$p_A(x^2) = \frac{\hbar}{a}, \qquad H_A(x^2) = \frac{\eta x}{a}, \qquad D_A\left(\frac{\eta x}{a}\right) = x^2 - \frac{\hbar}{a}. \tag{8.7}$$

At a positive real specialization $A > 0$, $\hbar > 0$, equation (8.6) is also the generating function of the normalized convergent Gaussian integral. Positivity is needed for that real integral, and is not a hypothesis of the algebraic contraction.

8.2.2 *Ordinary and transported products*

The ordinary product on $\mathcal{G}_A$ is not a cochain operation: (8.3) gives

$$D_A(\eta x) - (D_A\eta)x = -\hbar.$$

One can transport a product along the heat isomorphism. It is useful to do this explicitly, since the transported square has a different expectation.

Proposition 8.4. *In one variable define*

$$f \diamond g = T_A^{-1}\big((T_A f)(T_A g)\big) = \sum_{r\geq 0} \frac{(-\hbar/a)^r}{r!}(\partial_x^r f)(\partial_x^r g). \tag{8.8}$$

This is a unital graded-commutative associative product for which D_A is a derivation. The map p_A is an augmentation for $\diamond$ and a quasi-isomorphism. In particular,

$$x \diamond x = x^2 - \hbar/a, \qquad p_A(x \diamond x) = 0, \qquad p_A(x^2) = \hbar/a.$$

Proof. The differential $d_0 = ax\partial_\eta$ is a derivation for the ordinary product. Transporting that differential and product by T_A^{-1} proves the algebra and differential identities. Evaluation at zero is an augmentation of the classical Koszul complex, so its transported map is p_A.

To obtain the sum in (8.8), apply the heat operators to two independent variables u, v before setting $u = v = x$. The second derivative of a product is induced by $(\partial_u + \partial_v)^2$. Its two individual square terms cancel those of T_A on the factors, leaving $\exp(-\hbar\partial_u\partial_v/a)$. All these series terminate on polynomials. The displayed square follows at once, and the quasi-isomorphism follows from Theorem 8.3. □

For the ordinary observables, the variance in (8.7) still has to be retained. Its higher analogues give a comparison of products without changing which polynomial is being observed.

8.3 Cumulants as operation maps

Let B be a unital graded-commutative algebra over a commutative $\mathbb{Q}$-algebra. Write

$$S^c(B) = \bigoplus_{n\geq 0} \mathrm{Sym}^n(B)$$

for the coaugmented conilpotent symmetric coalgebra of its underlying graded module, with the unshuffle coproduct. A coaugmentation-preserving coalgebra map out of $S^c(B)$ is determined by its *Taylor coefficients*: its composites from each symmetric length to the length-one component of the target. The scalar in length zero and the algebra unit $1_B \in B$, which lies in length one, are distinct.

The products of B determine a coalgebra automorphism τ_B whose nth Taylor coefficient is

$$(\tau_B)_n(b_1 \odot \cdots \odot b_n) = b_1 \cdots b_n, \qquad n \geq 1.$$

On an input of length n, it sums over partitions of $\{1, \dots, n\}$: multiply within each block and take the symmetric tensor of the block products. Within a block the indices are increasing; blocks are ordered by their least indices. Reordering homogeneous inputs into this order gives the Koszul sign. The term with singleton blocks is the identity, and every other term lowers length. Thus the inverse is finite on each input.

Lemma 8.5. *The nth Taylor coefficient of τ_B^{-1} is*

$$(\tau_B^{-1})_n = (-1)^{n-1}(n-1)!\,\mu_n,$$

where μ_n is multiplication of n elements.

Proof. Corestrict the proposed composite $\tau_B^{-1}\tau_B$ to length one. All partitions into k blocks have the same coefficient $(-1)^{k-1}(k-1)!$ after their products are multiplied. The resulting scalar is

$$\sum_{k=1}^{n} \left\{ {n \atop k} \right\} (-1)^{k-1}(k-1)! = \begin{cases} 1, & n = 1, \\ 0, & n > 1, \end{cases}$$

where the braces count set partitions into k nonempty blocks. This identity is the coefficient identity of $\log(e^z) = z$. The same regrouping supplies the Koszul signs of the graded products. A coalgebra map is determined by these corestrictions, proving the claimed inverse. □

8.3.1 The deviations of a differential

Suppose $D : B \to B$ has degree one, $D^2 = 0$ and $D1_B = 0$. It need not be a derivation. Let $\widetilde{D}$ be the unary coderivation on $S^c(B)$: it applies D to one factor at a time, with the tensor signs. Put

$$\mathscr{D}_B = \tau_B^{-1}\widetilde{D}\tau_B. \tag{8.9}$$

This is again a degree-one square-zero coderivation. Such a coderivation with zero constant Taylor coefficient is an $L_\infty[1]$ structure; its operations are graded symmetric and have degree one.

Proposition 8.6. *The first Taylor coefficients of* (8.9) *are*

$$q_1(b) = Db, \qquad q_2(b,c) = D(bc) - (Db)c - (-1)^{|b|}b(Dc).$$

If D has differential-operator order at most two, all Taylor coefficients of arity at least three vanish.

Proof. In length two, the product term in τ_B followed by D contributes $D(bc)$. The length-two part of τ_B^{-1} subtracts the two unary terms, giving q_2. More generally, the Taylor coefficient is the iterated graded commutator of D with the multiplication operators of the inputs, evaluated at 1_B. By definition, an operator of order at most two has zero commutators with three or more multiplication operators. The square-zero identities for the remaining operations follow from conjugation in (8.9). □

For D_A, the binary deviation is divisible by $\hbar$. Equation (8.3) says $q_2(\eta, x) = -\hbar$ in one variable. This second-order operation is part of the Ward complex.

8.3.2 The higher components of a normalized map

Let $P : (B, D_B) \to (C, D_C)$ be a degree-zero cochain map between unital graded-commutative algebras, with $P1_B = 1_C$. The linear map need not preserve products. Define

$$\mathscr{K}(P) = \tau_C^{-1}S^c(P)\tau_B. \tag{8.10}$$

Its nth Taylor coefficient is the *cumulant*

$$\kappa_n^P(b_1, \dots, b_n) = \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1}(|\pi|-1)!\,\epsilon(\pi; b) \prod_{E \in \pi} P\left(\prod_{j \in E} b_j \right). \tag{8.11}$$

Here Π_n is the set of partitions into nonempty blocks and $\epsilon(\pi; b)$ is the sign of the grouping specified above. In particular,

$$\kappa_1^P = P, \qquad \kappa_2^P(b,c) = P(bc) - P(b)P(c).$$

This coalgebra construction and its compatibility with Koszul brackets are described by Bandiera [7, §1.2, Lemma 1.16 and Proposition 1.21].

Theorem 8.7. *The map $\mathscr{K}(P)$ is an $L_\infty[1]$ morphism between the multiplicative deviations of D_B and D_C:*

$$\mathscr{D}_C \mathscr{K}(P) = \mathscr{K}(P)\mathscr{D}_B.$$

If P is a quasi-isomorphism, this is an $L_\infty[1]$ quasi-isomorphism. Every cumulant of arity greater than one vanishes on an algebra-unit input. For a further normalized cochain map $Q : C \to E$,

$$\mathscr{K}(QP) = \mathscr{K}(Q)\mathscr{K}(P). \tag{8.12}$$

Proof. Inserting (8.10) and cancelling $\tau_C \tau_C^{-1}$ reduces the morphism equation to $\widetilde{D}_C S^c(P) = S^c(P)\widetilde{D}_B$. This is precisely the cochain equation for P, applied to each tensor factor. The linear Taylor coefficient is P; being a quasi-isomorphism in that coefficient is the definition of an L_∞ quasi-isomorphism.

The unit assertion can be seen by adding independent marked variables of the degrees that make a sum F of marked inputs have total degree zero. The cumulants are the coefficients of $\log P(e^F)$. Adding a scalar mark $z1_B$ changes this expression to $z + \log P(e^F)$ because $P1_B = 1_C$. All mixed coefficients involving z therefore vanish. This is a finite coefficient identity in any specified arity.

Finally the two adjacent multiplication automorphisms in $\mathscr{K}(Q)\mathscr{K}(P)$ cancel. The remaining composite is (8.10) for QP, proving (8.12) in every arity. □

At arity two, (8.12) is the law of total covariance:

$$\kappa_2^{QP}(b,c) = Q\big(\kappa_2^P(b,c)\big) + \kappa_2^Q(Pb, Pc). \tag{8.13}$$

The second term measures fluctuations of the variables left after the first elimination.

8.3.3 A Gaussian state has higher operations on composites

Take one variable of stiffness a and write $p = p_A$. For even polynomials $f, g_2, \dots, g_n$, the morphism equation of Theorem 8.7 on $\eta f, g_2, \dots, g_n$ gives, for $n \geq 2$,

$$\kappa_n^p(axf - \hbar f', g_2, \dots, g_n) = \hbar \sum_{j=2}^{n} \kappa_{n-1}^p(f g_j', g_2, \dots, \widehat{g_j}, \dots, g_n). \tag{8.14}$$

Indeed the only binary input terms are $q_2(\eta f, g_j) = -\hbar f g_j'$, and the target differential and its deviations are zero. For $n = 2$, $f = 1$, $g_2 = x$, the equation forces $a\kappa_2^p(x,x) = \hbar$. The variance is required by the cochain equation itself.

Although the higher cumulants of the linear variable vanish above arity two, those of its square do not. Wick's formula gives the formal generating function

$$p(e^{tx^2}) = (1 - 2t\hbar/a)^{-1/2}.$$

Taking its logarithm and comparing coefficients of $t^n/n!$ proves

$$\kappa_n^p(x^2, \dots, x^2) = 2^{n-1}(n-1)!\left(\frac{\hbar}{a}\right)^n, \qquad n \geq 1. \tag{8.15}$$

Thus the state on the full polynomial algebra has nonzero components in arbitrarily high arity. Its Gaussian character does not truncate the operations of composite observables.

8.4 Formal interactions and conditional elimination

Introduce a degree-zero coupling parameter g. We work in

$$\big(R_{\mathrm{o}}[\hbar, q] \otimes \Lambda\eta\big)[[g]].$$

This means a polynomial in the even and odd coordinates at each coupling order. Polynomial degree may grow with that order; there is no completion in the field coordinates themselves. The positive g-adic filtration makes the following perturbation construction well defined.

8.4.1 A filtered perturbation identity

Lemma 8.8. *Let (i, p, h) be a contraction of a complex (C, d), complete and separated for a descending filtration, onto (B, d_B), with*

$$dh + hd = \mathrm{I} - ip, \qquad pi = \mathrm{I}, \qquad h^2 = hi = ph = \mathrm{o}.$$

Assume all these maps preserve filtration. Let δ have degree one, strictly raise filtration, and satisfy $(d + \delta)^2 = \mathrm{o}$. Put

$$F = (\mathrm{I} + \delta h)^{-1}\delta = \delta(\mathrm{I} + h\delta)^{-1}.$$

Then a contraction for $d + \delta$ is given by

$$\begin{aligned} d'_B &= d_B + pFi, & i' &= i - hFi = (\mathrm{I} + h\delta)^{-1} i, \\ p' &= p - pFh = p(\mathrm{I} + \delta h)^{-1}, & h' &= h - hFh = h(\mathrm{I} + \delta h)^{-1}. \end{aligned} \tag{8.16}$$

In particular, $(d'_B)^2 = \mathrm{o}$, $p'i' = \mathrm{I}$ and

$$(d + \delta)h' + h'(d + \delta) = \mathrm{I} - i'p', \qquad (h')^2 = h'i' = p'h' = \mathrm{o}.$$

Proof. All inverses are geometric series. At any fixed filtration precision only finitely many perturbation factors occur. On endomorphisms write $\partial T = dT - (-\mathrm{I})^{|T|} Td$. The hypotheses give $\partial\delta = -\delta^2$ and $\partial h = \mathrm{I} - ip$. Apply ∂ to $F = \delta - \delta hF$. The graded product rule gives

$$(\mathrm{I} + \delta h)\partial F = -\delta ipF, \qquad \text{hence} \qquad dF + Fd = -FipF.$$

It follows that

$$(d'_B)^2 = p(dF + Fd + FipF)i = \mathrm{o}.$$

Substitution in the inclusion equation gives

$$(d + \delta)i' - i'd'_B = h(dF + Fd + FipF)i = \mathrm{o}.$$

Using both $F = \delta - \delta hF$ and $F = \delta - Fh\delta$, the homotopy equation expands to

$$\begin{aligned} (d + \delta)h' + h'(d + \delta) &= \mathrm{I} - ip + ipFh + hFip + h(dF + Fd)h \\ &= \mathrm{I} - (i - hFi)(p - pFh). \end{aligned}$$

The original side conditions give $p'i' = \mathrm{I}$ and $(h')^2 = h'i' = p'h' = \mathrm{o}$ directly. Commuting the homotopy equation with $d + \delta$ and using the inclusion equation gives $i'(d'_B p' - p'(d + \delta)) = \mathrm{o}$. Apply p' to obtain the projection equation. This proves all the contraction identities. □

8.4.2 An interacting state over the integral quantum ring

Let

$$S(q) = \tfrac{1}{2} q^{\mathrm{t}} A q + V(q), \qquad V \in g R_0[\hbar, q][[g]].$$

The potential is independent of the odd variables. Its Ward differential is

$$D_S = \sum_i (S_i - \hbar \partial_{q_i}) \partial_{\eta_i}, \qquad S_i = \partial_{q_i} S. \tag{8.17}$$

The operators in parentheses commute, since

$$[S_i - \hbar \partial_{q_i}, S_j - \hbar \partial_{q_j}] = \hbar(\partial_{q_j} S_i - \partial_{q_i} S_j) = 0.$$

Thus $D_S^2 = 0$. The perturbation $\delta_V = \sum_i V_i \partial_{\eta_i}$ has positive coupling order and kills constants.

Theorem 8.9. *The Gaussian maps of Theorem 8.3 give an interacting contraction with*

$$p_V = p_A(1 + \delta_V H_A)^{-1}, \qquad i_V = i, \qquad H_V = H_A(1 + \delta_V H_A)^{-1}. \tag{8.18}$$

Its target is $R_0[\hbar][[g]]$ with zero differential. All coefficients are polynomial in $\hbar$ at each coupling order. The normalized exponential formula

$$p_V(f) = \frac{p_A(f e^{-V/\hbar})}{p_A(e^{-V/\hbar})} \tag{8.19}$$

agrees with this projection, with negative cohomological degrees sent to zero. The quotient is first interpreted in $R_0[\hbar, \hbar^{-1}, q][[g]]$; its normalized value lies in the integral quantum ring.

Proof. Apply Lemma 8.8 to the Gaussian contraction. Since $\delta_V i = 0$, the inclusion and target differential remain unchanged. The other formulas are (8.18). Each coupling coefficient contains finitely many operations on polynomials. None of these operations divides by $\hbar$.

In the Laurent coefficient ring, the denominator of (8.19) is $1 + O(g)$ and therefore a unit. Gaussian integration by parts says

$$p_A\big((Aq)_i f - \hbar \partial_{q_i} f\big) = 0.$$

Apply this identity to $f e^{-V/\hbar}$ at each coupling order. The normalized quotient L then satisfies

$$L\big(S_i f - \hbar \partial_{q_i} f\big) = 0, \qquad L(1) = 1.$$

These equations give the cochain identity on inputs with one odd generator. With two or more odd generators, the image under D_S still has negative cohomological degree and is killed by L.

The normalized cochain functional is unique. For an even polynomial f, the free contraction gives $f - i\,p_A f = D_A H_A f$. Hence any such functional satisfies

$$L(f) = p_A f - L(\delta_V H_A f).$$

Solving recursively in coupling order gives $L = p_A(1 + \delta_V H_A)^{-1}$. This identifies the quotient with the already integral projection. □

For the formal cubic potential $S = a x^2/2 + g x^3/6$, the first moments are

$$\begin{aligned} p_V(x) &= -\frac{g\hbar}{2a^2} - \frac{5 g^3 \hbar^2}{8 a^5} + O(g^5), \\ p_V(x^2) &= \frac{\hbar}{a} + \frac{5 g^2 \hbar^2}{4 a^4} + O(g^4). \end{aligned} \tag{8.20}$$

These follow by expanding (8.19) and using $p_A(x^{2n}) = (2n-1)!!(\hbar/a)^n$. They obey $a\,p_V(x) + (g/2)\,p_V(x^2) = 0$, the Ward equation on the odd generator. All series in this example are formal in g.

8.4.3 The differential of the retained variables

Split the even variables into $x = (x_1, \dots, x_r)$ and $y = (y_1, \dots, y_s)$, with respective odd variables η_x, η_y. Fix symmetric invertible matrices A_x, A_y over R_0, and put

$$S(x, y) = \tfrac{1}{2} x^{\mathrm{t}} A_x x + \tfrac{1}{2} y^{\mathrm{t}} A_y y + V(x, y), \qquad V \in g R_0[\hbar, x, y][[g]]. \tag{8.21}$$

Write P_x^0 for the free Wick expectation in x. It kills every term containing an η_x and preserves the y, η_y variables. Define

$$\begin{aligned} Z_x(y) &= P_x^0(e^{-V/\hbar}), & P_x(f) &= Z_x^{-1} P_x^0(f e^{-V/\hbar}), \\ S_{\mathrm{eff}}(y) &= \tfrac{1}{2} y^{\mathrm{t}} A_y y - \hbar \log Z_x(y). \end{aligned} \tag{8.22}$$

These exponential and logarithmic expressions initially have Laurent quantum coefficients and are complete in g. The odd-variable rule is the same as for P_x^0.

For any potential W in a tuple of even variables q, write

$$\mathcal{B}_W = \big(R_0[\hbar, q] \otimes \Lambda \eta_q\big)[[g]]$$

with differential (8.17). Its binary multiplicative deviation q_2^W is divisible by $\hbar$; denote the quotient by $\overline{q}_2^W$. The operations $(D_W, \overline{q}_2^W)$ form an $L_\infty[1]$ structure. Indeed the original square-zero coderivation has only unary and binary components. Its differential identity for the binary operation has one factor of $\hbar$, and its Jacobi identity has two. Multiplication by $\hbar$ is injective in these polynomial coefficient modules, even if R_0 has zero divisors, so both identities can be divided.

Theorem 8.10 (Integral conditional elimination)**.** *The effective potential belongs to $R_0[\hbar, y][[g]]$. The normalized map in* (8.22) *is a unital cochain quasi-isomorphism*

$$P_x : (\mathcal{B}_S, D_S) \longrightarrow (\mathcal{B}_{S_{\mathrm{eff}}}, D_{S_{\mathrm{eff}}})$$

over $R_0[\hbar][[g]]$. It is the projection in the filtered perturbation of the free Gaussian contraction. For every $n \geq 1$, its cumulant is divisible by $\hbar^{n-1}$. The integral maps

$$\Phi_n^{P_x} = \hbar^{1-n} \kappa_n^{P_x} \tag{8.23}$$

give an $L_\infty[1]$ quasi-isomorphism for the divided operations $(D_S, \overline{q}_2^S)$ and $(D_{S_{\mathrm{eff}}}, \overline{q}_2^{S_{\mathrm{eff}}})$. They are continuous in g, polynomial in $\hbar$ at each coupling order, and vanish on a unit input when $n > 1$.

Proof. First work in Laurent quantum coefficients. The hidden Ward equations are

$$P_x\big(S_{x_i} f - \hbar \partial_{x_i} f\big) = 0.$$

Differentiating the normalized quotient in a retained variable gives

$$\begin{aligned} \partial_{y_j} P_x(f) &= P_x(\partial_{y_j} f) - \hbar^{-1}\big(P_x(f V_{y_j}) - P_x(f) P_x(V_{y_j})\big), \\ (S_{\mathrm{eff}})_{y_j} &= (A_y y)_j + P_x(V_{y_j}). \end{aligned} \tag{8.24}$$

Combining these equations yields

$$P_x(S_{y_j} f - \hbar \partial_{y_j} f) = (S_{\mathrm{eff}})_{y_j} P_x(f) - \hbar \partial_{y_j} P_x(f). \tag{8.25}$$

The hidden identity deals with the last eliminated odd generator; (8.25) deals with every retained odd generator, with its left-exterior sign. Terms still containing an eliminated odd generator vanish. This proves the cochain equation in every exterior degree.

Let $(\iota_{\mathrm{o}}, P_x^{\mathrm{o}}, H_{\mathrm{o}})$ be the Gaussian contraction in x tensored with the retained free Ward complex. Apply Lemma 8.8 with $\delta = \sum V_i \partial_{\eta_i}$. Its projection is $P_x^{\mathrm{o}}(1 + \delta H_{\mathrm{o}})^{-1}$. Each H_{o} adds one hidden odd generator. A term surviving the final P_x^{o} must start without a hidden odd generator, and each subsequent δ must remove the hidden generator just introduced. Thus this projection acts on retained coefficients by a normalized functional for the hidden Ward differential. The uniqueness recursion used in Theorem 8.9, now with y as coefficients, identifies it with P_x.

In the transferred differential, one more δ occurs than H_{o}. A surviving correction therefore removes exactly one retained odd generator and otherwise only hidden ones. It has the form $\sum_j F_j(y)\partial_{\eta_{y_j}}$. Evaluating the cochain equation on η_{y_j} identifies $F_j = \partial_{y_j}(S_{\mathrm{eff}} - \frac{1}{2} y^{\mathrm{t}} A_y y)$. The retained second-order term is unchanged. The filtered lemma now gives the asserted contraction and quasi-isomorphism. Its projection, inclusion and homotopy are all integral in $\hbar$.

It remains to prove integrality of the full effective potential and the stated powers in its cumulants. Expand Wick expectation with independent marked sources for the observables. The logarithm selects connected graphs. A graph with n marked vertices, v interaction vertices and E edges has

$$E \geq n + v - 1.$$

Every edge contributes a factor $\hbar A_x^{-1}$ and each interaction vertex contributes at worst $\hbar^{-1}$. The resulting power is at least $E - v \geq n - 1$. At coupling order g^k, the positive coupling orders of the vertices sum to k, so $v \leq k$. For fixed polynomial inputs there are only finitely many graphs at that order. Retained variables are coefficients in this count, including their exterior signs. An eliminated odd input gives zero.

Vacuum components cancel from normalized cumulants. For $\log Z_x$, a connected vacuum graph satisfies $E \geq v - 1$; the factor $-\hbar$ in the effective potential removes its one possible negative quantum power. An interaction independent of x contributes its original potential term, so this argument also includes zero-valent vertices. It proves both remaining integrality assertions.

Finally apply Theorem 8.7 to P_x. In its arity-n morphism equation, a unary term has the factor $\hbar^{n-1}$. A source binary term has $\hbar\hbar^{n-2}$, while a target binary term with input arities $r + s = n$ has $\hbar\hbar^{r-1}\hbar^{s-1}$. Dividing these common factors gives the equations for (8.23). The division is valid in the integral polynomial modules. The linear component remains P_x; the unit assertion follows from the corresponding assertion for cumulants. □

The quotients in (8.23) are constructed before reducing modulo any power $\hbar^M$. This order gives maps at every finite quantum precision. Division by $\hbar^{n-1}$ would not be well defined on an arbitrary class in the truncated ring alone.

8.4.4 *The determinant force*

For units $a, b \in R_{\mathrm{o}}$, take

$$S(x, y) = \tfrac{1}{2}(a + g y)x^2 + \tfrac{1}{2} b y^2.$$

The conditional Gaussian formulas are

$$\begin{aligned} Z_x &= (1 + g y/a)^{-1/2}, \\ S_{\mathrm{eff}} &= \tfrac{1}{2} b y^2 + \tfrac{\hbar}{2} \log(1 + g y/a), \\ P_x(x^2) &= \frac{\hbar}{a + g y}. \end{aligned} \tag{8.26}$$

All inverses and the square root are their g-adic expansions with the displayed constant terms. The logarithmic determinant changes the retained force by $\hbar g/(2(a+gy))$. It is visible in a single cochain equation: on $\eta_y x^2$, both $P_x D_S$ and $D_{S_{\mathrm{eff}}} P_x$ have value

$$\frac{by\hbar}{a+gy} + \frac{3g\hbar^2}{2(a+gy)^2}. \tag{8.27}$$

On the source side use $P_x(x^4) = 3\hbar^2/(a+gy)^2$. On the target side differentiate $\hbar/(a+gy)$ and include the determinant force. Omitting that force leaves the nonzero defect $g\hbar^2/(2(a+gy)^2)$.

8.4.5 A retained odd generator also changes

The projection is accompanied by the corrected inclusion $i' = (1 + H_0\delta)^{-1} i_0$ from (8.16). For the formal potential

$$S(x,y) = \tfrac{1}{2}ax^2 + \tfrac{1}{2}by^2 + g\,\frac{yx^3}{6},$$

this correction is already nonzero on η_y:

$$\begin{aligned} i'(\eta_y) = \eta_y &- g\eta_x\left(\frac{x^2}{6a} + \frac{\hbar}{3a^2}\right) \\ &+ g^2 y\eta_x\left(\frac{x^3}{12a^2} + \frac{5\hbar x}{12a^3}\right) + O(g^3). \end{aligned} \tag{8.28}$$

Indeed $\delta\eta_y = gx^3/6$, and the Gaussian homotopy sends x^3 to $\eta_x(x^2/a + 2\hbar/a^2)$. Applying $H_0\delta$ a second time gives the displayed quadratic-coupling term.

The same model has

$$\begin{aligned} Z_x &= 1 + \frac{5g^2\hbar y^2}{24a^3} + \frac{385g^4\hbar^2 y^4}{1152a^6} + O(g^6), \\ S_{\mathrm{eff}} &= \tfrac{1}{2}by^2 - \frac{5g^2\hbar^2 y^2}{24a^3} - \frac{5g^4\hbar^3 y^4}{16a^6} + O(g^6). \end{aligned} \tag{8.29}$$

The Gaussian sixth and twelfth moments give Z_x. Subtracting half the square of its order-two term in $\log Z_x$ gives the fourth-order effective interaction. Thus the state, the retained force and the lift of an odd observable change together under elimination.

8.5 Sewing the state

Conditional elimination provides maps on observables and an effective weight for the next integration. Both enter sewing. The quadratic calculation separates the covariance that crosses the cut from the determinant that normalizes the result.

8.5.1 Covariance and determinant

Order the coordinates as retained y followed by eliminated x, and let

$$A = \begin{pmatrix} B & L \\ L^{\mathrm{t}} & D \end{pmatrix}, \qquad S = B - LD^{-1}L^{\mathrm{t}}.$$

Assume that D and S are invertible. The matrix S is the *Schur complement* of the eliminated block. Completing the square gives

$$\tfrac{1}{2}(y,x)A(y,x)^{\mathrm{t}} = \tfrac{1}{2}y^{\mathrm{t}}Sy + \tfrac{1}{2}(x + D^{-1}L^{\mathrm{t}}y)^{\mathrm{t}}D(x + D^{-1}L^{\mathrm{t}}y). \tag{8.30}$$

Put

$$E = \begin{pmatrix} \mathrm{I} \\ -D^{-1}L^{\mathrm{t}} \end{pmatrix}, \qquad C_{\mathrm{o}} = \begin{pmatrix} \mathrm{o} & \mathrm{o} \\ \mathrm{o} & D^{-1} \end{pmatrix}.$$

The map E extends a retained configuration to the critical point in its eliminated fibre.

Proposition 8.11. *One has*

$$A^{-1} = C_{\mathrm{o}} + ES^{-1}E^{\mathrm{t}}, \qquad \det A = \det D \det S. \tag{8.31}$$

The normalized Wick functional of covariance $\hbar A^{-1}$ equals successive Wick integration in $\xi = x + D^{-1}L^{\mathrm{t}}y$ and then in y, with covariances $\hbar D^{-1}$ and $\hbar S^{-1}$. Successive Schur complements agree with elimination of the combined block whenever the intermediate complements are invertible.

Proof. The triangular matrix

$$M = \begin{pmatrix} \mathrm{I} & \mathrm{o} \\ -D^{-1}L^{\mathrm{t}} & \mathrm{I} \end{pmatrix}$$

has determinant one and satisfies $M^{\mathrm{t}}AM = \operatorname{diag}(S, D)$. Taking inverses gives the first identity in (8.31); taking determinants gives the second. The exponential generating function (8.6) for two independent Gaussian blocks then becomes the generating function of covariance $\hbar(C_{\mathrm{o}} + ES^{-1}E^{\mathrm{t}})$. Equality of its polynomial coefficients proves the state identity.

To compare successive block eliminations, solve the linear critical-point equations in the eliminated variables. Solving them one block at a time and solving the combined system give the same solution. Substitution in the quadratic form therefore gives the same retained form. The determinant identity and the generating-function identity then compose in the same order. This proves the assertion for any finite number of admissible blocks. □

The change of odd generators is also fixed by this calculation. For the coordinates (y, ξ), write the odd generators as $\vartheta_y, \vartheta_\xi$. The old ones become

$$\eta_{x_a} \longmapsto \vartheta_{\xi_a}, \qquad \eta_{y_i} \longmapsto \vartheta_{y_i} + \sum_a (D^{-1}L^{\mathrm{t}})_{ai}\vartheta_{\xi_a}.$$

These are the inverse-Jacobian transformations of coordinate vector fields. Substitution in the Ward differential gives the block-diagonal differential for (8.30). Thus the conditional comparison also applies to all odd observables.

At a positive real specialization, the unnormalized generating integral is

$$\begin{aligned} Z_A(J) &= \int_{\mathbb{R}^{r+s}} e^{(-q^{\mathrm{t}}Aq/2 + J^{\mathrm{t}}q)/\hbar}\, dq \\ &= (2\pi\hbar)^{(r+s)/2}(\det A)^{-1/2} e^{J^{\mathrm{t}}A^{-1}J/(2\hbar)}. \end{aligned} \tag{8.32}$$

The two identities in (8.31) preserve respectively its moments and its normalization. Over a general coefficient ring, an unnormalized state requires a specified square-root normalization in addition to the algebraic determinant identity. The normalized Wick functional itself needs no such choice.

8.5.2 The intermediate weight

Return to the block potential (8.21). Let P_y° denote the free Wick expectation in the next block. Its effective conditional expectation is

$$P_y^{\mathrm{eff}}(b) = \frac{P_y^{\circ}(Z_x b)}{P_y^{\circ}(Z_x)}.$$

Thus the factor Z_x, which may depend on y, is part of the next integration.

Theorem 8.12. *Successive normalized elimination equals joint elimination:*

$$P_y^{\mathrm{eff}} P_x(f) = \frac{P_y^{\circ} P_x^{\circ}(f e^{-V/\hbar})}{P_y^{\circ} P_x^{\circ}(e^{-V/\hbar})}. \tag{8.33}$$

For these maps, and for further admissible blocks, the divided cumulants of (8.23) *compose by*

$$\Phi_n^{QP}(b_1, \dots, b_n) = \sum_{\pi \in \Pi_n} \epsilon(\pi; b) \Phi_{|\pi|}^{Q}\Big(\Phi_{|E|}^{P}(b_j : j \in E) : E \in \pi\Big). \tag{8.34}$$

These equalities hold over the integral quantum ring, and iterated sewing is associative in every arity.

Proof. Insert the normalized numerator of P_x into the next expectation. Its denominator Z_x cancels the intermediate weight, giving (8.33). Free Wick operations on separate blocks commute. At each coupling order only finitely many polynomial contractions occur, so this cancellation and Gaussian Fubini are defined coefficientwise. Theorem 8.10 shows that the resulting normalized maps are integral.

Equation (8.12) gives the partition composition law before division. In the term indexed by π, the quantum exponents add to

$$(|\pi| - 1) + \sum_{E \in \pi} (|E| - 1) = n - 1.$$

Dividing that common power gives (8.34). Composition of the underlying coalgebra maps is associative; the divided formula preserves that equality. This argument is also the composition mechanism for derived BV morphisms in Bandiera [7, §2.2, Proposition 2.8]. □

For example, let $X = Y + \xi$, where the first expectation eliminates a centred Gaussian ξ of variance σ and leaves Y. Its generating function on the square is

$$P(e^{tX^2}) = (1 - 2\sigma t)^{-1/2} \exp\left(\frac{tY^2}{1 - 2\sigma t}\right).$$

The logarithm gives, in every arity,

$$\kappa_n^{P}(X^2, \dots, X^2) = 2^{n-1}(n-1)!\big(\sigma^n + nY^2\sigma^{n-1}\big). \tag{8.35}$$

If the next expectation eliminates an independent centred Gaussian Y of variance τ, the joint variable X has variance $\sigma + \tau$. Equation (8.12) therefore combines all the conditional terms into $2^{n-1}(n-1)!(\sigma + \tau)^n$. Keeping only the conditional mean of X^2 would lose these higher terms.

8.6 Coordinate descent of the whole Ward complex

A change of coordinates transforms the integration density and the odd observables at the same time. The logarithm of the Jacobian enters the potential, while the inverse Jacobian transforms the odd generators. Their two contributions are required by the cochain equation.

Let $S(x) = \frac{1}{2}x^{t}Ax + V(x)$ be a finite potential as above. Take a formal change of variables $x = \phi(q) = q + O(g)$, polynomial in q at each coupling order. Its Jacobian $J = D_q\phi$ is invertible g-adically. Define

$$\Phi_\phi(f) = f \circ \phi, \qquad \Phi_\phi(\eta_i) = \sum_a (J^{-1})_{ai}\vartheta_a, \qquad S_\phi = S \circ \phi - \hbar \log\det J. \tag{8.36}$$

Here ϑ_a are the new degree-minus-one generators. The logarithm is the unique formal logarithm with zero constant coupling term, since $\det J = 1 + O(g)$.

Theorem 8.13. *The map Φ_ϕ is a unital graded-algebra and cochain isomorphism over $R_0[\hbar][[g]]$, and it preserves the normalized formal state:*

$$D_{S_\phi}\Phi_\phi = \Phi_\phi D_S, \qquad p_{S_\phi}\Phi_\phi = p_S. \tag{8.37}$$

For a second formal change $q = \psi(r)$,

$$(S_\phi)_\psi = S_{\phi\circ\psi}, \qquad \Phi_{\phi\circ\psi} = \Phi_\psi\Phi_\phi.$$

Proof. Regard the odd generators as coordinate vector fields and their products as polyvectors. For the weighted volume form $\rho = e^{-S/\hbar}dx_1 \cdots dx_m$, divergence on polyvectors has the coordinate formula

$$\mathrm{Div}_\rho = \sum_i \left(\partial_{x_i} - \hbar^{-1}S_i\right)\partial_{\eta_i}.$$

Thus $D_S = -\hbar\,\mathrm{Div}_\rho$. This notation abbreviates the polynomial Ward operator; the displayed product has no negative quantum power. Contraction with a volume form identifies this divergence with the exterior differential, with the exterior signs fixed by left odd differentiation. It is therefore natural under coordinate changes. In the present coordinates,

$$\phi^*\rho = e^{-S(\phi(q))/\hbar}\det J\, dq = e^{-S_\phi(q)/\hbar}dq,$$

and vector fields pull back by J^{-1}. These two facts give the first equation in (8.37) in every exterior degree.

For the state, interpolate by $\phi_u(q) = q + u(\phi(q) - q)$, $0 \le u \le 1$. The derivative of its pulled-back density times any pulled-back observable is a total derivative in the integration variables. At each coupling order, it is a polynomial times the free Gaussian weight. Formal Gaussian integration by parts annihilates that derivative. Both the unnormalized numerator and denominator are consequently constant in u. Their quotient proves the state equation on even observables; negative cohomological degrees vanish on both sides. The argument is formal in g and uses no global real invertibility of ϕ.

Finally the chain rule gives

$$\det D(\phi\circ\psi) = ((\det D\phi)\circ\psi)\det D\psi.$$

The factors are commuting units with constant term one, so their formal logarithms add. The inverse-Jacobian rule has the same composition law on vector fields. This proves the two cocycle identities. □

For $S = ax^2/2$ and $\phi(q) = q + gq^2$,

$$S_\phi = \tfrac{a}{2}(q + gq^2)^2 - \hbar\log(1 + 2gq), \qquad \Phi_\phi(\eta) = \frac{\vartheta}{1 + 2gq}. \tag{8.38}$$

If the logarithmic term is omitted, the cochain defect on this odd observable is $2g\hbar/(1+2gq)^2$. The inverse coordinate begins $q = x - gx^2 + 2g^2x^3 - 5g^3x^4 + \cdots$; its fifth-power term is $14g^4x^5$, which has zero Gaussian expectation. Consequently

$$p_{S_\phi}(q) = -\frac{g\hbar}{a} - \frac{15g^3\hbar^2}{a^2} + O(g^5), \qquad p_{S_\phi}(\phi(q)) = 0.$$

Without the Jacobian in the density, the last expectation instead begins with $-2g\hbar/a$. The same missing term is detected by a cochain and by an ordinary observable.

8.6.1 A flat family of state operations

Let ϕ_s depend on finitely many parameters s_i with commuting derivatives ∂_i. Its coefficients may be formal, or smooth on a specified parameter domain; all identities remain coefficientwise in g. Put

$$S_s = S_{\phi_s}, \qquad w_i = (D_q\phi_s)^{-1}\partial_i\phi_s, \qquad \Psi_i = \sum_a w_i^a \vartheta_a.$$

For a vector field w, its Lie derivative on polyvectors acts by $\mathcal{L}_w f = w(f)$ on functions and $\mathcal{L}_w\vartheta_a = -\sum_b(\partial_{q_a}w^b)\vartheta_b$ on coordinate vector fields. Extend it as a derivation. Define the connection

$$\nabla_i = \partial_i - \mathcal{L}_{w_i}.$$

Proposition 8.14. *The coordinate family satisfies*

$$\begin{gathered}\partial_i S_s = D_{S_s}\Psi_i = w_i(S_s) - \hbar\,\mathrm{div}\,w_i,\\ [\nabla_i, D_{S_s}] = 0, \qquad [\nabla_i,\nabla_j] = 0, \qquad \partial_i p_s(F) = p_s(\nabla_i F).\end{gathered} \tag{8.39}$$

Every cumulant is horizontal:

$$\partial_i \kappa_n^{p_s}(F_1,\dots,F_n) = \sum_j \kappa_n^{p_s}(F_1,\dots,\nabla_i F_j,\dots,F_n). \tag{8.40}$$

The same statement holds for the integral divided cumulants.

Proof. Differentiation of pullback gives $\partial_i\Phi_s = \mathcal{L}_{w_i}\Phi_s$. Differentiating the cochain equation in Theorem 8.13 proves $[\nabla_i, D_{S_s}] = 0$. Commuting two parameter derivatives of Φ_s gives

$$\partial_i w_j - \partial_j w_i - [w_i, w_j] = 0,$$

which is the curvature equation for ∇. Differentiating the pulled-back density gives the first identity in (8.39).

On a scalar polynomial F, the Ward equation gives

$$p_s(FD_{S_s}\Psi_i) = \hbar\, p_s(w_i F), \qquad p_s(D_{S_s}\Psi_i) = 0.$$

Indeed $D_{S_s}(F\Psi_i) = FD_{S_s}\Psi_i - \hbar w_i F$, and its expectation is zero. The derivative of the normalized state is the covariance with $\partial_i S_s$ divided by $-\hbar$, together with $p_s(\partial_i F)$. Substituting these Ward identities gives $p_s(\partial_i F - w_i F)$. The expression is integral; both sides vanish on negative cohomological degrees.

Apply this state equation to every block of (8.11). Since $\mathcal{L}_{w_i}$ is a derivation, each input is differentiated once with the same partition sign. This proves (8.40). Dividing the already integral cumulants by their known quantum powers gives the final assertion. □

For example,

$$\phi_{u,v}(q) = q + ugq^2 + vgq^3, \qquad w_u = \frac{gq^2}{1+2ugq+3vgq^2}, \quad w_v = \frac{gq^3}{1+2ugq+3vgq^2}$$

satisfies the displayed curvature equation as a rational identity, before its g-adic expansion. On parameter forms of degree p, the total differential is $d_\nabla + (-1)^p D_{S_s}$. Flatness and the commutator equation make its square zero. Restriction to parameter faces preserves the cochain and cumulant equations.

8.6.2 Coordinate changes within an eliminated fibre

For a conditional reduction, take $(x, y) = (\phi(q, y), y)$ and write $J_x = D_q\phi, B = D_y\phi$. The full inverse Jacobian transforms the old odd generators as

$$\begin{aligned} \eta_{x_i} &\longmapsto \sum_a (J_x^{-1})_{ai}\vartheta_{q_a}, \\ \eta_{y_j} &\longmapsto \vartheta_{y_j} - \sum_a (J_x^{-1}B)_{aj}\vartheta_{q_a}. \end{aligned} \tag{8.41}$$

The density changes by $\det J_x$.

Proposition 8.15. *The fibre substitution preserves the conditional partition factor $Z_x(y)$ and effective potential. If P_q is the transformed conditional Ward map, then*

$$P_q\Phi_\phi = P_x$$

on the entire complex. The same square commutes on every cumulant and divided cumulant, and with subsequent eliminations.

Proof. Apply the formal change-of-variables argument of Theorem 8.13, with y held as coefficients. It preserves the conditional numerator and its normalization, hence also S_{eff}. Projection kills every term containing a ϑ_q. In the second line of (8.41), only ϑ_{y_j} survives; the same holds term by term in products of retained odd generators. This proves the cochain square in every exterior degree. Substitution in the partition formula proves its cumulant versions, and Theorem 8.12 proves compatibility with successive blocks. □

8.7 Canonical families and their marked homotopies

Coordinate descent transports a state by an algebra isomorphism. A canonical variation of a BV weight gives a more general comparison: the normalized elimination commutes with parallel transport up to a specified cochain homotopy. Its higher components have one additional marked insertion.

8.7.1 The integration domain and the master weight

Let A, B be graded-commutative coefficient algebras over a characteristic-zero field with central parameter $\hbar$. Initially allow Laurent coefficients in $\hbar$; integral restrictions will be specified for the finite Gaussian sector. Let Δ_A, Δ_B have degree one, square zero, differential-operator order at most two, and value zero on the unit. In the potential model our convention is

$$\Delta_A = -\sum_i \partial_{q_i}\partial_{\eta_i}.$$

Choose a degree-zero linear integration map $\mathcal{P}$ on a specified domain of weights in extensions of A, B, satisfying the Stokes equation

$$\Delta_B\mathcal{P} = \mathcal{P}\Delta_A. \tag{8.42}$$

Multiplication by the weights and division by the effective weights below are required to be defined on this domain.

In the finite potential model, the domain consists of Gaussian weights times polynomial observables, expanded coefficientwise in the positive coupling filtration. Integration means formal Wick integration in the eliminated variables. Gaussian integration by parts proves (8.42) there. For an analytic integral the integration cycle, decay and absence of boundary terms are hypotheses of the same equation.

A weight U_A of cohomological degree zero satisfying $\Delta_A U_A = 0$ is a *master weight*. Put $U_B = \mathcal{P}U_A$ and assume that U_A and U_B are invertible on the chosen domain. Define

$$D_A = \hbar U_A^{-1}\Delta_A U_A, \qquad D_B = \hbar U_B^{-1}\Delta_B U_B, \qquad p(f) = U_B^{-1}\mathcal{P}(U_A f). \tag{8.43}$$

Conjugation in the two differential formulas denotes operators: for example $D_A f = \hbar U_A^{-1}\Delta_A(U_A f)$. For $U_A = e^{-S/\hbar}$ with S independent of odd variables, this is exactly the Ward differential (8.17).

Proposition 8.16. *The effective weight is a master weight, and p is a unital cochain map. Successive normalized integrations compose when each integration uses the effective weight from the preceding one.*

Proof. Stokes gives $\Delta_B U_B = 0$. Conjugation therefore gives $D_A 1 = D_B 1 = 0$ and $D_A^2 = D_B^2 = 0$. Moreover

$$D_B p = \hbar U_B^{-1}\Delta_B \mathcal{P}U_A = \hbar U_B^{-1}\mathcal{P}\Delta_A U_A = pD_A,$$

and $p1 = 1$ by definition. If $U_1 = \mathcal{P}_1 U_A$ and $U_2 = \mathcal{P}_2 U_1$, the composite normalized map is

$$U_2^{-1}\mathcal{P}_2\big(U_1 U_1^{-1}\mathcal{P}_1(U_A f)\big) = U_2^{-1}\mathcal{P}_2\mathcal{P}_1(U_A f).$$

This is normalization of the composite integration, with the entire intermediate weight retained. □

Theorem 8.10 supplies the quasi-isomorphism and integral formulas in the finite potential sector. Proposition 8.16 uses only Stokes and the stated weight domain; those hypotheses alone assert a cochain map.

8.7.2 Transporting a canonical variation

Let t be a parameter and $R_t \in \mathcal{A}$ have degree -1. Suppose

$$\partial_t U_A = \Delta_A(R_t U_A), \qquad \Delta_A U_A = 0. \tag{8.44}$$

The integration map and the two BV operators remain fixed. A varying cycle can be treated in this form only after its pullback to a fixed domain has been specified. Write m_R for multiplication by R and put

$$L_R = [\Delta_A, m_R] = \Delta_A m_R + m_R\Delta_A, \qquad r_t = p_t(R_t).$$

By Stokes, the effective family obeys

$$\partial_t U_B = \Delta_B(r_t U_B). \tag{8.45}$$

Set $w_A = U_A^{-1}\partial_t U_A$ and $w_B = U_B^{-1}\partial_t U_B$. Define the degree-zero transports

$$v_A = U_A^{-1}L_R U_A - m_{w_A}, \qquad v_B = U_B^{-1}[\Delta_B, m_r]U_B - m_{w_B}. \tag{8.46}$$

A commutator of a second-order operator with multiplication has order at most one. Its value on the unit here is w. Subtracting that value therefore makes each v a derivation. Since $[\Delta, L_R] = 0$, differentiation gives

$$\partial_t D_A = [v_A, D_A], \qquad \partial_t D_B = [v_B, D_B]. \tag{8.47}$$

Thus the connections $\partial_t - v_A$ and $\partial_t - v_B$ commute with their respective differentials.

Theorem 8.17. *The degree-minus-one map*

$$h_t(f) = \frac{p_t(R_t f) - p_t(R_t)\,p_t(f)}{\hbar} \tag{8.48}$$

satisfies

$$\partial_t p_t + p_t v_A - v_B p_t = D_B h_t + h_t D_A, \qquad h_t(1) = 0. \tag{8.49}$$

The formula holds on the declared integration domain. For polynomial marked inputs in a formal Gaussian presentation with no negative quantum powers, the quotient in (8.48) *is integral.*

Proof. On unnormalized weights define the odd operator $K = \mathcal{P} m_R - m_r \mathcal{P}$. Equation (8.42) gives

$$\mathcal{P} L_R - L_r \mathcal{P} = \Delta_B K + K \Delta_A. \tag{8.50}$$

Differentiate $p_t = U_B^{-1} \mathcal{P} U_A$ and add $p_t v_A - v_B p_t$. The two logarithmic-derivative multiplication terms cancel, leaving

$$U_B^{-1}(\mathcal{P} L_R - L_r \mathcal{P}) U_A.$$

Conjugating (8.50) by the weights and using $h_t = \hbar^{-1} U_B^{-1} K U_A$ proves (8.49). Normalization gives $K(U_A) = 0$, hence $h_t 1 = 0$. In the formal Gaussian sector, the numerator is the two-input cumulant $\kappa_2^{p_t}(R_t, f)$. Its connected graphs have at least one quantum power by the argument of Theorem 8.10. □

When parallel transports $T_A(t, 0)$, $T_B(t, 0)$ for these connections exist in the specified parameter domain,

$$\widehat{p}_t = T_B(t, 0)^{-1} p_t T_A(t, 0)$$

has derivative homotopic to zero. Its homotopy is $T_B(t, 0)^{-1} h_t T_A(t, 0)$. Integrating this expression along a defined interval gives an endpoint cochain homotopy. This may be an ordinary analytic interval or coefficientwise integration in a positive formal filtration; the evolution and endpoint evaluation are part of the chosen domain.

8.7.3 One marked input in every arity

Apply the coalgebra construction of (8.10) to p_t. Let $S^c(p_t; h_t)$ be its unary tangent map: it applies h_t to exactly one symmetric factor and p_t to every other factor, with the graded tensor signs. It is a degree-minus-one coderivation along $S^c(p_t)$. Define

$$\mathcal{H}_t = \tau_B^{-1} S^c(p_t; h_t) \tau_A.$$

For a derivation v, write $\widetilde{v}$ for its unary coderivation. Derivations commute with multiplication in the sense $\widetilde{v}\tau = \tau \widetilde{v}$.

Theorem 8.18. *The full operation comparison satisfies*

$$\partial_t \mathscr{K}(p_t) + \mathscr{K}(p_t)\widetilde{v}_A - \widetilde{v}_B \mathscr{K}(p_t) = \mathscr{D}_B \mathcal{H}_t + \mathcal{H}_t \mathscr{D}_A. \tag{8.51}$$

Its homotopy has Taylor coefficients

$$(\mathcal{H}_t)_n(b_1, \dots, b_n) = \hbar^{-1} \kappa_{n+1}^{p_t}(R_t, b_1, \dots, b_n). \tag{8.52}$$

The marked input R_t is placed first; the partition sign convention in (8.11) *determines all signs. In the formal Gaussian sector, $(\mathcal{H}_t)_n$ is divisible by $\hbar^{n-1}$. Dividing gives a homotopy for the divided state operations without inverting $\hbar$.*

Proof. Differentiation of $S^c(p_t)$ differentiates one factor at a time. Equation (8.49) and the cochain equation for p_t therefore give the unary-coderivation version of (8.51). Conjugate it by τ_A, τ_B, and use the derivation identity for v_A, v_B. This proves the full equation.

To compute its Taylor coefficient, differentiate the partition formula for κ_n by replacing one factor $p(\prod_{j\in E} b_j)$ by

$$h\left(\prod_{j\in E} b_j\right) = \hbar^{-1}\left[p\left(R\prod_{j\in E} b_j\right) - p(R)\,p\left(\prod_{j\in E} b_j\right)\right].$$

The first terms enumerate partitions in which the block of R is not a singleton. For a partition of the original inputs into k blocks, the second term occurs k times. Its coefficient is $-k(-1)^{k-1}(k-1)! = (-1)^k k!$, precisely the coefficient of the added singleton block. Moving R to its block produces the Koszul sign prescribed by placing it first. This proves (8.52).

A connected graph with $n+1$ marked inputs, v interaction vertices and E edges has $E \geq n+v$. Its quantum power after the interaction factors is at least n. Division by the explicit $\hbar$ in (8.52) leaves $\hbar^{n-1}$. The same arity count as in Theorem 8.10 permits division of the entire homotopy equation. □

These homotopies also sew. For successive eliminations $p_t : A \to B$ and $q_t : B \to C$ with the same intermediate transport, differentiation gives

$$h_t^{\mathrm{tot}} = q_t h_t^p + h_t^q p_t, \qquad \mathcal{H}^{\mathrm{tot}} = \mathscr{K}(q)\mathcal{H}^p + \mathcal{H}^q\mathscr{K}(p). \tag{8.53}$$

For normalized integration of the same master family, the intermediate generator is $r_t = p_t(R_t)$. The two covariance expressions in the first formula then cancel their intermediate $q_t(r_t\,p_t(f))$ term, leaving the marked covariance for $q_t\,p_t$ itself. Three eliminations give the same three homotopy terms in either parenthesization. Thus the path comparison and the state sewing have a common composition law.

8.7.4 The total master equation on parameter faces

Let T be a finite-dimensional parameter simplex. Place parameter forms to the left of coefficient elements, and extend Δ_A by

$$\Delta_A(\alpha\otimes a) = (-1)^{|\alpha|}\alpha\otimes\Delta_A a.$$

Suppose a total-degree-zero weight has been supplied with

$$\mathcal{U}_A \in \Omega^\bullet(T)\widehat{\otimes}A, \qquad (d_T + \hbar\Delta_A)\mathcal{U}_A = 0, \tag{8.54}$$

and invertible zero-form weight on the integration domain. Its pushforward $\mathcal{U}_B = \mathcal{P}\mathcal{U}_A$ satisfies the same total equation by Stokes. The full weights are invertible wherever their zero-form parts are, because positive form degree is nilpotent.

Proposition 8.19. *For a supplied family* (8.54)*, the operators*

$$\mathbb{D}_A = \mathcal{U}_A^{-1}(d_T + \hbar\Delta_A)\mathcal{U}_A, \qquad \mathbb{D}_B = \mathcal{U}_B^{-1}(d_T + \hbar\Delta_B)\mathcal{U}_B$$

and the normalized map

$$\mathbb{p}(F) = \mathcal{U}_B^{-1}\mathcal{P}(\mathcal{U}_A F)$$

are compatible with restriction to every parameter face. They give square-zero unital differentials and a cochain map. Their cumulants supply the higher parameter homotopies and their boundary equations.

Proof. The form differential and the extended BV operator anticommute. Hence their sum squares to zero, and conjugation preserves that identity. The total master equation makes the conjugated differentials kill the unit. Stokes gives $\mathbb{D}_B\mathbb{p} = \mathbb{p}\mathbb{D}_A$ by the calculation of Proposition 8.16. All these operations commute with pullback to a face. Apply the finite-length cumulant conjugation to this cochain equation over the ground coefficient ring. Its coefficients in parameter form degree are the successive homotopy equations; integration of their exterior derivatives gives their face boundaries by Stokes on the parameter simplex. □

On an interval, (8.44) gives the actual total weight

$$\mathcal{U}_A = U_A + dt\, R_t U_A/\hbar.$$

The coefficient of dt in its normalized map is h_t, and its total differential is $D_A + dt(\partial_t - v_A)$ on zero-form inputs. For two or more parameters, independently chosen one-parameter generators need not extend to a total master weight. If $\mathcal{U}_1$ is its proposed one-form part, a two-form part must solve the degree-two equation of (8.54). The existence of that part is an additional mathematical condition. The flat coordinate families of Proposition 8.14 give one explicit class with all parameter compatibilities.

8.7.5 A canonical shear with nonzero homotopy

Take even variables x, y and odd variables η, ζ of degree -1, with

$$\Delta = -\partial_x\partial_\eta - \partial_y\partial_\zeta.$$

For invertible stiffnesses a, b, set

$$S_t(x,y) = \tfrac{1}{2}ax^2 + \tfrac{1}{2}b(y-tx)^2, \qquad d = a + bt^2, \qquad k = bt/d. \tag{8.55}$$

Work formally near $t = 0$, or with real $a, b > 0$ and real t. The odd generator $R = \zeta x$ satisfies

$$[\Delta, m_R] = -x\partial_y + \zeta\partial_\eta.$$

Since $\partial_t S_t = -bx(y - tx)$, the weight $U_A = e^{-S_t/\hbar}$ obeys (8.44).

Complete the square in x, and normalize the eliminated Gaussian relative to its value at $t = 0$. This gives

$$\begin{aligned} p_t(x) &= ky, & \operatorname{Var}_{p_t}(x) &= \hbar/d, \\ S_t^{\mathrm{eff}}(y) &= \frac{ab}{2d}y^2 + \frac{\hbar}{2}\log(d/a), & r_t &= \zeta k y. \end{aligned} \tag{8.56}$$

The effective differential and normalized transports are

$$\begin{aligned} D_B &= \left(\frac{ab}{d}y - \hbar\partial_y\right)\partial_\zeta, \\ v_A &= -x\partial_y + \zeta\partial_\eta, \qquad v_B = -ky\partial_y + k\zeta\partial_\zeta. \end{aligned} \tag{8.57}$$

The determinant term is independent of y, so it does not appear in D_B. It is nevertheless required by the effective master-family equation (8.45): differentiating the logarithmic determinant supplies its scalar term.

The map p_t kills η and retains ζ. For an even polynomial f, Gaussian integration by parts gives

$$h_t(f) = \frac{\zeta}{\hbar}\big(p_t(xf) - ky\, p_t(f)\big) = \frac{\zeta}{d}p_t(\partial_x f). \tag{8.58}$$

The homotopy is zero on terms containing either odd generator. In particular $h_t(x) = \zeta/d$ is nonzero and integral in $\hbar$.

On x, both sides of (8.49) are

$$\partial_t(ky) - v_B(ky) = \frac{ab}{d^2}y = D_B(\zeta/d).$$

On the eliminated odd generator η, the left side is $p_t(v_A\eta) = \zeta$, and the right side is

$$h_t(D_A\eta) = h_t(dx - bty) = \zeta.$$

This equation sees a component that cannot be tested on ordinary even observables alone.

The higher homotopy also acts on composites. If $m = ky$ and $\sigma = \hbar/d$, then

$$\kappa_2(x^2, x^2) = 4m^2\sigma + 2\sigma^2, \qquad \kappa_3(x, x^2, x^2) = \sigma\partial_m\kappa_2(x^2, x^2) = 8m\sigma^2.$$

The derivative identity follows by differentiating the Gaussian generating function with respect to its mean. Equation (8.52) therefore gives

$$(\mathcal{H}_t)_2(x^2, x^2) = \frac{8\zeta\, ky\,\hbar}{d^2}. \tag{8.59}$$

The linear homotopy and this higher component are parts of the same normalized comparison.

8.8 Comparing normalized Gaussian choices

A fixed projection onto retained variables does not determine Gaussian elimination. For linear observables x, y, the covariance remains in

$$R(xy) = p(x)p(y) + \hbar P(x, y).$$

Two choices with the same projection therefore require a comparison on products. In a finite paired complex, a contraction of the discarded coefficients constructs that comparison and all its relative parameter extensions.

8.8.1 A contractible complex of choices

Let (V, d) and (H, d_H) be bounded complexes of finitely generated projective R_0-modules. Fix a contraction

$$H \underset{p}{\overset{i}{\rightleftarrows}} V, \qquad pi = 1, \qquad dh + hd = 1 - ip, \qquad hi = ph = h^2 = 0.$$

The maps i, p are cochain maps and h has degree -1. Put $t = ip$ and $\mathsf{E} = t \otimes t$ on $V \otimes V$. Every tensor of maps uses the graded rule $(f \otimes g)(x \otimes y) = (-1)^{|g||x|}f(x) \otimes g(y)$. For a homogeneous endomorphism M of $V \otimes V$, define

$$\delta M = d_{V\otimes V}M - (-1)^{|M|}M d_{V\otimes V}.$$

The symmetrized tensor contraction is

$$A_0 = \frac{1}{2}\big(h \otimes 1 + t \otimes h + 1 \otimes h + h \otimes t\big). \tag{8.60}$$

The two unsymmetrized sums are exchanged by the graded interchange of the tensor factors. Each has differential $1 - \mathsf{E}$, so

$$\delta A_0 = 1 - \mathsf{E}, \qquad \mathsf{E} A_0 = A_0 \mathsf{E} = 0. \tag{8.61}$$

Let $\mathcal{B}$ be the graded endomorphisms commuting with that interchange, with differential δ, and set

$$\mathcal{N} = (1 - \mathsf{E})\mathcal{B}(1 - \mathsf{E}).$$

A *normalized two-coefficient homotopy* is an element $A \in \mathcal{N}^{-1}$ with $\delta A = 1 - \mathsf{E}$. This condition imposes no equation on A^2.

Lemma 8.20. *Left multiplication $S(M) = A_0 M$ contracts $(\mathcal{N}, \delta)$:*

$$\delta S + S\delta = 1.$$

In particular, two normalized homotopies A, B have the specified comparison $L = A_0(B - A)$ with $\delta L = B - A$.

Proof. The graded Leibniz rule gives

$$\delta(A_0 M) = (1 - \mathsf{E})M - A_0 \delta M = M - A_0 \delta M.$$

Normalization is preserved: both $\mathsf{E} A_0 M$ and $A_0 M \mathsf{E}$ vanish. Composition also preserves commutation with the graded interchange. Thus S takes values in $\mathcal{N}$ and the identity holds there. Apply it to the closed element $B - A$ to obtain the last assertion. □

8.8.2 The covariance homotopy

Suppose a graded symmetric pairing $b : V \otimes V \to R_0$ of degree 1 is closed in the Hom complex. Write $b_H = b(i \otimes i)$ for its retained pairing. For any graded symmetric pairing P, let ∂_P denote unordered pair contraction on $\mathrm{Sym}(V)$, normalized by $\partial_P(xy) = P(x, y)$. On a product of homogeneous linear observables, select two factors and move them to the front with the Koszul sign. Apply P to that pair and retain the other factors in their order, then sum over all selected pairs.

Extend d to $\mathrm{Sym}(V)$ as a derivation. The two paired differentials are

$$D_b = d + \hbar \partial_b, \qquad D_H = d_H + \hbar \partial_{b_H}.$$

Closedness of b gives $[d, \partial_b] = 0$; constant coefficient pair contractions supercommute, so $\partial_b^2 = 0$. Thus both displayed differentials square to zero. For a normalized homotopy A, put

$$P_A = -bA, \qquad R_A = \mathrm{Sym}(p) e^{\hbar \partial_{P_A}}. \tag{8.62}$$

The covariance has degree zero. Its exponential is a finite sum on every polynomial: each contraction lowers the number of linear factors by two. These maps are defined over $R_0[\hbar]$ and extend by scalar extension to $R_0[[\hbar]]$ without completing polynomial degree.

Proposition 8.21. *Each R_A is a cochain map from D_b to D_H. For two normalized homotopies A, B, define*

$$L = A_0(B - A), \qquad Q = bL, \qquad P_u = (1 - u)P_A + uP_B.$$

Then the degree-minus-one map

$$K_{B,A} = \hbar \, \mathrm{Sym}(p) \partial_Q \int_0^1 e^{\hbar \partial_{P_u}} \, du \tag{8.63}$$

satisfies

$$D_H K_{B,A} + K_{B,A} D_b = R_B - R_A.$$

It vanishes on constants and on polynomials whose coefficients all lie in $i(H)$.

Proof. The Hom differential obeys the graded composition rule. Since $|b| = 1$ and $\delta b = 0$,

$$\delta P_A = b(1 - \mathsf{E}) = b - p^* b_H.$$

Here $p^* b_H = b_H(p \otimes p)$. The identity $[d, \partial_{P_A}] = \partial_{\delta P_A}$ then gives

$$e^{\hbar \partial_{P_A}} D_b = (d + \hbar \partial_{p^* b_H}) e^{\hbar \partial_{P_A}}.$$

There are no further commutator terms, because the constant coefficient contractions commute with the degree-zero operator ∂_{P_A}. Application of $\mathrm{Sym}(p)$ proves the cochain equation for R_A.

By Lemma 8.20, $\delta L = B - A$. The same graded composition rule gives

$$\delta Q = -b(B - A) = P_B - P_A.$$

Place parameter forms before coefficient maps. The total-degree-zero covariance $P_u + du\, Q$ therefore satisfies

$$(d_u + \delta)(P_u + du\, Q) = b - p^* b_H.$$

Its exponential gives a total cochain map by the preceding calculation. Its coefficient of du is the integrand in (8.63). The coefficient cochain equation, integrated from 0 to 1, is the claimed homotopy identity. This integration is polynomial integration over the rational ground ring. Finally $L\mathsf{E} = 0$, so Q vanishes on two retained coefficients; pair contraction also kills constants. □

The factors left after contraction matter. A quadratic input has homotopy $\hbar Q(x, y)$; a cubic input has terms $\hbar Q(x, y)\, p(z)$ and the other two pairings, with their Koszul signs. A five-dimensional complex makes both values nonzero. Take a basis

$$|u| = -1, \quad |v| = |r| = |e| = 0, \quad |s| = 1, \qquad du = v, \quad dr = s,$$

with all other differentials zero. Retain e, and set $h(v) = u$, $h(s) = r$, with h zero on the other basis vectors. Let $b(r, u) = b(u, r) = 1$ and let its other basis values vanish. This is a closed degree-one pairing.

Define the degree-minus-two endomorphism J of $V \otimes V$ by

$$J(s \otimes v) = J(v \otimes s) = \tfrac{1}{2}(r \otimes u + u \otimes r),$$

and zero on other tensor basis vectors. It is normalized and commutes with graded interchange. Thus $A = A_0$ and $B = A_0 + \delta J$ are normalized homotopies. Their covariance data are

$$P_A(r, v) = -\tfrac{1}{2}, \qquad P_B(r, v) = \tfrac{1}{2}, \qquad Q(s, v) = 1. \tag{8.64}$$

Indeed (8.60) gives $A_0(r \otimes v) = r \otimes u/2$, while $\delta J(r \otimes v) = -(r \otimes u + u \otimes r)/2$. Also $A_0 \delta J(s \otimes v) = (r \otimes u + u \otimes r)/2$, proving the third value. Since $D_b(rv) = sv$ and $b_H = 0$, the homotopy equation reads

$$R_B(rv) - R_A(rv) = \hbar = K_{B,A}(sv).$$

After multiplication by the retained coefficient, $K_{B,A}(sve) = \hbar e$.

8.8.3 *Prescribed parameter faces*

For the cube I^m, use polynomial parameter forms

$$\Omega^\bullet_{\mathrm{poly}}(I^m) = \mathbb{Q}[t_1, \dots, t_m, dt_1, \dots, dt_m],$$

where the dt_i are exterior generators of degree 1. A parameter family of normalized homotopies is a total-degree-minus-one element $\mathcal{A}$ with coefficients in $\mathcal{N}$ satisfying

$$(d_{\mathrm{par}} + \delta)\mathcal{A} = 1 - \mathsf{E}. \tag{8.65}$$

At a vertex, it is an ordinary normalized homotopy. Restriction to a face sets its fixed coordinates to 0 or 1 and their differentials to zero.

Theorem 8.22. *A family satisfying* (8.65) *on a union of codimension-one faces, with agreeing restrictions on intersections, extends to the cube. Two extensions admit such a family between them relative to the prescribed faces. The same assertion holds after adjoining any finite number of parameters.*

Proof. First extend the forms without their differential equation. For a face $t_i = \varepsilon$, use its coordinate projection and the scalar factor $1 - t_i$ if $\varepsilon = 0$, or t_i if $\varepsilon = 1$. Sum these extensions and subtract repeated extensions from intersections by inclusion–exclusion. Opposite faces have disjoint restriction conditions, since the factor for one vanishes on the other. On each prescribed face, terms from every other compatible face cancel against their intersection terms. The resulting polynomial form $\widetilde{\mathcal{A}}$ has the given restrictions and coefficients in $\mathcal{N}$.

Write $D = d_{\mathrm{par}} + \delta$ and set

$$Z = D\widetilde{\mathcal{A}} - (1 - \mathsf{E}).$$

This is closed and restricts to zero on the prescribed faces. Extend the contraction of Lemma 8.20 by

$$S_{\mathrm{par}}(\alpha \otimes M) = (-1)^{|\alpha|}\alpha \otimes A_0 M.$$

The parameter differential anticommutes with this degree-minus-one map, and the coefficient identity gives $DS_{\mathrm{par}} + S_{\mathrm{par}}D = 1$. Consequently

$$\mathcal{A} = \widetilde{\mathcal{A}} - S_{\mathrm{par}}Z \tag{8.66}$$

solves (8.65) and keeps all the prescribed face values. To compare two extensions, prescribe them at the ends of one more interval and keep their common boundary family constant along that interval. Apply the same correction. Repeating this procedure gives the higher comparisons. □

Each family gives the total Gaussian map $\mathrm{Sym}(p)\exp(\hbar\partial_{-b\mathcal{A}})$. Composition with b includes its sign past parameter forms. The covariance calculation in Proposition 8.21 proves its total cochain equation. Thus the construction supplies relative homotopies whenever the prescribed Gaussian comparisons have these normalized coefficient lifts. An arbitrary prescribed homotopy of observables need not come with such a lift.

8.9 From elimination to supported field operations

The finite constructions retain three kinds of data: the relative support class in the normal direction, the normalized state and all its cumulants, and the parameter homotopies transporting that state. Conditional elimination and coordinate changes construct these data over integral quantum coefficients. A supplied total master weight extends them over parameter faces, while the normalized Gaussian choices above provide an explicit class for which every relative extension exists.

The corresponding field-theoretic comparison requires a common supported operation complex containing the regular, meromorphic and contact coefficients of the previous chapters. Its normal reduction must intertwine the differential and every allowed collision operation; its state map must carry the local vacuum and the prescribed sewing and parameter homotopies. A regulated master family must make its boundary terms and parameter face equations meaningful on that same coefficient domain. The finite endomorphism contraction above does not yet act on those distributional products. These are the missing constructions for a state-preserving geometric comparison.

There is a second source of higher information when observables obey algebraic constraints. Before applying elimination to matrix observables, one must retain the relations among their constraint equations. The resulting complex is the starting point for Hamiltonian actions on derived matrix spaces.

Chapter 9

Hamiltonian actions on derived matrix spaces

Imposing a matrix equation has two different effects on an action. A bracket defect can vanish on its zero set, while the complex that imposes the equation retains a homotopy for that defect. The homotopies must themselves satisfy identities. For the commuting-matrix equation, a calculation with cubic Hamiltonians exhibits the first of these identities. A contraction of a free associative resolution then solves them in every arity.

The resulting action preserves a cyclic orientation coherently. Its matrix realization retains both the relation classes and the gauge stabilizers. Requiring the homotopies to be Hamiltonian adds a further condition: the scalar charges remember constants that their vector fields forget. Quantizing the coefficient plane introduces another distinction. Its finite-dimensional representations and the quantization of matrix coordinates have different equations, and consequently different recipients.

9.1 The cubic defect and the relation complex

Work over $\mathbb{C}$, with cohomological differentials of degree one. All commutators are graded: $[a, b] = ab - (-1)^{|a||b|} ba$ for homogeneous elements. Put $A = \mathbb{C}[q, p]$ and give it the Poisson bracket

$$\{f, g\} = f_q g_p - f_p g_q.$$

For a monomial $q^a p^b$, let $\sigma(q^a p^b)$ be the average of the distinct words with a occurrences of q and b occurrences of p. Thus $\sigma(qp) = (qp + pq)/2$. The map σ takes values in the free associative algebra; it is a linear section of abelianization.

Fix a positive integer N. Let X, Y be $N \times N$ matrices of independent commuting variables, with

$$\{X_{ij}, Y_{kl}\} = \delta_{il}\delta_{jk}, \qquad \{X_{ij}, X_{kl}\} = \{Y_{ij}, Y_{kl}\} = 0.$$

The scalar charge associated to $f \in A$ is

$$J_N(f) = \operatorname{Tr} \sigma f(X, Y). \tag{9.1}$$

In particular $J_N(1) = N$. Write $\mu = [X, Y] = XY - YX$ for the commuting-matrix equation. The brackets of cubic charges give

$$\{J_N(q^3), J_N(p^3)\} - J_N(9q^2p^2) = -\frac{3}{2} \operatorname{Tr}(\mu^2). \tag{9.2}$$

Indeed, the first term is $9 \operatorname{Tr}(X^2Y^2)$. Of the six words in $\sigma(q^2p^2)$, four have cyclic trace $\operatorname{Tr}(X^2Y^2)$ and two have cyclic trace $\operatorname{Tr}(XYXY)$. Expanding $\operatorname{Tr} \mu^2$ proves the formula. At rank two, take

$$X = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $\mu^2 = -I_2$, so the defect is 3.

The defect vanishes after imposing $\mu = 0$. To retain how it vanishes, introduce an odd matrix Ψ of degree -1 and the commutative differential graded algebra

$$R_N = \mathbb{C}[X_{ij}, Y_{ij}] \otimes \Lambda(\Psi_{ij}), \qquad QX = QY = 0, \qquad Q\Psi = [X, Y]. \tag{9.3}$$

Here and below, matrices with graded coefficients are multiplied in their displayed order. The entries of Ψ impose the equations by a differential, including dependencies among them. For example $\operatorname{Tr}\Psi$ is a cycle. It is not a boundary: evaluation at $X = Y = 0$ is a cochain map to $\Lambda(\Psi_{ij})$ with zero differential and sends this cycle to a nonzero element. At $N = 1$ the differential vanishes entirely. Thus the relation complex contains information beyond the ordinary commuting scheme.

9.2 A free resolution of the plane

The homotopies can be constructed before choosing the matrix rank. Consider the free differential graded associative algebra

$$R = \mathbb{C}\langle q, p, t\rangle, \qquad |q| = |p| = 0, \quad |t| = -1, \qquad dq = dp = 0, \quad dt = qp - pq. \tag{9.4}$$

The projection $\pi : R \to A$ kills t and makes q, p commute. Its linear section i sends $q^a p^b$ to the normal word $p^b q^a$. Both are cochain maps, but i is not multiplicative: $i(q)i(p) - i(qp) = dt$.

Replace the differential temporarily by $d_0 t = qp$. Expand every t in a word into qp, marking the occurrence from which it came. Two occurrences of qp cannot overlap. For a fixed expanded word, its marked subsets form a tensor product of two-term complexes. The differential d_0 removes a mark, with sign $(-1)^a$ if a marks precede it.

Define h_0 by adding the first mark when that occurrence is unmarked, and by zero when it is already marked or no occurrence exists. Let p_0 send an unmarked normal word $p^b q^a$ to $q^a p^b \in A$ and every other word to zero. The two-term contraction gives

$$d_0 h_0 + h_0 d_0 = 1 - i p_0, \qquad h_0^2 = h_0 i = p_0 h_0 = 0. \tag{9.5}$$

To see the signs, removing a later mark anticommutes with adding the first. Removing the first mark contributes the identity. When no occurrence exists, the word is already $p^b q^a$.

Write $d = d_0 + \delta$, where $\delta t = -pq$. Exchanging an expanded qp for pq decreases the number of pairs in which a q precedes a p. The operator δh_0 is therefore locally nilpotent. Define

$$h = h_0 \sum_{j \geq 0} (-\delta h_0)^j, \qquad p' = p_0 \sum_{j \geq 0} (-\delta h_0)^j. \tag{9.6}$$

These are finite sums on every word.

Proposition 9.1. *The projection p' equals π, and*

$$\pi i = 1, \qquad dh + hd = 1 - i\pi, \qquad h^2 = hi = \pi h = 0.$$

Consequently $R \to A$ is a quasi-isomorphism and h contracts $J = \ker \pi$.

Proof. Substitute $(1 + \delta h_0)^{-1}$ in (9.5). Multiplying the finite series by $1 + \delta h_0$ cancels consecutive perturbation terms. Since $\delta i = 0$, the inclusion and the zero differential on normal words remain unchanged. This gives $dh + hd = 1 - ip'$ and the side identities.

The surviving reductions replace qp by pq. A degree-zero word therefore projects to its commutative monomial; a word containing t has negative degree and projects to zero. Thus $p' = \pi$. On J the homotopy identity becomes $dh + hd = 1$. □

For a commutative differential graded algebra B, a morphism $R \to \mathrm{Mat}_N(B)$ amounts to matrices X, Y, Ψ with the differential in (9.3). This is the universal property represented by R_N. The resolution of A does not make R_N acyclic in negative degrees; the trace class already distinguishes the two assertions.

9.3 The first action and its binary homotopy

The Lie algebra of polynomial Hamiltonian vector fields is $\mathfrak{l} = A/\mathbb{C}1$, with the induced Poisson bracket. For a word $a_1 \cdots a_m$ in q, p, define its cyclic derivative by

$$\partial_q^{\mathrm{cyc}}(a_1 \cdots a_m) = \sum_{a_j = q} a_{j+1} \cdots a_m a_1 \cdots a_{j-1},$$

and similarly for p. Removing an occurrence makes the remainder start just after that occurrence. Cyclic rotation therefore does not change the derivative.

For $f \in \mathfrak{l}$, define a degree-zero derivation of R by

$$F_1(f)q = -\partial_p^{\mathrm{cyc}} \sigma f, \qquad F_1(f)p = \partial_q^{\mathrm{cyc}} \sigma f, \qquad F_1(f)t = 0. \tag{9.7}$$

A constant has zero cyclic derivatives, so this is well defined. The identity

$$[q, \partial_q^{\mathrm{cyc}} W] + [p, \partial_p^{\mathrm{cyc}} W] = 0 \tag{9.8}$$

follows by cancelling consecutive cyclic rotations of W. It implies $[d, F_1(f)] = 0$. Abelianization gives $\pi F_1(f) = X_f \pi$, where $X_f = \{f, -\}$.

Set $\mu_R = qp - pq$. Cyclic differentiation gives

$$\big([F_1(q^3), F_1(p^3)] - F_1(9q^2p^2)\big)(q, p, t) = \big(3[\mu_R, q], 3[\mu_R, p], 0\big). \tag{9.9}$$

The derivation

$$F_2(q^3, p^3)(q, p, t) = \big(3[t, q], 3[t, p], 3t^2\big) \tag{9.10}$$

has degree -1 and has this defect as its differential $[d, F_2]$. On q, p the assertion follows from $dt = \mu_R$. On t, the identity

$$d(t^2) = \mu_R t - t\mu_R$$

cancels the term $F_2(\mu_R)$. The value on t is therefore necessary even though the original defect vanishes on t.

The failure of strictness cannot be removed by another polynomial choice of cyclic words. Let

$$\mathcal{N} = \mathbb{C}\langle q, p\rangle / \big(\mathbb{C}1 + [\mathbb{C}\langle q, p\rangle, \mathbb{C}\langle q, p\rangle]\big),$$

where the commutators span a vector subspace. Its elements are cyclic words modulo constants. The necklace bracket is

$$\{[U], [V]\}_{\mathcal{N}} = \big[\partial_q^{\mathrm{cyc}} U \, \partial_p^{\mathrm{cyc}} V - \partial_p^{\mathrm{cyc}} U \, \partial_q^{\mathrm{cyc}} V\big].$$

This bracket has a useful derivation description. Send a cyclic word U to $D_U q = -\partial_p^{\mathrm{cyc}} U$ and $D_U p = \partial_q^{\mathrm{cyc}} U$. Differentiating the displayed bracket gives $[D_U, D_V] = D_{\{[U],[V]\}_{\mathcal{N}}}$: terms selecting two positions in one cyclic word cancel in pairs, while terms selecting one position in each word give the two derivation composites. This also proves that the bracket is well defined under cyclic rotation. The map to derivations is injective modulo constants, since for a homogeneous word of length $r > 0$,

$$[q \, \partial_q^{\mathrm{cyc}} U + p \, \partial_p^{\mathrm{cyc}} U] = r[U].$$

Jacobi follows from Jacobi for derivations and this injectivity. Abelianization is a Lie map $\mathcal{N} \to \mathfrak{l}$.

Proposition 9.2. *The abelianization $\mathcal{N} \to \mathfrak{l}$ has no linear Lie section, including sections that do not preserve polynomial degree.*

Proof. In degrees at most three there is only one cyclic word for each commutative monomial. A lift of a cubic therefore differs from its cyclic cubic by terms of degree at least four. Bracketing such a correction with a cubic has degree at least five.

Put $C = [q^2 p^2]$ and $D = [q\,p q\,p]$. They are distinct cyclic words, and direct differentiation gives

$$\{[q^3], [p^3]\}_{\mathcal{N}} = 9C, \qquad \{[q^2 p], [q p^2]\}_{\mathcal{N}} = C + 2D.$$

The commutative brackets are respectively $9q^2p^2$ and $3q^2p^2$. The degree-four part of a Lie section applied to q^2p^2 would have to equal both C and $(C + 2D)/3$. Higher-degree corrections cannot change this contradiction. □

This obstruction concerns a strict polynomial necklace section. The relation generator in (9.4) provides the larger differential graded carrier on which (9.10) solves the first defect.

9.4 Solving the action equations in every arity

Give $\mathrm{Der}(R)$ its graded commutator and differential $\partial D = [d, D]$. The derivations with zero projection form a differential graded Lie algebra

$$K = \{D : \pi D = 0\}.$$

A degree-j derivation in K is specified by $U = Dq, V = Dp \in J^j$ and $W = Dt \in J^{j-1}$. Put $L(U,V) = [U,p] + [q,V]$. Then

$$(\partial D)(q,p,t) = \big(dU, dV, dW - (-1)^j L(U,V)\big). \tag{9.11}$$

The contraction of words yields a contraction of this entire derivation complex, including its nonclosed elements.

Lemma 9.3. *The degree-minus-one map*

$$(HD)(q,p,t) = \big(hU, hV, h(W - (-1)^j L(hU, hV))\big) \tag{9.12}$$

preserves K and satisfies $\partial H + H\partial = 1_K$. Every value of H is a finite polynomial.

Proof. The ideal J contains all arguments of h in the formula. Since $\pi h = 0$, the resulting derivation again lies in K. On q, p the contraction identity is $dh + hd = 1_J$. Use $dL(U,V) = L(dU, dV)$ to substitute the formula on t. The sum of the third components is

$$(dh + hd)W + (-1)^j hL\big((dh + hd - 1)U, (dh + hd - 1)V\big) = W.$$

The word contraction is finite, so the formula has the stated polynomial meaning. □

We specify all higher signs by a coalgebra convention. For a differential graded Lie algebra $\mathfrak{g}$, let s be the degree-minus-one shift to $\mathfrak{g}[1]$. The cofree conilpotent symmetric coalgebra $S^c(\mathfrak{g}[1]) = \bigoplus_{n \geq 0} S^n(\mathfrak{g}[1])$ has the unshuffle coproduct. Its Lie codifferential has corestrictions

$$b_1(sa) = -s\partial a, \qquad b_2(sa\,sb) = (-1)^{|a|} s[a,b]. \tag{9.13}$$

An L_∞ morphism is a coaugmented coalgebra map commuting with these codifferentials. On degree-zero source inputs, its arity-n corestriction is sF_n, where $F_n : \Lambda^n \mathfrak{l} \to \mathrm{Der}(R)$ has degree $1 - n$.

Suppose $F_1, \dots, F_{n-1}$ are defined. Their corestrictions extend by set partitions to a coalgebra map $\mathcal{F}$ with arity-n corestriction zero. Define Ω_n by

$$\mathrm{pr}_1(b\mathcal{F} - \mathcal{F}b)\big|_{S^n} = s\Omega_n, \qquad F_n = H\Omega_n.$$

For the first two residuals this says

$$\Omega_2(f,g) = [F_1(f), F_1(g)] - F_1(\{f,g\}), \tag{9.14}$$

$$\Omega_3(f,g,k) = \sum_{\mathrm{cyc}} \big([F_1(f), F_2(g,k)] - F_2(\{f,g\},k)\big). \tag{9.15}$$

At arity four, the brackets between two binary components contribute

$$\begin{aligned} &- [F_2(f_1,f_2), F_2(f_3,f_4)] + [F_2(f_1,f_3), F_2(f_2,f_4)] \\ &\qquad - [F_2(f_1,f_4), F_2(f_2,f_3)]. \end{aligned} \tag{9.16}$$

They accompany the unary–ternary and source-bracket terms.

Theorem 9.4. *The specified first action and the recursion $F_n = H\Omega_n$ define an L_∞ morphism*

$$F : \mathfrak{l} \longrightarrow \mathrm{Der}(R).$$

Every component has finite polynomial values on polynomial inputs. For every $N \geq 1$, matrix substitution gives a GL_N-equivariant L_∞ action on R_N, and hence on $R_N^{GL_N}$.

Proof. The first component commutes with d by (9.8). Let $E = b\mathcal{F} - \mathcal{F}b$. The identity $bE + Eb = 0$ follows from the square-zero codifferentials. If E vanishes below arity n, its arity-n corestriction therefore satisfies $\partial\Omega_n = 0$.

The first action projects to the strict Hamiltonian action on A, so Ω_2 projects to zero. Moreover F_1 normalizes K, and every constructed component of arity at least two belongs to K. The same is consequently true of each subsequent residual. Lemma 9.3 gives $\partial H\Omega_n = \Omega_n$. In the coalgebra equation, the new term is $-s\partial F_n$, by (9.13); it cancels $s\Omega_n$. Induction proves all the identities.

At each arity there are finitely many partitions. The word contraction terminates on every term, proving polynomial finiteness. A word derivation acts on matrix entries by substituting X, Y, Ψ for q, p, t in its generator values. The graded Leibniz rule makes this substitution preserve differentials and brackets. It also commutes with simultaneous conjugation, which proves the two matrix assertions. □

The contraction gives explicit higher terms beyond the binary homotopy. For $\mathbf{f} = (q^3, p^3, qp^3)$ and $\mathbf{g} = (q^3, p^3, qp^3, q^2p^2)$, one obtains

$$\begin{aligned} F_3(\mathbf{f})q &= -\frac{9}{2}(tpt + pt^2), \\ F_4(\mathbf{g})q &= \frac{33}{5}t^3p + \frac{39}{5}t^2pt + \frac{69}{5}tpt^2 + \frac{99}{5}pt^3. \end{aligned} \tag{9.17}$$

The other generator values are $F_3(\mathbf{f})p = F_3(\mathbf{f})t = 0$ and $F_4(\mathbf{g})p = F_4(\mathbf{g})t = 0$. These formulas follow by applying (9.12) to the residuals. They can also be checked by substituting $dt = qp - pq$ in $\partial F_n = \Omega_n$ on each generator. The four-input identity uses (9.16). Thus this action does not truncate at its binary term. The displayed values depend on the contraction; a nonzero chosen component alone does not establish an invariant obstruction to changing the action's presentation.

9.5 Choices and pointed formal Hamiltonians

Let $\mathrm{Der}_\pi(R)$ be the derivations preserving $J = \ker \pi$. Projection gives an exact sequence of differential graded Lie algebras

$$0 \longrightarrow K \longrightarrow \mathrm{Der}_\pi(R) \xrightarrow{\pi_*} \mathrm{Der}(A) \longrightarrow 0. \tag{9.18}$$

Surjectivity is degreewise. A derivation of A, concentrated in degree zero, lifts by choosing words for its values on q, p and giving value zero on t. Such a lift preserves J, although it need not commute with d. The contracted kernel shows that π_* is a quasi-isomorphism.

For a differential graded Lie algebra L, retain homotopies between actions by using the arity-completed convolution differential graded Lie algebra

$$\mathcal{G}(L) = \prod_{n\geq 1} \mathrm{Hom}(S^n(\mathfrak{l}[1]), L), \qquad \mathcal{F}^r\mathcal{G}(L) = \prod_{n\geq r} \mathrm{Hom}(S^n(\mathfrak{l}[1]), L).$$

The differential combines the source Chevalley codifferential with that of L; the bracket uses the reduced coproduct. For a degree-one element α, the Maurer–Cartan equation $\partial\alpha + [\alpha, \alpha]/2 = 0$ is the L_∞ morphism equation in the convention (9.13). Polynomial differential forms on a simplex give parameter families of such equations. Tensoring arity by arity defines their simplicial space.

The *derived space of lifts* is the homotopy fibre, over $f \mapsto X_f$, of the map of these spaces induced by π_*. This fixes the action on A and the comparison to it; it does not yet impose a cyclic or Hamiltonian condition on the lifts.

Proposition 9.5. *The derived space of lifts of X to $\mathrm{Der}_\pi(R)$ is nonempty and contractible. The word contraction selects one point of this space.*

Proof. On the arity-n associated graded, the kernel is

$$\mathrm{Hom}(S^n(\mathfrak{l}[1]), K),$$

contracted by postcomposition with H, with the tensor sign when parameter forms are present. Polynomial forms on a simplex restrict degreewise onto compatible forms on a boundary or horn. One proves this by extending faces successively: coefficients vanishing on earlier faces lie in the corresponding ideals (x_i, dx_i), and the barycentric relation $\sum x_i = 1$ handles the last face. No cochain primitive in the parameter direction is required.

Suppose the lower arities and a compatible boundary family are already lifted. Choose an arity-n component with that boundary and the prescribed projection. Its curvature lies in the kernel above, tensored with forms relative to the boundary. The arity-n Bianchi identity makes the curvature closed. The contraction gives a relative primitive; subtracting it solves the equation without changing the boundary or projection. The same argument fills horns. Induction and arity completeness give a trivial fibration of simplicial spaces. Its nonempty fibre is contractible. □

There is also a completion in polynomial weight. Give q, p weight one and t weight two; write R_w for the homogeneous piece and put $\widehat{R} = \prod_{w\geq 0} R_w$. The product topology has discrete homogeneous pieces. Assign the same weights to X, Y, Υ to define $\widehat{R}_N$. A pointed Hamiltonian has no constant or linear term.

Proposition 9.6. *On homogeneous inputs of degrees $d_1, \ldots, d_n$, the derivation F_n shifts weight by*

$$\sum_{i=1}^{n} (d_i - 2).$$

At each arity, the action therefore extends continuously from pointed polynomial Hamiltonians to

$$\widehat{\mathfrak{l}}_{\mathrm{pt}} = \prod_{d\geq 2} \mathrm{Sym}^d(\mathbb{C}q \oplus \mathbb{C}p)$$

on $\widehat{R}$, and to the corresponding equivariant action on $\widehat{R}_N$.

Proof. Both d and h preserve weight. Cyclic differentiation gives the claimed shift for $F_{\mathfrak{l}}$. Brackets add shifts, and the derivation contraction preserves them, so induction proves the formula for every component.

For pointed inputs, each $d_i - 2$ is nonnegative. At a fixed arity and output weight, only finitely many source weights and input degrees can contribute. Their homogeneous spaces are finite dimensional. The operation and each of its identities are thus finite in every output weight, giving the continuous extension. Matrix substitution preserves these weights. □

9.6 Gauge ghosts and Hamiltonian primitives

The matrices also carry simultaneous conjugation. Introduce an odd matrix c of degree 1 and the polynomial graded commutative algebra

$$\mathcal{B}_N = \mathbb{C}[X_{ij}, Y_{ij}] \otimes \Lambda(c_{ij}, \Psi_{ij}).$$

Extend the even Poisson bracket by

$$\{c_{ij}, \Psi_{kl}\} = \{\Psi_{kl}, c_{ij}\} = \delta_{il}\delta_{jk},$$

with all other new generator brackets zero. The symmetry on these two odd generators is the graded skew-symmetry of an even Poisson bracket. The degree-one Hamiltonian

$$\Theta = \mathrm{Tr}(c[X, Y] - \Psi c^2)$$

defines $Q = \{\Theta, -\}$, with

$$QX = -[c, X], \quad QY = -[c, Y], \quad Qc = -c^2, \quad Q\Psi = [X, Y] - [c, \Psi]. \tag{9.19}$$

Here $[c, \Psi] = c\Psi + \Psi c$. Applying the graded Leibniz rule once more to the four expressions gives $Q^2 = 0$. This is the classical BFV, or Hamiltonian BRST, complex. Equivalently it is the Chevalley–Eilenberg algebra $C^*_{\mathrm{CE}}(\mathfrak{gl}_N; R_N)$ with the displayed gauge sign. Its positive ghosts retain the infinitesimal gauge action in addition to the negative relation generators.

Proposition 9.7. *Every component of the matrix action extends to $\mathcal{B}_N$ by giving value zero on c. These extensions define an L_∞ action of $\mathfrak{l}$ on $\mathcal{B}_N$.*

Proof. Let D be a word derivation of degree j with $Dc = 0$. For a homogeneous matrix word M, the graded Leibniz rule gives $D[c, M] = (-1)^j[c, DM]$. Thus the extended derivation commutes in the graded sense with the gauge part of (9.19). Extension preserves both the differential and the bracket of derivations, so it preserves every action identity. □

The first derivation is Hamiltonian: it equals $H_{J_N(f)} = \{J_N(f), -\}$. Each $J_N(f)$ is Q-closed by conjugation invariance. The cubic scalar defect has a Hamiltonian primitive

$$K_N = -\frac{3}{2}\,\mathrm{Tr}(\Psi\mu), \qquad QK_N = -\frac{3}{2}\,\mathrm{Tr}(\mu^2). \tag{9.20}$$

Graded cyclicity cancels the gauge terms in QK_N; (9.2) identifies its remaining term with the bracket defect. Its Hamiltonian vector field is

$$\begin{aligned} H_{K_N}X &= \tfrac{3}{2}[\Psi, X], \quad & H_{K_N}Y &= \tfrac{3}{2}[\Psi, Y], \\ H_{K_N}c &= -\tfrac{3}{2}\mu, & H_{K_N}\Psi &= 0. \end{aligned} \tag{9.21}$$

The extended word homotopy, in contrast, has

$$DX = 3[\Psi, X], \quad DY = 3[\Psi, Y], \quad Dc = 0, \quad D\Psi = 3\Psi^2. \tag{9.22}$$

Both satisfy

$$[Q, D] = [Q, H_{K_N}] = [H_{J_N(q^3)}, H_{J_N(p^3)}] - H_{J_N(9q^2p^2)}.$$

The ghost component in (9.21) is part of its Hamiltonian solution.

For $N \geq 2$, D is not Hamiltonian for the displayed bracket. Indeed, $Dc = 0$ would force a polynomial Hamiltonian for D to be independent of every Ψ_{ij}. Such a polynomial has no component of degree -1, while D is a nonzero degree-minus-one derivation. A coherent derivation action therefore does not by itself supply Hamiltonian representatives for its homotopies.

There is a prior scalar obstruction to a homotopy moment map on the quotient $\mathfrak{l}$. Choose representatives with constant term zero. Their bracket can acquire a constant: the scalar cocycle is $\kappa(f, g) = \{f, g\}(0)$.

Proposition 9.8. *For $N > 0$, no L_∞ morphism $\mathfrak{l} \to (\mathcal{B}_N, Q, \{-,-\})$ has first component $f \mapsto J_N(f)$ on those representatives. The obstruction persists if the first components are changed by Q-boundaries.*

Proof. The classes of q, p have zero bracket in $\mathfrak{l}$, whereas

$$\{\operatorname{Tr} X, \operatorname{Tr} Y\} = N.$$

The binary morphism equation would make N a Q-boundary. Evaluation of X, Y, c, Ψ at zero is a cochain map to $\mathbb{C}$, since every value in (9.19) is quadratic. It sends the alleged boundary to the nonzero scalar N. Exact changes of closed charges leave their cohomology bracket unchanged, so they cannot remove this obstruction. □

Retaining the central extension $A \to \mathfrak{l}$, with $J_N(1) = N$, removes this particular obstruction. The cocycle also vanishes on the classical pointed subalgebra $(q, p)^2$. On vector fields constants already act by zero, which explains why Theorem 9.4 is consistent with the proposition.

The central extension contains a strict finite sector. Write $e = q^2/2$, $f = p^2/2$ and $h_{\mathrm{quad}} = qp$. Besides skew versions, its nonzero brackets are

$$\{q, p\} = 1, \quad \{e, p\} = q, \quad \{f, q\} = -p, \quad \{e, f\} = h_{\mathrm{quad}},$$

$$\{h_{\mathrm{quad}}, q\} = -q, \quad \{h_{\mathrm{quad}}, p\} = p, \quad \{h_{\mathrm{quad}}, e\} = -2e, \quad \{h_{\mathrm{quad}}, f\} = 2f.$$

The map J_N sends $1, q, p, e, h_{\mathrm{quad}}, f$ to N, $\operatorname{Tr} X$, $\operatorname{Tr} Y$, $\operatorname{Tr}(X^2)/2$, $\operatorname{Tr}(XY)$, $\operatorname{Tr}(Y^2)/2$. Canonical matrix differentiation gives precisely the same list of brackets. Thus scalar quadratics act by a strict Hamiltonian Lie map of closed functions. Higher scalar Hamiltonians require the homotopies whose first instance is (9.20).

9.7 A cyclic orientation of the resolution

The symplectic plane has the volume form $dq \wedge dp$. To preserve it on a resolution, the action must act on a complex that remembers cyclic order as well as the differential. Use normalized Hochschild chains

$$C_n(R) = R \otimes \overline{R}^{\otimes n}, \qquad \overline{R} = R/\mathbb{C}1,$$

written $a_0[a_1|\cdots|a_n]$. A unit in any bar slot represents zero. A chain of internal degree $m = \sum_i |a_i|$ has total cohomological degree $m - n$. Let $d_\otimes$ be the tensor differential with its ordinary Koszul signs. In the raw-chain convention the Hochschild differential is

$$d_{\mathrm{H}} = (-1)^n d_\otimes + b, \tag{9.23}$$

where

$$\begin{aligned} b(a_0[a_1|\cdots|a_n]) = {} & a_0a_1[a_2|\cdots|a_n] \\ & + \sum_{j=1}^{n-1} (-1)^j a_0[a_1|\cdots|a_ja_{j+1}|\cdots|a_n] \\ & + (-1)^{n+|a_n|\sum_{i<n}|a_i|} a_na_0[a_1|\cdots|a_{n-1}]. \end{aligned}$$

For $n = 0$ set $b = 0$. The normalized Connes operator is

$$B(a_0[a_1|\cdots|a_n]) = \sum_{j=0}^{n} (-1)^{nj+(\sum_{i<j}|a_i|)(\sum_{i\geq j}|a_i|)} 1[a_j|\cdots|a_n|a_0|\cdots|a_{j-1}]. \tag{9.24}$$

Adjacent face cancellations and cyclic rotation give $d_{\mathrm{H}}^2 = B^2 = d_{\mathrm{H}}B + Bd_{\mathrm{H}} = 0$. Thus $(C_\bullet(R), d_{\mathrm{H}}, B)$ is a mixed complex, with the second operator of degree -1.

Let ζ have degree two. The negative-cyclic complex is

$$M_R = \big(C_\bullet(R)[[\zeta]], d_{\mathrm{H}} + \zeta B\big).$$

Its degree-minus-two orientation is

$$\omega_0 = 1[q|p] - 1[p|q] - 1[t]. \tag{9.25}$$

The first two terms have Hochschild boundary $-1[qp - pq]$. The internal differential of the last term, with the sign in (9.23), is $+1[qp - pq]$. Its remaining Hochschild boundary is zero. Every Connes term has a unit in a bar slot. Hence $d_{\mathrm{H}}\omega_0 = B\omega_0 = 0$.

The sign of $1[t]$ is tied to this totalization. Multiplication of a chain of bidegree (m, n) by $(-1)^{mn}$ changes the differential to $d_\otimes + (-1)^m b$ and the Connes operator to $(-1)^m B$; in that convention the last sign is positive. The explicit formulas below use the raw convention throughout.

Let Ω^a_A denote polynomial differential a-forms with de Rham differential d_{dR}, and place it in cohomological degree $-a$. The normalized Hochschild–Kostant–Rosenberg projection is

$$p_{\mathrm{H}}(a_0[a_1|\cdots|a_n]) = \frac{1}{n!}\pi(a_0)\, d_{\mathrm{dR}}\pi(a_1) \wedge \cdots \wedge d_{\mathrm{dR}}\pi(a_n). \tag{9.26}$$

Substitution in the face and cyclic formulas gives

$$p_{\mathrm{H}}d_{\mathrm{H}} = 0, \qquad p_{\mathrm{H}}B = d_{\mathrm{dR}}p_{\mathrm{H}}, \qquad p_{\mathrm{H}}\omega_0 = dq \wedge dp.$$

The factorial is essential to the last normalization.

Proposition 9.9. *There is a weight-preserving contraction* $(p_{\mathrm{H}}, i_{\mathrm{H}}, H_{\mathrm{H}})$ *from* $(C_\bullet(R), d_{\mathrm{H}})$ *to* $(\Omega^\bullet_A, 0)$, *with* $i_{\mathrm{H}}(dq \wedge dp) = \omega_0$ *and*

$$H_{\mathrm{H}}^2 = H_{\mathrm{H}} i_{\mathrm{H}} = p_{\mathrm{H}} H_{\mathrm{H}} = 0.$$

Its negative-cyclic perturbation has unchanged projection and target differential $\zeta\, d_{\mathrm{dR}}$. *More precisely,*

$$\begin{aligned} i_\zeta &= (1 + \zeta H_{\mathrm{H}} B)^{-1} i_{\mathrm{H}}, \\ H_\zeta &= H_{\mathrm{H}}(1 + \zeta B H_{\mathrm{H}})^{-1}, \quad p_\zeta = p_{\mathrm{H}}. \end{aligned} \tag{9.27}$$

The choices can be made by finite rational linear algebra at each polynomial weight.

Proof. Give q, p, t weights 1, 1, 2 and each bar chain the sum of its generator weights. Normalized bar slots have positive weight, so the bar-length filtration is finite in each fixed weight. The quasi-isomorphism $R \to A$ therefore gives a quasi-isomorphism of Hochschild chains weight by weight: on each associated-graded piece the comparison is a tensor product of quasi-isomorphisms over the field.

Write $A^e = A \otimes A^{\mathrm{op}}$, and use subscripts L, R for its two factors. The diagonal of A has the bimodule Koszul resolution

$$A^e \otimes \Lambda(e_q, e_p), \qquad de_q = q_L - q_R, \qquad de_p = p_L - p_R.$$

The two differences form a regular sequence, since they become independent polynomial variables after a linear change of coordinates. Tensoring with A over A^e kills this differential and identifies e_q, e_p with dq, dp. The bar resolution is another free resolution of the diagonal; its augmented contraction inserts the unit. The comparison sends $e_q \wedge e_p$ to the antisymmetrized bar $1[q|p] - 1[p|q]$. Consequently (9.26) is a surjective quasi-isomorphism with the stated normalization.

Choose lifts of a basis of forms to cycles, using ω_0 for the volume. These give i_{H}. In the acyclic kernel, choose degreewise splittings

$$(\ker p_{\mathrm{H}})^r = B^r \oplus L^r, \qquad d_{\mathrm{H}} : L^r \xrightarrow{\sim} B^{r+1}.$$

Take H_{H} to be the inverse on B^{r+1} and zero on L and the chosen cycle lifts. This proves the contraction and side identities. Every space in a fixed weight is finite dimensional; lexicographic word bases and row reduction make the choices rational.

Apply perturbation by ζB. Its series converge in the ζ-adic topology. The relation

$$p_{\mathrm{H}} \zeta B H_{\mathrm{H}} = \zeta\, d_{\mathrm{dR}} p_{\mathrm{H}} H_{\mathrm{H}} = 0$$

keeps the projection fixed. It also kills every higher term in the transferred differential $p_{\mathrm{H}} \zeta B (1 + \zeta H_{\mathrm{H}} B)^{-1} i_{\mathrm{H}}$, leaving exactly $\zeta\, d_{\mathrm{dR}}$. Since $B\omega_0 = 0$, the perturbed inclusion still sends the volume to ω_0. □

There is no unbounded filtration exchange in this comparison. At fixed weight the bar length and the negative internal degree are bounded. Fixing total degree then bounds the exponent of ζ. The completed identities can therefore be checked in finite-dimensional pieces as well as in finite parameter quotients.

9.8 Coherent preservation of the volume

A derivation D of degree r acts on raw chains by

$$\mathcal{L}_D(a_0[a_1|\cdots|a_n]) = \sum_{j=0}^{n} (-1)^{nr + r\sum_{i<j}|a_i|} a_0[\cdots|D(a_j)|\cdots|a_n], \tag{9.28}$$

where the $j = 0$ term acts on the outer coefficient. The factor $(-1)^{nr}$ is required by the raw totalization. The tensor signs give

$$[d_{\mathrm{H}}, \mathcal{L}_D] = \mathcal{L}_{[d,D]}, \qquad [B, \mathcal{L}_D] = 0, \qquad [\mathcal{L}_D, \mathcal{L}_E] = \mathcal{L}_{[D,E]}.$$

Thus the action F makes M_R an L_∞ module.

For explicit totalization, write $C = S^c(\mathfrak{l}[1])$ and let b_C be its Chevalley codifferential. The twisting cochain $\alpha : C \to \mathrm{Der}(R)$ has degree one and

$$\alpha_n(sf_1 \cdots sf_n) = F_n(f_1, \dots, f_n).$$

Its equation is $\partial\alpha + \alpha b_C + [\alpha, \alpha]/2 = 0$, where convolution uses the tensor Koszul sign. On the completed coefficient complex

$$C_R = \prod_{n \geq 0} \mathrm{Hom}(S^n(\mathfrak{l}[1]), M_R),$$

the differential of a homogeneous map a is

$$D_F a = d_{M_R} a - (-1)^{|a|} a b_C + \mathcal{L}_\alpha \star a. \tag{9.29}$$

Here $\star$ denotes action convolution with the unshuffle coproduct. The twisting-cochain equation implies $D_F^2 = 0$.

The projection to forms intertwines the first action and kills the higher ones:

$$p_{\mathrm{H}} \mathcal{L}_{F_1(f)} = \mathcal{L}_{X_f} p_{\mathrm{H}}, \qquad p_{\mathrm{H}} \mathcal{L}_{F_n} = 0 \quad (n \geq 2). \tag{9.30}$$

The first equation is naturality of the differential-form formula. For the second, a negative-degree value of a derivation contains t and is killed by π.

A coherent lift is a D_F-closed element of total degree -2 whose projection is the volume, regarded as a constant arity-zero cochain. Its arity-n component is an alternating map $\omega_n : \Lambda^n \mathfrak{l} \to M_R^{-n-2}$. Polynomial parameter forms retain homotopies over this fixed projection and define the derived space of such lifts.

Theorem 9.10. *The orientation ω_0 extends to a D_F-closed element ω_{eq} of total degree -2. Over the fixed Hamiltonian action and fixed projection to $dq \wedge dp$, the derived space of coherent negative-cyclic lifts is contractible. The construction is continuous for pointed Hamiltonians in the positive-weight completion.*

Proof. Extend the contraction of Proposition 9.9 to the convolution complex, with its total signs. Let $\mathcal{H}_\zeta$ denote the extended homotopy and let δ_F denote the terms in (9.29) beyond the coefficient differential d_{M_R}. They raise arity. Perturbation in the arity filtration gives

$$\omega_{\mathrm{eq}} = (1 + \mathcal{H}_\zeta \delta_F)^{-1} i_\zeta(dq \wedge dp). \tag{9.31}$$

Equation (9.30) and $p_{\mathrm{H}} \mathcal{H}_\zeta = 0$ imply $p_{\mathrm{H}} \delta_F \mathcal{H}_\zeta = 0$. The transferred differential is therefore precisely the strict Hamiltonian action on forms together with $\zeta\, d_{\mathrm{dR}}$. The volume is closed and its Lie derivative under X_f is zero, since $\partial_q(-f_p) + \partial_p(f_q) = 0$. Its perturbed inclusion is consequently closed. Its arity-zero component is ω_0.

The same perturbation contracts the kernel of the fixed projection. The homotopy fibre over the specified closed volume is an affine translate of this acyclic complex. Polynomial parameter forms and its contraction fill relative families, so its derived mapping space is contractible.

At fixed arity only finitely many arity-raising factors occur. At fixed ζ precision the mixed perturbation is finite. Finally, the weight estimate in Proposition 9.6 makes every output weight a finite sum for pointed inputs. These three finiteness statements prove the completion assertion. □

Two coherence levels can be written without a matrix inversion. Abbreviate $\mathrm{I}[a_1|\cdots|a_n]$ by $\langle a_1|\cdots|a_n\rangle$. The arity-one equation is

$$d_{\mathrm{H}}\omega_1(f) = -\mathcal{L}_{F_1(f)}\omega_0.$$

Choose

$$\omega_1(q^3) = -3\langle q|q|q\rangle, \qquad \omega_1(p^3) = -3\langle p|p|p\rangle. \tag{9.32}$$

For instance, $d_{\mathrm{H}}\langle q|q|q\rangle = \langle q|q^2\rangle - \langle q^2|q\rangle$. A compatible quartic prescription is

$$\begin{aligned}\omega_1(q^2p^2) = {} & -\tfrac{2}{3}\langle tt\rangle \\ & + \tfrac{2}{3}\big(\langle q|pt\rangle - \langle q|tp\rangle - \langle p|qt\rangle + \langle p|tq\rangle\big) \\ & + \tfrac{2}{3}\big(\langle qt|p\rangle - \langle pt|q\rangle - \langle tq|p\rangle + \langle tp|q\rangle\big) \\ & - 2\big(\langle q|p|qp\rangle + \langle p|qp|q\rangle + \langle qp|q|p\rangle\big).\end{aligned} \tag{9.33}$$

All three chains have zero Connes boundary and zero HKR projection. For $f = q^3$, $g = p^3$, the second equation in the same convention is

$$\begin{aligned}d_{\mathrm{H}}\omega_2(f,g) = {} & \omega_1(\{f,g\}) - \mathcal{L}_{F_1(f)}\omega_1(g) + \mathcal{L}_{F_1(g)}\omega_1(f) \\ & - \mathcal{L}_{F_2(f,g)}\omega_0.\end{aligned} \tag{9.34}$$

Its solution

$$\begin{aligned}\tfrac{1}{9}\omega_2(q^3,p^3) = {} & \langle t|t\rangle + \langle q|p|t\rangle - \langle p|q|t\rangle \\ & - \langle q|t|p\rangle + \langle p|t|q\rangle + \langle t|q|p\rangle - \langle t|p|q\rangle \\ & - \langle q|q|p|p\rangle + \langle q|p|q|p\rangle + \langle q|p|p|q\rangle \\ & + \langle p|q|q|p\rangle - \langle p|q|p|q\rangle - \langle p|p|q|q\rangle\end{aligned} \tag{9.35}$$

also has zero Connes boundary and HKR projection. Substitution in the raw differential verifies these identities: internal terms replace t by $qp - pq$, and adjacent mergers cancel the cyclic faces. In particular the $\langle t|t\rangle$ term is required in the second equation. The contracted projection kernel extends these prescribed lower components to all arities, even if a particular row-reduction rule chooses different representatives.

9.9 The finite matrix tangent pairing

The diagonal Koszul resolution also explains the nondegeneracy of the orientation. Its top exterior product pairs the resolution with its dual and gives

$$\mathrm{RHom}_{A^e}(A, A^e) \simeq A[-2].$$

The volume $dq \wedge dp$ fixes this identification. Together with its negative-cyclic lift, this is the smooth 2-Calabi–Yau orientation of the plane: the diagonal is a perfect bimodule, and its bimodule dual is the displayed shift. Extension and restriction along the quasi-isomorphism $R \to A$ transport this duality to R. Properness would require finite total cohomology over $\mathbb{C}$, which A does not have. The orientation is a bimodule duality, distinct from a nondegenerate scalar trace on a proper algebra.

Take the quotient by GL_N of the derived affine scheme represented by R_N. At a commuting pair (X, Y) its tangent complex is

$$\underbrace{\mathfrak{gl}_N}_{-1} \xrightarrow{d_0} \underbrace{\mathrm{Mat}_N \oplus \mathrm{Mat}_N}_{0} \xrightarrow{d_1} \underbrace{\mathfrak{gl}_N}_{1}, \tag{9.36}$$

with

$$d_0(a) = ([a, X], [a, Y]), \qquad d_1(U, V) = [U, Y] + [X, V].$$

The first arrow differentiates conjugation; the second differentiates the relation. Their composite is $[a, [X, Y]] = 0$. Put

$$\omega((U, V), (U', V')) = \mathrm{Tr}(UV' - VU') \tag{9.37}$$

on the middle term. Pair the terms in degrees -1 and 1 by $\mathrm{Tr}(ab)$, symmetrically under reversal as required by graded skew-symmetry between two odd degrees. Trace invariance gives

$$\omega(d_0 a, (U, V)) = \mathrm{Tr}(a\, d_1(U, V)).$$

Thus these pairings form a degree-zero cochain pairing. They are perfect term by term, including at points with stabilizers. Discarding either outer term would discard part of this self-dual tangent complex. This finite tangent calculation and coherent cyclic invariance do not require every chosen derivation homotopy to be Hamiltonian for one fixed chain-level symplectic form.

9.10 Deforming the coefficient relation

Let $k_\varepsilon = \mathbb{C}[\varepsilon]$, with $|\varepsilon| = 0$, and define

$$\begin{aligned} W_\varepsilon &= k_\varepsilon\langle q, p\rangle/(qp - pq - \varepsilon), \\ R_\varepsilon = k_\varepsilon\langle q, p, t\rangle, \qquad |t| &= -1, \qquad dt = qp - pq - \varepsilon, \qquad dq = dp = 0. \end{aligned} \tag{9.38}$$

The projection π_ε kills t. Normal words $p^b q^a$ give a k_ε-basis of W_ε: the replacement $qp \mapsto pq + \varepsilon$ terminates, and the leading pattern qp has no overlap ambiguity. They define a linear section i_ε.

Use the same marked-word contraction h_0 as before, with $d_0 t = qp$, but now put $\delta t = -pq - \varepsilon$. Order expanded words first by length and then by the number of pairs in which q precedes p. Exchanging qp and pq decreases the second number; the constant term decreases the first. Every term of δh_0 therefore decreases this well-founded order.

Proposition 9.11. *The finite series*

$$h_\varepsilon = h_0(1 + \delta h_0)^{-1}, \qquad p' = p_0(1 + \delta h_0)^{-1}$$

satisfy

$$dh_\varepsilon + h_\varepsilon d = 1 - i_\varepsilon \pi_\varepsilon, \quad h_\varepsilon^2 = h_\varepsilon i_\varepsilon = \pi_\varepsilon h_\varepsilon = 0, \quad p' = \pi_\varepsilon.$$

They extend continuously to the ε-adic completions, allowing word support to grow with parameter order.

Proof. The marked-word proof of Proposition 9.1 applies with the decreasing order just described. Each surviving reduction is precisely the PBW rule $qp \mapsto pq + \varepsilon$, so the perturbed projection is π_ε. No division by ε occurs.

An element of the completed source is a series $\sum_{r \geq 0} \varepsilon^r a_r$ with each a_r a finite word polynomial. Every perturbation has nonnegative parameter order. At a fixed output order, only finitely many earlier coefficients contribute, and each has a terminating word calculation. This defines the continuous extensions. The identities hold modulo every power of ε, hence in the separated inverse limit. □

Let $m(f \otimes g) = fg$ be ordinary polynomial multiplication. On $k_\varepsilon[q, p]$, the Moyal product is

$$f \star_\varepsilon g = m \exp\Big[\frac{\varepsilon}{2}(\partial_q \otimes \partial_p - \partial_p \otimes \partial_q)\Big](f \otimes g). \tag{9.39}$$

It is finite on polynomial inputs. On three tensor factors, the three constant bidifferential operators commute; both parenthesizations are the exponential of their sum. This proves associativity. Symmetrization identifies this product with multiplication in W_ε. For linear forms u, v, centrality of their Weyl commutator gives

$$e^{\sigma u} e^{\sigma v} = e^{\varepsilon\{u,v\}/2} e^{\sigma(u+v)}.$$

Differentiation in a formal scalar parameter proves this exponential identity. Taking coefficients in u, v gives the polynomial product identity.

The commutator is divisible by ε in the polynomial ring. Write

$$B_\varepsilon(f, g) = (f \star_\varepsilon g - g \star_\varepsilon f)/\varepsilon, \qquad \mathfrak{l}_\varepsilon = k_\varepsilon[q, p]/k_\varepsilon 1.$$

Division here means the polynomial quotient; the parameter is not inverted. The quotient is a Lie algebra. The cyclic derivative formula (9.7) still defines a closed derivation of R_ε. Under π_ε it induces the strict Moyal derivation $B_\varepsilon(f, -)$. Indeed its values on q, p are the Weyl-ordered derivatives $-f_p, f_q$, and those generators determine a derivation of the Weyl algebra.

Theorem 9.12. *There is an L_∞ morphism over k_ε*

$$F^\varepsilon : \mathfrak{l}_\varepsilon \longrightarrow \mathrm{Der}_{k_\varepsilon}(R_\varepsilon)$$

with the specified first component. Each Taylor component is polynomial and finite at fixed arity on polynomial inputs. The action extends continuously to the parameter completions and gives, by matrix substitution, a GL_N-equivariant action on the derived representation algebra for every N.

Proof. Let K_ε be the derivations whose generator values lie in $\ker \pi_\varepsilon$. The differential of a degree-j derivation with values (U, V, W) is still (9.11): derivations over k_ε kill the added scalar. Replacing h by h_ε in (9.12) therefore contracts K_ε in every degree.

Use the coalgebra signs (9.13) and replace the source Poisson bracket by B_ε. The first action projects to a strict action on W_ε. Every higher residual is consequently closed and lies in K_ε. Applying its contraction supplies the next component, exactly as in Theorem 9.4. Finitely many partitions and the terminating word order prove polynomial finiteness. The maps never decrease parameter order, giving the continuous completed identities. Matrix substitution preserves differentials, brackets and conjugation over the same base ring. □

The first parameter-dependent correction is already visible on a cubic and a quartic:

$$\begin{aligned} B_\varepsilon(q^3, p^4) &= 12q^2p^3 + 6\varepsilon^2 p, \\ F_2^\varepsilon(q^3, p^4)q &= 6(-pqt + ptq - qtp + tpq - \varepsilon t), \\ F_2^\varepsilon(q^3, p^4)p &= 6(-p^2t + tp^2), \\ F_2^\varepsilon(q^3, p^4)t &= 6(pt^2 + tpt). \end{aligned} \tag{9.40}$$

The first and third odd Moyal terms give the bracket in the first line. Applying $\partial = [d, -]$ to the remaining lines gives the binary defect on all three generators. In particular $d(-6\varepsilon t)$ contributes $+6\varepsilon^2$

to the q-component, matching the contribution of $-F_1(6\varepsilon^2 p)$ to the defect. The third component on the ordered triple $(q^3, p^4, q^2 p)$ is

$$F_3^\varepsilon q = -12tqt + 12t^2 q, \qquad F_3^\varepsilon p = -12pt^2, \qquad F_3^\varepsilon t = -6t^3. \tag{9.41}$$

Substitution of $dt = qp - pq - \varepsilon$ verifies its ternary equation with the binary terms in (9.40).

The classical positive-weight completion and this parameter completion use different estimates. Higher Moyal terms can lower polynomial degree, as the term $\varepsilon^2 p$ shows. The parameter-adic theorem does not assert that the classical pointed polynomial subspace remains closed under the deformed bracket.

There is a relative negative-cyclic orientation over the parameter ring. Form normalized chains over k_ε, so that a parameter multiple of 1 is zero in a bar slot. In the raw totalization,

$$\omega_\varepsilon = 1[q|p] - 1[p|q] - 1[t] \tag{9.42}$$

satisfies

$$d_{\mathrm{H}}\omega_\varepsilon = -\varepsilon 1[1] = 0, \qquad B\omega_\varepsilon = 0.$$

This is a relative mixed-complex assertion. It uses the parameter in the base ring; an absolute de Rham variation of that parameter is additional data. The cycle itself also does not supply an equivariant cyclic comparison to a matrix-quantized recipient.

9.11 Matrix quantization and finite representations

Matrix quantization starts with the canonical matrix coordinates, using an independent parameter $\hbar$. Its even Weyl algebra $W_{N,\hbar}$ over $\mathbb{C}[\hbar]$ has

$$[X_{ij}, Y_{kl}] = \hbar\delta_{il}\delta_{jk}, \qquad [X_{ij}, X_{kl}] = [Y_{ij}, Y_{kl}] = 0.$$

Weyl ordering is symmetrization of all coordinate factors. The Moyal formula on this finite-dimensional symplectic vector space identifies its multiplication with the corresponding constant-tensor star product. The commutator of two scalar quadratic polynomials, divided by $\hbar$, is exactly their Poisson bracket: the third and higher odd Moyal terms differentiate each quadratic at least three times and vanish. Thus the strict scalar quadratic sector constructed above remains strict under matrix quantization.

The ordering of the matrix moment is part of this assertion. As operator-valued matrices, set

$$\widehat{\mu} = XY - YX - N\hbar I_N. \tag{9.43}$$

The Weyl symbols of XY and YX differ from their commuting symbols by $N\hbar I_N/2$ and $-N\hbar I_N/2$, respectively. Consequently $\widehat{\mu}$ has Weyl symbol $[X, Y]$ in commuting coordinates. In particular

$$\operatorname{Tr}(XY - YX) = N^2\hbar, \qquad \operatorname{Tr}\widehat{\mu} = 0.$$

This scalar ordering correction is fixed by the coordinate commutators. A specified nonzero value of a moment map is a separate datum.

The full quantum BRST algebra also retains both ghost degrees. Choose a basis a_i of $\mathfrak{gl}_N$ with $[a_i, a_j] = \sum_k f_{ij}^k a_k$, and set $\ell_i = \operatorname{Tr}(a_i\widehat{\mu})$. These are the Weyl-ordered quadratic comoments, with

$$[\ell_i, \ell_j] = \hbar \sum_k f_{ij}^k \ell_k.$$

This follows either by canonical matrix differentiation and exact quadratic quantization, or by realizing $X_{ij} = x_{ij}$ and $Y_{ij} = -\hbar\partial_{x_{ji}}$: $\ell_a = \hbar\tau(a)$, where $\tau(a)X = [X, a]$ is the infinitesimal pullback action.

Adjoin odd generators c^i, b_i of degrees 1, −1, commuting with the even matter generators and satisfying

$$c^i c^j + c^j c^i = 0, \qquad b_i b_j + b_j b_i = 0, \qquad b_i c^j + c^j b_i = \hbar\delta_i^j.$$

Write $\mathcal{B}_{N,\hbar}$ for the resulting graded tensor product with $W_{N,\hbar}$. Define

$$\Omega = \sum_i c^i \ell_i - \frac{1}{2}\sum_{i,j,k} f_{ij}^k c^i c^j b_k, \qquad Q_\hbar = \hbar^{-1}[\Omega, -]_{\mathrm{gr}}. \tag{9.44}$$

Modulo $\hbar$ the algebra is supercommutative, so the commutator is divisible by $\hbar$ and the formula is integral. Under the trace identification of the ghost and its dual, its classical limit is (9.19).

Proposition 9.13. *The element Ω satisfies $\Omega^2 = 0$. Consequently $Q_\hbar$ is a degree-one derivation with $Q_\hbar^2 = 0$ over $\mathbb{C}[\hbar]$.*

Proof. Normal words, with all c generators before all b generators, span the ghost algebra. Represent c^i by exterior multiplication and b_i by $\hbar\partial_{c^i}$, with left odd derivatives. After inverting $\hbar$, their normal words form a basis of the endomorphisms of the exterior space. They are therefore linearly independent over $\mathbb{C}[\hbar]$ as well. Together with the Weyl basis this proves parameter-torsion freeness and faithfulness on the tensor product of the matter regular module and the ghost exterior module.

In this representation,

$$\Omega = \sum_i c^i \ell_i + \hbar d_{\mathrm{CE}}, \qquad d_{\mathrm{CE}} c^k = -\frac{1}{2}\sum_{i,j} f_{ij}^k c^i c^j.$$

Jacobi gives $d_{\mathrm{CE}}^2 = 0$. The square of the first term is $\frac{\hbar}{2}\sum f_{ij}^k c^i c^j \ell_k$; its mixed term with $\hbar d_{\mathrm{CE}}$ is the negative of this expression. Thus $\Omega^2 = 0$ in a faithful representation and in the algebra. For odd Ω, the double graded commutator with Ω is the commutator with Ω^2. Hence $\hbar^2 Q_\hbar^2 = 0$, and torsion freeness gives the result. □

The full gauge complex has extra states even at rank one. Scalar conjugation is trivial, so $\ell = 0$, the Lie bracket vanishes and $Q_\hbar = 0$. In its degree-zero ghost algebra, 1 and $u = cb$ are independent and obey

$$u^2 = \hbar u.$$

Thus the full BRST cohomology is not obtained by simply replacing it with the ordinary one-pair Weyl reduction. Removing the ineffective gauge direction changes the input.

In contrast, a finite-dimensional representation of the *coefficient* Weyl algebra has commuting coordinate entries and the relation $[X, Y] = \varepsilon I_N$. Its derived representation algebra is

$$R_{N,\varepsilon} = \mathbb{C}[X_{ij}, Y_{ij}, \varepsilon] \otimes \Lambda(\Psi_{ij}), \qquad Q\Psi = [X, Y] - \varepsilon I_N, \qquad QX = QY = Q\varepsilon = 0. \tag{9.45}$$

This is the recipient supplied by matrix substitution in Theorem 9.12.

Proposition 9.14. *For $N > 0$, multiplication by ε is null-homotopic on $R_{N,\varepsilon}$. Its cohomology is annihilated by ε, and its localization at ε is contractible.*

Proof. Commuting entries give $\mathrm{Tr}[X, Y] = 0$. The element $v = -\mathrm{Tr}\,\Psi/N$ has degree −1 and satisfies $Qv = \varepsilon$. Multiplication by v obeys

$$Q(va) + vQa = \varepsilon a$$

for every homogeneous a. After inverting ε, multiplication by v/ε contracts the complex. □

This contraction is integral before localization. It does not contract the special fibre. At rank one the complex is

$$\mathbb{C}[x, y, \varepsilon]\psi \xrightarrow{-\varepsilon} \mathbb{C}[x, y, \varepsilon], \qquad |\psi| = -1.$$

Its cohomology is $\mathbb{C}[x, y]$ in degree zero, annihilated by ε. Derived specialization at $\varepsilon = 0$ is computed by the displayed free complex and retains the degree-minus-one class ψ. Simply specializing its degree-zero cohomology would miss that class.

The equations (9.43) and (9.45) therefore cannot be interchanged by naming their parameters alike. One quantizes the matrix coordinate algebra; the other asks for matrix representations of a quantized coefficient algebra. A comparison between them must specify a map and its effect on relation classes.

9.12 Framed and infinite-dimensional recipients

Framing changes the finite-dimensional equation. Add a column $\mathbf{i} \in \mathbb{C}^N$ and a row $\mathbf{j} \in (\mathbb{C}^N)^*$, and use

$$[X, Y] + \mathbf{ij} = \varepsilon I_N. \tag{9.46}$$

Simultaneous conjugation extends by $\mathbf{i} \mapsto g\mathbf{i}$ and $\mathbf{j} \mapsto \mathbf{j}g^{-1}$. The extra rank-one term permits a nonzero trace of the prescribed moment value.

Proposition 9.15. *On the ordered chart with pairwise distinct x_i, the formulas*

$$\begin{aligned} X &= \operatorname{diag}(x_i), \qquad \mathbf{i}_i = 1, \qquad \mathbf{j}_i = \varepsilon, \\ Y_{ii} &= y_i, \qquad Y_{ij} = -\frac{\varepsilon}{x_i - x_j} \quad (i \neq j) \end{aligned} \tag{9.47}$$

give a free framed orbit and a smooth local quotient of relative dimension $2N$ over the parameter. At fixed ε, the reduced symplectic form is $\sum_i dx_i \wedge dy_i$ on this ordered cover.

Proof. The off-diagonal entries of the left side of (9.46) are $-\varepsilon + \varepsilon = 0$; its diagonal entries are ε. A stabilizer commuting with X is diagonal, and fixing the column with all entries one makes it the identity. The infinitesimal action is injective.

On the open set with simple X and nonzero components of $\mathbf{i}$ in its eigenbasis, diagonalize X and use diagonal gauge transformations to set those components equal to one. The moment equation then forces the displayed values of $\mathbf{j}$ and the off-diagonal entries of Y. Thus (x_i, y_i) actually parametrize the local quotient.

With the canonical form $\operatorname{Tr}(dX \wedge dY) + \sum_i d\mathbf{i}_i \wedge d\mathbf{j}_i$, the moment differential is the symplectic transpose of the infinitesimal action, so it is surjective. The quotient is therefore smooth of relative dimension $(2N^2 + 2N) - 2N^2 = 2N$ near this orbit. On the slice, $d\mathbf{i} = d\mathbf{j} = 0$ at fixed parameter, and dX is diagonal. The pullback of the form is exactly $\sum_i dx_i \wedge dy_i$. Permuting the eigenvalues gives the remaining finite identification. □

An infinite-dimensional unframed recipient keeps the original coefficient equation. Complete W_ε in the ε-adic topology and write it as $\widehat{W}_\varepsilon$. Use complete left modules with continuous maps and strict exact sequences, resolved by topologically free modules. The completed polynomial Fock module

$$\mathcal{P} = \mathbb{C}[p][[\varepsilon]], \qquad q \cdot f = \varepsilon \partial_p f, \qquad p \cdot f = pf$$

satisfies $[q, p] = \varepsilon$. A coefficient at each parameter order is a polynomial in p, with no uniform degree bound.

Proposition 9.16. *Continuous derived Hom in this exact category gives*

$$\operatorname{Ext}^0_{\widehat{W}_\varepsilon}(\mathcal{P},\mathcal{P}) = \mathbb{C}[[\varepsilon]], \qquad \operatorname{Ext}^1_{\widehat{W}_\varepsilon}(\mathcal{P},\mathcal{P}) = \mathbb{C}[p],$$

with all other degrees zero. Multiplication by ε annihilates the degree-one group.

Proof. In the completed PBW basis $p^b q^a$, right multiplication by q increases a by one. It is injective with closed image, and its cokernel is $\mathcal{P}$. As a map of left modules it gives the strict topologically free resolution

$$0 \longrightarrow \widehat{W}_\varepsilon \xrightarrow{R_q} \widehat{W}_\varepsilon \longrightarrow \mathcal{P} \longrightarrow 0.$$

Continuous Hom into $\mathcal{P}$ is the two-term complex

$$\mathcal{P} \xrightarrow{\varepsilon\partial_p} \mathcal{P}$$

in degrees zero and one. Coefficientwise polynomial integration is a continuous right inverse to ∂_p. The kernel is $\mathbb{C}[[\varepsilon]]$, the image is $\varepsilon\mathcal{P}$, and the cokernel is $\mathcal{P}/\varepsilon\mathcal{P} = \mathbb{C}[p]$. □

The degree-one self-extension disappears when the parameter is inverted. Its integral presence records information that generic specialization loses. The framed chart and the Fock module thus avoid the finite unframed contraction in different ways: the first changes the equation, and the second changes the dimension and completion of the module.

9.13 Hamiltonian and geometric comparison maps

The construction has produced an action on an associative resolution, an action on derived matrix representations, and a coherent classical negative-cyclic orientation. The quotient-by-constants obstruction fixes the source of the stronger Hamiltonian problem. Retain either the full classical Poisson algebra A with $J_N(1) = N$, or its pointed Lie subalgebra, and keep the scalar quadratic map fixed.

Problem 9.17. Construct a homotopy moment map to the full classical BFV algebra whose first component is J_N, whose cubic binary component is $-\frac{3}{2}\operatorname{Tr}(\Psi[X,Y])$ up to specified homotopies, and whose choices respect block-diagonal sums of matrices. Compare its Hamiltonian derivation action with Theorem 9.4, retaining the cyclic orientation.

For matrix quantization, construct a filtered derived comparison on the full quantum BRST complex that agrees with the ordinary quantum trace action in degree zero and preserves the relation classes, stabilizer states and cyclic operations. Require the prescribed classical limit and cubic primitive, and extend the comparison across the matrix discriminant. Determine whether the coefficient-Weyl action compares to this recipient, to a framed recipient, or to an infinite-dimensional module by maps preserving those data.

The positive gauge ghosts in (9.19) already belong to the action constructed here. Supplying them is not the missing Hamiltonian comparison. Likewise a contraction of one finite coefficient representation does not obstruct every framed, completed or quantum-reduced recipient.

For mixed holomorphic–topological theory there is a further geometric map to construct. The polynomial cyclic operations must be realized by supported operations of the independently defined field theory. Such a comparison must preserve the boundary polarization, parameter normalization, descendant operations and sewing. At four inputs it must include the simultaneous collision support $\Delta_{13} \cap \Delta_{24}$ from Chapter 7. The universal word contraction solves the algebraic action equations; the geometric comparison must carry their solutions to these specified supports and operations.

Chapter 10

Relative quantum reduction across the discriminant

Diagonalizing commuting matrices suggests that their invariant functions should be functions of eigenvalue pairs. At repeated eigenvalues, diagonalization no longer supplies a coordinate chart. The relation complex also has odd functions that an ordinary quotient cannot see. A comparison across this locus must therefore be a map of complexes, with its product and coefficient ring specified.

In rank two the odd functions form a particularly small module. It has rank one away from a quadratic cone's vertex, but needs two generators at the vertex. We first compute this module and its restriction map. A terminating differential operator then deforms the restriction over $\mathbb{C}[\hbar]$. The result is a strict quasi-isomorphism, although it has no strict algebra section. Its higher inverse makes this distinction explicit. The final comparison with ordinary matrix reduction retains the quantum parameter and trace rank at every matrix size.

10.1 The relation module at a singular pair

Work over $\mathbb{C}$ with cohomological differentials of degree one. Identify the adjoint representation of $\mathfrak{sl}_2$ with the three-dimensional representation of $G = SO_3(\mathbb{C})$. Normalize its invariant form and the equation generators so that the commuting equation is the cross product. For even vectors x, y and an odd vector ψ, put

$$P = \mathbb{C}[x_1, x_2, x_3, y_1, y_2, y_3], \qquad K = P \otimes \Lambda(\psi_1, \psi_2, \psi_3), \qquad |\psi_i| = -1,$$
$$dx = dy = 0, \qquad d\psi = \mu, \qquad \mu = x \times y. \tag{10.1}$$

The orientation is $\epsilon_{123} = 1$. Exterior products are written in the order ψ_1, ψ_2, ψ_3, and odd derivatives act on the left. The relative complex is K^G, the invariants of the rational group action. Taking these invariants imposes no positive gauge ghosts.

Define

$$a = x \cdot x, \qquad b = x \cdot y, \qquad c = y \cdot y, \qquad S = \mathbb{C}[a, b, c], \qquad \Delta = ac - b^2,$$

and

$$\xi = x \cdot \psi, \qquad \eta = y \cdot \psi, \qquad z = \mu \cdot \psi.$$

The elements ξ, η are closed because μ is orthogonal to both x and y. To exhibit their relations, introduce

$$p = x_1\psi_2\psi_3 - x_2\psi_1\psi_3 + x_3\psi_1\psi_2,$$
$$q = y_1\psi_2\psi_3 - y_2\psi_1\psi_3 + y_3\psi_1\psi_2,$$
$$w = \mu_1\psi_2\psi_3 - \mu_2\psi_1\psi_3 + \mu_3\psi_1\psi_2, \qquad r = \psi_1\psi_2\psi_3.$$

Here p, q denote elements of exterior degree two, independently of the plane coordinates used in Chapter 9. Their cohomological degrees are -2.

Lemma 10.1. *The differential satisfies*

$$dz = \Delta, \qquad dp = b\xi - a\eta, \qquad dq = c\xi - b\eta, \qquad dr = w = \xi\eta, \qquad dw = 0. \tag{10.2}$$

Proof. Lagrange's identity gives $\mu \cdot \mu = ac - b^2$. Identify a vector v with the two-form $v_1\psi_2\psi_3 - v_2\psi_1\psi_3 + v_3\psi_1\psi_2$. Its differential is $(v \times \mu) \cdot \psi$. The vector identities

$$x \times (x \times y) = bx - ay, \qquad y \times (x \times y) = cx - by$$

give the middle two formulas. Expanding $d(\psi_1\psi_2\psi_3)$ gives w, and expanding $(x \cdot \psi)(y \cdot \psi)$ gives the same two-form. The last equation follows from $d^2 r = 0$. □

The invariant complex has eight coefficient generators. A polynomial vector covariant is a polynomial map $v(x, y)$ satisfying $v(gx, gy) = gv(x, y)$ for $g \in G$. The following calculation proves the completeness of these generators, including on $\Delta = 0$.

Lemma 10.2. *The scalar invariant ring is S. The polynomial vector covariants are the free S-module generated by x, y, μ. Consequently*

$$\begin{aligned} (K^G)^0 &= S, & (K^G)^{-1} &= S\xi \oplus S\eta \oplus Sz, \\ (K^G)^{-2} &= Sp \oplus Sq \oplus Sw, & (K^G)^{-3} &= Sr. \end{aligned} \tag{10.3}$$

Proof. Give a polynomial its separate degrees in x and y. The map $(x, y) \mapsto (a, b, c)$ is dominant: every nondegenerate two-dimensional symmetric form embeds in the standard form on $\mathbb{C}^3$. Thus a, b, c are algebraically independent. The determinant of the three columns x, y, μ is Δ, a nonzero polynomial. Hence these columns are linearly independent over the fraction field and, in particular, over S.

It remains to prove that these explicit inclusions exhaust the invariants. Let V_ℓ be the irreducible SO_3-representation of spin ℓ, of dimension $2\ell + 1$. The elementary $\mathfrak{sl}_2$ character formulas give

$$\operatorname{Sym}^m(V_1) = \bigoplus_{j=0}^{\lfloor m/2 \rfloor} V_{m-2j}, \qquad V_\ell \otimes V_k = \bigoplus_{n=|\ell-k|}^{\ell+k} V_n.$$

For example, the first formula follows by comparing the coefficients of u^m in

$$\frac{1}{(1-uz)(1-u)(1-uz^{-1})} = \frac{1}{1-u^2} \sum_{\ell \geq 0} u^\ell (z^\ell + z^{\ell-1} + \cdots + z^{-\ell}).$$

The second follows by multiplying the two finite weight sums and subtracting their successive highest weights.

The multiplicity of V_0 in $V_\ell \otimes V_k$ is one precisely when $\ell = k$. The multiplicity of V_1 is one precisely when $|\ell - k| \leq 1 \leq \ell + k$. Their bivariate generating series in the polynomial ring are therefore

$$\frac{1}{(1-u^2)(1-uv)(1-v^2)}, \qquad \frac{u+v+uv}{(1-u^2)(1-uv)(1-v^2)}.$$

These are exactly the series of the already embedded ring S and module $Sx \oplus Sy \oplus S\mu$. Each bidegree is finite-dimensional, so equality of dimensions makes those inclusions surjective. The invariant form identifies vectors and covectors. The invariant volume identifies their second exterior powers, and the top exterior power is trivial. This gives (10.3). □

Set

$$B = S/(\Delta), \qquad M = (B\xi \oplus B\eta)/(b\xi - a\eta,\ c\xi - b\eta), \qquad |\xi| = |\eta| = -1.$$

Let the involution on $\mathbb{C}[s,t]\otimes\Lambda(\theta)$ change the signs of all three generators, with $|\theta|=-1$. Restriction to a Cartan axis defines

$$\rho : K^G \longrightarrow \big(\mathbb{C}[s,t]\otimes\Lambda(\theta)\big)^{\mathbb{Z}/2}, \qquad x=(s,0,0), \quad y=(t,0,0), \quad \psi=(\theta,0,0). \tag{10.4}$$

The target has zero differential.

Theorem 10.3. *The map ρ is a surjective quasi-isomorphism of commutative differential graded algebras. It induces*

$$H^0(K^G)=B, \qquad H^{-1}(K^G)=M, \qquad H^j(K^G)=0 \quad (j\le -2),$$

with all products of negative-degree classes zero. It sends (a,b,c,ξ,η) to $(s^2,st,t^2,s\theta,t\theta)$.

This is the traceless rank-two instance of the derived Harish–Chandra comparison of Berest, Felder, Patotski, Ramadoss and Willwacher (2015), Theorem 4.1 of [11]. Here the coefficient presentation gives a direct proof.

Proof. The map on Sz is multiplication by Δ, hence injective. On $Sp\oplus Sq$ the differential has matrix

$$\begin{pmatrix} b & c \\ -a & -b \end{pmatrix}$$

with determinant Δ. It too is injective over the polynomial domain S. Finally $Sr\to Sw$ is an isomorphism. Thus the displayed four-term complex has no cohomology below degree -1. Its degree-zero quotient is B, and its odd quotient is $S^2/\langle(b,-a),(c,-b)\rangle$. The two column relations imply $\Delta\xi=\Delta\eta=0$, so this quotient is precisely M. The primitive $dr=\xi\eta$ proves the product assertion.

The target's even part is $\mathbb{C}[s^2,st,t^2]\cong B$. Its odd part consists of odd polynomials in s,t, multiplied by θ. It is generated by $s\theta,t\theta$ over the even ring. If even polynomials F,G satisfy $Fs+Gt=0$, coprimality gives $F=tH$ and $G=-sH$ with H odd. Write $H=sP+tQ$ with P,Q even. The relation is then $P(st,-s^2)+Q(t^2,-st)$, exactly the relation module above. Restriction is surjective on the displayed generators and induces these cohomology isomorphisms. □

On $a\neq 0$, the odd module is generated by ξ, since $\eta=(b/a)\xi$. The same conclusion holds on $c\neq 0$ with the roles reversed. These two open sets cover the punctured cone. At its vertex,

$$M/(a,b,c)M=\mathbb{C}\xi\oplus\mathbb{C}\eta. \tag{10.5}$$

This is the ordinary fibre of a cohomology module. It is not an assertion that cohomology commutes with derived base change. Replacing M by a free line would already give the wrong ordinary fibre. The restriction theorem uses no discriminant localization.

For $\mathfrak{gl}_2$, the scalar matrix coordinates supply a polynomial factor $\mathbb{C}[u,v]$. The trace equation is identically zero, so its relation generator ζ, of degree -1, is closed. Tensoring ρ with $\mathbb{C}[u,v]\otimes\Lambda(\zeta)$ gives the full rank-two invariant-function comparison. Its degree -2 includes $\mathbb{C}[u,v]\zeta M$. This extra odd direction cannot be discarded. The statement concerns invariant functions, rather than an equivalence of quotient stacks or of their module categories.

10.2 The quantum differential and its coefficient operator

Use the matrix quantum parameter $\hbar$, independently of the coefficient-plane parameter ε in Section 9.10. On polynomial symbols in the three canonical pairs (x_i, y_i) set

$$F \star G = m \exp\left[\frac{\hbar}{2} \sum_{i=1}^{3} \Big(\partial_{x_i} \otimes \partial_{y_i} - \partial_{y_i} \otimes \partial_{x_i} \Big)\right](F \otimes G). \tag{10.6}$$

Here m is ordinary multiplication. Each product is polynomial in $\hbar$. The ψ_i still multiply exteriorly. Define

$$\mathcal{K}_\hbar = \big(\mathbb{C}[x, y] \otimes \Lambda(\psi_1, \psi_2, \psi_3)\big)^G \otimes \mathbb{C}[\hbar]$$

with this product and

$$\delta_\hbar = \sum_i \mu_i \partial_{\psi_i} + \frac{\hbar^2}{4} \sum_{i,j,k} \epsilon_{ijk} \partial_{x_j} \partial_{y_k} \partial_{\psi_i}. \tag{10.7}$$

The first summand uses ordinary multiplication of symbols. The second summand is needed even though the moment functions μ_i themselves are unchanged.

Proposition 10.4. *The operator $\delta_\hbar$ is a square-zero derivation of $\mathcal{K}_\hbar$ over $\mathbb{C}[\hbar]$.*

Proof. Adjoin ghosts γ^i of degree 1 with $\psi_i \gamma^j + \gamma^j \psi_i = \hbar \delta_i^j$. The BRST element for the rotation moment is

$$\Theta = \sum_i \gamma^i \mu_i - \frac{1}{2} \sum_{i,j,k} \epsilon_{ijk} \gamma^i \gamma^j \psi_k.$$

The moment commutators are $[\mu_i, \mu_j]_\star = \hbar \sum_k \epsilon_{ijk} \mu_k$. Together with Jacobi they give $\Theta^2 = 0$, by the Weyl–Clifford construction of Section 9.11. Its polynomial inner derivation $[\Theta, -]/\hbar$ is therefore square-zero.

Use symmetric symbols for the ghosts. On an expression independent of γ, the terms that still contain γ^i are its infinitesimal simultaneous rotations, including the rotation of ψ. They vanish on G-invariants. The remaining term is

$$\frac{1}{2} \sum_i \Big(\mu_i \star \partial_{\psi_i} F + \partial_{\psi_i} F \star \mu_i \Big).$$

Since μ_i is quadratic, the Moyal expansion stops at its second derivative. The first derivatives cancel in the average, while the second derivatives give exactly (10.7). The cubic ghost term has no higher contraction on a γ-independent input. Thus this invariant subalgebra is preserved by the inner derivation, proving both assertions. □

The presentation (10.3) remains a presentation of polynomial symbols over $\mathbb{C}[\hbar]$. It is a statement about coefficient modules, not about ordinary multiplication being the quantum product. In particular $\xi \star \eta = w$: the possible first contraction is proportional to $\sum_i \psi_i^2$, hence vanishes.

Define operators on $S[\hbar]$ by

$$L = \partial_a \partial_c - \frac{1}{4} \partial_b^2, \qquad T = 1 + \hbar^2 L, \qquad T^{-1} F = \sum_{j \geq 0} (-\hbar^2 L)^j F. \tag{10.8}$$

The inverse sum terminates because L lowers polynomial degree in a, b, c by two. Neither $\hbar$ nor Δ is inverted.

Lemma 10.5. *For every $F \in S[\hbar]$,*

$$\begin{aligned} \delta_\hbar(Fz) &= T(\Delta F), \\ \delta_\hbar(Fp) &= T(bF)\xi - T(aF)\eta, \\ \delta_\hbar(Fq) &= T(cF)\xi - T(bF)\eta, \\ \delta_\hbar(Fr) &= T(F)w, \\ \delta_\hbar(F\xi) &= \delta_\hbar(F\eta) = \delta_\hbar(Fw) = 0. \end{aligned} \tag{10.9}$$

All products inside the arguments of T are ordinary coefficient products.

Proof. The chain rule gives $\partial_{x_i} F = 2x_i F_a + y_i F_b$ and $\partial_{y_i} F = x_i F_b + 2y_i F_c$. Therefore

$$\sum_{j,k} \epsilon_{ijk}\, \partial_{x_j}\partial_{y_k} F = 4\mu_i L F.$$

Applying the derivatives also to μ_i gives

$$\frac{1}{4}\sum_{i,j,k} \epsilon_{ijk}\, \partial_{x_j}\partial_{y_k}(F\mu_i) = \Delta L F + (a\partial_a + b\partial_b + c\partial_c)F + \frac{3}{2}F = L(\Delta F).$$

This proves the first equation. On Fp, the quantum coefficients of ξ, η are $bLF - F_b/2$ and $-aLF - F_c$. On Fq, they are $cLF + F_a$ and $-bLF + F_b/2$. These equal $L(bF), -L(aF), L(cF), -L(bF)$, respectively. The top exterior differential has quantum correction $LF\,w$. Antisymmetric contraction with x or y kills the corrections on $F\xi, F\eta$. Finally $F \mapsto T(F)$ is surjective, so every Fw is the differential of $T^{-1}(F)r$. Square-zero then gives its stated differential. □

Let U apply T^{-1} to coefficients in $S, S\xi, S\eta, Sw$ and act as the identity on Sz, Sp, Sq, Sr. The coefficient formulas give

$$U\delta_\hbar = \delta_0 U. \tag{10.10}$$

Thus U is an integral chain isomorphism. Its inverse uses T on the same four summands. This identity already computes the underlying cohomology modules. To identify their product, the restriction itself must be multiplicative.

10.3 Strict restriction over the quantum ring

Let $W_\hbar(s,t)$ be the one-pair Weyl algebra in symmetric symbols, so that $[s,t]_\star = \hbar$. Put

$$\mathcal{H}_\hbar = \big(W_\hbar(s,t) \otimes \Lambda(\theta)\big)^{\mathbb{Z}/2}, \qquad |\theta| = -1,$$

with zero differential and the simultaneous sign involution. For a scalar symbol write $\rho_0 F = F(s^2, st, t^2)$, where st in this substitution is the commuting symbol. Define

$$\begin{aligned} \rho_\hbar(F) &= \rho_0(T^{-1}F), \\ \rho_\hbar(F\xi + G\eta + Jz) &= \big(s\rho_0(T^{-1}F) + t\rho_0(T^{-1}G)\big)\theta, \\ \rho_\hbar(\mathcal{K}_\hbar^{\le -2}) &= 0. \end{aligned} \tag{10.11}$$

Theorem 10.6. *The map $\rho_\hbar : \mathcal{K}_\hbar \to \mathcal{H}_\hbar$ is a surjective strict quasi-isomorphism of differential graded algebras over $\mathbb{C}[\hbar]$. Its kernel has a $\mathbb{C}[\hbar]$-linear contraction. Consequently every base change to a commutative $\mathbb{C}[\hbar]$-algebra preserves the quasi-isomorphism, including every scalar specialization of $\hbar$.*

Proof. As a chain map, $\rho_\hbar = \rho U$. The classical theorem therefore proves surjectivity and the cohomology comparison. We prove multiplicativity by the operators of multiplication by the three invariant quadratics.

In d canonical pairs, left multiplication on invariant scalar symbols has the following operators:

$$A_d = a + \hbar(a\partial_b + 2b\partial_c) + \frac{\hbar^2}{4}(a\partial_b^2 + 4b\partial_b\partial_c + 4c\partial_c^2 + 2d\partial_c),$$
$$B_d = b + \hbar(c\partial_c - a\partial_a) - \frac{\hbar^2}{4}(2a\partial_a\partial_b + b\partial_b^2 + 4b\partial_a\partial_c + 2c\partial_b\partial_c + d\partial_b),$$
$$C_d = c - \hbar(2b\partial_a + c\partial_b) + \frac{\hbar^2}{4}(4a\partial_a^2 + 4b\partial_a\partial_b + c\partial_b^2 + 2d\partial_a).$$

They follow by applying the two derivatives in the Moyal product to a, b, c. For instance, $a \star F = aF + \hbar x \cdot \partial_y F + \hbar^2 \sum_i \partial_{y_i}^2 F/4$, and the chain rule gives A_d.

These differential operators satisfy

$$A_3 T = T A_1, \qquad B_3 T = T B_1, \qquad C_3 T = T C_1. \tag{10.12}$$

Here is the coefficient verification. The commutators of L with multiplication by a, b, c are $\partial_c, -\partial_b/2, \partial_a$. It commutes with the three first-order coefficients of $\hbar$. Its commutators with the coefficients of $\hbar^2$ in A_1, B_1, C_1 are, respectively, $L\partial_c, -L\partial_b/2, L\partial_a$. Thus $[T, A_1] = \hbar^2\partial_c T = (A_3 - A_1)T$, and the other two equations follow with the displayed constants.

Evaluation by ρ_0 changes A_1, B_1, C_1 into actual one-pair left multiplication by s^2, st, t^2. Equation (10.12) therefore proves that $\rho_\hbar$ intertwines multiplication by a, b, c on the scalar sector. These three quadratics generate its invariant Weyl algebra: a star monomial has its ordinary monomial as leading symbol, and subtraction lowers polynomial degree. Induction proves generation over $\mathbb{C}[\hbar]$.

On the coefficient column (F, G) of $F\xi + G\eta$, the corresponding left multiplication operators are

$$\begin{pmatrix} A_d & \hbar + \frac{\hbar^2}{2}\partial_b \\ 0 & A_d + \hbar^2\partial_c \end{pmatrix}, \qquad \begin{pmatrix} B_d - \frac{\hbar}{2} - \frac{\hbar^2}{4}\partial_b & -\frac{\hbar^2}{2}\partial_a \\ -\frac{\hbar^2}{2}\partial_c & B_d + \frac{\hbar}{2} - \frac{\hbar^2}{4}\partial_b \end{pmatrix},$$

$$\begin{pmatrix} C_d + \hbar^2\partial_a & 0 \\ -\hbar + \frac{\hbar^2}{2}\partial_b & C_d \end{pmatrix}.$$

For example, multiplying the coefficient G and the single y in η accounts for the two off-diagonal terms of the first matrix. The other entries follow from the same product rule. Their extra entries have constant coefficients, so they commute with T. The scalar intertwiners prove the column intertwiners. Replacing $\hbar$ by $-\hbar$ gives right multiplication as well.

Give x, y weight one and $\psi, \hbar$ weight two. The scalar and Sz sectors have even weight, while $S\xi, S\eta$ have odd weight. Scalar star multiplication therefore preserves the killed Sz sector. Products of total cohomological degree at most -2 have zero image, and two odd target factors have product zero because $\theta^2 = 0$. This exhausts the products and proves strict multiplicativity.

Choose a contraction of the classical kernel in each finite weight. Such a contraction exists by the classical quasi-isomorphism: split each vector space into boundaries and chosen preimages, and send boundaries back to their preimages. Extend it $\mathbb{C}[\hbar]$-linearly and conjugate by U. This contracts the quantum kernel. Together with a chain section of the surjection, it gives a contraction of the whole comparison. Tensoring these identities with any commutative coefficient algebra proves the base-change assertion without a flatness assumption. □

The correction is already visible in three small values:

$$\rho_\hbar(ac) = s^2t^2 - \hbar^2, \qquad \rho_\hbar(b^2) = s^2t^2 + \frac{1}{2}\hbar^2, \qquad \delta_\hbar z = \Delta + \frac{3}{2}\hbar^2. \tag{10.13}$$

The naive substitution ρ_0 would send the last boundary to $3\hbar^2/2$, so it is not the quantum chain map. The strict comparison instead identifies the even cohomology relation

$$\frac{1}{2}(a \star c + c \star a) - b \star b = \frac{3}{4}\hbar^2.$$

Its odd relations retain their primitives:

$$(b + \tfrac{3}{2}\hbar) \star \xi - a \star \eta = \delta_\hbar p, \qquad c \star \xi - (b - \tfrac{3}{2}\hbar) \star \eta = \delta_\hbar q.$$

Tensoring with the independent scalar Weyl pair and its closed central antighost gives the relative $\mathfrak{gl}_2$ comparison. The absolute complex still has its positive ghosts.

10.4 An inverse with higher components

A strict projection need not have a strict section. To construct an inverse of $\rho_\hbar$ with its homotopies, choose the classical section on the Cartan monomial basis by

$$i_0(s^m t^n) = \begin{cases} a^{m/2}c^{n/2}, & m, n \text{ even}, \\ a^{(m-1)/2}bc^{(n-1)/2}, & m, n \text{ odd}, \end{cases}$$

$$i_0(s^m t^n \theta) = \begin{cases} a^{(m-1)/2}c^{n/2}\xi, & m \text{ odd},\ n \text{ even}, \\ a^{m/2}c^{(n-1)/2}\eta, & m \text{ even},\ n \text{ odd}. \end{cases}$$

In each finite weight split the classical complex into these cohomology representatives, boundaries and chosen preimages of boundaries. Let H_0 send a boundary to its preimage and vanish on the other two summands. Choose $H_0\Delta = z$ and $H_0 w = r$. Then

$$\delta_0 H_0 + H_0\delta_0 = 1 - i_0\rho, \qquad H_0^2 = H_0 i_0 = \rho H_0 = 0.$$

Extend these maps $\mathbb{C}[\hbar]$-linearly and put

$$i_\hbar = U^{-1}i_0, \qquad H_\hbar = U^{-1}H_0 U.$$

They give the same normalized contraction identities for $(\rho_\hbar, i_\hbar, H_\hbar)$.

Let s lower cohomological degree by one. On the tensor coalgebras of the suspended underlying nonunital algebras use the codifferentials determined by

$$b_1(sa) = -s\delta_\hbar a, \qquad b_2(sa, sb) = (-1)^{|a|}s(a \star b).$$

The target algebra has only b_2. These coalgebras have the deconcatenation coproduct; their maps are determined by their components into tensors of length one. No scalar augmentation of a Weyl algebra is required. Put $H_s = -sH_\hbar s^{-1}$ and $I_1 = si_\hbar s^{-1}$. Given the components below arity n, set

$$O_n = (b_{\mathcal{K}} I_{<n} - I_{<n} b_{\mathcal{H}})_n, \qquad I_n = -H_s O_n. \tag{10.14}$$

Here $I_{<n}$ denotes the coalgebra map with the prescribed lower components and zero arity-n component.

Theorem 10.7. *This recursion defines a strictly unital A_∞ morphism $I : \mathcal{H}_\hbar \to \mathcal{K}_\hbar$ whose first component is a quasi-isomorphism and for which $\rho_\hbar I$ is the strict identity. Its only possible components are I_1, I_2, I_4. For the commuting target symbol $B = st$,*

$$I_2(sB, sB) = -sz, \qquad I_4(sB, sB, sB, sB) = -\hbar sr. \tag{10.15}$$

Proof. The equation $b^2 = 0$, together with the lower morphism equations, makes O_n closed for the unary differential. Strictness of $\rho_\hbar$ and $\rho_\hbar i_\hbar = 1$, $\rho_\hbar H_\hbar = 0$ put O_n in the kernel of the suspended projection. The contraction identity makes $b_1(-H_s O_n) = -O_n$. Thus (10.14) solves the next morphism equation. Every map is polynomial on its input.

The special vanishing follows from the weights and the four cohomological degrees. On inputs with k odd factors, the unsuspended third residual has degree $-1-k$ and weight parity k. For $k = 0, 1, 2$ it lies in $S[\hbar]z$, $S[\hbar]p \oplus S[\hbar]q$, and $S[\hbar]r$, respectively. The differential is injective on each of these sectors, by the coefficient formulas and the determinant Δ of the middle matrix. Closedness forces the residual to vanish. For larger k its degree is absent. Hence $I_3 = 0$.

The fourth component has unsuspended degree $-3-k$, so it can occur only on four even inputs and takes values in $S[\hbar]r$. All components of arity at least five would have degree below -3. The fifth residual on even inputs lies in $S[\hbar]r$ and is closed, hence zero by injectivity. Every later residual has absent degree. This proves all morphism equations, including those beyond the last nonzero component. Since the section preserves the unit and the homotopy annihilates its image, the normalized recursion vanishes on a unit input in every higher arity.

For the stated example,

$$b \star b = b^2 - \frac{3}{4}\hbar^2, \qquad i_\hbar(B \star B) = ac + \frac{3}{4}\hbar^2.$$

Their difference is $-\delta_\hbar z$, giving $I_2 = -sz$. The moment commutator and exterior signs give

$$z \star z = \hbar w, \qquad \delta_\hbar r = w.$$

Consequently the fourth residual is $b_2(-sz, -sz) = -\hbar sw$, and the recursion gives $I_4 = -\hbar sr$. □

For even inputs F, G, the binary component is particularly explicit. Set

$$D = i_\hbar(F) \star i_\hbar(G) - i_\hbar(F \star G).$$

Then $T^{-1}D$ is divisible by Δ in the polynomial ring. The unsuspended binary value is $(T^{-1}D/\Delta)z$. This is exact polynomial division, not localization at the discriminant. The fourth value is obtained by multiplying two binary values and applying $-T^{-1}$ to the resulting w coefficient. The target still has its ordinary binary product and no higher multiplications.

10.5 Changing a section

The nonzero fourth value belongs to a chosen inverse. There are two separate questions: whether the projection has a strict section on the whole algebra, and whether a particular higher value survives a change of section.

Proposition 10.8. *The map $\rho_\hbar$ has no $\mathbb{C}[\hbar]$-linear dg-algebra section.*

Proof. A section would specialize at $\hbar = 0$ to an algebra section of $S \to S/(\Delta)$ in degree zero. Its lifts of a, b, c would have the form $A = a + \Delta\alpha$, $B = b + \Delta\beta$, $C = c + \Delta\gamma$. The relation $AC = B^2$ would imply

$$0 = \Delta\big(1 + a\gamma + c\alpha - 2b\beta + \Delta(\alpha\gamma - \beta^2)\big).$$

Since S is a domain, the parenthesis would be zero. Its constant term is one, a contradiction. □

Proposition 10.9. *There is a unital weight-preserving chain section i' of $\rho_\hbar$ whose restriction to the polynomial star-subalgebra $\mathbb{C}[\hbar][B]$, $B = st$, is the algebra map*

$$i'(B^{\star n}) = b^{\star n}.$$

A normalized contraction with this section has a recursive A_∞ inverse whose higher components all vanish when every input belongs to this subalgebra.

Proof. The leading symbols $(st)^n$ show that the star powers $B^{\star n}$ are linearly independent over $\mathbb{C}[\hbar]$. Multiplicativity of $\rho_\hbar$ makes the proposed lift a section on their polynomial algebra. An even basis of the entire target is given by

$$A^{\star i} \star B^{\star j}\ (i \geq 1), \qquad B^{\star j}, \qquad B^{\star j} \star C^{\star k}\ (k \geq 1), \qquad A = s^2, \quad C = t^2.$$

Their leading symbols partition all even monomials in s, t. The lower terms are triangular by polynomial degree, so these star words are a basis over the parameter ring. Lift them by the same star words in a, b, c. Choose closed lifts of the odd monomial basis using the surjective restriction on $S\xi \oplus S\eta$. This gives a unital weight-preserving chain section i'.

From the previous normalized contraction put $H' = H_\hbar(1 - i'\rho_\hbar)$. The chain-section identity gives

$$\delta_\hbar H' + H'\delta_\hbar = 1 - i'\rho_\hbar.$$

Moreover $H'i' = \rho_\hbar H' = 0$. The identities $\rho_\hbar H_\hbar = H_\hbar^2 = 0$ give $(H')^2 = 0$. The binary defect on the polynomial subalgebra is zero because i' is multiplicative there. Inductively every higher residual consists of lower higher components on that same subalgebra, so the recursion chooses zero. □

The change already appears on the square:

$$i_\hbar(B \star B) = ac + \frac{3}{4}\hbar^2, \qquad i'(B \star B) = b^2 - \frac{3}{4}\hbar^2.$$

Their difference is $-\delta_\hbar z$. The target product has not changed. In particular the repeated-B fourth value in (10.15) can be removed.

Higher data must nevertheless occur somewhere for the weight-preserving recursive inverses. A classical section cannot be multiplicative by Proposition 10.8, so some even pair F, G has binary defect $\Delta\phi_0$ with $\phi_0 \neq 0$. Its quantum binary primitive has the form $\phi_\hbar z$, where $\phi_\hbar \bmod \hbar = \phi_0$. Invariant coefficients Poisson-commute with each μ_i. It follows that

$$(\phi_\hbar z) \star (\phi_\hbar z) = \big(\hbar\phi_0^2 + O(\hbar^2)\big)w \neq 0.$$

Here $O(\hbar^2)$ denotes a polynomial divisible by $\hbar^2$. The degree and parity proof above still makes $I_3 = 0$. On (F, G, F, G) the fourth residual is this nonzero product, so injectivity of $\delta_\hbar : S[\hbar]r \to S[\hbar]w$ forces a nonzero fourth value.

These facts are compatible with formality. A dg algebra is *formal* if a sequence of dg-algebra quasi-isomorphisms identifies it with its cohomology algebra with zero differential. The strict map $\rho_\hbar$ already proves that $\mathcal{K}_\hbar$ is formal. Failure of a section in its reverse direction does not contradict that conclusion. Neither a chosen fourth value nor its physical interpretation is an invariant obstruction merely by being nonzero.

10.6 The odd coefficient bimodule

The factor θ makes the product of two odd cohomology classes zero. Removing that factor gives a different operation worth retaining. Write

$$A_\hbar = W_\hbar(s,t)^{\mathrm{even}}, \qquad M_\hbar = W_\hbar(s,t)^{\mathrm{odd}}.$$

Parity here refers to the sign involution on the even coordinates, not to cohomological degree. Multiplication gives a bimodule map $M_\hbar \otimes_{A_\hbar} M_\hbar \to A_\hbar$.

Proposition 10.10. *Over $\mathbb{C}[\hbar]$, the image of this multiplication is the ideal $(s^2, st, t^2, \hbar)$ in $A_\hbar$, with cokernel $\mathbb{C}$ at $s = t = \hbar = 0$. After inverting $\hbar$, multiplication is an isomorphism, with inverse*

$$a \longmapsto \hbar^{-1}(s \otimes t - t \otimes s)a.$$

Proof. The four basic products $s \star s, s \star t, t \star s, t \star t$ generate $s^2, st, t^2, \hbar$. Every even ordered monomial of positive degree splits into two odd monomials. Their products span all the other even monomials modulo lower terms, by induction on degree. Thus their image is the stated ideal. Its quotient leaves only the scalar origin.

In the localized ring, multiplication of the displayed inverse gives a because $s \star t - t \star s = \hbar$. For the reverse composite on $m \otimes n$, use that $t \star m$ and $s \star m$ are even. Balancing them across the tensor product gives numerator

$$s \otimes (t \star m) \star n - t \otimes (s \star m) \star n = (s \star t - t \star s) \star m \otimes n = \hbar m \otimes n.$$

This proves the second identity. □

Only this last invertibility assertion inverts the parameter. The strict relative comparison was integral. An interpretation of the coefficient bimodule product as composition of physical lines requires a realization map; the algebraic calculation alone does not construct one.

10.7 The ordinary reduction at every matrix size

The rank-two complex has retained the relation classes. There is also an ordinary quantum reduction at every matrix size, defined across the discriminant. Its global trace action is an input to a derived comparison, rather than a replacement for that comparison.

Fix $N \geq 1$ and set $R = \mathbb{C}[\hbar]$. Let $W_{N,\hbar}$ be generated by two N by N matrices X, Y with

$$[X_{ij}, X_{kl}] = [Y_{ij}, Y_{kl}] = 0, \qquad [X_{ij}, Y_{kl}] = \hbar \delta_{il} \delta_{jk}.$$

The group $GL_N(\mathbb{C})$ acts by simultaneous conjugation. Use the ordered moment matrix from Section 9.11,

$$\widehat{\mu} = XY - YX - N\hbar I_N, \qquad I_{N,\hbar} = \sum_{i,j} W_{N,\hbar} \widehat{\mu}_{ij}.$$

This is a left ideal. Its invariant intersection is two-sided in the invariant algebra, since invariant operators commute with the infinitesimal conjugation operators. Define the ordinary reduction

$$\mathcal{A}_{N,\hbar} = W_{N,\hbar}^{GL_N} / (I_{N,\hbar} \cap W_{N,\hbar}^{GL_N}). \tag{10.16}$$

Replacing the full left ideal by the two-sided ideal it generates would define a different problem.

Let $\mathfrak{g} = \mathfrak{gl}_N$ and let $D(\mathfrak{g})$ be the polynomial differential operators. Its order filtration F_m allows arbitrary polynomial coefficients and at most m derivatives, with $F_{-1} = 0$. The Rees algebra is

$$\mathrm{Rees}_F D(\mathfrak{g}) = \bigoplus_{m \geq 0} \hbar^m F_m D(\mathfrak{g}) \subset D(\mathfrak{g})[\hbar].$$

The assignments $X_{ij} \mapsto x_{ij}$ and $Y_{ij} \mapsto -\hbar \partial_{x_{ji}}$ identify it with $W_{N,\hbar}$. The Weyl relations span the monomials with all X before all Y. Their differential-operator images are independent after inverting $\hbar$, so they have no nonzero R-linear relation before inversion either. They span the Rees algebra together with powers of $\hbar$. Thus X has Rees order zero, while Y and $\hbar$ have order one. These are not cohomological degrees.

Let $\tau(a)$ be the differentiated pullback conjugation action, normalized by $\tau(a)X = [X, a]$ on coordinate functions. The Weyl relations give

$$\operatorname{Tr}(a\widehat{\mu}) = \hbar\tau(a).$$

In particular $\operatorname{Tr}\widehat{\mu} = 0$. Put $J_D = (D(\mathfrak{g})\tau(\mathfrak{g}))^{GL_N}$. Let J_{cl} be the ideal generated by the entries of the commutator of a pair in $\mathfrak{g} \oplus \mathfrak{g}$. The trace pairing identifies the second factor with cotangent momentum.

Write $\mathfrak{t}$ for diagonal matrices and S_N for its Weyl group. On the distinct-eigenvalue locus, the raw radial operator $\operatorname{rad}(D)$ acts on restrictions of invariant functions. Its conjugated form is

$$\Phi(D) = \delta\, \operatorname{rad}(D)\, \delta^{-1}, \qquad \delta = \prod_{i<j}(\lambda_i - \lambda_j).$$

The radial definition begins on this open set. The following theorem is the extension statement that allows it to be used across the discriminant.

Theorem 10.11 (Filtered Harish–Chandra comparison)**.** *For $\mathfrak{g} = \mathfrak{gl}_N$, the conjugated radial map induces an algebra isomorphism*

$$D(\mathfrak{g})^{GL_N}/J_D \xrightarrow{\sim} D(\mathfrak{t})^{S_N}.$$

It is strict for the order filtrations. Its associated graded map is restriction of commuting pairs to diagonal pairs, and

$$\operatorname{gr}_F J_D = (\sqrt{J_{\mathrm{cl}}})^{GL_N} = J_{\mathrm{cl}}^{GL_N}.$$

The algebra isomorphism and associated graded assertions are the Harish–Chandra results recorded by Etingof and Ginzburg (2001), Theorems 6.8–6.10 of [36]. The equality of the invariant ideal with its radical is Gan–Ginzburg (2005), Theorem 1.2.1 and Section 2.8 of [39]. Strictness follows by lifting a leading symbol and subtracting: the differential order decreases at each step. The equality of invariant ideals does not assume that the whole commuting scheme is reduced.

Lemma 10.12. *Every $D \in J_D \cap F_m D(\mathfrak{g})$ is a finite invariant sum of terms $P_a\tau(a)$ with $P_a \in F_{m-1}D(\mathfrak{g})$. For $m = 0$ the only such D is zero.*

Proof. The leading symbol $\sigma_m(D)$ belongs to $J_{\mathrm{cl}}^{GL_N}$. Each moment generator has momentum degree one. Its homogeneous ideal expression can therefore be written

$$\sigma_m(D) = \sum_a p_a\, \sigma_1(\tau(a)), \qquad \deg_{\mathrm{momentum}} p_a = m - 1.$$

Only finitely many coefficients occur. The map from coefficient vectors to the ideal is equivariant. Rational GL_N-invariants are exact in characteristic zero: on a finite-dimensional rational module, average over the unitary compact form. Applying this to a degree-bounded module containing the coefficients gives an invariant coefficient vector with the same image.

Normal quantization, placing coordinates before derivatives, is equivariant under the linear action. It lifts the p_a to P_a of order at most $m - 1$. The invariant operator $\sum_a P_a\tau(a)$ has the same leading symbol as D. Subtract it and repeat at smaller order. This process terminates. The degree-zero momentum part of J_{cl} is zero, so the last order-zero remainder vanishes. □

An ideal J is $\hbar$-saturated if $\hbar u \in J$ implies $u \in J$. Let $W_{N,\hbar}^{\mathrm{Cartan}}$ be the Weyl algebra on N pairs λ_i, π_i with $[\lambda_i, \pi_j] = \hbar\delta_{ij}$ and all other generator commutators zero.

Theorem 10.13. *The conjugated radial map has an R-algebra realization*

$$\Phi_{N,\hbar} : \mathcal{A}_{N,\hbar} \xrightarrow{\sim} (W_{N,\hbar}^{\mathrm{Cartan}})^{S_N}.$$

Moreover

$$I_{N,\hbar} \cap W_{N,\hbar}^{GL_N} = \mathrm{Rees}_F J_D.$$

This invariant ideal is $\hbar$-saturated, the quotient is R-free, and its special fibre is

$$\mathcal{A}_{N,\hbar}/\hbar\mathcal{A}_{N,\hbar} = \mathbb{C}[\lambda_1, \pi_1, \dots, \lambda_N, \pi_N]^{S_N}.$$

Proof. The conjugation action preserves every filtration step, so invariants give $W_{N,\hbar}^{GL_N} = \mathrm{Rees}_F(D(\mathfrak{g})^{GL_N})$. In each Rees degree an element of the left moment ideal has coefficient in $D(\mathfrak{g})\tau(\mathfrak{g})$. Its invariant part therefore belongs to $\mathrm{Rees}_F J_D$.

Conversely, let $\hbar^m D$ have $D \in F_m J_D$. The preceding lemma expresses $D = \sum_a P_a \tau(a)$ with $P_a \in F_{m-1} D(\mathfrak{g})$. Hence

$$\hbar^m D = \sum_a (\hbar^{m-1} P_a)(\hbar\tau(a)) \in I_{N,\hbar}.$$

This element is invariant, proving equality of the ideals. The strict filtered Harish–Chandra isomorphism now identifies the quotient with the Rees algebra of $D(\mathfrak{t})^{S_N}$. Its Weyl generators are the displayed λ_i, π_i.

Weyl symmetrization identifies its underlying R-module equivariantly with the ordinary polynomial module. A complex basis of the S_N-invariant polynomial space therefore gives an R-basis of the quotient. It has no $\hbar$-torsion. Thus if $\hbar u$ maps to zero, u maps to zero too, which is saturation. The Rees special fibre is its associated graded algebra, giving the final formula. □

Saturation here concerns the invariant intersection. No assertion about saturation of the entire left ideal was required. At $\hbar = 0$, π_i remains a momentum symbol in the intrinsic special fibre. Substituting $\hbar = 0$ into the external representation $\pi_i = -\hbar\partial_{\lambda_i}$ would instead annihilate it. Specializing that representation does not compute the Rees special fibre.

10.8 The global ordinary trace action

For $f, g \in R[q, p]$, let $B_\hbar(f, g) = (f \star_\hbar g - g \star_\hbar f)/\hbar$ be the scalar Moyal bracket. Its numerator is polynomial and divisible by $\hbar$. For a monomial define the matrix-word symmetrized trace

$$J_N(q^a p^b) = \binom{a+b}{a}^{-1} \sum_v \mathrm{Tr}\, v(X, Y),$$

where v runs over distinct ordered products with a occurrences of X and b of Y. Extend J_N R-linearly. In particular $J_N(1) = N$. Let $\mathrm{Op}_W(f)(\lambda, \pi)$ denote one-pair Weyl symmetrization.

Theorem 10.14. *For every matrix size N and every polynomial f,*

$$\Phi_{N,\hbar}([J_N(f)]) = \sum_{i=1}^{N} \mathrm{Op}_W(f)(\lambda_i, \pi_i). \tag{10.17}$$

Consequently

$$\frac{[J_N(f), J_N(g)]}{\hbar} - J_N(B_\hbar(f, g)) \in I_{N,\hbar} \cap W_{N,\hbar}^{GL_N}.$$

The induced map $R[q, p]/R1 \to \mathcal{A}_{N,\hbar}/R1$ is injective and preserves the normalized Lie brackets.

Proof. Multiplication by $\operatorname{Tr} X^d$ restricts to $\sum_i \lambda_i^d$. The radial image of the quadratic momentum trace is

$$\operatorname{rad}(\operatorname{Tr} Y^2) = \hbar^2 \left(\sum_i \partial_{\lambda_i}^2 + 2 \sum_{i<j} \frac{\partial_{\lambda_i} - \partial_{\lambda_j}}{\lambda_i - \lambda_j} \right).$$

To see the mixed derivative coefficient, perturb entries (i, j) and (j, i) by u, v. Their two eigenvalues satisfy $(z - \lambda_i)(z - \lambda_j) - uv = 0$. Their mixed second derivatives are $\pm(\lambda_i - \lambda_j)^{-1}$, yielding the displayed first-order terms. The Vandermonde polynomial δ is harmonic: its Laplacian is alternating of lower degree, hence divisible by δ and therefore zero. Conjugation by δ removes the first-order terms. Thus $\Phi_{N,\hbar}([\operatorname{Tr} Y^2]) = \sum_i \pi_i^2$.

Put $H = -\operatorname{Tr} Y^2/2$. The polynomial derivation $D = [H, -]/\hbar$ satisfies $DX = Y$ and $DY = 0$. Its exponential terminates on any polynomial in the matrix entries and gives

$$e^{tD} \operatorname{Tr} X^d = \operatorname{Tr}(X + tY)^d.$$

Multiplicativity of $\Phi_{N,\hbar}$ and torsion-freeness of its target show that it intertwines D with $\lambda_i \mapsto \pi_i$, $\pi_i \mapsto 0$. The coefficient of t^b, with $a + b = d$, is the sum of the matrix products defining $J_N(q^a p^b)$. The same coefficient on the Cartan side is the sum of the corresponding one-pair Weyl products. Dividing by the same nonzero binomial coefficient proves (10.17).

Different Cartan pairs commute, and one-pair Weyl quantization intertwines Moyal products. The stated bracket defect has zero radial image. The exact kernel in Theorem 10.13 puts it in the invariant moment ideal. The division by $\hbar$ is the polynomial quotient of a divisible numerator throughout.

Finally suppose $[J_N(f)]$ is scalar. Taking Cartan Weyl symbols gives $\sum_i f(\lambda_i, \pi_i) = \alpha$ for some $\alpha \in R$. Set all pairs except the first equal to zero. Then $f(q, p) + (N - 1) f(0, 0) = \alpha$, so f is constant. This proves injectivity on the quotient by constants. □

The theorem constructs a global ordinary trace action and ideal membership, not a unique expression of every defect in moment generators. The strict order argument supplies finite expressions after choices. Compatible primitives in a resolution must still satisfy their higher equations. The scalar rank $J_N(1) = N$ also remains present before passing to the quotient by constants.

The parameter-adic completion has an exact description. Writing $J = I_{N,\hbar} \cap W_{N,\hbar}^{GL_N}$, saturation gives $J \cap \hbar^m W_{N,\hbar}^{GL_N} = \hbar^m J$. The R-free quotient makes reduction of the defining short exact sequence modulo $\hbar^m$ exact. The transition maps $J/\hbar^{m+1} J \to J/\hbar^m J$ are surjective, so compatible lifts corrected in these kernels prove exactness after inverse limit. Thus $\Phi_{N,\hbar}$ completes to an isomorphism, and the completed kernel is the closure of J. Every product at fixed $\hbar$-order uses finitely many polynomial coefficients. Their degrees may grow with the order.

10.9 Relative reduction and the remaining comparison

There are now two positive comparisons. The relative rank-two dg algebra maps strictly to its Weyl–Cartan cohomology algebra, retaining the odd relation module and every specialization. The ordinary reduction at arbitrary rank has its integral trace action across the discriminant. The maps have different sources and preserve different data. Neither identifies the absolute BRST complex merely by taking rational group invariants.

For rank three, an extension requires a specified relative quantum relation complex and a map to its derived Cartan recipient. Its relation cohomology and possible invariant Massey or extension classes must be computed. A value of one inverse component would not decide whether such a comparison exists.

The full Hamiltonian question also retains the positive gauge ghosts. It asks for a coherent moment comparison on the absolute rank-two quantum recipient and its higher-rank extensions, strict on the scalar quadratics and with the cubic primitive of Section 9.6. The comparison must carry the negative relation classes, stabilizer states and cyclic operations across the discriminant. The coefficient-plane parameter ε is still independent of matrix $\hbar$. These requirements are additional to the strict relative restriction and the ordinary trace theorem. They are the operations to compare after the absolute complex and its higher model have been constructed.

Chapter II

Absolute quantum reduction and higher products

The relative reduction retains relations among the moment equations. The absolute reduction must also retain the gauge directions. Even a trivial coefficient module for $\mathfrak{so}_3$ has a degree-three Lie cohomology class. Multiplying this class by the relative degree-minus-one relation class produces a degree-two class. Both disappear if one keeps only invariant coefficients and discards the positive ghosts.

We compute the full polynomial traceless rank-two complex over $R = \mathbb{C}[\hbar]$. Its cohomology is supported in degrees -1, 0, 2, 3. A contraction constructs this answer across the discriminant and specifies every higher product. Its projection cannot be a unital graded algebra map to the cohomology ring. That obstruction has a fixed direction: whether a zigzag of dg-algebra quasi-isomorphisms exists is a separate question.

11.1 The positive ghosts

Use the traceless coordinates x_i, y_i and antighosts ψ_i of Chapter 10, for $1 \le i \le 3$. Their degrees are 0, 0, -1. Adjoin positive ghosts γ^i of degree 1 and impose

$$[x_i, y_j] = \hbar\delta_{ij}, \qquad \psi_i\gamma^j + \gamma^j\psi_i = \hbar\delta_i^j, \qquad \gamma^i\gamma^j = -\gamma^j\gamma^i, \quad \psi_i\psi_j = -\psi_j\psi_i. \tag{11.1}$$

The even coordinates commute with the ghosts. The x_i commute among themselves, as do the y_i. The odd squares are zero. Write $\mathrm{Cl}_\hbar(\gamma, \psi)$ for this graded Clifford algebra and put

$$\mathcal{B}_\hbar = W_\hbar(x, y) \otimes_R \mathrm{Cl}_\hbar(\gamma, \psi).$$

Use symmetric Weyl symbols in x, y and place all positive ghosts to the left of all antighosts. The ordered products $\gamma^I\psi_J$, for increasing subsets $I, J \subset \{1, 2, 3\}$, form an R-basis of the ghost algebra. Indeed the relations span it by these products. After inverting $\hbar$, their action on $\Lambda(\gamma)$ by multiplication and $\psi_i = \hbar\partial_{\gamma^i}$ is the full matrix algebra, whose dimension is 2^6. Thus they are independent already over R.

For the rotation moment define

$$\mu_i = \sum_{j,k} \epsilon_{ijk} x_j y_k, \qquad \Theta = \sum_i \gamma^i\mu_i - \frac{1}{2}\sum_{i,j,k} \epsilon_{ijk}\gamma^i\gamma^j\psi_k, \qquad Q = \frac{[\Theta, -]_{\mathrm{super}}}{\hbar}. \tag{11.2}$$

Here $\epsilon_{123} = 1$ is alternating, and the bracket is the graded commutator for the full product. Division means the polynomial quotient of a divisible numerator. The algebra modulo $\hbar$ is graded commutative, so every such commutator is divisible. The Weyl–Clifford basis makes the quotient unique.

The moment relation is $[\mu_i, \mu_j] = \hbar\sum_k \epsilon_{ijk}\mu_k$. Together with Jacobi it gives $\Theta^2 = 0$, as in Proposition 9.13. One can also represent the ghosts on $\Lambda(\gamma)$ after inverting $\hbar$. The charge becomes the Chevalley operator for the representation $\mu_i/\hbar$, multiplied by $\hbar$. Its square vanishes. Faithfulness and polynomial

torsion-freeness give the same identity before inversion. Consequently $Q^2 = 0$, and Q is a degree-one R-linear derivation. These statements remain valid at $\hbar = 0$ by specializing the polynomial formulas.

The distinction from invariants already appears in the scalar Chevalley complex. Its differential is

$$d\gamma^1 = -\gamma^2\gamma^3, \qquad d\gamma^2 = \gamma^1\gamma^3, \qquad d\gamma^3 = -\gamma^1\gamma^2. \tag{11.3}$$

It is an isomorphism from exterior degree one to degree two and is zero in every other degree. Thus its cohomology is $\Lambda(\chi)$, where

$$\chi = \gamma^1\gamma^2\gamma^3, \qquad |\chi| = 3.$$

For the full complex, the positive ghosts also act on the polynomial coefficients. We next contract that action without removing the scalar and top-ghost classes.

11.2 Normal ordering and the two differentials

Let $K_{\mathrm{all}} = W_\hbar(x, y) \otimes \Lambda(\psi)$ be the space of coefficients with no positive ghost. For the basis e_i of $\mathfrak{so}_3$, use $[e_i, e_j] = \sum_k \epsilon_{ijk} e_k$. The simultaneous coefficient action ρ_i is

$$\rho_i x_j = \sum_k \epsilon_{ijk} x_k, \qquad \rho_i y_j = \sum_k \epsilon_{ijk} y_k, \qquad \rho_i \psi_j = \sum_k \epsilon_{ijk} \psi_k. \tag{11.4}$$

It extends by the derivation rule. On Weyl coefficients it is $[\mu_i, -]/\hbar$. It does not act on γ in this notation. Each polynomial lies in a finite-dimensional rational rotation module, so the coefficient representation is locally finite.

For $X \in \mathfrak{so}_3$, let μ_X and ψ_X denote the corresponding linear combinations. Define a degree-one operator on the antighost complex by

$$\begin{aligned} \partial_N(a\psi_{X_1}\cdots\psi_{X_q}) = {} & \sum_{i=1}^{q} (-1)^{i-1} (a \star \mu_{X_i}) \psi_{X_1} \cdots \widehat{\psi_{X_i}} \cdots \psi_{X_q} \\ & + \hbar \sum_{i<j} (-1)^{i+j} a \psi_{[X_i, X_j]} \psi_{X_1} \cdots \widehat{\psi_{X_i}} \cdots \widehat{\psi_{X_j}} \cdots \psi_{X_q}. \end{aligned} \tag{11.5}$$

The bracket factor in the second line stands first. For example,

$$\partial_N(a\psi_i\psi_j) = a \star \mu_i\, \psi_j - a \star \mu_j\, \psi_i - \hbar a \psi_{[e_i, e_j]}.$$

Applying ∂_N again gives the moment commutator minus the same Lie bracket. It is zero. In higher antighost degree, the terms involving two moments cancel in the same way; the terms involving three indices cancel by Jacobi. This proves $\partial_N^2 = 0$ in every degree. The moment map and the Lie bracket are equivariant, so ∂_N commutes with each ρ_i.

Let $\iota_i = \partial_{\gamma^i}$ be the left exterior derivative. Define

$$d_{\mathrm{CE}} = \sum_i \gamma^i \rho_i - \frac{1}{2} \sum_{i,j,k} \epsilon_{ijk} \gamma^i \gamma^j \iota_k. \tag{11.6}$$

This is the Chevalley cochain differential with coefficients in K_{all}. The Lie representation identity and Jacobi give $d_{\mathrm{CE}}^2 = 0$. On positive-ghost degree p, put $\partial^{\mathrm{tot}} = (-1)^p \partial_N$. Equivariance gives $d_{\mathrm{CE}}\partial^{\mathrm{tot}} + \partial^{\mathrm{tot}} d_{\mathrm{CE}} = 0$.

Lemma 11.1. *Normal ordering identifies the underlying complex with*

$$(\mathcal{B}_\hbar, Q) = \big(\Lambda(\gamma) \otimes K_{\mathrm{all}}, d_{\mathrm{CE}} + \partial^{\mathrm{tot}}\big). \tag{11.7}$$

On invariant positive-ghost-free coefficients, Q is the relative differential $\delta_\hbar$ of (10.7).

Proof. Commute each positive ghost produced by $[\Theta, -]$ to the left. The terms retaining one new ghost give the simultaneous coefficient action. A contraction with an existing positive ghost gives the second term of (11.6). Removing one antighost gives the moment term in (11.5). Contracting the cubic ghost term against two antighosts gives its bracket term. Moving this degree-one operator past p positive ghosts gives $(-1)^p$.

On generators, for example,

$$Q\gamma^1 = -\gamma^2\gamma^3, \qquad Q\psi_1 = \mu_1 - \gamma^2\psi_3 + \gamma^3\psi_2, \qquad Qx_j = \sum_{i,k} \gamma^i \epsilon_{ijk} x_k.$$

The same formula holds for Qy_j. The ordered computation above applies to an arbitrary product and hence proves the identity on the whole basis.

For invariant coefficients, $\sum \gamma^i \rho_i$ vanishes. Invariance combines the ordered moment and bracket terms into

$$\frac{1}{2}\sum_i \Big(\mu_i \star \partial_{\psi_i} F + \partial_{\psi_i} F \star \mu_i\Big).$$

The first Moyal derivatives cancel. The second derivatives of the bilinear moments give $\hbar^2 \sum \epsilon_{ijk}\partial_{x_j}\partial_{y_k}\partial_{\psi_i}/4$, and all higher derivatives vanish. This is the stated relative differential. □

Equation (11.7) is an equality of complexes after choosing an ordered basis. Multiplication still uses (11.1). Replacing it by the exterior tensor product would remove terms needed for both the ring comparison and the higher products.

11.3 Contracting the coefficient action

The scalar Chevalley complex should remain. To contract its nontrivial coefficient representations, define

$$\mathsf{C} = \sum_i \rho_i^2, \qquad K = \sum_i \rho_i \iota_i. \tag{11.8}$$

Here C acts only on K_{all}. The diagonal rotation action on both coefficients and positive ghosts has a different Casimir.

Lemma 11.2. *The operators satisfy*

$$d_{\mathrm{CE}}K + K d_{\mathrm{CE}} = \mathsf{C}, \qquad \partial^{\mathrm{tot}}K + K\partial^{\mathrm{tot}} = 0.$$

Moreover C commutes with both differentials.

Proof. Use $\iota_j\gamma^i + \gamma^i\iota_j = \delta^i_j$. The anticommutator of K with $\sum \gamma^i\rho_i$ is

$$\mathsf{C} + \sum_{i,j,k} \epsilon_{ijk}\gamma^i\rho_k\iota_j.$$

Its anticommutator with the bracket part of d_{CE} is the negative of the second summand. Indeed commuting ι_ℓ through $\gamma^i\gamma^j$ gives $\delta^i_\ell\gamma^j - \delta^j_\ell\gamma^i$; antisymmetry then gives the displayed cancellation.

The coefficient Casimir is central for the rotation action, so it commutes with d_{CE}. Equivariance of ∂_N gives its commutation with ρ_i and C. Finally K lowers positive-ghost degree by one, reversing the totalization sign of ∂_N. This proves the second anticommutator identity. □

On an irreducible spin-ℓ coefficient representation, $\mathsf{C} = -\ell(\ell+1)$, for $\ell \geq 0$. These are integer spins because the representation is rational for $SO(3)$. To check the normalization, put $E = \rho_1 + i\rho_2$, $F = -\rho_1 + i\rho_2$ and $H_{\mathrm{wt}} = 2i\rho_3$. They satisfy $[H_{\mathrm{wt}}, E] = 2E$, $[H_{\mathrm{wt}}, F] = -2F$, $[E, F] = H_{\mathrm{wt}}$, and

$$\mathsf{C} = -\frac{1}{4}H_{\mathrm{wt}}^2 - \frac{1}{2}(EF + FE).$$

On a highest-weight vector of weight 2ℓ, this is $-\ell^2 - \ell$. Complete reducibility in each finite coefficient module follows by averaging a Hermitian form over the compact real rotation group and taking invariant orthogonal complements. Thus the eigenvalue decomposition used below is an actual direct sum on every finite input module. For an input contained in spins at most L, define

$$P_L(X) = \prod_{\ell=1}^{L}\left(1 + \frac{X}{\ell(\ell+1)}\right), \qquad G_L(X) = \frac{1 - P_L(X)}{X} + P_L'(0)P_L(X). \tag{11.9}$$

Both expressions are polynomials. The second has value zero at $X = 0$ and value $1/X$ at every nonzero Casimir eigenvalue in this range. Thus

$$P_0 = P_L(\mathsf{C}), \qquad G = G_L(\mathsf{C})$$

are, respectively, projection to coefficient invariants and the inverse Casimir on their complement, extended by zero. Enlarging L gives the same operators on the original input. This constructs them on the union of finite coefficient modules. They commute with equivariant maps, including ∂_N. No power of $\hbar$ or of the discriminant is inverted.

Set $H_0 = GK$. Then

$$QH_0 + H_0Q = 1 - P_0, \qquad P_0H_0 = H_0P_0 = 0.$$

To impose the square-zero side condition, replace H_0 by

$$H_1 = H_0 d_{\mathrm{CE}} H_0. \tag{11.10}$$

On the complement of P_0, write $d = d_{\mathrm{CE}}$ and $h = H_0$. The identity $dh + hd = 1$ gives $dh^2 = h^2 d$. Hence

$$d(hdh) + (hdh)d = dh + hd = 1, \qquad (hdh)^2 = h(dh^2d)h = 0.$$

The total chain differential anticommutes with h, d, h in succession, and therefore with hdh. Consequently

$$QH_1 + H_1Q = 1 - P_0, \qquad H_1^2 = H_1P_0 = P_0H_1 = 0. \tag{11.11}$$

The image of P_0 is $\mathrm{CE}^*(\mathfrak{so}_3, \mathbb{C}) \otimes \mathcal{K}_\hbar$, with the tensor differential. The next step contracts its first factor explicitly, then uses the relative comparison.

11.4 The absolute contraction

Let $j_\gamma : \Lambda(\chi) \to \Lambda(\gamma)$ include the scalar and top-ghost monomials, and let p_γ project onto them. The homotopy for (11.3) is

$$h_\gamma(\gamma^2\gamma^3) = -\gamma^1, \quad h_\gamma(\gamma^1\gamma^3) = \gamma^2, \quad h_\gamma(\gamma^1\gamma^2) = -\gamma^3, \tag{11.12}$$

and zero on other exterior degrees. Direct substitution gives $dh_\gamma + h_\gamma d = 1 - j_\gamma p_\gamma$ and the normalized side conditions.

Write $(\rho_\hbar, i_{\mathrm{rel}}, H_{\mathrm{rel}})$ for the relative contraction in Section 10.4. Its target is

$$\mathcal{H}_\hbar = \big(W_\hbar(s,t) \otimes \Lambda(\theta)\big)^{\mathbb{Z}/2}, \qquad [s,t] = \hbar, \quad |\theta| = -1.$$

The involution negates s, t, θ and fixes $\hbar$. Choose its displayed section so that $i_{\mathrm{rel}}(s\theta) = \xi = x \cdot \psi$, $i_{\mathrm{rel}}(t\theta) = \eta = y \cdot \psi$ and $H_{\mathrm{rel}}(\xi \star \eta) = r = \psi_1\psi_2\psi_3$. Its formulas use finite classical weight splittings and the terminating coefficient operator T^{-1}.

Put $Z = \Lambda(\chi) \otimes \mathcal{H}_\hbar$, initially as a graded R-module with zero differential. Compose the three contractions: first (11.11), then $(j_\gamma, p_\gamma, h_\gamma)$ tensored with the relative complex, and finally the relative contraction tensored with $\Lambda(\chi)$. On the last tensor product, for $e \in \{0,1\}$, the homotopy is

$$H'(\chi^e \otimes u) = (-1)^{3e}\chi^e \otimes H_{\mathrm{rel}} u.$$

For two successive contractions, the composite homotopy is $H_A + i_A H_B p_A$. Expansion of this expression proves the contraction identity; the mixed terms in its square vanish by the side conditions. This specifies the final homotopy H on every polynomial input.

Theorem 11.3. *There is a normalized contraction over R*

$$(\mathcal{B}_\hbar, Q) \underset{i}{\overset{\Pi}{\rightleftarrows}} (Z, 0), \qquad QH + HQ = 1 - i\Pi, \qquad \Pi i = 1, \quad H^2 = Hi = \Pi H = 0.$$

It is polynomial on each input and remains a contraction after every commutative coefficient base change $R \to R'$. For $F = \sum_I \gamma^I F_I$ in normal order,

$$\Pi(F) = \rho_\hbar(P_0 F_\varnothing) + \chi\,\rho_\hbar(P_0 F_{123}), \qquad i(u + \chi v) = i_{\mathrm{rel}} u + \chi\, i_{\mathrm{rel}} v. \tag{11.13}$$

Proof. The three contractions just constructed have precisely these inclusion and projection maps. Composition proves all displayed identities. The Casimir operators use finite-dimensional coefficient representations and only nonzero complex integer denominators. The relative operator T^{-1} terminates on each polynomial, as do the finite weight splittings. Every constituent is R-linear with polynomial output. Tensoring the identities with R' therefore preserves them, without a flatness assumption on R'. The resulting homotopy also proves that the projection kernel is contractible, rather than merely giving a list of proposed cohomology representatives. □

Let $E = W_\hbar(s,t)^{\mathrm{even}}$ and $M = W_\hbar(s,t)^{\mathrm{odd}}$, where parity means total polynomial degree modulo two. The cohomology groups are

$$H^{-1} = M\theta, \qquad H^0 = E, \qquad H^2 = \chi M\theta, \qquad H^3 = \chi E, \tag{11.14}$$

and all other degrees vanish. The odd module and its singular fibre from the relative calculation remain present. The absolute operation adds the two positive degrees. This is a polynomial statement. Extending the contraction to a completion requires its own continuity and convergence conditions.

11.5 Products and the failure of a strict projection

Give $Z = \Lambda(\chi) \otimes \mathcal{H}_\hbar$ the graded tensor product. Thus $u\chi = (-1)^{|u|}\chi u$ and $\chi^2 = 0$. The product on $\mathcal{H}_\hbar$ is its Weyl–exterior product from the relative restriction.

Proposition II.4. *Under the contraction, the cohomology algebra is*

$$H^*(\mathcal{B}_\hbar, Q) \cong \Lambda(\chi) \otimes \mathcal{H}_\hbar.$$

This is an isomorphism of cohomology rings. The linear projection Π *is not an algebra map on the full complex.*

Proof. Every relative class has an invariant closed representative of antighost degree zero or one. Products of representatives with no top ghost have the relative cohomology product. For a product with one top ghost, normal ordering preserves the term with three positive ghosts, with exactly the graded tensor sign. All contraction terms have positive-ghost degree two and are killed by Π. The relative projection of the remaining coefficient is its relative product.

Products with two top ghosts and at most two antighosts are zero. Six positive ghosts cannot survive among three exterior generators unless at least three are removed by contractions. Only two antighosts are available. This proves all products in the asserted ring. Since the representatives are closed and Π is the projection of a contraction, these projected products compute their actual cohomology classes. □

The graded commutation with χ is a cohomology identity. Its cochain defect has an explicit primitive:

$$\chi \star \xi + \xi \star \chi = \hbar Q\Big(\sum_i x_i \gamma^i\Big), \qquad \chi \star \eta + \eta \star \chi = \hbar Q\Big(\sum_i y_i \gamma^i\Big). \tag{II.15}$$

For the first equation, normal ordering gives the left side as

$$\hbar\big(x_1\gamma^2\gamma^3 - x_2\gamma^1\gamma^3 + x_3\gamma^1\gamma^2\big).$$

The coefficient-action part of $Q(\sum x_i\gamma^i)$ gives twice the expression in parentheses, and the scalar Chevalley part subtracts it once. The antighost differential does not enter. This proves the first identity, including its sign; replacing x by y proves the second.

Proposition II.5. *There is no graded unital R-algebra map $\mathcal{B}_\hbar \to Z$.*

Proof. Equation (II.14) shows $Z^1 = 0$. Such a map would send every γ^i to zero. The Clifford relation $\psi_i\gamma^i + \gamma^i\psi_i = \hbar$ would then send $\hbar$ to zero. A unital R-linear map must preserve it, and $\hbar \neq 0$ in the R-free target. This is a contradiction. □

The assertion fixes the direction of the map and the coefficient ring. At the classical specialization $\hbar = 0$, its Clifford obstruction disappears. It also says nothing about a sequence of quasi-isomorphisms whose arrows may point in either direction. A dg algebra is *formal* if such a sequence connects it to its cohomology ring with zero differential. Determining formality of this absolute algebra requires an invariant of that comparison problem.

II.6 The higher model

The contraction provides the operation data beyond the ring. Let s denote suspension of degree -1, distinct from the Cartan coordinate in $W_\hbar(s,t)$, and put

$$V = s\mathcal{B}_\hbar, \qquad W = sZ, \qquad d_V = -sQs^{-1}, \qquad h_V = -sHs^{-1}, \qquad i_s = sis^{-1}, \quad p_s = s\Pi s^{-1}.$$

For a graded module U, use the tensor coalgebra $T^c(U) = \bigoplus_{n\geq 1} U^{\otimes n}$ with the deconcatenation coproduct. Tensor products of maps use the Koszul sign. On $T^c(V)$ let d_T be the tensor differential and let the multiplication coderivation be determined by

$$b_2(sa, sb) = (-1)^{|a|} s(a \star b). \tag{11.16}$$

The derivation property of Q and associativity give $(d_T + b_2)^2 = 0$. This uses the underlying algebra with its unit forgotten; it assumes no scalar augmentation of a Weyl algebra.

On tensor length n extend the contraction by

$$i_T = i_s^{\otimes n}, \quad p_T = p_s^{\otimes n}, \qquad \mathbf{H} = \sum_{j=1}^{n} (i_s p_s)^{\otimes (j-1)} \otimes h_V \otimes 1^{\otimes (n-j)}. \tag{11.17}$$

The identity $d_V h_V + h_V d_V = 1 - i_s p_s$ telescopes to

$$d_T \mathbf{H} + \mathbf{H} d_T = 1 - i_T p_T, \qquad p_T i_T = 1, \qquad \mathbf{H}^2 = \mathbf{H} i_T = p_T \mathbf{H} = 0.$$

For instance, in a mixed term of $\mathbf{H}^2$ the earlier homotopy is followed by $i_s p_s$ or the later one is preceded by it. The factors $h_V i_s$ and $p_s h_V$ kill these terms; the equal-position term contains h_V^2.

Theorem 11.6. *The finite-at-each-length formulas*

$$\begin{aligned} b_Z &= p_T b_2 (1 + \mathbf{H} b_2)^{-1} i_T, \\ \mathbf{I} &= (1 + \mathbf{H} b_2)^{-1} i_T, \qquad\qquad \mathbf{P} = p_T (1 + b_2 \mathbf{H})^{-1} \end{aligned} \tag{11.18}$$

give a minimal A_∞ structure on Z and A_∞ comparison maps with $(\mathcal{B}_\hbar, Q)$. The binary product is the ring in Proposition 11.4. All components are polynomial over R on polynomial inputs and persist under coefficient base change. The construction has a strictly unital normalized presentation.

Proof. Each occurrence of b_2 reduces tensor length by one. Thus every inverse in (11.18) is a finite geometric sum on a tensor of fixed length n, with at most $n - 1$ occurrences of b_2. Apply the perturbation identity of Lemma 8.8 to the tensor contraction. Its finite sums give

$$b_Z^2 = 0, \quad (d_T + b_2)\mathbf{I} = \mathbf{I} b_Z, \quad \mathbf{P}(d_T + b_2) = b_Z \mathbf{P}, \quad \mathbf{P}\mathbf{I} = 1.$$

The comparison homotopy is $\mathbf{H}' = (1 + \mathbf{H} b_2)^{-1} \mathbf{H}$.

We also need the coalgebra structures, not just these complex identities. Cutting a tensor in (11.17) on either side of its homotopy factor gives

$$\Delta \mathbf{H} = \big(\mathbf{H} \otimes 1 + i_T p_T \otimes \mathbf{H}\big)\Delta. \tag{11.19}$$

The maps i_T, p_T are coalgebra maps, and b_2 is a coderivation. Insert these three identities into the finite series. Induction on the number of multiplications gives

$$\Delta \mathbf{I} = (\mathbf{I} \otimes \mathbf{I})\Delta, \qquad \Delta \mathbf{P} = (\mathbf{P} \otimes \mathbf{P})\Delta, \qquad \Delta b_Z = (b_Z \otimes 1 + 1 \otimes b_Z)\Delta.$$

To describe the induction explicitly, cut each term at one of its surviving tensor gaps. The multiplication is on the left or right of that gap, so the two choices give the two tensor factors of the preceding identities. A homotopy crossing the cut splits as the two terms of (11.19). Terms with $\mathbf{H} i_T$, $p_T \mathbf{H}$ or $\mathbf{H}^2$ vanish. The remaining left and right strings are exactly the shorter geometric series. Their tensor signs are inherited from the displayed coproduct identity. This proves the induction at every length.

There is no unary component of b_Z, since the cohomology differential is zero. The binary component is $p_s b_2 (i_s \otimes i_s)$, which is the computed cohomology product. Projection to length one gives all higher components.

Each is a finite sum of products and the polynomial contraction, so it has the asserted coefficient and specialization properties.

Finally $i(1) = 1$, $\Pi(1) = 1$ and $H(1) = 0$. In a higher tree with a unit input, the first multiplication meeting that input removes the unit. An adjacent homotopy then gives $Hi = 0$ or $H^2 = 0$, or the final projection gives $\Pi H = 0$. This is the normalized strict-unit construction: the binary unit laws remain, while higher operations and higher comparison components with a unit input vanish. It removes the tensors degenerate by a unit insertion, without introducing an algebra augmentation. □

The formulas specify every operation once the relative weight splittings are chosen. They do not assert that the higher operations vanish. For the next calculation, keep the relative section and homotopy fixed in Section 11.4.

11.6.1 *A ternary operation from two trees*

The chosen full homotopy has

$$H(\xi \star \eta) = r, \qquad H(\eta \star \chi) = \hbar \sum_i y_i \gamma^i. \tag{11.20}$$

The first equality is the relative choice. For the second, write

$$\eta \star \chi = -\chi\eta + \hbar\omega_y, \qquad \omega_y = y_1\gamma^2\gamma^3 - y_2\gamma^1\gamma^3 + y_3\gamma^1\gamma^2.$$

The first term is a chosen cohomology representative. On the coefficient spin-one summand, $G = -1/2$. Direct rotation gives $K\omega_y = -2\sum_i y_i\gamma^i$ and $d_{\mathrm{CE}}(\sum_i y_i\gamma^i) = \omega_y$. Thus the normalized Casimir homotopy sends ω_y to $\sum_i y_i\gamma^i$, proving the second equality.

Proposition 11.7. *For this contraction, the suspended ternary coefficient is*

$$(b_Z)_3\big(s(s\theta), s(t\theta), s\chi\big) = s\Big(\hbar^2 st + \frac{1}{2}\hbar^3\Big). \tag{11.21}$$

Inside the parentheses, s, t are the Cartan Weyl symbols. The outer s is suspension.

Proof. The three suspended inputs have even parity. The length-three term of (11.18) is $-p_T b_2 \mathbf{H} b_2 i_T$. Using (11.20) and the minus sign in $h_V = -sHs^{-1}$, its two unsuspended contributions have a plus sign and are

$$\Pi(r \star \chi), \qquad \Pi\left(\xi \star \hbar \sum_i y_i \gamma^i\right).$$

The scalar part of $\psi_1\psi_2\psi_3\gamma^1\gamma^2\gamma^3$ is $-\hbar^3$. All other terms have one or two positive ghosts, or have three positive ghosts and the relative factor r, whose projection is zero. Hence

$$\Pi(r \star \chi) = -\hbar^3.$$

In the second tree, only the contractions $\psi_i\gamma^j \mapsto \hbar\delta_i^j$ contribute to positive-ghost degree zero. The Weyl product gives $x_i \star y_i = x_i y_i + \hbar/2$. The invariant sum $b = \sum_i x_i y_i$ restricts to st, so this tree has projection

$$\hbar^2 st + \frac{3}{2}\hbar^3.$$

Their sum proves the formula. □

Both the complete ghost contraction and the Weyl scalar term are necessary. The coefficient is a property of this specified minimal model. To use it as a formality obstruction, one would have to show that its Hochschild class cannot be removed by an A_∞ change of coordinates. A single displayed value does not prove that assertion. The polynomial-subalgebra section in Proposition 10.9 already shows how particular relative higher values can change without changing the underlying quasi-isomorphism type.

II.7 The central matrix direction

The traceless calculation uses $\mathfrak{sl}_2$. For $\mathfrak{gl}_2$, split off the scalar matrix direction using the trace pairing. Normalize its coordinates x_0, y_0 so that $[x_0, y_0] = \hbar$. Its moment function and adjoint action are zero. The corresponding ghosts therefore have zero differential, but their product is

$$\psi_0\gamma^0 + \gamma^0\psi_0 = \hbar, \qquad (\gamma^0)^2 = \psi_0^2 = 0, \qquad |\gamma^0| = 1, \quad |\psi_0| = -1. \tag{II.22}$$

In this split normalization, the full complex is

$$(\mathcal{B}_\hbar, Q) \otimes_R W_\hbar(x_0, y_0) \otimes_R \big(\mathrm{Cl}_\hbar(\gamma^0, \psi_0), 0\big).$$

Both additional factors are R-free with zero differential. Tensoring the constructed contraction computes its cohomology ring as the same two factors tensored with Z.

At $\hbar = 0$ the central ghost product specializes to the exterior product. Over R it retains (II.22). After inverting $\hbar$, the central Clifford algebra acts as the full endomorphism algebra of the two-dimensional graded space $\Lambda(\gamma^0)$. This may be used for a Morita comparison of module categories; it does not identify the algebra with an exterior algebra by a dg-algebra quasi-isomorphism.

II.8 Absolute formality and the Hamiltonian comparison

The contraction determines the absolute complex, its ring and one complete higher model over $\mathbb{C}[\hbar]$. The formality question asks for a zigzag to the displayed cohomology ring, or an obstruction invariant under every allowed change of higher model. The first possible higher product must be treated as a Hochschild class, including its indeterminacy under such changes. The strict-map obstruction and the chosen ternary value have their narrower conclusions.

The Hamiltonian comparison requires additional structure. Its first component must agree with the global ordinary trace map, remain strict on the scalar quadratic sector, and retain the cubic primitive of Section 9.6. The higher equations must be solved on this full absolute recipient, with the negative relation classes, positive gauge classes and cyclic operations present. The matrix parameter $\hbar$ is independent of the coefficient-plane parameter ε. A linear contraction does not supply those equivariances.

Nor does the rank-two calculation give a higher-rank absolute contraction or a field-theoretic realization. Those comparisons must specify their own relation complexes, supported operations and coefficient completions. The explicit recipient constructed here makes their required maps precise, including the classes that an ordinary degree-zero reduction would lose.

Chapter 12

Cyclic traces and wheel corrections

A matrix trace remembers the size of the matrix: $\mathrm{Tr}(I_N) = N$. Invariant quantization also remembers a scalar. For the quadratic $\mathfrak{sl}_2$ invariant, the Duflo operator adds $\hbar^2/2$ to the symmetrized Casimir. These scalars matter when matrices are stabilized or Lie algebras are completed. Removing them can destroy compatibility with brackets or continuity.

We first construct trace maps on cyclic chains and specify the rank transitions they preserve. We then compute the finite wheel corrections and examine two different infinite-dimensional coefficient spaces. The cyclic orientation of Chapter 9 remains a chain-level structure throughout. A scalar state requires its own functional.

12.1 Matrix trace on the mixed complex

Let B be a unital associative complex algebra and put $\overline{B} = B/\mathbb{C}1$. Its normalized Hochschild chains are

$$C_n(B) = B \otimes \overline{B}^{\otimes n}, \qquad a_0[a_1|\cdots|a_n] \in C_n(B).$$

Place $C_n(B)$ in cohomological degree $-n$. For $n > 0$ the Hochschild boundary is

$$\begin{aligned} b(a_0[a_1|\cdots|a_n]) = {} & a_0a_1[a_2|\cdots|a_n] \\ & + \sum_{j=1}^{n-1}(-1)^j a_0[a_1|\cdots|a_ja_{j+1}|\cdots|a_n] \\ & + (-1)^n a_na_0[a_1|\cdots|a_{n-1}], \end{aligned}$$

and $b = 0$ on $C_0(B)$. The Connes operator is

$$\mathsf{B}(a_0[a_1|\cdots|a_n]) = \sum_{j=0}^{n}(-1)^{nj}1[a_j|\cdots|a_n|a_0|\cdots|a_{j-1}].$$

Units in bar slots are zero. The identities $b^2 = \mathsf{B}^2 = b\mathsf{B} + \mathsf{B}b = 0$ give the mixed complex. With $|\zeta| = 2$, its negative-cyclic complex is

$$CC^-(B) = \big(C_\bullet(B)[[\zeta]], b + \zeta\mathsf{B}\big).$$

Here the completion is in ζ. In particular, it does not complete the coefficients of B.

For a second unital algebra D, define

$$\mathsf{T}_N\big(M_0[M_1|\cdots|M_n]\big) = \sum_{i_0,\dots,i_n=1}^{N}(M_0)_{i_0i_1}[(M_1)_{i_1i_2}|\cdots|(M_n)_{i_ni_0}], \tag{12.1}$$

from matrix chains to chains of D. For $n = 0$ the formula is the ordinary matrix trace. The indices form a closed path, so a change of the first index does not change the sum.

Proposition 12.1. *The map T_N is a morphism of mixed complexes $C_\bullet(\mathrm{Mat}_N D) \to C_\bullet(D)$. It therefore extends continuously to negative-cyclic complexes. It is natural in D and additive under block sums of representations. Its value on the unit in degree zero is $N\,1$.*

Proof. A scalar identity in a bar slot forces the two adjacent indices to agree. The corresponding output has a scalar unit in that bar slot and is zero. Thus the formula is defined on normalized chains.

An internal Hochschild face multiplies two adjacent matrices. Expanding the product sums precisely over the shared index in (12.1). The first face has the same interpretation at the outer coefficient. For the last face, reindex the closed path cyclically. These identities prove $b\mathsf{T}_N = \mathsf{T}_N b$, with the displayed face signs.

In a Connes term the leading identity matrix forces its adjacent indices to coincide. After deleting that repeated index, the remaining sum is the cyclic rotation of the same closed path. The rotation has the same sign $(-1)^{nj}$ on both sides. Hence $\mathsf{BT}_N = \mathsf{T}_N\mathsf{B}$. Coefficientwise application now commutes with $b + \zeta\mathsf{B}$ and with reduction modulo every power of ζ.

For block-diagonal matrices, a nonzero closed path lies entirely in one block. The trace of a block sum is consequently the sum of the two chain traces. Naturality follows by applying an algebra map to every matrix entry. Finally the ordinary trace of the identity is N. □

For dg algebras use the raw totalization and graded cyclic signs of (9.23) and (9.24). The same formula is a mixed-complex map: matrix indices have degree zero, the internal differential acts entrywise, and every cyclic permutation has the same Koszul sign before and after the index sum.

This gives a concrete image of the orientation already constructed on the free resolution. Let

$$Q = \mathbb{C}\langle q, p, t\rangle, \qquad |t| = -1, \qquad dt = qp - pq,$$

and let R_N be its derived representation algebra: the graded-commutative algebra on the entries of X, Y, Ψ, with $dX = dY = 0$ and $d\Psi = XY - YX$. Universal substitution is a dg algebra map $Q \to \mathrm{Mat}_N(R_N)$. The trace of its negative-cyclic orientation is

$$\omega_N = \sum_{i,j} \big(1[X_{ij}|Y_{ji}] - 1[Y_{ij}|X_{ji}]\big) - \sum_i 1[\Psi_{ii}]. \tag{12.2}$$

It is closed for both mixed differentials, by Proposition 12.1. The relation term has total degree -2, just as the two-bar terms do. The matrix trace also commutes with entrywise derivations, so it transports the coherent classical action and orientation of Theorem 9.10. These are constructions on derived representation coefficients. The full quantum BRST and supported field-theoretic comparisons retain their additional maps.

12.2 Rank and scalar reduction

The degree-zero cyclic space of B is

$$\mathfrak{c}_B = HC_0(B) = B/[B, B],$$

where $[B, B]$ is the linear span of commutators. Let $\mathrm{Rep}_N B$ represent unital algebra maps $B \to \mathrm{Mat}_N(D)$ for commutative complex algebras D, and put

$$\mathcal{R}_N = \mathcal{O}(\mathrm{Rep}_N B)^{GL_N}.$$

The ordinary trace is the linear map

$$\mathrm{Tr}_N : \mathfrak{c}_B \longrightarrow \mathcal{R}_N, \qquad \mathrm{Tr}_N([b])(\rho) = \mathrm{Tr}(\rho(b)).$$

The trace of a commutator is zero and conjugation preserves the trace, proving that this map has the stated source and target. It is generally not multiplicative. Even on scalars, its value at 1 is N.

Choose an augmentation $\epsilon : B \to \mathbb{C}$. It specifies a rank transition

$$s_N : \mathrm{Rep}_N B \longrightarrow \mathrm{Rep}_{N+1} B, \qquad \rho \longmapsto \rho \oplus \epsilon.$$

Define

$$\overline{\mathfrak{c}}_B = B/(\mathbb{C}1 + [B, B]), \qquad t_N(\overline{b})(\rho) = \mathrm{Tr}(\rho(b)) - N\epsilon(b).$$

Let $\mathfrak{m}_N$ be the ideal in $\mathcal{R}_N$ of functions vanishing at $b \mapsto \epsilon(b) I_N$, and set

$$\widehat{\mathcal{R}}_N = \varprojlim_{r \geq 1} \mathcal{R}_N/\mathfrak{m}_N^r.$$

If J is the ideal generated by $\overline{\mathfrak{c}}_B$ in $\mathrm{Sym}_{\mathbb{C}} \overline{\mathfrak{c}}_B$, write

$$\widehat{\mathrm{Sym}}_J \overline{\mathfrak{c}}_B = \varprojlim_{r \geq 1} \mathrm{Sym}_{\mathbb{C}} \overline{\mathfrak{c}}_B/J^r.$$

Proposition 12.2. *The maps t_N are well defined, have image in $\mathfrak{m}_N$, and satisfy $s_N^* t_{N+1} = t_N$. They extend to a continuous homomorphism of commutative algebras*

$$\widehat{\mathrm{Sym}}_J \overline{\mathfrak{c}}_B \longrightarrow \varprojlim_N \widehat{\mathcal{R}}_N.$$

All maps are natural for augmentation-preserving algebra maps.

Proof. Both terms defining t_N vanish on commutators, and their values on a scalar multiple of 1 cancel. Their values also cancel at the augmentation representation. Moreover,

$$s_N^* t_{N+1}(\overline{b}) = \mathrm{Tr}_N([b]) + \epsilon(b) - (N+1)\epsilon(b) = t_N(\overline{b}).$$

Since s_N takes the augmentation representation to the augmentation representation, $s_N^*(\mathfrak{m}_{N+1}^r) \subseteq \mathfrak{m}_N^r$. The symmetric-algebra extension of t_N sends J^r into $\mathfrak{m}_N^r$. It therefore defines a map on each pair of quotients and hence on their inverse limits. The stabilization identity makes these completed maps compatible in N. These formulas also prove naturality. □

The subtraction exists on cyclic chains as well. For a representation $\rho : B \to \mathrm{Mat}_N(D)$, subtract N times the map induced by $B \xrightarrow{\epsilon} \mathbb{C} \to D$ from $\mathsf{T}_N \rho_*$. Both summands are mixed-complex maps. Their difference vanishes on the scalar subcomplex and is unchanged by $\rho \mapsto \rho \oplus \epsilon$. This chain-level reduction has a chosen augmentation as part of its data.

A Poisson comparison requires more. Suppose $\mathfrak{c}_B$ is supplied with a Lie bracket and $\mathcal{R}_N$ with Poisson brackets satisfying

$$\{\mathrm{Tr}_N \beta, \mathrm{Tr}_N \gamma\}_N = \mathrm{Tr}_N [\beta, \gamma]_{\mathrm{cyc}}. \tag{12.3}$$

Then the ordinary trace is a Lie map. A Hamiltonian action through it also requires a Lie map $j : \mathfrak{h} \to \mathfrak{c}_B$. Neither datum is attached to an arbitrary associative algebra by its definition. For the free algebra on two generators, the necklace bracket and trace moment map provide (12.3); the polynomial-to-necklace comparison still has the cubic defect discussed in Chapter 9.

Scalar reduction need not preserve this bracket. For $B = \mathbb{C}[x, y]$ with $\{x, y\} = 1$ and augmentation at the origin, evaluation on N points gives

$$\{\mathrm{Tr}_N x, \mathrm{Tr}_N y\} = N, \qquad t_N(\overline{1}) = 0.$$

Thus Proposition 12.2 asserts a commutative algebra map. A reduced Poisson map needs a bracket-compatible reduction, for example a selected Hamiltonian sector on which the augmentation of every bracket vanishes.

12.3 The stable multisymmetric algebra

The rank dependence can be retained explicitly. For N points of the affine symplectic plane put

$$S_N = \mathbb{C}[x_1, y_1, \dots, x_N, y_N]^{\mathfrak{S}_N}, \qquad p_{ab}^{(N)} = \sum_{i=1}^{N} x_i^a y_i^b.$$

Give x_i, y_i degree one and, for $a + b > 0$, give the formal generator p_{ab} degree $a + b$. Evaluation at an extra point $(0, 0)$ defines $S_{N+1} \to S_N$.

Theorem 12.3. *In each fixed degree d, evaluation gives an isomorphism*

$$\mathbb{C}[p_{ab} \mid a, b \geq 0,\ a + b > 0]_d \xrightarrow{\sim} (S_N)_d \qquad (N \geq d).$$

Writing P for this graded polynomial algebra, one has

$$\widehat{P}_{\mathrm{wt}} = \prod_{d \geq 0} P_d \cong \varprojlim_N \prod_{d \geq 0} (S_N)_d$$

as complete commutative algebras.

Proof. A monomial orbit is specified by a finite multiset λ of nonzero pairs (a, b), one pair for each occupied point. Let $m_\lambda^{(N)}$ be the sum of the distinct monomials in this orbit. These sums form a basis when λ has at most N members. In total degree d, it has at most d members. The basis is therefore unchanged for $N \geq d$, and evaluation at the added origin preserves every basis element.

For $\lambda = \{\alpha_1, \dots, \alpha_r\}$ expand $p_{\alpha_1}^{(N)} \cdots p_{\alpha_r}^{(N)}$. Terms using distinct point indices give

$$\left(\prod_{\alpha} m_\alpha! \right) m_\lambda^{(N)},$$

where m_α is the multiplicity of α in λ. Every other term uses fewer than r points, merging the pairs assigned to the same point. Ordering orbit sums by the number of occupied points gives a triangular change of basis with nonzero diagonal. It proves both generation and independence in degree d.

Multiplication adds total degrees. Completing degreewise therefore gives $\widehat{P}_{\mathrm{wt}}$. A compatible family in the right-hand limit determines, in each degree, the unique stable coefficient just constructed. Conversely those coefficients determine the family. These inverse maps preserve multiplication and the inverse-limit topologies. □

The completion in this theorem is specified by total degree. It differs from the J-adic completion in which every p_{ab} has filtration one. For example, $\sum_{a \geq 1} p_{a0}$ belongs to $\widehat{P}_{\mathrm{wt}}$, but does not define an element of the J-adic completion: its reduction modulo J^2 would be an infinite linear combination in a vector space whose elements are finite linear combinations. Nor does the theorem replace the target by the inverse limit of the uncompleted rings S_N. The series $\sum_{r \geq 0} p_{10}^r$ already has a nonpolynomial image at $N = 1$.

These degree-zero trace functions do not record the relation and gauge cochain degrees of a derived matrix quotient. A comparison with a derived centre requires representation and trace maps on the specified complexes.

For the Poisson bracket, retain a central generator $r = p_{00}$ and put

$$P^{\mathrm{rk}} = \mathbb{C}[r, p_{ab} \mid a + b > 0].$$

Define on generators

$$\{p_{ab}, p_{cd}\} = (ad - bc)\, p_{a+c-1,b+d-1}, \qquad \{r, -\} = 0. \tag{12.4}$$

A term with a negative index is zero; in that case its coefficient is already zero.

Proposition 12.4. *Equation* (12.4) *defines a Poisson bracket. The evaluations $P^{\mathrm{rk}} \to S_N$, with $r \mapsto N$, are Poisson. Block sum is described by the Poisson coproduct*

$$\Delta p_{ab} = p_{ab} \otimes 1 + 1 \otimes p_{ab}, \qquad \Delta r = r \otimes 1 + 1 \otimes r.$$

Origin stabilization is not Poisson on the full S_N. On the subalgebras generated by $p_{ab}^{(N)}$ with $a + b \geq 2$, it is Poisson, and their stable algebra is $\mathbb{C}[p_{ab} \mid a + b \geq 2]$ with the same bracket.

Proof. The polynomial Poisson algebra $\mathbb{C}[x, y]$, regarded only as a Lie algebra, has basis $x^a y^b$ including 1. Its symmetric algebra is P^{rk}, where the Lie element 1 becomes the generator r, not the multiplicative unit. The Lie–Poisson construction extends its bracket by Leibniz and proves Jacobi.

Direct differentiation in S_N gives

$$\{p_{ab}^{(N)}, p_{cd}^{(N)}\} = (ad - bc)\, p_{a+c-1,b+d-1}^{(N)}, \qquad p_{00}^{(N)} = N.$$

This proves the evaluation assertion. The primitive coproduct preserves a Lie bracket on generators, so its extension is Poisson. It also equals the power-sum formula for a block sum.

In rank $N + 1$ the bracket of p_{10} and p_{01} is $N + 1$. Its pullback remains $N + 1$, whereas the bracket of the pullbacks is N. This proves the failure. If both input degrees are at least two, the output degree is at least two. The central rank never occurs, and setting the last point to zero preserves every displayed bracket. The triangular basis argument of Theorem 12.3, restricted to these generators, proves the final assertion. □

12.4 Stable matrix Lie homology

There is a second stabilization, with a different direction. Let

$$\mathfrak{gl}_\infty B = \varinjlim_N \mathfrak{gl}_N B$$

under block-zero Lie embeddings, and use trivial coefficients $\mathbb{C}$ for its Chevalley homology. This construction requires no augmentation of B. For clarity, ordinary cyclic homology is the homology of the cyclic complex

$$CC(B) = \big(C_\bullet(B)((\zeta))/\zeta C_\bullet(B)[[\zeta]], b + \zeta \mathsf{B}\big), \qquad HC_n(B) = H^{-n} CC(B).$$

The Loday–Quillen–Tsygan theorem identifies its primitive part:

$$\operatorname{Prim} H_n^{\mathrm{CE}}(\mathfrak{gl}_\infty B; \mathbb{C}) \cong HC_{n-1}(B).$$

Equivalently, if $HC_{n-1}(B)$ is placed in homological degree n, it gives an isomorphism of graded Hopf algebras

$$H_*^{\mathrm{CE}}(\mathfrak{gl}_\infty B; \mathbb{C}) \cong \operatorname{Sym}_{\mathrm{gr}}\left(\bigoplus_{n \geq 1} HC_{n-1}(B)\right). \tag{12.5}$$

Here $\mathrm{Sym}_{\mathrm{gr}}$ uses the Koszul sign rule. This is the characteristic-zero theorem for unital associative algebras of Loday and Quillen [70, Theorem 6.2 and Proposition 6.6]. The product comes from block sum and the coproduct from the diagonal Lie map.

The matrix-chain trace in (12.1) is also the trace appearing in this theorem. The invariant-theory description separates a permutation into its cycles: one cycle gives a primitive cyclic chain, and disjoint cycles give its graded symmetric products. The theorem concerns this stable Lie homology, not the inverse system of representation-coordinate rings.

For $B = \mathbb{C}$, normalized Hochschild chains are concentrated in degree zero. Thus $HC_{2j}(\mathbb{C}) = \mathbb{C}$ and $HC_{2j+1}(\mathbb{C}) = 0$, and

$$H_*^{\mathrm{CE}}(\mathfrak{gl}_\infty \mathbb{C}; \mathbb{C}) \cong \Lambda(u_1, u_3, u_5, \ldots), \qquad |u_{2j+1}| = 2j + 1.$$

These classes remain although the scalar-reduced cyclic space in Proposition 12.2 is zero.

12.5 Cyclic functionals on the Weyl algebra

Quantization further separates cyclic chains from a scalar trace. Let $R = \mathbb{C}[\hbar]$ and let $W_\hbar$ be the polynomial Weyl algebra generated by x, y with $[x, y] = \hbar$. Denote by $[W_\hbar, W_\hbar]_R$ the R-linear span of commutators.

Proposition 12.5. *One has*

$$[W_\hbar, W_\hbar]_R = \hbar W_\hbar, \qquad HH_0^R(W_\hbar) = W_\hbar/\hbar W_\hbar.$$

Every R-linear cyclic functional $W_\hbar \to R$ is zero. Every continuous $\mathbb{C}[[\hbar]]$-linear cyclic functional on its $\hbar$-adic completion with values in $\mathbb{C}[[\hbar]]$ is also zero.

Proof. Reduction modulo $\hbar$ is commutative, so every commutator belongs to $\hbar W_\hbar$. For every normal monomial,

$$[x, x^a y^{b+1}] = \hbar(b + 1) x^a y^b.$$

Since $b + 1$ is invertible in $\mathbb{C}$, these commutators span $\hbar W_\hbar$. The Hochschild boundary in degree one is a commutator, which gives the quotient formula.

If τ is cyclic and R-linear, then $\hbar\tau(x^a y^b) = 0$ for every a, b. The target has no $\hbar$-torsion, so all these values vanish. Continuity extends the same argument from the dense polynomial algebra to its completion. □

The quotient in the proposition is nonzero and has $\hbar$ acting as zero. It is the universal cyclic target in degree zero, rather than a torsion-free scalar ring. The negative-cyclic orientation of the resolved plane has a different degree and universal property. Neither it nor a prepared boundary state is removed by the proposition.

12.6 Finite formality and the Duflo map

Let $\mathfrak{g}$ be a finite-dimensional complex Lie algebra. Its linear Poisson tensor on $\mathfrak{g}^*$ is $\pi_\mathfrak{g}$. The polynomial polyvector complex, in unshifted cochain degrees, is

$$\mathcal{P}_\mathfrak{g} = \big((\mathrm{Sym}\,\mathfrak{g} \otimes \Lambda^\bullet \mathfrak{g}^*)[[\hbar]], \hbar d_{\mathrm{CE}}\big).$$

The differential is $[\hbar\pi_\mathfrak{g}, -]_{\mathrm{Sch}}$. Let $A_\hbar$ be the $\hbar$-adic completion of the Rees enveloping algebra with relations $xy - yx = \hbar[x, y]_\mathfrak{g}$. Coefficients remain polynomial at each parameter order.

Fix the affine formality morphism $\mathcal{U}$ with first component the normalized Hochschild–Kostant–Rosenberg map:

$$X_1 \wedge \cdots \wedge X_p \longmapsto \frac{1}{p!} \sum_{\sigma \in \mathfrak{S}_p} \operatorname{sgn}(\sigma) X_{\sigma(1)} \otimes \cdots \otimes X_{\sigma(p)}.$$

We use Kontsevich's affine theorem [64, Section 6.4]. Twisting by $\hbar\pi_{\mathfrak{g}}$ gives a product $\star$ on $\operatorname{Sym}\mathfrak{g}[[\hbar]]$. There is an algebra isomorphism

$$T : (\operatorname{Sym}\mathfrak{g}[[\hbar]], \star) \longrightarrow A_\hbar$$

fixing the linear generators. Indeed their star commutator is $\hbar[-,-]_{\mathfrak{g}}$. The induced enveloping-algebra map has associated graded the identity, and its filtered inverse is T. This algebra map includes the change from the star product to the PBW product. Raw PBW symmetrization identifies the modules and does not by itself identify these products.

The first component of the twisted comparison is

$$\Phi_{\mathfrak{g}}(v) = T_* \sum_{n \geq 0} \frac{1}{n!} \mathcal{U}_{n+1}((\hbar\pi_{\mathfrak{g}})^{\otimes n}, v). \tag{12.6}$$

It maps $\mathcal{P}_{\mathfrak{g}}$ to relative continuous Hochschild cochains of $A_\hbar$ over $\mathbb{C}[[\hbar]]$. The target differential is the Hochschild differential of the quantized product. All terms are defined at each parameter order. The affine theorem proves the twisted L_∞ identities on the complexes shifted by $[1]$.

The linear component is a quasi-isomorphism. Modulo $\hbar$ it is the polynomial HKR map, whose comparison with the full Hochschild complex is given by the polynomial diagonal Koszul resolution. Choose a contracting homotopy of its cone over $\mathbb{C}$ and extend it coefficientwise. The difference between the complete cone differential and this undeformed one raises $\hbar$ order. The convergent perturbation series of Lemma 8.8 therefore contracts the complete cone. This argument also explains why the scaled Chevalley differential cannot be omitted.

Invariant polynomials have a more explicit comparison. For $x \in \mathfrak{g}$, define the formal power series

$$j_{\mathfrak{g}}^{1/2}(x) = \det_{\mathfrak{g}}{}^{1/2}\left(\frac{\sinh(\operatorname{ad}_x/2)}{\operatorname{ad}_x/2}\right), \tag{12.7}$$

with constant term one. Interpret its Taylor coefficients as constant-coefficient differential operators on $\operatorname{Sym}\mathfrak{g}$. The Duflo theorem gives the algebra isomorphism

$$\operatorname{Duf}_\hbar = \operatorname{sym}_\hbar \circ j_{\mathfrak{g}}^{1/2}(\hbar\partial) : (\operatorname{Sym}\mathfrak{g})^{\mathfrak{g}}[[\hbar]] \xrightarrow{\sim} Z(A_\hbar); \tag{12.8}$$

see [64, Section 8.3.4]. On a polynomial, only finitely many differential orders act nontrivially. Its polynomial Rees formula may consequently be specialized at $\hbar = 1$. This specialization is not defined on arbitrary formal power series in $\hbar$.

For $\mathfrak{g} = \mathfrak{h}_{\leq d}$, these results apply to the finite Hamiltonian jets of Chapter 3. The relative bar–Chevalley comparison, the L_∞ formality map and the invariant algebra isomorphism in (12.8) preserve different specified operations. Their conclusions must be used with those operations.

The Hochschild brace action on $(C^\bullet(A_\hbar), A_\hbar)$ can be transported when the required operadic comparison is supplied. In detail, a homotopy-E_2 refinement is a quasi-isomorphism or a zigzag of quasi-isomorphisms from the selected complete homotopy-E_2 structure on $\mathcal{P}_{\mathfrak{g}}$ to the relative Hochschild complex, refining the cochain comparison. A mixed refinement also specifies the open operations and the coherences between the two colors.

Corollary 12.6 (Open–closed structure with a refinement)**.** *Given these refinements, the pair* $(\mathcal{P}_{\mathfrak{g}}, A_{\hbar})$ *has the transported homotopy Swiss-cheese action. Its open product is the enveloping product. Its closed and mixed operations are those carried by the specified operadic comparison.*

Proof. Transport the two-colored operations through the supplied equivalence of complete operadic algebras. The comparison includes the homotopies expressing substitution and the open–closed identities, so those identities hold on the transported pair. □

12.7 The quadratic and quartic corrections

For $\mathfrak{sl}_2$, use generators e, f, h with

$$[h,e] = 2e, \qquad [h,f] = -2f, \qquad [e,f] = h.$$

In the symmetric algebra set $q = 2ef + h^2/2$. In the Rees enveloping algebra write the generators as E, F, H and put $\Omega = EF + FE + H^2/2$. The relations have an additional factor $\hbar$. They imply that Ω is central: for example, $[E, EF + FE] = \hbar(EH + HE)$ and $[E, H^2/2] = -\hbar(EH + HE)$.

Proposition 12.7. *With these normalizations,*

$$\mathrm{Duf}_{\hbar}(q) = \Omega + \frac{\hbar^2}{2}.$$

At $\hbar = 1$*, its eigenvalue on the irreducible highest-weight-*n *module is* $(n+1)^2/2$*.*

Proof. For $x = ue + vf + wh$, the adjoint matrices give

$$\mathrm{Tr}(\mathrm{ad}_x^2) = 8uv + 8w^2.$$

The first nonconstant term of (12.7) is $\mathrm{Tr}(\mathrm{ad}_x^2)/48$. Thus its differential operator on q contributes

$$\frac{\hbar^2}{6}(\partial_e\partial_f + \partial_h^2)(2ef + h^2/2) = \frac{\hbar^2}{2}.$$

Higher differential orders annihilate q, and symmetrization sends q to Ω. On a highest-weight vector at $\hbar = 1$, EF acts by n, FE by zero and $H^2/2$ by $n^2/2$. Adding $1/2$ gives $(n+1)^2/2$. Centrality gives this value on the full irreducible module. □

The next invariant distinguishes a multiplicative quantization from raw symmetrization. Use complexified Hermitian generators J_i with

$$[J_i, J_j] = i\hbar\epsilon_{ijk}J_k, \qquad p = x_1^2 + x_2^2 + x_3^2, \qquad C = \sum_i J_i^2.$$

Here x_i are commuting symbols. The change $E = J_1 + iJ_2$, $F = J_1 - iJ_2$, $H = 2J_3$ gives $q = 2p$ and $\Omega = 2C$. Write $\Delta = \sum_i \partial_{x_i}^2$. The adjoint eigenvalues are $0, \sqrt{p}, -\sqrt{p}$, so the differential factor becomes

$$\frac{\sinh(\hbar\sqrt{\Delta}/2)}{\hbar\sqrt{\Delta}/2} = 1 + \frac{\hbar^2}{24}\Delta + \frac{\hbar^4}{1920}\Delta^2 + \cdots.$$

The square root denotes the displayed power series in Δ.

Proposition 12.8. *In the universal Rees algebra,*

$$\operatorname{sym}(p^2) = C^2 - \frac{\hbar^2}{3}C, \tag{12.9}$$

$$\operatorname{Duf}_\hbar(p) = C + \frac{\hbar^2}{4}, \tag{12.10}$$

$$\operatorname{Duf}_\hbar(p^2) = \left(C + \frac{\hbar^2}{4}\right)^2. \tag{12.11}$$

Proof. Antisymmetry of ϵ_{ijk} proves that C is central. The commutation relations give

$$\sum_j [J_j, [J_j, J_i]] = 2\hbar^2 J_i, \qquad \sum_j J_j J_i J_j = (C - \hbar^2) J_i.$$

The second identity follows by expanding the first. Symmetrizing the two repeated indices in p^2 gives

$$\operatorname{sym}(p^2) = \frac{1}{3}\sum_{i,j}\left(J_i^2 J_j^2 + J_i J_j J_i J_j + J_i J_j^2 J_i\right).$$

The three sums are C^2, $C^2 - \hbar^2 C$ and C^2, respectively. This proves (12.9).

Direct differentiation gives $\Delta p = 6$, $\Delta p^2 = 20p$ and $\Delta^2 p^2 = 120$. The quadratic correction is therefore $\hbar^2/4$. For p^2, the full expression is

$$C^2 - \frac{\hbar^2}{3}C + \frac{5\hbar^2}{6}C + \frac{\hbar^4}{16} = \left(C + \frac{\hbar^2}{4}\right)^2.$$

□

Thus the product of two operators assigned by $p \mapsto C$ differs from raw symmetrization of p^2 by $\hbar^2 C/3$. This is an operator identity before any state is chosen. In spin j, the eigenvalues of C and $\operatorname{Duf}_\hbar(p)$ are $\hbar^2 j(j+1)$ and $\hbar^2(j+1/2)^2$. At $\hbar = 1$, evolution by the second operator divided by two differs from evolution by $C/2$ by $e^{-t/8}$. Both factors obey the semigroup law, so that law alone does not choose the vacuum energy.

A multiplicative invariant map with $p \mapsto C$ is instead

$$Q(F(p)) = \operatorname{Duf}_\hbar\big(F(p - \hbar^2/4)\big).$$

The shift is an automorphism of the invariant polynomial algebra. This formula makes the ordering choice explicit; its relation to an independently prepared state requires the state comparison as well.

12.8 Half-densities and matrix wheels

The same formal determinant has a finite-dimensional geometric meaning. Let G be a local complex Lie group with unimodular Lie algebra $\mathfrak{g}$. For a dimensionless exponential coordinate X near zero, put $L = \operatorname{ad}_X$. Left trivialization gives

$$d\exp_X = \int_0^1 e^{-tL}\,dt = e^{-L/2}\frac{\sinh(L/2)}{L/2}.$$

Normalize the tangent volume and left-invariant volume to agree at the identity. Since $\operatorname{Tr} L = 0$, the exponential Jacobian and its normalized square root are

$$j(X) = \det\frac{\sinh(L/2)}{L/2}, \qquad \mathfrak{j}(X) = \exp\left(\frac{1}{2}\operatorname{Tr}\log\frac{\sinh(L/2)}{L/2}\right).$$

Expanding the scalar series yields

$$\log \mathfrak{j}(X) = \frac{\operatorname{Tr} L^2}{48} - \frac{\operatorname{Tr} L^4}{5760} + \frac{\operatorname{Tr} L^6}{362880} + O(X^8). \tag{12.12}$$

Each trace is the finite contraction of the adjoint indices around a wheel.

For $\mathfrak{g} = \mathfrak{gl}_N$, let $p_a = \operatorname{tr}(X^a)$, including $p_0 = N$.

Proposition 12.9. *For every $m \geq 0$,*

$$\operatorname{Tr}_{\mathfrak{gl}_N}(\operatorname{ad}_X)^m = \sum_{a=0}^{m} (-1)^a \binom{m}{a} p_{m-a}\, p_a.$$

In particular,

$$\operatorname{Tr} L^2 = 2N p_2 - 2p_1^2, \qquad \operatorname{Tr} L^4 = 2N p_4 - 8 p_1 p_3 + 6 p_2^2.$$

For matrices X, Y of sizes N, M, the quadratic part of $\log \mathfrak{j}(X \oplus Y) - \log \mathfrak{j}(X) - \log \mathfrak{j}(Y)$ is

$$\frac{1}{24}(M \operatorname{tr} X^2 + N \operatorname{tr} Y^2 - 2 \operatorname{tr} X \operatorname{tr} Y).$$

Proof. On $\operatorname{End}(\mathbb{C}^N)$, left and right multiplication commute and their difference is ad_X. The matrix-unit basis gives

$$\operatorname{Tr}(L_X^a R_X^b) = \operatorname{tr}(X^a) \operatorname{tr}(X^b).$$

Expand $(L_X - R_X)^m$ by the binomial theorem. The cases $m = 2, 4$ give the stated polynomials. Their derivation requires no diagonalizability. Finally the quadratic logarithm is $(N \operatorname{tr} X^2 - (\operatorname{tr} X)^2)/24$; subtracting its two block values gives the last formula. □

The off-diagonal matrix blocks are responsible for this failure of additivity. An adjoint wheel density is therefore different from a primitive matrix trace.

There is also a convergent product for this finite-dimensional half-density. On a sufficiently small neighborhood of zero,

$$\mathfrak{j}(X) = \lim_{K \to \infty} \prod_{n=1}^{K} \det\left(1 + \frac{L^2}{4\pi^2 n^2}\right)^{1/2}, \tag{12.13}$$

with branches equal to one at zero. The logarithms are locally bounded by a constant times n^{-2}, so the product converges locally uniformly. The Euler product for $\sinh(z/2)/(z/2)$, applied to the eigenvalues or its matrix power series, identifies the limit with the determinant above. If L is real diagonalizable with real eigenvalues, the partial product $\mathfrak{j}_K$ satisfies

$$0 \leq \log \mathfrak{j} - \log \mathfrak{j}_K \leq \frac{\operatorname{Tr} L^2}{8\pi^2 K}.$$

Indeed $0 \leq \log(1 + u) \leq u$ for $u \geq 0$, and $\sum_{n>K} n^{-2} \leq 1/K$. For imaginary eigenvalues the scalar factors are sine factors; the normalized local branch is taken before their first zeros. This product has a fixed finite-dimensional adjoint space. It does not sum traces over infinitely many independent Lie algebra blocks.

12.9 A two-step colour algebra

A family in which every wheel vanishes distinguishes the correction from the underlying comparison. Let W be finite dimensional and let $C \in \Lambda^3 W^*$. In dual bases define

$$\mathfrak{n}_C = W \oplus W^*, \qquad [x_i, x_j] = \sum_k C_{ijk} y^k, \qquad [x_i, y^j] = [y^i, y^j] = 0.$$

The image of every bracket is central, so Jacobi holds and $\operatorname{ad}_a \operatorname{ad}_b = 0$. The pairing $\kappa(x_i, y^j) = \delta_i^j$, zero on the other two pairs of summands, is invariant because C is alternating.

Over $R = \mathbb{C}[\hbar]$, the Rees enveloping algebra A_C has the symmetric PBW product

$$f *_C g = m \exp\left(\frac{\hbar}{2} \sum_{i,j,k} C_{ijk} Y^k \, \partial_{X_i} \otimes \partial_{X_j} \right)(f \otimes g). \tag{12.14}$$

Every polynomial pair gives a finite sum. The three bidifferential operators acting on pairs of slots in a triple tensor product commute, since their coefficients are central Y's and their derivatives act only on X's. Both bracketings are therefore the exponential of the same sum. The generator commutators are $[X_i, X_j]_{*_C} = \hbar C_{ijk} Y^k$, as required.

The adjoint coefficient Chevalley complex is

$$P_C = R[X, Y] \otimes \Lambda(\xi, \eta), \qquad |\xi_i| = |\eta^k| = 1,$$

with differential

$$Q\xi_i = QY^k = 0, \qquad Q\eta^k = -\frac{1}{2} C_{ijk} \xi_i \xi_j, \qquad QX_j = C_{ijk} Y^k \xi_i.$$

Repeated indices are summed. These are the Chevalley formulas, and Jacobi gives $Q^2 = 0$. The relative Hochschild cohomology of A_C is computed by

$$Z^p = \operatorname{Hom}_R(\Lambda_R^p(\mathfrak{n}_C \otimes R), A_C).$$

Its differential uses the commutator action and $\hbar$ times the bracket faces. The unnormalized alternating Chevalley-to-bar map of Chapter 3 gives this comparison.

Symmetrization is equivariant: $[T_a, \operatorname{sym}_\hbar f] = \hbar \operatorname{sym}_\hbar(a \cdot f)$. Consequently

$$\beta_p : P_C^p \longrightarrow Z^p, \qquad \beta_p = \hbar^p \operatorname{sym}_\hbar \tag{12.15}$$

is a cochain map. The target differential contributes the extra factor $\hbar$ needed to pass from degree p to degree $p+1$.

Proposition 12.10. *Symmetrization identifies the invariant polynomial algebra with the centre of A_C over R. After extending to $R[\hbar^{-1}]$, β induces an isomorphism of graded cup-product algebras*

$$H^*(P_C)[\hbar^{-1}] \xrightarrow{\sim} HH^*_R(A_C)[\hbar^{-1}].$$

The second assertion need not be an integral isomorphism in positive degree.

Proof. If f is invariant, then $\sum_{i,k} C_{jik} Y^k \partial_{X_i} f = 0$ for every j. The bidifferential operator in (12.14) therefore kills $f \otimes g$ for every g, as do all its positive powers. Hence $f *_C g = fg$. Conversely the PBW commutator of a linear generator with f is $\hbar$ times its adjoint derivative. The PBW module has no $\hbar$-torsion, so centrality is equivalent to invariance. This proves the integral degree-zero assertion.

After inverting $\hbar$, identify A_C with $U(\mathfrak{n}_C) \otimes R[\hbar^{-1}]$ by $T_a \mapsto \hbar a$. Evaluate a degree-p cochain on the rescaled arguments $\hbar^{-1} T_a$; this contributes $\hbar^{-p}$ and cancels the factor in β_p. The target becomes the ordinary adjoint Chevalley complex. Under these identifications, β is symmetrization composed with the source automorphism $f(X, Y) \mapsto f(\hbar X, \hbar Y)$. This automorphism commutes with Q and preserves the cup product. Every adjoint product of length at least two is zero, so the determinant (12.7) equals one. The cohomological Duflo theorem of Pevzner and Torossian [85, Theorem 2.5] then proves the cup-product assertion in every degree. The polynomial argument extends over the Laurent coefficient ring.

For the last assertion take $C = 0$. Both cochain differentials are zero, and β_p is multiplication by $\hbar^p$ on a nonzero free module whenever $0 < p \leq \dim \mathfrak{n}_C$. Its cokernel has $\hbar$-torsion and its specialization at $\hbar = 0$ is zero. □

There are algebraic product homotopies after this localization, although the theorem does not choose a field-theoretic one. Here is an explicit way to specify the first. Let $X = P_C \otimes P_C$ and $Y = Z$, now over $R[\hbar^{-1}]$, and write

$$D = \beta\, m_{P_C} - m_Z(\beta \otimes \beta) : X \longrightarrow Y.$$

The cup product on Z uses the usual Chevalley shuffle diagonal and the coefficient product. Choose contractions (i_X, p_X, h_X) and (i_Y, p_Y, h_Y) onto cohomology. They exist here: the preceding rescalings identify the complexes with extensions of complexes of complex vector spaces, which split into cohomology and contractible pairs. Multiplicativity on cohomology says $p_Y D i_X = 0$. Then

$$H = h_Y D + i_Y p_Y D h_X \quad \text{satisfies} \quad d_Y H + H d_X = D. \tag{12.16}$$

To verify it, use that D is a cochain map and expand the two contraction identities. The first summand gives $(1 - i_Y p_Y) D$. The second gives $i_Y p_Y D$, since $p_Y D d_X = 0$ and $p_Y D i_X = 0$. Their sum is D. This construction depends on the splittings. Compatibility with braces, geometric collisions and a prescribed two-point operation is further structure.

12.10 A mode completion that excludes the finite normalization

Finite Duflo maps do not automatically assemble along projections that discard Lie algebra blocks. For each $i \geq 1$, let $\mathfrak{g}_i$ be a copy of $\mathfrak{sl}_2$ with generators E_i, F_i, H_i. Different blocks commute. Put $\mathfrak{g}_{\leq N} = \bigoplus_{i=1}^{N} \mathfrak{g}_i$ and define

$$A_{\mathrm{mode}} = \varprojlim_{N,M} \mathrm{Sym}(\mathfrak{g}_{\leq N}) \otimes \mathbb{C}[\hbar]/(\hbar^M),$$

$$U_{\mathrm{mode}} = \varprojlim_{N,M} U_\hbar(\mathfrak{g}_{\leq N})/(\hbar^M).$$

The transition maps kill the omitted blocks. Each finite quotient has the discrete linear topology, and the limits have their inverse-limit topologies.

Proposition 12.11. *There is no continuous linear map $A_{\mathrm{mode}} \to U_{\mathrm{mode}}$ whose restriction to invariant polynomials in every finite set of blocks is the Duflo map with the normalization of* (12.8).

Proof. The invariant $c_i = H_i^2/2 + 2E_iF_i$ tends to zero in A_{mode}: every fixed block projection eventually kills it. Proposition 12.7 prescribes its image as

$$\mathrm{sym}_\hbar(c_i) + \frac{\hbar^2}{2}.$$

Project to the first block and reduce modulo $\hbar^3$. For every $i > 1$ the PBW term is zero, while $\hbar^2/2$ is nonzero. The proposed images do not tend to zero, contradicting continuity. □

The obstruction fixes both the topology and the normalization on every finite block. It does not exclude a different coefficient domain or a comparison with a specified renormalization. For example, using a parameter q_i of increasing positive weight as the ith Rees parameter replaces the scalar correction by $q_i^2/2$. Only finitely many such corrections survive at any fixed weight. The next construction makes that coefficient domain precise.

12.11 Positive weights and mixed coefficients

Let $A = \mathbb{C}[x_1, x_2, \dots]$, where each polynomial uses finitely many variables. Write T_{fin}^{p-1} for finite sums of polynomial p-vectors and D_{fin}^{p-1} for finite sums of operators

$$(f_1, \dots, f_p) \longmapsto a(x)\, \partial^{\alpha_1} f_1 \cdots \partial^{\alpha_p} f_p.$$

Every multi-index has finite support. In degree -1, both spaces are A. These are the shifted deformation complexes: a p-vector or p-ary cochain has degree $p - 1$. Use the Gerstenhaber insertion bracket on D_{fin}. If m is ordinary multiplication, its differential is $b_0 = [m, -]_G$. For the Schouten bracket use

$$[X_1 \wedge \cdots \wedge X_p, f]_{\mathrm{Sch}} = \sum_i (-1)^{i-1} X_i(f) X_1 \wedge \cdots \widehat{X}_i \cdots \wedge X_p.$$

Take a mixed coordinate domain $U = I^a \times D^b$, with I an open real interval and D a complex disk. Let $K(U)$ be the algebra of smooth real de Rham forms and complex $(0, *)$-forms on this product, with $d_K = d_{\mathbb{R}^a} + \bar\partial$ and the total-complex signs. Its coefficients are arbitrary smooth functions on the product, not just finite sums of separated functions. Give it the compact-open C^∞ topology.

Give each finite polynomial operator space the fine bornology: a bounded set is contained and bounded in a finite-dimensional subspace. In $K(U)\widehat{\otimes}T_{\mathrm{fin}}$ and $K(U)\widehat{\otimes}D_{\mathrm{fin}}$, this means the direct sum of copies of $K(U)$ indexed by a polynomial operator basis; a bounded set uses only finitely many basis directions. Field derivatives act on the x_i only. No spacetime differential operator or product of distributions is included in this definition.

Choose degree-zero central parameters q_i with positive integer weights w_i, only finitely many of which lie below any fixed bound. Set

$$\mathrm{wt}(q^\alpha) = \sum_i w_i \alpha_i, \qquad \mathscr{R}_{<M} = \mathbb{C}[q_1, q_2, \dots]/(q^\alpha : \mathrm{wt}(q^\alpha) \geq M), \qquad \mathscr{R} = \varprojlim_M \mathscr{R}_{<M}.$$

Each $\mathscr{R}_{<M}$ is finite dimensional. For the parameter-completed tensor define $V\widehat{\otimes}\mathscr{R} = \varprojlim_M (V \otimes \mathscr{R}_{<M})$ and put

$$\widehat{T}(U) = (K(U)\widehat{\otimes}T_{\mathrm{fin}})\widehat{\otimes}\mathscr{R}, \qquad \widehat{D}(U) = (K(U)\widehat{\otimes}D_{\mathrm{fin}})\widehat{\otimes}\mathscr{R}.$$

The filtration F^M is the kernel of projection to parameter precision M. Every coefficient has finite field support, although that support can increase with weight.

Finite polynomial differential graphs act on these spaces. On a bounded input set, only finitely many coefficient and derivative directions occur. A graph consequently has finitely many index assignments. Its spacetime coefficients are products, and Leibniz's rule bounds their derivatives on each compact subset by finitely many products of input seminorms. Thus the graph map is bounded. A series whose nth term raises weight by at least n has only finitely many terms at precision M, provided that each term has finitely many graphs. Its compatible finite-precision maps define a continuous map on the completion.

Extend the affine Taylor maps over the mixed coefficients by

$$\mathcal{U}_n^K(k_1\gamma_1,\dots,k_n\gamma_n) = (-1)^{\sum_{i<j}|\gamma_i||k_j|+(1-n)\sum_i|k_i|}k_1\cdots k_n\,\mathcal{U}_n(\gamma_1,\dots,\gamma_n). \tag{12.17}$$

The first exponent moves the coefficients to the left. The second is the tensor sign for a map of degree $1-n$. After suspension, the Taylor maps have degree zero. Their coefficient extensions commute with the total coderivations because d_K is a derivation. Undoing suspension proves all the extended L_∞ identities with the displayed sign. The finite graph formulas are stable under adjoining unused field coordinates.

Let $\Pi \in F^1\widehat{T}(U)$ be a bivector of mixed-form degree zero satisfying $d_K\Pi = 0$ and $[\Pi,\Pi]_{\mathrm{Sch}} = 0$. Independent finite Poisson blocks π_i give the example $\Pi = \sum_i q_i\pi_i$.

Theorem 12.12. *The formulas*

$$B_\Pi = \sum_{n\ge 1}\frac{1}{n!}\mathcal{U}_n^K(\Pi^{\otimes n}), \qquad m_\star = m + B_\Pi, \tag{12.18}$$

$$\mathcal{F}_r(v_1,\dots,v_r) = \sum_{n\ge 0}\frac{1}{n!}\mathcal{U}_{n+r}^K(\Pi^{\otimes n},v_1,\dots,v_r), \qquad r\ge 1 \tag{12.19}$$

define an associative product commuting with d_K and a filtered continuous L_∞ quasi-isomorphism

$$(\widehat{T}(U), d_K + [\Pi,-]_{\mathrm{Sch}}) \longrightarrow (\widehat{D}(U), d_K + [m_\star,-]_G).$$

The maps commute with restriction in U. The assertion is about the unreduced mixed complexes.

Proof. At precision M, the twisting sums have $n < M$. The input coefficients and the relevant coefficients of Π use a common finite set of field variables. The calculation is therefore the finite affine formality calculation with coefficients in $K(U)$ and signs (12.17). The Maurer–Cartan equation gives

$$(d_K + [m,-]_G)B_\Pi + \frac{1}{2}[B_\Pi,B_\Pi]_G = 0.$$

Since B_Π is binary and has mixed-form degree zero, this equation gives separately $d_K B_\Pi = 0$ and $[m_\star,m_\star]_G = 0$.

For the higher maps, conjugate the source symmetric coalgebra codifferential by multiplication with $\exp(\Pi)$ and the target one by $\exp(B_\Pi)$. The coefficient with r free inputs is (12.19). All coefficients and identities are finite at the specified precision. Compatibility in M proves them on the inverse limit.

The untwisted finite-direction HKR map is a quasi-isomorphism by the filtered colimit of the finite polynomial HKR comparisons. Its cone is an acyclic complex of complex vector spaces. Choose a square-zero contracting homotopy H_0 by splitting it into contractible pairs. Every linear map between fine bornological spaces is bounded. Extend H_0 over $K(U)$ with its degree-minus-one tensor sign. The two terms involving d_K cancel, so this contracts the unreduced mixed cone.

Write the cone differential of $\mathcal{F}_1$ as $d_0 + \delta$. Every term of δ raises parameter weight. The homotopy

$$H = H_0(1+\delta H_0)^{-1} = H_0\sum_{j\ge 0}(-\delta H_0)^j$$

is a finite sum at every precision. The perturbation identity gives $(d_0+\delta)H + H(d_0+\delta) = 1$. This proves the quasi-isomorphism and continuity. Finally the graph maps use only field derivatives and pointwise products of mixed coefficients, which commute with restriction in U. □

Both sign contributions in (12.17) can occur. On an interval with coordinate t, take $\pi = \partial_u \wedge \partial_v$ and $\Pi = h\pi$, where h has weight one. The constant-bivector normalization gives $\mathcal{U}_2(\pi,\pi)(u^2,v^2) = 1$. The order-$h$ term of $\mathcal{F}_1(t\pi)$ on these two functions is ht. Its mixed differential is $h\,dt$. In $\mathcal{U}_2^K(\pi, dt\,\pi)$, the input-crossing sign and the degree-minus-one map sign cancel, giving precisely this value.

The target remains the completed finite-direction polydifferential complex. For example, $\sum_i x_i\partial_{x_i}$ acts on every polynomial and is bounded for the fine bornology. It is not a finite-direction derivation. It is not a Hochschild boundary either, since the nullary Hochschild differential of a commutative algebra is zero. Thus a comparison with the full bounded Hochschild complex requires a larger source.

12.12 A variable Poisson coefficient

A small calculation shows why both the target differential and the derivative corrections matter. On $\mathbb{C}[x,y]$, take $\pi = x\partial_x \wedge \partial_y$ and use one parameter h. The first coefficient of the star product is

$$B_1(f,g) = \frac{x}{2}(f_x g_y - f_y g_x).$$

The Schouten and Hochschild formulas give

$$[\pi, x]_{\mathrm{Sch}} = x\partial_y, \qquad [\pi, \partial_x]_{\mathrm{Sch}} = -\partial_x \wedge \partial_y,$$
$$[m_\star, x]_G(y) = hx + O(h^2), \qquad [m_\star, \partial_x]_G(x,y) = -h/2 + O(h^2).$$

For the last expression, expand $(\partial_x x)\star y + x\star(\partial_x y) - \partial_x(x\star y)$. The first-order term is $-h/2$.

There cannot be a cochain map reducing to HKR from $(T_{\mathrm{fin}}[[h]], h[\pi,-]_{\mathrm{Sch}})$ to the undeformed target $(D_{\mathrm{fin}}[[h]], b_0)$. Such a map sends x to a nullary cochain, on which b_0 is identically zero. Its order-h chain equation would be $0 = h\ \mathrm{HKR}([\pi,x]) = hx\partial_y$. Higher coefficients cannot change that equation.

For the second product term, choose the affine gauge with

$$\begin{aligned} B_2(f,g) = \frac{x^2}{8}&(f_{xx}g_{yy} - 2f_{xy}g_{xy} + f_{yy}g_{xx}) \\ &+ \frac{x}{12}(f_x g_{yy} + f_{yy}g_x - f_{xy}g_y - f_y g_{xy}). \end{aligned} \tag{12.20}$$

This is the second-order affine formula [64, Section 1.4.2], normalized so that the first commutator is $h\{-,-\}$.

Proposition 12.13. *The product $f\star g = fg + hB_1(f,g) + h^2B_2(f,g)$ is associative modulo h^3. In particular,*

$$(x\star y)\star y = x\star(y\star y) = xy^2 + hxy + \frac{h^2x}{6} \pmod{h^3}.$$

Omitting the $x/12$ term in (12.20) leaves associator $h^2x/4$ on (x,y,y).

Proof. Leibniz's rule cancels the first-order associator. The second coefficient on (f,g,k) is

$$\begin{aligned} &B_2(f,g)k + B_1(B_1(f,g),k) + B_2(fg,k) \\ &\qquad - fB_2(g,k) - B_1(f,B_1(g,k)) - B_2(f,gk). \end{aligned}$$

Using only the $x^2/8$ term and expanding each derivative by Leibniz gives $x(f_xk_y - f_yk_x)g_y/4$. The four terms with coefficient $x/12$ give its negative. This proves the assertion for arbitrary polynomials. For the displayed triple, $B_2(xy,y) = -x/12$ and $B_2(x,y^2) = x/6$. The iterated B_1 term adds $x/4$ on the left, giving $x/6$ on each side. Without the derivative correction, the left side has $x/4$ and the right side has zero in order two. □

The parameter also affects Lie compatibility. After inverting h, multiplication by h^{p-1} on p-vectors identifies the differentials $[\pi,-]$ and $h[\pi,-]$ and preserves the Schouten bracket: on a bracket of a p-vector and a q-vector both powers are h^{p+q-2}. It sends the function 1 to h^{-1}. The unit-preserving scaling h^p instead preserves the bracket if the source bracket is multiplied by h. The two rescalings answer different comparison questions.

12.13 Comparing quantizations and continuous operations

Two quantizations on the same coefficient module must be compared in their allowed operator space. Suppose associative products $\mu = m + O(h)$ and $\mu' = m + O(h)$ agree modulo h^N, for $N \geq 1$. Assume their coefficient domains and differentials have been identified, and put $\Delta_N = h^{-N}(\mu' - \mu) \bmod h$. Subtracting associativity in order h^N gives $b_0\Delta_N = 0$. A permitted degree-zero unary operator D gives

$$\mu' = (1 + h^N D)^{-1}\mu\big((1 + h^N D) \otimes (1 + h^N D)\big) \pmod{h^{N+1}}$$

if and only if

$$\Delta_N = b_0 D. \tag{12.21}$$

Indeed expansion of the right side gives

$$\mu + h^N\big(m(D \otimes 1) + m(1 \otimes D) - Dm\big).$$

If the mixed differential d_K is fixed, the same comparison also requires $[d_K, D] = 0$. A primitive in a larger space of operators does not solve the equation in the prescribed domain. For a commutative product $b_0 D$ is symmetric in its inputs, so a nonzero antisymmetric first-order difference cannot be removed this way. For example, $b_0(\partial_x^2)(f, g) = -2f_x g_x$.

The Hamiltonian jet system has a different completion from the finite-direction weighted model. The two-index construction of Chapter 4 gives $\mathcal{P}_{\mathrm{uc}}$ and its cochain comparison with $C^\bullet_{\mathrm{uc}}(A/R, A)$. A geometric polyvector realization must first be compared with that complex in its stated topology. The compatible-jet criterion of Theorem 4.8 records the additional transitions and limit identifications needed for a proposed $\mathcal{P}_\infty$.

The further question is a continuous homotopy-E_2 quasi-isomorphism into this two-cutoff Hochschild recipient, with an invariant Duflo map and a specified renormalization of the wheels. A negative answer would require a brace obstruction that persists under every allowed renormalization. The fixed-normalization counterexample of Proposition 12.11 does not have that quantifier.

At the binary product level a proposed cochain map Φ already needs a homotopy satisfying

$$dH(u, v) + H(du, v) + (-1)^{|u|}H(u, dv) = \Phi(uv) - \Phi(u) \smile \Phi(v).$$

The brace substitutions require compatible higher homotopies on the same completed coefficient domain. A finite invariant algebra isomorphism does not specify them. The mixed-chain traces, cyclic orientation and finite wheel operators constructed above provide separate parts of such a comparison. To identify their action with an independently defined field theory, one must also preserve insertion supports, singular coefficients, collision substitutions and the chosen state.

Chapter 13

Infinite-mode states and local stabilization

A current commutator does not determine a state. For one oscillator, $[a, a^*] = 1$, the vacuum has expectation zero on a^*a. A thermal state at inverse temperature T has expectation $(e^T - 1)^{-1}$ on the same operator. Both give expectation one on the commutator. The singular current coefficient can therefore survive a change of preparation while its regular correlations change.

Chapter 8 constructed finite Gaussian states and their higher operations. Passing to infinitely many modes requires a domain, an energy estimate and a normalization. We construct these data on a Heisenberg Fock space. A local cubic interaction then separates two possible conclusions: every perturbative coefficient exists and sews, although the numerical cubic heat operator is unbounded. A local quartic correction gives positive states. Its coefficient determines whether those states approach the free state continuously.

13.1 The common Hilbert space

Fix a numerical level $\lambda > 0$. Positivity in this chapter concerns complex Hilbert spaces with this fixed real parameter. It is separate from an identity over a formal quantum coefficient ring. Let

$$\mathcal{H}_1 = \ell^2(\mathbb{N}_{>0}), \qquad \mathcal{F} = \Gamma_s(\mathcal{H}_1) = \widehat{\bigoplus}_{r\geq 0} \operatorname{Sym}^r \mathcal{H}_1.$$

The inner product is conjugate linear in its first argument. The occupation vectors $|\mathbf{m}\rangle$, indexed by finite-support sequences $\mathbf{m} = (m_1, m_2, \ldots)$, form an orthonormal basis. Define

$$a_n|\mathbf{m}\rangle = \sqrt{m_n}\,|\mathbf{m} - \mathbf{e}_n\rangle, \qquad a_n^*|\mathbf{m}\rangle = \sqrt{m_n + 1}\,|\mathbf{m} + \mathbf{e}_n\rangle.$$

Thus $[a_m, a_n^*] = \delta_{mn}$. Write

$$\alpha_n = \sqrt{n}\, a_n, \qquad \alpha_{-n} = \sqrt{n}\, a_n^*, \qquad J_n = \sqrt{\lambda}\, \alpha_n \quad (n \neq 0), \qquad \alpha_0 = J_0 = 0.$$

The current relations are

$$[\alpha_m, \alpha_n] = m\delta_{m+n,0}, \qquad J_n^* = J_{-n}, \qquad [J_m, J_n] = \lambda m \delta_{m+n,0}.$$

The vacuum $\Omega = |\mathbf{0}\rangle$ has energy zero. The energy operator and particle-number operator are

$$H_0 = \sum_{n\geq 1} n a_n^* a_n = \sum_{n\geq 1} \alpha_{-n}\alpha_n, \qquad N = \sum_{n\geq 1} a_n^* a_n \leq H_0.$$

These are diagonal self-adjoint operators on their maximal domains. For example,

$$\operatorname{Dom}(H_0) = \left\{ \sum_{\mathbf{m}} v_{\mathbf{m}}|\mathbf{m}\rangle : \sum_{\mathbf{m}} \left(\sum_n n m_n \right)^2 |v_{\mathbf{m}}|^2 < \infty \right\}.$$

Let $\mathcal{F}_E$ be the energy-E eigenspace and P_E its orthogonal projection. Its dimension is the partition number $p(E)$. The finite-energy space $\mathcal{F}_{\mathrm{fin}} = \bigoplus_{E\geq 0} \mathcal{F}_E$ is a common core for the finite-mode operators. Their closed extensions are used whenever an infinite vector is an input.

For $f \in \mathcal{H}_1$, the smeared oscillators are

$$a(f) = \sum_n \overline{f_n} a_n, \qquad a^*(f) = \sum_n f_n a_n^*.$$

They act on the algebraic direct sum of the particle-number spaces, which permits arbitrary square-summable one-particle vectors. On that domain,

$$\|a(f)v\| \leq \|f\|_2 \|N^{1/2}v\|, \qquad \|a^*(f)v\| \leq \|f\|_2 \|(N+1)^{1/2}v\|.$$

Successive application bounds a polynomial of degree d in finitely many such oscillators by a constant times $(N+1)^{d/2}$. This estimate specifies the unbounded observables whose thermal expectations will be used below.

13.2 A prepared trace and its current kernel

Choose a cylinder length $T > 0$ and put $q = e^{-T}$. Its evolution operator is $K_T = e^{-TH_0}$. This choice supplies the preparation before any comparison with a bulk theory.

Theorem 13.1. *The operator K_T is positive and trace class. If $E_1 e_n = n e_n$ on $\mathcal{H}_1$, then*

$$Z_0(T) = \operatorname{Tr}_{\mathcal{F}} K_T = \prod_{n\geq 1} (1 - e^{-Tn})^{-1} = \det_{\mathcal{H}_1}(1 - e^{-TE_1})^{-1}. \tag{13.1}$$

The density $\rho_T = K_T / Z_0(T)$ defines a positive normalized state on the bounded operators on $\mathcal{F}$. Every polynomial in finitely many square-summably smeared oscillators also has a thermal expectation

$$\varphi_T(A) = Z_0(T)^{-1} \operatorname{Tr}(AK_T).$$

For the radially ordered current $J(z) = \sum_n J_n z^{-n-1}$,

$$\varphi_T(J(z)J(w)) = \frac{\lambda}{zw} \sum_{n\geq 1} \frac{n\big((w/z)^n + q^n (z/w)^n\big)}{1 - q^n}, \qquad q < |w/z| < 1. \tag{13.2}$$

The series and every fixed holomorphic derivative converge normally on compact subannuli. Near $z = w \neq 0$, its singular part is $\lambda/(z-w)^2$.

Proof. At a mode cutoff M, independent occupation sums give $Z_{0,M}(T) = \prod_{n=1}^{M} (1-q^n)^{-1}$. Moreover,

$$0 \leq \log \frac{Z_0(T)}{Z_{0,M}(T)} \leq \frac{q^{M+1}}{(1-q)(1-q^{M+1})}. \tag{13.3}$$

Indeed $-\log(1-x) \leq x/(1-x)$, and summing the resulting geometric bound proves convergence of the logarithmic product. Monotone summation over all occupation vectors gives its equality with the trace. The eigenvalues q^n of e^{-TE_1} are summable, so its Fredholm determinant is this same convergent product.

For every $r \geq 0$, the operator $(H_0+1)^r e^{-TH_0}$ is trace class: the polynomial factor is bounded by a constant times $e^{TE/2}$ on energy E, and the remaining trace is $Z_0(T/2)$. The oscillator bounds therefore make AK_T trace class. They also make $AK_{T/2}$ Hilbert–Schmidt. Consequently the expectation of A^*A is its squared Hilbert–Schmidt norm divided by $Z_0(T)$, and is nonnegative. This extends the positive state to the stated polynomial domain.

The one-mode geometric sums give

$$\varphi_T(J_nJ_{-n}) = \frac{\lambda n}{1-q^n}, \qquad \varphi_T(J_{-n}J_n) = \frac{\lambda n q^n}{1-q^n} \quad (n>0).$$

They imply (13.2). On a compact subannulus, both $|w/z|$ and $q|z/w|$ are bounded strictly below one. The denominator is bounded below by $1-q$, and derivatives add only polynomial factors in n. Split the first numerator into its vacuum contribution and its q^n-weighted remainder. The vacuum sum $\sum_{n\ge 1} n(w/z)^n$ gives $\lambda/(z-w)^2$; the two remaining sums extend across $z=w$. □

At a cutoff, the normalized-oscillator contractions are bounded by $(1-q)^{-1}\|f\|_2\|h\|_2$. Wick expansion therefore gives the same polynomial expectations by a mode limit. The estimates prove convergence of an actual density and determinant, as well as convergence of its correlations.

A fixed charge $b \in \mathbb{R}$ can be retained by setting $J_0 = b$ and adding $b^2/(2\lambda)$ to the energy. The partition function becomes $e^{-Tb^2/(2\lambda)} Z_0(T)$, the mean current is b/z, and the connected two-point function is unchanged. A sum over charges additionally requires a measure or lattice and a convergence domain.

13.3 Sewing before normalization

For $z \in \mathcal{H}_1$, define the exponential vector

$$\mathrm{e}(z) = \bigoplus_{r\ge 0} \frac{z^{\otimes r}}{\sqrt{r!}}.$$

Its norm squared is $e^{\|z\|_2^2}$. The cylinder matrix element is

$$K_T(z,\overline{w}) = \langle \mathrm{e}(w), K_T \mathrm{e}(z)\rangle = \exp\left(\sum_{n\ge 1} e^{-Tn} z_n \overline{w_n}\right). \tag{13.4}$$

Cauchy–Schwarz makes the exponent convergent.

Proposition 13.2. *For $T_1, T_2 > 0$, coherent cylinder kernels sew by $K_{T_1} \circ K_{T_2} = K_{T_1+T_2}$. Closing the composite by the trace gives $Z_0(T_1+T_2)$.*

Proof. For a bounded one-particle contraction B, second quantization is $\Gamma(B) = \bigoplus_{r\ge 0} \mathrm{Sym}^r B$. It gives

$$\Gamma(e^{-T_1E_1})\Gamma(e^{-T_2E_1}) = \Gamma(e^{-(T_1+T_2)E_1}).$$

The finite-dimensional coherent Gaussian integral computes this identity at each mode cutoff. Its limit is Fock-space composition. The high-mode vacuum projections commute with the heat operators, and the omitted positive trace tends to zero by (13.3). Thus taking the trace of the composite gives the stated determinant. Associativity is already the associativity of operator composition. □

The infinite-mode pairing here is a limit of cylindrical Wick pairings. No translation-invariant measure on ℓ^2 is used. Normalization is performed after the required trace closure:

$$\rho_{T_1}\rho_{T_2} = \frac{Z_0(T_1+T_2)}{Z_0(T_1)Z_0(T_2)} \rho_{T_1+T_2}.$$

The scalar factor is part of sewing data.

The thermal state also satisfies, for finite-mode oscillator polynomials A, B,

$$\varphi_T(AB) = \varphi_T\big(Be^{-TH_0}Ae^{TH_0}\big). \tag{13.5}$$

This follows by cyclicity of the trace on the common domain, or directly from the occupation sums. The conjugation multiplies α_n by e^{Tn}. For general infinite smears it therefore requires a further domain condition. Equation (13.5) is the thermal, or KMS, identity. Ordinary traciality fails already on $a_n a_n^*$ and $a_n^* a_n$. A cyclic chain class from Chapter 12 and this scalar state are consequently different constructions.

13.4 The stress tensor and the cylinder constant

The vacuum has contraction $\langle \Omega, J(z)J(w)\Omega\rangle = \lambda/(z-w)^2$ in $|w| < |z|$. Subtract this contraction to define

$$T(z) = \frac{1}{2\lambda}{:}J(z)^2{:} = \frac{1}{2}{:}\alpha(z)^2{:} = \sum_{m\in\mathbb{Z}} L_m z^{-m-2}, \qquad L_m = \frac{1}{2}\sum_{j\in\mathbb{Z}} {:}\alpha_j\alpha_{m-j}{:}.$$

Normal ordering puts negative-index creators to the left. Terms of index zero vanish. Each sum is finite on a finite-energy vector, and $L_0 = H_0$.

Proposition 13.3. *On $\mathcal{F}_{\mathrm{fin}}$,*

$$[L_m, \alpha_n] = -n\alpha_{m+n}, \qquad [L_m, L_n] = (m-n)L_{m+n} + \frac{m^3-m}{12}\delta_{m+n,0}.$$

If $z = f(u)$ is a local holomorphic coordinate change with $f' \neq 0$, the stress tensor defined by subtraction in the u-coordinate is

$$T_u(u) = f'(u)^2 T_z(f(u)) + \frac{1}{12}\{f,u\}, \qquad \{f,u\} = \frac{f'''}{f'} - \frac{3}{2}\left(\frac{f''}{f'}\right)^2. \tag{13.6}$$

In particular $f(u) = e^u$ gives cylinder Hamiltonian $H_0 - 1/24$.

Proof. Commuting a current through the two factors of L_m gives the first identity. The Jacobi identity then shows that $[L_m, L_n] - (m-n)L_{m+n}$ commutes with every current. An operator with this property is scalar: it preserves the common kernel of the positive modes, which is $\mathbb{C}\Omega$, and then has the same scalar action on all created vectors. Its energy degree forces that scalar to vanish unless $m + n = 0$. For $m > 0$, its vacuum expectation is

$$\|L_{-m}\Omega\|^2 = \frac{1}{2}\sum_{j=1}^{m-1} j(m-j) = \frac{m^3-m}{12}.$$

The two contractions retain the factor $1/2$, including when $j = m - j$. Antisymmetry gives the other signs of m.

The current transforms as a one-form, so its pulled-back vacuum contraction is $\lambda f'(u)f'(v)/(f(u)-f(v))^2$. Taylor expansion at $v = u$ gives

$$\frac{f'(u)f'(v)}{(f(u)-f(v))^2} = \frac{1}{(u-v)^2} + \frac{1}{6}\{f,u\} + O(v-u).$$

For completeness, writing $a = f''/f'$, $b = f'''/f'$, the constant term is $b/2 - a^2 + 3a^2/4 - b/3 = b/6 - a^2/4$. Subtracting the u-coordinate pole and dividing by 2λ proves (13.6). Since $\{e^u, u\} = -1/2$, its constant is $-1/24$. □

The conformal cylinder convention consequently has partition function

$$\operatorname{Tr} e^{-T(H_0 - 1/24)} = e^{T/24} Z_0(T).$$

The scalar factor respects sewing and cancels from normalized expectations. It arises from the specified stress subtraction. Identifying it with a determinant in another field theory requires that theory's regulator, finite term, boundary condition and sewing normalization to be compared.

13.5 Formal interactions on two coefficient domains

Let V be a finite linear combination of normally ordered polynomials in $a(f_j)$, $a^*(f_j)$, for finitely many $f_j \in \ell^2$. Define a formal coupling series by

$$U_T(g) = \sum_{k\ge 0} g^k U_T^{(k)}, \qquad U_T^{(0)} = K_T,$$
$$U_T^{(k)} = (-1)^k \int_{0\le t_1\le\cdots\le t_k\le T} e^{-(T-t_k)H_0} V e^{-(t_k-t_{k-1})H_0}\cdots \tag{13.7}$$
$$\cdots V e^{-t_1 H_0}\, dt_1\cdots dt_k.$$

For this coefficient domain, the definition first uses Wick matrix elements and traces at a mode cutoff, followed by their limits.

Proposition 13.4. *Every thermal trace and coherent-vector matrix element of each coefficient in* (13.7) *has a mode-cutoff limit. The same holds with a fixed polynomial smeared insertion. In these coefficientwise sewn matrix elements,*

$$U_{T_1}(g)U_{T_2}(g) = U_{T_1+T_2}(g). \tag{13.8}$$

Dividing observable traces by $\operatorname{Tr} U_T(g)$ *defines a normalized formal functional.*

Proof. A time-ordered thermal contraction of normalized modes contains one of

$$\frac{e^{-n\tau}}{1-e^{-nT}}, \qquad \frac{e^{-n(T-\tau)}}{1-e^{-nT}}, \qquad 0\le\tau\le T.$$

Both are bounded by $C_T = (1-e^{-T})^{-1}$. A smeared contraction is therefore at most $C_T\|f\|_2\|h\|_2$, uniformly in the mode cutoff and time. At fixed order there are finitely many monomial choices and Wick pairings. Their bounds, multiplied by the simplex volume $T^k/k!$, are integrable. The contraction tails converge by Cauchy–Schwarz, so dominated convergence gives the mode limits.

For coherent matrix elements, each external contraction contains a one-particle heat contraction e^{-sE_1}, with $s\ge 0$. The free coherent factor is bounded by $e^{\|z\|_2\|w\|_2}$. These bounds prove the same assertion with external vectors. A fixed insertion only adds finitely many contractions.

At a cutoff, split the time simplex at T_2. Its pieces are indexed by the number of insertions before the cut and give the coefficient of g^k in (13.8). The bounds pass this identity to the limit. The denominator has constant term $Z_0(T) > 0$, and hence a formal inverse. □

This domain permits infinitely many oscillator modes in a smear, but only finitely many input directions. It does not permit a point-current smear: its regularized one-particle norm contains

$$\sum_{n\ge 1} n e^{-2\varepsilon n} = \frac{e^{-2\varepsilon}}{(1-e^{-2\varepsilon})^2} \sim \frac{1}{4\varepsilon^2}.$$

Local spatial Wick polynomials require a different estimate.

For fixed $n\in\mathbb{Z}$, $d\ge 1$ and polynomial P, define the local mode

$$W_n = \sum_{n_1+\cdots+n_d=n} P(n_1,\dots,n_d){:}\alpha_{n_1}\cdots\alpha_{n_d}{:}. \tag{13.9}$$

Zero indices give zero terms. The polynomial P includes factors from angular derivatives. Finite Fourier combinations of these operators, together with the identity, form the second domain.

Lemma 13.5. *The sum* (13.9) *is finite on $\mathcal{F}_E$, maps it to $\mathcal{F}_{E-n}$, and satisfies*

$$\|W_n P_E\| \le C_{n,d,P}(1 + E + |n|)^{3d/2-1+\deg P}.$$

The same estimate holds after discarding any set of its summands.

Proof. The annihilators act first. Their total index is at most E whenever the monomial is nonzero. The creators' total absolute index is that sum minus n, hence at most $E + |n|$. Each contributing index is bounded by $E + |n|$, and the sum constraint determines the last from the first $d - 1$. There are at most $C_d(1 + E + |n|)^{d-1}$ tuples.

Intermediate energies first decrease and then increase to $E - n$. The norm of each oscillator factor is at most $(1 + E + |n|)^{1/2}$: its square is bounded by the energy before annihilation or after creation. Multiply the d bounds and the polynomial bound for P, then sum over tuples. The exponent is $(d - 1) + d/2 + \deg P$. Since this argument bounds each summand separately, it also bounds every partial sum. □

For example $W_0 = 2H_0$ when $d = 2$ and $P = 1$. Its one-particle compression has eigenvalues $2, 4, 6, \ldots$. A polynomial in finitely many square-summable smears instead has one-particle compression equal to a scalar plus a bounded finite-rank operator. The two domains therefore do not coincide.

Theorem 13.6. *Let V be a finite Fourier combination of local Wick modes. For every $T > 0$ and $k \ge 0$, the coefficient $U_T^{(k)}$ has a unique trace-class extension from finite-energy vectors. Mode-cutoff coefficients, embedded by the high-mode vacuum, converge in trace norm. A fixed finite Fourier Wick-polynomial insertion A also makes $AU_T^{(k)}$ trace class. The formal operators obey* (13.8)*, and their normalized observable traces are well defined.*

Proof. There are C, M, R such that

$$V\mathcal{F}_E \subset \bigoplus_{|r|\le R} \mathcal{F}_{E+r}, \qquad \|V P_E\| \le C(1 + E)^M.$$

At order k, every intermediate energy lies between $E - kR$ and $E + kR$. For a fixed sequence of energy projections, the heat factors are bounded by $e^{-TE+TkR}$, uniformly over the closed time simplex. The operator factors contribute at most $C^k(1 + E + kR)^{Mk}$. Absorb the finite number of energy-shift choices into C^k. A block with source $\mathcal{F}_E$ has rank at most $p(E)$, so

$$\|U_T^{(k)}\|_1 \le \frac{T^k}{k!} C^k e^{TkR} \sum_{E\ge 0} p(E) e^{-TE}(1 + E + kR)^{Mk} < \infty. \tag{13.10}$$

The sum is finite because a polynomial energy weight is dominated by $e^{TE/2}$. Finite block sums are trace-norm continuous in the simplex variables; the uniform summable tail gives a trace-norm Bochner integral, including coincident-time faces.

For fixed E, k, all contributing oscillator indices occur below $E + kR$, after enlarging this bound by the fixed finite shift if necessary. Those blocks stabilize at a mode cutoff. Lemma 13.5 gives the same tail bound at every cutoff, proving trace-norm convergence. An insertion only adds fixed shifts and a polynomial factor to this estimate.

Subdivision of the simplex proves sewing on finite blocks. The trace-norm convergence passes the identity to the full space, using $\|BC\|_1 \le \|B\|_1\|C\|$ and boundedness of each trace-class coefficient. The free partition function is the nonzero constant denominator. Uniqueness follows from the density of finite-energy vectors. □

Both constructions are coefficientwise in g. Neither estimate asserts convergence as the coupling order tends to infinity. The local-mode theorem concerns spatially local boundary operators on the stated Fock carrier. A local interacting half-space theory also requires its supported counterterms and quantum master equation.

13.6 The cubic mode and its spectral obstruction

Consider the self-adjoint finite-energy matrices of the cubic mode

$$\begin{aligned} V &= \frac{1}{6}\operatorname{Res}_{z=0} z^2{:}\alpha(z)^3{:}\, dz \\ &= \frac{1}{2}\sum_{m,n\geq 1}\left(\alpha_{-m}\alpha_{-n}\alpha_{m+n} + \alpha_{-(m+n)}\alpha_m\alpha_n\right). \end{aligned} \tag{13.11}$$

It commutes with H_0. The factor $1/2$ comes from the three placements of the uniquely signed mode in the residue. In the polynomial realization $\alpha_{-n} = p_n$, $\alpha_n = n\partial_{p_n}$, it becomes

$$V = \frac{1}{2}\sum_{m,n\geq 1}\left((m+n)\,p_m p_n \partial_{p_{m+n}} + mn\,p_{m+n}\partial_{p_m}\partial_{p_n}\right). \tag{13.12}$$

This is the cut-and-join operator, with the normalization of Mironov–Morozov–Natanzon [78, equation (51)].

Lemma 13.7. *On energy E, the restriction V_E has norm $E(E-1)/2$, and both extreme signs occur.*

Proof. For a partition $\mu = (1^{m_1}2^{m_2}\cdots)$ of E, put $p_\mu = \prod_n p_n^{m_n}$ and $z_\mu = \prod_n n^{m_n} m_n!$. The Fock pairing is $\langle p_\mu, p_\nu\rangle = \delta_{\mu\nu} z_\mu$. The characteristic map on class functions of S_E,

$$\operatorname{ch}(f) = \sum_{\mu\vdash E}\frac{f(\mu)}{z_\mu}p_\mu,$$

is an isometry for normalized counting measure. Convolution by the sum of all transpositions preserves class functions. A transposition joins two cycles or splits one. Counting its marked entries gives the coefficients mn and $m+n$ in (13.12), with the factor $1/2$ for the two orders. Thus this sum corresponds to V_E.

Every transposition acts unitarily and self-adjointly by translation on functions. The sum has norm at most $\binom{E}{2}$. The trivial and sign characters attain the positive and negative values. Their characteristic images are the complete and elementary symmetric polynomials h_E, e_E, defined by

$$\sum_{E\geq 0} h_E t^E = \exp\!\left(\sum_{n\geq 1}\frac{p_n t^n}{n}\right), \qquad \sum_{E\geq 0} e_E t^E = \exp\!\left(\sum_{n\geq 1}\frac{(-1)^{n-1}p_n t^n}{n}\right).$$

They give $V h_E = \binom{E}{2} h_E$ and $V e_E = -\binom{E}{2} e_E$, proving equality in the norm bound. □

For nonzero real g, one of these families has energy

$$(H_0 + gV)\big|_{\text{extreme},E} = E - \frac{|g|}{2}E(E-1) \longrightarrow -\infty.$$

The self-adjoint direct sum $H_0 + gV$ is therefore unbounded below, and $e^{-T(H_0+gV)}$ is unbounded for every $T > 0$. It cannot prepare a trace-class Gibbs state. The formal coefficients from Theorem 13.6 nevertheless exist. Since $[H_0, V] = 0$, they are $U_T^{(k)} = (-T)^k e^{-TH_0}V^k/k!$.

The first response already sees the interaction. With $\langle A\rangle_q = \varphi_T(A)$, one has

$$\langle V^2\rangle_q = \sum_{m,n\geq 1}\frac{mn(m+n)q^{m+n}}{(1-q^m)(1-q^n)(1-q^{m+n})}. \tag{13.13}$$

To prove the formula, split V into its joining and splitting terms. Only a join followed by a split, or the reverse order, has complete thermal contractions. The two orders and the two pairings of the m, n legs cancel the factor $1/4$. The same multiplicity remains when $m = n$. Substitution of the one-mode expectations proves the formula. Its convergence follows from the exponential numerator and denominators bounded below by $1 - q$. Its first coefficients are

$$2q^2 + 16q^3 + 60q^4 + 160q^5 + 360q^6 + O(q^7).$$

Oscillator parity sends V to $-V$ and preserves the free density, so all odd moments vanish. Hence the formal partition function obeys

$$\log \frac{\operatorname{Tr} U_T(g)}{Z_0(T)} = \frac{g^2 T^2}{2} \langle V^2 \rangle_q + O(g^4).$$

13.7 Stabilization by the total energy

A square of the total energy provides a first positive preparation. For real ε, g, define the self-adjoint block operator

$$H_{\varepsilon,g} = H_0 + \varepsilon H_0^2 + gV.$$

The term H_0^2 is a prescribed nonlocal interaction on the circle. It changes the Hamiltonian whose state is being constructed.

Theorem 13.8. *For $T > 0$, the operator $e^{-TH_{\varepsilon,g}}$ is positive and trace class exactly when*

$$\varepsilon \geq |g|/2. \tag{13.14}$$

On this region it gives a positive normalized state on bounded operators, and its unnormalized heat kernels sew by addition of cylinder lengths. For fixed $T > 0$, the partition function is holomorphic in the complex tube $\operatorname{Re}\varepsilon > |\operatorname{Re} g|/2$.

Proof. Lemma 13.7 gives, on energy E,

$$H_{\varepsilon,g} \geq (\varepsilon - |g|/2)E^2 + (1 + |g|/2)E. \tag{13.15}$$

At equality in (13.14), this is still a positive linear lower bound. Summation against $p(E)$ proves trace class by (13.1). Below the threshold, choose the extreme eigenvector with the sign opposite to g. If $g = 0$, either extreme vector suffices. Its energy equals the right side of (13.15) and tends to $-\infty$, so the heat operator is unbounded.

Spectral calculus gives positivity and sewing for the allowed parameters. In a common eigenbasis of H_0, V, each exponential eigenvalue is entire in (ε, g). On a compact subset of the tube, its absolute value is at most $e^{-cE^2 - TE}$ for some $c > 0$. Every parameter derivative adds only a polynomial energy factor. Uniform summability proves holomorphy of the trace. □

This convergence has an explicit tail. Put $c = \varepsilon - |g|/2 \geq 0$. For an integer $M \geq 0$,

$$\sum_{E>M} \operatorname{Tr}_{\mathcal{F}_E} e^{-TH_{\varepsilon,g}} \leq Z_0(T/2) \frac{e^{-T(M+1)/2 - Tc(M+1)^2}}{1 - e^{-T/2 - Tc(2M+3)}}. \tag{13.16}$$

Indeed $p(E) \leq Z_0(T/2) e^{TE/2}$. The remaining dominating series has first term as displayed and a decreasing ratio, bounded by its first ratio. The estimate includes the equality case $c = 0$. At fixed nonzero real g, the stabilizer cannot be removed within this family while retaining trace class.

13.8 The unchanged cubic series and its Borel boundary

The original cubic formal trace has coefficients

$$Z(T,g)=\sum_{n\geq 0}a_n(T)g^n,\qquad a_n(T)=\frac{(-T)^n}{n!}\operatorname{Tr}(e^{-TH_0}V^n). \tag{13.17}$$

Its odd coefficients vanish and its even coefficients are positive. Define its ordinary Borel transform by

$$\widehat{Z}_T(\zeta)=\sum_{n\geq 0}\frac{a_n(T)}{n!}\zeta^n.$$

The Borel variable ζ is a complex scalar independent of the current coordinate and of the coupling.

Theorem 13.9. *For every $T>0$, (13.17) has radius zero in g. Its Borel transform has exact radius $T/2$ and diverges at both real endpoints $\zeta=\pm T/2$.*

Proof. Fix $0<\nu<T$ and write $t=T-\nu$. Since $p(E)\leq Z_0(\nu)e^{\nu E}$, the norm bound gives

$$\frac{|a_n(T)|}{n!}\leq Z_0(\nu)\frac{T^n}{2^n(n!)^2}\sum_{E\geq 0}E^{2n}e^{-tE}\leq Z_0(\nu)e^{t}\frac{T^n(2n)!}{2^n(n!)^2t^{2n+1}}.$$

For the last step, integration over $[E,E+1]$ bounds the summand by $e^t\int_E^{E+1}x^{2n}e^{-tx}\,dx$. The central binomial coefficient has nth root tending to four. Consequently

$$\limsup_n\left(\frac{|a_n(T)|}{n!}\right)^{1/n}\leq\frac{2T}{(T-\nu)^2}.$$

Letting ν decrease to zero gives the upper bound $2/T$.

For even n, retain one extreme eigenvector at energy $E_n=\lfloor 2n/T\rfloor$, with n sufficiently large. It contributes

$$\frac{a_n(T)}{n!}\geq\frac{T^ne^{-TE_n}}{(n!)^2}\left(\frac{E_n(E_n-1)}{2}\right)^n.$$

Stirling's formula makes the nth root of this expression tend to $2/T$. This proves the exact Borel radius. Multiplication by $n!$ makes the corresponding coefficient roots unbounded, proving radius zero for the original series.

For endpoint divergence, retain both extreme vectors at every energy $E\geq 2$. For $\zeta>0$, their combined contribution to the Borel sum is

$$e^{-TE}\big(I_0(X_E)+J_0(X_E)\big),\qquad X_E=2\sqrt{T\zeta E(E-1)/2},$$

where

$$I_0(X)=\sum_{n\geq 0}\frac{(X^2/4)^n}{(n!)^2},\qquad J_0(X)=\sum_{n\geq 0}\frac{(-1)^n(X^2/4)^n}{(n!)^2}.$$

Their sum retains precisely the even powers. At $\zeta=T/2$, $X_E=TE-T/2+O(E^{-1})$. The elementary integral representation

$$I_0(X)=\frac{1}{\pi}\int_0^{\pi}e^{X\cos\theta}\,d\theta$$

gives $I_0(X)\geq ce^X/\sqrt{X}$ for large positive X, by restricting to $0\leq\theta\leq X^{-1/2}$. The oscillatory integral for J_0 gives $|J_0(X)|\leq 1$. Thus the retained contribution is bounded below by a positive multiple of $E^{-1/2}$ for large E. Its sum diverges. All coefficients of the even Borel series are nonnegative, so this subseries comparison is valid. Evenness gives the other endpoint. □

A fixed vacuum-energy shift multiplies the entire scalar series by a nonzero constant and changes neither radius. Division by the free partition function has the same property. The theorem does not construct continuation outside the disk or choose a Laplace contour. A positive resummation requires a state functional on an observable star-algebra, with positivity and sewing for every admitted observable. Those requirements are stronger than a continuation of the partition function alone.

13.9 Fermionic charges and a local quartic density

The nonlocal square H_0^2 stabilizes the spectrum, but it does not express the correction as the integral of a local circle density. For that purpose, a second realization of the same Fock space exposes a useful positive charge.

Let r run over half-integers. A fermionic occupation set differs from the filled negative sea $S_0 = \{-1/2, -3/2, \dots\}$ in finitely many places. The corresponding wedges form an orthonormal basis. Creation c_r^* inserts the mode r, and annihilation c_r removes it, with the exterior signs. They satisfy $\{c_r, c_s^*\} = \delta_{rs}$. Normal ordering subtracts the vacuum occupation:

$$:c_r^* c_s: = c_r^* c_s - \delta_{rs}\mathbf{1}_{r<0}.$$

The charge is the number of added positive particles minus the number of holes in the negative sea. Charge-zero occupations are indexed by partitions μ, through

$$S_\mu = \{\mu_i - i + 1/2 : i \geq 1\}.$$

Their energy is $|\mu|$. Write $A(\theta) = \sum_{n\neq 0} \alpha_n e^{-in\theta}$ for the angular current, with each Fourier coefficient acting on finite-energy vectors.

Proposition 13.10. *The charge-zero wedge space is unitarily isomorphic to $\mathcal{F}$, with vacuum sent to the filled sea and*

$$\alpha_n = \sum_{r\in\mathbb{Z}+1/2} :c_{r-n}^* c_r: \qquad (n \neq 0). \tag{13.18}$$

Define diagonal operators on the partition basis by

$$Q_j(\mu) = \sum_{i\geq 1}\Big((\mu_i - i + 1/2)^j - (-i + 1/2)^j\Big). \tag{13.19}$$

Then $Q_1 = H_0$, $Q_2 = 2V$, and

$$Q_3 = \frac{1}{4}\int_0^{2\pi} \frac{d\theta}{2\pi}\Big(:A(\theta)^4: + :(\partial_\theta A(\theta))^2: - \frac{1}{2}:A(\theta)^2:\Big). \tag{13.20}$$

All these identities hold on finite-energy vectors.

Proof. For a fixed occupation set, only finitely many summands of (13.18) act. The anticommutation relations give $[\alpha_m, \alpha_n] = m\delta_{m+n,0}$: off-diagonal matrix units cancel, and for $m = -n > 0$ the vacuum boundary contributes the m modes crossing zero. The positive modes kill the filled sea. The vectors obtained by applying products of negative modes therefore have the same orthogonal norms z_μ as the bosonic occupation monomials. They are linearly independent. In energy E, both spaces have dimension $p(E)$, so they span. This proves the isometry and its extension to the Hilbert completions.

The diagonal charges and their vacuum scalar can be calculated through a generating field. Put $\varsigma(s) = 2\sinh(s/2)$ and

$$\mathcal{E}_n(s) = \sum_r e^{s(r-n/2)} :c^*_{r-n}c_r: + \frac{\delta_{n0}}{\varsigma(s)}.$$

Direct multiplication of the fermionic matrix units gives

$$[\alpha_m, \mathcal{E}_n(s)] = \varsigma(ms)\mathcal{E}_{m+n}(s). \tag{13.21}$$

When $m + n = 0$, the boundary sum is $\varsigma(ms)/\varsigma(s)$, which accounts for the scalar in this equation. These are the charge and operator conventions of Okounkov–Pandharipande [83, Sections 2.1–2.2]. Here the vacuum scalar is subtracted from every Q_j.

The normally ordered exponential

$$\mathcal{B}(z,s) = \frac{1}{\varsigma(s)} : \exp\left(\sum_{n\neq 0} \frac{\varsigma(ns)}{n} \alpha_n z^{-n} \right) :$$

has commutator $[\alpha_m, \mathcal{B}] = \varsigma(ms) z^m \mathcal{B}$. The field $\sum_n \mathcal{E}_n(s) z^{-n}$ has the same commutators by (13.21). Both have vacuum matrix element $1/\varsigma(s)$. Commuting positive modes from the left and negative modes from the right reduces any matrix element between created vacuum vectors to that scalar matrix element. Induction on the total number of external modes therefore proves equality of the two fields. This is an identity of formal Laurent series in s, evaluated between finite-energy vectors.

Let CT_z denote the angular constant term. Taking it in the field identity and subtracting the vacuum gives

$$\sum_{j\geq 0} \frac{s^j}{j!} Q_j = \frac{1}{\varsigma(s)} \left[\mathrm{CT}_z : \exp\left(\sum_{n\neq 0} \frac{\varsigma(ns)}{n} \alpha_n z^{-n} \right) : - 1 \right]. \tag{13.22}$$

The left side follows directly from the finite occupation difference in (13.19). In charge zero, $Q_0 = 0$.

Write $A(z) = \sum_{n\neq 0} \alpha_n z^{-n}$ and $B(z) = \sum_{n\neq 0} n^2 \alpha_n z^{-n}$. The exponent is $sA + s^3 B/24 + O(s^5)$, while $\varsigma(s)^{-1} = s^{-1} - s/24 + O(s^3)$. Since $\mathrm{CT}\, A = 0$, extraction gives

$$\begin{aligned} Q_1 &= \tfrac{1}{2}\,\mathrm{CT} :A^2: = H_0, \\ Q_2 &= \tfrac{1}{3}\,\mathrm{CT} :A^3: = 2V, \\ Q_3 &= \tfrac{1}{4}\,\mathrm{CT} :A^4: + \tfrac{1}{4}\,\mathrm{CT} :AB: - \tfrac{1}{8}\,\mathrm{CT} :A^2:. \end{aligned}$$

Finally,

$$\mathrm{CT} :AB: = 2\sum_{n>0} n^2 \alpha_{-n}\alpha_n = \mathrm{CT} :(\partial_\phi A)^2:.$$

This proves (13.20), including its derivative correction and quadratic subtraction. □

The isometry in the proof sends the orthonormal Schur basis

$$s_\mu(p) = \sum_{\nu \vdash |\mu|} \frac{\chi_\mu(\nu)}{z_\nu} p_\nu$$

to S_μ. One can check this directly on finite alternants: multiplication by p_n raises an occupied exponent by n, and reordering supplies the exterior sign. Character orthogonality gives the stated normalization. Formula (13.19) also gives $V s_\mu = (\sum_{(i,j)\in\mu}(j-i)) s_\mu$, consistent with the two extreme partitions above.

13.10 A positive local preparation

Let d be the number of diagonal boxes of μ, and put

$$p_i = \mu_i - i + \tfrac{1}{2}, \qquad h_i = \mu'_i - i + \tfrac{1}{2} \qquad (1 \le i \le d).$$

These positive half-integers are the added particle energies and the absolute energies of the holes. Separating the two occupation differences in (13.19) gives

$$E = \sum_i (p_i + h_i), \qquad Q_2 = \sum_i (p_i^2 - h_i^2), \qquad Q_3 = \sum_i (p_i^3 + h_i^3). \tag{13.23}$$

In particular $Q_3 \ge 0$, and Cauchy–Schwarz yields

$$Q_2^2 \le E Q_3. \tag{13.24}$$

The two vectors used here have entries $(\sqrt{p_i}, \sqrt{h_i})$ and $(p_i^{3/2}, -h_i^{3/2})$.

Theorem 13.11. *For real g, define*

$$H_g = H_0 + gV + \frac{g^2}{8} Q_3. \tag{13.25}$$

It is essentially self-adjoint on finite-energy vectors, with its real diagonal closure in the partition basis, and $H_g \ge H_0/2$. For every $T > 0$,

$$1 \le Z(T, g) = \operatorname{Tr} e^{-T H_g} \le Z_0(T/2) < \infty.$$

The normalized trace is a positive state on bounded operators. Finite Fourier Wick polynomials also have finite expectations. The unnormalized heat kernels sew exactly, and their dependence on real g is smooth in trace norm.

Proof. All charges are real diagonal in an orthonormal basis. Their sum has maximal domain

$$\left\{ (v_\mu) : \sum_\mu |H_g(\mu)|^2 |v_\mu|^2 < \infty \right\}.$$

Finite-support vectors are a core for this multiplication operator. By (13.24),

$$|gV| \le \frac{|g|}{2} \sqrt{E Q_3} \le \frac{E}{2} + \frac{g^2 Q_3}{8}.$$

The last inequality is the nonnegativity of $(\sqrt{E/2} - |g|\sqrt{Q_3/8})^2$. It proves the lower bound and hence trace class by summing $e^{-T H_g(\mu)} \le e^{-T|\mu|/2}$. The vacuum contributes one. Polynomial energy bounds extend the trace domain to the stated unbounded insertions.

The spectral calculus gives $e^{-T_1 H_g} e^{-T_2 H_g} = e^{-(T_1+T_2) H_g}$. On compact real coupling intervals, each derivative of order k of a diagonal exponential is bounded by a constant times $(1 + E)^{3k} e^{-TE/2}$. Indeed $|Q_2| \le E^2$ and $Q_3 \le E^3$ follow from (13.23). This dominating sequence is summable against $p(E)$. Termwise differentiation is therefore valid in trace norm, at every order. □

The correction is the integral of the local density (13.20). It first appears at order g^2. The derivatives of the resulting state agree with the formal expansion of this modified Hamiltonian.

The real smoothness does not imply a holomorphic trace-class family near $g = 0$. For $g = i\gamma \neq 0$, the real part of H_g is $H_0 - \gamma^2 Q_3/8$. On a one-row partition of size E, $Q_3 = (E - 1/2)^3 + 1/8$, so the corresponding exponential is unbounded. Any holomorphic trace-class extension agreeing on a real interval would have these diagonal matrix entries, by uniqueness of scalar holomorphic continuation. Such an extension is impossible.

There is also a convergent product expression for the positive trace. For $r \in \mathbb{Z}_{\geq 0} + 1/2$, set

$$\varepsilon_+(r) = r + \frac{g}{2}r^2 + \frac{g^2}{8}r^3, \qquad \varepsilon_-(r) = r - \frac{g}{2}r^2 + \frac{g^2}{8}r^3.$$

Each is at least $r/2$. The grand-canonical product

$$\mathcal{Z}_{T,g}(y) = \prod_{r>0} \bigl(1 + y e^{-T\varepsilon_+(r)}\bigr)\bigl(1 + y^{-1} e^{-T\varepsilon_-(r)}\bigr) \tag{13.26}$$

converges absolutely and uniformly on $|y| = 1$. The sum of the magnitudes of its nonconstant factors is bounded by $2\sum_{r>0} e^{-Tr/2}$. Consequently

$$Z(T, g) = [y^0]\mathcal{Z}_{T,g}(y) = \frac{1}{2\pi}\int_0^{2\pi} \mathcal{Z}_{T,g}(e^{i\theta})\, d\theta. \tag{13.27}$$

Each factor records one particle or one hole. The exponent of y records their charge difference, so its zero coefficient retains exactly the partition basis of $\mathcal{F}$. Termwise extraction is justified by absolute convergence and gives the spectral trace. At $g = 0$, the expression reduces to $Z_0(T)$. Omitting the charge projection would change the Hilbert space.

13.11 The sharp coefficient and the critical state

To determine what the coefficient $1/8$ achieves, consider the larger real family

$$H_{g,c} = H_0 + \frac{g}{2}Q_2 + c\,g^2 Q_3.$$

It is defined by the same diagonal closure and local density.

Proposition 13.12. *For $c > 0$, the largest constant a_c such that $H_{g,c} \geq a_c H_0$ for every real g is*

$$a_c = 1 - \frac{1}{16c}.$$

Nonnegativity for every coupling holds exactly for $c \geq 1/16$. The uniform half-energy bound holds exactly for $c \geq 1/8$. For $c > 1/16$, the heat operators and normalized densities are continuous in g in trace norm.

Proof. For a nonempty partition, complete the square:

$$H_{g,c} = cQ_3\left(g + \frac{Q_2}{4cQ_3}\right)^2 + E - \frac{Q_2^2}{16cQ_3} \geq \left(1 - \frac{1}{16c}\right)E.$$

The vacuum charges vanish. On the row partition (N), put $p = N - 1/2$. Then

$$E = N, \qquad Q_2 = p^2 - \tfrac{1}{4} = N(N-1), \qquad Q_3 = p^3 + \tfrac{1}{8}.$$

The ratio $Q_2^2/(EQ_3)$ tends to one. At $g = -Q_2/(4cQ_3)$, the ratio $H_{g,c}/E$ therefore tends to a_c, proving optimality. For $c < 0$, the energy of a nonempty partition becomes negative at large $|g|$. For $c = 0$, a row of length at least two has a nonzero cubic charge and becomes negative for one sign of g. This proves both coefficient thresholds.

If $c > 1/16$, the diagonal heat entries are bounded uniformly by e^{-Ta_cE}. Their sum is $Z_0(Ta_c)$. For $g_j \to g$, differences converge entrywise and are dominated by twice this summable sequence. Their sum of absolute values is precisely the trace norm. The traces also converge and are at least one, so division gives continuity of the densities. □

At the critical value, fixed-coupling trace class survives but this continuity fails.

Theorem 13.13. *Let $c = 1/16$, $K_g(T) = e^{-TH_{g,c}}$, and $\rho_g(T) = K_g(T)/\operatorname{Tr} K_g(T)$. For every real g and $T > 0$, the heat operator is positive and trace class. As $g \to 0$, it converges strongly to e^{-TH_0}. Nevertheless, along*

$$g_N = -\frac{4}{N - 1/2}, \qquad N \geq 4,$$

one has

$$\begin{aligned} \operatorname{Tr} K_{g_N}(T) &\longrightarrow \infty, \\ \|K_{g_N}(T) - K_0(T)\|_1 &\longrightarrow \infty, \\ \|\rho_{g_N}(T) - \rho_0(T)\|_1 &\longrightarrow 2. \end{aligned}$$

Proof. The particle and hole contributions are now

$$\varepsilon_\pm(r) = r\left(1 \pm \frac{gr}{4}\right)^2 \geq 0.$$

For fixed $g \neq 0$ and $r \geq 8/|g|$, both are at least r. Each lower particle or hole mode can occur only once. Thus $H_{g,c} \geq E - C_g$, where the finite constant is

$$C_g = \sum_{0<r<8/|g|} \big((r - \varepsilon_+(r))_+ + (r - \varepsilon_-(r))_+\big), \qquad x_+ = \max(x, 0).$$

It follows that $\operatorname{Tr} K_g(T) \leq e^{TC_g} Z_0(T)$. At $g = 0$, trace class holds directly. Nonnegativity bounds $\|K_g(T)\|$ by one. Its entries converge on every fixed partition vector, so density of their finite span proves strong convergence on the Hilbert space.

Set $p = N - 1/2$. For the row partitions $(N + k)$, with $0 \leq k \leq \lfloor \sqrt{p} \rfloor$, the energy at g_N is

$$H_{g_N,1/16}(N + k) = \frac{(p + k)k^2}{p^2} + \frac{1}{2}\left(1 + \frac{1}{2p}\right)^2 < 3. \tag{13.28}$$

The first term is at most $1 + 1/\sqrt{p} < 2$, and the second is less than one for $p \geq 7/2$. These distinct states give

$$\operatorname{Tr} K_{g_N}(T) \geq (1 + \lfloor \sqrt{p} \rfloor)e^{-3T} \longrightarrow \infty.$$

The trace difference is a lower bound on the trace-norm difference, which proves the second limit.

For a fixed finite energy projection F, $\operatorname{Tr}(F\rho_{g_N}(T)) \to 0$, since its numerator is bounded by rank F and the denominator diverges. Given $\eta > 0$, choose F with $\operatorname{Tr}(F\rho_0(T)) > 1 - \eta$. Both densities are diagonal and have trace one. Summing absolute differences on F and its complement gives

$$\|\rho_{g_N}(T) - \rho_0(T)\|_1 \geq 2\big|\operatorname{Tr}(F\rho_{g_N}(T)) - \operatorname{Tr}(F\rho_0(T))\big|.$$

The lower limit is at least $2(1 - \eta)$, and the norm is at most two. Let η decrease to zero. □

The critical example distinguishes strong convergence of heat operators from convergence of the prepared states. Above the critical coefficient, uniform energy decay prevents the mass from escaping to partitions of increasing size. The coefficient $1/8$ gives the particularly simple bound $H_g \geq H_0/2$.

13.12 The interacting current and the remaining state map

Return to the local Hamiltonian (13.25). Its state preserves energy, so a two-current expectation vanishes unless the two indices sum to zero. The fermionic occupations give its remaining coefficients explicitly. For $r > 0$, put

$$f_r(y) = \frac{y e^{-T\varepsilon_+(r)}}{1 + y e^{-T\varepsilon_+(r)}}, \qquad f_{-r}(y) = \frac{1}{1 + y^{-1} e^{-T\varepsilon_-(r)}}.$$

These functions record occupancy of a positive mode and occupancy of a negative mode in the filled sea. Their denominators do not vanish on $|y| = 1$. Define, for $n > 0$,

$$A_n(T, g) = \frac{[y^0]\Big(\mathcal{Z}_{T,g}(y)\, \Sigma_{r \in \mathbb{Z}+1/2}\, f_{r-n}(y)(1 - f_r(y))\Big)}{Z(T, g)}. \tag{13.29}$$

For fixed n, the tails decay exponentially and uniformly on the unit circle. The finitely many modes crossing the sea boundary are harmless.

Proposition 13.14. *The coefficient $A_n(T, g)$ is $\varphi_{T,g}(\alpha_n \alpha_{-n})$. If $c_n = A_n - n$, then $c_n \geq 0$ and*

$$c_n \leq C_T e^{-Tn/4}, \qquad C_T = \sum_{E \geq 0} E\, p(E) e^{-TE/4} < \infty. \tag{13.30}$$

The current two-point function is

$$\varphi_{T,g}(J(z)J(w)) = \frac{\lambda}{(z - w)^2} + \frac{\lambda}{zw} \sum_{n \geq 1} c_n(T, g)\big((w/z)^n + (z/w)^n\big). \tag{13.31}$$

Its radially ordered domain includes $e^{-T/4} < |w/z| < 1$. The remainder and all fixed holomorphic derivatives extend across $z = w \neq 0$.

Proof. In the grand-canonical occupation measure, a raising move from $r - n$ to r has weight $f_{r-n}(1 - f_r)$. By (13.18), the squared norm of this move is the corresponding contribution to $\alpha_n \alpha_{-n}$. Different moves from a fixed occupation vector give orthogonal vectors. Summing their expectations therefore gives the numerator in (13.29). The insertion is charge neutral, so taking the zero-charge coefficient commutes with it. Absolute convergence justifies the extraction.

The commutator gives $c_n = \varphi_{T,g}(\alpha_{-n} \alpha_n) \geq 0$. This positive operator is at most H_0, and it kills energy spaces below n. The half-energy bound and $Z(T, g) \geq 1$ give

$$c_n \leq \sum_{E \geq n} E\, p(E) e^{-TE/2} \leq e^{-Tn/4} \sum_{E \geq 0} E\, p(E) e^{-TE/4}.$$

The constant is finite by the free partition estimate. Inserting $A_n = n + c_n$ into the current series yields (13.31). The vacuum sum supplies the pole; the exponential bound makes the two remainder series normally convergent in a neighborhood of $|w/z| = 1$. □

The modified preparation retains the current pole while changing the regular coefficients. It need not be Gaussian as a state of the bosonic current algebra. Its operator sewing and scalar normalization are fixed by the spectral preparation, including the charge projection. A conformal shift $-1/24$ multiplies its unnormalized kernels by $e^{T/24}$ and leaves normalized correlations unchanged.

For the unchanged cubic interaction, the outstanding construction is a state-valued Borel continuation with a specified Laplace prescription on a chosen observable star-algebra and sewing completion. It

must give a normalized positive functional on every allowed observable and preserve the unnormalized cylinder composition. The scalar Borel boundary does not decide whether such a prescription exists. The positive states of the modified local Hamiltonian differ from that interaction at order g^2.

A bulk realization has a further requirement. It must construct the boundary preparation from independently defined bulk fields, counterterms and a boundary condition, then intertwine the current, collision and sewing operations on their declared domains. An equality of characters or a matching determinant alone does not supply that state map.

Chapter 14

Native fields and the boundary action

A boundary current can be obtained by integrating fields in a normal direction. The integration also determines how an interior insertion acts on that current. These two outputs contain different information. An eliminated field may have zero projection and still contract nontrivially with a boundary observable. The resulting operation depends on the propagator, its supports and its quantum normalization.

We construct this comparison for the Gaussian theory on a three-dimensional half-space. Its Wilson source then gives an independent incoming junction, including a charged vacuum and a sewn operation. Finite regular jets can be retained by explicit changes of representatives. The completions used for these constructions will be specified separately: a smooth smearing, a polynomial state and a formal series of states have different pole bounds.

14.1 The half-space action

Put $M = \mathbb{R}_{\geq 0,t} \times \mathbb{C}_z$, with $z = x + iy$ and orientation $dt \wedge dx \wedge dy$. The partial forms on M are generated by $dt, d\bar{z}$. Write

$$D = dt\, \partial_t + d\bar{z}\, \partial_{\bar{z}}, \qquad \partial = \partial_z, \qquad \partial_{\bar{z}} = \tfrac{1}{2}(\partial_x + i\partial_y).$$

The differential D has degree one, and ∂ has degree zero. They commute. Fields are smooth up to the boundary. Variations and test fields have compact support, which may meet that boundary.

For partial forms of compact support use the traces

$$\tau_M(a) = \frac{1}{2\pi i} \int_M dz \wedge a, \qquad \tau_\partial(b) = \frac{1}{2\pi i} \int_{\mathbb{C}} dz \wedge b. \tag{14.1}$$

They select partial form degree two and antiholomorphic degree one, respectively. If $i : \mathbb{C} \hookrightarrow M$ is the boundary inclusion, these conventions give $\tau_M(Da) = \tau_\partial(i^* a)$. The sign includes both the outward normal and the position of dz.

Let $A = \mathbb{C}[u_n^a : 1 \leq a \leq r,\ n \geq 0]$ be a differential polynomial algebra, with $\partial u_n^a = u_{n+1}^a$. A finite-order matrix differential operator

$$H^{ab}(\partial) = \sum_{j=0}^{N} H_j^{ab}(u, \partial u, \ldots)\partial^j$$

specifies generator brackets by $\{u_\lambda^a u^b\} = H^{ba}(\lambda)$. Here λ records a holomorphic derivative of a contact. The extension to all differential polynomials is

$$\{f_\lambda g\} = \sum_{a,b,m,n} \frac{\partial g}{\partial u_n^b}(\lambda + \partial)^n H^{ba}(\lambda + \partial)(-\lambda - \partial)^m \frac{\partial f}{\partial u_m^a}. \tag{14.2}$$

Every operator acts on the expression to its right. The sum is finite on each pair of polynomials.

We call H Hamiltonian when this bracket is skew and satisfies Jacobi. Equivalently, $H^{ab} = -(H^{ba})^*$, where the formal adjoint reverses composition and sends ∂ to $-\partial$, and Jacobi holds on generator triples. The master formula then propagates the identities to all polynomials [8, Theorem 1.15 and Proposition 1.16]. In particular it gives

$$\{\partial f_\lambda g\} = -\lambda\{f_\lambda g\}, \qquad \{f_\lambda \partial g\} = (\lambda + \partial)\{f_\lambda g\}.$$

The scalar operator $H = \kappa\partial$ is the current example.

Introduce superfields Φ^a and η_a of total degrees zero and one. In their expansions in 1, dt, $d\bar z$, $dt\, d\bar z$, their coordinate degrees are

$$\deg \Phi^a = (0, -1, -1, -2), \qquad \deg \eta_a = (1, 0, 0, -1).$$

The degree-one zero-form in η_a is the gauge ghost. The negative degrees are antifields. The BV two-form and action are

$$\omega_{\mathrm{BV}} = \tau_M(\delta\Phi^a\, \delta\eta_a), \qquad S_H = \tau_M\Big(\eta_a D\Phi^a + \tfrac{1}{2}\eta_a H^{ab}(\partial;\Phi)\eta_b\Big). \tag{14.3}$$

The two-form has degree -1, and the action has degree zero. Its inverse pairing defines the degree-one BV bracket on local functionals. This is the local Poisson vertex sigma model [62, §§2.2 and 4.1].

Proposition 14.1. *For a Hamiltonian operator H, the action* (14.3) *satisfies the classical master equation on the boundary condition*

$$i^*\eta = 0.$$

This is a Lagrangian condition on the whole boundary superfield and is preserved by the BV differential.

Proof. The kinetic square vanishes by $D^2 = 0$. Its bracket with the interaction is a total partial derivative. Its boundary trace contains $i^*\eta$, and vanishes. Pair the interaction bracket with three independent odd test fields and integrate every holomorphic derivative by parts. The remaining coefficient is the generator Jacobi expression

$$\{u^a_\lambda\{u^b_\mu u^c\}\} - \{u^b_\mu\{u^a_\lambda u^c\}\} - \{\{u^a_\lambda u^b\}_{\lambda+\mu}u^c\}.$$

It vanishes by the Hamiltonian-operator identity. This proves $\{S_H, S_H\}_{\mathrm{BV}} = 0$, and the graded Jacobi identity gives $Q^2 = 0$ for $Q = \{S_H, -\}_{\mathrm{BV}}$.

Variation has boundary term $-\tau_\partial(\eta_a\delta\Phi^a)$. The condition sets one complementary summand of this non-degenerate pairing to zero and leaves the other free. It is Lagrangian. The interaction part of $Q\eta$ is quadratic in η and its holomorphic derivatives. Its tangential pullback vanishes on the condition, and $\bar\partial$ preserves the same condition. □

For $H = \kappa\partial$, write the physical fields as a scalar ϕ and a partial one-form a. Then

$$S^{\mathrm{cl}}_\kappa = \tau_M\big(aD\phi + \tfrac{\kappa}{2}a\,\partial a\big), \qquad D\phi + \kappa\partial a = 0, \qquad Da = 0.$$

The gauge variations are $\delta\phi = \kappa\partial\epsilon$ and $\delta a = -D\epsilon$, with ϵ zero at the boundary. For $\kappa \neq 0$, the one-form $\mathcal{A} = a - \kappa^{-1}\phi\, dz$ identifies this action with $-\kappa(4\pi i)^{-1}\int \mathcal{A}\wedge d\mathcal{A}$. Indeed expansion and one integration by parts give the displayed scalar action, with boundary term zero under $i^*a = 0$. Thus the current model has a pure abelian Chern–Simons realization with this normalization.

The finite Hamiltonian Lie algebra $\mathfrak{p}_N$ of Chapter 3 supplies another classical example. If $[e_a, e_b] = f^{\,c}_{ab}e_c$, set $\{u^a_\lambda u^b\} = f^{\,c}_{ab}u^c$. This has differential order zero and satisfies Jacobi by the Lie identity. Its action is the shifted cotangent action with those coefficients. The finite-dimensional quantum measure of that chapter concerns its algebraic model. Quantizing all spacetime modes and identifying a physical boundary algebra remain additional comparisons.

14.2 Regular Gaussian observables

Keep κ as an indeterminate and put $R = \mathbb{C}[\kappa, \hbar]$. For the scalar Gaussian model use the tangent complex

$$E = \left(\Omega^{\bullet}_{\mathrm{HT}} e_\Phi \oplus \Omega^{\bullet}_{\mathrm{HT}} e_\eta\right) \otimes \mathbb{C}[\kappa], \qquad |e_\Phi| = 0, \quad |e_\eta| = -1.$$

For a partial form α of degree p, its differential is

$$\begin{aligned} Q(\alpha e_\Phi) &= D\alpha e_\Phi, \\ Q(\alpha e_\eta) &= D\alpha e_\eta + (-1)^{p+1} \kappa \partial \alpha e_\Phi. \end{aligned} \tag{14.4}$$

The two off-diagonal terms in Q^2 have opposite signs. Impose the relative boundary condition on every tangential e_η component. The coordinate and tangent degrees are dual conventions.

The degree-minus-one pairing is

$$B(\alpha e_\Phi, \beta e_\eta) = \tau_M(\alpha\beta), \qquad B(\alpha e_\eta, \beta e_\Phi) = -(-1)^{\deg \beta} \tau_M(\alpha\beta),$$

and it is zero on equal summands. Let $E_c(U)$ denote the compactly supported fields on an open U, with the boundary condition when needed. On $C(U) = E_c(U)[1]$, write

$$d(s\xi) = -sQ\xi, \qquad b(s\xi, s\zeta) = (-1)^{|\xi|+1} B(\xi, \zeta).$$

Thus b is a graded-symmetric form of degree one.

At fixed field number, a regular observable is a compact smooth joint kernel in the corresponding graded symmetric tensor power of $C(U)$. Complete that tensor at fixed compact support in its smooth seminorms, then take the inductive limit over compact supports. Denote the result by $\mathrm{Sym}^m_b C(U)$. Equivalently one can use the completed bornological tensor of the compact smooth spaces. The direct sum over field number remains algebraic. Define

$$\mathcal{O}^q(U) = \left(\bigoplus_{m \geq 0} \mathrm{Sym}^m_b C(U)[\hbar], d + \hbar \Delta_b\right). \tag{14.5}$$

Here Δ_b contracts an unordered pair with b, with its Koszul sign. Restriction of a smooth joint kernel to a diagonal and compact integration define this operation continuously.

Compatibility of the pairing with Q gives $[d, \Delta_b] = 0$, and odd pair contractions give $\Delta_b^2 = 0$. Consequently (14.5) is a cochain complex. Disjoint opens have products by extension by zero and concatenation of kernels. There are no cross BV contractions on disjoint supports, so these products commute with the differential and compose associatively. The constant observable is their unit.

The descent statement uses the free-field theorem for an elliptic complex of finite total rank, topological in the normal direction, with a local Lagrangian elliptic boundary condition [51, Definitions 2.1–2.3 and Theorems 4.1–4.2]. Our tangential differential is the triangular extension of two Dolbeault complexes, and the normal factor is de Rham. These are precisely the normally topological Gaussian data. The theorem gives a factorization algebra of these regular observables and its boundary restriction. The maps below construct the collar comparison explicitly.

Point evaluation has a delta-distribution kernel and is absent from (14.5). Its extension to local composite observables is a renormalization problem. Likewise, the formal completion $\mathcal{O}^q(U)^\wedge_\hbar$ allows the field-number bound to grow with the coefficient order. It is larger than completing each fixed field number and then taking their algebraic sum.

14.3 The normal contraction and its quantum correction

Choose a smooth profile ρ of integral one supported in $[0, \varepsilon)$, and put $F(t) = \int_0^t \rho(s)\, ds$. For a tangential open V, split its compact fields into

$$L = \Omega_c^{0,\bullet}(V) e_\Phi \otimes \mathbb{C}[\kappa], \qquad G = \Omega_c^{0,\bullet}(V) e_\eta \otimes \mathbb{C}[\kappa].$$

The tangential differential is $d_\partial(\ell, g) = (d_L \ell + Q_{\mathrm{rel}} g, d_G g)$, where

$$Q_{\mathrm{rel}}(\alpha e_\eta) = (-1)^{\deg \alpha + 1} \kappa\, \partial \alpha e_\Phi, \qquad d_L Q_{\mathrm{rel}} + Q_{\mathrm{rel}} d_G = 0.$$

Write a collar field as $\gamma(t) + dt\, \beta(t)$, with $\gamma_G(0) = 0$. Its differential is $d_\partial \gamma + dt(\partial_t \gamma - d_\partial \beta)$.

Define maps to and from $(G[-1], -d_G)$ by

$$P(\gamma + dt\, \beta) = \int_0^\infty \beta_G(s)\, ds, \qquad I g = dt\, \rho g + (F - 1) Q_{\mathrm{rel}} g. \tag{14.6}$$

The second term in I is required. Its normal derivative cancels the unwanted term $-dt\, \rho Q_{\mathrm{rel}} g$ in the differential of the first term. In particular $QI = -I d_G$ and $PI = 1$.

Lemma 14.2. *The degree-minus-one map*

$$h(\gamma + dt\, \beta) = -\int_t^\infty \beta(s)\, ds + (1 - F(t)) \int_0^\infty \beta_G(s)\, ds$$

satisfies

$$Qh + hQ = 1 - IP, \qquad hI = Ph = h^2 = 0.$$

These identities hold continuously on compact smooth joint kernels and retain compact tangential support. No inverse of κ is used.

Proof. The correction sets the G-component of h to zero at $t = 0$. All tail integrals vanish above the supports of the input and ρ. Differentiating the tail gives $\beta(t)$, while differentiating $1 - F$ gives $-\rho$. Substitution in the collar differential proves $Qh + hQ = 1 - IP$. Since I has normal component ρg, its two terms under h cancel. The output of h has no normal one-form, so $Ph = h^2 = 0$. Integrals in a compact normal interval are bounded in each smooth tangential seminorm. The same estimates with other kernel variables as parameters prove the completed-kernel statement. □

Let $\mathfrak{a}(V) = \Omega_c^{0,\bullet}(V) \otimes \mathbb{C}[\kappa]$. Fix $\zeta^2 = -1$. The map $\alpha \mapsto \zeta(-1)^q \alpha e_\eta$, for Dolbeault degree q, identifies $(\mathfrak{a}(V), \bar\partial)$ with $(G[-1], -d_G)$. After suspension, (14.6) and $-sh$ give a contraction (i, p, h_C) from C to $\mathfrak{a}(V)[1]$.

The pairing induced by i is determined by

$$\mu_\kappa(\alpha, \beta) = -\kappa \tau_\partial(\alpha \partial \beta), \qquad b(i(s\alpha), i(s\beta)) = (-1)^q \mu_\kappa(\alpha, \beta). \tag{14.7}$$

To check the normalization, only $q + r = 1$ contributes. The two terms of I pair to $\kappa \tau_\partial(\alpha \partial \beta)$, since $2 \int \rho(1 - F)\, dt = 1$ and $\tau_M(dt\, \omega) = -\int \tau_\partial(\omega)\, dt$. The two phase factors give $\zeta^2 (-1)^{q+r} = 1$, and suspension gives $(-1)^{q+1}$. This proves the formula.

Write $\mu^s = i^* b$ for the suspended pairing and set

$$\mathcal{A}_\kappa(V) = \Big(\mathrm{Sym}_b(\mathfrak{a}(V)[1])[\hbar], d + \hbar \Delta_{\mu^s} \Big).$$

Integration by parts makes μ_κ a graded-skew cocycle. The inclusion $i^q = \mathrm{Sym}(i)$ therefore preserves the quantum differential. The projection needs a correction.

Theorem 14.3. *Regular Gaussian collar observables contract onto $\mathcal{A}_\kappa(V)$ over R. Both chain maps preserve the unit and products on tangentially disjoint regions. They are continuous and polynomial in $\hbar$ at fixed field number, and extend coefficientwise to the $\hbar$-adic completion.*

Proof. Put $e = i\,p$. Average $h_C \otimes 1 + e \otimes h_C$ with its graded transpose to obtain H_2. Then $dH_2 + H_2 d = 1 - e \otimes e$. The even pairing $P_2 = -bH_2$ satisfies

$$\delta P_2 = b - p^* \mu^s, \qquad [d, \partial_{P_2}] = \Delta_b - \Delta_{p^* \mu^s},$$

where ∂_{P_2} contracts an unordered pair. Constant contractions graded-commute. Hence for $W = \exp(\hbar \partial_{P_2})$ the conjugation has only one commutator:

$$W(d + \hbar\Delta_b)W^{-1} = d + \hbar\Delta_{p^* \mu^s}.$$

Take

$$i^q = \mathrm{Sym}(i), \qquad p^q = \mathrm{Sym}(p)W. \tag{14.8}$$

The pairing P_2 is zero on $i \otimes i$, so $W i^q = i^q$ and $p^q i^q = 1$.

Split $C = \mathrm{im}\, i \oplus \ker p$. Extend h_C as a derivation of the symmetric algebra and divide by the positive number of $\ker p$ factors. Set the resulting H_0 to zero on the retained factors alone. Then $[d, H_0] = 1 - \mathrm{Sym}(e)$. Only retained factors enter $\Delta_{p^* \mu^s}$, so it graded-commutes with H_0. Consequently

$$H^q = W^{-1} H_0 W, \qquad (d + \hbar\Delta_b)H^q + H^q(d + \hbar\Delta_b) = 1 - i^q p^q.$$

Each contraction lowers field number by two. All exponentials therefore terminate on every polynomial. Normal integration and diagonal restriction are continuous on smooth joint kernels. Their tangential supports remain those of the uncontracted variables. In particular P_2 has no cross term on disjoint tangential supports, proving the product assertion. At a fixed coefficient of $\hbar$, only finitely many coefficients of a formal input contribute. This proves the completed assertion as well. □

14.4 Currents and supported interior insertions

The finite-circle-spectrum sector on a centered disk is the algebraic sum of its rotation eigenspaces. Choose a radial annular cutoff χ, equal to one near the inner circle and zero beyond the outer circle. The compact forms $\alpha_n = (\bar\partial \chi) z^{-n}$, $n \geq 1$, obey

$$\tau_\partial(z^m \alpha_n) = \delta_{m,n-1}, \qquad -\partial \alpha_n = n\alpha_{n+1} - \bar\partial(z^{-n} \partial \chi).$$

Stokes' formula proves the first identity. Differentiation proves the second. They fix the creation modes and translation, including their normalization.

Proposition 14.4. *The Gaussian boundary has polynomial vacuum states*

$$V_\kappa = R[x_1, x_2, \ldots], \qquad J_{-n} = x_n, \quad J_n = \hbar\kappa n \partial_{x_n}\ (n > 0), \quad J_0 = 0.$$

Its current satisfies $J(z)J(w) \sim \hbar\kappa(z - w)^{-2}$. Translation is $T = \sum_{n>0} n x_{n+1} \partial_{x_n}$. These formulas include the nonsimple zero-level specialization.

Proof. The cocycle (14.7) on Laurent test functions is $-\kappa\,\mathrm{Res}(f\,\partial g\,dz)$. Thus the contour modes obey $[J_m, J_n] = \hbar\kappa m \delta_{m+n,0}$. The compact representatives supply the polynomial creation states and $[T, J_m] = -mJ_{m-1}$. Normal-ordered products of derivatives of J then give every state field, with the vacuum and translation identities. Wick contraction proves locality and the stated double pole.

The action is quadratic. If its level term is treated as an insertion, each kinetic propagator joins Φ to η, while that insertion has two η legs. Two insertions cannot join each other. A boundary pair receives exactly one insertion, and its two attachments cancel the factor $1/2$. Hence the current level has no interaction correction. At $\kappa = 0$, the positive modes vanish and the same polynomial carrier is a commutative differential vertex algebra. □

The current algebra does not select the Hilbert-space preparation of Chapter 13. Even in this Gaussian theory, normalized polynomial correlations leave determinant and framing factors undetermined. They are additional data of a physical state.

For an interior open $U \subset V \times (a,b)$ with $0 < \varepsilon < a < b$, the collar inclusion i^q has normal support disjoint from U. Define

$$\beta_{U;V}(f)(a_0) = p^q\big(\mathrm{ext}_U(f)\, i^q(a_0)\big). \tag{14.9}$$

Its target is the complex of continuous operations on $\mathcal{A}_\kappa(V)$, with Hom differential $T \mapsto dT - (-1)^{|T|}Td$. Both comparison maps are cochain maps. Disjointness removes the cross BV term in the middle product. Thus $\beta_{U;V}$ is a cochain map.

This action detects a contraction that linear restriction misses. Take a unit-integral normal profile $\rho_{\mathrm{int}}dt$ above the boundary profile, and compact tangential tests ϕ, β of Dolbeault degrees zero and one with $\tau_\partial(\phi\beta) = 1$. Set

$$u = \zeta^{-1}\kappa s(\rho_{\mathrm{int}}dt\, \phi e_\Phi), \qquad v = s\beta.$$

Then $p(u) = 0$, but $h_C u = \zeta^{-1}\kappa s(\phi e_\Phi)$ on the boundary profile. Substitution in $P_2 = -bH_2$ gives

$$P_2(u, i(v)) = \kappa, \qquad \beta(u)(v) = \hbar\kappa, \qquad \beta(u)(1) = 0.$$

The cochain u is not assumed closed. This is an operation on the supported complexes, rather than a new bulk cohomology class.

For a homogeneous linear input, the complete formula is

$$\begin{aligned}\beta(u)(v_1 \cdots v_r) &= p(u)v_1 \cdots v_r \\ &\quad + \hbar \sum_{j=1}^{r} (-1)^{|u|\sum_{\ell<j}|v_\ell|} P_2(u, i(v_j))v_1 \cdots \widehat{v}_j \cdots v_r.\end{aligned}$$

Its quantum factor remains over every truncated or completed coefficient ring. There is no division by $\hbar$.

If f is supported above g, the normal homotopy of g remains below f. Inserting $1 - i^q p^q = dH^q + H^q d$ gives the ordered composition homotopy

$$K(f,g,a_0) = (-1)^{|f|}\, p^q\big(\mathrm{ext}(f)H^q(\mathrm{ext}(g)i^q(a_0))\big). \tag{14.10}$$

Its Hom differential is the joint projected product minus the successive actions. The sign moves the degree-minus-one homotopy past f. Arbitrary motion of supports requires parameter-dependent operations. The forest coefficient homotopies of Chapter 7 retain their contact terms, but their decoration by native field operations is a further comparison.

14.5 Smooth periods and polynomial states

A smooth compact smearing need not have finitely many holomorphic moments. For $0 < r < R_0$, define

$$E_r = \left\{ \sum_{n\geq 1} a_n w^{-n} : \sum_{n\geq 1} |a_n| r^{-n} < \infty \right\}, \qquad \mathscr{P}(D_{R_0}) = \operatorname*{colim}_{r<R_0} E_r.$$

The norm is the displayed weighted sum. For $r < r'$, the inclusion has diagonal coefficients $(r/r')^n$ in normalized bases and is nuclear and compact. A countable cofinal sequence of radii defines the complete locally convex inductive limit. Its elements are exterior holomorphic germs vanishing at infinity. Cauchy estimates identify this description with the coefficient description above.

Put

$$\mathscr{F}(D_{R_0}) = \bigoplus_{m\geq 0} \mathrm{Sym}^m_\pi \mathscr{P}(D_{R_0})[\hbar].$$

Here the projective tensor is completed at each fixed m, and the field-number sum is algebraic. A finite union of disjoint disks uses the direct sum of their germ spaces. Common inner radii and compact support bounds are required locally on parameter families.

The exterior Cauchy map on a compact $(0,1)$-form is

$$C(sa)(w) = \tau_\partial\left(\frac{a(z)}{w-z}\right) = \sum_{n\geq 0} \tau_\partial(z^n a) w^{-n-1}. \tag{14.11}$$

It is zero on shifted functions. The expansion is taken outside the support disk. A radial cutoff χ provides a section $j_\chi(f) = s(\bar\partial\chi)f$, on any germ stage lying inside its transition annulus.

Lemma 14.5. *At every fixed field number, the symmetric Cauchy map is a continuous quasi-isomorphism from the compact Dolbeault current complex to the corresponding symmetric germ space. Its cohomology is in degree zero, with its Hausdorff quotient topology, and its differential has closed range.*

Proof. The Cauchy convolution on the whole plane is a smooth function $\widetilde{C}a$ with $\bar\partial\widetilde{C}a = -a$. If its exterior germ is zero, it is a compact primitive. There are no nonzero compact holomorphic functions. Choose radii $r_0 < r < r' < r'' < R_0$, with the source supported in D_{r_0} and $\chi = 1$ through $D_{r'}$, supported in $D_{r''}$. Then

$$h_\chi(sa) = s\Big(\widetilde{C}a - (1-\chi)C(a)\Big), \qquad h_\chi(s\phi) = 0$$

satisfies $d_1 h_\chi + h_\chi d_1 = 1 - j_\chi C$, where $d_1(s\alpha) = -s\bar\partial\alpha$. The second term is defined outside D_r and extended by zero where $1-\chi = 0$. Stokes' formula gives $C\, j_\chi = 1$.

Cauchy estimates on the annular gap bound every derivative by finitely many source seminorms and the E_r norm. Apply this contraction in each variable of a joint kernel. The tensor homotopy applies h_χ to one variable, the projection to earlier variables and the identity to later variables, with Koszul signs. The estimates are uniform in the remaining smooth parameters. Thus the identity holds on completed projective tensors. Averaging over the finite permutation group gives symmetric powers. The ranges are kernels of continuous maps to Hausdorff spaces, and the stagewise sections make the inverse on cohomology continuous. □

The Gaussian and source terms can also be removed explicitly. Let $k_1(z) = 1/z$ as a locally integrable distribution and $k_2 = -\partial_z k_1$. On shifted top forms define

$$P_C(sa, sb) = \kappa\tau_z\tau_w\big(k_2(z-w)a(z)b(w)\big), \qquad \gamma(sa) = \tau_z\left(\frac{a(z)}{z}\right).$$

They vanish on other form degrees. Their Hom differentials are $\delta P_C = \mu^s$ and $\delta\gamma = \epsilon_0$, where $\epsilon_0(s\phi) = \phi(0)$ and ϵ_0 is zero on top forms. Indeed $\partial_{\bar z}(1/z) = \pi\delta_0$ and $\tau(f\, d\bar z) = -\pi^{-1}\int f$ give

$$\tau\left(\frac{\bar\partial\phi}{z}\right) = \phi(0), \qquad \tau_z\big(k_2(z-w)\bar\partial\phi(z)\big) = \partial_w\phi(w).$$

For a scalar Wilson charge q, the boundary differential is $d_1 + \hbar\Delta_{\mu^s} + \hbar q\epsilon_0$. The exponential

$$N_q = \exp\big(\hbar\partial_{P_C} + \hbar q\,\partial_\gamma\big)$$

conjugates it to d_1. The proof is the same one-commutator calculation as in Theorem 14.3. Every contraction removes factors. Products of contracted distributions use disjoint pairs of variables, and therefore exist as distributional tensor products. They are paired with smooth joint kernels.

The line source on bulk observables is the constant derivation

$$\ell(s\xi) = \zeta^{-1} \int_0^\infty \beta_\eta^{(0)}(t,0)\, dt, \qquad \xi = \gamma + dt\, \beta.$$

Its degree is one. The relative boundary condition gives $\ell d = 0$, and it graded-commutes with constant contractions. Thus $d + \hbar\Delta_b + \hbar q\ell$ squares to zero. Moreover $\ell = \epsilon_0 p$, and the quantum collar maps still give a contraction onto the charged boundary complex. The Gaussian identity $e^{-q\lambda}(d + \hbar\Delta_b)e^{q\lambda} = d + \hbar\Delta_b + \hbar q\ell$ expresses the Wilson insertion, where λ is its distributional linear representative. The identity defines the operation on regular observables without placing λ inside the smooth observable algebra.

Consequently

$$\Pi = \mathrm{Sym}(C) N_q p^q \tag{14.12}$$

is a continuous quasi-isomorphism onto $\mathscr{F}(D_{R_0})$, with zero target differential. The statement follows by composing the three explicit contractions. It is a comparison with the germ completion. For example $e^{c/w} - 1$, for $c \neq 0$, is the Cauchy transform of $s(\bar\partial\chi)(e^{c/z} - 1)$ and has infinitely many moments.

On a centered disk, rotation gives w^{-n} weight $-n$. At a fixed total negative weight there are finitely many partitions into positive mode numbers. Hence the finite-circle-spectrum target is exactly $R[x_1, x_2, \ldots]$, with $x_n = w^{-n}$. The restricted period map remains a quasi-isomorphism. To see this, choose a primitive of a finite-spectrum cocycle in its full acyclic cone, then apply the finite Fourier projector onto the weights of that cocycle. All rotations lie in a common compact support and germ stage, so the integral exists and commutes with the differential. This gives a finite-spectrum primitive.

The same map identifies separated current insertions with

$$J(z) = \sum_{n\geq 1} x_n z^{n-1} + \frac{\hbar q}{z} + \hbar\kappa \sum_{n\geq 1} n z^{-n-1} \partial_{x_n}.$$

A compact radial current smearing centered at z, disjoint from the inner support, has Cauchy transform $(w - z)^{-1}$. Its charge contraction is $\hbar q/z$, and its covariance with an inner period f is $-\hbar\kappa f'(z)$. These three terms give the formula. Contour integration extracts each mode. Changing the contour radius within the separated annulus is a cochain homotopy by Stokes' formula.

14.6 An incoming junction from the Gauss constraint

We now set $\hbar = \kappa = 1$ and retain a numerical charge q. Consider the incoming Wilson ray in complex abelian Chern–Simons theory

$$S = -\frac{1}{4\pi i} \int \mathcal{A} \wedge d\mathcal{A}, \qquad W_q = \exp\left(q \int_\gamma \mathcal{A}\right).$$

On an oriented boundary disk with coordinate w, orient the incoming line opposite to the outward normal. Variation of the normal multiplier then imposes

$$d_D a = -2\pi i q\, \delta_0, \qquad a = a_q + \alpha, \qquad a_q = -q\, \frac{dw}{w}, \tag{14.13}$$

where α is smooth and δ_0 has integral one. The singularity class is fixed in this statement. Gauge functions vanish on the boundary circle. There is no quotient by large gauge transformations and no added monopole sector.

The fluctuation complex is

$$\Omega^0_\partial(D) \xrightarrow{d} \Omega^1(D) \xrightarrow{d} \Omega^2(D)$$

in degrees -1, 0, 1. It retracts to $C^\infty(S^1)/\mathbb{C}$ in degree zero. One explicit contraction uses radial Poincaré maps h_1, h_2, with $dh_1 + h_2 d = 1$, $dh_2 = 1$, $h_1 h_2 = 0$, and an extension E of mean-zero boundary functions. Put

$$P\alpha = [\operatorname{tr} h_1 \alpha], \qquad Ib = dEb, \qquad H_0\alpha = h_1\alpha - E(P\alpha) - \operatorname{mean}(\operatorname{tr} h_1\alpha), \quad H_1 = h_2.$$

The Poincaré identities give $PI = 1$ and $dH + Hd = 1 - IP$. For a gauge function vanishing at the circle, the mean term removes the constant in $h_1 df = f - f(0)$. Thus the contraction respects the relative domain, not only the unrestricted disk complex.

The reduced form and currents are

$$\Omega_{\mathrm{red}} = \sum_{n>0} \frac{\delta J_{-n} \wedge \delta J_n}{n}, \qquad J_n = -\frac{1}{2\pi i} \oint w^n a, \qquad J_0 = q. \tag{14.14}$$

Indeed the smooth solutions are $a_q + df$, modulo zero-boundary gauge functions. Stokes' formula gives $\Omega(\delta_1 f, \delta_2 f) = (2\pi i)^{-1} \oint \delta_1 f \, d\delta_2 f$, which gives the nonzero modes in (14.14). The source fixes the zero mode.

To retain the constraints before reduction, choose Dirichlet functions ϵ_j orthonormal for the energy pairing and put

$$u_j = d\epsilon_j, \qquad v_j = 2\pi i * d\epsilon_j.$$

For $\Omega_D(\alpha, \beta) = (2\pi i)^{-1} \int_D \alpha \wedge \beta$, these obey $\Omega_D(u_j, v_k) = \delta_{jk}$ and zero equal-type pairings. Harmonic gradients are symplectically orthogonal to these modes by Green's formula. Solving the two Dirichlet Poisson equations gives the exact, coexact and harmonic splitting.

Write the constraint part of α as $\sum_j (s_j u_j + p_j v_j)$. Its moment is

$$-\frac{1}{2\pi i} \int_D \epsilon_j (d_D a + 2\pi i q \delta_0) = p_j.$$

The constraint $p_j = 0$ translates s_j. Keep finitely many gauge and boundary modes, quantize in the incoming polarization $x_n = J_{-n}$, s_j, and take the union of the resulting polynomial state spaces. With odd ghosts ν_j of degree one this defines

$$\mathcal{J}_q = \left(\mathbb{C}[x_1, x_2, \ldots, s_1, s_2, \ldots] \otimes \Lambda(\nu_1, \nu_2, \ldots), \quad Q_{\mathrm{BFV}} = \sum_j \nu_j \partial_{s_j} \right). \tag{14.15}$$

All polynomials are finite. The differential is the quantized commuting moment constraint $\sum_j \nu_j p_j$. No constant gauge ghost occurs for the chosen zero-boundary gauge group.

Theorem 14.6. *Evaluation at zero gauge coordinates and ghosts gives a quasi-isomorphism*

$$\rho_q : \mathcal{J}_q \longrightarrow F_q = \mathbb{C}[x_1, x_2, \ldots], \qquad \rho_q(f) = f(x, 0, 0).$$

Its inverse up to homotopy is the inclusion $\iota_q(f) = f(x)$. These maps and their contraction commute with

$$J_{-n} = x_n, \qquad J_n = n\partial_{x_n}, \qquad J_0 = q, \qquad T_q = \sum_{n>0} n x_{n+1}\partial_{x_n} + q x_1.$$

Thus they preserve every current field, the rooted residues and the identity action of the local vacuum.

Proof. Let N count total polynomial and exterior degree in s, ν, excluding x. With left odd derivatives set

$$H = \frac{1}{N}\sum_j s_j \partial_{\nu_j} \quad (N>0), \qquad H = 0 \quad (N=0).$$

The anticommutator of its numerator with Q_{BFV} is the gauge Euler derivation. Therefore $Q_{\mathrm{BFV}}H + HQ_{\mathrm{BFV}} = 1 - \iota_q\rho_q$. The sum is finite on each state and passes to the algebraic union of cutoffs. Every displayed current and T_q acts only on x, so it commutes with the contraction. Normal ordering and differentiation give all current fields. Their Laurent coefficients, including coefficients tested by multipliers before taking principal parts, give the rooted residues. The vacuum acts as the identity throughout. □

For the unnormalized disk function $\epsilon = 1 - r^2$, its energy is 2π. Thus $u = d\epsilon$, $v = i * d\epsilon$ form one canonical pair and give the differential $\nu\partial_s$. Removing the Wilson source leaves this contractible gauge complex unchanged but sets $J_0 = 0$. It then has zero-mode defect two on the unit of the charge-two recipient. Gauge contraction alone does not recover the missing charge.

14.7 The charged operation and its collision lattice

Place a second unit charge at separation $z \neq 0$. The source in (14.13) changes to $-dw/w - dw/(w-z)$. Contour integration gives the current Ward identity

$$J_m^{\mathrm{out}}\mathcal{V}(z) - \mathcal{V}(z)J_m^{\mathrm{in}} = z^m\mathcal{V}(z), \qquad m \in \mathbb{Z}.$$

In the incoming polarization, its oscillator solution with highest coefficient one is

$$K(z)f(x) = \exp\left(\sum_{n>0}\frac{x_n z^n}{n}\right) f(x_n - z^{-n}). \tag{14.16}$$

The creation exponential supplies the positive-mode Ward term, and the shift supplies the negative-mode term. On a polynomial input, the shift is finite. At a fixed output weight, only finitely many creation modes contribute. Values lie in the product of oscillator-weight spaces.

Translation fixes the remaining scalar. Direct substitution gives

$$T_2K - KT_1 = \partial_z K + z^{-1}K.$$

For a charged operation $h(z)K(z)$, translation covariance requires $h' = h/z$. Normal ordering removes each endpoint's self-contraction, and normalization at $z = 1$ gives

$$\mathcal{V}(z) = zK(z). \tag{14.17}$$

Insertion of negative current modes gives every descendant of the first input. Explicitly, if $I(u,z)$ denotes an endpoint operation, then

$$I(J_{-n}u, z) = \frac{\partial_z^{n-1}J^-(z)}{(n-1)!}I(u,z) + I(u,z)\frac{\partial_z^{n-1}J^+(z)}{(n-1)!}.$$

Here J^{-} contains negative modes, while J^{+} contains nonnegative modes. The indicated left and right operations commute for different n, so this gives a well-defined extension to every polynomial state. Its current commutator follows from $m\binom{m-1}{n-1} = n\binom{m}{n}$. Thus the operation retains descendants, not just the highest coefficient.

Let $\mathcal{L}^{-}$ be the operation line with normalized frame $k = K$. The connection on operations is $\nabla^{\mathrm{op}} I = dI - T_2 I\, dz + I T_1\, dz$, and hence

$$\nabla^{-} k = -\frac{dz}{z} k, \qquad \mathcal{L}^{+} = (\mathcal{L}^{-})^{\vee}, \qquad \nabla^{+} k^{\vee} = \frac{dz}{z} k^{\vee}.$$

The fused state recipient is $F_2 \otimes \mathcal{L}^{+}$. To verify its representing property on finite sums of scalar-charge Fock modules, apply the zero-mode Ward identity first: the output charge must be two. Removing the creation exponential and shift leaves a current-linear map of polynomial Fock modules. Such a map is determined by a scalar on the highest vector. The descendant recursion supplies every first-input state. The resulting operation space is $\mathcal{L}^{-} \otimes \mathrm{Hom}_{\mathrm{cur}}(F_2, W)$, which is the Hom space from the stated dual-line recipient, with its Hom connection.

The map ι_2 therefore gives a quasi-isomorphism of flat families

$$F_2 \otimes \mathcal{L}^{+} \longrightarrow \mathcal{J}_2 \otimes \mathcal{L}^{+}.$$

The physical Gaussian operation is carried to $\mathcal{V}$ by ρ_2. Counterclockwise half-transport is -1 on the state line and on its dual operation line. For general scalar charges a, b at level κ, the same calculation gives opposite connection forms $\pm(ab/\kappa)d\log z$. Their transports are dual, and need not be equal when ab/κ is nonintegral.

There is a specific extension through collision for the recipient. Extend the normalized frames over $\mathbb{C}[z]$, defining

$$\overline{\mathcal{E}}_F = \mathbb{C}[z] \otimes F_2\, k^{\vee}, \qquad \overline{\mathcal{E}}_{\mathrm{phys}} = \mathbb{C}[z] \otimes \mathcal{J}_2\, k^{\vee}.$$

The connection is logarithmic: $\nabla_{z\partial_z}(a(z)m) = (za'(z) + a(z))m$. The contraction is independent of z, so $1 \otimes \iota_2$ is a quasi-isomorphism over $\mathbb{C}[z]$, preserves this connection and has residue $+1$. Its derived fibre at zero is the quasi-isomorphism $F_2 \to \mathcal{J}_2$. The horizontal frame $z^{-1}k^{\vee}$ specifies a different lattice with residue zero. A meromorphic change of frame cannot be evaluated at zero to identify their chosen fibres.

The operation has a different limit. If f has oscillator weight w, then its output of weight d has the form

$$\mathrm{pr}_d\, \mathcal{V}(z) f = z^{1-w+d} v_d(f).$$

An annihilation removes weight a with factor z^{-a}, a creation adds weight c with factor z^{c}, and $d = w - a + c$. Consequently, separately in every output weight,

$$\lim_{z \to 0} z^{w-1} \mathcal{V}(z) f = f(-1, -1, \ldots)|2\rangle.$$

There is no common finite-power normalization for all inputs:

$$\langle 2|\mathcal{V}(z) x_1^{m} = (-1)^{m} z^{1-m}. \tag{14.18}$$

The first three values are z, -1, z^{-1}. Removing the negative shift would remove these poles and violate the J_{-1} Ward identity. The framed recipient fibre and the weight-dependent operation limit are therefore distinct constructions.

14.8 Regular Wilson jets and corrected cutoff maps

Return to a nonzero numerical κ, keeping $\hbar = 1$. For the incoming face use the pairing $b_- = -b$ and the linear inclusion $i_- = \zeta^{-1} i$. These changes preserve the boundary pairing: $b_-(i_- \otimes i_-) = \mu^s$. The ray functional then has no phase factor. For $\xi = \gamma + dt\,\beta$, define constant derivations

$$\begin{aligned} \ell_n(s\xi) &= \frac{1}{n!}\int_0^\infty \partial_z^n \beta_\eta^{(0)}(t,0)\,dt, \quad |\ell_n| = 1, \\ r_n(s\xi) &= \frac{1}{n!}\int_0^\infty \partial_z^n \beta_\Phi^{(0)}(t,0)\,dt, \quad |r_n| = 0, \\ e_n(s\xi) &= \frac{1}{n!}\partial_z^n \gamma_\Phi^{(0)}(0,0), \qquad\qquad |e_n| = 1. \end{aligned} \tag{14.19}$$

The superscript selects Dolbeault degree zero. Each derivation evaluates one variable of a smooth kernel. No product of point distributions is introduced.

Lemma 14.7. *The ray and endpoint derivations satisfy*

$$[d, \ell_n] = [d, e_n] = 0, \qquad [d, r_n] = \kappa(n+1)\ell_{n+1} - e_n. \tag{14.20}$$

They graded-commute with each other and with Δ_{b_-}.

Proof. On a linear test field, suspension and normal integration give

$$r_n(d(s\xi)) = e_n(s\xi) - \kappa(n+1)\ell_{n+1}(s\xi).$$

The endpoint term uses $\int_0^\infty \partial_t \gamma_\Phi^{(0)}\,dt = -\gamma_\Phi^{(0)}(0)$. The other term is the off-diagonal part of (14.4), including the derivative factorial. Since r_n has degree zero, its commutator with d is $-r_n d$ on a linear input. The relative condition kills the possible endpoint terms for $\ell_n d$ and $e_n d$. Constant contractions graded-commute. Continuity extends these calculations to smooth joint kernels. □

For a finite set $F \subset \mathbb{Z}_{>0}$, introduce the regular coefficient algebra

$$Y_F = \mathbb{C}[B_0, B_n : n \in F] \otimes \Lambda(c_{n-1} : n \in F), \quad Q_Y B_n = \kappa n c_{n-1}, \quad Q_Y B_0 = Q_Y c_j = 0.$$

The B's have degree zero and the c's degree one. Tensors with Y_F are algebraic in the coefficients. Operator expressions use graded left multiplication, while the retained module structure is the regular right action. Set

$$D_0 = d + \Delta_{b_-} + Q_Y + B_0\ell_0, \qquad L_F = \sum_{n\in F} \frac{B_n}{\kappa n} r_{n-1}.$$

The bare descendant source would add $\sum_{n\in F}(B_n\ell_n + c_{n-1}r_{n-1})$ to D_0. Its square is $\sum_{n\in F} c_{n-1}e_{n-1}$. For the first jet it takes the value c_0 on a suspended test function equal to one at the endpoint. This is a nonzero boundary defect.

Define the corrected differential by

$$D_F = D_0 + \sum_{n\in F}\left(B_n\ell_n + c_{n-1}r_{n-1} - \frac{B_n}{\kappa n}e_{n-1}\right). \tag{14.21}$$

The correction is forced by the endpoint transgression.

Proposition 14.8. *On regular smooth observables with coefficients in Y_F,*

$$D_F = e^{-L_F} D_0 e^{L_F}, \qquad D_F^2 = 0.$$

The conjugation is right Y_F-linear, fixes the unit and is finite on every input of finite field number.

Proof. The term $Q_Y B_n = \kappa n c_{n-1}$ cancels the interior part of $[d, c_{n-1} r_{n-1}]$. Its endpoint part is the bare defect above. Differentiating $-B_n e_{n-1}/(\kappa n)$ cancels that defect. More precisely,

$$[D_0, L_F] = D_F - D_0, \qquad [L_F, [D_0, L_F]] = 0.$$

The second identity follows from constant contractions and commuting even coefficients. The conjugation formula follows. Each r_n lowers field number by one and kills constants, so its exponential terminates and fixes $1 \otimes Y_F$. □

Let (i_0, p_0, H_0) be the scalar-source quantum collar contraction in the incoming phase, extended in the remaining coefficients. Then

$$I_F = i_0, \qquad P_F = p_0 e^{L_F}, \qquad H_F = e^{-L_F} H_0 e^{L_F}$$

contract $(O(U) \otimes Y_F, D_F)$ onto

$$\mathcal{A}_F(V) = \Big(\mathrm{Sym}_b(\Omega_c^{0,\bullet}(V)[1]) \otimes Y_F, d_1 + \Delta_{\mu^s} + B_0 \epsilon_0 + Q_Y\Big).$$

Indeed $L_F i_0 = 0$: the normal inclusion has no normal e_Φ-component, and its Wick contractions commute with L_F. Conjugation now transports every strong contraction identity. The maps preserve the unit, extension of compact tangential supports and products on tangentially disjoint regions. In particular all B_n, c_j continue to act regularly.

Adding a jet changes the differential of a raw smooth observable. For $F \subset G$, coefficient inclusion $\iota_{F,G}$ is therefore replaced by

$$j_{F,G} = e^{-(L_G - L_F)} \iota_{F,G}. \tag{14.22}$$

The commuting L's give $D_G j_{F,G} = j_{F,G} D_F$ and $j_{G,H} j_{F,G} = j_{F,H}$. The maps are injective. With $Y_\infty = \mathrm{colim}_F Y_F$, their direct limit has the explicit untwisting

$$\Upsilon : \mathrm{colim}_F (C_F, j_{F,G}) \xrightarrow{\sim} (O(U) \otimes Y_\infty, D_0), \qquad [f]_F \longmapsto e^{L_F} f. \tag{14.23}$$

For its inverse, choose a finite stage containing an input's coefficient indices and send a to $[e^{-L_F} a]_F$. Enlarging the stage gives the same class. This proves the isomorphism and defines the action of every coefficient. It is continuous for the final topology of finite coefficient, field-number and compact smooth stages.

The normal contractions commute with these transitions, since $P_G j_{F,G} = P_F$, $j_{F,G} I_F = I_G$ and $H_G j_{F,G} = j_{F,G} H_F$ after coefficient extension. They consequently retract the direct limit onto $\mathcal{A}_\infty$. No inverse-limit cohomology assertion is used.

This construction does not apply the literal infinite source to an unchanged smooth kernel. On a disk of radius less than one choose $\chi = 1$ near zero and an interior normal profile of integral one. The linear observable

$$u = s\left(dt\ \rho(t) \frac{\chi(z)}{1 - z} e_\Phi\right)$$

has $r_n(u) = 1$ for every n, and $\ell_n(u) = e_n(u) = 0$. For $F = \{1, \dots, K\}$,

$$D_F u = du + \sum_{j=0}^{K-1} c_j, \qquad e^{L_F} u = u + \sum_{n=1}^{K} \frac{B_n}{\kappa n}.$$

Neither infinite sum belongs to Y_∞. The representative of $[u]_\varnothing$ at stage F is instead $u - \sum_{n\in F} B_n/(\kappa n)$. Quotient cutoff maps also define an inverse-limit carrier, but that carrier admits coefficient series absent from the direct limit above.

14.9 The incoming cap and its state domain

Geometric rotations act on the incoming test fields by

$$R_\theta(s\alpha e_\eta) = s\rho_\theta^*\alpha e_\eta, \qquad R_\theta(s\alpha e_\Phi) = e^{i\theta} s\rho_\theta^*\alpha e_\Phi, \quad \rho_\theta(t,z) = (t, e^{i\theta}z),$$

and give B_n, c_{n-1} weight n, with B_0 of weight zero. Pullback includes the form factors. The extra factor on e_Φ makes the off-diagonal ∂_z equivariant. Direct evaluation shows that every source term, L_F, cutoff map and normal contraction commutes with this circle action.

Define $\mathcal{J}_{\mathrm{rot}}(D_{R_0})$ as the algebraic sum of the circle eigenspaces in the direct-limit source. This is a geometric condition before taking periods. Combining (14.12) with Ψ gives a continuous right Y_∞-linear quasi-isomorphism

$$\Pi_{\mathrm{rot}} : \mathcal{J}_{\mathrm{rot}}(D_{R_0}) \longrightarrow (\mathbb{C}[x_1, x_2, \dots] \otimes Y_\infty, Q_Y). \tag{14.24}$$

For clarity, the period at stage F is $\mathrm{Sym}(C)\exp(\partial_{P_C} + B_0\partial_\gamma)\,p_0 e^{L_F}$. All exponentials are finite on each source input.

To prove the assertion about finite circle spectrum, each coefficient element uses finitely many monomials of Y_∞. Separating their weights reduces the target to the finite-weight germ calculation above. The finite Fourier projector argument then restricts the full quasi-isomorphism. Separated current insertions give

$$J_{-n} = x_n, \qquad J_n = \kappa n \partial_{x_n}, \qquad J_0 = B_0.$$

The same Cauchy, covariance and charge contractions prove this on cochains, before passing to cohomology.

The state produced by an incoming cap is a further object. For fixed finite F, write the boundary superfields as $\Phi_\Sigma = \phi + d\bar z\,\phi_b$ and $\eta_\Sigma = c + d\bar z\, a_b$. The boundary jet is $x_n = \partial_z^{n-1}\phi(0)/(n-1)!$. The incoming boundary charge, with its endpoint, is

$$\begin{aligned} S_{\mathrm{in}}^F = -\frac{1}{\pi}\int c(\partial_{\bar z}\phi + \kappa\partial_z a_b)\,dx\,dy + B_0 c(0) \\ + \sum_{n\in F}\left(\frac{B_n\partial_z^n c(0)}{n!} + c_{n-1}x_n\right). \end{aligned} \tag{14.25}$$

The line coefficients are Casimirs for the boundary pairing. Their differential Q_Y is retained externally. The incoming form and potential have the opposite signs to those of the outward face. In the polarization with η_Σ as momentum directions,

$$\widehat{a}_b = \pi\frac{\delta}{\delta\phi}, \qquad \widehat{c} = \pi\frac{\delta}{\delta\phi_b}.$$

Derivatives are placed to the right.

The relative master identity can be checked on smeared fields. For a test function g, the induced derivation has

$$q_F\phi(g) = -\kappa c(\partial_z g), \qquad q_F c(g) = 0,$$
$$q_F a_b(g) = c(\partial_{\bar z} g) - \pi \sum_{n\in F} c_{n-1}\frac{\partial_z^{n-1} g(0)}{(n-1)!},$$
$$q_F \phi_b(g) = \phi(\partial_{\bar z} g) + \kappa a_b(\partial_z g) + \pi B_0 g(0) + \pi \sum_{n\in F} B_n \frac{\partial_z^n g(0)}{n!}.$$

Together with $q_F|_{Y_F} = Q_Y$, its square vanishes. On $\phi_b(g)$, the two bulk derivatives cancel, and the endpoint cancellation is

$$-\pi\kappa \sum_n c_{n-1}\frac{\partial_z^n g(0)}{(n-1)!} + \pi \sum_n \kappa n c_{n-1}\frac{\partial_z^n g(0)}{n!} = 0.$$

Omitting the line-ghost term $c_{n-1}x_n$ leaves the second sum.

Let $\mathcal{P}_{\mathrm{sm}}$ be polynomial functionals of ϕ, ϕ_b with compact smooth kernels. The state domain is

$$\mathcal{H}_{\mathrm{in}}^F = \operatorname*{colim}_{N\geq \max(F\cup\{1\})} (\mathcal{P}_{\mathrm{sm}} \otimes_{\mathbb{C}} Y_F[[x_1,\dots,x_N]]). \tag{14.26}$$

The series are completed by jet weight, and each weight coefficient is polynomial in Y_F. The tensor with smooth functionals is algebraic. Completion precedes the union over the finite set of jet variables.

Proposition 14.9. *On the domain* (14.26)*, the normal-ordered quantization* $Q_Y + \Omega_{\mathrm{in}}^F$ *of* (14.25) *squares to zero. On the subspace of junction-jet functionals alone it is*

$$D_G = Q_Y + \sum_{n\in F} x_n c_{n-1}. \tag{14.27}$$

Proof. At a fixed jet stage, differentiation of a series lowers weight by a bounded amount. Contraction with a smooth kernel takes finitely many derivatives at the junction, while its remaining contractions restrict smooth kernels to diagonals. The operator is therefore defined on the stated domain.

The charge has no ϕ_b-dependent coefficient. Among its terms, only the a_b-momentum can contract with a ϕ-coefficient. Thus a commutator has at most one such contraction, and equals the quantization of the relative Poisson bracket. The classical calculation above gives $Q_Y S_{\mathrm{in}}^F + \frac{1}{2}\{S_{\mathrm{in}}^F, S_{\mathrm{in}}^F\}_{\mathrm{in}} = 0$. There is no higher-contraction correction to its quantization. On a junction-jet functional, every momentum term has $\delta/\delta\phi_b$ on the right and vanishes. The endpoint line-ghost multiplication remains, giving (14.27). □

The larger completion in all jet variables together with smooth functionals fails this domain test. Choose g smooth and compactly supported, equal to $(1-z)^{-1}$ near zero. The sequence $u_M = \phi_b(g)\sum_{n=1}^M x_n$ converges in that larger jet-weight completion. The scalar part of its quantum image is

$$\Omega_{\mathrm{in}}^F u_M\Big|_{\phi=\phi_b=0} = \pi\kappa \sum_{n=1}^M \frac{\partial_z^n g(0)}{(n-1)!} = \pi\kappa\frac{M(M+1)}{2}.$$

It does not stabilize even at output weight zero. This is why (14.26) retains a finite jet stage for each mixed smooth state.

The incoming zero-momentum Lagrangian has constant flat prequantum section away from the junction. A normalized exponential cap is e^f, where f is a polynomial in finitely many x_n, has coefficients in Y_F^0, and has zero constant term in the jets. Equation (14.27) says that it is closed exactly when

$$f = -\sum_{n\in F} \frac{B_n x_n}{\kappa n} + h(x), \qquad h \in \mathbb{C}[B_0][x_1, x_2, \dots].$$

Indeed the coefficient of c_{n-1} is $\kappa n \partial_{B_n} f + x_n$. Rotation gives x_n weight $-n$, while B_n has weight n. Every nonconstant monomial of h has negative weight. The unique rotation-invariant cap in this declared exponential class is therefore

$$G_F = \exp\left(-\sum_{n\in F} \frac{B_n x_n}{\kappa n}\right). \tag{14.28}$$

The constant functional 1 is not closed when $F \neq \varnothing$.

Gluing a linear junction functional $\sum_n y_n x_n$ by the incoming BV pairing gives $\sum_n y_n e_{n-1}$ on regular source observables. For (14.28), this is exactly the endpoint correction in (14.21). The sign is essential: the opposite exponential leaves twice the bare endpoint defect. Nonlinear junction functionals leave point observables after contraction and require a larger observable domain.

On the degreewise oscillator-weight product, the all-jet expression

$$G = \exp\left(-\sum_{n>0} \frac{x_n B_n}{\kappa n}\right)$$

is well-defined: each weight coefficient is a finite polynomial in the line coefficients. Define its polynomial descendant image $\mathsf{P}_\kappa = G(\mathbb{C}[x_1, x_2, \dots] \otimes Y_\infty)$. Then

$$D_G(Gf) = GQ_Y f, \qquad (\kappa n \partial_{x_n} + B_n)(Gf) = G\kappa n \partial_{x_n} f.$$

Multiplication by G is a chain isomorphism onto this image, with inverse G^{-1}. Composing it with (14.24) gives the regular incoming cap map, sending 1 to G and retaining every line coefficient and separated current. It identifies the cap image. It does not identify the whole incoming state completion with a polynomial Fock module.

14.10 Formal states and their Laurent fields

The preceding completions control different variables. Completing the coefficient ring introduces a further distinction, even for the constant Heisenberg theory. Fix a numerical $\kappa \neq 0$, specialize the quantum parameter to one, and introduce an independent formal parameter ε. In this section put $R_\varepsilon = \mathbb{C}[[\varepsilon]]$ and $R_{\varepsilon,r} = R_\varepsilon/(\varepsilon^r)$. The parameter ε is neither a spacetime coordinate nor the quantum parameter used to define the current commutator.

Let $H = \mathbb{C}[x_1, x_2, \dots]$ be the polynomial vacuum space, with $J = x_1$ and translation $Tx_n = nx_{n+1}$. Its current has singular product $Y(J, z)J = \kappa z^{-2} +$ regular terms. Every finite-precision state space $H \otimes R_{\varepsilon,r}$ has the usual Laurent fields. The compatible inverse limit of the state spaces is $H[[\varepsilon]]$.

Proposition 14.10. *There is a state in $H[[\varepsilon]]$ whose field on J has no pole bound independent of coefficient precision. Consequently the finite-precision fields do not take all completed state pairs into $H[[\varepsilon]]((z))$.*

Proof. The state

$$a = \sum_{n\ge 0} \frac{(-\varepsilon)^n}{(n+1)!} T^n J$$

is well-defined coefficientwise. Translation covariance gives $Y(T^n J, z) = \partial_z^n Y(J, z)$. Therefore its singular value is

$$Y(a, z)J\big|_{\mathrm{sing}} = \kappa \sum_{n\ge 0} \varepsilon^n z^{-n-2}. \tag{14.29}$$

Modulo ε^r, the pole has order $r+1$. All its coefficients are forced by the compatible reductions. An element of the displayed uniform Laurent space has only finitely many negative powers of z, so it cannot have these reductions. □

This state is excluded by a fixed finite-weight restriction. Thus a finite-weight state core and the total formal state completion answer different questions. A natural field recipient for the latter is the restricted Laurent module

$$\mathcal{L}_\varepsilon(M) = \varprojlim_r (M/\varepsilon^r M)((z)) = \left\{ \sum_{j\in\mathbb{Z}} m_j z^j : \forall r\ \exists N_r \quad m_j \in \varepsilon^r M \text{ for } j < -N_r \right\}, \tag{14.30}$$

where M is separated and complete. The equality follows by lifting each compatible coefficient and applying the finite pole bound at each precision. Formula (14.29) belongs to this module. Coefficientwise Wick operations extend here because, at every precision, they are the original finite-precision operations. This is a formal recipient; it asserts no numerical convergence of the series.

The same distinction occurs in a separated Wilson junction. At $\kappa = 1$, take rank-one coefficient modules with zero differential and

$$B'_0 = 0, \quad B'_1 = \varepsilon, \quad B'_n = 0\ (n > 1), \qquad B''_0 = 1, \quad B''_n = 0\ (n > 0), \qquad c'_j = c''_j = 0.$$

They satisfy the line relation $[d, B_n] = n c_{n-1}$. The separated scalar twist is $s(z) = \varepsilon/z$; the two connections are

$$\nabla_0 = d, \qquad \nabla_1 = d - \varepsilon z^{-2} dz.$$

At precision r, the horizontal comparison and its inverse are

$$G_r(z) = \sum_{n=0}^{r-1} \frac{(-\varepsilon)^n}{n! z^n}, \qquad G_r(z)^{-1} = \sum_{n=0}^{r-1} \frac{\varepsilon^n}{n! z^n}. \tag{14.31}$$

Their product is one by the binomial identity, and $G_r' - \varepsilon z^{-2} G_r = 0$.

Over $R_\varepsilon((z))$ there is no nonzero horizontal comparison. If $g = \sum_{j\ge j_0} g_j z^j \neq 0$, with $g_{j_0} \neq 0$, then the coefficient of z^{j_0-2} in $g' - \varepsilon z^{-2} g$ is $-\varepsilon g_{j_0} \neq 0$. A current-linear endomorphism of the polynomial Fock factor is scalar: its value on the vacuum is killed by every positive mode, hence is a constant, and negative modes generate all other states. It follows that

$$\mathrm{Hom}^{\mathrm{flat,current}}_{R_\varepsilon((z))} = 0, \qquad \varprojlim_r \mathrm{Hom}^{\mathrm{flat,current}}_{R_{\varepsilon,r}((z))} = R_\varepsilon.$$

The comparison $e^{-\varepsilon/z}$ does exist in (14.30). These are comparisons of separated flat current families. Extending them through coincidence requires the supported collision operation as additional data.

There is a useful sufficient bound. If all twist coefficients lie in an ideal whose m-th power is zero, and their largest inverse-separation degree is k, the exponential has pole order at most $k(m-1)$. The exponent must be uniform. In (14.31), the nilpotence exponent grows with r, and so does the pole order.

14.11 Normal support after coefficient completion

The analogous support issue is visible on one normal coordinate u. For this section abbreviate $R = R_\varepsilon$, $R_r = R_{\varepsilon,r}$, and use right modules over the Weyl algebra

$$D_R = R\langle u, \partial_u\rangle/(\partial_u u - u\partial_u - 1).$$

The parameter ε commutes with both operators. The principal-part module and its density action are

$$P = R[u, u^{-1}]/R[u] = \bigoplus_{m\geq 0} R\beta_m, \qquad \beta_0 u = 0, \quad \beta_m u = \beta_{m-1}\ (m > 0), \quad \beta_m \partial_u = (m+1)\beta_{m+1}. \tag{14.32}$$

Right differentiation is minus ordinary differentiation of the density. In particular $P = D_R/uD_R$, with right-module resolution

$$0 \longrightarrow D_R \xrightarrow{L_u} D_R \longrightarrow P \longrightarrow 0.$$

Here L_u is left multiplication by u, and is right linear.

The coefficient completion is the restricted mode product

$$\widehat{P} = \left\{ \sum_{m\geq 0} a_m \beta_m : a_m \in R, \quad \forall r \text{ all but finitely many } a_m \in \varepsilon^r R \right\}. \tag{14.33}$$

It is derived complete because the reductions have surjective transitions and P is scalar free. Its ordinary normal support has changed.

Proposition 14.11. *For $Q = \widehat{P}/P$, both u and ε act invertibly, and*

$$Q \neq 0, \qquad \widehat{P}[u^{-1}] = Q, \qquad Q \otimes^{\mathbb{L}}_R R_r = 0.$$

Thus P_r has no lift that is both derived complete with those compatible reductions and ordinarily supported at $u = 0$.

Proof. Lowering the modes is surjective on $\widehat{P}$. An element killed by u^N has no modes above β_{N-1}, so the union of these kernels is exactly P. Localization is therefore the quotient Q, on which u is invertible. The element $p = \sum_{m\geq 0} \varepsilon^m \beta_m$ has nonzero image there and satisfies $pu = \varepsilon p$.

Subtracting a finite representative modulo ε and dividing its remaining coefficients by ε proves surjectivity of ε on Q. If εx has finite mode support, so does x, since R has no ε-torsion. This proves injectivity. Derived reduction is the cone of multiplication by ε^r, and hence vanishes on Q. Finally, a derived complete lift of the compatible tower is its derived inverse limit $\widehat{P}$, whose localization is nonzero. □

The scalar vacuum R still exists. The proposition concerns its normal coefficient, and the perfect test D_R detects the nonzero open quotient by $\mathrm{Hom}_{D_R}(D_R, Q) = Q$.

For a complex M, write

$$M_r = M \otimes^{\mathbb{L}}_R R_r, \qquad \Lambda M = R\varprojlim_r M_r.$$

Call M *derived adically supported* at $u = 0$ when

$$M_r[u^{-1}] \simeq 0 \quad \text{for every } r. \tag{14.34}$$

Open restriction is flat localization of the underlying $R[u]$-module with its induced differential action. It does not treat u as central in D_R. Every cohomology module of every derived reduction is tested in (14.34). For a scalar-flat module this is equivalent to

$$\forall v\, \forall r\, \exists N : v u^N \in \varepsilon^r M.$$

For a general module the displayed condition controls only ordinary quotients, and misses the negative cohomology of derived reduction.

Here is an explicit instance of that difference. Let M consist of families in $\prod_{n\geq 1, m\geq 0} (R/\varepsilon^n)\beta_m^{(n)}$ whose coefficients are nonzero modulo ε^r at only finitely many pairs (n, m), for each r. Give each summand the normal action (14.32). Division of coefficients shows that

$$M/\varepsilon^r M = \bigoplus_{n\geq 1} P/\varepsilon^{\min(n,r)} P.$$

These quotients are supported, and their inverse limit is M. The negative cohomology of M_r is $T_r = M[\varepsilon^r]$, with transition multiplication by ε. Its inverse limit is zero by separatedness. Moreover, for $y_r \in T_r$, the convergent series $x_r = \sum_{j\geq 0} \varepsilon^j y_{r+j}$ belongs to T_r and satisfies $x_r - \varepsilon x_{r+1} = y_r$. Thus $\varprojlim^1 T_r = 0$, proving derived completeness of M.

Nevertheless, in $N = M[u^{-1}]$ the elements

$$w_r = \sum_{n\geq r} \varepsilon^{n-r} \beta_{n-1}^{(n)} \tag{14.35}$$

give nonzero compatible classes in $N[\varepsilon^r]$. Indeed $\varepsilon^r w_r = 0$, and $w_r - \varepsilon w_{r+1} = \beta_{r-1}^{(r)}$ vanishes after localization. No power of u kills w_r: choose an index $n \geq r$ larger than that power. Hence the compatible class in $\varprojlim_r N[\varepsilon^r]$ is nonzero, and $H^{-1}(\Lambda N) \neq 0$. In fact the Milnor exact sequence in degree -1 identifies these two groups, since the reductions have no degree -2 cohomology. This proves that the complete module M fails (14.34) despite its supported ordinary quotients.

14.12 Reconstruction on the normal Weyl carrier

The derived support condition has a positive reconstruction theorem. It applies to normal coefficients; identifying them with all native fields or their physical fusion requires additional comparisons. Let $\mathcal{G}_\varepsilon$ be the category of derived complete scalar complexes, and let C_ε be the category of derived complete right D_R-complexes satisfying (14.34).

First, completion preserves every derived reduction:

$$(\Lambda X)_r \simeq X_r. \tag{14.36}$$

To see this without a boundedness assumption, tensor the inverse limit with the perfect complex $[R \xrightarrow{\varepsilon^r} R]$. For $n \geq r$, the resulting two-term model of $X_r \otimes_R^{\mathbb{L}} R_n$ has zero differential between its two copies of X_r. The transitions are multiplication by ε on the negative copy and identity on the other copy. The first tower has zero homotopy limit: its shift on the product has r-th power zero, so $1 -$ shift has a finite geometric-series inverse. The second tower is constant. This proves (14.36) with its natural map. The fibre of completion is an inverse limit of multiplication by ε, hence has invertible ε. Maps from such an object to a complete object vanish, by computing them into its reductions. Consequently Λ is an exact reflection onto complete complexes.

Localization commutes with scalar reduction. Therefore (14.34) can also be tested by $M_1[u^{-1}] \simeq 0$, or by $\Lambda(M[u^{-1}]) \simeq 0$. Indeed the first condition says that ε is invertible on the open restriction, which is precisely the kernel of completion.

Theorem 14.12. *The functors*

$$I(E) = \Lambda(P \otimes^{\mathbb{L}}_R E), \qquad K(M) = R\operatorname{Hom}_{D_R}(P, M)$$

give inverse equivalences $\mathcal{G}_\varepsilon \simeq \mathcal{C}_\varepsilon$. No scalar flatness or cohomological boundedness is required.

Proof. Over a commutative complex algebra B, let V be a right Weyl module with locally nilpotent u. Define the finite Taylor projector

$$\pi(v) = \sum_{j \geq 0} \frac{(-1)^j}{j!} (vu^j)\partial_u^j.$$

The identity $\partial_u^j u = u\partial_u^j + j\partial_u^{j-1}$ gives $\pi(v)u = 0$ by consecutive cancellation. The binomial identity gives the inverse formula

$$v = \sum_{m \geq 0} \frac{\pi(vu^m)\partial_u^m}{m!}.$$

Thus $P_B \otimes_B \ker u \to V$, sending $\beta_m \otimes e$ to $e\partial_u^m/m!$, is surjective. Applying the largest power of u in a finite relation extracts its highest coefficient, so the map is injective. It also proves surjectivity of u on V.

The resolution of P_B computes

$$K_B(V) = [V \xrightarrow{u} V]_{[0,1]}.$$

This is a complex of scalar modules; its differential is not claimed to be right Weyl linear. For a complex with supported cohomology, the long exact sequence identifies $H^q K_B(V) = \ker(u : H^q V \to H^q V)$, since u is surjective on that cohomology. Scalar freeness of P_B now proves that the evaluation counit is an isomorphism on all cohomology groups. The unit is an equivalence because lowering on P_B is split surjective with kernel $B\beta_0$.

The finite free resolution makes K commute with derived scalar reduction and completion. Tensor–Hom adjunction and the reflection property give $I \dashv K$ on complete objects. At precision r, the unit and counit are the equivalences just proved over $B = R_r$. Their limits give the assertion. □

For example, this equivalence supplies an intrinsic supported product. It need not be the ambient product. The latter fails already for $W = \prod_{i \geq 0} \widehat{P}$. It is scalar flat and complete, and $W_1 = \prod_i P_1$. The element $e = (\beta_0, \beta_1, \ldots)$ has a nonzero $i = N$ coordinate after multiplication by u^N, for every N. Thus W_1 is not supported. The intrinsic product is instead $I(\prod_i R)$: at each coefficient precision it imposes a mode bound uniform over i. The example concerns all indices; every finite product is supported.

There is also an actual geometric operation compatible with the equivalence. Let $a : \mathbb{A}^1_x \times \mathbb{A}^1_y \to \mathbb{A}^1_t$ be addition, $t = x + y$, and define

$$M *_+ N = \Lambda a_+(M \boxtimes^{\mathbb{L}}_R N),$$

where a_+ is right differential-module direct image, computed by the relative de Rham complex. For complete scalar complexes put $E\widehat{\otimes}_R F = \Lambda(E \otimes^{\mathbb{L}}_R F)$.

Proposition 14.13. *There is a natural equivalence*

$$I(E) *_+ I(F) \simeq I(E\widehat{\otimes}_R F).$$

It is compatible with associativity, symmetry and the convolution unit $\widehat{P}$. Relative convolution over an algebra $I(B)$ corresponds to $I(\Lambda(E \otimes^{\mathbb{L}}_B F))$, with its specified right and left module actions.

Proof. At precision r, change variables to $t = x + y$, $s = y$. Since $\partial_x = \partial_t$ and $\partial_y = \partial_t + \partial_s$, the density modes transform by

$$\beta_i(x)\beta_j(y) \longmapsto \sum_{k=0}^{j} \binom{i+k}{k} \beta_{i+k}(t)\beta_{j-k}(s).$$

The relative de Rham differential on P_s, in degrees $-1, 0$, is ∂_s; it raises the mode by one with coefficient $m + 1$. It is split injective over R_r, with cokernel $R_r\beta_0$. Projection to that cokernel sends the displayed expression to $\binom{i+j}{i}\beta_{i+j}(t)$. This remains a contraction after tensoring with any derived scalar ground. The three-input coefficients agree by the multinomial identity. Addition and iterated relative de Rham direct image provide the coherent comparisons, symmetry and unit.

Scalar reduction commutes with this construction. Equation (14.36) then gives the completed comparison. Finally its two-sided bar has the actual action maps and terms $E\widehat{\otimes}B^{\widehat{\otimes}n}\widehat{\otimes}F$. Its realization is the completed relative tensor in the statement, because completion is a monoidal localization: its kernel of ε-invertible objects is a tensor ideal. □

This additive normal convolution retains the coordinate t. The balanced Weyl tensor consumes the normal coordinate instead:

$$R\operatorname{Hom}_{D_R}(P, P) = R, \qquad P \otimes^{\mathbb{L}}_{D_R} P^t = R[1].$$

Here transpose sends u to u and ∂_u to $-\partial_u$. The same lowering operator computes both formulas, in degrees $0, 1$ for Hom and $-1, 0$ for tensor. Neither operation alone is the physical boundary fusion constructed from separated insertions and their collision maps.

Nor is scalar flatness preserved by every relative tensor. For $B = \widehat{R[t]}^{(\varepsilon)}$, take the two rank-one modules with $t = 0$ and $t = \varepsilon$. The free resolution for the first gives their relative tensor as $[R \xrightarrow{\varepsilon} R]_{[-1,0]} \simeq R/(\varepsilon)$. At precision r, both its kernel and cokernel are nonzero. Derived normal support retains both; ordinary scalar reduction loses one of them.

14.13 The interacting comparison

Return to a nonconstant Hamiltonian operator H. The classical master identity constructs its local bulk action. The Gaussian calculation constructs a quantum action on supported boundary observables, a charged junction and an incoming cap on their declared domains. Extending all of them to the interacting theory is a coupled problem.

To formulate its perturbative obstruction, suppose a regulator and a local subtraction prescription give a BV complex of bulk, boundary and defect functionals, with the boundary terms included in its differential. Let S_0 be its classical solution and let $s = Q + \{S_0, -\}$. Write a proposed quantum action as $S = S_0 + \sum_{n\geq 1} \hbar^n S_n$. In a regularized BV model its equation is

$$QS + \tfrac{1}{2}\{S, S\} + \hbar\Delta S = 0. \tag{14.37}$$

At order n, the equation for the degree-zero correction is

$$sS_n = -O_n, \qquad O_n = \Delta S_{n-1} + \tfrac{1}{2}\sum_{i=1}^{n-1}\{S_i, S_{n-i}\}.$$

The BV identities and the lower-order equations give $sO_n = 0$, by taking the order-n term of the identity that the differential twisted by S annihilates its curvature. Thus the obstruction is a class in degree one

of this joint local complex. A class in the bulk-only complex does not test endpoint counterterms or the boundary Ward identities.

The required local prescription must also respect restrictions to nested regions. Its annulus amplitudes must sew with the chosen cap, and collision extensions must obey the same current and defect Ward identities. The forest contractions of Chapter 7 provide explicit Gaussian homotopies. They do not solve the interacting master equation on all these supports.

An algebraic Hochschild action poses a different comparison. To identify it with the physical action, one needs a map from the renormalized bulk observables to the coherent operations on the boundary category, preserving the unit, translation, collision maps and their homotopies. If a centre represents those operations, the map must factor through that representing object. Equivalence further requires that the chosen boundary objects detect and reconstruct the bulk. Neither a shared current bracket nor a solution of the master equation supplies these maps by itself.

The Gaussian normal contraction shows why the operation is the relevant datum: a field removed by linear projection can still act through its contraction with a boundary insertion. Formal support and Laurent completion then specify where that action lives. The general interacting bulk-to-boundary evaluator, and its identification with a universal holomorphic centre, remain open beyond the constructions proved here.

Chapter 15

Marked reconstruction, descent and sewing

A boundary junction can detect some boundary objects and miss others. Its endomorphism algebra determines the objects generated by the junction. Recovering a larger category requires a generation theorem; recovering local fields also requires mapping objects that respect quasi-isomorphisms and geometric composition. These requirements have different tests.

The simplest test uses the complex

$$P_0 = \big(k[x]\theta \xrightarrow{x} k[x]\big), \qquad |\theta| = -1.$$

It reconstructs the sector supported at $x = 0$, but it cannot detect the nonzero module $k[x, x^{-1}]$. For the Ramond junction, the failure persists without localization: nonzero polynomial boundary modules with nonnegative energy have zero derived Hom from the junction. A local residue gives a second obstruction. Its cone belongs to the stable category generated by the junction but admits no compatible action of the original differential graded quartet.

We first establish the positive reconstruction statement and retain its marking. We then prove these obstructions on their full coefficient categories. The final sections specify the comparisons needed for collision descent, holomorphic globalization and sewing with a prepared state. Throughout, complexes are cohomological, $M[1]^n = M^{n+1}$, and k is a field of characteristic zero.

15.1 The sector detected by one object

Put $A = k[x]$ and regard P_0 as a left A-module complex. Its augmentation to $A/(x)$ is a quasi-isomorphism. The derived category $D(A)$ is obtained from complexes of left A-modules by inverting quasi-isomorphisms. Define its x-torsion sector by

$$D_{x\text{-tor}}(A) = \{M \in D(A) : A[x^{-1}] \otimes^{\mathbb{L}}_{A} M \simeq 0\}. \tag{15.1}$$

This condition is imposed on the derived localization. It does not impose x-adic completion. An object is compact when its derived Hom functor preserves coproducts. It generates a stable sector when the smallest full stable subcategory containing it and closed under coproducts is that sector.

Proposition 15.1. *The object P_0 is a compact generator of $D_{x\text{-tor}}(A)$. It does not generate $D(A)$.*

Proof. Let $\mathrm{Loc}(P_0)$ be the smallest full stable subcategory of $D(A)$ containing P_0 and closed under coproducts. Localization preserves triangles and coproducts and kills P_0, so $\mathrm{Loc}(P_0) \subset D_{x\text{-tor}}(A)$. The localization fibre

$$\Gamma_x A = [A \longrightarrow A[x^{-1}]]$$

has its terms in degrees 0, 1. It is quasi-isomorphic to $(A[x^{-1}]/A)[-1]$, and

$$A[x^{-1}]/A = \varinjlim_{n} (A/x^n, \times x).$$

Each A/x^n is a finite extension of copies of A/x. After tensoring with M, each successive factor is a complex of k-vector spaces with the quotient A-action. Such a complex belongs to $\mathrm{Loc}(A/x)$: over a field it is a direct sum of shifts of its cohomology. Consequently $\Gamma_x A \otimes^{\mathbb{L}}_A M$ belongs to $\mathrm{Loc}(P_o)$. The localization triangle identifies this object with M whenever M satisfies (15.1). This proves equality of the two subcategories. The two-term finite free model makes P_o compact.

On the other hand,

$$\mathrm{RHom}_A(P_o, A[x^{-1}]) = [A[x^{-1}] \xrightarrow{-x} A[x^{-1}]]$$

is contracted by multiplication by $-x^{-1}$. Its nonzero target is therefore invisible to P_o, which excludes generation of $D(A)$. □

The general statement uses the same evaluation map. Let A now be a dg algebra over a commutative dg ring R, and let $P \in D(A)$. Choose a cofibrant representative and write

$$E = \mathrm{REnd}_A(P).$$

We use right E-modules. Precomposition makes $\mathrm{RHom}_A(P, M)$ a right E-module; the evaluation action makes P a left E-module commuting, with the graded sign, with its left A-action. The adjunction is

$$G(N) = N \otimes^{\mathbb{L}}_E P, \qquad F(M) = \mathrm{RHom}_A(P, M), \qquad G \dashv F. \tag{15.2}$$

Let $\mathrm{thick}(P)$ denote closure under finite extensions, shifts and retracts. Let $\mathrm{Perf}(E)$ denote the corresponding closure of the free right module E.

Theorem 15.2 (Reconstruction of the generated sector)**.** *The unit and counit of* (15.2) *give inverse equivalences*

$$\mathrm{thick}(P) \simeq \mathrm{Perf}(E).$$

If P is compact in $D(A)$, they extend to

$$\mathrm{Loc}(P) \simeq D(E^{\mathrm{op}}).$$

Here $D(E^{\mathrm{op}})$ means the category of right E-modules. The image of P is the free module E; all specified arrows between objects of the sector are carried with their full derived mapping spaces.

Proof. There are canonical identifications $F(P) = E$ and $G(E) = P$. The adjunction unit on E and counit on P are identities under these identifications. Both functors are exact and preserve retracts. The subcategories on which their unit and counit are equivalences are therefore closed under finite extensions, shifts and retracts. They contain the respective generators, proving the first assertion and full faithfulness.

If P is compact, F also preserves coproducts; G does so as a left adjoint. The same argument now applies to localizing subcategories. The free right module E generates its whole derived category: every module admits a semifree resolution formed from coproducts of shifts of E, with successive free attachments. Thus the unit is an equivalence on every right E-module. The image of G is $\mathrm{Loc}(P)$, where the counit is an equivalence. The equivalence is enriched, since the adjunction was formed from derived tensor and Hom; it therefore carries arrows and all their homotopies. □

This is the module form of derived Morita theory [60]. Its domain is part of the theorem. If $N \neq 0$ and $F(N) = 0$, the counit at N is $0 \to N$. Such an N is a *right orthogonal* to P. It excludes reconstruction on any category containing N, without affecting reconstruction on $\mathrm{thick}(P)$.

15.1.1 The unit and the coefficient action

An equivalence of module categories identifies their centres when it preserves the specified coefficient action. For ordinary dg algebras over R, the centre is the Hochschild cochain E_2-algebra

$$Z_{\mathrm{Mor},R}(A) = \mathrm{RHom}_{A^e}(A, A), \qquad A^e = A \otimes^{\mathbb{L}}_R A^{\mathrm{op}}.$$

Its E_2 operations arise from composition and the coherent interchange of endomorphisms of the identity functor.

Proposition 15.3. *An R-linear derived Morita equivalence identifies these centres and their actions on modules. If it is represented by a B–A bimodule U with inverse A–B bimodule V, let K be an R-complex. The image of a bimodule map $z : K \otimes^{\mathbb{L}}_R A \to A$ is*

$$K \otimes^{\mathbb{L}}_R B \simeq U \otimes^{\mathbb{L}}_A (K \otimes^{\mathbb{L}}_R A) \otimes^{\mathbb{L}}_A V \xrightarrow{1 \otimes z \otimes 1} U \otimes^{\mathbb{L}}_A V \simeq B.$$

The comparison uses both inverse bimodule equivalences and their triangle homotopies.

Proof. A colimit-preserving R-linear functor on module categories is determined by its value on the free module. Applying it to the free bar resolution identifies it with derived tensor by that value, including the right algebra action and all natural transformations. In particular, identity-functor endomorphisms are diagonal-bimodule endomorphisms. Conjugation by the equivalence and its inverse preserves composition, the identity functor and the interchange of transformations. It therefore preserves the centre's E_2 structure. The displayed formula is this conjugation on evaluation; tensoring it with a module proves compatibility with its action. □

For example, take $B = \mathrm{Mat}_2(k)$, $U = k^2$ and $V = (k^2)^*$. The outer product and evaluation give

$$U \otimes_k V \xrightarrow{\sim} B, \qquad V \otimes_B U \xrightarrow{\sim} k.$$

The first map sends the four standard tensors to the matrix units. In the second tensor product, balancing kills the off-diagonal tensors and identifies the two diagonal tensors. The inverse images of the units are $\sum_i e_i \otimes e_i^*$ and $[e_1^* \otimes e_1]$. The triangle composites are $\sum_i e_i e_i^*(u) = u$ and $\sum_i v(e_i) e_i^* = v$. There is nevertheless no unital algebra map $B \to k$: both square-zero off-diagonal units would map to zero, and so would their products and their sum 1_B. Morita units do not supply scalar augmentations.

15.2 Reconstruction with a marked arrow

An action on an object and an action preserving a chosen preparation are different data. Fix a closed arrow $\nu : Q \to P$ of A-bimodules, with cofibrant endpoints. The marking is the entire arrow, including its domain. Write Hom for the bimodule mapping complex in this subsection. A concrete model for its derived end algebra is

$$E_\nu = \mathrm{End}(Q) \oplus \mathrm{End}(P) \oplus \mathrm{Hom}(Q, P)[-1]. \tag{15.3}$$

For a homogeneous triple of degree n, its third component has degree $n - 1$. Set

$$d(a, b, h) = (da, db, \nu a - b\nu - dh), \tag{15.4}$$

$$(a, b, h)(a', b', h') = (aa', bb', ha' + (-1)^n bh'), \tag{15.5}$$

$$1 = (1_Q, 1_P, 0).$$

A closed degree-zero triple records two endomorphisms and the homotopy making the marked square commute.

Proposition 15.4. *Equations* (15.4) *and* (15.5) *define a unital dg algebra model of the endomorphisms of ν in the derived arrow category.*

Proof. Put $F(a,b) = \nu a - b\nu$. The differential is the homotopy-fibre differential of F, so it squares to zero. The identity

$$F(aa', bb') = F(a,b)a' + bF(a',b')$$

and the Hom differential's Leibniz rule verify the dg Leibniz identity with the sign in (15.5). The third component of either triple product, for degrees n, m, is

$$ha'a'' + (-1)^n bh'a'' + (-1)^{n+m} bb'h'',$$

which proves associativity.

To identify the multiplication in the derived category, replace ν by the cofibrant arrow $i : Q \to T$, where

$$T = P \oplus Q \oplus Q[1], \qquad i(q) = (0, q, 0),$$
$$d_T(p,q,s) = (d_P p - \nu s, d_Q q + s, -d_Q s).$$

The source Q and the cokernel $\mathrm{Cone}(-\nu)$ are cofibrant, which verifies the assertion about the arrow. The map $\pi : T \to P$, $\pi(p,q,s) = p + \nu q$, is a quasi-isomorphism with contractible kernel and $\pi i = \nu$. For a degree-n triple, define

$$\Phi(a,b,h) = (a, B), \qquad B(p,q,s) = (bp + (-1)^n hs,\ aq,\ (-1)^n as).$$

Substitution into the differentials and composition proves that Φ is a strict dg algebra map to $\mathrm{End}(i)$. Postcomposition with $(1, \pi)$ identifies its source complex with $\mathrm{Hom}(i, \nu)$: the three free coefficients are b, νa, $(-1)^n h$. Since i is cofibrant, $\mathrm{End}(i) \to \mathrm{Hom}(i, \nu)$ is a quasi-isomorphism. Thus Φ is a quasi-isomorphism, as required. □

To prescribe the scalar action on the domain of the marking, form the homotopy pullback of unital E_1-algebras

$$Z_\nu = E_\nu \times^h_{\mathrm{REnd}_{A^e}(Q)} R. \tag{15.6}$$

The map from R is scalar multiplication; the projection $Z_\nu \to R$ is an augmentation.

Theorem 15.5. *Let B be an independently specified augmented unital E_1-algebra over R, with augmentation ε. Then*

$$\mathrm{Map}_{\mathrm{Alg}_R/R}(B, Z_\nu) \simeq \mathrm{Act}^{E_1}_\varepsilon(B; \nu).$$

The right side is the full derived space of actions on the arrow ν, together with an identification of the action on Q with $\varepsilon\, 1_Q$. It retains the marked square's homotopy and every higher identification.

Proof. Maps from a cofibrant resolution of B to the derived end algebra of an object classify homotopy coherent associative actions on that object. Applying this construction in the arrow category gives $\mathrm{Map}(B, E_\nu)$. Mapping into the homotopy pullback (15.6), and taking the fibre over the prescribed augmentation, gives

$$\mathrm{Map}(B, E_\nu) \times^h_{\mathrm{Map}(B, \mathrm{REnd}(Q))} \{\varepsilon\, 1_Q\}.$$

This is exactly the space in the statement. The construction uses derived mapping spaces, so it also identifies their homotopies in every degree. □

15.2.1 Two actions on the same marked object

The Hochschild centre $\mathcal{H}_A = \mathrm{REnd}_{A^e}(A)$ has two natural unital E_1 maps to $\mathrm{REnd}_{A^e}(P)$:

$$\lambda_P(z) = z \otimes^{\mathbb{L}}_A 1_P, \qquad \rho_P(z) = 1_P \otimes^{\mathbb{L}}_A z. \tag{15.7}$$

They use the two unit identifications with P. Derived tensor is an enriched functor, so it preserves composition and all homotopies. It acts naturally on a supplied arrow ν, giving lifts to E_ν. A lift through Z_ν additionally requires the prescribed scalar action on Q. No E_2 structure on the end algebra of an arbitrary object is implied by these E_1 maps.

Example 15.6. Let $A = k[x]$, $P = A \oplus A^e$, and let $\nu : A \to P$ include the diagonal summand. The class $x \in HH^0(A)$ acts identically by λ_P and ρ_P on that summand. On $A^e = k[x_L, x_R]$, their difference sends 1 to $x_L - x_R \neq 0$. Since P is concentrated in degree zero, this is a nonzero class in $H^0\,\mathrm{REnd}_{A^e}(P)$. Hence the two algebra maps are not homotopic, even though the marking is split.

The example also explains why the comparison paths in Chapter 5 cannot be discarded. An equivalence preserving its polynomial-root object must preserve the full Massey coset $-1 + pB$. An equivalence over the level ring must preserve the arrow to the operator complex, its nonsplit cohomology extension and its derived special fibre. Retaining only an ordinary scalar centre loses these data. A fixed-mode gauge equivalence must also retain its residual automorphisms; equality of orbit sets is weaker.

15.3 The Ramond junction and its boundary categories

We now test generation on a junction defined by its fields. Let $R = \mathbb{C}[\![q]\!]$ and $S_\ell = R/(q^\ell)$, $\ell \geq 1$. All states over S_ℓ are algebraic. Completion means $V[\![q]\!]$, with a finite algebraic state at each coefficient; no product over oscillator energy or internal shifts is taken.

Fix a Ramond coordinate and the normalized coefficient lines. The boundary algebras have zero differential and regular self-products:

$$A_0 = S_\ell[\partial^j \gamma : j \geq 0], \qquad A_1 = \Lambda_{S_\ell}(\partial^j C : j \geq 0).$$

The physical odd field is $c = qC$. The normalized action of C is specified independently, without inverting q. The junction state complex is

$$P_\ell = S_\ell[x_i, y_n] \otimes \Lambda(\theta_i, \eta_n), \qquad D = \sum_{i \geq 0} x_i \partial_{\theta_i} - \sum_{n \geq 1} \eta_n \partial_{y_n}. \tag{15.8}$$

The degrees of x, y, θ, η are 0, 0, −1, 1. Exterior derivatives are left derivatives. Its two internal gradings W, E are

$$(W,E)(x_i) = (W,E)(\theta_i) = (1, 2i+1), \qquad (W,E)(y_n) = (W,E)(\eta_n) = (-1, 2n-1).$$

Here E is the specified oscillator grading.

The independent boundary actions are

$$X_i = x_i, \quad G_n = -\partial_{y_n}, \qquad K_i = \partial_{\theta_i}, \quad H_n = \eta_n.$$

The even modes commute, and the odd modes supercommute. Their generating fields are

$$\Gamma(z) = \sum_{i \geq 0} X_i z^{i-1/2} + \sum_{n \geq 1} G_n z^{-n-1/2}, \tag{15.9}$$

$$C(z) = \sum_{n \geq 1} H_n z^{n-1/2} + \sum_{i \geq 0} K_i z^{-i-1/2}. \tag{15.10}$$

The even action is originally a right action. Its commutativity allows left-module notation here; no odd opposite-algebra identification is made.

Define $C_{0,\ell}$ to consist of internally bigraded dg modules with this unital even field action such that, for every vector v, there are N, h with

$$(G_n : n > N)v = 0, \qquad (G_1, \dots, G_N)^h v = 0. \tag{15.11}$$

Equivalently, $J_{N,h}v = 0$, where $J_{N,h} = (G_n : n > N) + (G_1, \dots, G_N)^h$. Define $C_{1,\ell}$ using the odd field action and the condition $K_i v = 0$ for sufficiently large i, for each v. A finite exterior family is already jointly nilpotent. All derivative, product and regular substitution identities are part of these field actions.

The restricted enriched mapping complex is the direct sum over internal shifts,

$$\underline{\mathrm{Hom}}^{\mathrm{res}}(M, N) = \bigoplus_{u,v\in\mathbb{Z}} \mathrm{Hom}_{\mathrm{fields,bigr}}(M, N\langle u, v\rangle).$$

The shift means that a map raises W, E by u, v. A degree-r odd-boundary map obeys $f(am) = (-1)^{r|a|} af(m)$; its differential is $d_N f - (-1)^r f d_M$. Derive these mapping complexes at each S_ℓ, then take $\mathbf{R}\lim_{\longleftarrow \ell}$. Weak equivalences are underlying quasi-isomorphisms.

Translation is additional structure: an even operator T_M of internal degree $(0, 2)$, commuting with d_M, and $[T_M, Y_M(a, z)] = \partial_z Y_M(a, z)$. Use a superscript T when it is retained. On P_ℓ,

$$\begin{aligned} T_P = &\sum_{i\geq 0} (i + \tfrac{1}{2})(x_{i+1}\partial_{x_i} + \theta_{i+1}\partial_{\theta_i}) \\ &+ \sum_{n\geq 1} (n + \tfrac{1}{2})(y_{n+1}\partial_{y_n} + \eta_{n+1}\partial_{\eta_n}) - \tfrac{1}{2} y_1 x_0 + \tfrac{1}{2}\eta_1\theta_0. \end{aligned} \tag{15.12}$$

All sums are finite on an input. The two final terms are multiplication operators. In particular,

$$T_P 1 = D(-y_1\theta_0/2) \neq 0. \tag{15.13}$$

Both boundary unit fields still act by the identity. Write $\mathcal{D}^T_{a,\ell}$ for the corresponding category localized at the underlying quasi-isomorphisms.

15.3.1 *A positive reverse mode comparison*

There is an exact reconstruction of the reverse mode algebra at the junction. Its recipient is the derived category of all bigraded modules over $\mathsf{B}_\ell = \Lambda_{S_\ell}(\kappa_i, \eta_n)$, where $\kappa_i = K_i$, with the restricted sum of internal shifts defined above. Write $P = \lim_{\longleftarrow \ell} P_\ell$ and $\mathsf{B} = \lim_{\longleftarrow \ell} \mathsf{B}_\ell$ for their coefficientwise completions. This paragraph does not impose the additional odd finite-tail condition on every object.

Put $a_i = 2i + 1$, $b_n = 2n - 1$. Define an independent algebra of coefficientwise continuous operators by

$$(\mathsf{A}_{\mathrm{res}})^{m,e} = \left(\prod_{|\beta|\leq m} \bigoplus_{\substack{|\alpha|=m-|\beta| \\ a\cdot\alpha - b\cdot\beta = e}} \mathbb{C}\, x^\alpha g^\beta \right)[\![q]\!], \qquad m \geq 0. \tag{15.14}$$

Take the coefficientwise direct sum over m, e; all multiindices have finite support. For fixed β, the sum over α is finite because the positive weights a_i tend to infinity. On an input, only finitely many β act through $g_n = -\partial_{y_n}$. Thus (15.14) is an actual operator algebra, with composition adding internal degrees.

Proposition 15.7. *The independent right action induces an equivalence*

$$\mathsf{A}_{\mathrm{res}} \xrightarrow{\sim} \underline{\mathrm{REnd}}^{\mathrm{res}}_{\mathsf{B}}(P)^{\mathrm{op}}.$$

It is a statement in this mode category, with the specified coefficient completion.

Proof. At coefficient level S_ℓ, take the semifree resolution

$$Q_{\mathsf{B}} = \mathsf{B}_\ell[t_i, s_n], \qquad d_Q = \sum_i \kappa_i \partial_{t_i} + \sum_n \eta_n \partial_{s_n}.$$

The new generators have degree zero and internal degrees $(-1, -a_i)$, $(-1, b_n)$. Use divided monomials. Its augmentation to S_ℓ is a quasi-isomorphism by the positive-degree Euler contraction. Filtration by total t, s-degree is semifree.

The restricted dual $\underline{\mathrm{Hom}}^{\mathrm{res}}_{\mathsf{B}}(Q_{\mathsf{B}}, S_\ell)$ is concentrated in degree zero with zero differential. A map of shift (m, e) can be nonzero on $t^{[\alpha]}s^{[\beta]}$ exactly when

$$|\alpha| + |\beta| = m, \qquad a \cdot \alpha - b \cdot \beta = e.$$

Its independent coefficient values give precisely (15.14) modulo q^ℓ. The primitive coproduct on divided powers identifies convolution with multiplication of the dual symbols.

To identify the actual action, define the B_ℓ-linear quasi-isomorphism

$$f(t^{[\alpha]}s^{[\beta]}) = \delta_{\alpha,0}(-y)^{[\beta]} \quad : Q_{\mathsf{B}} \longrightarrow P_\ell.$$

It preserves the constant cohomology. Define a degree-minus-one map by

$$H_i(t^{[\alpha]}s^{[\beta]}) = \theta_i \delta_{\alpha,0}(-y)^{[\beta]}, \qquad H_i(bv) = (-1)^{|b|} b H_i(v).$$

The Clifford identity gives

$$d_{\mathrm{Hom}} H_i = x_i f - f \partial_{t_i}, \qquad g_n f = f \partial_{s_n}.$$

For $F = x_{i_1} \cdots x_{i_r} g^\beta$, telescope:

$$H_F = \sum_{j=1}^{r} x_{i_1} \cdots x_{i_{j-1}} H_{i_j} \partial_{t_{i_{j+1}}} \cdots \partial_{t_{i_r}} \partial_s^\beta, \qquad dH_F = F_P f - f F_Q.$$

The finite-input bounds of (15.14) make this homotopy valid for every admitted coefficient sum. Composing with the augmentation of Q_{B} identifies the actual action with the corresponding dual coefficient. It induces an isomorphism on cohomology and was already an associative derived-algebra map, proving the equivalence. The coefficient reductions of these Hom models are surjective, so the ordinary inverse limit computes the prescribed derived limit. All maps commute with reduction. □

Even this equivalence does not reconstruct all local fields. The bidegree $(2, 0)$ consists of operators $\sum_i \lambda_i x_i g_{i+1}$. Modes of finite differential-polynomial fields in γ, and finite products of such modes, have coefficient sequences in

$$\mathbb{C}^{(\mathbb{N}_0)} + \mathbb{C}[i(i+1)]$$

at each coefficient of q. Indeed a quadratic derivative field contributes

$$(-1)^{a+b}\big[(\tfrac{1}{2} - i)_a (i + \tfrac{3}{2})_b + (i + \tfrac{3}{2})_a (\tfrac{1}{2} - i)_b\big],$$

where $(z)_a$ is the rising factorial. These polynomials are invariant under $i \mapsto -i - 1$; the cases $(a, b) = (2r, 0)$ have degree $2r$ and leading coefficient 2, so they span $\mathbb{C}[i(i+1)]$. Products of two linear modes supply exactly the finitely supported terms. Higher polynomial degree cannot contribute at weight two. The sequence $\lambda_i = 2^i$ is absent: every sufficiently high finite difference kills a polynomial but preserves 2^i. It therefore gives a nonzero action-cone class, not killed by any power of q. The mode completion and the local field algebra retain different information.

15.4 Polynomial right orthogonals

Consider the zero-differential single-boundary modules

$$N_{0,\ell} = S_\ell[x_i, y_n], \qquad N_{1,\ell} = \Lambda_{S_\ell}(\theta_i, \eta_n).$$

Their modes and gradings are inherited from P_ℓ. Their translations are the even and odd parts, respectively, of (15.12), including the multiplication terms $-y_1 x_0/2$ and $+\eta_1\theta_0/2$. They satisfy (15.11) or the odd finite-tail condition. Each has nonnegative energy and finite free energy components, because all its state generators have positive energy. Their constant classes are nonzero.

Theorem 15.8. *In both boundary categories, with translation either retained or forgotten,*

$$\underline{\mathrm{RHom}}^{\mathrm{res}}(P_\ell, N_{0,\ell}) = 0, \qquad \underline{\mathrm{RHom}}^{\mathrm{res}}(P_\ell, N_{1,\ell}) = 0.$$

The same holds after the prescribed derived coefficient completion. Thus the junction does not generate either full category, even on the objects with nonnegative energy and finite free energy components.

The proof needs the infinite set of modes. A finite cutoff has a surviving top Koszul class, so its vanishing cannot serve as a substitute.

15.4.1 Cofree reduction

Put $B_0 = S_\ell[x_i]$ and $B_1 = \Lambda_{S_\ell}(\eta_n)$. Let C_+ be the coalgebra with finite-support divided-power basis e_α and

$$\Delta e_\alpha = \sum_{\beta+\gamma=\alpha} e_\beta \otimes e_\gamma.$$

Let $C_\theta = \Lambda_{S_\ell}(\theta_i)$ be the exterior coalgebra with primitive generators. There are exact presentations

$$\begin{aligned} C_{0,\ell} &\simeq C_+\text{-}\,\mathrm{Comod}(B_0\text{-}\,\mathrm{Mod}_{\mathrm{bigr}}), \\ C_{1,\ell} &\simeq C_\theta\text{-}\,\mathrm{Comod}(B_1\text{-}\,\mathrm{Mod}_{\mathrm{bigr}}). \end{aligned} \tag{15.15}$$

Indeed, the even coaction $\rho(v) = \sum_\alpha e_\alpha \otimes G^\alpha v$ is finite precisely by (15.11). The odd coaction uses finite subsets and the exterior signs. Conversely these modes define the fields of every differential polynomial. Move annihilators to the right: their indices and powers are bounded on an input, and a fixed Laurent coefficient then bounds the sum of the remaining creation indices. Every coefficient is finite. Zero self-contractions give the product and substitution identities. Intertwining all modes is equivalent to intertwining these fields, proving (15.15).

The forgetful functor U is exact. Its cofree right adjoint $F(V) = C \otimes_{S_\ell} V$ is exact because either coalgebra is a direct sum of shifts of free coefficient modules. For a K-injective base resolution I of V and an acyclic comodule Z,

$$\mathrm{Hom}_C(Z, FI) = \mathrm{Hom}_B(UZ, I)$$

is acyclic. Hence FI computes the derived cofree object, including for unbounded complexes. In every internal shift,

$$\mathrm{RHom}_C(M, FV) \simeq \mathrm{RHom}_B(UM, V). \tag{15.16}$$

Taking the exact direct sum gives the restricted enrichment.

The targets are $F(B_0)$ and $F(B_1)$; on the even side use $e_\alpha = (-y)^\alpha/\alpha!$. The appropriate Koszul or polynomial de Rham factor of UP_ℓ contracts by the Euler homotopy, leaving S_ℓ. For instance, over B_0

set y, η to zero and retain $B_0 \otimes \Lambda(\theta_i)$, $d\theta_i = x_i$; over B_1 set x, θ to zero and retain $B_1[y_n]$, $dy_n = -\eta_n$. Thus it remains to prove

$$\underline{\mathrm{RHom}}^{\mathrm{res}}_{B_0}(S_\ell, B_0) = 0, \qquad \underline{\mathrm{RHom}}^{\mathrm{res}}_{B_1}(S_\ell, B_1) = 0. \tag{15.17}$$

15.4.2 A finite head in every cohomological degree

For B_0, use the semifree resolution $B_0 \otimes \Lambda(\theta_i)$ with $d\theta_i = x_i$. Its positive-degree Euler contraction proves exactness. Semifreeness gives K-projectivity: against an acyclic target, a primitive of a closed Hom map is chosen successively on free generators in each internal shift.

A degree-p Hom cochain is an alternating family of finite polynomials indexed by p-element subsets; the family itself need not have finite support. With dual exterior symbols ϵ_i, its raw differential is $(-1)^{p+1} \sum_i x_i \epsilon_i \wedge -$. Conjugation in degree p by $(-1)^{p(p+1)/2}$ changes it to

$$\partial = \sum_i x_i \epsilon_i \wedge -.$$

Each output coefficient uses only the finitely many indices in its indexing subset. Define

$$Q_j f = \frac{f - f|_{x_j=0}}{x_j}, \qquad h_j = \iota_{\epsilon_j} Q_j.$$

Let π_j retain terms containing ϵ_j and set $x_j = 0$ in their coefficients. Polynomial division gives

$$\partial h_j + h_j \partial = 1 - \pi_j. \tag{15.18}$$

On a term without ϵ_j, use $Q_j(x_j f) = f$. On a term with it, the two sides subtract the constant coefficient in x_j. Cross-index terms anticommute.

For a head of length m, put

$$H_m = h_0 + \pi_0 h_1 + \cdots + \pi_0 \cdots \pi_{m-2} h_{m-1}, \qquad \Pi_m = \pi_0 \cdots \pi_{m-1}.$$

The projections commute with ∂, so telescoping gives

$$\partial H_m + H_m \partial = 1 - \Pi_m. \tag{15.19}$$

On a degree-p cocycle choose $m = p + 1$. The image of Π_m requires m exterior symbols and therefore vanishes. This constructs a primitive of every cocycle. Each primitive still has finite polynomial output on each input and preserves the two internal shifts.

For B_1, use

$$Q = B_1 \otimes S_\ell[s_1, s_2, \ldots], \qquad d_Q = \sum_n \eta_n \partial_{s_n}, \qquad |s_n| = 0,$$

with divided monomials and $(W, E)(s_n) = (-1, 2n - 1)$. The positive-degree Euler contraction and filtration by total s-degree prove that this is a semifree resolution. A degree-p Hom cochain is a family $F_\alpha \in B_1^p$. Its raw differential is $-\sum_{\alpha_n > 0} \eta_n F_{\alpha - e_n}$: the Hom sign and graded left-linearity cancel their $(-1)^p$ factors. Conjugating by $(-1)^p$ gives

$$(\partial F)_\alpha = \sum_{\alpha_n > 0} \eta_n F_{\alpha - e_n}.$$

Now set

$$(h_j F)_\alpha = \iota_{\eta_j} F_{\alpha+e_j}, \qquad (\pi_j F)_\alpha = \begin{cases} \eta_j \wedge \iota_{\eta_j} F_\alpha, & \alpha_j = 0, \\ 0, & \alpha_j > 0. \end{cases}$$

The exterior contraction identity proves (15.18), separately for $\alpha_j = 0$ and $\alpha_j > 0$. The same telescoping proof applies to the head $1, \dots, m$. Its top projection forces m distinct exterior output generators, hence vanishes for $m = p + 1$. These homotopies query finitely many input coefficients and preserve both internal shifts. This proves (15.17).

15.4.3 *Translation and completion*

Let δ_T be the derivation of boundary modes induced by translation. Adjoin T by $Ta - aT = \delta_T(a)$. Induction from the category forgetting translation has underlying module $S_\ell[T] \otimes M$, and

$$a(T^r \otimes v) = \sum_{j=0}^{r} (-1)^j \binom{r}{j} T^{r-j} \otimes \delta_T^j(a) v.$$

It remains inside the admitted support class. If $J_{N,h}$ kills v, then $J_{N+r,h+r}$ kills $T^r \otimes v$: at most r annihilators can be differentiated, leaving at least h original ones. Their indices decrease by at most r, and the even creation modes at the seam commute with them. The odd tail bound follows from the same finite index change.

For a translated module, the relative sequence

$$0 \longrightarrow \mathrm{Ind}(UM)^{(E+2)} \xrightarrow{\partial} \mathrm{Ind}(UM) \longrightarrow M \longrightarrow 0,$$
$$\partial(a \otimes m) = aT \otimes m - a \otimes T_M m$$

is exact by division by the monic operator $T - T_M$. The first term's generators have energy increased by two. The covariance identity makes the map balanced; commutation with the differential makes it a chain map. Derived adjunction therefore computes translated Hom by a two-term totalization of underlying derived Hom objects. Both terms vanish by the preceding calculation. Finally $\mathbf{R}\varprojlim_\ell$ of the zero mapping objects is zero, while N_a retains its constant class. This completes the proof of Theorem 15.8.

The proof excludes generation in the stated categories. It does not calculate a dual for an unspecified native tensor. Regular ambient mode modules and supported boundary modules also have different admissibility conditions. An assertion about one of their derived duals cannot be transferred to the other without a comparison.

15.5 A local residue whose cone has no quartet lift

The polynomial right orthogonals have zero differential. They cannot carry the original dg quartet, since its relations would require nonzero boundary actions to be nullhomotopic. Imposing quartet liftability on each object might therefore appear to select a reconstructible sector. The next calculation shows that this condition is not stable under the boundary cones already generated by P_ℓ.

On P_ℓ, the original conjugate fields are

$$\widehat{\beta}(z) = \sum_{n \geq 1} y_n z^{n-1/2} + \sum_{i \geq 0} \partial_{x_i} z^{-i-1/2},$$
$$b(z) = \sum_{i \geq 0} \theta_i z^{i-1/2} + \sum_{n \geq 1} \partial_{\eta_n} z^{-n-1/2}.$$

A compatible quartet extension on another boundary object means fields with the same normalized Weyl and Clifford relations, all original zero mixed supercommutators, and

$$[d, b(z)] = \Gamma(z), \qquad [d, \widehat{\beta}(z)] = -C(z).$$

In particular its mode b_0 is even-boundary-linear and satisfies $[d, b_0] = X_0$. Thus the X_0-action vanishes in the derived even-boundary category after forgetting translation. This necessary condition is invariant under boundary quasi-isomorphism.

Define the degree-two operator

$$\Omega = \sum_{i \geq 0} (2i + 1)\eta_{i+1}\partial_{\theta_i}. \tag{15.20}$$

It is finite on each input, has internal degree $(-2, 0)$, and commutes with D and the even boundary modes.

Lemma 15.9. *The operator Ω is a translation-invariant local residue:*

$$\Omega = -\operatorname{Res}_{z=0} C(z)\partial_z C(z), \qquad [T_P, \Omega] = 0.$$

Within the family $\sum_i a_i \eta_{i+1}\partial_{\theta_i}$, translation invariance is equivalent to $a_i = (2i + 1)a_0$.

Proof. A residue contains one creation mode and one annihilation mode. The pair with index i contributes

$$(i + \tfrac{1}{2})\partial_{\theta_i}\eta_{i+1} - (i + \tfrac{1}{2})\eta_{i+1}\partial_{\theta_i} = -(2i + 1)\eta_{i+1}\partial_{\theta_i}.$$

The field C has zero self-contraction, so this is a mode of its differential-polynomial local field. Translation gives

$$[T_P, \partial_{\theta_0}] = \eta_1/2, \qquad [T_P, \partial_{\theta_i}] = (\tfrac{1}{2} - i)\partial_{\theta_{i-1}} \quad (i \geq 1),$$
$$[T_P, \eta_n] = (n + \tfrac{1}{2})\eta_{n+1}.$$

The coefficient of $\eta_{i+2}\partial_{\theta_i}$ in the commutator is $a_i(i + \frac{3}{2}) - a_{i+1}(i + \frac{1}{2})$. These operators are independent, as evaluation on θ_i shows. Their vanishing is the stated recurrence. The remaining seam is $a_0\eta_1^2/2 = 0$. □

Let $a \in S_\ell$, and adjoin one exterior variable ϵ with degree -1 and internal degree $(2, 0)$. Set

$$\mathcal{E}_{\ell,a} = P_\ell \otimes \Lambda(\epsilon), \qquad d_a = D + a\epsilon\Omega, \qquad T_{\mathcal{E}} = T_P \otimes 1. \tag{15.21}$$

The even boundary modes are unchanged. Equivalently,

$$d_a\theta_i = x_i + a(2i + 1)\epsilon\eta_{i+1}, \qquad d_a y_n = -\eta_n, \qquad d_a x_i = d_a\eta_n = d_a\epsilon = 0.$$

The differential squares to zero, commutes with translation, and preserves the support and internal gradings. The short exact sequence

$$0 \longrightarrow \epsilon P_\ell \longrightarrow \mathcal{E}_{\ell,a} \longrightarrow P_\ell \longrightarrow 0$$

has connecting morphism $a\Omega$. Its first term is a cohomological shift of P_ℓ, with weight increased by two. Thus $\mathcal{E}_{\ell,a}$ belongs to the stable closure of the admitted shifts of P_ℓ. The new variable has energy zero and is exterior, so each energy component is the direct sum of two finite free components with different internal shifts. Energy remains nonnegative.

Its underlying coefficient cohomology is $\Lambda_{S_\ell}(\epsilon)$. Indeed, the homogeneous polynomial change

$$x_i' = x_i + a(2i+1)\epsilon\eta_{i+1}$$

leaves independent pairs $d\theta_i = x_i'$ and $dy_n = -\eta_n$. The Euler homotopy contracts all positive pair-degree, with ϵ a spectator. This change is not linear for the original boundary action. To test that action, one must retain a boundary coordinate.

Proposition 15.10. *Forget translation and restrict to the two even modes $x = X_0$ and $g = G_1$. The derived functor*

$$\mathcal{L}_g(M) = M \otimes^{\mathbb{L}}_{S_\ell[g]} S_\ell, \qquad g \mapsto 0,$$

retains the x-action and satisfies

$$H^*(\mathcal{L}_g \mathcal{E}_{\ell,a}) \cong e\,\Lambda_{S_\ell}(\epsilon, \eta_1), \qquad |e| = -1.$$

In the ordered basis $(e, \epsilon e, \eta_1 e, \epsilon\eta_1 e)$, x has matrix

$$X_a = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -a & 0 & 0 & 0 \end{pmatrix}. \tag{15.22}$$

It is nonzero whenever $a \neq 0$.

Proof. The bounded free resolution $[S_\ell[g]e \xrightarrow{g} S_\ell[g]]$, with $(W, E)(e) = (1, -1)$, computes the functor even on unbounded complexes. On $\mathcal{E}_{\ell,a}$, $g = -\partial_{y_1}$ is degreewise surjective and commutes with d_a. Its two-term Koszul tensor is therefore quasi-isomorphic to its kernel in the indicated shift. This follows from the short exact sequence of complexes; polynomial integration need not be a chain map.

The kernel contains precisely the states independent of y_1. The coordinates x_i' contract every (θ_i, x_i') pair, and the remaining $(y_n, -\eta_n)$ pairs for $n \geq 2$. What survives is $\Lambda(\epsilon, \eta_1)$. Since $x_0 = x_0' - a\epsilon\eta_1$, its action is exactly (15.22).

For an explicit last contraction, put

$$B = \Lambda_{S_\ell}(\epsilon, \eta), \qquad K_a = B[x] \otimes \Lambda(\theta), \qquad d\theta = x + a\epsilon\eta.$$

The multiplier $x + a\epsilon\eta$ is monic, hence injective on $B[x]$, including over the nonreduced ring S_ℓ. Polynomial division gives

$$Q_a(u) = \frac{u - u|_{x=-a\epsilon\eta}}{x + a\epsilon\eta}, \qquad h(u) = \theta Q_a(u), \qquad h(\theta u) = 0.$$

With ρ setting $\theta = 0, x = -a\epsilon\eta$ and ι including exterior representatives, $dh + hd = 1 - \iota\rho$. Thus the four surviving classes are free over S_ℓ, and a nonzero entry $-a$ cannot vanish by coefficient torsion. □

Theorem 15.11. *No full replete stable subcategory of $\mathcal{D}^T_{0,\ell}$, containing P_ℓ and closed under its admitted internal shifts, can consist entirely of objects admitting the original dg quartet extension, even up to boundary quasi-isomorphism. The assertion survives coefficient completion and any further support restrictions on the extra quartet operators.*

Proof. Fullness admits the translated local morphism Ω between shifts of P_ℓ; stability admits $\mathcal{E}_{\ell,1}$. Any quartet lift would make its x_0-action zero by the boundary-linear homotopy b_0. Restriction to two modes and derived tensor send a zero morphism to zero. Proposition 15.10 contradicts this. No support condition on b_0 was used. The coefficient -1 survives already modulo q, so reduction proves the completed assertion. □

More precisely, $\mathcal{E}_{\ell,a}$ is quartet-liftable if and only if $a = 0$. At $a = 0$ the original operators act on the first tensor factor. Every nonzero a is excluded by the same matrix. The underlying vector complex still has the primitive

$$d_a(\theta_0 - a\epsilon y_1) = x_0, \qquad [g_1, \theta_0 - a\epsilon y_1] = a\epsilon.$$

The second equation is the failure of boundary-linearity.

The integral physical residue is

$$-\operatorname{Res} c(z)\partial_z c(z) = q^2\Omega.$$

Its cone is obstructed for $\ell \geq 3$; for $\ell = 1, 2$ it splits. As a map of complex vector spaces, $\operatorname{rank} X_{q^2} = \max(\ell - 2, 0)$. Thus the complete lattice is obstructed already modulo q^3, without cancelling or inverting q.

15.5.1 The extension class and its local contact

The two-mode class has a direct Ext interpretation. Over $A = S_\ell[x, g]$, let

$$P^{(2)} = S_\ell[x, y] \otimes \Lambda(\theta, \eta), \qquad D\theta = x, \quad Dy = -\eta, \quad g = -\partial_y.$$

The finite free resolution of its constant cohomology is

$$Q = A \otimes \Lambda(u, v), \qquad |u| = |v| = -1, \quad du = x, \quad dv = g.$$

The A-linear map $f : Q \to P^{(2)}$ given by $f(1) = 1, f(u) = \theta, f(v) = f(uv) = 0$ is a quasi-isomorphism. For $\omega = \eta\partial_\theta$, define the degree-one map H by $H(u) = -y$ and zero on the other generators. Then

$$(d_{\mathrm{Hom}}H)(u) = \eta, \qquad (d_{\mathrm{Hom}}H)(uv) = -1.$$

Consequently $\omega f - d_{\mathrm{Hom}}H$ sends uv to 1 and the other basis elements to zero. It is the image under f of the degree-two endomorphism of Q with that same formula. Compose this endomorphism with the augmentation $Q \to S_\ell$. The resulting Hom complex has zero differential, and the value 1 on uv is its top generator. Thus the class generates $\operatorname{Ext}^2_A(S_\ell, S_\ell) = S_\ell$. The discarded all-mode residue tail has the explicit two-mode-linear homotopy

$$\left[D, -\sum_{i\geq 1}(2i+1)\, y_{i+1}\partial_{\theta_i}\right] = \sum_{i\geq 1}(2i+1)\eta_{i+1}\partial_{\theta_i}.$$

Every sum is finite on an input. Thus the Ext class is the restriction of the actual all-mode residue.

There is a local correction of the field differential, but it changes a mixed contact. Put

$$b_a(z) = b(z) - 2a\epsilon\partial_z\widehat{\beta}(z).$$

Oscillator commutators give

$$[\Omega, b(z)] = 2\partial_z C(z), \qquad [\Omega, \widehat{\beta}(z)] = 0.$$

The tensor sign gives $[d_a, \epsilon\partial_z\widehat{\beta}] = \epsilon\partial_z C$. Hence

$$[d_a, b_a] = \Gamma, \qquad [d_a, \widehat{\beta}] = -C, \qquad [T_{\mathcal{E}}, b_a] = \partial_z b_a.$$

However, if $\Delta(z,u)=\sum_{r\in\mathbb{Z}} z^{r-1/2}u^{-r-1/2}$, then

$$[\Gamma(u),b_a(z)]=2a\epsilon\,\partial_z\Delta(z,u). \tag{15.23}$$

For example its modes are

$$[g_n,b_{a,-i}]=a(2i+1)\epsilon\delta_{n,i+1},\qquad [x_i,b_{a,n}]=-a(2n-1)\epsilon\delta_{i,n-1}.$$

This is a local derivative contact. It is not the original zero mixed bracket, and Theorem 15.11 excludes every correction restoring all original relations on this derived boundary object.

15.6 Local fields must respect derived equivalence

The mapping object is as consequential as the object class. Local polynomial endomorphism fields at a fixed junction do not automatically extend to a derived bifunctor on all pairs. The obstruction is visible on the acyclic localization

$$Q_\ell=P_\ell[x_{\mathrm{o}}^{-1}].$$

The variables retain their gradings, with $(W,E)(x_{\mathrm{o}}^{-1})=(-1,-1)$. Laurent differentiation extends all quartet operators and translation. The boundary support conditions still hold: inverting x_{o} changes neither finite y-degree nor finite θ-support. The multiplication operator $h=m_{\theta_{\mathrm{o}}/x_{\mathrm{o}}}$ satisfies

$$Dh+hD=1$$

as an operator anticommutator. Thus $Q_\ell\to 0$ is a quasi-isomorphism preserving all specified structures. The contracting homotopy need not preserve them.

Fix the following polynomial field prescription. A normal symbol uses the original creation and annihilation variables, with polynomial creation coefficients and a uniform bound on total symbol degree at each coefficient of q. It can contain infinitely many mode monomials, provided its value on each input is a field with finite lower Laurent order and honest state coefficients. Inverse creation variables are not among its generators. Impose quartet locality, translation covariance and exact linearity for the selected boundary. For an endomorphism, locality means the first equation below for each quartet field f and some finite order K; covariance means the second:

$$(u-z)^K[f(u),F(z)]=0,\qquad [T,F(z)]=\partial_z F(z),$$

together with supercommutation with every boundary mode. For $F:P_\ell\to Q_\ell((z^{1/2}))$, use the corresponding rectangular equations and differential $DF-(-1)^{|F|}FD$. Call these complexes $\mathcal{E}^T_{a,\mathrm{poly}}(M)$ and $\mathcal{H}^T_{a,\mathrm{poly}}(P_\ell,Q_\ell)$.

Proposition 15.12. *Extension and restriction identify*

$$\mathcal{E}^T_{a,\mathrm{poly}}(Q_\ell)\cong\mathcal{E}^T_{a,\mathrm{poly}}(P_\ell)\cong\mathcal{H}^T_{a,\mathrm{poly}}(P_\ell,Q_\ell)\qquad(a=0,1).$$

The identity of Q_ℓ and inclusion $\iota:P_\ell\to Q_\ell$ represent nonzero degree-zero classes in their respective complexes.

Proof. A normal expression has bounded order in ∂_{x_0}. On a fixed Laurent input, only finitely many other annihilator rows can act, by the input's finite support. Each such row has a lower Laurent bound in the field variable and finite polynomial coefficients. The expression therefore extends to Q_ℓ.

Successive evaluation on polynomial monomials recovers each normal row: subtract the proper subrows already recovered, then divide by the monomial factorial. This proves faithfulness over S_ℓ. Every polynomial expression on Q_ℓ preserves P_ℓ; every rectangular polynomial expression lands in P_ℓ. The field equations are identities of these bounded-degree polynomial operators, coefficient by coefficient, so they extend and restrict faithfully.

Let $\varepsilon : P_\ell \to S_\ell$ extract the scalar coefficient. Then $D1 = 0$ and $\varepsilon D = 0$. If a degree-minus-one polynomial field had $dF = \iota$, its value on 1 would lie in $P_\ell((z^{1/2}))$. Applying ε to $D(F(z)1) = 1$ would give $0 = 1$. The diagonal identity is excluded in the same way by restriction. □

Theorem 15.13. *No reduced bifunctor invariant under underlying quasi-isomorphisms can be quasi-isomorphic to the unchanged polynomial field complex on (P_ℓ, Q_ℓ), or on every diagonal. Here reduced means that its value is zero if either input is zero. The obstruction holds with translation and after coefficient completion.*

Proof. Derived invariance and $Q_\ell \simeq 0$ force both cross-object and diagonal values involving Q_ℓ to vanish. Proposition 15.12 contradicts the proposed comparisons. A completed primitive would reduce to an impossible primitive modulo q. □

This does not exclude a derived replacement agreeing with the polynomial actor only at P. It also does not apply to a recipient imposing nonnegative energy on every object: Q_ℓ contains x_0^{-m}, and its energy-zero part contains the independent states $x_i x_0^{-(2i+1)}$. The polynomial right orthogonals of Theorem 15.8 require neither of these features.

15.6.1 An inverse field and a remaining odd obstruction

An enlarged local field class supplies an even-boundary primitive explicitly. Write

$$a(z) = \sum_{i \ge 0} x_i z^i, \qquad p(z) = \sum_{n \ge 1} \partial_{y_n} z^{-n}, \qquad \Gamma(z) = z^{-1/2}(a(z) - p(z)).$$

Since x_0 is invertible, set

$$I(z) = z^{1/2} \sum_{r \ge 0} a(z)^{-r-1} p(z)^r, \qquad H(z) = b(z) I(z). \tag{15.24}$$

Proposition 15.14. *These are fields with finite Laurent-polynomial state coefficients on every input. They satisfy*

$$\Gamma I = I\Gamma = 1, \qquad [D, H] = 1.$$

The field H is translation-covariant, local with the quartet and exactly even-boundary-linear. There is no degree-minus-one strictly odd-boundary-linear finite-output field $P_\ell \to Q_\ell((z^{1/2}))$ with differential ι, nor such an endomorphism field of Q_ℓ with differential 1.

Proof. On a vector of y-degree d, only $r \le d$ contributes. At a fixed coefficient of z,

$$a(z)^{-s} = x_0^{-s} \left(1 + \sum_{i \ge 1} (x_i / x_0) z^i \right)^{-s}$$

has finitely many partition terms. The geometric-series telescoping is therefore finite on every input and proves the inverse identities. The factors b, I have zero mutual contractions; applying inverse-field coefficients does not increase y-degree or η-support. Their product is a field, with $[D, I] = 0$, and $[D, H] = [D, b]I = \Gamma I = 1$.

Put $J = (a - p)^{-1}$. Modewise differentiation gives

$$[\partial_{x_i}, J] = -z^i J^2, \qquad [y_n, J] = -z^{-n} J^2.$$

With $\Delta(u, z) = \sum_{r \in \mathbb{Z}} u^{r-1/2} z^{-r-1/2}$, the nonzero commutators are

$$[\widehat{\beta}(u), H(z)] = -\Delta(u, z) b(z) I(z)^2,$$
$$[C(u), H(z)] = \Delta(u, z) I(z).$$

Each mode coefficient is finite on an input, and $(u - z)\Delta(u, z) = 0$ proves locality. Exact commutation with Γ proves even-boundary linearity. Differentiating $\Gamma I = 1$, and commuting it with T, gives $[T, I] = -I(\partial_z \Gamma) I = \partial_z I$, then the covariance of H.

For the odd obstruction, every K_i kills 1. Strict graded linearity of a degree-minus-one field gives $K_i F(z)1 = 0$ for every i. The common kernel of the $K_i = \partial_{\theta_i}$ is

$$S_\ell[x_i, x_0^{-1}, y_n] \otimes \Lambda(\eta_n),$$

which has no negative cohomological degree. Thus $F(z)1 = 0$, and its differential cannot send 1 to 1. No locality or translation assumption is needed for this last argument. □

The enlargement in (15.24) is substantial. On the constant state,

$$H(z)1 = \frac{\sum_i \theta_i z^i}{\sum_i x_i z^i},$$
$$H_0(1) = \theta_0 / x_0, \quad H_1(1) = \theta_1 / x_0 - \theta_0 x_1 / x_0^2.$$

Already the θ_0, x_0, x_1 terms have pole order $m + 1$ at coefficient z^m. They are honest finite states coefficientwise, with no uniform pole or symbol-degree bound. Enlarging symbols can repair the even primitive; it cannot repair strict odd-boundary linearity.

15.7 What full quartet linearity does reconstruct

There is a positive reconstruction with a different mapping category. Let $\mathscr{W}_\ell$ be the algebra of finite polynomial oscillator operators

$$x_i, y_n, \theta_i, \eta_n, \quad \xi_i = \partial_{x_i}, \quad \zeta_n = \partial_{y_n}, \quad \kappa_i = \partial_{\theta_i}, \quad \pi_n = \partial_{\eta_n},$$

with the normalized Weyl and Clifford relations. All other supercommutators vanish. Its differential $\delta = [D, -]$ satisfies

$$\delta \theta_i = x_i, \qquad \delta y_n = -\eta_n, \qquad \delta \xi_i = -\kappa_i, \qquad \delta \pi_n = -\zeta_n,$$

and kills the other generators.

Proposition 15.15. *The unit inclusion $S_\ell \to \mathscr{W}_\ell$ and evaluation $\mathscr{W}_\ell \to P_\ell$, $w \mapsto w1$, are quasi-isomorphisms. For full quartet-linear morphisms,*

$$D(\mathscr{W}_\ell) \simeq D(S_\ell), \qquad \mathrm{REnd}_{\mathscr{W}_\ell}(P_\ell) \simeq S_\ell.$$

In the bigraded category, the internal shifts of P_ℓ form the compact generating family.

Proof. Normal ordering gives a free superpolynomial symbol basis. The differential preserves this order and has exactly the four displayed pair differentials. In positive total symbol degree N, the homotopy is

$$h = N^{-1}\left(\sum_i \theta_i \partial_{x_i} - \sum_n y_n \partial_{\eta_n} - \sum_i \xi_i \partial_{\kappa_i} - \sum_n \pi_n \partial_{\zeta_n}\right).$$

Its anticommutator with δ is the identity in positive degree. It is finite on each symbol and preserves internal degrees. Only the identity class survives; evaluation on 1 identifies it with the constant cohomology of P_ℓ.

Tensoring the coefficient contraction with any complex V shows that $V \to \mathscr{W}_\ell \otimes_{S_\ell} V$ is a quasi-isomorphism. For a dg $\mathscr{W}_\ell$-module M, the multiplication map $\mathscr{W}_\ell \otimes_{S_\ell} M \to M$ is a module map whose composite with the coefficient unit is the identity. It too is a quasi-isomorphism. These are the derived unit and counit for extension and restriction along $S_\ell \to \mathscr{W}_\ell$. The assertions about the end and compact generators follow. □

The scalar symbol projection used in the proof is not an algebra augmentation. Indeed, $\xi_i x_i = x_i \xi_i + 1$ has scalar part 1, while ξ_i and x_i each have scalar part zero. The multiplicative quasi-isomorphism is the unit inclusion.

One may also impose polynomial joint support: on every vector all but finitely many annihilators vanish, and the remaining even annihilators are jointly locally nilpotent. For one even pair $[a, z] = 1$, the finite-on-input operator

$$\Pi_{z,a} = \sum_{r \geq 0} \frac{(-1)^r}{r!} z^r a^r$$

projects to $\ker a$; for an odd pair $az + za = 1$, the projector is $1 - za$. The projectors of distinct pairs commute. Their product Π_{vac} is finite on an input and projects to the common annihilator kernel $\mathrm{Vac}(M)$.

Taylor expansion gives

$$m = \sum_{\alpha, I} \frac{z^\alpha z_I}{\alpha!} \Pi_{\mathrm{vac}}(a_I a^\alpha m),$$

$$P_{\ell,\mathrm{gr}} \otimes_{S_\ell} \mathrm{Vac}(M) \xrightarrow{\sim} M.$$

Odd annihilators are ordered opposite to the creation monomial. Applying projected derivatives proves uniqueness. Projecting a lift of a vacuum vector proves exactness of Vac. The differential preserves its kernel and becomes $D_P \otimes 1 + 1 \otimes d_{\mathrm{vac}}$. This gives the same derived coefficient category on the jointly supported modules. Fixing oscillator translation makes the comparison translation-compatible. An arbitrary extra covariant translation is instead an extra degree-zero chain operator on the vacuum complex and must be retained.

This positive theorem uses full quartet-linear maps. Forgetting to the boundary category is not full: $[\Omega, \theta_0] = \eta_1 \neq 0$, and the scalar end has no degree-two class mapping to the boundary residue. Nor does primitive diagonal tensoring give a normalized quartet tensor:

$$[\xi \otimes 1 + 1 \otimes \xi,\ x \otimes 1 + 1 \otimes x] = 2\,1.$$

The construction therefore does not identify a native relative sewing product.

15.8 Endomorphisms under coefficient and collision maps

The endomorphisms of a marked object need not commute with restriction of its coefficients. The comparison is induced by evaluation and exists before it is known to be invertible. Let $R \to S$ be a map

of commutative dg rings, B a dg R-algebra, and P, Q left B-modules. Write $B_S = S \otimes^{\mathbb{L}}_R B$ and similarly for modules. Extending evaluation and taking its adjoint gives

$$S \otimes^{\mathbb{L}}_R \mathrm{RHom}_B(P, Q) \longrightarrow \mathrm{RHom}_{B_S}(P_S, Q_S). \tag{15.25}$$

Proposition 15.16. *If P is perfect over B, the comparison (15.25) is an equivalence for every Q. For $Q = P$, it is an equivalence of unital E_1-algebras. For two perfect modules over R-algebras B_1, B_2, the external-product comparison*

$$\mathrm{REnd}_{B_1}(P_1) \otimes^{\mathbb{L}}_R \mathrm{REnd}_{B_2}(P_2) \longrightarrow \mathrm{REnd}_{B_1 \otimes^{\mathbb{L}}_R B_2}(P_1 \otimes^{\mathbb{L}}_R P_2)$$

is also an equivalence. Both assertions retain a finite diagram of marking arrows when all its source objects are perfect.

Proof. For $P = B$, (15.25) is the identity on $S \otimes^{\mathbb{L}}_R Q$. Both sides, as functors of P, take finite extensions to finite limits and preserve retracts. Scalar extension preserves finite limits in a stable category. Induction on finite cells proves the assertion for every perfect P. For the external product, replace the target copies of P_1, P_2 by arbitrary Q_1, Q_2. The comparison is the identity for the two free source modules; apply the same finite-cell argument in each source variable. The maps are adjoint to evaluation, so they preserve composition, units and successive base changes. Cross Hom has the same proof, and the arrow end model uses a finite homotopy fibre. This proves the marked assertion. □

Perfectness cannot simply be omitted. Let $R = k[t]$, $S = R[t^{-1}]$ and $P = \bigoplus_{n \geq 0} Re_n$. The endomorphism $e_n \mapsto t^{-n} e_n$ of P_S has no common denominator, so it is absent from $S \otimes_R \mathrm{End}_R(P)$. There is a separate tensor defect. On $V = \bigoplus_{i \geq 0} ke_i$, the endomorphism

$$T(e_i \otimes e_j) = \delta_{ij} e_i \otimes e_j$$

does not belong to $\mathrm{End}(V) \otimes \mathrm{End}(V)$. A sum of r elementary tensors would make every finite diagonal coefficient matrix have rank at most r, contradicting its $(r+1)$-square identity corner.

15.8.1 *The collision comparisons*

Let X be a smooth curve and C a connected coaugmented factorizing coalgebra on its ordered collision carrier. Write $\mathscr{R}_I$ for the carrier with a finite nonempty set of roots I. Suppose a marked bicomodule category $\mathcal{M}_I$ and its regular object $\mathbb{V}_I$ have been specified, with a closed action of the normalized coefficient category. On a smooth chart Y of dimension d, and for a collision map $f : Y' \to Y$, use

$$L_Y(M) = \omega_Y \otimes_{\mathcal{O}_Y} M[d], \qquad f^\dagger = L^{-1}_{Y'} f^! L_Y.$$

This fixes the left-to-right normalization; in particular $f^\dagger \mathcal{O}_Y = \mathcal{O}_{Y'}$. Define the internal end

$$E_I = \underline{\mathrm{RHom}}_{\mathcal{M}_I}(\mathbb{V}_I, \mathbb{V}_I).$$

Its enrichment, rather than a global scalar Hom, determines the coefficient object.

If collision restriction takes the regular object to the regular object and is linear over normalized pullback, evaluation induces

$$c_\alpha : \delta^\dagger_\alpha E_I \longrightarrow E_{I'} \qquad (\alpha : I \twoheadrightarrow I'). \tag{15.26}$$

For a partition $\mathcal{P} = (I_1 | \cdots | I_r)$, write $j_{\mathcal{P}} : U_{\mathcal{P}} \to \mathscr{R}_I$ for the separated open. Let $E^{\mathrm{loc}}_{I,\mathcal{P}}$ be the end of the regular object in the restricted marked category. Restriction and external-product evaluation give the span

$$j^\dagger_{\mathcal{P}} E_I \xrightarrow{b_{\mathcal{P}}} E^{\mathrm{loc}}_{I,\mathcal{P}} \xleftarrow{e_{\mathcal{P}}} \boxtimes_a E_{I_a}\big|_{U_{\mathcal{P}}}. \tag{15.27}$$

Neither arrow is inverted by its construction.

Proposition 15.17. *Suppose the entire marked category on each root chart is presented as* Mod_{B_I}*, carrying* $\mathbb{V}_I$ *to a perfect* P_I*. Suppose each collision and separated-open restriction is the corresponding derived extension of scalars, and the separated categories and regular objects are the external tensor products of their blocks. If these identifications preserve coefficient actions, chart transitions, promotions of labels and every ordered collision diagram, then* c_α, $b_{\mathcal{P}}$, $e_{\mathcal{P}}$ *are coherent equivalences preserving composition and evaluation.*

Proof. Apply Proposition 15.16 on each chart. The maps obtained there are the maps adjoint to the same evaluations as (15.26) and (15.27). The supplied coherent presentations therefore identify them on overlaps and on composite restrictions. Equivalences of underlying coefficient complexes detect equivalences of the normalized differential modules. They descend with their algebra and evaluation structures. □

Perfectness at each fixed number of ordinary labels is weaker than the premise. It neither identifies the restriction functor of the entire marked category with scalar extension nor commutes an infinite homotopy end with pullback.

Let C^f be the category of factorizing coalgebras on this carrier, with transverse tensor $\otimes^!$ and local unit J. A transverse monoid is a unital associative algebra object for this tensor. The space $\mathrm{Act}(F; C)$ consists of its unital associative actions on C, with all their comparison homotopies.

Theorem 15.18 (Conditional factorizing recognition)**.** *Suppose the comparisons* (15.26) *and* (15.27) *are equivalences with all mixed collision and factorization coherences. Suppose their evaluations descend, preserve the local unit, and commute with promotion of ordinary labels to roots. Then*

$$Z_I = L_{\mathscr{R}_I}(E_I), \qquad Z_\varnothing = k,$$

defines a factorizing transverse monoid with an action on C*. It represents the full transverse action functor if, in addition, evaluation induces equivalences*

$$\mathrm{Map}_{\mathrm{Alg}(C^f)}(F, Z) \xrightarrow{\sim} \mathrm{Act}(F; C) \tag{15.28}$$

for every actor F *in the specified category* C^f*.*

Proof. The maps $b_{\mathcal{P}}^{-1} e_{\mathcal{P}}$ give the separated-product equivalences, and the c_α give collision equivalences. Their stipulated coherences are the descent diagrams. Composition and the unit descend because the comparisons preserve them; the evaluation hypothesis supplies the action. Composing a map $F \to Z$ with this action constructs (15.28) on all simplices, naturally in F. Its additional equivalence assertion is exactly representability of the full action functor. □

The additional assertion is substantive. The full pointed and ordinary multiplicative action categories considered in Chapter 6 have explicit nonfinality obstructions. A proposed restriction must explain which actions or maps it excludes and why that restriction is geometrically appropriate. The marked associative classifier of Theorem 15.5 remains valid in its arrow category. Identifying that category with the supported factorizing action category is a separate map. The two-object question of a pair of actions with no common recipient also retains its own quantifier; an unequalizable parallel pair does not by itself supply that pair of objects.

15.9 Complete filtrations and formal comparison

A comparison of finite coefficients becomes a comparison of formal objects only when the limit and its operations are controlled. Let C be a complex with descending subcomplex filtration $C = F^0C \supset F^1C \supset \cdots$. Call it complete and separated here when the natural map

$$C \longrightarrow \varprojlim_n C/F^nC$$

is an isomorphism degree by degree. The differential is continuous for this filtration. Set $\mathrm{gr}^n C = F^nC/F^{n+1}C$.

Lemma 15.19. *Let $f : C \to D$ preserve complete separated filtrations of this form. If every associated-graded map $\mathrm{gr}^n f$ is a quasi-isomorphism, then f is a quasi-isomorphism. The conclusion is an equivalence of structured algebras if f was already a continuous map of those structured algebras.*

Proof. Give the cone of f the componentwise filtration. Its associated graded is acyclic. For a cone cocycle z, acyclicity of gr^0 gives b_0 with $z - db_0 \in F^1$. This error is closed. Inductively, acyclicity of gr^n gives $b_n \in F^n$ such that

$$z - d(b_0 + \cdots + b_n) \in F^{n+1}.$$

Completeness defines $b = \sum_{n\geq 0} b_n$, and continuity with separatedness gives $db = z$. Thus the cone is acyclic. No new operation is supplied by this argument; it proves equivalence for the continuous structured map assumed in the statement. □

The existence of a compatible map is a different problem. For example, let $\mathfrak{g}$ be a complete dg Lie algebra with $[F^i\mathfrak{g}, F^j\mathfrak{g}] \subset F^{i+j}\mathfrak{g}$. A deformation is a Maurer–Cartan element $\alpha \in F^1\mathfrak{g}^1$:

$$d\alpha + \tfrac{1}{2}[\alpha, \alpha] = 0.$$

If this equation holds modulo F^n, $n \geq 1$, its error $r_n \in F^n\mathfrak{g}^2$ is closed modulo F^{n+1}. Indeed the Bianchi identity $dr_n + [\alpha, r_n] = 0$ makes the second term one filtration step higher. A correction $u_n \in F^n\mathfrak{g}^1$ changes the error by du_n modulo F^{n+1}. Thus the next obstruction lies in $H^2(\mathrm{gr}^n\,\mathfrak{g})$. When it vanishes, choices differ by degree-one classes; their gauge identifications involve degree-zero classes. Completeness sums actual compatible corrections. It does not force these obstruction or automorphism groups to vanish.

This distinction preserves the Hamiltonian comparisons of the preceding chapters. A complete cochain map can be a quasi-isomorphism while its proposed brace refinement remains unconstructed. A selected higher coefficient can change under transfer. A full Massey coset, a nonzero action-cone class or a residual gauge stabilizer requires its own invariant calculation.

15.10 Holomorphic descent with a specified comparison

The Hamiltonian Lie-cochain model and the symplectic polyvector model have different carriers. To formulate the second, let (S, ω) be a holomorphic symplectic surface and $R = \mathbb{C}[\![\hbar]\!]$. Write $\pi = \omega^{-1}$. A chosen formal Poisson tensor $\pi_\hbar = \hbar\pi + O(\hbar^2)$, with $[\pi_\hbar, \pi_\hbar] = 0$, defines

$$\mathcal{P}_S = \big(\mathrm{PV}_S[\![\hbar]\!], [\pi_\hbar, -]\big).$$

Polyvector degree is its cohomological degree; the Schouten bracket has degree -1, so this differential has degree one. Here $\mathrm{PV}_S^p = \wedge^p T_S$. The choice $\pi_\hbar = \hbar\pi$ is a particular formal class, not the only quantization datum.

A Weyl algebroid $\mathcal{W}$ is specified by local formal Weyl algebras, equivalences on overlaps and their coherent cocycle data. A Fedosov presentation uses formal fibre coordinates and a flat resolution differential, locally written

$$D_F = \nabla - \partial + \hbar^{-1}[\gamma, -].$$

Here ∇ is the chosen local symplectic connection, γ is a Weyl-valued one-form, and ∂ is the fibre Koszul differential. The commutator is divisible by $\hbar$, so the displayed operator need not invert the coefficient parameter. Flatness is an equation on γ and the prescribed curvature data. In the holomorphic setting these resolution and descent data must be supplied, for example through compatible Dolbeault models; they are not obtained by assuming a global holomorphic symplectic connection exists.

Dolgushev's smooth formality construction uses Fedosov resolutions and gives an L_∞ comparison of polyvectors with polydifferential operators [31, Theorems 2–4]. That result alone does not give a holomorphic homotopy-E_2 comparison with every marking and cyclic structure. We state precisely the data used below.

Fix a cover, all its multiple intersections, and continuous homotopy-E_2 resolutions of $\mathcal{P}_S$ and $C_{\mathcal{W}}$, the relative Hochschild cochain sheaf of $\mathcal{W}$. Suppose a homotopy coherent map of their descent diagrams

$$\Phi_\bullet : \mathcal{P}_\bullet \longrightarrow C_{\mathcal{W},\bullet} \tag{15.29}$$

has been constructed. Its leading maps are required to be quasi-isomorphisms for complete separated positive filtrations as in Lemma 15.19. The coefficient completion, chosen formality refinement and formal Poisson class are fixed throughout.

Theorem 15.20 (Descent of a supplied formality comparison)**.** *Under these hypotheses, $\Phi_\bullet$ descends to an equivalence of the complete homotopy-E_2 sheaves and induces*

$$\mathbf{R}\Gamma(S, \mathcal{P}_S) \simeq \mathbf{R}\Gamma(S, C_{\mathcal{W}}).$$

If the chosen continuous R-linear Morita theory identifies the latter with the centre of $\mathrm{Perf}(\mathcal{W})$*, this is the corresponding global centre comparison. Coherent markings, cyclic maps and state maps descend whenever $\Phi_\bullet$ is supplied in their diagram category.*

Proof. The filtered lemma makes the map an equivalence on every member of the descent diagram. A homotopy limit sends an objectwise equivalence of diagrams to an equivalence. Taking the descent totalization therefore gives the sheaf equivalence, and derived global sections give the displayed map. Limits of algebra objects carry their operations objectwise, so the homotopy-E_2 structure is preserved. The Morita assertion uses its stated centre model. The same limit argument in a diagram category retains markings and all their comparison homotopies. □

A change of Fedosov data gives the same conclusion when a compatible gauge equivalence of the chosen resolutions and its descent homotopies are supplied. An arbitrary change of the Fedosov one-form is not such a hypothesis: it can change the formal curvature class. Likewise, equivariance requires a coherent equivariant choice of (15.29). These qualifications apply on a K3 surface as well. Its holomorphic symplectic form supplies π, not all of the displayed comparison data.

There is also a formal algebraic realization in two topological directions. For each holomorphic open, the specified E_2-algebra has its factorization realization on framed two-disks. The framed structure may be promoted to an oriented one when the required rotation-equivariant structure is supplied. If these assignments satisfy the required cosheaf descent in the holomorphic variable, they define a mixed factorization theory on $\mathbb{R}^2_{\mathrm{top}} \times S$. Theorem 15.20 then realizes the supplied centre equivalence in those

topological directions. Neither this realization nor Morita invariance alone identifies a universal physical boundary category.

For the Hamiltonian BF action

$$S_{\mathrm{BF}}(A,B)=\int_{\mathbb{R}^2\times S}\left\langle B,(d+\bar\partial)A+\tfrac{1}{2}[A,A]\right\rangle,$$

the first comparison is a map from its Hamiltonian Chevalley observables to the chosen symplectic polyvector complex. It must preserve the operations and coefficient completion. Gauge fixing and a coupled quantum master equation are additional requirements for a quantum factorization theory. When S is a complex surface, this spacetime has six real dimensions. The three-dimensional half-space theory of Chapter 14 has a different geometric input.

15.11 Sewing with a prepared state

A reconstruction can preserve an algebra and lose its state. Let B be a dg R-algebra, M a B-module, $\nu : R \to M$ a chosen closed preparation in R-complexes, and $s : M \to R$ a closed R-linear state map with $s\nu = 1$. An equivalence of such data consists of the algebra and module comparisons, a homotopy carrying the preparation, a homotopy carrying s, and compatibility of these homotopies with $s\nu = 1$. Neither the unit of B nor an augmentation of B specifies s.

To express sewing, fix boundary algebras B_0, B_1, B_2, a B_0–B_1 bimodule M_{01}, and a B_1–B_2 bimodule M_{12}. The sewn object M_{02} is a B_0–B_2 bimodule. A sewing map has domain

$$\mu_{012} : M_{01}\widehat{\otimes}^{\mathbb{L}}_{B_1}M_{12}\longrightarrow M_{02},$$

where the completed balanced tensor has been independently defined on the allowed objects. Preparations must obey $\mu_{012}(\nu_{01}\otimes\nu_{12})\simeq\nu_{02}$. If σ_{012} is the supplied contraction functional on that balanced tensor, state compatibility is the specified homotopy

$$s_{02}\mu_{012}\simeq\sigma_{012}. \tag{15.30}$$

For three regions these homotopies must agree with the associativity comparison; higher regions require their higher coherence. A tensor of scalar functionals descends to a balanced tensor only when it satisfies the actual balancing relation. This must be checked in the state model.

Proposition 15.21. *Suppose two such systems are related by equivalences of their full diagrams of boundary algebras, bimodules, markings, state maps and sewing maps, including every coherence. Then all finite sewn states agree under those equivalences. If the systems are diagrams of derived complete objects and their comparisons commute with the coefficient quotients, the same is true after the prescribed derived completion.*

Proof. For two regions the assertion is (15.30) transported through the given square. Inductively compose this square with the next sewing map. The associativity homotopies identify the different parenthesizations, and their higher coherences identify the resulting paths of identifications. Thus the full finite diagram, including its final state map, is carried by the equivalence. At the complete level, apply the derived inverse limit to the compatible coefficient diagrams. It preserves their equivalences and homotopies. □

This proposition requires states on the completed domain in question. Finite Gaussian integration constructs the normalization and balanced contractions in Chapter 8; the determinant cannot be removed from that state comparison. The infinite-mode and stabilized states have the specific analytic domains of Chapter 13. A formal coefficient equivalence by itself supplies neither positivity nor a Laplace prescription, and coefficientwise existence does not establish continuity on a larger state completion.

Problem 15.22 (Local mapping and physical reconstruction)**.** Construct a stable boundary category with homotopy-invariant cross-object local mapping objects, independently defined relative sewing and evaluation against both boundary algebras. The construction must retain the coefficient lines, local units, preparations and regular, localized, supported and vacuum collision terms. Determine its comparison with the marked associative action classifier and with independently defined bulk observables.

For the Ramond junction, this requires deciding how the local residue $q^{2}\Omega$ and its cone occur in that category, including the derivative contact (15.23). A category containing the acyclic localization Q_{ℓ} cannot have unchanged polynomial local actors on every diagonal and still be derived invariant. Within the specified even-boundary derived category, no full stable sector containing the junction and its shifts can require every object to lift to the original quartet. An admissible alternative must construct its mapping and sewing operations and prove its comparison, rather than infer them from the end algebra of one object.

Chapter 16

Quantum symmetries and geometric comparisons

An algebra of boundary operators determines how those operators compose. It does not determine how two representations combine. The latter operation uses a coproduct, or a more general tensor construction on the category of representations. Nor does an exchange operator determine the coefficient lattice on which multiplication acts. These distinctions persist in explicit polynomial examples.

We begin with one algebra carrying two different Hopf structures. We then compare ordinary bars, polynomial shuffle lattices and matrix-coefficient reconstruction. The final examples concern geometric and gravitational comparisons. A tangent complex does not determine its nonlinear equations. The spectrum of a connection does not identify its holonomy with its coefficients.

Throughout, k is a field of characteristic zero. All tensors are algebraic until a different completion or localization is specified.

16.1 One operator algebra and two representation tensors

Put $A = k[x, y, z]$, with augmentation $\varepsilon(x) = \varepsilon(y) = \varepsilon(z) = 0$. For $\eta = 0, 1$, define algebra maps $\Delta_\eta : A \to A \otimes_k A$ by

$$\begin{aligned} \Delta_\eta x &= x \otimes 1 + 1 \otimes x, \\ \Delta_\eta y &= y \otimes 1 + 1 \otimes y, \\ \Delta_\eta z &= z \otimes 1 + 1 \otimes z + \eta x \otimes y. \end{aligned} \tag{16.1}$$

A Hopf structure requires coassociativity, the two counit identities, and an antipode S with $m(S \otimes 1)\Delta = m(1 \otimes S)\Delta = \mathbf{1}\varepsilon$. Here m is multiplication and $\mathbf{1} : k \to A$ is the unit.

Proposition 16.1. *Both coproducts in* (16.1) *define Hopf algebras. Their antipodes are*

$$S_\eta x = -x, \qquad S_\eta y = -y, \qquad S_\eta z = -z + \eta x y.$$

They have the same augmented algebra and the same ordinary bar coalgebra, but they are not isomorphic as Hopf algebras.

Proof. It suffices to check the generators. The only nonprimitive generator is z. Both iterated coproducts give its three single-factor terms and

$$\eta\bigl(x \otimes y \otimes 1 + x \otimes 1 \otimes y + 1 \otimes x \otimes y\bigr).$$

This proves coassociativity.

The augmentation kills each extra tensor factor, giving the counits. On z, either antipode composite is $(-z + \eta x y) + z - \eta x y = 0$. The checks on x, y are immediate, and the identities extend multiplicatively.

The coproduct Δ_0 is cocommutative, whereas

$$\Delta_1 z - \Delta_1^{\mathrm{op}} z = x \otimes y - y \otimes x \neq 0.$$

Cocommutativity is preserved by a Hopf isomorphism, so none exists. The ordinary augmented bar uses multiplication, augmentation and unit, with word deconcatenation as its coalgebra structure. Those data are identical for the two algebras. □

The difference is visible on one-dimensional modules. Let M have character $(x, y, z) = (1, 0, 0)$ and let N have character $(0, 1, 0)$. On $M \otimes_k N$, the action induced by Δ_0 has $z = 0$. The action induced by Δ_1 has $z = 1$. Thus even the ordinary tensor of representations is extra information relative to the augmented algebra and its bar.

Assigning weights 1, 1, 2 to x, y, z makes both coproducts graded. Their common Hilbert series

$$\frac{1}{(1-t)^2(1-t^2)}$$

does not distinguish them either.

16.2 Ordinary bars and geometric coefficients

Let V be finite-dimensional and concentrated in degree zero. Give its generators weight one. The tensor algebra and the square-zero algebra are

$$T(V) = \bigoplus_{r \geq 0} V^{\otimes r}, \qquad A_{\mathrm{sq}} = k \oplus V, \qquad V^2 = 0.$$

Both are ordinary augmented algebras over k.

We use a cohomological bar convention: a word of bar length n has degree $-n$, and $sV = V[1]$ has degree -1. For an augmentation ideal $\bar{A}$, the bar has terms $(s\bar{A})^{\otimes n}$ and differential given by adjacent products. Its coproduct cuts a word at every position, including both empty cuts.

Proposition 16.2. *There is a quasi-isomorphism of coaugmented coalgebras*

$$k \oplus sV \longrightarrow B(T(V)),$$

where sV is primitive. Consequently the derived augmentation end algebra is

$$\mathrm{RHom}_{T(V)}(k, k) \simeq k \oplus V^*[-1]$$

with square-zero reduced multiplication. For the square-zero algebra,

$$B(A_{\mathrm{sq}}) = T^c(sV),$$

with zero differential. Its continuous weight dual is $\widehat{T}(V^*[-1]) = \prod_{n \geq 0} (V^*[-1])^{\otimes n}$, with concatenation product.

Proof. Choose a basis of V. A bar element of $T(V)$ is a nonempty word in basis elements, together with a set of cuts separating its nonempty algebra factors. Fix an underlying word of length r. Its possible cuts form subsets $C \subset \{1, \dots, r-1\}$, and its bar length is $|C| + 1$.

Choose the bar sign for which

$$dC = \sum_{j=1}^{|C|} (-1)^{j-1} \big(C \setminus \{c_j\}\big), \qquad c_1 < \cdots < c_{|C|}.$$

For $r \geq 2$, define $hC = \{1\} \cup C$ when $1 \notin C$, and $hC = 0$ otherwise.

If $1 \notin C$, the term deleting the new first cut is C. The other terms cancel those of hdC. If $1 \in C$, only the term of dC deleting that cut contributes to hdC, again giving C. Hence $dh + hd = 1$ on this word. Only words of length one and the empty word survive. Their inclusion respects deconcatenation and proves the first assertion.

The ordinary two-sided free bar resolution computes the derived Hom. Dualizing the displayed coalgebra quasi-isomorphism gives the square-zero multiplication, since the surviving reduced classes are primitive. Dualization is exact over k. For A_{sq}, every product of two augmentation-ideal elements is zero, so the multiplication differential vanishes. Each weight is finite-dimensional. Dualizing the algebraic weight sum therefore gives the indicated product and the dual concatenation law. □

These formulas require an ordinary scalar augmentation. An empty word, a local vacuum and a prepared state have different definitions. None can replace that augmentation without a map. Likewise, the coefficient dual $(\bigoplus_n W_n)^* = \prod_n W_n^*$ changes a sum to a product. It does not by itself identify a field dual, an internal line dual or the boundary-junction construction of Chapter 15.

16.2.1 Componentwise tensor algebras and factorization

For a factorization coefficient F, restriction to two distinct points identifies $F_{\{1,2\}}$ with $F_{\{1\}} \otimes F_{\{2\}}$. Suppose these three coefficients are k. The componentwise sum $F \oplus F$ has coefficient k^2 at both one and two points, whereas factorization would require k^4 at two points.

The same difficulty applies to the naive componentwise tensor algebra. For the coefficient F just described, it has $k[t]$ at every nonempty configuration. The inherited separated map is

$$k[t] \longrightarrow k[t_1, t_2], \qquad t^n \longmapsto t_1^n t_2^n.$$

It misses t_1 and is not a factorization equivalence. Weight completion changes the map to $k[[t]] \to k[[t_1, t_2]]$ with the same missing coefficient.

The free and square-zero calculation remains valid in its ordinary algebraic category. Placing it in a factorization category requires the actual free construction there, or a comparison functor that preserves the required structure. Pointwise tensor closure alone does not give closure under the coproducts used to build a tensor algebra.

Proposition 16.3. *Let $F : C \to \mathcal{D}$ be a strong monoidal functor between monoidal categories in which the augmented simplicial bars and their geometric realizations exist. Suppose F preserves those realizations. Then for every augmented associative algebra A there is a natural equivalence*

$$F(B_C A) \simeq B_{\mathcal{D}}(FA).$$

It preserves the specified unit, augmentation and all simplicial maps.

Proof. In simplicial degree n, the strong monoidal structure identifies $F(A^{\otimes n})$ with $(FA)^{\otimes n}$. It carries adjacent-multiplication faces to faces, augmentation faces to augmentation faces, and unit insertions to degeneracies. The coherence axioms make these identifications a map of simplicial objects. Apply preservation of geometric realization. □

An ordered collision atlas and a category of consecutive word cuts have different diagrams. Applying the proposition to them first requires a functor on those diagrams and a comparison of their coefficient tensors. It also requires preservation of any support condition and completion used in the realization. An ordinary bar–cobar equivalence on either carrier does not supply this functor. The Lie-to-commutative-coalgebra chiral duality has yet another operadic source and target. It cannot replace the associative comparison.

16.2.2 *A finite Chevalley model is not a finite full bar*

For $A = U(\mathfrak{sl}_2)$, PBW gives the basis $e^a h^b f^c$, excluding the constant when forming the augmentation ideal I. Thus $I^{\otimes n}$ is countably infinite-dimensional for every $n \geq 1$. Its finite associated-graded pieces satisfy

$$\sum_{d \geq 0} \dim \mathrm{gr}_d^{\mathrm{PBW}}(I^{\otimes n})\, t^d = \big((1-t)^{-3} - 1\big)^n.$$

The numbers 3^n occur only in the least possible PBW degree $d = n$.

The Chevalley complex instead has dimensions 1, 3, 3, 1. With $d(x \wedge y) = [x, y]$,

$$d(e \wedge h) = -2e, \qquad d(e \wedge f) = h, \qquad d(h \wedge f) = -2f,$$

while

$$d(e \wedge h \wedge f) = -2e \wedge f - h \wedge h - 2f \wedge e = 0.$$

Its homology has dimensions 1, 0, 0, 1. The bar–Chevalley comparison identifies homology, not these different chain spaces.

16.3 Zeros, poles and polynomial shuffle lattices

Let $K(u)$ be a rational kernel. For symmetric functions in m and n variables, sum over the $\binom{m+n}{m}$ ordered two-block partitions. Insert one factor K for each cross-block pair. This defines their shuffle product. There is no factorial divisor.

An iterated product in three blocks has one summand for each ordered three-block partition and the same cross factors in either bracketing. This proves associativity wherever the stated coefficient space is closed under the formula.

Consider first the meromorphic kernel

$$K_{\mathrm{mer}}(u) = \frac{(u+\epsilon_1)(u+\epsilon_2)(u+\epsilon_3)}{u^3}, \qquad \epsilon_1 + \epsilon_2 + \epsilon_3 = 0.$$

Put $\sigma_2 = \sum_{i<j} \epsilon_i \epsilon_j$ and $\sigma_3 = \epsilon_1\epsilon_2\epsilon_3$. Then

$$K_{\mathrm{mer}}(u) = 1 + \frac{\sigma_2}{u^2} + \frac{\sigma_3}{u^3}, \qquad K_{\mathrm{mer}}(u) - K_{\mathrm{mer}}(-u) = \frac{2\sigma_3}{u^3}. \tag{16.2}$$

If the parameters are nonzero and pairwise distinct, $u = -\epsilon_a$ is a zero, not a pole. The reciprocal has residue

$$\mathrm{Res}_{u=-\epsilon_a} K_{\mathrm{mer}}(u)^{-1}\, du = \frac{(-\epsilon_a)^3}{\prod_{b \neq a}(\epsilon_b - \epsilon_a)}. \tag{16.3}$$

Both statements follow by factoring the numerator at that point.

Degenerate parameter fibers must be calculated before cancelling coincident zeros and poles.

For the convention using $K_{\mathrm{mer}}(z_1 - z_2)$, put $u = z_1 - z_2$ and $w = z_2$. The commutator of the degree-one inputs z^r, z^s is

$$\frac{2\sigma_3}{u^3}\big((w+u)^r w^s - (w+u)^s w^r\big).$$

For $(r, s) = (1, 0)$ it is $2\sigma_3/u^2$. For $(r, s) = (2, 0)$ it is $4\sigma_3 w/u^2 + 2\sigma_3/u$. The latter has residue $2\sigma_3$ at the true diagonal.

Neither expression has a pole at a generic nonzero shifted zero $u = -\epsilon_a$. To assign shifted residues to a geometric operation one must first specify the meromorphic coefficient of that operation.

16.3.1 *The coefficient lattice retained by multiplication*

Now use the distinct polynomial Hall carriers

$$\begin{aligned} R &= \mathbb{Q}[h_1, h_2], \qquad h_3 = -h_1 - h_2, \\ a &= -(h_1^2 + h_1 h_2 + h_2^2), \qquad b = -h_1 h_2 (h_1 + h_2), \\ N(v) &= v^3 + av + b, \qquad K_0(v) = N(v)/v, \qquad K_1(v) = vN(v). \end{aligned} \tag{16.4}$$

For $j = 0, 1$, independently set

$$\mathcal{A}^{(j)} = \bigoplus_{n \geq 0} A_n^{(j)}, \qquad A_n^{(j)} = R[x_1, \ldots, x_n]^{\mathfrak{S}_n}.$$

Every cochain degree and parity is zero. Charge is the separate grading n. Each element has finite charge support and polynomial coefficients. No parameter or root is inverted.

The product is

$$(F *_j G)(X) = \sum_{\substack{I \sqcup J = [d+e] \\ |I| = d}} F(X_I) G(X_J) \prod_{\substack{i \in I \\ k \in J}} K_j(x_k - x_i). \tag{16.5}$$

The unit is $1 \in A_0^{(j)} = R$. The constant polynomial $e_0 = 1 \in A_1^{(j)}$ has charge one and is not that unit.

Proposition 16.4. *Formula* (16.5) *gives a polynomial, unital associative algebra. Both kernels have exchange ratio*

$$r(v) = \frac{K_j(-v)}{K_j(v)} = \frac{v^3 + av - b}{v^3 + av + b}.$$

Proof. For K_1 polynomiality is immediate. For K_0, each possible denominator is a single root difference in a summand. Pair two summands by exchanging the labels of that pair between the two blocks. On their diagonal all other factors agree and the two residues are opposite. Every possible denominator therefore cancels in the polynomial unique factorization domain over R.

Symmetry follows by permuting the summands. The ordered-partition argument proves associativity in every charge. An empty block proves the unit identities. Finally $K_1 = v^2 K_0$, and this even factor cancels in the ratio. □

The ordinary zero-potential Hall product has precisely the numerator Euler factors from moving arrows and denominator factors from the flag tangent space [66, Section 2.4, Theorem 2]. Three moving arrow weights $v + h_i$ give K_0. Adding two zero-weight loops multiplies this kernel by v^2. This geometric interpretation does not identify these ordinary polynomial algebras with a critical Hall algebra or a native field theory.

In charge two put

$$c = (x + y)/2, \qquad v = y - x, \qquad t = v^2, \qquad B = R[c, t].$$

The entire input tensor is $R[x, y] = B \oplus vB$. For $p = A + vC$, its linearized product is

$$\mu_0(p) = 2\big((t + a)A + bC\big), \qquad \mu_1(p) = 2t\big((t + a)A + bC\big). \tag{16.6}$$

This follows by adding $K_j(v)p(x, y)$ and $K_j(-v)p(y, x)$. Hence the complete binary images are $J = (t + a, b)B$ and tJ.

Theorem 16.5. *There is no charge-preserving R-linear algebra isomorphism $\mathcal{A}^{(0)} \to \mathcal{A}^{(1)}$. The conclusion still holds for maps semilinear under one automorphism of R. Nevertheless the maps*

$$\Phi_n(F) = D_n F, \qquad D_n = \prod_{i<k} (x_k - x_i)^2, \qquad D_0 = D_1 = 1, \tag{16.7}$$

give a unital injective algebra map. They identify the two subalgebras generated by charge one. They also give an isomorphism after inverting D_n in each charge, and commute with simultaneous root translation.

Proof. An isomorphism would identify the cokernels of the two full binary products. But

$$B/J = R[c]/(b), \qquad B/tJ \twoheadrightarrow R[c], \quad F \longmapsto F(c, 0).$$

The first has parameter annihilator (b) and the second has annihilator zero. These cannot be isomorphic as R-modules. Under a coefficient automorphism, the nonzero annihilator becomes $(\sigma(b))$, so the semilinear case has the same contradiction.

For the positive map, the discriminant identity

$$D_{d+e}(X) = D_d(X_I) D_e(X_J) \prod_{\substack{i \in I \\ k \in J}} (x_k - x_i)^2$$

holds for every ordered partition. Replace K_1 by $v^2 K_0$ in each summand of $\Phi_d(F) *_1 \Phi_e(G)$. The discriminants combine to D_{d+e}, independent of the partition. This proves multiplicativity.

The polynomial rings are domains, so each Φ_n is injective. It is the identity in charges zero and one. Its image on the charge-one-generated subalgebra is exactly the corresponding generated subalgebra. Hence the restricted map is an isomorphism. Its inverse divides by D_n only on this specified image.

After inverting D_n, the same division is defined on the whole charge-n carrier. Products remain in this localization: every internal and cross root difference divides a power of the total discriminant. Finally $\sum_i \partial_{x_i} D_n = 0$, which proves translation compatibility. □

This is a distinction between the full polynomial realization and the smaller algebra generated by its currents. Equality of exchange ratios decides neither.

The quotient $Q_n = A_n^{(1)} / D_n A_n^{(1)}$ retains the discarded collision coefficients. It is an $\mathcal{A}^{(0)}$-bimodule through

$$a \cdot [z] = [\Phi(a) *_1 z], \qquad [z] \cdot a = [z *_1 \Phi(a)].$$

These are well-defined because Φ is multiplicative.

The quotient is not an algebra quotient: the image of Φ contains the unit and is a proper subalgebra. In charge two its cone is $[B \xrightarrow{t} B]$, with cohomology $B/(t)$ in degree zero. On the ordered cover this is the length-two normal quotient $R[c, v]/(v^2)$.

16.4 Cooperation depends on the tensor

The product on a polynomial shuffle carrier does not automatically admit a coproduct into the ordinary tensor product. The obstruction uses only charge one and charge two. Write $e_m = x^m \in A_1^{(j)}$. Formula (16.6) gives

$$[e_0, e_3]_{*_j} - 3[e_1, e_2]_{*_j} + a[e_0, e_1]_{*_j} = b(e_0 *_j e_0). \tag{16.8}$$

Indeed, the three commutators on the left are $2bt^j(x^2 + xy + y^2)$, $2bt^j xy$ and $2bt^j$, while $e_0 *_j e_0 = 2t^j(t + a)$.

Theorem 16.6. *Neither $\mathcal{A}^{(j)}$ admits a charge-preserving unital R-algebra map*

$$\Delta : \mathcal{A}^{(j)} \longrightarrow \mathcal{A}^{(j)} \otimes_R \mathcal{A}^{(j)}$$

with the ordinary tensor multiplication and with two-sided counit equal to projection onto charge zero. The same obstruction applies to the current-generated subalgebras, including a derived algebra map with the ordinary Koszul tensor. Coassociativity is not needed for this conclusion.

Proof. Charge preservation leaves only the splits $(1,0)$ and $(0,1)$ for each e_m. The two counits force $\Delta e_m = e_m \otimes 1 + 1 \otimes e_m$. Commutators of even primitive elements are primitive, since their mixed terms cancel. Thus the reduced coproduct of the left side of (16.8) is zero. Multiplicativity makes the reduced coproduct of its right side

$$2b\, e_0 \otimes e_0 \neq 0 \quad \text{in } A_1 \otimes_R A_1 = R[x, y].$$

This is a contradiction over the original coefficient ring.

For the ambient algebras all charge components are R-free. For a current-generated subalgebra, taking degree-zero cohomology of a proposed derived map still gives the same identity in the free $(1,1)$ component. Hence the derived formulation has the same contradiction. □

The detector can vanish on a special parameter fiber. The theorem asserts no uniform fibrewise obstruction when $b = 0$. At $(h_1, h_2, h_3) = (1, 1, -2)$, however, it becomes $-4e_0 \otimes e_0$, so the obstruction is already visible over $\mathbb{Q}$. Neither this result nor its proof concerns every possible cut-convolution coalgebra or every shifted coefficient tensor.

16.4.1 A rational cooperation with a different interchange

There is an explicit positive construction after localizing the spectral coefficients and changing the middle interchange. For ordered root lists X, Y, put

$$Q^{(j)}_{d,e}(X,Y) = \prod_{x \in X,\ y \in Y} K_j(y - x), \qquad \Delta^{\mathrm{rat}}_{d,e} F = \frac{F(X,Y)}{Q^{(j)}_{d,e}(X,Y)}.$$

The recipient consists of block-symmetric polynomials localized at all pair differences and their translates by $\pm h_i$. Parameters themselves are not inverted.

Let F have bidegree (d,e) and G have bidegree (d',e'). Define their product by summing over $X = X_1 \sqcup X_2$, $Y = Y_1 \sqcup Y_2$ of these sizes, with summand

$$\begin{aligned} &F(X_1,Y_1)G(X_2,Y_2) \prod_{\substack{x \in X_1 \\ x' \in X_2}} K_j(x' - x) \\ &\cdot \prod_{\substack{y \in Y_1 \\ y' \in Y_2}} K_j(y' - y) \prod_{\substack{y \in Y_1 \\ x' \in X_2}} r(y - x'). \end{aligned} \tag{16.9}$$

The last product is the middle interchange. Its factors would be absent from the ordinary tensor multiplication.

Proposition 16.7. *The multiplication* (16.9) *is associative and unital. The sum of the maps $\Delta^{\mathrm{rat}}_{d,e}$ is multiplicative into this recipient and is rationally coassociative. It has both charge-zero counits.*

Proof. For multiplicativity, consider a term with four ordered groups X_1, Y_1, X_2, Y_2. Divide its source shuffle product by the cross kernel between $X_1 \sqcup X_2$ and $Y_1 \sqcup Y_2$. The factors from X_1 to Y_2 cancel. The remaining middle ratio is

$$\frac{K_j(x' - y)}{K_j(y - x')} = r(y - x'), \qquad y \in Y_1,\ x' \in X_2.$$

The other factors are exactly those in (16.9), together with ΔF and ΔG. Summing over all four groups proves the identity.

For three inputs, either bracketing includes the same internal kernel factors and the same factors from an earlier right block to a later left block. This proves associativity. For three output lists, either iterated cooperation divides by $Q_{12}Q_{13}Q_{23}$ in the common localization. This proves coassociativity. The empty lists give the unit and both counits. Every input has finite charge support, so these sums are finite. □

The four-current relation is now compatible with

$$(1 - r(v))v(v^2 + a) = b(1 + r(v)),$$

which follows directly from $r(v) = (v^3 + av - b)/(v^3 + av + b)$. The mixed contribution excluded by the ordinary tensor is present. The construction still leaves the polynomial carrier:

$$\Delta^{\mathrm{rat}}_{1,1}(1 \in A_2) = \frac{v}{N(v)} \quad (j = 0), \qquad \Delta^{\mathrm{rat}}_{1,1}(1 \in A_2) = \frac{1}{vN(v)} \quad (j = 1).$$

It also removes the closed loci at its inverted denominators. An equivalence on this open coefficient domain is not yet a supported comparison across those loci.

16.4.2 Shifted collisions retain their excess

Keep the same coefficient ring R. On three position variables consider

$$f_{12} = z_1 - z_2 + h_1, \qquad f_{23} = z_2 - z_3 + h_1.$$

Their intersection is $R[z_3]$, with $z_2 = z_3 - h_1$ and $z_1 = z_3 - 2h_1$. The composed endpoint shift is $-2h_1$. Thus the finite list of shifts $0, \pm h_1, \pm h_2, \pm h_3$ is not closed under composition of these shifted collisions.

For an ordered list of equations, use the Koszul complex with degree-minus-one generators ξ_{ij} and $d\xi_{ij} = f_{ij}$. Adding the induced endpoint equation gives

$$f_{13} = z_1 - z_3 + 2h_1 = f_{12} + f_{23}.$$

The change of exterior generator $\eta = \xi_{13} - \xi_{12} - \xi_{23}$ yields

$$K(f_{12}, f_{23}, f_{13}) \simeq R[z_3] \otimes \Lambda(\eta), \qquad |\eta| = -1, \qquad d\eta = 0. \tag{16.10}$$

Indeed, f_{12}, f_{23}, z_3 are polynomial coordinates over R, so their first two Koszul pairs contract. η remains. Discarding the redundant equation would discard its nonzero negative-degree class.

The true endpoint diagonal instead restricts to $-2h_1$. Its derived restriction is

$$[R[z_3] \xrightarrow{-2h_1} R[z_3]]$$

in degrees $-1, 0$. Before specialization its only cohomology is $R[z_3]/(h_1)$ in degree zero. Derived specialization to $h_1 = 0$ makes the differential zero and retains both degrees.

There is a separate obstruction to viewing the scalar quotient as the unchanged supported differential module. For $P = R[z_1, z_2]$ and $f = z_1 - z_2 - \lambda$, with $\lambda \in R$, a relative differential-module structure on $P/(f)$ would satisfy $[\partial_{z_1}, f] = 1$. But f acts by zero on that module, so its commutator with every operator is zero. The quotient is nonzero, giving a contradiction.

The same argument applies to a differential-module complex whose underlying P-complex is quasi-isomorphic to $P/(f)$. Its degree-zero cohomology would have that impossible action. A supported differential-module realization must retain its normal derivatives and the maps attaching them. The scalar Koszul calculation does not supply those maps.

16.5 Matrix coefficients reconstruct a specified tensor category

The preceding examples distinguish a product, its cooperation and its coefficient domain. Tensor reconstruction begins with all of them encoded in a category. Let C be an essentially small k-linear monoidal category, and let

$$\omega : C \longrightarrow \mathrm{Vect}_{\mathrm{fd}}(k)$$

be a k-linear strong monoidal functor. Write $V_M = \omega(M)$ and use its monoidal identifications $V_M \otimes V_N \simeq V_{M\otimes N}$ and $V_{\mathbf{1}} \simeq k$. Choose a small skeleton when forming the direct sum below.

The matrix-coefficient coend is the vector space

$$H_\omega = \left(\bigoplus_{M \in C} V_M^* \otimes V_M \right) \Big/ \big\langle c_M(\lambda\omega(f), v) - c_N(\lambda, \omega(f)v) \big\rangle. \tag{16.11}$$

Here $f : M \to N$, $v \in V_M$, $\lambda \in V_N^*$, and $c_M(\lambda, v)$ denotes the class of the indicated tensor. The relations remember every morphism of C.

Proposition 16.8. *The formulas*

$$\begin{aligned} \Delta c_M(\lambda, v) &= \sum_i c_M(\lambda, e_i) \otimes c_M(e^i, v), \\ \varepsilon c_M(\lambda, v) &= \lambda(v) \end{aligned} \tag{16.12}$$

are independent of a basis (e_i) and its dual (e^i). They define a counital coassociative coalgebra. Its universal coaction is

$$\delta_M(v) = \sum_i e_i \otimes c_M(e^i, v).$$

The monoidal structure supplies the multiplication

$$c_M(\lambda, v) c_N(\mu, w) = c_{M\otimes N}(\lambda \otimes \mu, v \otimes w), \tag{16.13}$$

with unit $c_{\mathbf{1}}(1, 1)$. Together these operations make H_ω a bialgebra. If C is rigid, it has an invertible antipode.

Proof. The tensor $\sum_i e_i \otimes e^i$ is the identity of V_M, so its insertion in (16.12) is basis-independent. For $f : M \to N$, inserting this identity before or after $\omega(f)$ gives the same linear map. The coend relation therefore makes Δ well-defined. The same relation respects evaluation, so it also defines ε.

Inserting the identity twice gives

$$\sum_{i,j} c_M(\lambda, e_i) \otimes c_M(e^i, e_j) \otimes c_M(e^j, v)$$

for either iterated coproduct. Evaluating either outer pair gives $c_M(\lambda, v)$, proving both counits. These same calculations prove the coaction identities. The coend relation says exactly that each $\omega(f)$ is colinear.

Naturality of the monoidal constraints makes (16.13) compatible with the relations. Their associativity and unit coherences give the algebra laws. Using the basis $e_i \otimes f_j$ of $V_M \otimes V_N$ shows that Δ preserves multiplication. Evaluation shows the same for the counit.

For a left dual $M^\vee$, identify $V_{M^\vee}$ with V_M^*. Let $\widetilde{v} \in V_M^{**}$ be evaluation at v. Then

$$S\, c_M(\lambda, v) = c_{M^\vee}(\widetilde{v}, \lambda).$$

The evaluation morphism $M^\vee \otimes M \to \mathbf{1}$ and the coevaluation morphism $\mathbf{1} \to M \otimes M^\vee$ are morphisms of C. Applying their coend relations gives the two antipode identities: the two sums contract to $\lambda(v)c_{\mathbf{1}}(1, 1)$. Right duals give the inverse antipode by the corresponding right-dual formula. □

Coevaluation gives the coproduct, whereas the tensor product of objects gives multiplication. Composition has a third role. The full linear dual H_ω^* is the algebra of linear natural endomorphisms of ω. An endomorphism T gives the functional $c_M(\lambda, v) \mapsto \lambda(T_M v)$, and

$$(\phi_T * \phi_U)c_M(\lambda, v) = \sum_i \lambda(T_M e_i)e^i(U_M v) = \lambda(T_M U_M v).$$

Thus convolution is composition.

Tensor-compatible natural automorphisms must additionally preserve the monoidal constraints and the unit. They are not all elements of this dual algebra.

Theorem 16.9 (Tensor reconstruction). *Suppose C is k-linear abelian, with finite-dimensional morphism spaces and finite-length objects. Suppose it is rigid, its tensor product is exact in both variables, and ω is exact and faithful. Then the comparison given by δ_M is a monoidal equivalence*

$$C \simeq \mathrm{Comod}_{\mathrm{fd}}(H_\omega).$$

Proof. The coalgebra reconstruction theorem for an exact faithful functor to finite-dimensional vector spaces identifies C with the finite-dimensional right comodules of the coend (16.11) [35, Theorem 1.10.1]. Its comparison is exactly the coaction (16.12). Equation (16.13) identifies the coaction on $V_{M\otimes N}$ with the tensor product of the two coactions. The given monoidal constraints therefore make the equivalence strong monoidal, including its unit. Proposition 16.8 supplies the antipode, as in the Hopf reconstruction theorem [35, Theorem 5.4.1]. □

For a braided C, write $\beta_{M,N} : V_M \otimes V_N \to V_N \otimes V_M$ for the induced linear exchange. Its coefficient functional is

$$\mathbf{r}\big(c_M(\lambda, v), c_N(\mu, w)\big) = (\mu \otimes \lambda)\beta_{M,N}(v \otimes w). \tag{16.14}$$

Naturality makes this well-defined on both coends. The inverse braiding gives its convolution inverse.

Taking matrix coefficients of the two hexagon diagrams gives the two coquasitriangular identities. Taking coefficients of the colinearity square gives compatibility with multiplication. Thus (16.14) is the coquasitriangular form.

Under the reconstruction equivalence it recovers the braiding by

$$v \otimes w \longmapsto \sum w_{(0)} \otimes v_{(0)}\mathbf{r}(v_{(1)}, w_{(1)}), \qquad \delta(v) = \sum v_{(0)} \otimes v_{(1)}.$$

When H_ω is infinite-dimensional, the coefficient functional lives in $(H_\omega \otimes H_\omega)^*$. Regard H_ω^* with the inverse-limit topology from its finite coefficient coalgebras, and define $H_\omega^* \widehat{\otimes} H_\omega^* = (H_\omega \otimes H_\omega)^*$ with that topology. The functional is then the corresponding completed R-operator. This does not place it in the ordinary algebraic tensor $H_\omega^* \otimes H_\omega^*$. A meromorphic version additionally needs specified expansion maps, domains and continuity for its spectral parameter.

16.5.1 An exchange relation is not a presentation

For a selected object, the naturality of its exchange gives the matrix relation usually written $RT_1T_2 = T_2T_1R$. These relations define an algebra with a comparison map to the coend. They form a presentation precisely when that map is an isomorphism. The category's morphisms can impose further relations.

For example, compare finite-dimensional vector spaces with finite-dimensional $\mathbb{Z}$-graded vector spaces, using the ordinary flip and the forgetful fiber functor in both cases. For vector spaces, every map $k \to V$ is allowed. The coend relations therefore give

$$c_V(\lambda, v) = \lambda(v)c_k(1,1), \qquad H_\omega = k.$$

For graded vector spaces, decompose into homogeneous lines. There are no morphisms between distinct degrees. There is one coefficient u_n in each degree, and the formulas give

$$\Delta u_n = u_n \otimes u_n, \qquad u_n u_m = u_{n+m}, \qquad u_0 = 1, \qquad S(u_n) = u_{-n}.$$

Hence $H_\omega = k[t, t^{-1}]$ with $u_n = t^n$.

The exchange on a chosen one-dimensional fiber is the scalar 1 in both examples. The one-by-one RTT relation imposes no further condition on its coefficient t. In the first category the morphisms additionally impose $t = 1$. The same scalar exchange therefore leads to different reconstructed Hopf algebras.

The geometric action algebra of a collection of kernels records the operators they generate on a state space. The coend instead records natural symmetries respecting every chosen kernel morphism. Identifying these two constructions requires a comparison of their actions and tensor structures. A list of root multiplicities, or a common exchange ratio, provides neither that map nor its inverse.

16.6 Geometric comparison preserves equations, not only counts

Two positively graded Lie algebras can have the same graded PBW character and different brackets. Give x, y weight one and z weight two. The abelian Lie algebra on these generators and the Heisenberg Lie algebra with $[x, y] = z$ have the same PBW Hilbert series $((1-t)^2(1-t^2))^{-1}$. One enveloping algebra is commutative and the other is not. The equality of their characters therefore supplies no algebra isomorphism.

Similarly, the zero super vector space and a space with one even and one odd basis vector both have superdimension zero. An index cannot recover the forgotten parity dimensions.

An equivalence of underlying tangent complexes is also weaker than a comparison of deformation equations. Let L have basis u in degree one and v in degree two, with zero differential. Consider the two dg Lie structures

$$L_{\mathrm{ab}} : [\,,\,] = 0, \qquad L_{\mathrm{q}} : [u,u] = 2v, \quad [u,v] = [v,v] = 0.$$

The graded bracket on the odd element u is symmetric. Every iterated bracket is zero because v is central, so both structures satisfy Jacobi. Their tangent complexes are identical.

Proposition 16.10. *The first-order deformation ϵu over $k[\epsilon]/(\epsilon^2)$ extends to $k[\epsilon]/(\epsilon^3)$ for L_{ab} and does not extend for L_{q}. Thus the identity of tangent complexes cannot extend to an L_∞ equivalence between these deformation problems.*

Proof. The Maurer–Cartan equation is $d\alpha + \frac{1}{2}[\alpha, \alpha] = 0$. It vanishes identically in the abelian case. In the second algebra every possible lift has the form $\alpha = (\epsilon + c\epsilon^2)u$. Its curvature modulo ϵ^3 is $\epsilon^2 v \neq 0$. There is no degree-zero element giving a gauge correction.

Equivalently, an L_∞ map with identity first component would have to preserve the binary bracket up to the differential of its second component. Both differentials are zero, so it would identify the zero bracket with $[u, u] = 2v$, which is impossible. □

A geometric field comparison must consequently carry the nonlinear operations as well as its tangent map. For a cotangent BV theory it must also carry the pairing, action and boundary term. The classical master equation of a supplied coupled algebra proves the consistency of that algebra. It does not construct the action of an independently specified gravitational or matter field theory on it. At the quantum level, the chosen regularization and the local quantum master equation remain additional comparisons, as in Chapter 14.

16.6.1 A projector must respect the actual transition maps

A finite decomposition model has a simple idempotent: on a sum indexed by pairs (d_1, d_2), retain precisely the summands with $d_1 = d_2$. Writing its inclusion and retraction as ι, π gives

$$\pi\iota = 1, \qquad \iota\pi = e_{\mathrm{al}}.$$

It is an equivalence on the full object exactly when the off-aligned summands vanish. These equations determine the invertibility of this canonical comparison. They do not rule out every possible noninvertible distributive law.

To descend the projector, equality of indices must be compatible with each transition correspondence. Preserving equality alone is insufficient. The map $\{0, 1\} \to \{*\}$ preserves equal pairs but sends the off-diagonal pair $(0, 1)$ to the diagonal pair $(*, *)$. Extension along this map does not commute with the two projectors.

In a six-functor construction, assume the actual squares with the diagonal index locus are Cartesian. Assume also that their base-change maps are isomorphisms. Under those hypotheses, base change identifies the pull–push idempotents across every transition. The triple-diagonal identity then descends as well. Finite combinatorics alone does not verify those geometric squares.

16.7 Gravitational spectra, holonomy and boundary data

Three-dimensional Cartan gravity begins with independent fields e, ω. For negative cosmological constant, the two connections are $A^{\pm} = \omega \pm \ell^{-1}e$. On the invertible-coframe locus, their flatness equations give torsion zero and constant curvature. This is a field construction with an action comparison. Specifying two boundary central charges or a list of current spins does not supply these fields or their action on the boundary.

16.7.1 The principal embedding and its spin spectrum

Let $V_m = \mathrm{Sym}^m(k^2)$ be the irreducible $\mathfrak{sl}_2$ module of highest weight m. The principal embedding in $\mathfrak{sl}_N$ acts on its defining representation as V_{N-1}. Its adjoint therefore decomposes as

$$\mathfrak{sl}_N \cong \bigoplus_{s=2}^{N} V_{2s-2}, \qquad \sum_{s=2}^{N}(2s-1) = N^2 - 1. \tag{16.15}$$

Indeed, the tensor product $V_{N-1} \otimes V_{N-1}^*$ contains once each highest weight $0, 2, \ldots, 2N-2$.

This follows by subtracting successively the characters of those highest weights from its weight character. Finite-dimensional $\mathfrak{sl}_2$ modules are semisimple. Removing the scalar endomorphisms removes V_0, proving (16.15).

For a specified principal Chern–Simons theory linearized about its gravitational $\mathfrak{sl}_2$ background, these summands are the frame-like fields of spins $2, \dots, N$. With principal Drinfeld–Sokolov boundary conditions, the reduced classical currents have the same spins [19, 52]. This matches the representations in that construction. Arbitrary supplied W_N boundary data need a comparison from this boundary reduction. It must preserve the Poisson operations and the bulk coupling. The dimension identity alone does not construct it.

16.7.2 Connection coefficients and holonomy are different functions

Work over $\mathbb{C}$ in this subsection, and let $L > 0$. Consider a constant traceless companion matrix $A(q)$ of size N:

$$A_{i,i+1} = 1 \quad (1 \le i < N), \qquad A_{N,j} = q_{N+1-j} \quad (1 \le j < N), \qquad A_{N,N} = 0,$$

with all remaining entries zero. These are the constant coefficients of the corresponding companion-gauge oper. Expansion of the characteristic determinant gives

$$\det(\lambda I - A(q)) = \lambda^N - \sum_{s=2}^{N} q_s \lambda^{N-s}. \tag{16.16}$$

One can also recover the same polynomial from the eigenvector recursion $v_{i+1} = \lambda v_i$ and the final row.

For a circle of length L and the parallel-transport convention $\partial_\theta \psi = A(q)\psi$, the holonomy is $H(q) = \exp(LA(q))$. It is an exponential of the matrix, not its characteristic polynomial.

Proposition 16.11. *Holonomy conjugacy does not in general determine the coefficients q_s. Suppose $A(q)$ is regular semisimple. Choose a domain containing its eigenvalues on which $\lambda \mapsto e^{L\lambda}$ is injective. The corresponding logarithm branch then recovers its coefficients from holonomy conjugacy.*

Proof. For $N = 2$,

$$A(q) = \begin{pmatrix} 0 & 1 \\ q & 0 \end{pmatrix}, \qquad A(q)^2 = qI,$$

so the exponential series gives

$$H(q) = \cosh(L\sqrt{q})\, I + \frac{\sinh(L\sqrt{q})}{\sqrt{q}}\, A(q), \qquad \operatorname{Tr} H(q) = 2\cosh(L\sqrt{q}). \tag{16.17}$$

Both coefficients are entire functions of q. The second has value L at $q = 0$. For $L = 2\pi$ and $q = -n^2$, every positive integer n gives $H(q) = I$. Distinct connection coefficients therefore have the same holonomy. This is already a counterexample among constant companion connections.

Under the additional hypothesis, the distinct eigenvalues of $H(q)$ give the unordered eigenvalues $\lambda_i = L^{-1} \log \lambda_i(H)$ of $A(q)$. They determine its characteristic polynomial. Equation (16.16) then recovers every q_s. □

For nonconstant boundary fields the holonomy is a path-ordered exponential. Its traces are global functions of those fields. The local Drinfeld–Sokolov current brackets are obtained from Hamiltonian reduction. They are not identified with the algebra of these holonomy traces by a matrix characteristic polynomial. Boundary conditions, a chosen logarithm or spectral family, and the actual Hamiltonian comparison are therefore part of any charge-identification theorem. The constant example does not decide which of its connections satisfies a separately imposed physical regularity condition.

16.7.3 Algebraic removal of a central curvature

Let $\mathfrak{g}$ be a dg Lie algebra and let $\kappa \in \mathfrak{g}^2$ be closed and central. Adjoin a central generator u of degree one with $du = \kappa$. The differential remains square-zero and a derivation: the only new derivation equation is $[\kappa, x] = 0$. The class of κ becomes exact in the larger algebra. If κ is the curvature term of the corresponding curved Maurer–Cartan equation, twisting by $-u$ changes that term to

$$\kappa + d(-u) + \tfrac{1}{2}[-u, -u] = 0.$$

This is an explicit algebraic construction.

It can change the deformation problem: for $\mathfrak{g} = k\kappa$ with zero differential, the original degree-two cohomology is nonzero, whereas the extension $[ku \xrightarrow{d} k\kappa]$ is acyclic.

A geometric anomaly cancellation requires more. Suppose an independently constructed invertible field theory has the opposite anomaly. A comparison must identify both classes in the same complex and normalization. Under these hypotheses, tensoring the theories cancels their sum. Adjoining u does not construct that invertible theory, its descent data or the comparison of anomaly complexes. The algebraic curvature, a determinant-line curvature and a local BV obstruction retain their separate meanings until such maps are supplied.

16.8 The remaining geometric comparisons

The constructions above preserve definite operations. Ordinary bars retain augmented multiplication. The discriminant map retains the shuffle product and translation on its stated image. Rational cooperation retains a changed middle interchange on an open coefficient domain. Tensor reconstruction retains a specified category, fiber functor and all its morphisms. Cartan gravity and companion-matrix holonomy begin with actual connection fields.

To compare any of these with native chiral quantum symmetry, the map must be defined before its equivalence is asserted. For the shuffle models it must retain the integral binary images, the shifted collisions and the excess class (16.10), with a specified tensor and counit. For boundary reconstruction it must retain the derived local mapping objects and relative sewing of Problem 15.22. For a gravitational realization it must carry the nonlinear field equations, pairing, action and boundary reduction, followed by the chosen quantum and state comparisons. These requirements leave room for restricted coefficient domains, larger mapping categories and interfaces between different realizations. Each possibility has its own comparison map and its own test of invertibility.

Hamiltonian jets and boundary quantization

Quantization of a symplectic boundary changes the product of observables while retaining the classical Poisson bracket to first order. The formal symplectic plane gives a concrete setting in which to compare Hamiltonian operations with operations on a quantized brane. Pointed Hamiltonians vanish to order at least two; translations belong to a separately adjoined affine sector. Finite Hamiltonian jets and their shifted cotangent fields provide the local algebraic models.

The universal boundary brane leads to a Rees enveloping algebra. Its relative Hochschild complex is compared with a Chevalley complex whose differential carries the same Rees parameter. Duflo quantization gives the separate invariant comparison, and the quadratic Moyal calculation determines the oscillator Casimir quotient. These finite constructions distinguish the underlying cochain comparison from higher operations on it.

The all-arity open–closed problem asks for compatible closed and mixed operations on the boundary algebra. Its stated refinement supplies the operadic maps and coherence homotopies. Passing through the Hamiltonian jet limit also requires actual cochain transition systems and their continuous limit identifications. A geometric brane realization must supply the boundary module and compare its centre action with the chosen observable or trace maps; global realization further requires descent and quantization data.

Book entrance: Mixed Holomorphic–Topological Deligne Theory

Hamiltonian jets, shifted cotangent fields, universal branes, Duflo quantization, and the all-arity open–closed map.

Book I

The finite-jet laboratory

Chapter 17

Moyal quantization in coordinates

17.1 The bidifferential formula

For polynomials on $\mathbb{C}^2$, the Moyal product is

$$f \star g = \sum_{n \geq 0} \frac{1}{n!} \left(\frac{\hbar}{2} \right)^n \sum_{r=0}^{n} (-1)^r \binom{n}{r} (\partial_1^{n-r} \partial_2^r f)(\partial_1^r \partial_2^{n-r} g). \tag{17.1}$$

Every polynomial pair gives a finite sum.

Theorem 17.1 (Quadratic exactness)**.** *For polynomials f, g of degree at most two,*

$$[f, g]_\star = \hbar \{f, g\}.$$

The quadratic generators therefore define a Lie representation of the Rees-scaled $\mathfrak{sl}_2$. Their associative algebra has an additional Casimir relation.

Proof. The commutator retains the odd powers of the Poisson bidifferential. Derivatives of order three vanish on quadratics, so the first-order term is the entire commutator. □

Proposition 17.2 (The oscillator quotient)**.** *Let $R_N = \mathbb{C}[\hbar]/(\hbar^{N+1})$ and let $W_N = R_N[q, p]$ have the Moyal product for $\{q, p\} = 1$. Let U_N have generators E, F, H with relations*

$$[H, E] = 2\hbar E, \qquad [H, F] = -2\hbar F, \qquad [E, F] = \hbar H.$$

Set $\Omega = EF + FE + H^2/2$. The assignments

$$E \longmapsto e = q^2/2, \qquad F \longmapsto f = -p^2/2, \qquad H \longmapsto h = -qp$$

induce an algebra isomorphism

$$U_N/(\Omega + 3\hbar^2/8) \xrightarrow{\sim} W_N^{(q,p) \mapsto (-q, -p)}.$$

These compatible isomorphisms also hold after $\hbar$-adic completion. If linear Hamiltonians are retained, the generated associative algebra is the full Weyl algebra.

Proof. Quadratic exactness proves the three commutator relations. The associative products are

$$\begin{aligned} e \star f &= -q^2 p^2/4 - \hbar qp/2 - \hbar^2/8, \\ f \star e &= -q^2 p^2/4 + \hbar qp/2 - \hbar^2/8, \\ h \star h &= q^2 p^2 - \hbar^2/4. \end{aligned}$$

Thus Ω maps to $-3\hbar^2/8$. Its commutator with each of E, F, H vanishes by the defining relations. Give these generators PBW degree one. Give an even monomial $q^a p^b$ degree $(a + b)/2$. The leading symbol of $\Omega + 3\hbar^2/8$ is

$C = 2EF + H^2/2$. It is a non-zero-divisor in $R_N[E, F, H]$, since $2C$ is monic in H. Centrality and this property give

$$\operatorname{gr}\big(U_N/(\Omega + 3\hbar^2/8)\big) = R_N[E, F, H]/(C).$$

Indeed, every element of the two-sided ideal is a multiple of the central generator. Its leading symbol is the product of the two leading symbols. The quotient has basis $E^a H^b F^c$ with $a, c \geq 0$ and $b = 0, 1$. Their leading symbols map to unit multiples of the distinct monomials $q^{2a+b} p^{2c+b}$, precisely the even monomials. Hence the associated-graded map is an isomorphism. Lifting highest-degree terms proves surjectivity and comparing them proves injectivity. Every element at fixed N has finite polynomial degree. The formulas commute with reduction in N, so their inverse limit gives the formal isomorphism. Finally q, p generate W_N and topologically generate its formal limit. □

The primitive coproduct on U_N does not descend to this quotient. Modulo $\hbar$, it sends the classical relation to

$$\Delta(C) = C \otimes 1 + 1 \otimes C + 2E \otimes F + 2F \otimes E + H \otimes H.$$

The cross-term is nonzero in the tensor square of $\mathbb{C}[E, F, H]/(C)$. For example, evaluation at $(E, F, H) = (1, 0, 0)$ in the first factor and $(0, 1, 0)$ in the second gives 2. Thus the oscillator quotient is an associative algebra, with no coproduct inherited from these primitive generators.

Calculation 17.3 (Cubic correction)**.** For $f = z_1^3$ and $g = z_2^3$,

$$f \star g - g \star f = 9\hbar z_1^2 z_2^2 + \frac{3}{2}\hbar^3.$$

The first term quantizes the quartic Hamiltonian channel. The constant is the first wheel-free ordering correction visible beyond the quadratic Lie algebra.

Proof. In (17.1), the odd orders are $n = 1$ and $n = 3$. The first contributes $9\hbar z_1^2 z_2^2$. The third contributes $2(\hbar/2)^3(3!) = 3\hbar^3/2$. □

17.2 Weyl generators

With the convention $\{q, p\} = 1$, the symbols satisfy $[q, p]_\star = \hbar$. Their Weyl-ordered differential operators are

$$\widehat{q} = q, \qquad \widehat{p} = -\hbar\partial_q, \qquad [\widehat{q}, \widehat{p}] = \hbar.$$

The abstract Weyl algebra is defined by this commutator relation. Weyl ordering transports its multiplication to the Moyal product. The differential-operator representation realizes these relations over $\mathbb{C}[\hbar]$ and its formal completion. The finite quotient argument above uses the abstract R_N-algebra, so it does not require a faithful differential-operator representation at nilpotent $\hbar$.

Chapter 18

The quadratic invariant

18.1 The $\mathfrak{sl}_2$ invariant

For the quadratic jet, identify the coordinate functions with e, f, h. The invariant

$$q = 2ef + \frac{1}{2}h^2$$

is d_{CE}-closed. Its formal Duflo image is $\Omega + \hbar^2/2$, where $\Omega = ef + fe + h^2/2$ in the Rees enveloping algebra. This is the degree-zero invariant comparison. Under the oscillator quotient of Proposition 17.2, its image is $\hbar^2/8$.

Chapter 19

All-arity open–closed operations

19.1 Hochschild braces

For Hochschild cochains $F \in C^p(A, A)$ and $G_i \in C^{q_i}(A, A)$, the brace

$$F\{G_1, \dots, G_r\}$$

is the signed sum over ordered insertions of the G_i into the inputs of F. The brace identities state that a nested insertion is the sum over all ways of distributing the inner insertions among the outer slots.

Theorem 19.1 (Brace realization with a refinement)**.** *For $A = A_d$, supply the homotopy-E_2 and mixed refinements of Corollary 12.6. They identify the selected homotopy Gerstenhaber operations on $\mathcal{P}_d$ with the relative Hochschild centre operations. The chosen mixed refinement also identifies their action on the boundary algebra.*

Proof. The relative Hochschild complex has its brace operations. The supplied operadic equivalence preserves their coherent homotopies. An equivalence of complexes alone would not imply this conclusion. □

19.2 The first mixed identities

A relative Hochschild 1-cocycle D of an algebra concentrated in degree zero is an R-linear derivation:

$$D(ab) = D(a)b + aD(b).$$

For two such cocycles D, E, their Gerstenhaber bracket acts by

$$[D, E](a) = D(E(a)) - E(D(a)).$$

For a graded algebra and homogeneous derivations, these formulas become

$$D(ab) = D(a)b + (-1)^{|D||a|}aD(b),$$
$$[D, E](a) = D(E(a)) - (-1)^{|D||E|}E(D(a)).$$

Higher Hochschild cochains act through multilinear evaluations and braces. They need not act by derivations or carry a coproduct defining a diagonal action. A compatible mixed refinement transports these operations, including their homotopies.

Corollary 19.2 (Continuous open–closed structure with compatible jets)**.** *Under the jet comparison and mixed refinement hypotheses of Theorem 4.8, the pair $(\mathcal{P}_\infty, \widehat{U}_\hbar(\mathfrak{h}))$ has the corresponding complete homotopy Swiss-cheese action. Its closed color is equivalent to the continuous relative derived centre.*

Proof. The supplied finite operations commute with the transition maps and converge in the specified completion. Their limit acts on the two limit objects. The comparison theorem identifies the closed complex with continuous relative Hochschild cochains. □

Chapter 20

Finite Chan–Paton ranks

20.1 Rank one

For $N = 1$, the commuting equation is automatic. The boundary coordinate algebra is $\mathbb{C}[x, y]$, and

$$J_1(f) = f(x, y).$$

The necklace bracket reduces to the ordinary Poisson bracket. Restriction to $f \in \mathfrak{m}^2$ gives the defining pointed-Hamiltonian representation. The two linear Hamiltonians act by translations and recover the affine center-of-mass extension.

20.2 Rank two and Cayley–Hamilton reduction

For a 2×2 matrix X,

$$X^2 - (\operatorname{Tr} X)X + (\det X)I = 0.$$

Hence every trace containing two consecutive powers of X reduces to traces with X-degree at most one after adjoining $\operatorname{Tr} X$ and $\det X$. The same statement holds for Y. On the commuting locus, simultaneous diagonalization identifies the invariant algebra with the symmetric polynomials in two ordered pairs.

Proposition 20.1 (First finite-rank trace relation)**.** *For commuting 2×2 matrices,*

$$p_{20} = p_{10}^2 - 2 \det X, \qquad p_{02} = p_{01}^2 - 2 \det Y,$$

and

$$p_{11}^2 - p_{20} p_{02} = -\big(x_1 y_2 - x_2 y_1\big)^2.$$

The stable generators remain independent until a relation forced by the rank is reached.

Proof. Write the joint eigenvalues as (x_1, y_1) and (x_2, y_2) and expand both sides. □

20.3 Stable passage

Adding the origin defines the rank transition. Theorem 12.3 identifies the stable graded algebra and its total-degree completion. Proposition 12.4 retains the central rank and identifies the stable Poisson sector of degree at least two. A quotient map from a derived centre requires a representation and trace comparison on the specified complexes; the stable ring calculation alone does not construct that map.

Book II

Global holomorphic symplectic descent

Chapter 21

Fedosov descent

21.1 The Weyl algebroid

All Hochschild cochains and Morita centres in this chapter are continuous and relative to $R = \mathbb{C}[\![\hbar]\!]$. Let (S, ω) be a holomorphic symplectic surface. Choose a formal Poisson tensor $\pi_\hbar = \hbar\omega^{-1} + O(\hbar^2)$ and a Weyl algebroid $\mathcal{W}_{S,\hbar}$, with specified holomorphic or Dolbeault Fedosov resolutions and coherent overlap maps. The formal Poisson class is part of these data. It is not determined by the finite Hamiltonian Lie-cochain comparison.

Theorem 21.1 (Global comparison with formality and descent data)**.** *Suppose the resolutions carry the coherent complete homotopy-*E_2 *comparison of Theorem 15.20, with quasi-isomorphic associated graded maps. Then there is an equivalence*

$$\Phi_S : (\mathrm{PV}_S[\![\hbar]\!], [\pi_\hbar, -]) \xrightarrow{\sim} C^\bullet_{\mathrm{HH}}(\mathcal{W}_{S,\hbar}, \mathcal{W}_{S,\hbar}),$$

and the corresponding equivalence on derived global sections. A compatible gauge equivalence of the supplied resolutions transports this comparison. Equivariance requires equivariant comparison and descent data.

Proof. The complete filtered comparison is an equivalence on every multiple intersection by Lemma 15.19. The homotopy limit of its descent diagram preserves that equivalence. Apply derived global sections and the supplied gauge or equivariance homotopies. □

Corollary 21.2 (Global Morita centre with a continuous model)**.** *Assume in addition that the chosen continuous Morita theory computes the centre of* $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$ *by derived global sections of these relative Hochschild cochains. Then*

$$Z_{\mathrm{Mor}}\big(\mathrm{Perf}(\mathcal{W}_{S,\hbar})\big) \simeq \mathbb{R}\Gamma(S, \mathrm{PV}_S[\![\hbar]\!], [\pi_\hbar, -]).$$

This applies to a K3 surface when the same comparison data are supplied. A Hamiltonian BF identification additionally requires a map from its Chevalley observables to this symplectic polyvector complex.

Proof. Use the stated Morita model and the preceding derived global equivalence. □

Chapter 22

Factorization and descent

22.1 The algebraic mixed-HT realization

Retain the complete homotopy-E_2 sheaf

$$\mathcal{P}_S = (\mathrm{PV}_S[\![\hbar]\!], [\pi_\hbar, -]).$$

For each holomorphic open, its factorization realization uses two framed topological directions. Suppose these realizations have the cosheaf descent needed in the holomorphic variable. They then define an algebraic mixed factorization theory

$$\mathcal{A}^{\mathrm{alg}}_{S,\mathrm{bulk}} \quad \text{on} \quad \mathbb{R}^2_{\mathrm{top}} \times S.$$

The supplied formality comparison realizes the corresponding equivalence with the relative Hochschild centre. An oriented realization requires rotation-equivariant structure. Identifying $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$ with a universal physical boundary category requires the independent module, action and generation comparisons of Chapter 15. Neither the factorization realization nor the scalar centre calculation supplies them.

22.2 BV realization

The cyclic Lie sheaf of formal Hamiltonians has the shifted-cotangent action

$$S_{\mathrm{BF}}(A, B) = \int_{\mathbb{R}^2 \times S} \left\langle B, (\mathrm{d} + \bar{\partial}) A + \frac{1}{2}[A, A] \right\rangle.$$

Here S is a complex symplectic surface, so $\mathbb{R}^2 \times S$ has six real dimensions. A comparison of the classical observables with $\mathcal{A}^{\mathrm{alg}}_{S,\mathrm{bulk}}$ must specify a map from Hamiltonian Chevalley cochains to the symplectic polyvector complex. The finite Lie algebra identification does not supply it. Gauge fixing, extensions of graph weights and a solution of the quantum master equation would construct a perturbative factorization theory. Identifying its bulk–boundary action with the algebraic map additionally requires a comparison preserving support, units and collision homotopies. The finite jet zero modes carry the exact quantum measure constructed above. The three-dimensional half-space construction and its interacting comparison problem are treated separately in Chapter 14.

Book III

Matrix branes and trace geometry

Chapter 23

The derived commuting stack

Let $X, Y \in \mathfrak{gl}_N$ and let $\psi \in \mathfrak{gl}_N[1]$. The Koszul differential

$$QX = QY = 0, \qquad Q\psi = [X, Y]$$

resolves the commuting equation. The simultaneous conjugation action gives the boundary cdga

$$\mathcal{A}_N = C^*_{\mathrm{CE}}\big(\mathfrak{gl}_N, \mathrm{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])\big).$$

Cyclic differentiation gives moment-preserving matrix derivations for cyclic Hamiltonians. A polynomial Hamiltonian requires a cyclic lift, and these lifts do not give a strict Lie map on all polynomials: the cubic defect is (9.2). Theorem 9.4 constructs the coherent action on the free resolution, and Proposition 9.7 extends its matrix realization through the gauge complex. Their higher relation homotopies are part of the action. A strict continuous representation of the completed enveloping algebra would require an additional comparison in that same recipient.

Chapter 24

Necklace Hamiltonians

Let

$$F = \mathbb{C}\langle x, y\rangle, \qquad F_{\mathrm{cyc}} = F/[F, F].$$

For a cyclic word f, define cyclic derivatives by cutting the word at every occurrence of the selected letter. Put

$$\delta_f(x) = -\partial_y^{\mathrm{cyc}} f, \qquad \delta_f(y) = \partial_x^{\mathrm{cyc}} f.$$

Theorem 24.1 (Necklace Lie algebra)**.** *The bracket*

$$\{f, g\}_{\mathrm{neck}} = \left[\partial_x^{\mathrm{cyc}} f\ \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f\ \partial_x^{\mathrm{cyc}} g\right]_{\mathrm{cyc}}$$

makes $F_{\mathrm{cyc}}/\mathbb{C}$ a Lie algebra. The assignment $f \mapsto \delta_f$ is a Lie morphism. It preserves the moment map:

$$\delta_f([x, y]) = 0.$$

Proof. The cyclic Euler identity is

$$[x, \partial_x^{\mathrm{cyc}} f] + [y, \partial_y^{\mathrm{cyc}} f] = 0.$$

It gives the moment-map statement. The canonical double Poisson bracket on the free algebra descends to the displayed necklace bracket; its Jacobi identity gives the first assertion. Evaluation on generators gives $[\delta_f, \delta_g] = \delta_{\{f,g\}_{\mathrm{neck}}}$. □

Theorem 24.2 (Trace moment map)**.** *For*

$$J_N(f) = \operatorname{Tr} f(X, Y),$$

one has

$$\{J_N(f), J_N(g)\}_{\mathcal{M}_N} = J_N\big(\{f, g\}_{\mathrm{neck}}\big),$$

where $\mathcal{M}_N = \mathfrak{gl}_N^2$ carries $\operatorname{Tr}(\mathrm{d}X \wedge \mathrm{d}Y)$.

Proof. The differential of a trace is the matrix evaluation of the cyclic derivative. Contracting the two differentials with the inverse trace symplectic form gives the necklace formula. □

Chapter 25

The two-brane cone

Separate center-of-mass and relative coordinates. Put

$$u = x_1 - x_2, \qquad v = y_1 - y_2, \qquad E = \frac{1}{2}u^2, \quad F = -\frac{1}{2}v^2, \quad H = -uv.$$

Then

$$H^2 + 4EF = 0.$$

Proposition 25.1 (The A_1 Kirillov–Kostant cone)**.** *With* $\{u, v\} = 2$*,*

$$\{H, E\} = 4E, \qquad \{H, F\} = -4F, \qquad \{E, F\} = 2H.$$

After the evident rescaling this is the $\mathfrak{sl}_2$ *Lie–Poisson bracket on the nilpotent cone.*

Proof. The three identities follow by direct differentiation. The quadratic equation is the determinant-zero equation for the traceless matrix

$$\begin{pmatrix} H & 2E \\ 2F & -H \end{pmatrix}.$$

□

Book IV

Wheel corrections and exact quantum structure

Chapter 26

The local centre with comparison data

Assume a compatible jet comparison and a complete homotopy-E_2 refinement as in Theorem 4.8. Put

$$\mathcal{A} = \widehat{U}_\hbar(\mathfrak{h}), \qquad \mathcal{A}^{\mathrm{alg}}_{\mathrm{bulk}} = \mathrm{Fact}_{\mathsf{E}_2}(\mathcal{P}_\infty).$$

The factorization envelope uses two framed topological directions. An oriented refinement requires the corresponding rotation-equivariant structure. The formal holomorphic coefficient remains at the Darboux point. Let C_∂ be the category of perfect complete $\mathcal{A}$-modules in the chosen continuous R-linear Morita theory. Assume its centre is computed by $C^\bullet_{\mathrm{HH,cont}}(\mathcal{A}/R, \mathcal{A})$.

Theorem 26.1 (Local derived-centre comparison)**.** *These data give an equivalence of complete homotopy-E_2 factorization algebras*

$$\mathcal{A}^{\mathrm{alg}}_{\mathrm{bulk}} \xrightarrow{\sim} Z^{\mathrm{der}}_{\mathrm{Mor},R}(C_\partial).$$

At the formal insertion point the map is the selected refinement of Φ_∞. Every continuous boundary module has the induced derived centre action.

Proof. The compatible jet theorem gives the complete homotopy-E_2 comparison. The factorization envelope preserves this equivalence. The assumed continuous Morita model identifies its target with the centre. A centre element is a derived natural endomorphism of the identity functor, so evaluation gives the action on each boundary module. □

Corollary 26.2 (Global K3 comparison with descent data)**.** *Let S be a K3 surface with holomorphic symplectic form. Suppose the local comparisons extend to a sheafwise equivalence with the relative Hochschild centre of* $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$*. Suppose further that the selected cyclic and mixed refinements extend and satisfy descent in the same completed category. Then derived global sections give the corresponding global centre, cyclic, and open–closed comparisons.*

Proof. Derived global sections preserve equivalences of sheaves. Apply them to the supplied sheafwise maps and their coherence homotopies. □

Chapter 27

Synthesis

At each fixed Hamiltonian jet, the Rees enveloping algebra has relative Hochschild complex equivalent to the Chevalley complex with differential $\hbar d_{\mathrm{CE}}$. PBW antisymmetrization proves this over every R_N and in the formal coefficient limit. The Duflo theorem gives the separate multiplicative comparison on invariant functions. Quadratic Moyal Hamiltonians generate the oscillator quotient of the enveloping algebra. Their associative Casimir relation prevents the primitive coproduct from descending.

The Hamiltonian jet limit requires the compatible cochain systems of Theorem 4.8. With its operadic refinement and continuous Morita model, the centre acts on every supplied continuous boundary module M through

$$\mathcal{P}_\infty \xrightarrow{\Phi_\infty} C^\bullet_{\mathrm{HH,cont}}(\mathcal{A}/R, \mathcal{A}) \longrightarrow \mathrm{RHom}_{\mathcal{A}}(M, M).$$

The last arrow evaluates a derived natural endomorphism at M. A matrix-brane realization must construct such a module and compare its centre action with the selected trace functions. A commuting square with scalar representation invariants requires these additional comparison maps. Likewise, Fedosov descent concerns symplectic polyvectors and a Weyl algebroid. Its relation to Hamiltonian Lie cochains requires the specified map between their carriers.

Book V

Technical appendices to the preceding book

Chapter 28

Bibliographic note

The finite affine formality map and the separate symplectic globalization use Kontsevich formality and its Fedosov resolution [18, 31, 64]. The relative Hochschild model is the Cartan–Eilenberg comparison, with both Chevalley terms scaled by $\hbar$. Its finite-base resolution proof is Theorem 3.8. Jet limits and operadic refinements require the additional data stated above. The all-arity centre action is the Deligne–Tamarkin brace action [65, 77]. The global deformation-quantization category is the holomorphic Weyl algebroid. The matrix trace and necklace formulas are the noncommutative symplectic moment-map construction.

Book entrance: Ordered and Associative Chiral Geometry

Collision on a complex curve, order on a transverse line, exchange in a plane, and the three corresponding bar constructions.

Book **VI**

The invariant object

Chapter 29

Two directions of locality

29.1 The local experiment

Let X be a smooth complex algebraic curve and let $I \subset \mathbb{R}$ be an oriented interval. Place two protected observables at (z_1, t_1) and (z_2, t_2) in $X \times I$. The limits $z_1 \to z_2$ and $t_1 \to t_2$ carry independent information. The first limit produces a singular distribution along the diagonal of X^2. The second composes operators in the order selected by the orientation of I.

Principle 29.1 (Two-direction principle). Holomorphic collision belongs to the factorization coefficient on X. Ordered fusion belongs to an internal E_1 multiplication. Exchange belongs to an additional two-dimensional transport structure.

This separation gives five geometric carriers:

carrier	operation
diagonal in X^2	singular chiral operation
oriented interval	ordered fusion and bar construction
oriented plane	braid transport
circle	Hochschild trace
stable curve	modular sewing.

Each later structure is introduced by the datum carried by its row.

29.2 Carrier and operation

For the diagonal embedding $\Delta : X \hookrightarrow X^2$ and its complement $j : X^2 \setminus \Delta(X) \hookrightarrow X^2$, a Beilinson–Drinfeld chiral Lie operation is a morphism

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

in the chiral pseudo-tensor category of right D-modules [9]. The D-module is the carrier. The morphism μ is the operation. A factorization object packages carriers on all powers X^I together with their disjoint-locus structure maps. The complete operator product is the pair consisting of that system and its multiplication maps.

In a formal coordinate z near the diagonal, the pole filtration of $\mathcal{O}_{X^2}(*\Delta)$ has

$$\operatorname{gr}^F_m \mathcal{O}_{X^2}(*\Delta) \cong \mathcal{O}_\Delta(m\Delta) \cong N_{\Delta/X^2}^{\otimes m}, \qquad m \geq 1.$$

A principal part of order m therefore has an intrinsic normal symbol. Coordinate changes act triangularly on the filtered object. Residues recover the first symbol; normal jets recover the higher symbols.

Chapter 30

The Ran coefficient category

30.1 The fixed six-functor setting

Fix a presentable stable category $\mathcal{D}_X$ of right D-modules on $\mathrm{Ran}(X)$ with the following operations:

(i) the pointwise tensor product $\otimes^!$;

(ii) chiral convolution $\otimes^{\mathrm{ch}}$ along disjoint union;

(iii) base change and projection formulae for the disjoint-union correspondences;

(iv) geometric realizations and totalizations preserved by the tensor products used below.

The standard renormalized D-module setting of chiral Koszul duality has these operations [38]. Every construction below is formed in this fixed setting.

Let

$$\mathrm{Ran}(X)\times\mathrm{Ran}(X) \xleftarrow{\ j\ } (\mathrm{Ran}(X)\times\mathrm{Ran}(X))_{\mathrm{disj}} \xrightarrow{\ u\ } \mathrm{Ran}(X)$$

be the disjoint-union correspondence. Define

$$F\otimes^{\mathrm{ch}} G = u_! j^!(F\boxtimes G).$$

The pointwise product is induced by the diagonal of $\mathrm{Ran}(X)$.

Definition 30.1 (Aligned-decomposition projector)**.** For a finite set I, let $\mathrm{Dec}(I)$ be the finite groupoid of ordered decompositions $I = A\sqcup B$ and let

$$a_I : \mathrm{Dec}(I)\longrightarrow \mathrm{Dec}(I)\times\mathrm{Dec}(I), \qquad d\longmapsto (d,d)$$

be the diagonal. In the linear species model define

$$e_I = a_{I,!}a_I^!.$$

Thus e_I projects a pair of decompositions onto the summands in which the two decompositions agree. In the six-functor Ran model the same formula denotes extraordinary pullback to the diagonal correspondence followed by the available renormalized pushforward.

Theorem 30.2 (Aligned sector in the finite decomposition model)**.** *Let $\mathcal{S}_I$ be a stable category indexed by ordered decompositions of a finite set I, and assume the direct-sum decomposition*

$$\mathcal{M}_I \simeq \bigoplus_{d_1,d_2\in\mathrm{Dec}(I)} \mathcal{M}_{d_1,d_2}.$$

The projection

$$e_I : \mathcal{M}_I\longrightarrow\mathcal{M}_I, \qquad e_I\big|_{\mathcal{M}_{d_1,d_2}} = \begin{cases}\mathrm{id}, & d_1 = d_2,\\ 0, & d_1\neq d_2,\end{cases}$$

is idempotent. Its image is the aligned sector

$$\mathcal{M}_I^{\mathrm{al}} = \bigoplus_{d \in \mathrm{Dec}(I)} \mathcal{M}_{d,d}.$$

The family e_I is natural under bijections and compatible with iterated disjoint union. A mixed distributive law in this model is a natural transformation defined on $\mathcal{M}_I^{\mathrm{al}}$ whose two restrictions to the triple diagonal $d_1 = d_2 = d_3$ agree.

Proof. Idempotence is checked on each direct summand. Naturality follows because a bijection carries an ordered decomposition to an ordered decomposition and preserves equality of the two indices. For three products, both iterated alignments project

$$\bigoplus_{d_1,d_2,d_3} \mathcal{M}_{d_1,d_2,d_3} \quad \text{onto} \quad \bigoplus_{d} \mathcal{M}_{d,d,d}.$$

The two projectors are therefore equal. This equality is precisely the triple-alignment coherence. □

Theorem 30.3 (Descent of an aligned projector)**.** *Suppose the common-decomposition inclusions are clopen on the chosen charts. Suppose every transition correspondence gives Cartesian squares with these inclusions and the corresponding Beck–Chevalley maps are isomorphisms. Then the functors*

$$e_I = a_{I,!} a_I^!$$

assemble to a natural idempotent e_{al} on the mixed correspondence object. Its image is the common-decomposition correspondence. The two iterated projectors on three decomposition indices coincide, and the assumed base-change maps transport this equality through the transitions.

Proof. Restriction and extension by zero along a clopen inclusion form a projector. The Cartesian-square hypothesis and base change identify these composites across transitions. For three indices, both iterated projectors select the same locus $d_1 = d_2 = d_3$, so the identification respects triple alignment. □

The Cartesian-square hypothesis must be checked on the actual correspondences. Preserving equal indices alone does not suffice: a map can also send distinct indices to equal ones, as the map $\{0, 1\} \to \{*\}$ does. On a common disjoint-union correspondence, the projection formula gives the aligned comparison. Its inclusion and retraction are inverse on the full object exactly when every off-aligned summand vanishes. This criterion concerns that canonical comparison. It does not exclude every noninvertible distributive law.

30.2 Factorization coalgebras

A chiral commutative coalgebra is a commutative coalgebra object for $\otimes^{\mathrm{ch}}$. It is factorizing when every coproduct becomes an equivalence on the relevant disjoint locus. Let

$$\mathrm{Fact}_D(X) \subset \mathrm{CoCAlg}_{\mathrm{ch}}(\mathcal{D}_X)$$

be the full subcategory of factorizing objects. Chiral Koszul duality identifies the corresponding Lie-theoretic and factorization descriptions on the completed Ran category [38, 53].

Proposition 30.4 (Pointwise closure)**.** *The componentwise tensor product of two factorization objects is a factorization object. Thus* $\mathrm{Fact}_D(X)$ *is symmetric monoidal under $\otimes^!$ whenever the fixed Ran enhancement is closed under that tensor product.*

Proof. For a decomposition $I = I_1 \sqcup I_2$, restrict to its disjointness locus. The factorization equivalences of F and G give

$$\begin{aligned} j^!(F_I \otimes^! G_I) &\simeq j^! F_I \otimes^! j^! G_I \\ &\simeq (F_{I_1} \boxtimes F_{I_2}) \otimes^! (G_{I_1} \boxtimes G_{I_2}) \\ &\simeq (F_{I_1} \otimes^! G_{I_1}) \boxtimes (F_{I_2} \otimes^! G_{I_2}). \end{aligned}$$

Compatibility under refinement and permutation follows componentwise from the corresponding compatibilities for F and G. □

Chapter 31

The two noncommutative chiral objects

31.1 Transverse order

Two operators at the same chiral support can compose in an independent oriented real direction. Their multiplication is internal to the pointwise coefficient category. This is the structure carried by a boundary or line defect in a holomorphic–topological theory on $X \times \mathbb{R}$.

Definition 31.1 (Transversely ordered chiral algebra)**.** A transversely ordered chiral algebra is an augmented object

$$A^{\perp} \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{Fact}_D(X), \otimes^{!}).$$

The superscript $\perp$ records the transverse oriented direction. The shorter phrase *ordered chiral algebra* will always mean this object.

31.2 Associativity inside chiral convolution

Chiral convolution combines disjoint finite subsets before allowing them to collide. Associativity for this product is intrinsic to the curve and has a different source and a different bar construction.

Definition 31.2 (Associative chiral algebra)**.** An associative chiral algebra is an augmented algebra

$$A^{\mathrm{ch}}_{\mathsf{Ass}} \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathcal{D}_X, \star^{\mathrm{ch}}).$$

A chiral Lie algebra is a Lie algebra object

$$\mathfrak{g}^{\mathrm{ch}} \in \mathrm{Alg}_{\mathsf{Lie}}(\mathcal{D}_X, \star^{\mathrm{ch}}).$$

The commutator of an associative chiral algebra is a chiral Lie algebra.

The operation $m^{\perp} : A \otimes^{!} A \to A$ composes transverse intervals. The operation $m^{\mathrm{ch}} : A \star^{\mathrm{ch}} A \to A$ composes chiral configurations. They have different domains, different units, and different deformation complexes.

Theorem 31.3 (Independence of the two associative structures)**.** *Let M be an object for which the split square-zero extensions below exist. A pointwise associative product $\mu_{\perp} : M \otimes^{!} M \to M$ defines a transversely ordered algebra on $\mathbf{1}_{\perp} \oplus M$, while the reduced chiral-convolution product may independently be zero. A chiral-convolution product $\mu_{\mathrm{ch}} : M \star^{\mathrm{ch}} M \to M$ defines an associative chiral algebra on $\mathbf{1}_{\mathrm{ch}} \oplus M$, while the reduced pointwise product may independently be zero. Hence neither associative structure determines the other.*

A common carrier equipped with both products is a mixed object after a mixed distributive datum has been supplied and its unit, binary, and triple coherences have been verified.

Proof. Adjoining the unit to an associative product on the augmentation ideal gives a unital split extension. The zero product is associative in either monoidal category. These two constructions vary independently. On a common carrier, compatibility is additional structure because the mixed composites have different a priori factorizations through the two monoidal products. □

Definition 31.4 (Mixed associative chiral datum)**.** A mixed associative chiral datum is a factorization object A equipped with

(i) a pointwise multiplication $m^{\perp} : A \otimes^{!} A \to A$ and a unit map $\eta_{\perp} : \mathbf{1}_{\perp} \to A$;

(ii) a chiral-convolution multiplication $m^{\mathrm{ch}} : A \star^{\mathrm{ch}} A \to A$ and a unit map $\eta_{\mathrm{ch}} : \mathbf{1}_{\mathrm{ch}} \to A$;

(iii) an aligned mixed distributive law identifying the two composites on $\mathrm{im}(e)$.

The coherence diagrams are the two color-3 faces of the mixed forest complex. A bare symbol $\mathbf{1}$ below denotes the unit of the monoidal category in which the displayed construction is formed.

31.3 Formal-disc realization

Fix a formal coordinate on a disc. Translation-equivariance turns a chiral-convolution product into a state–field operation $Y(a, z)b$ with coefficients in Laurent series. The following theorem packages the exact relationship with the vertex-algebra language.

Theorem 31.5 (Formal-disc realizations)**.** *Let V be a translation-equivariant state–field system with vacuum and injective state–field map.*

(i) *If its fields are weakly associative in the sense of Li, then V is a nonlocal vertex algebra.*

(ii) *If, in addition, V is topologically free over $\mathbb{k}[[\hbar]]$, carries a unitary rational quantum Yang–Baxter operator $S(z)$, and satisfies S-locality and the hexagon compatibility, then V is a quantum vertex algebra.*

(iii) *A translation-equivariant associative chiral-convolution algebra produces such a state–field system whenever its formal-disc restriction satisfies the reconstruction, lower-truncation, and weak-associativity hypotheses. Conversely, a nonlocal or quantum vertex algebra produces the corresponding formal-disc factorization system by its products of fields.*

Proof. The first assertion is Li's reconstruction theorem for quasi-compatible fields [68]. The second is the quantum-vertex reconstruction theorem: unitarity gives inverse exchange, the quantum Yang–Baxter equation compares the two three-field exchange paths, and the hexagon relates exchange to the vertex operation [29, 69]. For the third assertion, the formal-disc chiral product supplies the iterated field products; the stated analytic-formal hypotheses are exactly those used by the reconstruction theorems. The converse assigns to a configuration of formal points the iterated field product and uses weak associativity, or braided associativity, for factorization under refinement. □

31.4 Three separation laws

The complete principal-part filtration, transverse order, and planar exchange form a strict hierarchy of supplied data. The mth normal symbol of a pole is recovered by the mth normal-jet map. An oriented line supplies contractible order chambers and hence E_1 fusion. A plane supplies braid paths and hence exchange. These positive constructions locate each operation in the first geometry that carries it.

Chapter 32

Transversely ordered chiral algebras

Definition 32.1 (Ordered chiral algebra)**.** An ordered chiral algebra on X is an augmented E_1-algebra object

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}\big(\mathrm{Fact}_D(X), \otimes^!\big).$$

Equivalently, A is a factorization object equipped with multiplication maps

$$m_I : A_I \otimes^! A_I \longrightarrow A_I$$

and a unit, compatible with all diagonal transition maps and all disjoint-locus factorization equivalences.

For every decomposition $I = I_1 \sqcup I_2$, locality is the componentwise identity

$$\Delta_{I_1,I_2} \circ m_I = (m_{I_1} \boxtimes m_{I_2}) \circ \sigma \circ (\Delta_{I_1,I_2} \otimes^! \Delta_{I_1,I_2}),$$

where σ reorders the middle two factors. This equation says that ordered fusion commutes with separation of chiral supports.

Theorem 32.2 (Locality recognition)**.** *To give an ordered chiral algebra is equivalent to giving a factorization family A_I, an associative unital multiplication m_I on every component, and the displayed locality identities, compatibly with refinements, permutations, and diagonal transitions.*

Proof. An E_1-algebra in the symmetric monoidal category $\mathrm{Fact}_D(X)$ is, by definition, an associative multiplication and unit whose structure maps are morphisms in $\mathrm{Fact}_D(X)$. A morphism in $\mathrm{Fact}_D(X)$ is exactly a componentwise morphism compatible with the transition and factorization maps. Expanding this compatibility for a binary decomposition gives the displayed identity; the higher decompositions follow by iteration. □

32.1 Product field theories

Let $\mathrm{Fact}_{\mathrm{hol,lc}}(X^{\mathrm{an}} \times \mathbb{R})$ denote factorization algebras that are holomorphic along X and locally constant along $\mathbb{R}$. Product additivity gives

$$\mathrm{Fact}_{\mathrm{hol,lc}}(X^{\mathrm{an}} \times \mathbb{R}) \simeq \mathrm{Alg}_{\mathsf{E}_1}\big(\mathrm{HolFact}(X^{\mathrm{an}})\big)$$

for product-generated objects. Algebraization by regular holonomic Riemann–Hilbert then gives ordered chiral algebras in $\mathrm{Fact}_D(X)$ whenever the coefficient system is regular holonomic and the comparison preserves the required colimits.

Theorem 32.3 (Holomorphic–topological recognition)**.** *Let $\mathbb{F}$ be a product-generated factorization algebra on $X^{\mathrm{an}} \times \mathbb{R}$ that is holomorphic on X and locally constant on $\mathbb{R}$. Then the restriction of $\mathbb{F}$ to a transverse point is an E_1-algebra in holomorphic factorization algebras on X. A regular-holonomic algebraization produces an ordered chiral algebra on X.*

Proof. An ordered family of disjoint intervals in $\mathbb{R}$ acts on the restriction by factorization. Contractibility of each ordered configuration chamber gives the coherent E_1 structure. Factorization in the X direction is preserved because product embeddings commute. Riemann–Hilbert transports the structure in the regular-holonomic subcategory. $\square$

Chapter 33

Block-ordered operations

The preceding definition is intrinsic. A geometric model arises after one chooses an infinity-operad Ch_X representing the selected chiral factorization theory. The Boardman–Vogt tensor product

$$\mathsf{OrdCh}_X = \mathsf{E}_1 \otimes \mathsf{Ch}_X$$

then represents compatible ordered and chiral operations.

Theorem 33.1 (Operadic exponential law)**.** *Let P and Q be infinity-operads whose Boardman–Vogt tensor product exists, and let $\mathcal{V}$ be a compatible presentable symmetric monoidal infinity-category. Then*

$$\mathrm{Alg}_{P\otimes Q}(\mathcal{V}) \simeq \mathrm{Alg}_P\big(\mathrm{Alg}_Q(\mathcal{V})\big).$$

In particular, after a chiral operad Ch_X has been constructed, the tensor product $\mathsf{E}_1 \otimes \mathsf{Ch}_X$ classifies E_1-algebra objects in Ch_X-algebras.

Proof. The tensor product of infinity-operads is characterized by the displayed universal property [75]. The second statement is the specialization $P = \mathsf{E}_1$ and $Q = \mathsf{Ch}_X$. This formal equivalence applies to each geometric model of Ch_X after that model and its operadic tensor product have been constructed. □

A profile is an ordered list $(I_1, \dots, I_r)$ of finite chiral label sets. Its geometric carrier is $X^{I_1} \times \cdots \times X^{I_r}$. Chiral morphisms refine and collide labels inside one block. Transverse morphisms merge adjacent blocks. The tensor-product operad imposes their mixed squares.

33.1 Mixed forests

Fix a cofibrant dg-operad $Q_X \to \mathsf{Ch}_X$. Apply the functorial W-construction to $\mathsf{E}_1 \otimes Q_X$. A cell is represented by a levelled forest with transverse and chiral internal edges. Put

$$\mathrm{or}(F) = \det(E_\perp(F)) \otimes \det(E_{\mathrm{ch}}(F)).$$

The differential is the sum of the screen differential and one-edge contractions.

Theorem 33.2 (Two-color forest differential)**.** *Let C_* be the chain complex freely generated by oriented two-color forests, with one determinant-line generator for every transverse edge and one for every chiral edge. Suppose contraction of an edge is functorial on the coefficient system and every codimension-two contraction square commutes before orientations are inserted. With the product orientation, the transverse and chiral contraction operators satisfy*

$$d_\perp^2 = 0, \qquad d_{\mathrm{ch}}^2 = 0, \qquad d_\perp d_{\mathrm{ch}} + d_{\mathrm{ch}} d_\perp = 0.$$

Thus $d = d_{\mathrm{int}} + d_\perp + d_{\mathrm{ch}}$ squares to zero whenever the internal differential anticommutes with both contraction operators.

Proof. Every term of $d_{\perp}^{2}$ contracts the same pair of transverse edges in two orders. The underlying coefficient maps agree by functoriality, while exchanging the two transverse determinant generators reverses the sign. The terms cancel in pairs. The same argument proves $d_{\mathrm{ch}}^{2} = 0$. For one edge of each color the coefficient maps again form a commuting square and exchanging the two determinant generators contributes the Koszul sign -1, proving the mixed anticommutator identity. The internal terms vanish by the chain-map assumption. □

This construction supplies a geometric model whenever a concrete Q_X and its screen coefficients are available. The intrinsic definition of ordered chiral algebra remains independent of that choice.

Book VII

Bar, cobar, and the boundary state

Chapter 34

Three bar constructions

The phrase *the chiral bar* denotes three functors once its monoidal product and algebraic operad have been named.

Definition 34.1 (Transverse, associative-chiral, and Lie-chiral bars)**.** For augmented objects of the indicated types define

$$\mathrm{Bar}^{\perp}(A^{\perp}) := \mathbf{1} \otimes^{\mathbb{L},!}_{A^{\perp}} \mathbf{1},$$

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}_{\mathsf{Ass}}) := \mathbf{1} \star^{\mathrm{ch}\,\mathbb{L}}_{A^{\mathrm{ch}}_{\mathsf{Ass}}} \mathbf{1},$$

and

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}^{\mathrm{ch}}) := C^{\mathrm{ch}}_{*}(\mathfrak{g}^{\mathrm{ch}}) = \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}[1])$$

with its chiral Chevalley differential. The first bar is formed with $\otimes^{!}$, the second with $\star^{\mathrm{ch}}$, and the third is the Lie-to-commutative-coalgebra Koszul functor.

Theorem 34.2 (Typed bar theorem)**.** *The three bars have the following chain models.*

(i) $\mathrm{Bar}^{\perp}(A^{\perp})$ *is the tensor coalgebra* $T^{c}_{\otimes^{!}}(s\bar{A})$ *with coderivation induced by* $m^{\perp}$*.*

(ii) $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}_{\mathsf{Ass}})$ *is the tensor coalgebra* $T^{c}_{\star^{\mathrm{ch}}}(s\bar{A})$ *with coderivation induced by* m^{ch}*.*

(iii) $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}^{\mathrm{ch}})$ *is the chiral symmetric coalgebra with differential induced by the chiral Lie bracket.*

Each differential squares to zero by the defining relation of its own operad. The outputs live in different coalgebra categories and are compared by a displayed monoidal or PBW morphism.

Proof. Normalize the two simplicial two-sided bars in their respective monoidal categories. Degeneracies remove units, leaving the two tensor coalgebras, and faces contract adjacent factors using the relevant multiplication. The Lie formula is the ordinary operadic bar of the Lie operad internal to chiral convolution. Associativity and Jacobi are exactly the quadratic identities that make the corresponding coderivations square to zero. □

Definition 34.3 (PBW-admissible chiral Lie algebra)**.** A chiral Lie algebra $\mathfrak{g}^{\mathrm{ch}}$ is PBW-admissible when its chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ has a complete separated PBW filtration, the canonical map

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr}_{\mathrm{PBW}}\, U^{\mathrm{ch}}(\mathfrak{g})$$

is an equivalence, and the bar filtration is strongly convergent in the chosen nilcomplete Ran category.

Theorem 34.4 (Filtered chiral PBW comparison)**.** *Let* $\mathfrak{g}$ *be a chiral Lie algebra in a complete filtered chiral category. Assume:*

(i) *its chiral enveloping algebra* $U^{\mathrm{ch}}(\mathfrak{g})$ *exists;*

(ii) *the PBW filtration is exhaustive, complete, and separated;*

(iii) *the chiral PBW morphism*

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr}\, U^{\mathrm{ch}}(\mathfrak{g})$$

is an isomorphism;

(iv) *the induced filtrations on the two bar complexes are strongly convergent.*

Then antisymmetrization induces a filtered quasi-isomorphism

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}\big(U^{\mathrm{ch}}(\mathfrak{g})\big) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}).$$

Proof. The associated graded of the associative chiral bar is the Koszul bar of $\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g})$. The usual Koszul antisymmetrization identifies this coalgebra with the chiral Chevalley coalgebra. Hypotheses (2)–(4) give a complete, separated, strongly convergent comparison spectral sequence. Since its associated-graded map is a quasi-isomorphism, the filtered map is a quasi-isomorphism [9, 38]. □

Definition 34.5 (Bar-admissible mixed datum)**.** A mixed associative chiral datum is bar-admissible when its aligned distributive law acts on the full bisimplicial two-sided bar and both geometric realizations commute with the tensor products occurring in that bisimplicial object.

Theorem 34.6 (Mixed Fubini theorem)**.** *Let A be a mixed algebra in a presentable stable category. Assume the mixed distributive datum extends to the bisimplicial two-sided bar, and that both monoidal products preserve its geometric realizations separately. Then the two iterated realizations and the diagonal realization are canonically equivalent:*

$$\Big|\, p \mapsto \big|q \mapsto B_{p,q}(A)\big|_{\mathrm{ch}}\Big|_{\perp} \simeq \big|\mathrm{diag}\, B_{\bullet,\bullet}(A)\big| \simeq \big|\, q \mapsto \big|p \mapsto B_{p,q}(A)\big|_{\perp}\big|_{\mathrm{ch}}.$$

Proof. The distributive datum makes the two simplicial directions commute, so $B_{\bullet,\bullet}(A)$ is a genuine bisimplicial object. Geometric realization is a colimit. Since the two tensor products preserve the relevant colimits, Fubini for colimits identifies both iterated realizations with the realization of the diagonal simplicial object. □

The notation used in the remainder of this part is

$$\mathrm{Bar}^{\mathrm{ord}}_{X} = \mathrm{Bar}^{\perp}.$$

It always refers to transverse order.

34.1 Realization and factorization

Factorization compares the coefficient at a disjoint configuration with an external tensor product of its coefficients at the separate configurations. This condition is nonlinear. For example, suppose those coefficients are k at one point and at two distinct points, where k is a field. The componentwise sum of two copies has coefficient k^2 at both configurations. Factorization would instead require $k^2 \otimes_k k^2 \cong k^4$ at two points. A componentwise shift also fails: its two-point coefficient is $k[1]$, whereas the tensor product of its singleton coefficients is $k[2]$. Thus an ambient stable coefficient category does not make its factorization objects closed under componentwise sums or shifts.

Geometric realization has a different colimit property. A small indexing ∞-category J is *sifted* when it is nonempty and its diagonal $J \to J \times J$ is cofinal. Here cofinal means that, for every diagram D on $J \times J$, the canonical map $\mathrm{colim}_J(D \circ \Delta) \to \mathrm{colim}_{J\times J} D$ is an equivalence whenever these colimits exist. Iteration gives the same property for $J \to J^m$, for every finite $m \geq 1$. The simplex category with arrows reversed is sifted, so simplicial geometric realization is a sifted colimit.

Proposition 34.7. *Fix coefficient categories on configuration charts, with specified restriction and collision functors and their coherent identifications. Suppose the coefficient categories admit J-indexed colimits and these functors preserve them. Suppose external tensor products preserve colimits separately in each variable. Let $(F_j)_{j\in J}$ be a diagram of factorization objects and morphisms preserving their factorization equivalences. If J is sifted, the componentwise colimit is a factorization object.*

Proof. Write $F_I = \operatorname{colim}_{j\in J}(F_j)_I$ on a chart indexed by I. The collision maps and their coherences pass to the colimit by hypothesis. For a decomposition $I = I_1 \sqcup \cdots \sqcup I_m$, let u denote restriction to its disjointness locus. Restriction, factorization, cofinality, and separate preservation give

$$\begin{aligned} uF_I &\simeq \operatorname*{colim}_{j\in J} u(F_j)_I \\ &\simeq \operatorname*{colim}_{j\in J} \boxtimes_{a=1}^{m} (F_j)_{I_a} \\ &\simeq \operatorname*{colim}_{(j_1,\dots,j_m)\in J^m} \boxtimes_{a=1}^{m} (F_{j_a})_{I_a} \simeq \boxtimes_{a=1}^{m} F_{I_a}. \end{aligned}$$

Each comparison is natural in the diagram and the decomposition. Refinements and permutations therefore preserve the given coherence diagrams. If an empty-configuration unit is specified, its constant diagram also has that unit as colimit, since a sifted category has contractible groupoid completion. □

For a simplicial two-sided bar, this proposition applies when every simplicial term and every face and degeneracy are already in the specified factorization category. It also requires the actual restriction and collision functors to preserve realization. Normalization may then be computed in an ambient stable category. Its individual shifted summands need not be factorization objects. A coproduct on the normalized complex requires its own compatibility with the factorization equivalences.

Chapter 35

The internal ordered bar

Let A be an augmented ordered chiral algebra. Assume the restriction and collision functors preserve geometric realizations, and the external and pointwise tensor products preserve them separately. The two-sided simplicial bar has

$$B_r(\mathbf{1}, A, \mathbf{1}) = \mathbf{1} \otimes^! A^{\otimes^! r} \otimes^! \mathbf{1}.$$

Every face and degeneracy is a factorization morphism by the algebra and augmentation axioms. The sifted-colimit proposition therefore constructs its realization $\mathrm{Bar}^{\mathrm{ord}}_X(A)$ as a factorization object.

Theorem 35.1 (Normalized bar). *Choose a symmetric monoidal dg model for the ambient coefficients in which the displayed tensors compute derived tensors and direct-sum normalization computes realization. Write $\bar{A}$ for the augmentation ideal in that model. The underlying bar complex is*

$$\mathrm{Bar}^{\mathrm{ord}}_X(A) \simeq \big(T^c(s\bar{A}), d_{\mathrm{int}} + b_m\big).$$

Here s decreases cohomological degree by one. The direct sum and its individual shifted summands are formed in the ambient coefficient category. Deconcatenation makes this complex a conilpotent coassociative coalgebra there:

$$\Delta_{\mathrm{dec}}[a_1|\cdots|a_n] = \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n].$$

Proof. Normalization removes words containing a unit and leaves the shifted augmentation ideal in every slot. The faces give the internal differential and adjacent multiplication. The simplicial identities give a square-zero differential. Two successive cuts enumerate the same three subwords in either order. An internal differential or multiplication lies on one side of any surviving cut, with the tensor sign of its prefix. Thus the differential is a coderivation. Every element has finite word support, so sufficiently many reduced cuts give zero. □

35.1 The bar comparison on separated configurations

The normalized word decomposition is not itself a decomposition in $\mathrm{Fact}_D(X)$. Its compatibility with factorization is expressed by a map between whole bar complexes. For augmented dg algebras P, Q in the chosen model, define

$$\mathrm{sh} : BP \otimes BQ \longrightarrow B(P \otimes Q)$$

by shuffling the letters $p_i \otimes 1$ and $1 \otimes q_j$, keeping their respective orders. The sign is the Koszul sign for the permutation of the suspended letters. For degree-zero letters,

$$\mathrm{sh}([p] \otimes [q]) = [p \otimes 1|1 \otimes q] - [1 \otimes q|p \otimes 1].$$

Proposition 35.2 (Shuffle and factorization). *The shuffle is a natural quasi-isomorphism and a coalgebra map, where its source has the tensor-product coalgebra structure. For the geometric coefficient models above,*

suppose these maps commute with the specified restriction and collision functors. Then they give factorization equivalences for the whole bar and make deconcatenation a morphism of factorization objects in the derived category.

Proof. In the bar differential of a shuffle, a merge of letters from different factors occurs twice with opposite suspended signs. The equality $(p \otimes 1)(1 \otimes q) = (-1)^{|p||q|}(1 \otimes q)(p \otimes 1)$ cancels these terms. Merges within one factor give the differential of that factor. Internal differentials commute with the same signed permutation. Thus the shuffle is a cochain map.

A cut of a shuffled word determines one cut in each input word, a shuffle of their prefixes, and a shuffle of their suffixes. Conversely these four choices determine exactly that cut shuffle. The signs on both sides come from the same permutation of suspended letters. This proves the coalgebra identity with the tensor-product coproduct.

To verify the quasi-isomorphism, form the bisimplicial object $B_p(k, P, k) \otimes B_q(k, Q, k)$, with k denoting the tensor unit. Its iterated realization is $BP \otimes BQ$. Its diagonal is the simplicial bar of $P \otimes Q$. Cofinality of the simplicial diagonal identifies these realizations. On normalized chains the map from the iterated realization to the diagonal is the shuffle just defined: the monotone paths from $(0, 0)$ to (p, q) triangulate the product of its two simplices. Each path inserts precisely the degeneracies of its shuffle. Adjacent paths have opposite orientations on a shared face, so their signed sum has exactly the product boundary. This is the realization comparison, hence an equivalence.

For $I = I_1 \sqcup I_2$ on the disjointness locus, the given algebra factorization identifies A_I with $A_{I_1} \boxtimes A_{I_2}$. The shuffle therefore gives the comparison $BA_{I_1} \boxtimes BA_{I_2} \to BA_I$. Its inverse in the derived category is the bar factorization map. The coalgebra identity just proved gives compatibility with deconcatenation. For several blocks, iterated shuffles enumerate the same ordered multishuffles, with the same Koszul signs. This proves refinement coherence. Naturality and the stated functor compatibility give the collision coherence. □

The restriction and collision compatibility in this proposition is a hypothesis on the chosen geometric model. It does not follow from naming an ambient stable category. Conilpotence here refers to finite normalized words in that model. No word-length completion is implicit.

35.2 Three coproducts

The theorem system uses three coproducts with distinct geometric meanings:

$$\Delta_{\text{fact}} : \text{separation of chiral support},$$
$$\Delta_{\text{dec}} : \text{cutting an ordered bar word},$$
$$\Delta_{\text{Hopf}} : \text{fusion of representations}.$$

The first two are present on the transverse bar. The third is additional bialgebra data on an acting algebra.

Proposition 35.3 (Coproduct separation)**.** *Factorization separation is carried by every ordered coefficient. Deconcatenation is generated functorially by the bar construction. A Hopf coproduct is supplied by an independent bialgebra structure. A coalgebra comparison morphism relates the Hopf coproduct to either geometric coproduct whenever a theory identifies their carriers.*

Proof. Factorization separation is part of the coefficient object, and deconcatenation is the canonical cut map on tensor words. A Hopf coproduct is a map from the algebra to a completed tensor square satisfying algebraic coassociativity. Its source, target, and defining universal property differ from the first two, so a relation among them is exactly the data of a comparison morphism. □

35.3 Interval factorization homology

For a right A-module R and a left A-module L, factorization homology of an interval with endpoint labels gives

$$\int_{([0,1],\{0,1\})} (A; R, L) \simeq R \otimes^{\mathbb{L}}_{A} L.$$

This statement holds internally in every presentable monoidal category supporting factorization homology [5, 75]. Taking $R = L = \mathbf{1}$ identifies the interval amplitude with the ordered bar.

Chapter 36

Cobar and twisting cochains

Let C be a coaugmented conilpotent differential graded coalgebra in the ambient coefficient model. The ordinary, direct-sum tensor cobar algebra is

$$\mathrm{Cobar}_X^{\mathrm{ord}}(C) = \big(T(s^{-1}\bar{C}),\, d_{\mathrm{int}} + d_{\Delta}\big).$$

A twisting cochain $\tau : C \to A$ has degree 1, vanishes on the coaugmentation, and has zero composite with the augmentation of A. It satisfies

$$d\tau + \tau d + m(\tau \otimes \tau)\bar{\Delta} = 0.$$

The universal twisting cochains are

$$\tau_A : \mathrm{Bar}_X^{\mathrm{ord}}(A) \to A, \qquad \tau_C : C \to \mathrm{Cobar}_X^{\mathrm{ord}}(C).$$

Theorem 36.1 (Bar–cobar adjunction)**.** *For ordinary augmented dg algebras and coaugmented conilpotent dg coalgebras in the specified coefficient model, tensor cobar is left adjoint to tensor bar. Their mapping set is the set of twisting cochains. An internal adjunction on factorization objects additionally requires both constructions to produce factorization objects and their twisting maps to preserve the geometric data.*

Proof. An algebra map from the tensor algebra $T(s^{-1}\bar{C})$ is determined by its restriction to $s^{-1}\bar{C}$. Compatibility with the cobar differential is the Maurer–Cartan equation above. The same equation describes coalgebra maps from C to the cofree conilpotent coalgebra $T^c(s\bar{A})$. □

Chapter 37

Complete bar–cobar reconstruction

Ordinary tensor cobar uses finite words. Completion changes that carrier. Let k be a commutative ring and $A_{\mathrm{alg}} = \bigoplus_{w\geq 0} A_w$ an augmented algebra in degree zero, with $A_0 = k$, augmentation given by projection onto A_0, and multiplication adding weights. Put $D = \Omega B(A_{\mathrm{alg}})$ using ordinary direct-sum tensors, and let D_w be its weight-w subcomplex. The complete algebra and its complete resolution are

$$\widehat{A} = \prod_{w\geq 0} A_w, \qquad \widehat{D} = \prod_{w\geq 0} D_w.$$

Products of complexes are taken degreewise, with the weight topology.

Theorem 37.1 (Complete reconstruction)**.** *The componentwise counit $\widehat{D} \to \widehat{A}$ is a quasi-isomorphism of complete differential graded algebras.*

Proof. A weight-w block word has at most w positive-weight entries. Filter by their number. Multiplication lowers this number and splitting preserves it. At a fixed list of $N \geq 2$ entries, the splitting complex is the tensor product of the coefficient module with $N-1$ copies of $[k \xrightarrow{1} k]$ in degrees $-1, 0$. Changing the first outer cut to an inner gap contracts it. The other differential terms cancel by their prefix signs. This identity survives tensoring with any coefficient module. Subtracting a primitive at the largest entry number terminates, so negative cohomology vanishes. In degree zero, the boundaries impose $c(a)c(b) = c(ab)$, leaving exactly A_w. Taking the degreewise product preserves kernels and images of these differentials, and therefore preserves the computed cohomology. □

For $k[x]$ with weight one on x, the ordinary counit recovers $k[x]$, whereas this weight completion recovers $k[[x]]$. A completed geometric bar–cobar equivalence needs a specified completed tensor product, coalgebra category, and comparison maps compatible with collision and inverse limits. An operadic nilcompletion or a boundary double centralizer is not identified with the ordinary tensor counit by this weight calculation. The coalgebra-unit and geometric duality statements require their own convergence and comparison arguments.

37.1 Quadratic recognition

Let $A = T(V)/(R)$ with $R \subset V \otimes V$. Its quadratic dual coalgebra $A^{\mathrm{q!}}$ is cogenerated by sV with corelations $s^2 R$. The canonical map $A^{\mathrm{q!}} \to \mathrm{Bar}(A)$ gives the Koszul complex.

Theorem 37.2 (Quadratic Koszul criterion)**.** *Assume positive weight and degreewise finite duals. The following conditions are equivalent:*

(i) $A^{\mathrm{q!}} \to \mathrm{Bar}(A)$ *is a quasi-isomorphism;*

(ii) $\mathrm{Cobar}(A^{\mathrm{q!}}) \to A$ *is a quasi-isomorphism;*

(iii) *the left and right Koszul complexes have homology concentrated in weight zero;*

(iv) $\mathrm{Tor}_i^A(\mathbf{1}, \mathbf{1})$ *is concentrated in internal weight i.*

Proof. The universal twisting morphism $A^{\mathrm{q}!} \to A$ is Koszul exactly when the associated twisted tensor products resolve the unit. The fundamental theorem of twisting morphisms identifies this condition with the bar and cobar quasi-isomorphisms. Weight purity is the homological form of the same statement [71, 86]. □

Chapter 38

Modules, comodules, and Morita theory

The ordinary tensor counit reconstructs an algebra. Reconstruction of its modules requires the unit and counit of the twisted module functors as well. The following weight conditions make these maps finite in every component.

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ be an associative algebra in cohomological degree zero. Assume $A_0 = k$, multiplication adds weights, and augmentation projects onto A_0. Put $C_{\mathrm{alg}} = B(A_{\mathrm{alg}})$, with its bar weight grading, and set

$$A = \prod_{w \geq 0} A_w, \qquad C = \prod_{w \geq 0} (C_{\mathrm{alg}})_w.$$

Products are degreewise in complexes. The universal twisting map $\tau : C \to A$ has degree one, sends $[a]$ to a, and vanishes on the empty word and all longer words. The bar differential has the earlier convention $b_2[a|b] = [ab]$.

A complete weight-graded left A-module here means a product $M = \prod_{w \geq w_0} M_w$ of complexes, for some integer w_0, with differential preserving weight and actions $A_u \otimes M_v \to M_{u+v}$. A complete weight-graded left C-comodule means such a product N with a counital coassociative coaction $\rho : N \to C \widehat{\otimes} N$ preserving weight and differential. The completed tensor product is defined componentwise by

$$(C \widehat{\otimes} N)_w = \bigoplus_{u+v=w} (C_{\mathrm{alg}})_u \otimes N_v.$$

This sum is finite since $u \geq 0$ and $v \geq w_0$. Each reduced coaction lowers the weight of its final N-component. It therefore vanishes after more than $w - w_0$ iterations on N_w. These comodules need no chosen coaugmentation.

Write $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$ for the categories obtained by inverting maps that are quasi-isomorphisms in every weight, respectively for these modules and comodules. This specifies the weak equivalences; no coderived localization is implicit. Define

$$F(M) = C \widehat{\otimes}_\tau M, \qquad G(N) = A \widehat{\otimes}_\tau N.$$

The first has the coaction induced by Δ_C and the second the left A-action. Writing $\Delta(c) = \sum c_{(1)} \otimes c_{(2)}$ and $\rho(n) = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$d_F(c \otimes m) = d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m,$$
$$d_G(a \otimes n) = a \otimes d_N n - \sum a \tau(n_{(-1)}) \otimes n_{(0)}.$$

All sums are finite in each weight. The twisting equation, associativity, and coassociativity make both differentials square to zero.

Theorem 38.1 (Koszul module equivalence in positive weights)**.** *These functors induce inverse equivalences*

$$D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C).$$

Their counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a\epsilon_C(c)m, \qquad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The differential formulas show that these are cochain maps, respectively A-linear and C-colinear. The twisting terms cancel in the unit equation; the counit equation uses the module action. Coassociativity and the counit identities give the two triangle identities.

The complex $GF(M)$ is the two-sided normalized bar resolution $A\widehat{\otimes}B(A_{\mathrm{alg}})\widehat{\otimes}M$. For its contraction, use the simplicial boundary convention $\partial = \sum_i (-1)^i d_i$. Multiplying a bar word of length r by $(-1)^r$ translates this convention to the displayed twisting differentials. Writing $\bar{a} = a - \epsilon_A(a)1$, the extra degeneracy is

$$h\big(a_0[a_1|\cdots|a_r]m\big) = 1[\bar{a}_0|a_1|\cdots|a_r]m.$$

The first face returns the original term minus its scalar first factor. The other faces cancel against applying h after the boundary, including the module-action face. The internal differential of M cancels by the suspension sign. Thus $\partial h + h\partial = 1 - i\epsilon_M$, with $i(m) = 1[\,]m$. This is a linear contraction; i need not be A-linear. It preserves total weight, so the counit is a quasi-isomorphism in every weight.

At fixed total weight, filter $FG(N)$ by the weight of its final N-factor. This filtration is finite. The reduced-coaction term lowers that weight and disappears in the associated graded. The associated graded unit is therefore the inclusion into $B(k, A_{\mathrm{alg}}, A_{\mathrm{alg}}) \otimes N_v$. The rightmost extra degeneracy contracts this augmented bar onto k:

$$h\big([a_1|\cdots|a_r]a\big) = (-1)^{r+1}[a_1|\cdots|a_r|\bar{a}]1$$

in the same simplicial convention. The last face gives $1 - i\epsilon$; all other faces cancel in pairs. Consequently the unit is a quasi-isomorphism on each associated graded component, and the finite filtration proves the same for each weight of N.

Both functors preserve weightwise quasi-isomorphisms. Filter by the weight of the final module or comodule component. The twisting terms strictly change that weight in one direction. On associated graded components, the functors tensor complexes of k-vector spaces, which preserves quasi-isomorphisms. The filtration is finite at fixed total weight. Hence F, G descend to the stated localizations, where their unit and counit are isomorphisms. □

This theorem uses an explicit weight decomposition and a lower weight bound on every module and comodule. Derived I-adic completion, with $I = \ker(A \to k)$, instead uses $M \to \varprojlim_p (A/I^p \otimes^{\mathbb{L}}_A M)$. The two completions require a comparison of their filtrations. A coderived category also has a separately specified class of coacyclic complexes. Neither identification follows from the algebra counit alone. For right and left modules, the two-sided bar computes their relative derived tensor product in the same finite weight components. An identification with a geometric derived cotensor product additionally requires the two-sided twisting comparison and its compatibility with the chosen completion and collision maps.

Let C be a presentable factorization category and let b be a compact factorization generator. The internal endomorphism object

$$A_b = \mathrm{RHom}_C(b, b)$$

has ordered multiplication by composition.

Theorem 38.2 (Factorization Morita theorem)**.** *Assume b generates each local category and the local Morita equivalences commute with factorization descent. Then*

$$C \simeq \mathrm{Mod}_{A_b}(\mathrm{Fact}_D(X)).$$

Proof. The compact-generator Morita theorem applies on every factorization chart. The descent hypothesis identifies the equivalences on intersections and hence on the Ran colimit. □

Chapter 39

Derived centre and bulk action

The ordered Hochschild cochains are

$$Z_{\mathrm{ord}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A).$$

Theorem 39.1 (Higher Deligne centre)**.** *Let the ambient coefficient category be presentable, stable, symmetric monoidal, closed, and let tensor product preserve geometric realizations separately. Then $Z_{\mathrm{ord}}(A)$ carries a canonical E_2-algebra structure. It is the universal E_2 algebra acting centrally on A.*

Proof. Hochschild cochains carry the brace operations, and the brace operad is equivalent to chains on little two-disks. This is the Deligne theorem [65, 77]. Universality is the identification of Hochschild cochains with endomorphisms of the identity A-bimodule. □

A boundary condition in a field theory gives a morphism

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A_{\partial}).$$

Locality makes the action central. A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

For a Calabi–Yau category, Hochschild cochains carry the framed E_2 refinement of the cyclic Deligne theorem [15].

Chapter 40

Cyclic closure and modular sewing

A cyclic trace becomes a surface theory after dualizability and nondegeneracy are supplied.

Definition 40.1 (Perfect Calabi–Yau ordered algebra)**.** A perfect Calabi–Yau ordered algebra is a dualizable ordered algebra A together with a trace

$$\mathrm{tr} : \mathrm{HH}_0(A) \longrightarrow \mathbf{1}[-d]$$

whose induced pairing $A \otimes A \to \mathbf{1}[-d]$ is nondegenerate.

Theorem 40.2 (Ribbon-graph realization of a Calabi–Yau algebra)**.** *Let A be a homologically finite, smooth and proper cyclic A_∞ algebra with a nondegenerate pairing of degree d. Then the cyclic bar construction of A defines an open topological conformal field theory. Its operations are indexed by chains on moduli spaces of bordered surfaces, equivalently by the corresponding ribbon-graph complexes. The value on the circle is Hochschild homology* $\mathrm{HH}_*(A)$.

Proof. The nondegenerate cyclic pairing identifies outgoing and incoming boundary labels and contracts the two flags of every internal ribbon edge. The A_∞ identities are exactly the boundary identities for metric ribbon graphs. Smoothness and properness give the dualizability and finite pairing required for all contractions. Costello's construction then produces the open TCFT and identifies the closed state complex with Hochschild chains [22]. □

Remark 40.3. The trace functional supplies the annular operation. Smoothness, properness, and the nondegenerate Calabi–Yau pairing extend it to the complete ribbon-surface theory of the theorem.

Stable-curve sewing is supplied by a modular operad action. Its contractions form the modular structure carried by the chiral coefficient. The two sewing geometries coexist when their contraction maps supercommute.

Book **VIII**

Exact noncommutative models

Chapter 41

Free and square-zero pairs

Work in a presentable stable linear closed monoidal coefficient category C whose tensor preserves coproducts separately and is exact. For $M \in C$, put

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus \cdots$$

This is the free augmented algebra in C. All sums and shifts are formed in C. Even when M is a factorization object, this componentwise construction need not be one, as the separated-locus calculation in Chapter 16 shows.

Theorem 41.1 (Free–square-zero duality)**.** *Assume M is dualizable and use the continuous dual for word weight. Let* Bar *in this statement denote the ordinary augmented bar in C. Then*

$$\mathrm{Bar}(T_X(M)) \simeq \mathbf{1} \oplus sM, \qquad \mathrm{RHom}_{T_X(M)}(\mathbf{1}, \mathbf{1}) \simeq \mathbf{1} \oplus M^\vee[-1]$$

with square-zero multiplication on the reduced summand. Conversely, for $A = \mathbf{1} \oplus M$ with $M^2 = 0$,

$$\mathrm{Bar}(A) = T^c(sM), \qquad A^! \simeq \widehat{T}(M^\vee[-1]).$$

Proof. In weight r, bar terms are copies of $M^{\otimes r}$ indexed by cuts of a word of length r. For $r \geq 2$, insertion of the first possible cut contracts the cut complex. The suspension signs give the total complex homotopy. In weight one the primitive class sM survives.

The explicit cut identity is proved in Proposition 16.2. Dualizing this primitive coalgebra gives square-zero multiplication. For the square-zero algebra all multiplication faces vanish, leaving the cofree tensor coalgebra. Its continuous weight dual is the completed tensor algebra. □

Chapter 42

The quantum plane

Let

$$A_q = \Bbbk\langle x, y\rangle/(yx - qxy), \qquad q \in \Bbbk^\times.$$

Give x and y weight one.

Theorem 42.1 (PBW basis of the quantum plane)**.** *The monomials $x^a y^b$, $a, b \geq 0$, form a basis of A_q.*

Proof. Use the rewriting rule $yx \mapsto qxy$. Every reduction decreases the inversion number of a word. The leading word yx has distinct initial and terminal letters, so every overlap decomposes into independent reductions. Bergman's diamond argument therefore gives the ordered monomials as normal forms. □

The quadratic dual is

$$A_q^! = \Bbbk\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

Theorem 42.2 (Koszulness)**.** *A_q is Koszul, and its Koszul complex is*

$$0 \to A_q \otimes \Bbbk(\xi\eta) \xrightarrow{d_2} A_q \otimes (\Bbbk\xi \oplus \Bbbk\eta) \xrightarrow{d_1} A_q \to \Bbbk \to 0,$$

with

$$d_1(a \otimes \xi + b \otimes \eta) = ax + by, \quad d_2(a \otimes \xi\eta) = ay \otimes \xi - qax \otimes \eta.$$

Proof. Direct substitution gives $d_1 d_2 = ayx - qaxy = 0$. The image of d_1 is the augmentation ideal (x, y), so the complex is exact at A_q and has cokernel $\Bbbk$. We now compute $\ker d_1$.

Write

$$u = \sum_{i,j\geq 0} u_{ij} x^i y^j, \qquad v = \sum_{i,j\geq 0} v_{ij} x^i y^j.$$

The relation $y^j x = q^j x y^j$ gives

$$ux + vy = \sum_{a,b\geq 0} \left(q^b u_{a-1,b} + v_{a,b-1}\right) x^a y^b,$$

where coefficients with a negative index are zero. Thus $ux + vy = 0$ implies $u_{i,0} = 0$, $v_{0,j} = 0$, and

$$v_{a,b} = -q^{b+1} u_{a-1,b+1}.$$

Define $c = \sum_{i,j\geq 0} u_{i,j+1} x^i y^j$. Then $u = cy$ and, using $x^i y^j x = q^j x^{i+1} y^j$,

$$v = -qcx.$$

Hence $(u, v) = d_2(c)$. Finally, $d_2(c) = 0$ implies $cy = 0$; the PBW basis and $q \neq 0$ show that A_q is a domain, so $c = 0$. The complex is exact, and the quadratic Koszul criterion proves that A_q is Koszul. □

Theorem 42.3 (Yoneda algebra of the quantum plane)**.** *Use the convention that a Yoneda product $\alpha\beta$ means first β and then α. The augmentation module has*

$$\operatorname{Ext}^{*}_{A_q}(\Bbbk,\Bbbk) \cong \Bbbk\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi) = A_q^{!}$$

as a bigraded algebra, where $|\xi| = |\eta| = 1$ in cohomological degree and each generator retains its internal weight one.

Proof. Apply $\operatorname{Hom}_{A_q}(-,\Bbbk)$ to the minimal Koszul resolution in the preceding theorem. Every matrix entry of d_1 and d_2 lies in the augmentation ideal, so the dual differential vanishes. The cohomology has basis

$$1, \quad \xi, \eta, \quad \xi\eta$$

in cohomological degrees 0, 1, 1, 2. Splicing the two degree-one extensions gives the degree-two class. The syzygy $y \otimes \xi - qx \otimes \eta$ identifies the two splicings by $\xi\eta = -q\eta\xi$, while the absence of degree-two x^2 and y^2 syzygies gives $\xi^2 = \eta^2 = 0$. These relations account for the complete four-dimensional cohomology, and therefore present the Yoneda algebra. □

The bar, quadratic dual and derived augmentation boundary of A_q are ordinary algebraic constructions. Tensoring with a coefficient object gives an ordered algebra in the ambient pointwise coefficient category when that tensor is defined. A factorization realization additionally requires its separated-locus equivalences and collision maps. These do not follow from componentwise tensoring with A_q.

Chapter 43

Quiver path algebras

Let Q be a finite quiver over a field $\Bbbk$ and $S = \bigoplus_{i \in Q_0} \Bbbk e_i$. Assign a vector space M_a to each arrow $a : i \to j$ and set

$$M = \bigoplus_{a \in Q_1} e_j M_a e_i, \qquad T_S(M) = S \oplus M \oplus M \otimes_S M \oplus \cdots .$$

Words are composable paths. These are ordinary S-bimodules. Placing the construction in geometric coefficients requires a functor preserving the relative tensors, bar realization and intended factorization structures, as in Proposition 16.3.

For the quiver $1 \xrightarrow{a} 2$, the path algebra has basis e_1, e_2, a and relation $a = e_2 a e_1$. The simple modules S_1, S_2 satisfy

$$\operatorname{Ext}^1(S_1, S_2) \cong \Bbbk, \qquad \operatorname{Ext}^1(S_i, S_i) = 0, \qquad \operatorname{Ext}^1(S_2, S_1) = 0.$$

The bar differential records concatenation of paths, and the quadratic dual reverses the arrow with cohomological shift.

Theorem 43.1 (Relative path-algebra bar)**.** *For a tensor path algebra $T_S(M)$ over semisimple S,*

$$S \otimes^{\mathbb{L}}_{T_S(M)} S \simeq S \oplus sM$$

as an S-bicomodule. A quadratic quotient $T_S(M)/(R)$ is Koszul precisely when its relative quadratic complex is exact away from weight zero.

Proof. The last-arrow contraction is S-bilinear and gives the same extra degeneracy as in the free case. The quotient statement follows from the relative twisting-morphism criterion. □

Chapter 44

Weyl and Clifford deformations

Let V be finite-dimensional and let $\omega : V \otimes V \to \Bbbk$ be alternating. The Weyl algebra is

$$W_\hbar(V, \omega) = T(V)[[\hbar]]/\langle uv - vu - \hbar\omega(u, v)\rangle.$$

For a purely odd space and symmetric ω, the same formula with the supercommutator $uv + vu$ gives a Clifford algebra.

Theorem 44.1 (Filtered PBW theorem)**.** *For alternating ω in the even Weyl case, or symmetric ω in the purely odd Clifford case, the associated graded algebra is* $\mathrm{Sym}(V)$ *in the Weyl case and* $\bigwedge(V)$ *in the Clifford case.*

Proof. Filter by word length. The constant term lowers filtration by two, so the associated graded relation is the commutative or skew-commutative quadratic relation. The cubic ambiguities reduce to the cocycle identity for a central scalar form, which holds because scalars commute. The ordered monomials give the PBW basis. □

The quadratic-linear-constant presentation has a curved Koszul dual. The constant term becomes the curvature functional

$$m_0 : s^2 R \longrightarrow \Bbbk[2].$$

This construction is the algebraic origin of the curvature; geometric anomaly curvature enters through a separate comparison morphism.

Chapter 45

Associative chiral algebras on the formal disc

Let V be a complete topological vector space with vacuum $\mathbf{1}$ and translation T. A state–field map

$$Y(-,z) : V \longrightarrow \mathrm{End}(V)((z))$$

is a coordinate presentation of an associative chiral-convolution product when its iterated products are expansions of one factorization kernel in the nested regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$.

Proposition 45.1 (Associative convolution from weak associativity)**.** *Suppose $Y(\mathbf{1},z) = \mathrm{id}$, $Y(a,z)\mathbf{1} \in a + zV[[z]]$, $[T, Y(a,z)] = \partial_z Y(a,z)$, and for every a, b, c there is N such that*

$$(z_1 - z_2)^N Y(a,z_1)Y(b,z_2)c = (z_1 - z_2)^N Y(Y(a,z_1 - z_2)b, z_2)c$$

in the common formal expansion ring. Then the fields define an associative chiral algebra on the formal disc. The converse holds for a vacuum-generated translation-equivariant associative chiral algebra admitting state–field reconstruction.

Proof. The displayed identity is the equality of the two nested restrictions of the three-point chiral product. Vacuum and translation give the unit and coordinate covariance. Factorization on higher configuration spaces is obtained by iterating the equality; coherence follows from associativity of the chiral-convolution product. □

Example 45.2 (Braided current fields)**.** Let $R(z)$ be an invertible solution of the spectral Yang–Baxter equation, normalized by $R^{21}(-z)R(z) = 1$. Generating fields $a_i(z)$ with exchange

$$a_i(z_1)a_j(z_2) = \sum_{k,l} R_{ij}^{kl}(z_1 - z_2)\, a_l(z_2)a_k(z_1)$$

form a quantum vertex algebra whenever the vacuum, translation, and weak associativity relations hold. The R-matrix supplies the comparison between the two formal expansion regions and records planar exchange in addition to transverse order.

Chapter 46

Zamolodchikov–Faddeev exchange

Let V be finite-dimensional and let $R(u) \in \operatorname{End}(V \otimes V) \otimes \Bbbk(u)$ be invertible. Introduce generators $Z_i(u)$ with exchange relation

$$Z_1(u) Z_2(v) = R_{12}(u - v) Z_2(v) Z_1(u).$$

Theorem 46.1 (Yang–Baxter confluence)**.** *The exchange algebra has a consistent normal ordering for three generators precisely when*

$$R_{12}(u-v) R_{13}(u-w) R_{23}(v-w) = R_{23}(v-w) R_{13}(u-w) R_{12}(u-v).$$

Proof. Reverse the word $Z_1(u) Z_2(v) Z_3(w)$. Exchanging positions in the order (12), (13), (23) gives the left side. Exchanging in the order (23), (13), (12) gives the right side. These are the two reductions of the unique length-three overlap. Equality is therefore the diamond condition. □

For the rational permutation solution

$$R(u) = 1 - \frac{\hbar P}{u},$$

the Yang–Baxter equation follows from $P_{ij}^2 = 1$ and $P_{12} P_{13} P_{23} = P_{23} P_{13} P_{12}$.

Chapter 47

The RTT Yangian of $\mathfrak{gl}_2$

Let

$$T(u) = 1 + \sum_{r \geq 1} T^{(r)} u^{-r}, \qquad R(u) = 1 - \frac{\hbar P}{u}.$$

The RTT Yangian $Y_\hbar(\mathfrak{gl}_2)$ is generated by the coefficients $t_{ij}^{(r)}$ subject to

$$R(u-v) T_1(u) T_2(v) = T_2(v) T_1(u) R(u-v).$$

Equivalently,

$$(u-v)[t_{ij}(u), t_{kl}(v)] = \hbar\big(t_{kj}(u) t_{il}(v) - t_{kj}(v) t_{il}(u)\big).$$

Proposition 47.1 (Coefficient relations)**.** *Put $t_{ij}^{(0)} = \delta_{ij}$. The RTT identity is equivalent to*

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar\big(t_{kj}^{(r)} t_{il}^{(s)} - t_{kj}^{(s)} t_{il}^{(r)}\big), \qquad r, s \geq 0.$$

In particular,

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj} t_{il}^{(1)} - \delta_{il} t_{kj}^{(1)}),$$

so the first modes form the enveloping algebra of the copy $\hbar\mathfrak{gl}_2$ inside the Yangian.

Proof. Multiply the generating-series relation by $u - v$ and compare the coefficient of $u^{-r-1} v^{-s-1}$. The two terms contributed by multiplication by $u - v$ produce the two commutators on the left, and the two ordered products on the right are the corresponding coefficients of the RTT numerator. Setting $r = 0$ gives the displayed first-mode relation. □

Theorem 47.2 (Hopf structure)**.** *The formulas*

$$\Delta(T(u)) = T(u) \dot{\otimes} T(u), \qquad \varepsilon(T(u)) = 1, \qquad S(T(u)) = T(u)^{-1}$$

define a Hopf algebra structure in the u^{-1}-adic completion. Coefficientwise,

$$\Delta(t_{ij}^{(r)}) = \sum_{a=1}^{2} \sum_{p=0}^{r} t_{ia}^{(p)} \otimes t_{aj}^{(r-p)}.$$

Proof. Apply the RTT relation in each tensor factor. Matrix multiplication in the auxiliary space gives the RTT relation for $\Delta(T)$. Coassociativity is associativity of matrix multiplication. Expanding the matrix product gives the coefficient formula. The inverse series exists uniquely because $T(u) = 1 + O(u^{-1})$; its defining identities give the antipode axioms. □

The evaluation map

$$\mathrm{ev}_a : T(u) \longmapsto 1 + \frac{\hbar E}{u - a}, \qquad E = (E_{ij}),$$

lands in $U(\mathfrak{gl}_2)[[u^{-1}]]$ and gives the standard evaluation modules. The PBW theorem identifies the associated graded with $U(\mathfrak{gl}_2[t])$ [32, 79].

This example contains three structures: ordered multiplication in the Yangian, coproduct on tensor products of modules, and exchange by the rational R-matrix.

Chapter 48

The Hall algebra of the A_2 quiver

Let $Q : 1 \to 2$ over $\mathbb{F}_q$. Write S_1, S_2 for the simples and M for the unique non-split extension

$$0 \to S_2 \to M \to S_1 \to 0.$$

In the twisted Hall algebra, multiplication counts subobjects with Euler-form normalization. Direct counting gives

$$[S_1] * [S_2] = v^{-1}[S_1 \oplus S_2] + v^{-1}[M], \qquad [S_2] * [S_1] = [S_1 \oplus S_2], \qquad v^2 = q.$$

Hence the root vector

$$E_{12} = E_1E_2 - v^{-1}E_2E_1$$

is represented by the indecomposable extension.

Theorem 48.1 (Quantum Serre relation)**.** *The Hall generators satisfy*

$$E_1^2E_2 - (v + v^{-1})E_1E_2E_1 + E_2E_1^2 = 0,$$

and the analogous relation with 1 *and* 2 *exchanged.*

Proof. Every representation of dimension vector $(2, 1)$ is a linear map $f : \mathbb{F}_q^2 \to \mathbb{F}_q$ of rank zero or one. Read a word from the top quotient to the bottom quotient. The complete-flag counts are

$\operatorname{rk} f$	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

For rank zero every one-dimensional subspace of $\mathbb{F}_q^2$ is admissible, giving $q + 1$ flags in each word. For rank one, the word 121 forces the unique kernel line, the word 211 is incompatible with the final subrepresentation, and the word 112 allows every line at the remaining stage.

The total Euler-form exponents of the three words are respectively $-1, 0, 1$. Their twisted coefficients are therefore $v^{-1}, 1, v$. On the rank-zero stratum the Serre combination has coefficient

$$(q + 1)\big(v^{-1} - (v + v^{-1}) + v\big) = 0.$$

On the rank-one stratum it has coefficient

$$v^{-1}(q + 1) - (v + v^{-1}) = v^{-1}(v^2 + 1) - (v + v^{-1}) = 0.$$

The two isomorphism classes exhaust the dimension vector $(2, 1)$, proving the relation. This is Ringel's realization of $U_v^+(\mathfrak{sl}_3)$ [45, 88]. □

Chapter 49

Matrix traces and the necklace bracket

Let $F = \Bbbk\langle x, y\rangle$ and $F_{\mathrm{cyc}} = F/[F, F]$. Cyclic derivatives are defined by

$$\partial_x^{\mathrm{cyc}}[a_1 \cdots a_n] = \sum_{a_i = x} a_{i+1} \cdots a_n a_1 \cdots a_{i-1}$$

and similarly for y. Define

$$\{f, g\}_{\mathrm{neck}} = \left[\partial_x^{\mathrm{cyc}} f \, \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f \, \partial_x^{\mathrm{cyc}} g\right]_{\mathrm{cyc}}.$$

Theorem 49.1 (Trace realization)**.** *For matrices $X, Y \in \mathfrak{gl}_N$ and the canonical symplectic bracket on matrix entries,*

$$\{\operatorname{Tr} f(X,Y), \operatorname{Tr} g(X,Y)\} = \operatorname{Tr}\{f, g\}_{\mathrm{neck}}(X,Y).$$

Proof. Differentiate the trace by moving the varied matrix entry cyclically to the first position. This gives the cyclic derivative. Contract the X derivative of one trace with the Y derivative of the other and subtract the reversed contraction. The matrix indices close into the trace of the cyclic product. □

In the stable commuting-matrix quotient, let

$$p_{ab} = \operatorname{Tr}(X^a Y^b).$$

Then

$$\{p_{ab}, p_{cd}\} = (ad - bc)\, p_{a+c-1,b+d-1}.$$

For two eigenvalues, the relative invariants

$$A = (x_1 - x_2)^2, \quad B = (x_1 - x_2)(y_1 - y_2), \quad C = (y_1 - y_2)^2$$

satisfy $B^2 = AC$ and carry the standard $\mathfrak{sl}_2$ Poisson bracket after linear rescaling.

Book IX

The nonlocal ladder

Chapter 50

Spectral exchange as an associative chiral operation

50.1 The coefficient field and the exchange operator

Let V be a finite-dimensional vector space and let K be a field of meromorphic functions in spectral differences. Fix an invertible operator

$$R(u) \in \operatorname{End}(V \otimes V) \otimes K$$

that satisfies the spectral Yang–Baxter equation

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v). \tag{50.1}$$

The equation states that two routes through the chamber complex of three labelled points give the same transport.

Definition 50.1 (Zamolodchikov–Faddeev algebra)**.** The exchange algebra $\mathcal{Z}_R$ is generated by the components of a current $A(u) \in V \otimes \mathcal{Z}_R$ with relation

$$A_1(u)A_2(v) = R_{12}(u-v)A_2(v)A_1(u). \tag{50.2}$$

A normal word has spectral labels in a fixed chamber $u_1 \succ \cdots \succ u_n$.

Theorem 50.2 (Confluence is Yang–Baxter)**.** *The quadratic rewriting system defined by* (50.2) *is confluent on words with pairwise distinct spectral labels precisely when* (50.1) *holds. In that case the normal ordered words form a K-basis of $\mathcal{Z}_R$.*

Proof. Every two-letter inversion is removed by one application of (50.2). The minimal overlap occurs in a word of length three. Moving the leftmost label through the other two gives the coefficient $R_{12}R_{13}R_{23}$, while moving the rightmost label in the opposite sequence gives $R_{23}R_{13}R_{12}$. Equality of the two coefficients is (50.1). Bergman's diamond lemma then gives the normal-word basis. $\square$

50.2 A diagonal exchange model

Choose a basis $e_1, \dots, e_N$ of V and functions $q_{ij}(u)$ satisfying

$$q_{ij}(u)q_{ji}(-u) = 1, \qquad q_{ii}(u) = 1.$$

The diagonal operator

$$R(u)(e_i \otimes e_j) = q_{ij}(u)e_i \otimes e_j$$

satisfies the Yang–Baxter equation because all three factors are scalars. The generators $a_i(u)$ obey

$$a_i(u)a_j(v) = q_{ij}(u-v)a_j(v)a_i(u).$$

Every ordered monomial is a basis vector. The quantum plane is the constant two-generator specialization $q_{12} = q$, $q_{21} = q^{-1}$.

Calculation 50.3 (Three-particle coefficient)**.** For $u \succ v \succ w$,

$$a_i(w)a_j(v)a_k(u) = q_{ij}(w-v)q_{ik}(w-u)q_{jk}(v-u)a_k(u)a_j(v)a_i(w).$$

The coefficient is independent of the sequence of adjacent exchanges. This scalar identity is the diagonal Yang–Baxter equation.

Proof. Each inversion of two adjacent letters contributes its scalar exchange factor. Both reduced words for the longest permutation invert the same three pairs. Scalar commutativity makes the products identical. □

Chapter 51

RTT and reflection algebras

51.1 The RTT bialgebra

Let $T(u) = (t_{ij}(u))$ be a matrix of formal currents. The RTT algebra $\mathcal{T}_R$ is defined by

$$R_{12}(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u-v). \tag{51.1}$$

Theorem 51.1 (RTT coproduct)**.** *The formula*

$$\Delta(T(u)) = T_{13}(u)T_{12}(u), \qquad \epsilon(T(u)) = I,$$

defines a coassociative bialgebra structure on $\mathcal{T}_R$.

Proof. Apply (51.1) in each tensor factor. The Yang–Baxter equation moves the three R-operators through the three matrix factors. The resulting identity is the RTT relation for $\Delta(T)$. Matrix multiplication is associative, hence the two iterated coproducts both equal $T_{14}T_{13}T_{12}$. □

For the Yang matrix $R(u) = 1 - \hbar P/u$, expansion at $u = \infty$ gives the RTT Yangian. The coefficient of $u^{-1}v^{-1}$ is

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar\big(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}\big),$$

so the first current layer is the Rees enveloping algebra of $\mathfrak{gl}_N$.

51.2 Boundary exchange

A reflecting boundary reverses the spectral path. For a constant invertible $R \in \mathrm{End}(V^{\otimes 2})$ satisfying Yang–Baxter, define the reflection algebra $\mathcal{K}_R$ by

$$R_{21}K_1R_{12}K_2 = K_2R_{21}K_1R_{12}. \tag{51.2}$$

Theorem 51.2 (Boundary confluence)**.** *The relation (51.2) gives a coherent exchange law for three boundary insertions. The two reductions of every length-three overlap agree by the Yang–Baxter equation and its inverse.*

Proof. A boundary exchange consists of an incoming crossing, reflection, and outgoing crossing. The two reductions of three insertions sweep the same braid in the doubled half-plane. Algebraically, one inserts (51.2) three times and uses Yang–Baxter to pass the middle R-matrix through the outer pair. The remaining factors cancel in inverse pairs. □

The RTT algebra governs bulk line exchange. The reflection algebra governs boundary line exchange. The ZF algebra governs charged fields in a fixed representation. Their common coefficient is the same R-matrix, while their products live in three different endomorphism objects.

Chapter 52

Quantum vertex realization

52.1 Two expansion regions

Let V be a complete topological vector space with vacuum $\mathbf{1}$, translation T, state-field map Y, and a unitary rational operator

$$S(z) : V \otimes V \longrightarrow V \otimes V((z)).$$

The two formal regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$ are related by $S(z_1 - z_2)$.

Definition 52.1 (Rational quantum vertex datum)**.** A rational quantum vertex datum satisfies vacuum and translation covariance together with

$$(z_1 - z_2)^N Y(a, z_1) Y(b, z_2) = (z_1 - z_2)^N Y^{21}(S(z_1 - z_2)(a \otimes b); z_2, z_1) \tag{52.1}$$

for a sufficiently large N, and weak associativity in every triple of states.

Theorem 52.2 (Exchange coherence)**.** *Suppose S is unitary,*

$$S^{21}(-z) S(z) = 1,$$

and satisfies the spectral Yang–Baxter equation. Then the pairwise continuation maps obtained from (52.1) *define a braid-group action on every n-point field product. Weak associativity makes the action compatible with iterated operator products.*

Proof. Unitarity identifies a crossing followed by its reverse with the identity. Yang–Baxter identifies the two continuations around every codimension-two triple-collision stratum. These are the defining relations of the braid group. Weak associativity identifies the collision limit in each chamber with the corresponding iterated field. □

When S is the graded flip, the datum is an ordinary local vertex algebra. A nontrivial S retains the ordered chiral product and its exchange monodromy. This is the formal-disc realization of an associative chiral algebra with braiding data in the sense of quantum vertex algebra theory [68, 69].

52.2 The Yangian field

Let $T(u)$ act on a family of evaluation modules. The RTT relation becomes S-locality for the matrix coefficients of $T(u)$. The normalized Yang operator

$$S(u) = f(u)\left(1 - \frac{\hbar P}{u}\right), \qquad f(u) f(-u) = \left(1 - \frac{\hbar^2}{u^2}\right)^{-1},$$

is unitary. Its expansion at infinity gives the quantum vertex exchange of Yangian currents. The scalar factor changes normalization and preserves the Yang–Baxter equation.

Chapter 53

Shuffle realization and the positive current algebra

53.1 Pairwise kernel

Let $K(z)$ be a rational function and put

$$\mathrm{Sh}_n = \mathbb{C}(z_1, \dots, z_n)^{\mathfrak{S}_n}.$$

Write Sym for the sum over ordered two-block partitions, with no factorial divisor. For $f \in \mathrm{Sh}_m$ and $g \in \mathrm{Sh}_n$ define

$$(f * g)(z_1, \dots, z_{m+n}) = \mathrm{Sym}\left(f(z_1, \dots, z_m)\, g(z_{m+1}, \dots, z_{m+n}) \prod_{i \le m < j} K(z_i - z_j) \right). \tag{53.1}$$

Theorem 53.1 (Associativity of the shuffle product)**.** *The product* (53.1) *is associative on every subspace closed under the displayed symmetrization and pole conditions.*

Proof. In a triple product, every ordered pair of variables belonging to different blocks contributes exactly one factor $K(z_i - z_j)$. The product of all pairwise factors is independent of the parenthesization. Final symmetrization gives the same function. □

For the tripled Jordan quiver, one uses

$$K(z) = \prod_{a=1}^{3} \frac{z + \epsilon_a}{z}, \qquad \epsilon_1 + \epsilon_2 + \epsilon_3 = 0.$$

A critical Hall or Yangian realization additionally specifies its coefficient lattice, wheel conditions and comparison map [89, 90]. The displayed meromorphic kernel alone does not identify that realization. The ordinary polynomial Hall kernels $N(v)/v$ and $vN(v)$ of Chapter 16 have their own distinct lattices.

Calculation 53.2 (Degree one against degree one)**.** For $f(z_1) = z_1^r$ and $g(z_1) = z_1^s$,

$$f * g = z_1^r z_2^s K(z_1 - z_2) + z_2^r z_1^s K(z_2 - z_1).$$

Put $u = z_1 - z_2$ and $\sigma_3 = \epsilon_1\epsilon_2\epsilon_3$. The antisymmetric part is

$$\frac{2\sigma_3}{u^3}(z_1^r z_2^s - z_1^s z_2^r).$$

For generic nonzero parameters, $u = -\epsilon_a$ is a zero of $K(u)$ and contributes no residue of this commutator. For $(r, s) = (2, 0)$, the residue at the true diagonal $u = 0$, with z_2 fixed, is $2\sigma_3$.

Proof. The two ordered partitions give the displayed product. Subtract the reversed product and use $K(u) - K(-u) = 2\sigma_3/u^3$. For $(r, s) = (2, 0)$, substitute $z_1 = z_2 + u$. The result is $4\sigma_3 z_2/u^2 + 2\sigma_3/u$. Shifted residues of the reciprocal kernel are instead given by (16.3). A geometric operation must specify which meromorphic coefficient it uses. □

53.2 Coproduct and double

A chamber expansion fixes a coefficient completion, but it does not prove compatibility of a proposed coproduct with multiplication. Theorem 16.6 gives a charge-two obstruction for the ordinary tensor on the polynomial Hall carriers. Proposition 16.7 constructs a positive rational cooperation with a specified changed middle interchange. A Drinfeld double additionally requires compatible positive and negative Hopf data, a continuous pairing and a completion in which its canonical tensor exists. Under those hypotheses the double construction gives the quasitriangular structure.

Chapter 54

The ladder in one diagram

The nonlocal structures form the chain

$$\text{ordered coefficient} \longrightarrow \mathcal{Z}_R \longrightarrow \mathcal{T}_R \longrightarrow (V, Y, S),$$

$$(V, Y, S) \longrightarrow \mathrm{Sh}^{+} \longrightarrow D\big(\mathrm{Sh}^{+}\big).$$

The arrows retain different data:

Object	Defining operation	Coherence
$\mathcal{Z}_R$	exchange of charged fields	Yang–Baxter equation
$\mathcal{T}_R$	matrix-current product and coproduct	RTT and coassociativity
$\mathcal{K}_R$	boundary exchange	reflection equation
(V, Y, S)	formal field product	S-locality and weak associativity
Sh^{+}	extension shuffle	pairwise-kernel associativity
$D(\mathrm{Sh}^{+})$	positive–negative pairing	quasitriangularity

This ladder gives a concrete generator-and-relation theory of doubly noncommutative chiral observables. The first noncommutativity is ordered composition. The second is exchange between distinct order chambers. Chiral singularity remains the diagonal coefficient throughout.

Book X

Field theory and master equations

Chapter 55

A finite cotangent BF model

Let L be a finite-dimensional dg Lie algebra. Its cotangent BV space is

$$\mathcal{E}_L = L[1] \oplus L^*[-2],$$

with the canonical degree -1 symplectic pairing. Write $A \in L[1]$ and $B \in L^*[-2]$. The cotangent action is

$$S_L(A, B) = \langle B, d_L A + \tfrac{1}{2}[A, A] \rangle.$$

Theorem 55.1 (Cotangent master equation)**.** *The Hamiltonian vector field of S_L is square-zero, and consequently $\{S_L, S_L\} = 0$.*

Proof. The A component is the Maurer–Cartan vector field $d_L A + \frac{1}{2}[A, A]$; the B component is its cotangent lift. The square of the first component is the sum of d_L^2, the derivation identity for d_L, and the Jacobi identity. The cotangent lift preserves commutators of vector fields, so the full Hamiltonian vector field is square-zero. □

Proposition 55.2 (Coordinate form of the modular term)**.** *Let L be an ordinary finite-dimensional Lie algebra with basis e_a and structure constants $[e_a, e_b] = f_{ab}^{c} e_c$. Let c^a be the odd coordinates on $L[1]$ and b_a their BV-conjugate coordinates. With*

$$S = \frac{1}{2} b_c f_{ab}^{c} c^a c^b, \qquad \Delta = -\sum_a \frac{\partial}{\partial c^a} \frac{\partial}{\partial b_a},$$

one has

$$\Delta S = \sum_b \operatorname{Tr}(\operatorname{ad}_{e_b}) c^b.$$

Thus the one-loop class of the finite cotangent theory is exactly the modular character.

Proof. Differentiation first in b_a gives $\frac{1}{2} f_{ij}^{a} c^i c^j$. Odd differentiation in c^a gives $f_{aj}^{a} c^j$, and the chosen sign in Δ converts this to $f_{ja}^{a} c^j = \operatorname{Tr}(\operatorname{ad}_{e_j}) c^j$. This is the displayed formula. □

A finite-dimensional dg Lie algebra is unimodular when the supertrace of every adjoint endomorphism vanishes. Every finite-dimensional nilpotent Lie algebra in degree zero is unimodular because each ad_x is nilpotent.

Theorem 55.3 (Finite-dimensional quantum master equation)**.** *Let L be unimodular and let Δ be the BV Laplacian determined by the translation-invariant Berezinian on $\mathcal{E}_L$. Then*

$$\Delta S_L = 0, \qquad \frac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L = 0,$$

and equivalently

$$\hbar \Delta\left(e^{S_L/\hbar}\right) = 0.$$

The quantum observable differential

$$\delta_{S_L} = \{S_L, -\} + \hbar\Delta$$

satisfies $\delta_{S_L}^2 = 0$.

Proof. The classical master equation gives $\{S_L, S_L\} = 0$. The scalar ΔS_L is the divergence of the Chevalley vector field on $L[1]$. Its linear coefficient is the modular character $x \mapsto \mathrm{str}(\mathrm{ad}_x)$, which vanishes by unimodularity. The BV exponential identity

$$\hbar^2 e^{-S_L/\hbar}\Delta(e^{S_L/\hbar}) = \frac{1}{2}\{S_L, S_L\} + \hbar\Delta S_L$$

proves the exponential form. The standard BV commutator calculation then gives

$$\delta_{S_L}^2 = \left\{\frac{1}{2}\{S_L, S_L\} + \hbar\Delta S_L, -\right\} = 0.$$

□

Now take the local dg Lie algebra

$$\mathcal{L}_X = \Omega^{0,*}(X) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes \mathfrak{l}$$

for a finite-dimensional nilpotent $\mathfrak{l}$, and form its shifted cotangent theory with compactly supported density dual. The preceding finite theorem governs every pairing-preserving finite-dimensional deformation retract. The local classical observables are Chevalley cochains and are holomorphic on X and locally constant on $\mathbb{R}$.

Choose the standard interval polarization in which the $L[1]$ variables are fixed at one end and the cotangent variables at the other. Canonical quantization of the boundary linear Poisson algebra gives the associative zero-mode algebra $U(\mathfrak{l})$. Gluing the two augmentation boundary states across an interval gives the derived tensor product

$$\mathbf{1} \otimes^{\mathbb{L}}_{U(\mathfrak{l})} \mathbf{1} \simeq \mathrm{Bar}(U(\mathfrak{l})) \simeq C_*(\mathfrak{l}),$$

the Chevalley–Eilenberg chains. A solution of the renormalized quantum master equation in the local BV deformation complex produces the corresponding infinite-dimensional quantum factorization algebra.

Chapter 56

BRST and ordered reduction

Let (A, Q) be a dg ordered algebra, with Q a derivation. Extend Q to the bar factorwise. Write b for the multiplication bar differential.

Proposition 56.1 (BRST–bar bicomplex)**.** *With the suspension sign convention,*

$$Q_{\mathrm{Bar}} b + b Q_{\mathrm{Bar}} = 0.$$

Hence $D = Q_{\mathrm{Bar}} + b$ satisfies $D^2 = 0$.

Proof. Apply Q to a bar face that multiplies adjacent entries. The derivation formula splits the result into the two terms obtained by applying Q before that face. The totalization sign makes their sum the anticommutator. □

The filtration by bar length gives

$$E_1 \cong \mathrm{Bar}(H^*(A, Q)) \Longrightarrow H^*(\mathrm{Bar}(A), D)$$

under boundedness and completeness. Quantum Drinfeld–Sokolov reduction is a current-algebra realization of this bicomplex.

Chapter 57

Stokes and algebraic identities

Let $C_*(K)$ be the oriented cellular chains of a regular manifold with corners. Suppose each codimension-one face F is assigned a degree-one operator ρ_F on a coefficient complex E, and the assignments are compatible with restrictions to corners.

Theorem 57.1 (Stokes-to-master)**.** *Assume that each ρ_F is a chain map of total degree one, that its restriction to a corner depends on the induced contraction, and that disjoint contractions act with the Koszul sign prescribed by the oriented normal bundle. Then the component of*

$$D^2, \qquad D = d_E + d_K + \sum_F \rho_F,$$

supported on a codimension-two corner S is the oriented sum

$$\sum_{F \supset S} \epsilon(F, S)\, \rho_{F/S} \rho_F.$$

Thus D is a differential precisely when these corner sums vanish.

Proof. Expand D^2 and group terms by support strata. The chain-map condition cancels the mixed internal terms. A term supported on S is obtained by crossing two incident faces. The two orders carry the boundary orientations of the two sides of the oriented normal square. Their signed sum is the displayed expression. □

Associahedra give Stasheff identities. Fulton–MacPherson collision strata give chiral and graph identities after a coefficient map is supplied. Stable-graph boundaries give BV and modular master equations.

Chapter 58

Two edge colors and two loop operators

Let $\mathsf{G}^{\perp,\mathrm{ch}}$ be the completed vector space spanned by stable graphs with transverse and chiral edges. Contracting an edge that joins distinct vertices gives brackets $[-,-]_{\perp}$ and $[-,-]_{\mathrm{ch}}$. Contracting a loop gives operators $\Delta_{\perp}$ and Δ_{ch}. Product orientation gives

$$\Delta_{\perp}\Delta_{\mathrm{ch}} + \Delta_{\mathrm{ch}}\Delta_{\perp} = 0.$$

Theorem 58.1 (Bicolored master equation). *Let $\mathsf{P}^{\perp,\mathrm{ch}}$ be a two-colored modular operad whose edge contractions carry the product determinant orientation, and let $\mathfrak{g}_{\mathsf{P}}$ be the completed convolution Lie algebra of its Feynman transform with an endomorphism modular operad. Algebra structures on the transform are Maurer–Cartan elements $\Theta \in \mathfrak{g}_{\mathsf{P}}^{1}$ satisfying*

$$d\Theta + \frac{1}{2}[\Theta,\Theta]_{\perp} + \frac{1}{2}[\Theta,\Theta]_{\mathrm{ch}} + \hbar_{\perp}\Delta_{\perp}\Theta + \hbar_{\mathrm{ch}}\Delta_{\mathrm{ch}}\Theta = 0.$$

Proof. The Feynman-transform differential contracts one edge. Separating edges give the two brackets and loop edges give the two BV operators. The product orientation makes mixed contractions anticommute. The condition that the structure map commute with the transform differential is exactly the displayed Maurer–Cartan equation [40]. □

Chapter 59

Typed anomalies

An anomaly is a cohomology class in the deformation complex that governs the attempted construction.

construction	complex	class
central extension	local Lie algebra cochains	cocycle
BRST quantization	BRST deformation complex	nilpotence obstruction
BV quantization	local BV cochains	quantum master anomaly
conformal-block family	Atiyah algebra of its determinant line	projective curvature
curved Koszul duality	curved convolution Lie algebra	curvature m_0

A comparison theorem consists of a morphism between two complexes and an equality of the transported classes. This formulation keeps the geometric origin of every scalar visible.

Book XI

Calabi–Yau and automorphic interfaces

Chapter 60

Calabi–Yau categories

Let C be a smooth linear category with a d-Calabi–Yau structure. Two universal consequences are available.

Theorem 60.1 (Cyclic and shifted outputs)**.** *The Hochschild cochains $CC^*(C)$ form a framed E_2 algebra. The derived moduli stack of objects carries a $(2-d)$-shifted symplectic structure whenever the representability hypotheses hold.*

Proof. The first assertion is the cyclic Deligne theorem [15]. The second is the PTVV–Brav–Dyckerhoff construction: the Calabi–Yau trace pairs the tangent complex $\mathrm{RHom}(E,E)[1]$ with itself in degree $2-d$ [14, 84]. □

A geometric BV theory or an open factorization category supplies an ordered chiral algebra through the recognition theorem of Chapter 3. This is the bridge from Calabi–Yau geometry to the present volume.

Chapter 61

Hall and BPS algebras

A Hall multiplication starts from the extension correspondence

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}_{\alpha,\beta}^{\mathrm{ext}} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

With a coefficient theory for which p^* and q_* are defined,

$$a * b = q_* p^*(a \boxtimes b).$$

Associativity follows from the stack of two-step filtrations.

For 2-Calabi–Yau categories with good moduli spaces, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra [27]. For quivers and surface resolutions, preprojective and surface CoHAs recover positive Yangian halves in established cases [30, 89].

The ordered chiral structure enters when Hall multiplication is promoted to a factorization coefficient over a curve or defect parameter. A coproduct, pairing, and completion then produce the corresponding double.

Chapter 62

The Igusa denominator as a bar character

Let $\mathfrak{g}_\Delta$ be the Gritsenko–Nikulin Borcherds–Kac–Moody superalgebra and let $\mathfrak{n}_+$ be its positive nilpotent part. Root spaces are finite-dimensional and the positive cone is locally finite. Fix a proper integral height h that is positive on the effective root cone, and set

$$F^{>H}\mathfrak{n}_+ = \bigoplus_{h(\alpha)>H} \mathfrak{g}_{\Delta,\alpha}, \qquad \mathfrak{n}_+^{(H)} = \mathfrak{n}_+/F^{>H}\mathfrak{n}_+.$$

Height additivity makes $F^{>H}\mathfrak{n}_+$ an ideal; properness makes $\mathfrak{n}_+^{(H)}$ finite-dimensional and nilpotent. Every bar, Chevalley, enveloping, and character formula in this chapter is interpreted coefficientwise in the separated height completion.

Theorem 62.1 (Bar–Chevalley identification)**.** *For each height H there is a natural quasi-isomorphism of conilpotent coalgebras*

$$\mathrm{Bar}\big(U(\mathfrak{n}_+^{(H)})\big) \simeq C_*(\mathfrak{n}_+^{(H)}).$$

The quotient maps are compatible with these quasi-isomorphisms, and therefore induce a quasi-isomorphism in the separated height-complete category

$$\mathrm{Bar}_h^\wedge\big(U(\mathfrak{n}_+)\big) := \varprojlim_H \mathrm{Bar}\big(U(\mathfrak{n}_+^{(H)})\big) \simeq C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

The coefficientwise Euler supercharacter of the completed Chevalley complex is

$$\chi\, C_{*,h}^\wedge(\mathfrak{n}_+) = \prod_{\alpha>0} (1-e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta,\alpha}}.$$

Proof. For every H, the PBW filtration on $U(\mathfrak{n}_+^{(H)})$ identifies the associated graded with $\mathrm{Sym}(\mathfrak{n}_+^{(H)})$. The bar complex of the enveloping algebra and the Chevalley complex have the same associated-graded Koszul resolution. The standard antisymmetrization map is a filtered quasi-isomorphism. Naturality for the quotient maps gives the inverse-limit comparison; proper height makes every homogeneous coefficient stabilize. For an even root vector the exterior Chevalley factor contributes $1-e^{-\alpha}$; for an odd root vector it contributes its inverse. Multiplication over the root spaces gives the product. □

With the positive-root convention $e^{-\rho} = q^{1/2}r^{1/2}s^{1/2}$, the denominator theorem reads

$$e^{-\rho}\,\chi\, C_{*,h}^\wedge(\mathfrak{n}_+) = D_5(2\Omega) = 2^{-6}\Delta_5(2\Omega)$$

[46, 48]. This identity is the exact meeting point of ordered bar theory and automorphic products.

Chapter 63

Finite-height currents and an exact physical model

Retain the finite-height quotients $\mathfrak{n}_+^{(H)}$ fixed in the preceding chapter.

The current Lie conformal superalgebra is

$$\operatorname{Cur}(\mathfrak{n}_+^{(H)}) = \Bbbk[\partial] \otimes \mathfrak{n}_+^{(H)}, \qquad [a_\lambda b] = [a, b].$$

Its universal enveloping vertex superalgebra is V_H. The current generators carry the cocommutative vertex-bialgebra coproduct

$$\Delta\big(a(-1)\mathbf{1}\big) = a(-1)\mathbf{1} \otimes \mathbf{1} + \mathbf{1} \otimes a(-1)\mathbf{1}, \qquad a \in \mathfrak{n}_+^{(H)},$$

extended uniquely as a vertex-algebra morphism.

Theorem 63.1 (Height-complete current realization)**.** *Let $\mathfrak{n} = \bigoplus_{\alpha>0} \mathfrak{n}_\alpha$ be a root-graded Lie superalgebra with finite-dimensional root spaces and a proper additive height function. Put*

$$\mathfrak{n}^{(H)} = \mathfrak{n} \Big/ \bigoplus_{h(\alpha)>H} \mathfrak{n}_\alpha.$$

Then every $\mathfrak{n}^{(H)}$ is finite-dimensional and nilpotent, the quotient maps form an inverse system, and

$$\widehat{U}_h(\mathfrak{n}) = \varprojlim_H U(\mathfrak{n}^{(H)}), \quad \widehat{C}_{*,h}(\mathfrak{n}) = \varprojlim_H C_*(\mathfrak{n}^{(H)}), \quad \widehat{V}_h(\mathfrak{n}) = \varprojlim_H U^{\mathrm{ch}}(\operatorname{Cur}_X \mathfrak{n}^{(H)})$$

are separated complete algebras, coalgebras, and current algebras. Their associated graded objects are the corresponding completed symmetric objects, and the bar–Chevalley comparison is compatible with the inverse system.

Proof. Properness gives finitely many roots below a fixed height. Additivity implies that the roots of height $> H$ form a Lie ideal and that the quotient is nilpotent. PBW and Chevalley antisymmetrization are natural under quotient maps. Taking inverse limits in each finite total height therefore gives the displayed complete objects and preserves the coefficientwise quasi-isomorphism. □

Apply the finite BF model of Chapter 19 to every $\mathfrak{n}_+^{(H)}$. The inverse system gives a height-complete holomorphic–topological classical field theory with completed boundary algebra and interval complex

$$U_h^\wedge(\mathfrak{n}_+) := \varprojlim_H U(\mathfrak{n}_+^{(H)}), \qquad C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

This gives an exact algebraic realization, a local classical holomorphic–topological realization, and an exact quantum realization of the finite-dimensional zero-mode sectors of the positive enveloping algebra and its denominator character. The compact $K3 \times E$ identification is the geometric comparison studied in the companion volumes.

Chapter 64

Character, bracket, parity, and orientation

A character is an Euler invariant. Faithful reconstruction combines it with the bracket, parity decomposition, and orientation line carried by the intended object.

Theorem 64.1 (Three reconstruction fibres)**.** *The following fibres contain distinct structured objects with the same scalar image.*

(i) *The PBW-character fibre of a graded vector space with basis x, y, z in degrees a, b, $a + b$ contains both the abelian Lie algebra and the Heisenberg Lie algebra $[x, y] = z$.*

(ii) *The superdimension-zero fibre contains both the zero super vector space and $\Bbbk^{1|1}$.*

(iii) *The squaring fibre of a line-valued section contains s and $s \otimes \chi$ for every sign local system χ.*

Consequently a faithful reconstruction datum consists respectively of a bracket, a parity-refined dimension, and an oriented square-root line.

Proof. PBW identifies the graded vector-space character of an enveloping algebra with the symmetric-superalgebra character of its underlying graded Lie space, so the two displayed brackets have the same PBW character. The two super vector spaces in the second item both have superdimension zero and have parity dimensions $(0, 0)$ and $(1, 1)$. Finally $\chi^{\otimes 2} \cong \mathbf{1}$, hence

$$(s \otimes \chi)^{\otimes 2} \cong s^{\otimes 2}.$$

The extra structures listed in the conclusion distinguish each pair. □

Recognition Theorem 64.2 (Minimal PBW recognition)**.** *Let $\phi : \mathfrak{g} \twoheadrightarrow \mathfrak{h}$ be a surjective morphism of positively graded Lie superalgebras with finite-dimensional homogeneous pieces. If the parity-refined Hilbert series of $U(\mathfrak{g})$ and $U(\mathfrak{h})$ agree, then ϕ is an isomorphism. Equality of signed PBW characters suffices when the parity of every primitive degree is prescribed in advance.*

Proof. If the kernel is nonzero, choose its first positive degree. In that degree all decomposable PBW contributions agree by minimality, while the primitive even or odd dimension in $U(\mathfrak{h})$ is strictly smaller. This contradicts equality of parity-refined Hilbert series. With prescribed parity, the formal logarithm of the PBW product recovers the signed primitive dimension inductively in height, and prescribed parity recovers the ordinary dimension. □

Chapter 65

Exact dependency theorem

Theorem 65.1 (Ordered-chiral closure)**.** *Fix:*

(i) *a six-functor Ran category and its factorization subcategory;*

(ii) *a transversely ordered algebra, an associative chiral algebra, or a chiral Lie algebra, as appropriate;*

(iii) *every mixed distributive, completion, duality, trace, or analytic datum named by the intended construction.*

Then the three bars, their complete cobars, their module/comodule Koszul transforms, the Hochschild centre, the formal-disc realization, the ribbon theory, and the finite-height current realization are defined functorially and satisfy the identities proved in this volume. Every passage between the pointwise and chiral-convolution monoidal products is carried by the displayed mixed distributive datum.

Proof. Each output is one of the functors constructed in the preceding chapters. The source and target monoidal categories are displayed in every definition. The comparison theorems invoke the explicitly supplied distributive, PBW, completion, duality, or trace datum. Composition of these functors gives the asserted functorial system. □

Chapter 66

Synthesis

The invariant language has three algebraic axes. Transverse order is an E_1 structure for $\otimes^!$, associative chirality an E_1 structure for $\star^{\mathrm{ch}}$, and chiral Lie theory a Lie structure for $\star^{\mathrm{ch}}$. Their bars are $\mathrm{Bar}^{\perp}$, $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}$ and $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}$. The aligned projector describes the canonical common-decomposition comparison under its stated descent hypotheses. A mixed distributive law and comparisons among the three bars retain their own maps and coherence equations.

The constructed operations form the two functorial rows

$$\text{HT boundary theory} \xrightarrow{\mathrm{Obs}_{\partial}} \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(X)) \xrightarrow{\mathrm{Bar}^{\mathrm{ord}}} \mathrm{Coalg}_{\mathrm{conil,cpl}}(\mathrm{Fact}_D(X)),$$

$$\mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(X)) \xrightarrow{Z_{\mathrm{ord}}} \mathrm{Alg}_{\mathsf{E}_2}(\mathrm{Fact}_D(X)).$$

A primitive Lie algebra enters when a separate Chevalley or Hall comparison identifies the bar coalgebra with the Chevalley chains of that Lie algebra. Hall geometry supplies this comparison through extension correspondences. Quantum groups require an additional coproduct and exchange structure. Automorphic forms arise from graded Euler characters after a grading and completion have been fixed. Each arrow therefore preserves the geometric operation named at its source.

Book XII

Technical appendices to the preceding book

Chapter 67

Sign calculation for the length-three bar

For homogeneous $a, b, c \in \bar{A}$, write $[a|b|c] = sa \otimes sb \otimes sc$. With the standard suspension convention,

$$b_m[a|b|c] = (-1)^{|a|}[ab|c] + (-1)^{|a|+|b|+1}[a|bc].$$

Applying b_m again gives

$$b_m^2[a|b|c] = (-1)^{|b|}\big[(ab)c - a(bc)\big],$$

which vanishes by associativity.

Chapter 68

Rational R-matrix verification

Let P_{ij} permute tensor factors i and j. Expanding

$$R_{12}(u)R_{13}(u+v)R_{23}(v)$$

and its reversed product shows equality coefficientwise. The cubic coefficient uses

$$P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12},$$

the quadratic coefficients use the symmetric-group relations, and the linear coefficients agree termwise.

Chapter 69

Carriers and comparison maps

The five-carrier geometry, transverse bar construction, mixed-forest signs, and noncommutative examples use four kinds of categorical data:

(i) a factorization carrier is paired explicitly with its operation;

(ii) the Ran category is fixed before its tensor products are used;

(iii) the block-ordered operad is introduced after a representing chiral operad is chosen;

(iv) ribbon and stable-curve sewing are derived from nondegenerate Calabi–Yau and modular data.

A comparison between these constructions therefore specifies its source and target categories and the operations preserved by its map.

Chapter 70

Enveloping bars and planar path sequences

The dimension of a bar chain space and the dimension of its homology are different invariants. For an enveloping algebra, the Chevalley comparison gives a Lie-theoretic model for the homology. This makes it possible to compare Motzkin and Riordan path counts with the ordered bar on a specified carrier.

Bar length and Chevalley degree are homological degrees: H_n denotes degree-n homology. The cohomology of the dual Chevalley complex is written H^n. For finite-dimensional weight pieces their dimensions agree.

70.1 Chevalley comparison

Theorem 70.1 (Bar–Chevalley comparison). *Let k be a field of characteristic zero and let $\mathfrak{g}$ be an ordinary Lie algebra over k. Give $U(\mathfrak{g})$ its augmentation killing $\mathfrak{g}$. There is a natural quasi-isomorphism of ordinary conilpotent differential graded coalgebras*

$$\kappa : C_*(\mathfrak{g}) \longrightarrow \mathrm{Bar}^{\perp}\big(U(\mathfrak{g})\big),$$

where both constructions use direct sums and ordinary tensors over k. It preserves the PBW filtration. If $\mathfrak{g} = \bigoplus_{w \geq 1} \mathfrak{g}_w$ and $[\mathfrak{g}_u, \mathfrak{g}_v] \subseteq \mathfrak{g}_{u+v}$, the map is a quasi-isomorphism in each total weight. It extends to the degreewise product over weights, using completed tensor products by total weight. The completed coalgebras are conilpotent in each weight; no global conilpotence is asserted for infinite sums.

Proof. Put $C_n(\mathfrak{g}) = \Lambda^n \mathfrak{g}$ in homological degree n, with differential

$$d(x_1 \wedge \cdots \wedge x_n) = \sum_{i<j} (-1)^{i+j+1} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_n.$$

Its coproduct splits the exterior factors between two ordered sublists, with the sign of the corresponding permutation. The ordinary normalized bar differential joins positions $i, i+1$ with sign $(-1)^{i-1}$. Define

$$\kappa(x_1 \wedge \cdots \wedge x_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) [x_{\sigma(1)} | \cdots | x_{\sigma(n)}].$$

At degree zero this map preserves the unit. At degree two its differential is $[x_1 x_2 - x_2 x_1] = [[x_1, x_2]]$. In every degree, pair permutations whose merged adjacent entries occur in opposite orders. Their difference is the bracket of those entries, with the displayed exterior sign. This proves the chain-map equation. Cutting an antisymmetrized word amounts to choosing an ordered subset and antisymmetrizing its two subwords. The two procedures give the same permutations and signs, proving the coalgebra identity.

Filter $U(\mathfrak{g})$ by PBW degree, the bar by total PBW degree, and $\Lambda^n \mathfrak{g}$ by degree n. PBW identifies the associated graded algebra with $S(\mathfrak{g})$. The associated graded comparison is the antisymmetrization $\Lambda(\mathfrak{g}) \to B(S(\mathfrak{g}))$, with zero differential on its source. To verify that it is a quasi-isomorphism, consider the free Koszul resolution $S(V) \otimes \Lambda(V)$ for $V = \mathfrak{g}$. For a basis (x_i) of V, its differential is $\partial = \sum_i x_i \iota_i$, where ι_i contracts the exterior factor x_i. On polynomial degree p and exterior degree q, with $p + q > 0$, the map

$$h = \frac{1}{p+q} \sum_i (x_i \wedge -) \, \frac{\partial}{\partial x_i}$$

satisfies $\partial h + h\partial = 1$. Only finitely many summands act on each element, even when V is infinite-dimensional. The identity follows by cancellation of cross terms and the Euler identity in the two degrees. Thus this free resolution has augmentation homology k only. The free normalized bar resolution has the same homology by its extra-degeneracy contraction. Antisymmetrization defines an augmentation-preserving comparison between these free resolutions. The simplicial bar sign convention is related to the displayed bar convention by multiplication by $(-1)^n$ in degree n. Free bases permit a reverse comparison and homotopies to be lifted inductively, since every positive-degree cycle in a resolution is a boundary. Tensoring these homotopies with k proves the asserted quasi-isomorphism on associated graded complexes.

Every cycle in the cone of κ has finite PBW filtration degree. A primitive for its highest associated graded part lowers that degree; iteration terminates because the filtration is bounded below. The cone is acyclic, proving the ordinary assertion. For a positive weight grading, PBW degree is at most total weight. The same argument is therefore finite within each weight. Finally, kernels and images of a degreewise product differential are products of kernels and images, so the componentwise quasi-isomorphisms extend to the stated completion. □

The ordinary comparison does not itself construct a morphism in $\mathrm{FactD}(X)$. For geometric coefficient systems, one must give compatible enveloping and Chevalley constructions on every chart and coherent factorization equivalences for the whole Chevalley complex. On each disjointness locus, with restriction denoted by $j^!$, the required equation is

$$f_{B,I}\, j^! \kappa_I = (\kappa_{I_1} \boxtimes \kappa_{I_2})\, f_{C,I}, \qquad I = I_1 \sqcup I_2.$$

Here $f_{C,I}$ and $f_{B,I}$ are the specified Chevalley and bar factorization maps. The bar map may use the signed shuffle comparison under its geometric functor hypotheses. The Chevalley factorization maps, collision compatibility, refinement coherence, and continuity of the completed maps remain additional geometric data to construct or verify. Internal tensor formulas and PBW alone do not provide them.

Calculation 70.2 (Finite-dimensional simple case)**.** For a simple Lie algebra of rank r with exponents $m_1, \dots, m_r$, the Chevalley cohomology is the exterior algebra on primitive classes of degrees $2m_i + 1$. The corresponding homology Poincaré polynomials are

$$\mathrm{Bar}^{\perp} U(\mathfrak{sl}_2) : 1 + t^3, \qquad \mathrm{Bar}^{\perp} U(\mathfrak{sl}_3) : 1 + t^3 + t^5 + t^8, \qquad \mathrm{Bar}^{\perp} U(\mathfrak{g}_2) : 1 + t^3 + t^{11} + t^{14}.$$

The corresponding Betti sequences for $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ are $1, 0, 0, 1$ and $1, 0, 0, 1, 0, 1, 0, 0, 1$. For a finite-dimensional Lie algebra, the Chevalley chain dimensions are $\binom{\dim \mathfrak{g}}{n}$.

For an ordinary augmented algebra A over $\mathbb{C}$, let $I = \ker(A \to \mathbb{C})$. Its normalized two-sided bar with trivial coefficients has homological chain spaces

$$B_0(A) = \mathbb{C}, \qquad B_n(A) = I^{\otimes n} \quad (n \geq 1).$$

Here n is bar length, independently of any internal weight or PBW degree. If I has finite dimension d, then $\dim B_n(A) = d^n$; in particular, an augmentation ideal of dimension three gives the chain counts 3^n. In a cohomological convention that places B_n in degree $-n$, its homology is denoted H^{-n} instead of H_n.

For $A = U(\mathfrak{sl}_2)$, PBW gives the basis $e^a h^b f^c$, with $a, b, c \geq 0$, where $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$. The augmentation ideal has all these basis elements except 1. Consequently $B_n(A)$ is countably infinite-dimensional for every $n \geq 1$. Its total PBW filtration has finite-dimensional associated graded pieces, with generating function

$$\sum_{d \geq 0} \dim \mathrm{gr}^{\mathrm{PBW}}_d B_n(A)\, z^d = \big((1 - z)^{-3} - 1\big)^n.$$

In particular, 3^n is the dimension in the least possible total PBW degree $d = n$. These least-degree dimensions are 3, 9, 27, 81, 243 for $n = 1, 2, 3, 4, 5$; the full bar chain spaces are infinite-dimensional. The quotient of indecomposables is zero: the perfectness of $\mathfrak{sl}_2$ implies $I/I^2 = 0$.

The Chevalley chain model is different: its degree-n space is $\Lambda^n \mathfrak{sl}_2$, of dimensions 1, 3, 3, 1 in degrees 0, 1, 2, 3. With the convention $d(x \wedge y) = [x, y]$, the map $d : \Lambda^2 \mathfrak{sl}_2 \to \mathfrak{sl}_2$ sends

$$e \wedge h \longmapsto -2e, \qquad e \wedge f \longmapsto h, \qquad h \wedge f \longmapsto -2f,$$

and is an isomorphism. The next differential vanishes, since

$$d(e \wedge h \wedge f) = -2e \wedge f - h \wedge h - 2f \wedge e = 0.$$

Together with $d : \mathfrak{sl}_2 \to \mathbb{C}$ equal to zero, this gives Chevalley homology dimensions 1, 0, 0, 1. The ordinary bar–Chevalley comparison identifies these homology groups with those of $U(\mathfrak{sl}_2)$; it does not identify the two chain spaces.

70.2 The Virasoro row

The transverse presentation of the Virasoro family is the enveloping algebra of the positive Witt algebra

$$L_1 = \bigoplus_{n \geq 1} \mathbb{C}\, e_n, \qquad [e_i, e_j] = (j - i) e_{i+j},$$

to which Theorem 70.1 applies: L_1 is positively graded with one-dimensional weight spaces, so every weight has finitely many decompositions.

Theorem 70.3 (Pentagonal rigidity)**.** *The bar homology of $U(L_1)$ satisfies*

$$\dim H_n\big(\mathrm{Bar}^{\perp} U(L_1)\big) = 2 \qquad \text{for every } n \geq 1,$$

one dimension in each of the weights $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Its graded Euler character is

$$\chi_q\big(\mathrm{Bar}^{\perp} U(L_1)\big) = \prod_{n \geq 1} (1 - q^n), \tag{70.1}$$

and (70.1) *is accounted for degree by degree with no residual freedom.*

Proof. By Theorem 70.1 the bar of $U(L_1)$ is quasi-isomorphic to $C_*(L_1)$. Each weight subcomplex is finite-dimensional, so its homology has the same dimensions as the weightwise dual cohomology. Goncharova's calculation gives two dimensions in each degree $n \geq 1$, in the weights $(3n^2 \mp n)/2$. Identity (70.1) is the denominator product applied to L_1, where every root space is one dimensional and even, so $\mathrm{smult}(L_1)_n = 1$. Euler's pentagonal number theorem expands the product as $\sum_{m \in \mathbb{Z}} (-1)^m q^{m(3m-1)/2}$, whose support is exactly $\{(3n^2 \pm n)/2\}$ with coefficient $(-1)^n$ at each of $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Comparing with $\sum_n (-1)^n \dim H_{n,w}$, each weight of the Euler character is saturated by a single class in the degree Goncharova predicts, so no further classes and no cancellation are available. $\square$

For $n = 1, 2, 3, 4$, the nonzero homology groups occupy the weight pairs $(1, 2)$, $(5, 7)$, $(12, 15)$, $(22, 26)$, one dimension in each. The corresponding nonzero coefficients of $\prod_{n \geq 1}(1 - q^n)$ occur at 0, 1, 2, 5, 7, 12, 15, 22, 26 with signs $+, -, -, +, +, -, -, +, +$.

70.3 Planar path sequences

A Motzkin path is a lattice path from $(0,0)$ to $(n,0)$, staying at nonnegative height, with steps $(1,1)$, $(1,0)$, and $(1,-1)$. Let $M(n)$ count all such paths of length n, and let $R(n)$ count those with no horizontal step at height zero. Both counts include the empty path: $M(0) = R(0) = 1$, while $M(1) = 1$ and $R(1) = 0$. For $n \geq 1$ consider the two sequences

$$a_n = M(n+1) - M(n) = 1, 2, 5, 12, 30, 76, \ldots, \qquad b_n = R(n+3) = 3, 6, 15, 36, 91, \ldots.$$

Proposition 70.4 (Motzkin and Riordan identities)**.** *For every $n \geq 0$,*

$$M(n) = R(n) + R(n+1), \qquad M(n+1) - M(n) = R(n+2) - R(n).$$

The ordinary generating functions of the two path counts, and hence those of a_n and b_n, belong to $\mathbb{Q}(z, \sqrt{1-2z-3z^2})$.

Proof. Work in the formal power-series ring $\mathbb{Q}[[z]]$, and write $M(z) = \sum_{n\geq 0} M(n)z^n$ and $R(z) = \sum_{n\geq 0} R(n)z^n$. A nonempty Motzkin path either starts with a horizontal step followed by an arbitrary Motzkin path, or has a unique first-return decomposition $UPDQ$, where U and D are the up and down steps, P and Q are arbitrary Motzkin paths, and P is drawn one unit above the horizontal axis. Thus

$$M = 1 + zM + z^2M^2.$$

A nonempty path counted by R has no initial horizontal step. Its first return similarly has the form $UPDQ$, with P arbitrary and Q counted by R. Consequently

$$R = 1 + z^2MR, \qquad R = (1 - z^2M)^{-1}.$$

The inverses here exist because their denominators have constant term one. Using $z^2M^2 = M - 1 - zM$, compute

$$(1 + zM)(1 - z^2M) = 1 + zM - z^2M - z^3M^2 = 1 + z.$$

It follows that $(1+z)R = 1 + zM$. Comparing coefficients of z^{n+1} gives $R(n+1) + R(n) = M(n)$ for all $n \geq 0$. Subtracting consecutive identities gives the difference formula.

Solving the quadratic equation for the formal series with constant term one gives

$$M(z) = \frac{1 - z - \sqrt{1-2z-3z^2}}{2z^2}, \qquad R(z) = \frac{1 + zM(z)}{1+z},$$

where the square root has constant term one. The apparent pole in the formula for M is removable as a formal series. Finite index shifts and differences preserve the indicated quadratic function field, proving the last assertion. These identities are consequences solely of the path decompositions. □

Remark 70.5 (Low-degree homology). Theorem 70.3 gives homology dimensions 2, 2, 2 in degrees 1, 2, 3, whereas a_1, a_2, a_3 are 1, 2, 5. These triples agree only in degree two. The degree-one discrepancy already excludes equality of the full homology dimension sequences.

Theorem 70.6 (Enveloping bars and planar path counts)**.** *Let $B_\bullet(A)$ be the ordinary normalized bar over $\mathbb{C}$, with trivial coefficients and homological degree given by bar length. With $a_n = M(n+1) - M(n)$ and $b_n = R(n+3)$ indexed by $n \geq 1$ as above,*

$$\dim H_1B_\bullet(U(\mathfrak{sl}_2)) = 0 \neq b_1 = 3, \qquad \dim H_1B_\bullet(U(L_1)) = 2 \neq a_1 = 1,$$

where $L_1 = \bigoplus_{m\geq 1} \mathbb{C}e_m$ and $[e_i, e_j] = (j-i)e_{i+j}$. Hence neither sequence is the bar homology dimension sequence of the corresponding enveloping algebra. Neither is its full bar chain dimension sequence: in both cases B_n is countably infinite-dimensional for every $n \geq 1$.

Proof. For any augmented algebra A with augmentation ideal I, the bar differential out of $B_1 = I$ is zero and the image from $B_2 = I \otimes I$ is I^2. Therefore

$$H_1 B_\bullet(A) = I/I^2.$$

For an ordinary Lie algebra $\mathfrak{g}$, the inclusion of generators induces an isomorphism

$$\mathfrak{g}/[\mathfrak{g},\mathfrak{g}] \xrightarrow{\sim} I/I^2, \qquad I = \ker(U(\mathfrak{g}) \to \mathbb{C}).$$

Indeed the generators span I/I^2, and $[x, y] = xy - yx$ vanishes there. For injectivity, equip $\mathbb{C} \oplus \mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ with the square-zero product on its second summand. The quotient map from $\mathfrak{g}$ to that summand is a Lie map into the commutator Lie algebra of this associative algebra. The universal property of $U(\mathfrak{g})$ extends it to an augmented algebra map, which kills I^2 and provides an inverse on the displayed quotient.

The brackets $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$ show that $\mathfrak{sl}_2$ equals its commutator algebra. For L_1, every nonzero bracket has weight at least three. Conversely, for $m \geq 3$,

$$e_m = \frac{1}{m-2}[e_1, e_{m-1}].$$

Thus $[L_1, L_1] = \bigoplus_{m\geq 3} \mathbb{C}e_m$, and its abelianization has basis the classes of e_1, e_2, in weights one and two. This proves both degree-one assertions without using any higher-degree calculation.

PBW gives a countably infinite basis of the augmentation ideal of either enveloping algebra. In the L_1 case the basis consists of nonempty ordered monomials in the countable set $e_1, e_2, \ldots$, each monomial having finite length. Finite tensor powers of either augmentation ideal are therefore countably infinite-dimensional. The path counts are finite in every degree, so they cannot count these full chain spaces. □

In particular, a complex whose homology dimensions are b_n cannot be quasi-isomorphic to $B_\bullet(U(\mathfrak{sl}_2))$, and one whose homology dimensions are a_n cannot be quasi-isomorphic to $B_\bullet(U(L_1))$, with the stated degree convention. A discrepancy of chain dimensions alone would not imply this conclusion; the degree-one homology discrepancy does.

The weight grading gives a different, finite comparison. An ordered PBW monomial of weight w specifies a partition of w, so for $n \geq 1$,

$$\sum_{w\geq 0} \dim B_n(U(L_1))_w q^w = \left(\prod_{m\geq 1}(1-q^m)^{-1} - 1\right)^n.$$

Each coefficient is finite. Thus a weightwise dimension, a dimension after a specified weight cutoff, and an ungraded dimension are distinct invariants. In a weight completion, dimensions must likewise be interpreted weightwise; the countable algebraic dimensions above refer to the ordinary direct-sum enveloping algebras.

Remark 70.7 (Chiral coefficients and transverse multiplication). Independently of these scalar calculations, a Virasoro chiral coefficient does not determine the transverse bar: a fixed coefficient may be combined with free, square-zero, enveloping, or quantum-plane transverse algebras whose bars differ, while the coefficient and the central charge are unchanged. A chain-level comparison with an ordered bar therefore also requires a specified transverse multiplication and a quasi-isomorphism from the proposed complex to that bar.

Chapter 71

The obstruction constitution

Each statement below names what is *not* determined, and therefore what must be supplied. Without them the preceding theorems admit false corollaries.

Proposition 71.1 (Invertibility of the aligned comparison)**.** *In the finite decomposition model, the canonical aligned inclusion and retraction are inverse on the full object exactly when its off-aligned summands vanish. Descent to geometric coefficients additionally requires compatibility with the actual transition correspondences.*

Proof. The canonical maps satisfy $\pi\iota = \mathrm{id}$ and $\iota\pi = e_{\mathrm{al}}$. The latter is the identity exactly when all off-aligned summands vanish. On geometric charts the required base-change squares must identify these projectors across transitions. The conditional descent theorem states those hypotheses. □

Failure of this canonical comparison to be invertible does not exclude every noninvertible distributive law. Such a law must be given as a natural transformation satisfying its own unit and composition diagrams. For Zamolodchikov–Faddeev algebras, the spectral variable and transverse word supply the two inputs to this separate construction problem.

Theorem 71.2 (No quantum group from dimension loss)**.** *The dimensions of the kernels of $V^{\otimes n} \to \mathrm{Sym}^n V$ do not determine a multiplication or a Hopf structure. Such a determination requires operations and their compatibility maps in addition to dimensions.*

Proof. The abelian and Heisenberg Lie algebras with generator weights 1, 1, 2 have enveloping algebras with the same PBW character $((1-t)^2(1-t^2))^{-1}$. The first algebra is commutative and the second is not. Their graded vector spaces, and hence the stated kernel dimensions, are nevertheless isomorphic. Even fixing the multiplication and augmentation is insufficient: Proposition 16.1 gives two nonisomorphic Hopf structures on the same polynomial algebra. □

Theorem 71.3 (No modular object from topology alone)**.** *The identity $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ assigns no coefficient operations to boundary strata. A modular master element exists only after cyclic contractions and clutching-compatible coefficient maps have been supplied.*

Theorem 71.4 (No bulk from a boundary centre)**.** *The derived centre is terminal among algebraic central actions on the boundary algebra. It does not detect a decoupled bulk sector, so a physical bulk is not determined by the boundary chiral algebra unless a bulk-to-centre comparison is supplied and proved to be an equivalence.*

Proof. Terminality is the universal property of the centre. It is a statement about the boundary category and is insensitive to any sector acting trivially on it, so a bulk with trivial boundary action is invisible. □

Theorem 71.5 (No classification by central charge)**.** *Central charge determines the Virasoro anomaly coefficient. It does not determine the chiral operation, the representation category, the braid monodromy, or the higher-genus conformal blocks. Distinct holomorphic theories at equal central charge exist.*

Theorem 71.6 (No identification of anomalies of different types)**.** *A current-algebra level, a Virasoro central charge, a determinant-line curvature, a BRST ghost-number anomaly, and a Borcherds-product weight are invariants in different theories, living in different deformation complexes. They may be related by a theorem in a particular family. They are not one universal scalar, and no identity among them follows from numerical agreement.*

Proof. Each is a class in a distinct complex: an Atiyah-algebra extension class, a Virasoro cocycle, the curvature of a determinant line, a class in H^1 or H^2 of a BRST convolution complex, and a weight in the character lattice of an automorphic form. A map between two of these complexes is additional data; in its absence, equality of numbers is a coincidence of values, not an identification of classes. □

Theorem 71.7 (No unnamed comparison of bars)**.** *An asserted equality* $\mathrm{Bar}^{\perp}(A) = \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A)$ *has no meaning without a functor carrying the pointwise multiplication problem to the chiral-convolution multiplication problem. The two have different underlying gradings, counting interval subdivisions and collision trees respectively. Agreement of Euler characteristics is not a comparison map.*

Theorem 71.8 (A scalar sequence is not homology data)**.** *A sequence of integers becomes bar homology only after a chain complex, a differential, a grading convention, and a rank computation or quasi-isomorphism are specified. Matching initial values, discriminants, or growth rates constructs no such map.*

Proof. Chain-space dimensions and homology dimensions differ by the ranks of the differentials, which are not functions of the chain dimensions. Calculation 70.2 exhibits an algebra whose positive bar chain spaces are countably infinite-dimensional and whose bar homology dimensions are 1, 0, 0, 1. The numbers 3^n count only the least-PBW-weight piece in bar length n. Proposition 70.4 exhibits two sequences with the same discriminant and growth rate that are transforms of one another and of neither homology. □

Theorem 71.6 together with the central-charge statement retracts any programme treating a tuple of scalars drawn from these sources as a single classifying invariant, and any trace identity asserted among them without a named comparison morphism.

Book entrance: The Igusa Square Root

The Jacobi seed, Borcherds denominator, root superalgebra, local–wrapped currents, quantum group, Hall model, and Fredholm realization.

Book XIII

The automorphic determinant

Chapter 72

The protected Jacobi input

72.1 The K3 elliptic genus

Let S be a K3 surface. The right-moving supersymmetric index of a unitary $\mathcal{N} = (4, 4)$ sigma model is the elliptic genus

$$Z_{K3}(\tau, z) = 2\phi_{0,1}(\tau, z).$$

The normalized weak Jacobi form has weight zero, index one, and Fourier expansion

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0} \sum_{\ell \in \mathbb{Z}} f(n, \ell) q^n r^\ell.$$

In Eichler–Zagier normalization,

$$\phi_{0,1}(\tau, z) = 4 \sum_{a=2}^{4} \frac{\vartheta_a(\tau, z)^2}{\vartheta_a(\tau, 0)^2} \in J^{\mathrm{w}}_{0,1}$$

[34]. The coefficient depends on the discriminant $4n - \ell^2$ and on ℓ modulo 2.

Theorem 72.1 (Exact low Fourier coefficients)**.** *The first four q-rows are*

$$\begin{aligned}\phi_{0,1} = {} & r^{-1} + 10 + r \\ & + q\left(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2\right) \\ & + q^2\left(r^{-3} + 108r^{-2} - 513r^{-1} + 808 - 513r + 108r^2 + r^3\right) \\ & + q^3\left(-64r^{-3} + 808r^{-2} - 2752r^{-1} + 4016 \right. \\ & \qquad \left. - 2752r + 808r^2 - 64r^3\right) + O(q^4).\end{aligned}$$

In particular,

$$f(0, 0) = 10, \qquad \frac{f(0, 0)}{2} = 5.$$

Proof. Insert the theta-series expansions into the preceding quotient formula. Work with $t = q^{1/8}$ and $x = r^{1/2}$; each quotient is an exact formal power series in t with Laurent-polynomial coefficients in x. The fractional powers cancel in the sum. Division through t^{24} gives the displayed rows. The arithmetic calculation is reproduced in Appendix 103. □

72.2 The virtual half-index

For every (n, ℓ) choose a finite-dimensional super vector space $U_{n,\ell}$ satisfying

$$\operatorname{sdim} U_{n,\ell} = f(n, \ell).$$

The integer $f(n, \ell)$ is its superdimension. In the ordinary Grothendieck group of finite-dimensional super vector spaces, the even and odd dimensions remain separate. The signed integer is their image under superdimension. The determinant depends only on that image.

Lemma 72.2 (Fock character and Berezinian)**.** *Let $V = \bigoplus_\gamma V_\gamma$ be a locally finite graded super vector space. Then*

$$\operatorname{sch}\big(\operatorname{Sym}(V_{\bar 0}) \otimes \bigwedge(V_{\bar 1})\big) = \prod_\gamma (1 - x^\gamma)^{-\operatorname{sdim} V_\gamma},$$

and

$$\operatorname{sdet}_V(1 - x) = \prod_\gamma (1 - x^\gamma)^{\operatorname{sdim} V_\gamma}.$$

Proof. An even one-dimensional summand contributes the geometric series $(1 - x^\gamma)^{-1}$. An odd one-dimensional summand contributes $1 - x^\gamma$. Multiplication over a homogeneous basis proves both identities. □

Chapter 73

The Borcherds product

73.1 The effective chamber

Define

$$\Gamma_{\mathrm{eff}} = \{(n,\ell,m) \in \mathbb{Z}^3 : m > 0,\ n \geq 0\} \cup \{(n,\ell,0) : n > 0\} \cup \{(0,\ell,0) : \ell < 0\}.$$

The ordering is lexicographic in $(m, n, -\ell)$ and is precisely the ordering selected by the cusp chamber. For $\gamma = (n,\ell,m)$ put

$$V_\gamma = U_{nm,\ell}, \qquad \operatorname{sdim} V_\gamma = f(nm,\ell).$$

Definition 73.1 (Monic virtual determinant)**.** The monic determinant of the half-index is

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \operatorname{sdet}_V(1 - x) = q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\mathrm{eff}}} (1 - q^n r^\ell s^m)^{f(nm,\ell)}.$$

Theorem 73.2 (Borcherds–Gritsenko–Nikulin product)**.** *The preceding product converges normally on the chosen cusp chamber and extends to a holomorphic genus-two modular form of weight five with a character of order two. It is the monic Igusa form D_5. The theta-normalized form is*

$$\Delta_5^\theta = 2^6 D_5.$$

Proof. The singular theta lift of the vector-valued form corresponding to $\phi_{0,1}$ produces the product, divisor, weight, and automorphic character. The weight is $f(0,0)/2 = 5$. The genus-two identification with the Igusa form and the product formula are the Gritsenko–Nikulin theorems [12, 13, 46, 48]. The product of the ten even theta constants has coefficient 2^6 at $q^{1/2} r^{1/2} s^{1/2}$, fixing the scalar. □

73.2 The theta constant normalization

Let $\mathfrak{E}$ be the ten even genus-two characteristics. Define

$$\Delta_5^\theta(Z) = \prod_{\mathfrak{m} \in \mathfrak{E}} \vartheta[\mathfrak{m}](Z, 0).$$

The cusp leading terms of the ten factors contribute six factors of 2 and total exponent $(1/2, 1/2, 1/2)$. Thus

$$[q^{1/2} r^{1/2} s^{1/2}] \Delta_5^\theta = 64.$$

This coefficient selects a canonical scalar normalization of the automorphic section.

Proposition 73.3 (Ten-factor leading-term calculation). *Group the ten even characteristics by the first half-characteristic $a \in (\mathbb{Z}/2)^2$. Their leading contributions in the chosen cusp chamber are*

a	#	*scalar*	*monomial*
$(0,0)$	4	1	1
$(1,0)$	2	2	$q^{1/8}$
$(0,1)$	2	2	$s^{1/8}$
$(1,1)$	2	2	$q^{1/8}r^{1/4}s^{1/8}$.

Multiplication gives

$$1^4\,2^2\,2^2\,2^2\,q^{1/2}r^{1/2}s^{1/2} = 64q^{1/2}r^{1/2}s^{1/2}.$$

Proof. Insert each characteristic into the genus-two theta series and retain the unique terms of least order in the lexicographic cusp valuation $(m, n, -\ell)$. For $a = (0,0)$ the zero lattice vector contributes one. For each nonzero a, the two shortest lattice representatives are exchanged by sign and contribute the scalar two. Their quadratic exponents are the entries in the final column. Summing the exponents and multiplying the six factors of two gives the formula. □

Proposition 73.4 (Boundary factor of the Borcherds product). *The $m = 0$ part of the monic product is*

$$\prod_{n>0,\ell\in\mathbb{Z}} (1-q^n r^\ell)^{f(0,\ell)} \prod_{\ell<0} (1-r^\ell)^{f(0,\ell)} = (1-r^{-1}) \prod_{n\geq 1} (1-q^n r^{-1})(1-q^n)^{10}(1-q^n r).$$

Consequently the first Fourier–Jacobi coefficient is

$$q^{1/2}r^{1/2}(1-r^{-1}) \prod_{n\geq 1} (1-q^n r^{-1})(1-q^n)^{10}(1-q^n r) = \eta(\tau)^9 \vartheta_{11}(\tau, z),$$

where the odd theta function is fixed by the exact convention

$$\vartheta_{11}(\tau, z) = -q^{1/8}r^{-1/2} \prod_{n\geq 1} (1-q^{n-1}r)(1-q^n r^{-1})(1-q^n).$$

Proof. The q^0 row of $\phi_{0,1}$ has exactly three nonzero coefficients: $f(0,-1) = 1$, $f(0,0) = 10$, and $f(0,1) = 1$. Substitution into the $m = 0$ factors gives the first formula. The Jacobi triple product and $\eta(\tau) = q^{1/24} \prod_{n\geq 1}(1-q^n)$ give the second, including its leading sign and monomial. □

Corollary 73.5 (The scalar square). *The monic weight-ten form is*

$$\chi_{10} = D_5{}^2 = 2^{-12}\Delta_5^{\theta 2}.$$

Its automorphic character is trivial.

Proof. The identity $\Delta_5^\theta = 2^6 D_5$ gives the scalar factor. The character of D_5 has order two, so its square is trivial. □

Chapter 74

Divisor, character, and the meaning of the square root

The diagonal Humbert divisor is

$$H_1 = \{Z \in \mathbb{H}_2 : z = 0\}/\mathrm{Sp}_4(\mathbb{Z}).$$

Theorem 74.1 (Diagonal divisor)**.** *The divisor of D_5 is the $\mathrm{Sp}_4(\mathbb{Z})$-orbit of H_1, with multiplicity one. The form transforms by the Maass character ν_5 of order two, and $\chi_{10} = D_5^2$ is scalar-valued.*

Proof. This is the defining diagonal-divisor property of the Igusa square root and the first row of the Cléry–Gritsenko classification [21, 55, 56]. □

Real roots of the denominator algebra define Weyl reflections. Imaginary roots contribute root spaces and product exponents. This division of labor is intrinsic to a Borcherds algebra: the Weyl group is generated by real simple roots, while the imaginary simple roots enter the correction term of the denominator identity.

74.1 Automorphic section, orientation line, and Pfaffian operator

An automorphic square root is a section of a line bundle with character. An orientation is a square root of a determinant line. A Pfaffian operator is a skew family whose regularized Pfaffian is a section of that orientation line. These form a sequence of successive refinements:

$$\text{automorphic section} \longrightarrow \text{determinant-line identification}$$
$$\longrightarrow \text{oriented square-root line} \longrightarrow \text{Fredholm family and Pfaffian section.}$$

Proposition 74.2 (Logical separation of the four levels)**.** *The product theorem constructs the first object in the sequence. A determinant-line comparison constructs the second. A coherent square root of that line constructs the third. A first-order operator requires a symbol, domain, ellipticity or Fredholm condition, and regularization, and thereby produces the fourth refinement.*

Proof. Each stage adds a specified geometric structure. A determinant realization adds an isomorphism from a geometric determinant line to the automorphic line. An oriented realization adds a coherent square-root line and its parity refinement. An operator realization adds a principal symbol, a Fredholm domain, a spectrum, and a regularization whose Pfaffian is the resulting section. □

The title names the monic automorphic section D_5, characterized by the exact identity $D_5^2 = \chi_{10}$. Part IV formulates an analytic Pfaffian realization as a further geometric refinement.

Book XIV

The denominator algebra

Chapter 75

The Lorentzian root lattice

Let

$$L = \mathbb{Z} f_2 \oplus \mathbb{Z} f_3 \oplus \mathbb{Z} f_{-2}$$

with bilinear form

$$(f_2, f_{-2}) = -1, \qquad (f_3, f_3) = 2,$$

and all remaining basis pairings zero. Its signature is $(2, 1)$. Define the type-II root lattice

$$L_{II} = 2\mathbb{Z} f_2 \oplus \mathbb{Z} f_3 \oplus 2\mathbb{Z} f_{-2} \cong U(4) \oplus \langle 2 \rangle.$$

The Gram-index map is

$$\alpha : \Gamma_{\mathrm{ind}} = \mathbb{Z}^3 \longrightarrow L_{II}, \qquad \alpha(n, \ell, m) = 2n f_2 - \ell f_3 + 2m f_{-2}.$$

Then

$$(\alpha(n, \ell, m), \alpha(n', \ell', m')) = -2\big(2nm' + 2n'm - \ell\ell'\big),$$

and

$$(\alpha(n, \ell, m), \alpha(n, \ell, m)) = -2(4nm - \ell^2).$$

75.1 Real simple roots and Weyl vector

Put

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

Proposition 75.1 (Type-II Cartan matrix)**.** *The Gram matrix of the real simple roots is*

$$\big((\delta_i, \delta_j)\big)_{i,j=1}^{3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The reflections

$$s_i(x) = x - (x, \delta_i)\delta_i$$

are integral involutions preserving L_{II} and its form.

Proof. Substitution into the basis pairing gives the matrix. Since $(\delta_i, \delta_i) = 2$, the reflection formula has integral coefficients. Direct expansion gives $s_i^2 = 1$ and $(s_i x, s_i y) = (x, y)$. □

The Weyl vector is

$$\rho = f_2 - \frac{1}{2} f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

In the Gram-index variables, $e^{-\rho} = q^{1/2} r^{1/2} s^{1/2}$.

Chapter 76

The Gritsenko–Nikulin Lie superalgebra

The Gritsenko–Nikulin construction specifies a generalized Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$, with real simple roots δ_1, δ_2, δ_3 and the imaginary correction attached to $\phi_{0,1}$. Its generators and relations supply the bracket whose denominator is compared with the Borcherds product.

Theorem 76.1 (Root supermultiplicities and denominator)**.** *On the active positive support,*

$$\operatorname{smult} \mathfrak{g}_{\Delta_5\,\alpha(n,\ell,m)} = f(nm,\ell).$$

In the Lorentzian coordinate convention of Gritsenko–Nikulin,

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 2^{-6}\Delta_5^{\theta}(2Z).$$

Proof. The real roots and Weyl chamber are the type-II case of the Gritsenko–Nikulin construction. The imaginary simple-root correction is chosen so that its Weyl–Kac–Borcherds product has exponent $f(nm,\ell)$. The resulting product is the doubled-variable monic Igusa form [46–48]. □

The theorem gives a Lie superalgebra with a bracket, Cartan, Weyl group, root parity, and denominator. The scalar product alone records the signed dimensions of the root spaces; the Gritsenko–Nikulin construction supplies the algebraic operations.

Chapter 77

The ordered bar as the denominator complex

Let $\mathfrak{n}_{\Delta_5,+}$ be the positive nilpotent subalgebra of $\mathfrak{g}_{\Delta_5}$. Fix a proper integral height h on the positive root cone and define

$$F^{>H}\mathfrak{n}_{\Delta_5,+} = \bigoplus_{h(\alpha)>H} \mathfrak{g}_{\Delta_5\,\alpha}, \qquad {\mathfrak{n}_{\Delta_5,+}}^{(H)} = \mathfrak{n}_{\Delta_5,+}/F^{>H}\mathfrak{n}_{\Delta_5,+}.$$

Height additivity makes $F^{>H}\mathfrak{n}_{\Delta_5,+}$ an ideal. Properness and local finiteness make ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$ finite-dimensional and nilpotent.

Theorem 77.1 (Height-complete Bar–Chevalley comparison)**.** *For every H, antisymmetrization gives a natural quasi-isomorphism of conilpotent dg coalgebras*

$$\mathrm{Bar}\big(U({\mathfrak{n}_{\Delta_5,+}}^{(H)})\big) \simeq C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)}).$$

The quotient maps preserve these comparisons and induce a quasi-isomorphism in the separated height-complete category

$$\mathrm{Bar}_h^{\wedge}\big(U(\mathfrak{n}_{\Delta_5,+})\big) := \varprojlim_H \mathrm{Bar}\big(U({\mathfrak{n}_{\Delta_5,+}}^{(H)})\big) \simeq C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) := \varprojlim_H C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)}).$$

Its root-graded Euler supercharacter is

$$\chi_{\mathrm{root}}\, C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha>0}(1-e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\,\alpha}}.$$

Proof. For each H, the antisymmetrization map sends $\mathrm{Sym}({\mathfrak{n}_{\Delta_5,+}}^{(H)}[1])$ to the normalized bar of $U({\mathfrak{n}_{\Delta_5,+}}^{(H)})$. The PBW filtrations have the same associated-graded Koszul resolution of $\mathrm{Sym}({\mathfrak{n}_{\Delta_5,+}}^{(H)})$, so antisymmetrization is a filtered quasi-isomorphism. Naturality gives a morphism of inverse systems. Every fixed root degree is visible in all sufficiently large height quotients and is unchanged thereafter; hence the inverse-limit map is a quasi-isomorphism coefficientwise. An even root vector becomes odd after the Chevalley shift and contributes $1-e^{-\alpha}$; an odd root vector becomes even and contributes its reciprocal. Multiplication over the finite-dimensional root spaces gives the product. □

Let $\mathbb{T}_{L_{II}}$ be the formal root torus and write Z_{root} for its logarithmic coordinate, normalized by

$$e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m, \qquad e^{-\rho} = Q^{1/2}R^{1/2}S^{1/2}.$$

Corollary 77.2 (Bar denominator identity)**.** *On the formal root torus,*

$$e^{-\rho}\chi_{\mathrm{root}} C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = D_5(Z_{\mathrm{root}}).$$

The Gritsenko–Nikulin embedding of the root torus into the Siegel period domain sends Z_{root} to $2Z$ [46, 48]. Hence

$$e^{-\rho}\chi_{\mathrm{root}} C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z)$$

in Siegel period coordinates.

Proof. Multiply the Euler product in the preceding theorem by the Weyl monomial $e^{-\rho}$. The denominator theorem identifies the result on the root torus, and the stated embedding gives the Siegel-coordinate formula. □

The denominator is therefore the Weyl-shifted Euler supercharacter of the height-complete interval bar of the positive enveloping algebra.

Chapter 78

Finite-height currents

Retain the finite-height quotients $\mathfrak{n}_{\Delta_s,+}{}^{(H)}$ of the preceding chapter. Define the current Lie conformal superalgebra

$$\mathrm{Cur}(\mathfrak{n}_{\Delta_s,+}{}^{(H)}) = \Bbbk[\partial] \otimes \mathfrak{n}_{\Delta_s,+}{}^{(H)}, \qquad [a_\lambda b] = [a, b],$$

and its universal chiral enveloping algebra

$$V_H = U^{\mathrm{ch}}\Big(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+}{}^{(H)})\Big).$$

Theorem 78.1 (Height-complete current envelope)**.** *The quotient maps induce a coefficientwise inverse limit*

$$\widehat{V}_{\Delta_s} = \varprojlim_H V_H.$$

Every V_H is PBW-admissible as an associative chiral algebra, the current generators are primitive for the standard current coproduct, and

$$\operatorname{sch} \widehat{V}_{\Delta_s} = \prod_{j\geq 1}\prod_{\alpha>0}(1 - t^j e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_{\Delta_s,\alpha}}.$$

Moreover,

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(V_H) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}\Big(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+}{}^{(H)})\Big),$$

and these equivalences pass to the separated height completion.

Proof. PBW orders the negative modes $a(-j)$, $j \geq 1$. Even modes contribute symmetric powers and odd modes contribute exterior powers, giving the product. Proper height makes every coefficient stationary. The current coproduct sends $a(z)$ to $a(z) \otimes 1 + 1 \otimes a(z)$. Chiral antisymmetrization identifies the associative-chiral bar of the chiral enveloping algebra with the chiral Chevalley coalgebra; the complete separated PBW filtration is coefficientwise finite. □

78.1 Classical local theory and exact quantum zero modes

For every H, the finite-dimensional nilpotent Lie superalgebra $\mathfrak{n}_{\Delta_s,+}{}^{(H)}$ defines a cotangent BF theory on $E \times \mathbb{R}$. Its local dg Lie algebra is

$$\mathcal{L}_H = \Omega^{0,*}(E) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes \mathfrak{n}_{\Delta_s,+}{}^{(H)}$$

and the classical action is

$$S_H(A, B) = \int_{E\times\mathbb{R}} \Omega_E \wedge \Big\langle B, (\bar\partial + \mathrm{d}_{\mathbb{R}})A + \frac{1}{2}[A, A]\Big\rangle.$$

Theorem 78.2 (Finite-height BF realization)**.** *The action S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra. Harmonic reduction along a compact interval produces the finite-dimensional cotangent BV theory of ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$. The positive-root height grading makes every even adjoint endomorphism strictly height-increasing and every odd adjoint endomorphism odd; hence the modular supercharacter vanishes and the reduced theory satisfies the quantum master equation.*

Choose the standard complementary Lagrangian boundary polarizations, fixing the ${\mathfrak{n}_{\Delta_5,+}}^{(H)}[1]$ coordinate at one endpoint and its cotangent coordinate at the other. The algebraic boundary quantization, the interval amplitude between the two augmentation states, and the holomorphic boundary current envelope are then

$$U({\mathfrak{n}_{\Delta_5,+}}^{(H)}), \qquad \mathbf{1} \otimes^{\mathbb{L}}_{U({\mathfrak{n}_{\Delta_5,+}}^{(H)})} \mathbf{1} \simeq C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)}), \qquad V_H.$$

The height quotient maps preserve the polarizations and yield the inverse systems

$$\widehat{U}_h(\mathfrak{n}_{\Delta_5,+}), \qquad C^{\wedge}_{*,h}(\mathfrak{n}_{\Delta_5,+}), \qquad \widehat{V}_{\Delta_5}.$$

Proof. The classical master equation is the Jacobi identity paired with invariance of the cotangent pairing. Elliptic descent gives the classical factorization algebra. Harmonic reduction along the interval gives a finite-dimensional BV space. If x is even of positive root height, ad_x raises height and is nilpotent on the finite quotient, so its restrictions to the even and odd summands both have trace zero; if x is odd, ad_x is odd and has supertrace zero. Thus the modular character vanishes and the finite BV quantum master equation follows.

The chosen boundary polarization quantizes the linear Kirillov–Kostant boundary bracket to $U({\mathfrak{n}_{\Delta_5,+}}^{(H)})$. Gluing the two augmentation states gives the two-sided derived bar, and the PBW antisymmetrization identifies it with $C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)})$. Holomorphic descendants generate $U^{\mathrm{ch}}(\mathrm{Cur}_E\, {\mathfrak{n}_{\Delta_5,+}}^{(H)}) = V_H$. All maps preserve root height, so they pass coefficientwise to the separated inverse limits. □

Book XV

The compact scalar character

Chapter 79

Reduced curve counting on $K3 \times E$

Let S be a nonsingular projective K3 surface and E an elliptic curve. For a primitive class $\beta_h \in H_2(S, \mathbb{Z})$ with $\beta_h^2 = 2h - 2$, let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant of class (β_h, d) with the standard incidence insertion. Define

$$\mathcal{Z}(u, q, s) = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q^{h-1} s^{d-1}.$$

Theorem 79.1 (Primitive Igusa cusp-form theorem)**.** *Under $r = e^{iu}$, the partition function is the Laurent expansion*

$$\mathcal{Z}(u, q, s) = -\frac{1}{\chi_{10}(r, q, s)}.$$

Equivalently, in theta normalization,

$$\mathcal{Z} = -2^{12} \Delta_5^{\theta^{-2}}.$$

Proof. Oberdieck and Pixton identify the primitive reduced partition function with the Laurent expansion of $-\chi_{10}^{-1}$ [81]; their normalization is the monic Borcherds product χ_{10}. The second formula is the identity $\chi_{10} = 2^{-12} \Delta_5^{\theta^2}$. □

The theorem identifies the numerical generating series in its stated primitive sector. The Borcherds root algebra supplies a separate algebraic character to compare with that series.

Corollary 79.2 (Exact character comparison)**.** *Let $\jmath : \mathbb{H}_2 \to \mathbb{H}_2$ be the half-coordinate map $\jmath(Z) = Z/2$, and let*

$$\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{Sym}(\mathfrak{n}_{\Delta_5,+_{\bar{0}}}) \otimes \bigwedge(\mathfrak{n}_{\Delta_5,+_{\bar{1}}}).$$

Then

$$\mathcal{Z}^{\mathrm{red}}_{S \times E} = -\jmath^* \Big(e^{2\rho} \operatorname{sch}\big(\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}\big) \Big) = -\chi_{10}^{-1}.$$

Proof. Put

$$P = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_{\Delta_5, \alpha}}.$$

The denominator theorem gives $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = e^{-\rho} P = D_5(2Z)$, while PBW gives $\operatorname{sch} \mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = P^{-1}$. Hence

$$e^{2\rho} \operatorname{sch}\big(\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}\big) = \mathrm{den}(\mathfrak{g}_{\Delta_5})^{-2}.$$

Pullback by $\jmath$ sends $\mathrm{den}(\mathfrak{g}_{\Delta_5})$ to $D_5(Z)$, whose square is χ_{10}. The reduced incidence convention supplies the overall minus sign in the Oberdieck–Pixton theorem. □

The equality compares the completed root-Fock supercharacter with the numerical reduced generating series. Its proof uses the denominator identity and the half-coordinate pullback. A map from compact geometric states to this Fock space, preserving operators and their trace, is an additional comparison.

Chapter 80

Extensions over the elliptic classifying stack

Work on locally noetherian complex schemes with the fppf topology. Let BE be the stack of right E-torsors. For a left E-stack M, an object of $[M/E](T)$ is a right torsor $P \to T$ and an equivariant map $P \to M$. Forgetting the map gives $[M/E] \to BE$. All products of stacks below are two-fibre products.

Proposition 80.1 (The common translation torsor)**.** *There is a natural equivalence*

$$[(M \times N)/E_{\mathrm{diag}}] \simeq [M/E] \times_{BE} [N/E].$$

The equivalences for any finite number of factors commute with regrouping and insertion of the terminal E-stack.

Proof. An object on the right is (P, m), (Q, n) together with a torsor isomorphism $\alpha : P \to Q$. Send it to $(P, m, n\alpha)$. Conversely, use a common torsor twice and its identity isomorphism. A pair of torsor maps a, b on the right satisfies $b\alpha = \alpha' a$, so b is determined by a. The same construction identifies the maps and their descent isomorphisms. It commutes with base change. For several factors, every regrouping gives one torsor with the same list of equivariant maps, proving all stated coherences. □

For $M = N = E$ with translation, the diagonal quotient is E, with coordinate $y - x$. Both separate quotients are points. Their fibre product over BE retains the relative translation, while their ordinary product loses it.

Let $\mathfrak{M}$ be the stack of coherent sheaves on $X = S \times E$, with families flat over the parameter scheme, and put $\mathfrak{Q} = [\mathfrak{M}/E]$. An object of $\mathfrak{Q}(T)$ is a torsor $P \to T$ and a flat sheaf on $S \times P$. Fix a polarization and write $\mathfrak{Q}_\alpha$ for a Hilbert-polynomial component. Translation preserves these polynomials. Let $\mathfrak{H}_{\alpha,\beta}$ classify short exact sequences $0 \to A \to F \to B \to 0$ on a common $S \times P$, with all terms flat over T and polynomials α, $\alpha + \beta$, β. Its span over BE is

$$\mathfrak{Q}_\alpha \times_{BE} \mathfrak{Q}_\beta \xleftarrow{p} \mathfrak{H}_{\alpha,\beta} \xrightarrow{q} \mathfrak{Q}_{\alpha+\beta}.$$

The category of spans of fppf stacks over BE has composition by fibre product and tensor product $\times_{BE}$. Its unit object is the identity map of BE.

Theorem 80.2 (Geometric Hall correspondence)**.** *For fixed α, β, q is representable by algebraic spaces and proper. The spans form a coherently associative unital Hall object in the category of spans over BE. Its unit is the zero-sheaf span $BE \leftarrow BE \to \mathfrak{Q}$.*

Proof. Over a fixed (P, F), the fibre of q classifies quotients $F \twoheadrightarrow B$ of polynomial β. Its kernel is flat, because F and B are flat. An automorphism fixing the quotient and F is the identity. Trivializing P fppf-locally identifies this fibre with the fixed-polynomial Quot scheme, projective over the parameter scheme by Grothendieck's theorem [49, Theorem 3.2]. The quotient functor descends to an algebraic space, and properness descends with it.

Both composites of three Hall spans classify $0 \subset A \subset D \subset F$ on the same torsor. One bracketing concatenates inclusions. The other starts with $A \subset F$ and a subobject $B \subset F/A$, and sets $D = F \times_{F/A} B$. These constructions are inverse on objects and isomorphisms. They preserve flatness and commute with base change. For longer flags all regroupings identify the same subobjects and quotients. Their comparison maps agree, proving the pentagon and higher coherences. Finally $\mathfrak{Q}_0 = BE$. The sequences with zero subobject or zero quotient give the unit and its coherences. □

Properness here concerns the full coherent-sheaf stack on a fixed pair of Hilbert-polynomial components. A stability restriction can replace its fibre by an open part of a Quot scheme, which need not be proper. A bounded charge window also needs closure under the extensions and flags used in its operations. A reduced critical-coefficient realization must supply pullback, pushforward, and orientation comparisons on these same flags. Its unit has stabilizer E. A coefficient theory for this quotient must include that nonaffine stabilizer, or work relative to BE, on the unit and every flag.

Chapter 81

Additive physical charge and quadratic automorphic degree

Let

$$\Lambda = \widetilde{H}(S, \mathbb{Z})$$

be the Mukai lattice of signature $(4, 20)$. The electric–magnetic charge lattice is

$$\Gamma_{\mathrm{phys}} = \Lambda \oplus \Lambda, \qquad c = (Q, P).$$

Its addition is the physical superselection law. Define the Gram map

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right) \in \Gamma_{\mathrm{ind}}.$$

Proposition 81.1 (Polarization identity)**.** *For $c_i = (Q_i, P_i)$,*

$$\Pi(c_1 + c_2) = \Pi(c_1) + \Pi(c_2) + B(c_1, c_2),$$

where

$$B(c_1, c_2) = \big(Q_1 \cdot Q_2, Q_1 \cdot P_2 + Q_2 \cdot P_1, P_1 \cdot P_2 \big).$$

The cross-term is a symmetric vector-valued two-cocycle:

$$B(c_1, c_2) + B(c_1 + c_2, c_3) = B(c_2, c_3) + B(c_1, c_2 + c_3).$$

Proof. Expand the three quadratic expressions. The cocycle identity is bilinearity of B. □

81.1 The polarized central extension

Define the abelian central extension

$$0 \longrightarrow \Gamma_{\mathrm{ind}} \longrightarrow \widetilde{\Gamma}_\Pi \longrightarrow \Gamma_{\mathrm{phys}} \longrightarrow 0, \qquad \widetilde{\Gamma}_\Pi = \Gamma_{\mathrm{phys}} \times \Gamma_{\mathrm{ind}},$$

with multiplication

$$(c, \lambda) \cdot (d, \mu) = \big(c + d, \lambda + \mu + B(c, d) \big).$$

The cocycle identity makes this multiplication associative. The polarization identity makes the graph map

$$j_\Pi : \Gamma_{\mathrm{phys}} \longrightarrow \widetilde{\Gamma}_\Pi, \qquad c \longmapsto (c, \Pi(c))$$

a homomorphism.

Let $A = \bigoplus_{c \in \Gamma_{\mathrm{phys}}} A_c$ be a charge-graded associative algebra, and let $\Bbbk[\Gamma_{\mathrm{ind}}]$ have monomials u^λ. Define the polarized central-extension algebra

$$\mathcal{R}_\Pi(A) = A \otimes \Bbbk[\Gamma_{\mathrm{ind}}]$$

with product

$$(a_c u^\lambda) \star (b_d u^\mu) = a_c b_d \, u^{\lambda+\mu+B(c,d)}.$$

Give $a_c u^\lambda$ degree $(c, \lambda) \in \widetilde{\Gamma}_\Pi$.

Theorem 81.2 (Graph-multiplicative polarization)**.** *The product $\star$ is associative and $\widetilde{\Gamma}_\Pi$-graded. The map*

$$\iota_\Pi : A \longrightarrow \mathcal{R}_\Pi(A), \qquad a_c \longmapsto a_c u^{\Pi(c)}$$

is an algebra homomorphism whose homogeneous degree is the graph element $j_\Pi(c)$.

Proof. The two parenthesizations differ by

$$B(c,d) + B(c+d,e) - B(d,e) - B(c,d+e),$$

which vanishes by the cocycle identity. The polarization formula gives

$$\iota_\Pi(a_c) \star \iota_\Pi(b_d) = a_c b_d \, u^{\Pi(c+d)}.$$

Thus the graph of the quadratic Gram map is multiplicative inside the central extension. □

The same construction applies to Lie superalgebras and their enveloping algebras. It retains additive physical charge, Gram degree, and the interaction cross-term in one grading group.

Chapter 82

Point-local and wrapped sectors

Let $p_E : S \times E \to E$ be projection.

Lemma 82.1 (Positive elliptic degree is wrapped)**.** *Let $C \subset S \times E$ be a proper one-dimensional subscheme whose pushforward $(p_E)_*[C]$ has positive degree. Then $p_E(C) = E$.*

Proof. Properness makes $p_E(C)$ closed. Positive pushforward degree implies that at least one irreducible component of C dominates a one-dimensional closed subset of E. Every one-dimensional closed subset of the irreducible curve E is E itself, so the image of that component, and hence the image of C, is all of E. □

Projection-finite objects define point-supported sectors on $\mathrm{Ran}(E)$. Positive elliptic degree defines wrapped sectors supported over the entire elliptic curve.

Definition 82.2 (The local–wrapped factorization operad)**.** Let Ch_E denote the chosen chiral/factorization operad on E and let Ass denote the ordered associative operad. Define the colored operad Hyb_E with colors

$$\mathsf{l}, \qquad \mathsf{w}_m \quad (m \geq 1)$$

by the following operation objects.

(i) Operations with local output are

$$\mathsf{Hyb}_E(\mathsf{l}^{\otimes k}; \mathsf{l}) = \mathsf{Ch}_E(k).$$

(ii) For $r \geq 1$ and $m = m_1 + \cdots + m_r$, operations with wrapped output are

$$\mathsf{Hyb}_E(\mathsf{l}^{\otimes k}, \mathsf{w}_{m_1}, \ldots, \mathsf{w}_{m_r}; \mathsf{w}_m) = \mathsf{Ch}_E(k) \otimes \mathsf{Ass}(r).$$

All other operation objects are initial.

Operadic composition uses chiral insertion and disjoint union in the first factor and ordered substitution in the second. The equality

$$(\gamma_1 \circ \gamma_2) \otimes (a_1 \circ a_2) = (\gamma_1 \otimes a_1) \circ (\gamma_2 \otimes a_2)$$

is the mixed interchange law. The arities $(k, r) = (k, 0), (0, 2), (k, 1)$ recover, respectively, local factorization, wrapped multiplication, and the action of local insertions on a wrapped sector.

Definition 82.3 (Hybrid local–wrapped algebra)**.** A hybrid local–wrapped algebra over E is an algebra over Hyb_E. It consists of a factorization algebra $\mathbb{F}_{\mathrm{loc}}$, a height- or charge-complete family $W = \prod_{m>0} W_m$, ordered products

$$\mu_{m_1,\ldots,m_r} : W_{m_1} \otimes \cdots \otimes W_{m_r} \longrightarrow W_{m_1+\cdots+m_r},$$

and factorization-compatible actions of the local operation objects $\mathsf{Ch}_E(k)$ on every wrapped product. The operad axioms state associativity, local factorization, and compatibility of the two operations.

A compact Hall theory of $S \times E$ produces a correspondence-valued Hyb_E-algebra when disjoint-support direct sums, wrapped extensions, and the mixed extension correspondences satisfy the same base-change identities.

Book XVI

The Borcherds algebra and its quantum group

Chapter 83

The local–wrapped root algebra

The positive cone is graded by $\gamma = (n, \ell, m)$, with elliptic degree m. Set

$$\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} = \bigoplus_{\substack{\alpha>0\\ m(\alpha)=0}} \mathfrak{g}_{\Delta_5\,\alpha}, \qquad \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}} = \bigoplus_{\substack{\alpha>0\\ m(\alpha)>0}} \mathfrak{g}_{\Delta_5\,\alpha}.$$

Proposition 83.1 (Semidirect local–wrapped decomposition)**.** *The space $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ is a Lie subsuperalgebra, $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ is a Lie ideal, and*

$$\mathfrak{n}_{\Delta_5,+} = \mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} \ltimes \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}.$$

The proper height filtration preserves both summands.

Proof. The bracket adds root degrees. The sum of two roots of elliptic degree zero again has degree zero, and addition of a nonnegative degree to a positive degree remains positive. Height is additive on every nonzero bracket. □

Define the complete enveloping Hopf algebra and current envelope

$$\mathbb{U}_{\Delta_5} = \widehat{U}_h(\mathfrak{n}_{\Delta_5,+}), \qquad \mathbb{V}_{\Delta_5,E} := \widehat{V}_{\Delta_5} = \varprojlim_H U^{\mathrm{ch}}\Big(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+}{}^{(H)})\Big).$$

The adjoint action of $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ on $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ extends by derivations to $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})$. Let $\#$ denote the corresponding completed smash product.

Definition 83.2 (Current-generated local–wrapped operad)**.** The colored operad $\mathsf{Hyb}^{\mathrm{cur}}_{E,\Delta}$ is generated by the chiral current operations of $\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}})$, the ordered multiplications of $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})$, and the adjoint derivation action of the former on the latter, modulo the chiral Jacobi, associative, derivation, and semidirect-product relations.

Theorem 83.3 (Local–wrapped Borcherds decomposition)**.** *Let $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ be the sum of positive root spaces of elliptic degree $m = 0$ and let $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ be the sum of those with $m > 0$. Then*

$$\mathfrak{n}_{\Delta_5,+} = \mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} \ltimes \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}},$$

where $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ is a Lie subsuperalgebra and $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ is a Lie ideal. For the proper-height completion,

$$\widehat{U}_h(\mathfrak{n}_{\Delta_5,+}) \cong \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}) \# \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}),$$

and

$$\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) \simeq \mathrm{Tot}\, C_*(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}; \widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})).$$

The primitive Lie superalgebra of the completed enveloping algebra is the height completion $\widehat{\mathfrak{n}_{\Delta_5,+}}_h = \varprojlim_H \mathfrak{n}_{\Delta_5,+}{}^{(H)}$, and the Weyl-shifted Euler supercharacter of the completed Chevalley complex is $D_5(2Z)$.

Proof. Root addition adds elliptic degree. Thus the $m = 0$ summand is closed under the bracket, and every bracket with an $m > 0$ root again has $m > 0$. PBW orders wrapped generators before local generators and gives the completed smash product. The Chevalley complex of a semidirect product is the Hochschild–Serre total complex. Primitive elements of a completed enveloping algebra are recovered height by height. The denominator theorem supplies the last character identity. □

Remark 83.4. The theorem constructs a Borcherds root algebra with local and wrapped colors. The word Hall is reserved for an algebra obtained from geometric extension correspondences.

Chapter 84

The formal Igusa quantum group

Let $\mathfrak{b}_+$ be the positive Borcherds Borel half. Form the restricted graded dual $\mathfrak{b}_- := \mathfrak{b}_+^{\vee,\mathrm{gr}}$ with bracket and cobracket transported by the invariant pairing, identify the two Cartan copies along their common radical quotient, and complete by a proper root height. The completed Drinfeld double of this paired Borel datum is denoted ${\mathfrak{g}_{\Delta_5}}^{\mathrm{pair}}$. Its positive and negative halves, diagonal Cartan, invariant form, and canonical tensor form a height-admissible Manin triple and hence a quasitriangular Lie superbialgebra.

Definition 84.1 (Height-admissible Lie superbialgebra)**.** A root-graded Lie superbialgebra is height-admissible when every root space is finite-dimensional, every bounded-height region contains finitely many roots, and the bracket, cobracket, invariant tensor, and all iterated operations are coefficientwise finite in the height completion.

Proposition 84.2 (Height admissibility)**.** *The paired Gritsenko–Nikulin Borcherds Lie superbialgebra* ${\mathfrak{g}_{\Delta_5}}^{\mathrm{pair}}$ *is height-admissible.*

Proof. The Borcherds root spaces are finite-dimensional and the chosen height is proper. The bracket and cobracket preserve total root degree. In a bounded total height, finitely many decompositions of a root occur. The canonical tensor is therefore a coefficientwise finite sum in every height quotient. □

Theorem 84.3 (Quantized Borcherds superalgebra)**.** *Choose the symmetrizable Borcherds–Cartan superdatum and multiplicity index set that present* $\mathfrak{g}_{\Delta_5}$*. Hong's construction gives a quantized enveloping Hopf superalgebra*

$$U_q(\mathfrak{g}_{\Delta_5})$$

over $\mathbb{Q}(q)$*, generated by its Cartan elements and the real and imaginary simple-root generators, with triangular decomposition*

$$U_q(\mathfrak{g}_{\Delta_5}) \cong U_q^-(\mathfrak{g}_{\Delta_5}) \otimes U_q^0(\mathfrak{g}_{\Delta_5}) \otimes U_q^+(\mathfrak{g}_{\Delta_5}).$$

Its PBW root spaces have the same graded ranks as the classical Borcherds superalgebra. The canonical pairing of the two Borel halves yields a universal R-matrix in the root-height completion of

$$U_q^+(\mathfrak{g}_{\Delta_5}) \mathbin{\widehat{\otimes}} U_q^-(\mathfrak{g}_{\Delta_5}).$$

Consequently the quantization preserves the root-multiplicity data that enter $D_5(2Z)$*.*

Proof. The Gritsenko–Nikulin algebra is a symmetrizable Borcherds–Kac–Moody Lie superalgebra. Apply the generators-and-relations construction of the quantized Borcherds superalgebra. The nondegenerate bilinear form, triangular decomposition, centre, and universal R-matrix are established by Hong [54]. The PBW theorem identifies the associated graded root spaces with their classical counterparts. Since the denominator exponents are the classical signed root multiplicities, they are unchanged by the flat root-graded deformation. □

Remark 84.4. The quantum group is attached to the Borcherds generalized Cartan superdatum. A geometric Hall quantum group is identified with it through a Hall coproduct, pairing, completion, and primitive comparison.

Chapter 85

A holomorphic–topological BF realization

For every height bound H, let ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$ be the finite-dimensional nilpotent quotient. On $E \times \mathbb{R}$ form the shifted cotangent theory of

$$\mathcal{L}_H = \Omega^{0,*}(E) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes {\mathfrak{n}_{\Delta_5,+}}^{(H)}$$

with action

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \left\langle B, (\bar{\partial} + \mathrm{d}_{\mathbb{R}})A + \frac{1}{2}[A, A] \right\rangle.$$

Theorem 85.1 (Classical local and exact zero-mode realization)**.** *For every H:*

(i) *S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra;*

(ii) *harmonic reduction along a compact interval gives the finite-dimensional cotangent BV theory of ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$, whose positive-root grading makes its modular supercharacter zero and hence gives the quantum master equation;*

(iii) *after choosing the complementary Lagrangian boundary polarizations, the boundary algebra is $U({\mathfrak{n}_{\Delta_5,+}}^{(H)})$, the interval amplitude is $\mathbf{1} \otimes^{\mathbb{L}}_{U({\mathfrak{n}_{\Delta_5,+}}^{(H)})} \mathbf{1} \simeq C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)})$, and the holomorphic boundary current envelope is V_H.*

The transition maps yield a pro-object with boundary algebra $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+})$, interval complex $C^{\wedge}_{*,h}(\mathfrak{n}_{\Delta_5,+})$, and current envelope $\widehat{V}_{\Delta_5}$.

Proof. The classical master equation is the Jacobi identity and invariance of the cotangent pairing. The classical observables satisfy factorization descent. Harmonic reduction is finite-dimensional. An even adjoint operator raises positive-root height and is nilpotent on the finite quotient; an odd adjoint operator is odd. Both have supertrace zero, so the modular term in the finite BV Laplacian vanishes. The chosen polarization quantizes the boundary Lie–Poisson bracket to the enveloping algebra; the two augmentation states give the two-sided bar, identified with Chevalley chains by PBW. Holomorphic descendants give the current algebra. All structure maps preserve height. □

The theorem gives the precise field-theoretic realization used in this volume: a classical local theory together with an exact quantum protected zero-mode sector.

Chapter 86

The determinant and Fredholm square-root operators

Let

$$\mathcal{V}_H = \bigoplus_{\substack{\alpha>0\\ h(\alpha)\le H}} \mathfrak{g}_{\Delta,\alpha}$$

and define

$$\mathscr{D}_H|_{\mathfrak{g}_{\Delta,\alpha}} = (1 - e^{-\alpha})\,\mathrm{id}\,.$$

Theorem 86.1 (Universal determinant section)**.** *The finite-height Berezinian is*

$$\mathrm{Ber}(\mathscr{D}_H) = \prod_{\substack{\alpha>0\\ h(\alpha)\le H}} (1 - e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta,\alpha}}.$$

The coefficientwise limit satisfies

$$e^{-\rho}\,\mathrm{Ber}(\mathscr{D}_\infty) = D_5(2Z).$$

After pullback by $\jmath(Z) = Z/2$,

$$\jmath^*(e^{-\rho}\,\mathrm{Ber}(\mathscr{D}_\infty)) = D_5(Z), \qquad D_5(Z)^2 = \chi_{10}(Z).$$

Proof. The Berezinian of a diagonal endomorphism is the product of its eigenvalues with exponent equal to the superdimension of the eigenspace. Proper height makes every coefficient stationary. The denominator theorem supplies the Weyl line and the root-to-Siegel coordinate map. □

86.1 Trace-class realization on the root Hilbert superspace

Choose homogeneous orthonormal bases of the finite-dimensional root spaces and set

$$\mathcal{H}_\Delta = \widehat{\bigoplus}_{\alpha>0} \mathfrak{g}_{\Delta,\alpha}$$

with its induced $\mathbb{Z}/2$ grading. Let $\mathfrak{C}_{\mathrm{abs}}$ be the subchamber on which

$$\sum_{\alpha>0} \dim \mathfrak{g}_{\Delta,\alpha} \left| e^{-\alpha(Z)} \right| < \infty.$$

For $Z \in \mathfrak{C}_{\mathrm{abs}}$, define the trace-class diagonal operator

$$K_Z|_{\mathfrak{g}_{\Delta,\alpha}} = e^{-\alpha(Z)}\,\mathrm{id}, \qquad D_Z = 1 - K_Z.$$

On each parity summand form the skew doubling

$$\mathbb{A}_{Z,\bar\epsilon} = \begin{pmatrix} 0 & D_{Z,\bar\epsilon} \\ -D_{Z,\bar\epsilon}^{\mathsf{t}} & 0 \end{pmatrix}.$$

Orient every finite-height truncation by the fixed root order and normalize its Pfaffian to 1 at $K_Z = 0$. The oriented Fredholm super-Pfaffian is the stabilized ratio

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{0}})}{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{1}})}.$$

Theorem 86.2 (Polarized Fredholm realization)**.** *Choose the protected parity refinement and the root-space Hilbert norms above. For every Z in the absolute-convergence chamber, the diagonal operator $K_Z|_{\mathfrak{g}_{\Delta_5\alpha}} = e^{-\alpha(Z)}\,\mathrm{id}$ is trace class on the even and odd root Hilbert spaces. For $D_Z = 1 - K_Z$,*

$$\mathrm{sdet}_F(D_Z) = \frac{\det_F(D_Z|_{\bar{0}})}{\det_F(D_Z|_{\bar{1}})} = \prod_{\alpha>0}(1 - e^{-\alpha(Z)})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}.$$

After choosing the standard polarization of $\mathcal{H}_\Delta \oplus \mathcal{H}_\Delta^*$ and its orientation, the paritywise skew block operators $\mathbb{A}_{Z,\bar{0}}$ and $\mathbb{A}_{Z,\bar{1}}$ have an oriented Fredholm super-Pfaffian

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{0}})}{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{1}})} = \mathrm{sdet}_F(D_Z).$$

Hence

$$e^{-\rho}\,\mathrm{sdet}_F(D_Z) = D_5(2Z).$$

Proof. The defining summability condition for $\mathfrak{C}_{\mathrm{abs}}$ is precisely the trace norm of the diagonal operator, summed in both parities. The Fredholm determinant of $1 - K_Z$ is the convergent product over eigenvalues; taking the even/odd ratio gives the superdeterminant. On each finite root-height truncation the Pfaffian of the oriented skew doubling equals the determinant of D_Z. Trace-norm convergence of the truncations passes this identity to the Fredholm limit. The final equality is the Gritsenko–Nikulin denominator formula. □

Remark 86.3. The polarization and orientation in this theorem belong to the root Hilbert space. A compact moduli-space orientation is a separate line bundle and is compared with this one by a specified intertwiner.

This constructs the square-root operator on the root Hilbert superspace where its stated summability condition holds. The nonemptiness estimate below concerns the minimal virtual superspace, whose dimensions are $|\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}|$. Applying that estimate to the full root superspace requires a bound on both parity dimensions. A compact geometric realization requires an operator intertwiner and a compatible map of determinant lines.

Chapter 87

What the automorphic character determines

Theorem 87.1 (Denominator data and reconstruction data)**.** *The Borcherds product determines:*

(i) *the signed root multiplicity* $\operatorname{sdim} \mathfrak{g}_{\Delta_{\mathrm{s}}\alpha}$ *in every active root degree;*

(ii) *the Weyl vector and denominator character;*

(iii) *the divisor and automorphic multiplier of the section.*

A Lie bracket is supplied by the Borcherds–Kac–Moody presentation, parity dimensions by a parity-refined root datum, Hall multiplication by extension correspondences, and an oriented square root by a square-root line with monodromy. These reconstruction data refine the scalar product independently.

Proof. The first three items are read from the exponent, prefactor, and automorphy law. Independence is witnessed by explicit pairs: two brackets on the same graded vector space have the same PBW character; an even–odd pair has signed dimension zero; distinct Hall correspondences may have the same Euler product; and twisting a square-root line by a sign local system preserves its square. Hence each refined object is reconstructed from the additional datum named in the statement. □

Recognition Theorem 87.2 (Complete primitive recognition)**.** *Let* $\mathfrak{p}$ *be a positive graded Lie superalgebra and let* $\Psi : \mathfrak{n}_{\Delta_{\mathrm{s}},+} \twoheadrightarrow \mathfrak{p}$ *be a graded Lie map. The map is an isomorphism if either:*

(i) *the parity-refined Hilbert series of* $U(\mathfrak{n}_{\Delta_{\mathrm{s}},+})$ *and* $U(\mathfrak{p})$ *agree; or*

(ii) *each primitive degree has prescribed pure parity and the signed PBW characters agree.*

The induced height completions, Chevalley complexes, and current envelopes are then isomorphic.

Proof. Choose the first height containing a homogeneous kernel vector. All decomposable PBW terms in that height are determined by lower heights and agree. The primitive parity dimension therefore drops, contradicting the first hypothesis. In the second case the logarithm of the PBW product recovers signed primitive dimensions inductively, and prescribed parity recovers ordinary dimensions. Naturality of completion, Chevalley chains, and current envelopes gives the final assertion. □

Book **XVII**

The universal Borcherds Hall and three-Calabi–Yau models

Chapter 88

Finite simple-root windows

Let I_Δ be the multiset of simple roots of the Gritsenko–Nikulin Borcherds superalgebra. A repeated imaginary simple root is written once for each copy. The parity map is

$$p : I_\Delta \longrightarrow \mathbb{Z}/2.$$

Choose a proper height function

$$h : I_\Delta \longrightarrow \mathbb{Z}_{>0}$$

whose sublevel sets are finite. Put

$$I_H = \{ i \in I_\Delta : h(i) \le H \}.$$

The restriction of the invariant form gives a finite symmetric Borcherds–Cartan matrix

$$\begin{aligned} &A_H = (a_{ij})_{i,j \in I_H}, \\ &a_{ii} = 2 \qquad\qquad \text{for real } i, \\ &a_{ii} \le 0 \qquad\qquad \text{for imaginary } i, \\ &a_{ij} \le 0 \qquad\qquad \text{for } i \ne j. \end{aligned}$$

The evenness of the root lattice makes every imaginary diagonal entry even.

Construction 88.1 (The Igusa Borcherds quiver)**.** Order I_H. For $i < j$ place $-a_{ij}$ arrows from i to j. At an imaginary vertex i place

$$g_i = 1 - \frac{a_{ii}}{2}$$

loops. A real vertex has loop number zero. Denote the resulting quiver by $Q_{\Delta,H}$.

Proposition 88.2 (Euler form)**.** *The symmetrized Euler form of $Q_{\Delta,H}$ is A_H:*

$$\langle e_i, e_j \rangle + \langle e_j, e_i \rangle = a_{ij}.$$

Proof. For $i \ne j$, the total number of arrows between the two vertices is $-a_{ij}$, so the symmetrized Euler entry is a_{ij}. At i the value is $2 - 2g_i = a_{ii}$. □

The inclusions $I_H \subset I_{H+1}$ identify $Q_{\Delta,H}$ with a full subquiver of $Q_{\Delta,H+1}$. Representations supported on the old vertices form an exact extension-closed subcategory. This gives compatible Hall embeddings.

Chapter 89

The real-root seed window

The three real simple roots are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with Gram matrix

$$A_{\mathrm{re}} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Choose the order $\delta_1 < \delta_2 < \delta_3$. The real-root quiver Q_{re} has two arrows from vertex i to vertex j for each $i < j$.

Proposition 89.1 (Real-root Hall seed)**.** *The symmetrized Euler form of Q_{re} is A_{re}. The Hall composition algebra generated by its three simple representations is the positive quantum Kac–Moody algebra with Cartan matrix A_{re}.*

Proof. The diagonal Euler entries are 2. The total number of arrows between distinct vertices is two, hence the symmetrized off-diagonal entries are -2. The Ringel–Green Hall theorem identifies the composition algebra with the positive quantum algebra defined by that symmetric Cartan matrix. □

The three vertices already contain the Weyl group and the reflective divisor. Imaginary vertices add the automorphic correction. The finite-height tower therefore begins with a completely explicit hyperbolic Hall algebra and enlarges it by one finite collection of imaginary simple roots at each stage.

89.1 The first Ginzburg differential

Let $a_{ij}^{(1)}, a_{ij}^{(2)} : i \to j$ be the two arrows for $i < j$. Add reverse arrows $(a_{ij}^{(r)})^* : j \to i$ of degree -1 and loops t_i of degree -2. The canonical three-Calabi–Yau completion has

$$d a_{ij}^{(r)} = 0, \qquad d(a_{ij}^{(r)})^* = 0,$$

and

$$d t_i = \sum_{a : s(a) = i} a a^* - \sum_{a : t(a) = i} a^* a. \tag{89.1}$$

The identity $d^2 = 0$ follows immediately. Formula (89.1) is the derived moment-map equation for the doubled real-root quiver.

Calculation 89.2 (Vertex one)**.** At vertex 1,

$$d t_1 = \sum_{r=1}^{2} \Big(a_{12}^{(r)} (a_{12}^{(r)})^* + a_{13}^{(r)} (a_{13}^{(r)})^* \Big).$$

At vertex 3,

$$dt_3 = -\sum_{r=1}^{2}\Big((a_{13}^{(r)})^* a_{13}^{(r)} + (a_{23}^{(r)})^* a_{23}^{(r)}\Big).$$

The sum over vertices is $\sum_a (aa^* - a^*a)$. It vanishes in the cyclic quotient and after matrix trace, but generally not in the path algebra.

Proof. The outgoing arrows at vertex 1 have targets 2 and 3. The incoming arrows at vertex 3 have sources 1 and 2. The displayed formulas follow from these incidence lists. Each arrow contributes one commutator to the sum over vertices. The trace of a commutator vanishes, giving the asserted trace identity. □

Chapter 90

Hall realization of the positive algebra

Let $\mathcal{A}_H = \mathrm{Rep}_{\mathbb{F}_q}(Q_{\Delta,H})$. Its twisted Hall product is

$$[M] * [N] = v^{\langle \dim M, \dim N\rangle} \sum_{[L]} |\{N' \subset L : N' \cong N,\ L/N' \cong M\}|[L], \qquad v^2 = q.$$

The class $[0]$ is the unit: the required subobject is unique on either side. Associativity follows by counting flags $N_1 \subset N_2 \subset L$ in two orders. The twist factors agree because the Euler form is bilinear:

$$\langle \alpha, \beta\rangle + \langle \alpha + \beta, \gamma\rangle = \langle \beta, \gamma\rangle + \langle \alpha, \beta + \gamma\rangle.$$

Thus this normalization defines the unital twisted Hall algebra on isomorphism classes.

Theorem 90.1 (Finite-window composition algebra)**.** *At each vertex i of $Q_{\Delta,H}$, choose the one-dimensional representation S_i supported at i, with every loop acting by zero. Use the generic composition algebra generated by these classes across finite fields, with the twisted subobject-count multiplication above. After extending coefficients from $\mathbb{Z}[v, v^{-1}]$ to $\mathbb{C}(v)$, this is the positive quantum generalized Kac–Moody algebra for the even Borcherds–Cartan matrix A_H. The parameter in Kang–Schiffmann is $v_{\mathrm{KS}} = v^{-1}$. Their specified semisimple perverse-sheaf construction on nilpotent representation varieties gives the corresponding canonical basis.*

Proof. This is the selected-simple composition construction of Kang–Schiffmann, Section 2.3 and Theorem 2.2, with one selected zero-loop simple at each vertex. Their geometric realization uses its split graded Grothendieck group over $\mathbb{Z}[v_{\mathrm{KS}}, v_{\mathrm{KS}}^{-1}]$ (Theorem 4.1), with the stated inversion of parameter [58]. The theorem does not use every simple object of the full representation category. A vertex with c loops has q^c one-dimensional simple representations over $\mathbb{F}_q$, whereas only the chosen zero-loop representation is the generator here. □

A semi-derived Hall realization of a full quantum group additionally specifies two-periodic complexes, the invertible acyclic classes and Cartan reduction [4, 73]. Its coproduct, pairing and completion are separate from the positive composition algebra used here.

90.1 Parity and the quantized superdatum

The Gritsenko–Nikulin denominator algebra carries a parity map on its simple-root multiset. It is additive on the free lattice of labelled simple generators. Projection to the Lorentzian degree can identify contributions of different parity. Both parity dimensions must therefore be retained in each projected root space. Hong's construction applied to this Borcherds superdatum gives the positive Borel

$$U_q^+(\mathfrak{g}_{\Delta_{\mathrm{s}}}{}^{(H)})$$

and the full quantized Borcherds superalgebra $U_q(\mathfrak{g}_{\Delta_{\mathrm{s}}}{}^{(H)})$.

Theorem 90.2 (Ordinary and super presentations)**.** *The ordinary quiver Hall presentation and Hong's quantum superalgebra use the same specified Cartan matrix and labelled simple generators. Their relation ideals and parity conventions are different inputs. Equality of their Cartan gradings alone does not identify their PBW characters.*

Proof. The two presentations are those of the preceding Hall construction and Hong's superalgebra construction. The difference already occurs for a single isotropic generator. The enveloping algebra of a one-dimensional even abelian Lie algebra is $\mathbb{C}[x]$. Declaring its generator odd only as a graded vector space leaves a nonzero vector x^2 in degree two. The enveloping algebra of a one-dimensional odd abelian Lie superalgebra is $\bigwedge(\xi)$, since its defining relation is $2\xi^2 = [\xi, \xi] = 0$. Its degree-two component is zero. Thus parity assignment to an ordinary PBW basis does not produce the super relation ideal or its character. A comparison must respect these relations degree by degree. □

Chapter 91

The three-Calabi–Yau completion

Let $kQ_{\Delta,H}$ be the path algebra. Its algebraic three-Calabi–Yau completion is

$$\Gamma_{\Delta,H} = \Pi_3(kQ_{\Delta,H}).$$

Write J for the ideal generated by all arrows in its graded quiver. Its completed version is the separate topological dg algebra $\widehat{\Gamma}_{\Delta,H} = \varprojlim_n \Gamma_{\Delta,H}/J^n$. The algebraic Calabi–Yau theorem is used for $\Gamma_{\Delta,H}$; a topological Calabi–Yau assertion for $\widehat{\Gamma}_{\Delta,H}$ requires its completed bimodule category. This is the Ginzburg model with zero potential: a degree-minus-one reverse arrow for every arrow and a degree-minus-two loop at each vertex. The loop differential is the signed sum of incident arrow–reverse-arrow products.

Theorem 91.1 (Borcherds–Ginzburg projections)**.** *For each finite quiver $Q_{\Delta,H}$, its undeformed three-Calabi–Yau completion $\Gamma_{\Delta,H}$ is the Ginzburg dg algebra with zero potential. If $H \leq H'$ and $Q_{\Delta,H}$ is the full subquiver on a subset of vertices, there is a surjective dg algebra map*

$$\pi_{H'H} : \Gamma_{\Delta,H'} \longrightarrow \Gamma_{\Delta,H}.$$

It kills the added vertices, every arrow incident to them, the corresponding reverse arrows, and the added degree-minus-two loops. These maps compose strictly and extend continuously to the completed algebras.

Proof. The finite Calabi–Yau assertion is the Ginzburg model of Keller's completion theorem [41, 61]. We concatenate paths from left to right, so aa^* is based at the source of a. Original and reverse arrows have zero differential. For the degree-minus-two loop t_i, the differential is

$$dt_i = \sum_{s(a)=i} aa^* - \sum_{t(a)=i} a^*a.$$

The projection deletes exactly the terms involving an added vertex, leaving the same sum for the full subquiver. It therefore commutes with the differential. It preserves path length and extends to the completed path algebra. Successive projections kill the union of their omitted vertices and arrows, proving strict composition. □

The evident inclusion of graded quivers does not give the reverse dg map. A single old vertex has loop differential zero. Adding an arrow $a : i \to j$ changes that differential to aa^* at i. The map fixing the old loop therefore fails to commute with the differential.

Write $\Gamma_\Delta = (\Gamma_{\Delta,H}, \pi_{H'H})_H$ for this pro-system of dg algebras. It is separate from the ordinary Hall embeddings obtained by extension by zero on representations of a full subquiver. A direct system of Calabi–Yau dg categories compatible with those Hall embeddings requires additional dg functors. It does not follow from the inclusion of the displayed Ginzburg generators.

The quantum group retains its separately specified root-height completion. Its transition maps must preserve the Cartan generators, coproduct, and pairing. Quotienting the positive nilpotent Lie algebra by height and truncating a full quantum group are different operations.

Chapter 92

Hall states and protected superstates

The representation varieties of $Q_{\Delta,H}$ carry the perverse-sheaf complexes used in the composition-algebra construction. The full equivariant derived category is a larger carrier. Already for one real vertex and dimension one, its quotient is $B\mathbb{G}_m$, with $H^*(B\mathbb{G}_m;\mathbb{Q}) = \mathbb{Q}[c_1]$, where $|c_1| = 2$. Its total equivariant cohomology is infinite-dimensional. Finite root multiplicities therefore refer to the specified composition algebra, not to total equivariant cohomology of this stack.

Definition 92.1 (Igusa Hall category)**.** For every dimension vector $\mathbf{d}$, let

$$\mathfrak{R}_{H,\mathbf{d}} = [\mathrm{Rep}_{\mathbf{d}}(Q_{\Delta,H})/G_{\mathbf{d}}].$$

The finite-height Hall category is the direct sum of the equivariant constructible derived categories on the $\mathfrak{R}_{H,\mathbf{d}}$, equipped with Hall convolution along the stack of short exact sequences. Denote it by $\mathcal{B}_{\Delta,H}$ and set

$$\mathcal{B}_{\Delta} = \varinjlim_{H} \mathcal{B}_{\Delta,H}.$$

Independently, the protected one-particle superspace is the root-graded superspace

$$\mathcal{H}_{\mathrm{ip},H} = \bigoplus_{\alpha>0} \mathfrak{g}_{\Delta_5,\alpha}^{(H)},$$

with the parity of the Borcherds superalgebra. Its multiparticle completion is the PBW/Fock superspace of $U(\mathfrak{n}_{\Delta_5,+}^{(H)})$.

Theorem 92.2 (Hall and protected-state comparison)**.** *The composition subalgebra in the finite-window Hall theorem is the ordinary positive quantum generalized Kac–Moody algebra. This statement does not identify the Grothendieck group of the full category $\mathcal{B}_{\Delta,H}$ with that composition algebra. The protected one-particle superspace has signed root multiplicities*

$$\operatorname{sdim} \mathcal{H}_{\mathrm{ip},H,\alpha} = \operatorname{sdim} \mathfrak{g}_{\Delta_5,\alpha}^{(H)}.$$

Its height-complete PBW character is

$$\operatorname{sch}\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha>0} (1-e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_{\Delta_5,\alpha}},$$

and its Weyl-shifted inverse is $D_5(2Z)$. The composition canonical basis and the protected root presentation use the specified simple-root indexing and Cartan grading; the parity refinement belongs to the Borcherds superdatum.

Proof. The composition-algebra assertion is the finite-window theorem. The protected assertion is PBW for the Borcherds Lie superalgebra. Proper height passes both constructions to the limit. The Gritsenko–Nikulin denominator theorem gives the final product identity. □

Chapter 93

Extension determinants and critical orientations

Let C be a three-Calabi–Yau dg category over $\mathbb{C}$, with its chosen trace. For families of objects A, B over a base T, let $C(A,B)$ denote the relative derived Hom complex on T. Suppose these complexes are perfect, and that the trace gives base-change-compatible duality

$$C(B,A) \simeq C(A,B)^{\vee}[-3].$$

Put $L(A,B) = \det C(A,B)$ and $D(A) = L(A,A)$. The determinant convention is $\det C = \otimes_i (\det C^i)^{(-1)^i}$, with symmetry sign specified by $\chi(C) \bmod 2$.

Theorem 93.1 (Extension determinant)**.** *For an exact triangle $A \to F \to B \to A[1]$, there is a base-change-compatible isomorphism*

$$D(F) \simeq D(A) \otimes D(B) \otimes L(A,B)^{\otimes 2}.$$

The determinant comparisons obtained by refining a flag agree, with the stated determinant symmetry signs.

Proof. Apply derived Hom to the triangle in each variable. Determinant additivity gives

$$D(F) \simeq L(B,A) \otimes D(A) \otimes D(B) \otimes L(A,B).$$

Dualization and the odd shift each invert the determinant line. Thus Calabi–Yau duality identifies $L(B,A)$ with $L(A,B)$. Reorder by determinant symmetry to obtain the formula. For a full flag, all iterated decompositions identify with the same ordered tensor product of $C(A_i, A_j)$, for its successive quotients A_i. Determinant additivity for refined filtrations proves coherence. Each operation commutes with base change. □

A square-root orientation consists of lines o_A and isomorphisms $o_A^{\otimes 2} \simeq D(A)$. Extension compatibility requires maps

$$o_A \otimes o_B \otimes L(A,B) \longrightarrow o_F$$

whose squares are the determinant comparisons. Choosing coordinates on a smooth critical chart trivializes its ambient canonical line. Hall convolution requires transition maps compatible with the cross line and the zero object. On three-step flags, the two orientation maps can differ by an element of μ_2, even when their squares agree. That sign must be trivialized coherently.

Example 93.2 (A nontrivial cross line)**.** Let X be a smooth projective Calabi–Yau threefold, with a chosen volume form, and let $x \in X$. On $X \times T$, where $T = \mathbb{P}^1$, take

$$A = \mathcal{O}_X \boxtimes \mathcal{O}_T(1), \qquad B = \mathcal{O}_x \boxtimes \mathcal{O}_T.$$

Then $C(A,B) = \mathcal{O}_T(-1)[0]$, and relative Serre duality gives $C(B,A) = \mathcal{O}_T(1)[-3]$. Thus $L(A,B) = L(B,A) = \mathcal{O}_T(-1)$. The self-Hom complexes of A and B have constant determinant lines, since the factors $\mathcal{O}_T(1)$ cancel in the first self-Hom. Nevertheless

$$D(A \oplus B) = D(A) \otimes D(B) \otimes \mathcal{O}_T(-2).$$

The orientation cross factor is $\mathcal{O}_T(-1)$. A sign change cannot replace this line of nonzero degree.

Corollary 93.3 (Oriented critical coefficients)**.** *Suppose the chosen critical charts have square-root orientation maps on overlaps and extension flags, with unit and flag coherences. Suppose their vanishing-cycle comparison maps preserve these data. Then the local vanishing-cycle complexes descend to the specified oriented coefficient object. Their Hall convolution is associative whenever its pullback and pushforward maps are defined and satisfy base change on the flag diagrams.*

Proof. The overlap cocycle identifies the local coefficient complexes by descent. On a three-step flag, orientation coherence identifies the two Thom–Sebastiani coefficient maps. Proper or compact-support base change, on the stipulated domain, identifies the two composites of convolution. The zero flag gives the unit equation. □

For the quiver charts, the remaining construction is an equivariant system of square roots and coefficient maps on the actual representation and extension stacks, compatible with changes of height. For compact $S \times E$, the reduced complexes and the common elliptic torsor must also be retained. A coordinate volume on an affine chart specifies neither descent system by itself.

Chapter 94

The colored elliptic Ran space

The elliptic degree is a global charge carried by the whole configuration. Define

$$\mathrm{Ran}^{\mathrm{wr}}(E) = \coprod_{m \geq 0} \mathrm{Ran}(E)_m$$

with monoidal operation

$$(I, m) \sqcup (J, n) = (I \sqcup J, m + n)$$

on disjoint supports. The component $m = 0$ is point-local. The components $m > 0$ are wrapped.

Let

$$\mathfrak{n}_{\Delta_s,+} = \bigoplus_{m \geq 0} \mathfrak{n}_{\Delta_s,+}[m]$$

be the elliptic-degree grading. The root bracket obeys

$$[\mathfrak{n}_{\Delta_s,+}[m], \mathfrak{n}_{\Delta_s,+}[n]] \subset \mathfrak{n}_{\Delta_s,+}[m + n].$$

Construction 94.1 (Hybrid factorization algebra)**.** Form the chiral Chevalley factorization coalgebra

$$\mathcal{F}^{\mathrm{hyb}}_{\Delta,E} = C^{\mathrm{ch}}_*\Big(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+})\Big).$$

Retain the elliptic-degree grading on every Chevalley word and place its degree-m summand on $\mathrm{Ran}(E)_m$. Disjoint union adds the degree labels.

Theorem 94.2 (Local–wrapped factorization)**.** *$\mathcal{F}^{\mathrm{hyb}}_{\Delta,E}$ is a graded factorization coalgebra on $\mathrm{Ran}^{\mathrm{wr}}(E)$. Its degree-zero color is the current theory of $\mathfrak{n}_{\Delta_s,+}[0]$. Every degree-$m$ color is a module for the degree-zero color. The factorization product of degrees m and n lands in degree $m + n$. Its chiral primitives are $\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+})$, and constant modes recover $\mathfrak{n}_{\Delta_s,+}$.*

Proof. The chiral Chevalley construction is a factorization coalgebra for every chiral Lie algebra. The bracket preserves elliptic-degree addition, so its differential and coproduct preserve the graded Ran placement. The primitive theorem for the Chevalley coalgebra recovers the current Lie algebra. □

This construction retains the nonnegative root-degree label on a point-supported current theory. It does not replace a sheaf projecting onto the whole elliptic curve by a point-supported sheaf. The compact wrapped comparison must map its common-torsor extension correspondences to these graded current operations, preserving support and charge. The bracket on the present carrier is the Borcherds bracket.

Chapter 95

Physical charge and Gram degree

Let Γ_{phys} be an additive physical charge lattice and let

$$\Pi(c) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right)$$

be the Gram map. Its polarization is

$$B(c,d) = \Pi(c+d) - \Pi(c) - \Pi(d).$$

Define the central extension

$$\widetilde{\Gamma}_{\Pi} = \Gamma_{\mathrm{phys}} \times \Gamma_{\mathrm{ind}}$$

with multiplication

$$(c,\lambda)(d,\mu) = (c+d, \lambda+\mu+B(c,d)).$$

Theorem 95.1 (Graph homomorphism)**.** *The map*

$$j_{\Pi} : \Gamma_{\mathrm{phys}} \longrightarrow \widetilde{\Gamma}_{\Pi}, \qquad c \longmapsto (c, \Pi(c))$$

is a group homomorphism. For every Γ_{phys}-graded algebra A, the Rees lift

$$a_c \longmapsto a_c u^{\Pi(c)}$$

turns multiplication into an additive $\widetilde{\Gamma}_{\Pi}$-grading with cocycle $u^{-B(c,d)}$.

Proof. The defining multiplication gives

$$j_{\Pi}(c)\, j_{\Pi}(d) = (c+d, \Pi(c)+\Pi(d)+B(c,d)) = j_{\Pi}(c+d).$$

The algebra identity follows immediately. □

The universal Igusa category is graded directly by the Lorentzian root lattice. A geometric $K3 \times E$ theory enters through a charge functor to $\widetilde{\Gamma}_{\Pi}$.

Chapter 96

The universal protected field theory

For each H take cotangent BF theory with gauge Lie superalgebra ${\mathfrak{n}_{\Delta_s,+}}^{(H)}$ on $\mathbb{R}^2_{\mathrm{top}} \times E$. Its fields are

$$\Omega^\bullet(\mathbb{R}^2) \mathbin{\widehat{\otimes}} \Omega^{0,\bullet}(E) \mathbin{\widehat{\otimes}} \left({\mathfrak{n}_{\Delta_s,+}}^{(H)}[1] \oplus ({\mathfrak{n}_{\Delta_s,+}}^{(H)})^\vee[2] \right).$$

The action is

$$S_H = \int_{\mathbb{R}^2 \times E} \langle B, (\mathrm{d} + \bar\partial) A + \frac{1}{2}[A, A] \rangle.$$

Theorem 96.1 (Classical local theory and exact quantum harmonic sector)**.** *For every finite height H the shifted-cotangent action defines a classical mixed holomorphic–topological factorization algebra on $\mathbb{R}^2_{\mathrm{top}} \times E$. Harmonic reduction on a compact product cell gives the finite-dimensional cotangent BV theory of ${\mathfrak{n}_{\Delta_s,+}}^{(H)}$, and this reduced theory satisfies the quantum master equation. Complementary Lagrangian boundary polarizations give the boundary enveloping algebra $U({\mathfrak{n}_{\Delta_s,+}}^{(H)})$, the two-sided bar complex $C_*({\mathfrak{n}_{\Delta_s,+}}^{(H)})$, and the current envelope on E. The proper-height inverse system is coefficientwise finite and defines a complete classical local theory together with an exact quantum harmonic sector.*

Proof. The invariant cotangent pairing and the Jacobi identity give the classical master equation and factorization descent. Harmonic reduction leaves a finite positive nilpotent Lie superalgebra. Every even adjoint endomorphism raises proper height and is strictly upper triangular; every odd adjoint endomorphism has vanishing supertrace. The reduced BV divergence therefore vanishes. Canonical quantization in the selected polarization gives the enveloping algebra, and the augmentation boundary states give the two-sided bar, identified with Chevalley chains by PBW. Each map preserves height, so the inverse system is defined coefficientwise. □

Chapter 97

Absolute convergence of the virtual root operator

For each positive root α, let V_α be the minimal parity realization of the signed multiplicity: it is purely even of dimension $\operatorname{sdim} \mathfrak{g}_{\Delta_5 \alpha}$ when this integer is nonnegative and purely odd of dimension $-\operatorname{sdim} \mathfrak{g}_{\Delta_5 \alpha}$ when it is negative. Set

$$\mathcal{V}_\Delta = \widehat{\bigoplus}_{\alpha>0} V_\alpha .$$

This virtual root superspace is the canonical linear carrier of the denominator character; the Lie bracket remains carried by $\mathfrak{g}_{\Delta_5}$.

Let $h(\alpha)$ be the proper height. Jacobi-form coefficient estimates give constants $C_0, C_1 > 0$ such that

$$\dim V_\alpha = \left|\operatorname{sdim} \mathfrak{g}_{\Delta_5 \alpha}\right| \le C_0 \exp(C_1 h(\alpha)) \tag{97.1}$$

for every positive root. The number of Gram triples of height H is polynomial in H.

Theorem 97.1 (Nonempty trace-class chamber)**.** *Let Z_0 lie in the interior of the positive Weyl chamber. For sufficiently large $t > 0$,*

$$\sum_{\alpha>0} \dim V_\alpha \left| e^{-\alpha(tZ_0)} \right| < \infty .$$

Hence the absolute-convergence chamber contains a conical tail in every interior direction.

Proof. Interior positivity gives a constant $c(Z_0) > 0$ with $\Re \alpha(Z_0) \ge c(Z_0) h(\alpha)$. Combine this with (97.1). The contribution of roots of height H is bounded by a polynomial in H times

$$\exp\bigl((C_1 - t c(Z_0)) H\bigr).$$

The series converges when $t c(Z_0) > C_1$. □

Corollary 97.2 (Trace-class root Hamiltonian)**.** *On the conical tail, the diagonal operator*

$$K_Z\big|_{V_\alpha} = e^{-\alpha(Z)} \,\mathrm{id}$$

is trace class on both parity summands. The Fredholm product is holomorphic and equals the absolutely convergent Borcherds product.

Proof. The trace norm is the summable series of the preceding theorem, separated by parity. Standard trace-class determinant theory gives holomorphy, and diagonalization gives the product over root eigenvalues. □

Chapter 98

The virtual root Fredholm operator

Let

$$\mathcal{H}_{\Delta}^{\mathrm{vir}} = \widehat{\bigoplus}_{\alpha>0} V_{\alpha}$$

with an orthonormal basis in each minimal parity realization. This is the virtual root Hilbert space used in the preceding absolute-convergence theorem. In the absolute-convergence chamber define

$$K_Z\big|_{V_\alpha} = e^{-\alpha(Z)}\,\mathrm{id}, \qquad D_Z = 1 - K_Z.$$

Theorem 98.1 (Oriented Fredholm square root)**.** *The operator K_Z is trace class on the even and odd summands of the virtual root Hilbert space. Its Fredholm Berezinian is*

$$\mathrm{sdet}_F(D_Z) = \prod_{\alpha>0} (1 - e^{-\alpha(Z)})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}.$$

Choose an order of the positive roots, first by height and then lexicographically. This choice orients every finite-height root space. The skew doubling

$$\mathbb{A}_Z = \begin{pmatrix} 0 & D_Z \\ -D_Z^{\mathrm{t}} & 0 \end{pmatrix}$$

has an oriented Fredholm super-Pfaffian satisfying

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \mathrm{sdet}_F(D_Z).$$

Consequently

$$e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

Proof. Absolute convergence of the Borcherds product is the trace-norm summability condition. The determinant formula follows from the spectrum. At finite height the Pfaffian of the oriented skew doubling is the determinant. Trace-norm convergence passes the identity to the limit. The denominator theorem supplies the Weyl shift. □

98.1 Automorphic orientation

The product defines a section of the automorphic line $\mathcal{L}^5 \otimes \nu_5$. On the absolute-convergence chamber, the oriented Fredholm Pfaffian is a trivialization of this line. Analytic continuation uses the automorphy factor of D_5 and therefore glues the chamberwise Fredholm models with transition character ν_5.

Corollary 98.2 (Orientation character)**.** *The automorphic orientation line of the root Dirac family is the sign local system ν_5. On every real simple reflection it has value -1.*

Proof. The Fredholm product equals D_5 on the convergence chamber, and D_5 transforms with automorphy factor ν_5. This transformation law is the gluing rule for the automorphic orientation line. A real simple root occurs with multiplicity one, so reflection reverses its oriented normal line. □

Chapter 99

The universal Igusa protected effective theory

Definition 99.1 (Universal Igusa completion)**.** The protected completion selected by the K3 half-index is the quintuple

$$\mathfrak{K}_{\Delta} = \left(\Gamma_{\Delta}, \mathcal{B}_{\Delta}, \mathcal{H}_{\mathrm{ip}}, \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \mathbb{A}_{Z}\right).$$

Its components are the pro-system of finite three-Calabi–Yau completions, the ordinary quiver Hall category, the Borcherds one-particle Lie superalgebra, the local–wrapped current factorization algebra, and the virtual root Dirac family.

Theorem 99.2 (Igusa protected completion)**.** *The quintuple $\mathfrak{K}_{\Delta}$ satisfies:*

(i) *its one-particle Lie superalgebra is $\mathfrak{g}_{\Delta_5}$;*

(ii) *its ordinary Hall presentation and Hong's quantum supergroup retain the stated Cartan grading; comparison of their relation ideals and parity-refined PBW spaces requires an additional map;*

(iii) *its chiral primitive algebra on E is the local–wrapped current algebra of $\mathfrak{g}_{\Delta_5}$;*

(iv) *its raw multiparticle supercharacter is P^{-1}, where $e^{-\rho}P = \mathrm{Den}_{\Delta}$; equivalently $e^{\rho}\,\mathrm{sch}\,\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{Den}_{\Delta}^{-1}$;*

(v) *the half-coordinate pullback of the doubled Weyl-normalized character is $-{\chi_{10}}^{-1}$ in the Oberdieck–Pixton normalization;*

(vi) *its oriented first-order determinant is Den_{Δ} and its orientation monodromy is ν_5.*

Proof. The first three statements are the Borcherds root, quiver Hall, Hong quantum-supergroup, Ginzburg, and colored-Ran theorems. PBW gives the fourth. The identity $\chi_{10} = {D_5}^2$ and the established reduced scalar theorem give the fifth. The Fredholm theorem gives the sixth. □

The tuple retains algebraic constructions associated with the Igusa presentation and their scalar characters. A geometric type-IIA or topological-string realization must construct compatible functors from its oriented extension category to these targets. Primitive states and extension amplitudes are inputs to those functors. Their action on morphisms, convolution, orientations, and chiral operations must also be specified and proved compatible.

Chapter 100

The primitive comparison and its extensions

Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra with finite-dimensional homogeneous pieces and a proper positive height. A primitive realization of the Igusa presentation is a parity-preserving homogeneous map from its labelled simple-root spaces to $\mathfrak{p}$, satisfying the defining relations.

Theorem 100.1 (Universal presentation map)**.** *Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra over $\mathbb{C}$, with finite-dimensional homogeneous pieces. Suppose a homogeneous parity-preserving map from every simple-root generating space of $\mathfrak{n}_{\Delta_5,+}$ to $\mathfrak{p}$ satisfies the defining Borcherds–Serre relations. It extends uniquely to a graded Lie-superalgebra map*

$$f : \mathfrak{n}_{\Delta_5,+} \longrightarrow \mathfrak{p}.$$

It induces continuous maps on height completions, enveloping algebras, Chevalley chains, and level-zero currents. The map f is an isomorphism if and only if it is surjective and the root-graded PBW series agree in each parity.

Proof. Extend the chosen maps to the free Lie superalgebra on the simple-root spaces. The relation hypothesis makes the defining ideal lie in its kernel, giving f. The generators prove uniqueness. On enveloping algebras the map sends $x_1 \cdots x_j$ to $f(x_1) \cdots f(x_j)$. On Chevalley chains it is the graded symmetric map on the shifted generators. It commutes with the Chevalley differential because f preserves brackets. The current map is $\partial^j x \mapsto \partial^j f(x)$, preserving $[x_\lambda y] = [x, y]$. Root degree preservation gives compatible maps at each height and hence on the stated limits.

If f is surjective, $U(f)$ is surjective. Proper positive height makes each root-graded PBW component finite-dimensional. Equality of its even and odd dimensions makes $U(f)$ bijective in every component. PBW embeds each Lie superalgebra into its enveloping algebra, so f is injective. The converse follows by applying the constructions to its inverse. □

A geometric comparison also requires functors on the actual Hall and critical categories, a compatible coproduct and Hopf pairing, and maps of determinant lines and operator domains. These categories and lines are not the domain of the primitive Lie map. The additional maps must be constructed and checked against that map. They are not consequences of its universal property.

Chapter 101

The eight diagonal-divisor forms

Cléry and Gritsenko classify eight genus-two modular forms whose divisor is the diagonal with order one [21]. Gritsenko–Nikulin and the CHL denominator constructions give the sourced reflective presentations used for the corresponding rows [42–44, 46–48]. Every row has a minimal virtual product superspace obtained from its Fourier exponents, together with a Fock determinant and a diagonal Fredholm operator. A sourced denominator presentation refines this product model by its Borcherds–Cartan bracket, Weyl group, and quantum supergroup. The half-integral rows are formed on their multiplier covers.

form	group	weight	character order
Δ_5	$\mathrm{Sp}_4(\mathbb{Z})$	5	2
Δ_2	Γ_2	2	4
Δ_1	Γ_3	1	6
$\Delta_{1/2}$	Γ_4	1/2	8
∇_3	$\Gamma_0^{(2)}(2)$	3	2
∇_2	$\Gamma_0^{(2)}(3)$	2	2
$\nabla_{3/2}$	$\Gamma_0^{(2)}(4)$	3/2	4
Q_1	$\Gamma_2(2)$	1	4

Theorem 101.1 (Eight-row algebraic realization)**.** *Each diagonal-divisor row has a proper-height virtual root superspace, a super-Fock determinant, a colored abelian current theory, and a trace-class Fredholm denominator operator whose character is the corresponding automorphic product. For every row equipped with its reflective denominator presentation, the finite-window Borcherds quiver, ordinary Hall category, three-Calabi–Yau completion, and parity-refined quantum supergroup refine this product model and recover the same denominator character.*

Proof. Local finiteness of the product exponents gives the graded superspace and its proper-height completion. The Fock and Fredholm constructions depend only on these signed multiplicities and reproduce the product coefficientwise. A reflective denominator presentation supplies a symmetrizable Borcherds superdatum; the quiver, Hall, Ginzburg, current, and Hong constructions then apply to that datum and preserve its PBW denominator. □

Chapter 102

Algebraic and physical closure

Theorem 102.1 (Universal Igusa realization theorem)**.** *The constructions have the following inputs and outputs. An arrow labelled by a construction records its dependence on that input:*

$$\phi_{0,1} \xrightarrow{\text{Borch}} D_5, \qquad \phi_{0,1} \xrightarrow{\text{root datum}} \mathfrak{g}_{\Delta_5}, \qquad A_\Delta \xrightarrow{\text{quiver}} Q_{\Delta,H} \xrightarrow{\text{Hall}} \mathcal{B}_\Delta,$$

$$kQ_{\Delta,H} \xrightarrow{\Pi_3} \Gamma_{\Delta,H}, \qquad \mathfrak{g}_{\Delta_5} \xrightarrow{\mathrm{Cur}_E} \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \qquad \mathfrak{g}_{\Delta_5} \xrightarrow{\text{Hong}} \widehat{U}_q(\mathfrak{g}_{\Delta_5}),$$

$$\mathfrak{g}_{\Delta_5} \xrightarrow{\text{root spectrum}} \mathbb{A}_Z, \qquad e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = \mathrm{Den}_\Delta(Z) = D_5(2Z),$$

$$\mathcal{Z}^{\mathrm{red}}_{K3\times E} = -(\jmath^*\,\mathrm{Den}_\Delta)^{-2} = -\chi_{10}{}^{-1}.$$

The system contains parity-refined one-particle spaces, an ordinary Hall extension product, the Borcherds root bracket, a local–wrapped collision law, a quantum supergroup, an automorphic orientation line, a first-order Fredholm determinant, and the reduced scalar square.

Proof. The Borcherds lift, Gritsenko–Nikulin presentation, and primitive reduced curve-counting theorem give the scalar identities. The enveloping and Chevalley maps are induced by the presentation. The quiver Hall algebra and finite Calabi–Yau completion have their respective quiver and path-algebra inputs. Their common grading specifies the degrees a comparison must preserve. A map between those constructions requires the additional functors, coefficient maps, and pairings described above. □

Book XVIII

Technical appendices to the preceding book

Chapter 103

Exact theta-series calculation

Set $t = q^{1/8}$ and $x = r^{1/2}$. Then

$$\vartheta_2(\tau, z) = \sum_{n \in \mathbb{Z}} t^{4n^2+4n+1} x^{2n+1},$$
$$\vartheta_3(\tau, z) = \sum_{n \in \mathbb{Z}} t^{4n^2} x^{2n},$$
$$\vartheta_4(\tau, z) = \sum_{n \in \mathbb{Z}} (-1)^n t^{4n^2} x^{2n}.$$

Squaring numerator and denominator, dividing in $\mathbb{Q}[x^{\pm 1}][[t]]$, and summing the three quotients gives the coefficient table in Chapter 1. The calculation uses integer convolution and formal-series inversion; fractional powers cancel before conversion from (t, x) to (q, r).

Chapter 104

The root reflection matrices

In the basis (f_2, f_3, f_{-2}), the form is

$$G = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

The three reflection matrices are

$$S_1 = \begin{pmatrix} 1 & 4 & 4 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & -1 & 0 \\ 4 & 4 & 1 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$S_i^2 = I, \qquad S_i^{\mathsf{T}} G S_i = G.$$

Chapter 105

Automorphic, homological, and Hall comparisons

The half-index determinant, theta normalization, Borcherds denominator, reduced scalar theorem, charge Gram map, and geometric Hall comparison datum belong to the following constructions:

(i) the automorphic square root is completed entirely within the Borcherds product theorem;

(ii) the BKM bracket comes from the Gritsenko–Nikulin algebra;

(iii) the bar–Chevalley comparison gives the denominator a homological source;

(iv) the current and BF constructions give an exact first-order algebraic model;

(v) orientation data refine determinant lines, while an operator Pfaffian requires an actual Fredholm family;

(vi) Weyl reflections are attached to real roots, and imaginary roots enter through multiplicities;

(vii) the compact Hall comparison uses additive Mukai charge, the polarized central extension, and hybrid local–wrapped sectors.

Book entrance: Calabi–Yau Quantum Groups

An extension of geometric objects defines a correspondence. Its action on cohomology depends on the coefficient theory and the allowed pushforward. Tensor products and exchange require a coproduct and compatible intertwiners. A trace on a specified representation then produces an amplitude. The reconstruction problem is to retain these operations through the comparisons between Calabi–Yau categories, boundary fields, Hall algebras, and quantum groups.

Book XIX

Categorical and field-theoretic foundations

Chapter 106

Calabi–Yau categories

106.1 Smoothness, properness, and the Calabi–Yau class

Let C be a small stable $\Bbbk$-linear category. Smoothness means that the diagonal bimodule is perfect. Properness means that the morphism complexes are perfect over $\Bbbk$. A d-Calabi–Yau structure is a nondegenerate negative-cyclic class

$$\theta_{\mathrm{CY}} \in HC^{-}_{d}(C)$$

whose underlying Hochschild trace identifies the diagonal bimodule with its dual shifted by d.

For $E \in C$, the tangent complex of the derived moduli stack of objects is

$$T_E \mathcal{M}_C \simeq \mathrm{RHom}_C(E, E)[1].$$

The Calabi–Yau trace pairs two tangent vectors by composition and trace.

Theorem 106.1 (The two universal categorical outputs)**.** *Let C be a smooth proper d-Calabi–Yau category.*

(i) *The Hochschild cochains $CC^*(C)$ carry a framed E_2-algebra structure.*

(ii) *Whenever the moduli stack of objects is representable in the PTVV setting, $\mathcal{M}_C$ carries a $(2-d)$-shifted symplectic form.*

Proof. The cyclic Deligne theorem upgrades the brace E_2 structure on Hochschild cochains by the circle action supplied by the cyclic trace [15]. The shifted form at E is

$$\mathrm{RHom}(E,E)[1]^{\otimes 2} \xrightarrow{\circ} \mathrm{RHom}(E,E)[2] \xrightarrow{\mathrm{Tr}} \Bbbk[2-d].$$

Nondegeneracy is the Calabi–Yau duality isomorphism, and closedness is the negative-cyclic lift [14, 84]. □

The integer d controls the shift of the moduli form. The Hochschild operation remains framed E_2. These structures meet through deformation theory while retaining their separate operadic degrees.

Chapter 107

Local BV theories and factorization observables

107.1 Cyclic elliptic local Lie algebras

Let X be a complex manifold. A classical local field theory is specified by a sheaf $\mathcal{L}$ of elliptic L_∞ algebras, a local invariant pairing, and an action functional S satisfying the classical master equation. For an open set $U \subset X$, the classical observables are

$$\mathrm{Obs}^{\mathrm{cl}}(U) = C^*_{\mathrm{CE,cont}}(\mathcal{L}(U)).$$

Here $\mathcal{L}(U)$ consists of unrestricted smooth fields with its nuclear smooth-section topology. Continuous cochains use the completed symmetric powers of its continuous dual, with the completed tensor products of the elliptic-observable construction [23, 24]. The dual consists of compactly supported distributions. For $U \subset V$, restriction of fields $\mathcal{L}(V) \to \mathcal{L}(U)$ induces the covariant map on observables. For pairwise disjoint $U_i \subset V$, restrict to $\bigoplus_i \mathcal{L}(U_i)$ and multiply the pulled-back cochains. These are the prefactorization maps.

Theorem 107.1 (Factorization observables)**.** *An elliptic classical BV theory produces a factorization algebra* $\mathrm{Obs}^{\mathrm{cl}}$*. A renormalized quantization satisfying the quantum master equation produces a factorization algebra* $\mathrm{Obs}^{\mathrm{q}}$ *over* $\Bbbk[[\hbar]]$.

Proof. Restriction is a morphism of local L_∞ algebras, so its contravariant continuous Chevalley map commutes with the differential. For disjoint opens, the completed cochains of the finite direct sum are the completed tensor product of the cochains. This constructs the displayed maps and their composition identities. The elliptic descent theorem in this completed coefficient category supplies Weiss descent [23, 24]. In the quantum theory, renormalized time-ordered products define the structure maps, and the quantum master equation makes their differential square-zero and compatible with nested ultraviolet contractions [23, 24]. □

107.2 Boundary and defect restriction

Let $Y = C \times \mathbb{R}_{\geq 0}$, with C a complex curve, and let a boundary condition be given at $C \times \{0\}$. Boundary observables are holomorphic along C and ordered along the inward normal line. A compact boundary object b in the boundary factorization category gives

$$A_b = \mathrm{RHom}(b, b).$$

Theorem 107.2 (Ordered boundary algebra)**.** *When boundary observables are holomorphic on C and locally constant along the oriented normal direction, A_b is an ordered chiral algebra*

$$A_b \in \mathrm{Alg}_{\mathsf{E}_1}\left(\mathrm{Fact}_D(C), \otimes^!\right).$$

If b is a local compact generator and the Morita equivalences obey factorization descent, the boundary category is equivalent to A_b-modules in factorization coefficients.

Proof. Composition of boundary endomorphisms gives the internal E_1 multiplication. Holomorphic collision gives the factorization coefficient on C. The componentwise locality identity makes multiplication compatible with every disjoint-support factorization map. Local compact-generator Morita theory descends over the Ran diagrams. These are the recognition and Morita theorems of Volume I. □

Chapter 108

Three curve-level algebraic outputs

Let C be a smooth curve and let $\mathcal{D}_C$ be the fixed Ran category. The same geometric source can produce three algebraic outputs, and their distinction is functorial.

Definition 108.1 (Curve-level faces)**.** A transverse face is

$$A^{\perp} \in \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(C), \otimes^!).$$

An associative chiral face is

$$A^{\mathrm{ch}}_{\mathsf{Ass}} \in \mathrm{Alg}_{\mathsf{E}_1}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

A chiral Lie face is

$$\mathfrak{g}^{\mathrm{ch}} \in \mathrm{Alg}_{\mathsf{Lie}}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

Their bars are $\mathrm{Bar}^{\perp}$, $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}$, and $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}$.

Theorem 108.2 (Source separation)**.** *A boundary normal direction determines the transverse face. A chiral-convolution multiplication determines the associative chiral face. A Lie bracket for chiral convolution determines the Lie face. The commutator functor sends the associative chiral face to the Lie face. A chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ satisfies*

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(U^{\mathrm{ch}}(\mathfrak{g})) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g})$$

on the nilcomplete PBW locus. A mixed source joins the transverse and associative chiral faces through an aligned distributive law.

Proof. Each statement is the corresponding operadic restriction or adjunction in the named monoidal category. The PBW equivalence is the bar comparison for chiral enveloping algebras. The mixed statement is the distributive-law theorem of Volume I. □

The source geometry selects the available directions. The E_n level records the operations and coherences constructed on them; dimension alone does not specify those structures.

Chapter 109

Holomorphic Chern–Simons theory

Let X be a complex threefold with holomorphic volume form Ω_X, and let $\mathfrak{a}$ be a quadratic Lie algebra. The BV fields are

$$\mathcal{E}_X = \Omega^{0,*}(X, \mathfrak{a})[1],$$

with degree -1 pairing

$$\omega(A, B) = \int_X \Omega_X \wedge \langle A, B \rangle.$$

The action is

$$S_{\mathrm{hCS}}(A) = \int_X \Omega_X \wedge \left(\frac{1}{2} \langle A, \bar{\partial} A \rangle + \frac{1}{6} \langle A, [A, A] \rangle \right).$$

Theorem 109.1 (Classical hCS factorization algebra)**.** *The action S_{hCS} satisfies the classical master equation. Its classical observables form a holomorphic factorization algebra on X.*

Proof. The Hamiltonian vector field is the Maurer–Cartan differential $\bar{\partial} A + \frac{1}{2}[A, A]$. Its square is the Dolbeault square, the derivation identity, and the Jacobi identity. The elliptic Dolbeault complex then gives factorization observables by the preceding theorem. □

A perturbative quantization is controlled by the local BV deformation complex. In complex dimension three, its one-loop gauge anomaly occupies the degree-four invariant-polynomial summand; mixed and gravitational components enter when the background fields carry the corresponding couplings. A quantization is a trivialization of the complete local anomaly class, compatible with restriction and factorization [25].

Chapter 110

Pushforward to a curve

Let $p : X \to C$ be a holomorphic map to a complex curve. For a factorization algebra $\mathbb{F}$ on X, define

$$(p_*\mathbb{F})(U) = \mathbb{F}(p^{-1}U).$$

Theorem 110.1 (Factorization pushforward)**.** *The assignment $p_*\mathbb{F}$ is a factorization algebra on C. Suppose in addition that p is proper and locally trivial in a topology compatible with the factorization theory, that $\mathbb{F}$ is locally constant in the fiber directions, and that product-chart excision and base change hold. Then on every trivializing open $U \subset C$ with fiber F_p there is a natural equivalence*

$$(p_*\mathbb{F})(U) \simeq \int_{F_p} \mathbb{F}|_{F_p \times U},$$

compatible on overlaps. If the resulting object is holomorphic and regular holonomic along C, Riemann–Hilbert algebraization gives a factorization D-module on C.

Proof. Disjoint opens in C have disjoint inverse images, so the factorization products push forward. Weiss descent is inherited from inverse images of Weiss covers. On a product chart, local constancy in the fiber and factorization excision identify the value with fiberwise factorization homology. The base-change hypothesis glues these local equivalences. Regular-holonomic Riemann–Hilbert transports the holomorphic factorization object to the algebraic D-module setting. □

Thus a curve shadow is attached to an actual map and an actual field theory. Different fibrations and boundary conditions produce different ordered chiral algebras from the same categorical source.

Chapter III

The derived centre and the physical bulk

For an ordered boundary algebra A, define

$$Z_{\mathrm{ord}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A).$$

It is a canonical E_2 algebra. A bulk theory acting on the boundary gives

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}\,|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A).$$

Definition III.1 (Open–closed completeness)**.** A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

The definition turns the phrase ‘‘bulk equals centre’’ into a comparison theorem. The centre exists algebraically for every ordered boundary algebra; the physical bulk is identified with it by the open–closed map.

Book XX

Hall and BPS algebras

Chapter 112

Extension correspondences

Let $\mathcal{M} = \coprod_{\gamma \in \Gamma} \mathcal{M}_\gamma$ be a moduli stack in an exact or derived category. Let $\mathcal{M}^{\mathrm{ext}}_{\alpha,\beta}$ classify extensions

$$0 \longrightarrow E_\beta \longrightarrow E \longrightarrow E_\alpha \longrightarrow 0.$$

The correspondence is

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}^{\mathrm{ext}}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Theorem 112.1 (Hall associativity)**.** *Assume a coefficient theory in which p^* and q_* are defined and satisfy base change for the two-step filtration stack. Then*

$$a * b = q_* p^* (a \boxtimes b)$$

defines an associative product.

Proof. The two parenthesizations are the two projections from the stack of filtrations $0 \subset E_1 \subset E_2 \subset E_3$. Base change identifies both composites with pullback to this stack followed by pushforward to E_3. □

For critical CoHAs, the coefficient theory includes vanishing cycles and a coherent orientation. For symplectic CoHAs, it includes the selected equivariant Borel–Moore theory. Completion is chosen so that each output degree receives a convergent sum.

Chapter 113

Orientation data and the threefold Hall product

For compact Calabi–Yau threefolds, natural orientation data on moduli of coherent sheaves and perfect complexes is constructed by Joyce and Upmeier [57]. A general cohomological Hall algebra for a 3-Calabi–Yau category uses strong orientation data, which carries the coherent multiplicativity required by the attractor and extension correspondences [63].

Theorem 113.1 (Threefold Hall algebra from strong orientation)**.** *Let C be a 3-Calabi–Yau category satisfying the geometric hypotheses of Kinjo–Park–Safronov, and assume strong orientation data. Then its cohomological Donaldson–Thomas complexes carry an associative cohomological Hall product. The product is the parabolic-induction correspondence supplied by the attractor Lagrangian correspondence.*

Proof. Kinjo–Park–Safronov prove the functoriality of the Donaldson–Thomas perverse sheaf under the attractor Lagrangian correspondence and use strong orientation to define the parabolic-induction morphism. Their base-change and associativity theorem gives the CoHA [63]. □

Definition 113.2 (Reduced $K3 \times E$ Hall datum)**.** A reduced $K3 \times E$ Hall datum is a strong-oriented threefold Hall datum together with descent along the elliptic translation action, a reduced deformation theory, and a separated completion in physical charge. A full quantum-group datum additionally includes a coproduct, Cartan extension, negative half, and nondegenerate continuous Hopf pairing.

Chapter 114

The geometric E-equivariant Hall source

Let S be a projective K3 surface, let E be an elliptic curve, and put

$$X = S \times E.$$

Write $\mathrm{Coh}_{\le 1}(X)$ for coherent sheaves whose support has dimension at most one. This is an extension-closed abelian category. Its numerical charge group is denoted Γ_X, and the degree of the projected one-cycle on E is the additive function

$$m : \Gamma_X \longrightarrow \mathbb{Z}_{\ge 0}.$$

For $\gamma \in \Gamma_X$, let $\mathcal{M}_\gamma$ be the moduli stack of objects of charge γ. Translation on the elliptic factor gives an action of E on every $\mathcal{M}_\gamma$ and on every extension stack.

Oberdieck and Shen construct the E-equivariant Grothendieck ring of stacks and its Hall convolution. In the present notation the completed algebra is

$$\mathcal{H}_X^{E,\mathrm{mot}} = \widehat{\bigoplus}_{\gamma \in \Gamma_X} K_0^E(\mathrm{St}/\mathcal{M}_\gamma),$$

where the completion is taken in effective curve class and Euler characteristic.

Theorem 114.1 (Geometric reduced Hall algebra)**.** *The extension correspondence in* $\mathrm{Coh}_{\le 1}(X)$ *defines an associative E-equivariant motivic Hall product on* $\mathcal{H}_X^{E,\mathrm{mot}}$*. The Oberdieck–Shen reduced integration morphism*

$$I_X^E : \mathcal{H}_X^{E,\mathrm{mot}} \longrightarrow \widehat{\mathbb{Q}[\Gamma_X]}[\epsilon]/(\epsilon^2)$$

is multiplicative on the regular subalgebra on which their integration theorem is defined. The coefficient of ϵ is the reduced contribution of the positive-dimensional E-orbits.

Proof. The stack of short exact sequences is E-equivariant because translation preserves exactness and numerical charge. The stack of two-step filtrations identifies the two composites of three Hall products, so equivariant base change proves associativity. The integration statement is the E-equivariant motivic Hall theorem of Oberdieck–Shen [82]. Their stabilizer stratification separates fixed contributions from moving elliptic orbits and packages the latter in the square-zero parameter ϵ. □

Chapter 115

Local and wrapped geometry inside the Hall algebra

Define the completed degree-zero and positive-degree pieces

$$\mathcal{H}_X^{\mathrm{loc}} = \widehat{\bigoplus}_{m(\gamma)=0} K_0^E(\mathrm{St}/\mathcal{M}_\gamma), \qquad \mathcal{H}_X^{\mathrm{wr}} = \widehat{\bigoplus}_{m(\gamma)>0} K_0^E(\mathrm{St}/\mathcal{M}_\gamma).$$

A pure one-dimensional support of degree zero projects to a finite subset of E. A support of positive degree projects onto E. The first sector is point-local on the elliptic curve; the second is wrapped.

Theorem 115.1 (Geometric local–wrapped decomposition)**.** *Hall multiplication respects elliptic degree:*

$$\mathcal{H}_{m_1} * \mathcal{H}_{m_2} \subseteq \mathcal{H}_{m_1+m_2}.$$

Consequently $\mathcal{H}_X^{\mathrm{loc}}$ is a subalgebra, $\mathcal{H}_X^{\mathrm{wr}}$ is a closed two-sided ideal, and

$$\mathcal{H}_X^{E,\mathrm{mot}} \cong \mathcal{H}_X^{\mathrm{loc}} \ltimes \mathcal{H}_X^{\mathrm{wr}}$$

as a complete vector space with multiplication determined by the Hall correspondences. The reduced integration morphism preserves this decomposition.

Proof. The cycle of the middle term of a short exact sequence is the sum of the cycles of its subobject and quotient. Projection to E therefore adds degrees. Degree zero is closed under addition, and positive degree absorbs addition by a nonnegative degree. The same grading is present on the target group algebra of I_X^E. □

This theorem supplies the geometric version of the local–wrapped splitting constructed from the Borcherds root algebra in the companion volume. The two constructions have the same additive elliptic degree and become comparable after a primitive root framing is chosen.

Chapter 116

Cohomological refinement and the Gram character

A cohomological refinement uses the Donaldson–Thomas sheaf on the derived moduli stack. The datum required for the convolution is the strong orientation coherence of Kinjo–Park–Safronov. A strong orientation enhancement of the natural Joyce–Upmeier orientation therefore produces

$$\mathcal{H}_X^{E,\mathrm{coh}} = \widehat{\bigoplus}_{\gamma\in\Gamma_X} H_*^{\mathrm{BM},E}(\mathcal{M}_\gamma, \Phi_\gamma),$$

with its cohomological Hall product.

Theorem 116.1 (Strong-oriented cohomological lift)**.** *For every strong orientation enhancement on the E-equivariant derived moduli problem, the attractor correspondence constructs an associative cohomological Hall algebra $\mathcal{H}_X^{E,\mathrm{coh}}$. Elliptic degree again gives a subalgebra in degree zero and a closed ideal in positive degree. Forgetting the cohomological coefficient system recovers the corresponding motivic Hall correspondences.*

Proof. The parabolic-induction theorem of Kinjo–Park–Safronov gives the cohomological product from the strong-oriented attractor Lagrangian correspondence [63]. Numerical charge and elliptic degree are constant on connected components of every correspondence, so the proof of the preceding decomposition applies verbatim. □

For the dyonic charge lattice

$$\Gamma_{\mathrm{phys}} = \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z}),$$

write

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right).$$

To compare sheaf charges with this lattice, choose an additive map

$$c : \Gamma_X \longrightarrow \Gamma_{\mathrm{phys}}.$$

Put $B_\Pi(a,b) = \Pi(a+b) - \Pi(a) - \Pi(b)$ and give $\widetilde{\Gamma}_{\mathrm{phys}} = \Gamma_{\mathrm{phys}} \times \mathbb{Z}^3$ the group law

$$(a,u)(b,v) = (a+b, u+v+B_\Pi(a,b)).$$

Then $j_\Pi(a) = (a, \Pi(a))$ is a homomorphism. Assume $j_\Pi c$ preserves the selected charge completions. On the regular subalgebra $\mathcal{H}_{\mathrm{reg}}$ of the reduced integration theorem, define

$$I^E_{X,\Pi,c} = (j_\Pi c)_* I^E_X : \mathcal{H}_{\mathrm{reg}} \longrightarrow \widehat{\mathbb{Q}[\widetilde{\Gamma}_{\mathrm{phys}}]}[\epsilon]/(\epsilon^2).$$

Write $I^E_{X,\Pi}$ for this map when c has been fixed.

Proposition 116.2 (Polarized reduced character). *The map $I^{E}_{X,\Pi,c}$ is multiplicative. Its coefficients are the reduced invariants supplied by I^{E}_{X}, regraded by $j_{\Pi}c$. If $g \in 1 + F^{1}H$ is group-like in a complete filtered Hall bialgebra and the logarithm converges, then* $\log g$ *is primitive.*

Proof. Expanding $\Pi(a+b+d)$ gives

$$B_{\Pi}(a,b) + B_{\Pi}(a+b,d) = B_{\Pi}(b,d) + B_{\Pi}(a,b+d).$$

This proves associativity of the group law. The definition gives $j_{\Pi}(a+b) = j_{\Pi}(a)\, j_{\Pi}(b)$. Composing its group-algebra map with reduced integration proves multiplicativity on $\mathcal{H}_{\mathrm{reg}}$. For the last assertion, the two factors in $\Delta g = (g \mathbin{\widehat{\otimes}} 1)(1 \mathbin{\widehat{\otimes}} g)$ commute. Taking the convergent logarithm gives $\Delta \log g = \log g \mathbin{\widehat{\otimes}} 1 + 1 \mathbin{\widehat{\otimes}} \log g$. □

The primitive reduced Gromov–Witten theorem supplies the separate character $-\chi_{10}^{-1}$ in classes primitive in the K3 factor [81]. Comparing this character with the Hall character requires the same charge map, sector, and normalization. A primitive root framing must also preserve the extension bracket and parity of the chosen BPS states.

Chapter 117

BPS primitives

A connected graded bialgebra has primitive Lie algebra

$$\operatorname{Prim}(H) = \{x : \Delta x = x \otimes 1 + 1 \otimes x\}.$$

In cohomological Donaldson–Thomas theory, the BPS sheaf isolates the primitive part of the Hall algebra.

Theorem 117.1 (Two-Calabi–Yau BPS algebra)**.** *For the geometric two-Calabi–Yau categories covered by the BPS-algebra theorem, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra. This scope includes semistable sheaves on K3 and other symplectic surfaces under the stated good-moduli hypotheses.*

Proof. The theorem is the Davison–Hennecart–Schlegel Mejia construction [27]. The simple BPS sheaves give the simple-root spaces, extension correspondences give the bracket, and the integrality/PBW theorem identifies the Hall algebra with the positive enveloping algebra. □

This result supplies a global K3 generalized Kac–Moody algebra at the two-Calabi–Yau Hall level. Yangian structures arise from additional equivariant correspondences and coproducts in the surface and quiver models described below.

Chapter 118

The integral Hall–Borcherds amalgam

Let $\mathfrak{p}_X^+$ be a complete charge-graded BPS Lie superalgebra supplied by a geometric Hall source, and let $\mathfrak{n}_{\Delta_s,+}$ be the positive Igusa Borcherds Lie superalgebra. The polarized charge extension places both gradings over one abelian Cartan algebra $\mathfrak{h}_\Pi$. Form the two complete Borels

$$\mathfrak{b}_X = \mathfrak{h}_\Pi \ltimes \mathfrak{p}_X^+, \qquad \mathfrak{b}_\Delta = \mathfrak{h}_\Pi \ltimes \mathfrak{n}_{\Delta_s,+}.$$

The Cartan acts through the physical charge on the first Borel and through Gram degree on the second.

Definition 118.1 (Integral protected Lie algebra)**.** The integral Hall–Borcherds algebra is the completed amalgamated coproduct

$$\mathfrak{b}_{X\boxtimes\Delta} = \mathfrak{b}_X \widehat{\amalg}_{\mathfrak{h}_\Pi} \mathfrak{b}_\Delta$$

in complete charge-filtered dg Lie superalgebras.

Theorem 118.2 (Hall–Borcherds amalgamation)**.** *The completed amalgam exists and represents pairs of continuous Lie morphisms from $\mathfrak{b}_X$ and $\mathfrak{b}_\Delta$ that agree on $\mathfrak{h}_\Pi$. The Cartan projections of the two Borels induce canonical retractions*

$$r_X : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_X, \qquad r_\Delta : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_\Delta.$$

Its completed enveloping algebra is the amalgamated free product

$$\widehat{U}(\mathfrak{b}_{X\boxtimes\Delta}) \cong \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The geometric Hall operations survive under r_X, while the Igusa bracket, PBW character, and automorphic determinant survive under r_Δ.

Proof. Construct the ordinary Lie coproduct from the free product of the two Borels and divide by the ideal identifying their two Cartan copies. Complete by total positive charge and proper root height. Every fixed filtered degree uses finitely many Lie words, so the completion is separated and has the stated mapping property. The projection $\mathfrak{b}_X \to \mathfrak{h}_\Pi \to \mathfrak{b}_\Delta$ together with the identity on $\mathfrak{b}_\Delta$ gives r_Δ by universality; the second retraction is symmetric. The enveloping functor preserves coproducts of Lie algebras, and PBW identifies the completed algebraic filtration with the displayed completed free product. □

Definition 118.3 (Comparison ideal)**.** A primitive comparison $\varphi : \mathfrak{n}_{\Delta_s,+} \to \mathfrak{p}_X^+$ preserving Cartan degree defines the closed ideal

$$\mathcal{J}_\varphi = \overline{\langle i_\Delta(x) - i_X(\varphi(x)) \mid x \in \mathfrak{n}_{\Delta_s,+} \rangle} \subset \mathfrak{b}_{X\boxtimes\Delta}.$$

The diagonal realization is

$$\mathfrak{b}_{X\simeq\Delta,\varphi} = \mathfrak{b}_{X\boxtimes\Delta} / \mathcal{J}_\varphi.$$

Proposition 118.4 (Graph quotient of a primitive comparison)**.** *Let $\varphi : \mathfrak{b}_\Delta \to \mathfrak{b}_X$ be a continuous Lie morphism restricting to the identity on $\mathfrak{h}_\Pi$. Use a completion in which the amalgam and closed quotient have their stated universal properties. Then*

$$\mathfrak{b}_{X\boxtimes\Delta}/\mathcal{J}_\varphi \cong \mathfrak{b}_X.$$

The two factor maps become id *and φ. If φ is an isomorphism, the quotient is also isomorphic to $\mathfrak{b}_\Delta$.*

Proof. The maps id and φ agree on $\mathfrak{h}_\Pi$. They give $q_\varphi : \mathfrak{b}_{X\boxtimes\Delta} \to \mathfrak{b}_X$, which kills the generators of $\mathcal{J}_\varphi$. The inclusion of $\mathfrak{b}_X$ gives a map s in the other direction. The composite $q_\varphi s$ is the identity. The other composite is the identity on the geometric factor. On the Borcherds factor, the defining relation identifies x with $s\varphi(x)$, so it is also the identity there. The two factor images topologically generate the quotient. Continuity proves that the maps are inverse on the completion. □

The graph quotient implements a specified primitive comparison. Constructing that comparison requires geometric simple states whose extension brackets satisfy the Borcherds relations. The quotient formula alone produces neither those states nor their orientation data.

Chapter 119

Coproducts, pairings, and doubles

A positive Hall algebra becomes a quantum group after additional structure is supplied. Let H^+ be a complete graded bialgebra, let H^0 be a Cartan algebra, and let H^- be the opposite or dual half. Let

$$\langle -, - \rangle : H^+ \mathbin{\widehat{\otimes}} H^- \longrightarrow \Bbbk$$

be a nondegenerate Hopf pairing.

Definition 119.1 (Completed Hall double)**.** The completed Drinfeld double is the topological algebra

$$\widehat{D}(H^+) = H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+$$

with internal products in the three factors and cross-relations determined by the Hopf pairing. Its representation category is monoidal by the coproduct and braided by the universal R-matrix whenever the canonical tensor converges.

Theorem 119.2 (Double construction)**.** *Let H^+ and H^- be paired complete graded bialgebras, let H^0 be a Cartan bialgebra acting by the stated weight grading, and let*

$$\langle -, - \rangle : H^+ \mathbin{\widehat{\otimes}} H^- \longrightarrow \Bbbk$$

be a continuous nondegenerate Hopf pairing. Assume the triangular multiplication map from $H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+$ is separated and complete. Then the pairing cross-relations determine a unique completed double with this triangular decomposition. On every continuous module category on which the canonical tensors converge, the completed canonical element gives the universal R-matrix.

Proof. The Hopf pairing defines the cross-relations between the paired halves and makes the three canonical embeddings bialgebra morphisms. The Hopf identities identify the two reductions of every triple product, proving associativity. Separated triangular decomposition gives uniqueness. Finite-degree canonical tensors form a compatible inverse system; convergence on the selected module category yields the quasitriangular element [33, 37]. □

Theorem 119.3 (Quantum Hall–Borcherds amalgam)**.** *Let H_X and H_Δ be complete Hopf algebras containing the same complete Cartan Hopf algebra H^0, with continuous Hopf retractions onto H^0. Their completed amalgamated free product*

$$H_{X \boxtimes \Delta} = H_X \mathbin{\widehat{*}_{H^0}} H_\Delta$$

has a unique complete Hopf structure extending the two coproducts, counits, and antipodes. It represents pairs of continuous Hopf morphisms agreeing on H^0 and has canonical Hopf retractions to both factors. A mixed universal R-matrix is equivalent to a continuous cross-pairing satisfying the two hexagon identities; the intrinsic R-matrices of the two retracts remain defined independently.

Proof. The algebraic amalgamated free product exists because complete filtered algebras are obtained degreewise from finite quotients. The two coproducts agree on H^0, so the universal property extends them to a coproduct on the free product. Coassociativity is checked on the two generating Hopf subalgebras. The same argument extends the counit and antipode. The Hopf retractions are induced by the factor retractions onto H^0. A quasitriangular structure extending both factors requires the additional commutation data between the two generating halves; writing this data as a cross-pairing turns the quasitriangular and Yang–Baxter equations into the two hexagon identities. □

The deconcatenation coproduct on a bar complex and the representation-theoretic coproduct on a quantum group occupy different structures. A geometric comparison between them is a theorem about a selected model.

Chapter 120

Ordered chiral bialgebras

A tensor product of representations requires compatible actions on threefold products and the unit. Retain the factorization coefficient category $(\mathrm{Fact}_D(C), \otimes^!)$ on the smooth complex curve C. Fix a category $\widehat{\mathcal{V}}_C$ of complete filtered objects with specified presentations $F \simeq \varprojlim_r F_r$, where $F_r \in \mathrm{Fact}_D(C)$. Use derived limits in the fixed Ran enhancement and define

$$F \mathbin{\widehat{\otimes}^!} G = \varprojlim_{r,s} (F_r \otimes^! G_s).$$

Assume these limits remain factorizing and give $\widehat{\mathcal{V}}_C$ a symmetric monoidal structure with unit $\mathbf{1}$. Its associativity, symmetry, and unit equivalences must preserve the presentations and the Ran structure. Morphisms are continuous maps compatible with these presentations. These are hypotheses on the chosen completion.

Definition 120.1 (Ordered chiral bialgebra)**.** An ordered chiral bialgebra is a counital E_1-coalgebra object in complete ordered chiral algebras:

$$A \in \mathrm{CoAlg}_{\mathsf{E}_1}\Big(\mathrm{Alg}_{\mathsf{E}_1}(\widehat{\mathcal{V}}_C, \mathbin{\widehat{\otimes}^!})\Big).$$

It has continuous multiplication m and unit η, and continuous morphisms of unital ordered chiral algebras

$$\Delta : A \longrightarrow A \mathbin{\widehat{\otimes}^!} A, \qquad \epsilon : A \longrightarrow \mathbf{1}.$$

The counit is the chosen augmentation. Write τ for the coefficient symmetry, including Koszul signs. Suppressing ambient associativity and unit equivalences, compatibility reads

$$\Delta m = (m \mathbin{\widehat{\otimes}^!} m)(\mathrm{id} \mathbin{\widehat{\otimes}^!} \tau \mathbin{\widehat{\otimes}^!} \mathrm{id})(\Delta \mathbin{\widehat{\otimes}^!} \Delta),$$
$$\Delta\eta = \eta \mathbin{\widehat{\otimes}^!} \eta, \qquad \epsilon m = \epsilon \mathbin{\widehat{\otimes}^!} \epsilon, \qquad \epsilon\eta = \mathrm{id}_{\mathbf{1}}.$$

Coassociativity and the counit identities read

$$(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)\Delta,$$
$$(\epsilon \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = \mathrm{id}_A = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \epsilon)\Delta.$$

For a homotopy-coherent presentation, these identities include the higher coherences of the displayed coalgebra object. Every operation and coherence preserves diagonal transitions and disjoint-locus factorization maps. An antipode is a continuous convolution inverse $S : A \to A$ of id_A:

$$m(S \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = \eta\epsilon = m(\mathrm{id} \mathbin{\widehat{\otimes}^!} S)\Delta.$$

A bialgebra with an antipode is a Hopf object in this coefficient category.

A continuous left A-module is an object $M \in \widehat{\mathcal{V}}_C$ with a unital E_1 action $\rho_M : A \mathbin{\widehat{\otimes}^!} M \to M$. Its action and coherences are morphisms in these factorization coefficients. Write $\mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$ for their category.

Theorem 120.2 (Representation fusion)**.** *An ordered chiral bialgebra makes* $\mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$ *monoidal. Its product is* $M \mathbin{\widehat{\otimes}^!} N$ *with action*

$$\begin{aligned} A \mathbin{\widehat{\otimes}^!} (M \mathbin{\widehat{\otimes}^!} N) &\xrightarrow{\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id}} (A \mathbin{\widehat{\otimes}^!} A) \mathbin{\widehat{\otimes}^!} (M \mathbin{\widehat{\otimes}^!} N) \\ &\xrightarrow{\mathrm{id} \mathbin{\widehat{\otimes}^!} \tau_{A,M} \mathbin{\widehat{\otimes}^!} \mathrm{id}} (A \mathbin{\widehat{\otimes}^!} M) \mathbin{\widehat{\otimes}^!} (A \mathbin{\widehat{\otimes}^!} N) \\ &\xrightarrow{\rho_M \mathbin{\widehat{\otimes}^!} \rho_N} M \mathbin{\widehat{\otimes}^!} N. \end{aligned}$$

The unit is $\mathbf{1}$ *with action induced by* ϵ*. The ambient associator and unit equivalences are module equivalences.*

Suppose a continuous universal R*-matrix is also given. It is an invertible morphism* $R : \mathbf{1} \to A \mathbin{\widehat{\otimes}^!} A$ *for the tensor-product algebra, satisfying*

$$R\Delta(a) = \Delta^{\mathrm{op}}(a) R, \qquad \Delta^{\mathrm{op}} = \tau_{A,A} \Delta,$$
$$(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})(R) = R_{13} R_{23}, \qquad (\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)(R) = R_{13} R_{12},$$
$$(\epsilon \mathbin{\widehat{\otimes}^!} \mathrm{id})(R) = \eta = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \epsilon)(R).$$

The first equation is an identity of morphisms from A *to its completed tensor square. Leg notation uses the unit and symmetry of* $\widehat{\mathcal{V}}_C$*. In a homotopy-coherent presentation, require compatible lifts of these identities. Then* $c_{M,N} = \tau_{M,N} \circ R_{M,N}$ *is a braiding, where* $R_{M,N}$ *acts through the external tensor-product module action.*

Proof. The symmetry combines the independent module actions into an $A \mathbin{\widehat{\otimes}^!} A$-action. Restriction along the unital algebra map Δ gives the displayed action. For three modules, the two actions use $(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta$ and $(\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)\Delta$. Coassociativity makes the ambient associator A-linear. The counit identities make the ambient unit equivalences A-linear. The coalgebra coherences give the pentagon and triangle coherences. The completion hypotheses keep these maps continuous and factorizing. Quasitriangularity makes $c_{M,N}$ an intertwiner and invertibility makes it an equivalence. The two coproduct identities for R give the hexagons. Its counit identities give compatibility with the unit. □

The tensor in this construction is the completed transverse tensor $\otimes^!$ on factorization coefficients. A chiral-convolution or Hall coproduct requires its own construction and a comparison in the specified category. A spectral version additionally requires a difference coordinate, a Laurent-series completion, and compatible domains for iterated expansions. Its identities must hold in those domains before defining module actions.

Example 120.3 (A counital algebra map without coassociativity)**.** In ordinary complex vector spaces, let $A = \mathbb{C}[x]$ and put

$$\Delta(x) = x \otimes 1 + 1 \otimes x + x \otimes x^2, \qquad \epsilon(x) = 0.$$

Polynomial substitution extends these assignments to unital algebra maps. Writing $F(u, v) = u + v + uv^2$, the equations $F(0, v) = v$ and $F(u, 0) = u$ prove both counit identities. But

$$\begin{aligned} F(F(u,v),w) - F(u,F(v,w)) &= -uv^2w^4 - uv^2w^2 - 2uvw^3 - 2uvw \\ &= -uvw(1 + w^2)(vw + 2) \neq 0. \end{aligned}$$

On three regular modules, the two actions differ already on $1 \otimes 1 \otimes 1$. Thus the canonical tensor associator is not A-linear. This ordinary example shows why multiplicativity and the counit identities must be supplemented by coassociativity. It makes no assertion about a geometric realization in factorization coefficients.

Book XXI

Exact model cases

Chapter 121

The A_2 Hall algebra

For the quiver $1 \to 2$ over $\mathbb{F}_q$, the twisted Hall algebra has generators E_1, E_2 and $v^2 = q$. Counting subrepresentations gives

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0.$$

For a representation of dimension vector $(2,1)$, the arrow has rank zero or one. The relevant flag counts are

rank	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

After the Hall twists v^{-1}, 1, v, both rows give zero. This is the elementary geometric origin of the quantum Serre relation [45, 88].

Chapter 122

The tripled Jordan quiver and $\mathbb{C}^3$

Let Q have one vertex and loops x, y, z, with potential

$$W = \operatorname{Tr} x[y, z].$$

The Jacobi algebra is $\mathbb{C}[x, y, z]$. The critical CoHA uses the vanishing cycles of $\operatorname{Tr} W$ on the representation spaces. Its positive currents are generated by the one-vertex dimension sectors.

Theorem 122.1 (Affine-Yangian positive half)**.** *After the standard equivariant localization and completion, the critical CoHA of the tripled Jordan quiver is identified with the positive half of the affine Yangian of* $\mathfrak{gl}_1$*. The independently defined affine Yangian carries its standard Cartan and negative current halves; the positive-half comparison embeds the CoHA into that triangular presentation.*

Proof. The CoHA product becomes the shuffle product under localization. Dimensional reduction identifies the tripled-quiver critical theory with the equivariant surface CoHA, and Davison's affine-BPS theorem identifies its completed positive algebra with the strictly positive affine Yangian. The shuffle kernel satisfies the affine-Yangian current relations with equivariant parameters ϵ_1, ϵ_2, ϵ_3 and Calabi–Yau relation $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$. PBW and current comparison identify the positive algebra. The independently presented full affine Yangian is the triangular algebra generated by its negative currents, Cartan currents, and this positive half [26, 28, 89, 90]. □

The vacuum representation produces a $W_{1+\infty}$-type vertex algebra through an evaluation homomorphism. The CoHA, its double, and the vacuum algebra occupy successive levels: positive algebra, paired quantum group, and distinguished representation.

Chapter 123

Surface CoHAs and ADE Yangians

Let $S_\Gamma \to \mathbb{C}^2/\Gamma$ be the minimal resolution for a finite subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$. The exceptional curves have ADE intersection matrix. Moduli of sheaves supported on these curves carry convolution correspondences.

Theorem 123.1 (Curve-supported surface CoHA)**.** *The completed positive CoHA of one-dimensional sheaves supported on the exceptional curves is identified with a nonstandard positive half of the affine Yangian of the corresponding ADE Lie algebra.*

Proof. The theorem is the curve-supported surface construction of Diaconescu–Porta–Sala–Schiffmann–Vasserot [30]. The simple curve sheaves give the simple generators, Hecke correspondences give the current operators, and the completed PBW comparison gives the affine Yangian presentation. □

This theorem is a local K3 model whenever an ADE configuration occurs on a K3 surface. The global K3 surface adds compact curve classes and monodromy beyond the local root lattice.

Chapter 124

Coulomb branches and shifted Yangians

For a three-dimensional $\mathcal{N} = 4$ gauge theory, the BFN Coulomb branch is defined by equivariant Borel–Moore homology of an affine-Grassmannian correspondence. Its convolution algebra quantizes the Coulomb branch [16]. In quiver gauge theories, shifted Yangians and their truncations provide presentations of these quantizations. Boundary conditions select the shift and truncation data.

The BFN construction supplies a second geometric source of Yangians. Its grading and coproduct arise from loop rotation and convolution, while the surface CoHA source arises from sheaf extensions. A comparison between the two is a model-specific equivalence.

Book XXII

K3 symmetries

Chapter 125

The Mukai lattice

For a K3 surface S, the Mukai lattice is

$$\widetilde{H}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z})$$

with pairing

$$((r,c,s),(r',c',s')) = c\cdot c' - rs' - r's.$$

It is even unimodular of signature $(4,20)$ and rank 24.

Chapter 126

The Nakajima Heisenberg algebra

Let

$$\mathbb{H}_S = \bigoplus_{n \geq 0} H^*(S^{[n]}).$$

For $m \in \mathbb{Z} \setminus \{0\}$ and $\alpha \in H^*(S)$, the Nakajima correspondence gives an operator $p_m(\alpha)$.

Theorem 126.1 (K3 Heisenberg action)**.** *The operators satisfy*

$$[p_m(\alpha), p_n(\beta)] = m\, \delta_{m+n,0} \int_S \alpha \cup \beta \ \mathrm{id}\,.$$

They generate the rank-twenty-four Heisenberg algebra acting irreducibly on $\mathbb{H}_S$ with vacuum $1 \in H^*(S^{[0]})$.

Proof. The commutator is the excess intersection of the two incidence correspondences. Opposite creation and annihilation correspondences meet along the diagonal with multiplicity m and intersection pairing $\int_S \alpha\beta$. Correspondences of the same sign commute. The vacuum construction gives the Fock representation [80]. □

This action is global, compact, and abelian at the Lie level. It is an exact K3 symmetry and supplies the rank-twenty-four Fock sector of any broader K3 algebra.

Chapter 127

The orthogonal LLV algebra

For a compact hyperkähler manifold, every Lefschetz class gives an $\mathfrak{sl}_2$ triple on cohomology. The Lie algebra generated by all such triples is the Looijenga–Lunts–Verbitsky algebra.

Theorem 127.1 (Orthogonal LLV symmetries)**.** *For a K3 surface S, the LLV algebra acting on $H^*(S)$ is*

$$\mathfrak{g}_{\mathrm{LLV}}(S) \cong \mathfrak{so}\big(\widetilde{H}(S), (-,-)_{\mathrm{Muk}}\big) \cong \mathfrak{so}(4, 20).$$

For an irreducible hyperkähler manifold M of $K3^{[n]}$ type with $n \geq 2$, one has $b_2(M) = 23$ and

$$\mathfrak{g}_{\mathrm{LLV}}(M) \cong \mathfrak{so}(4, 21).$$

Moduli spaces of stable objects on S carry their own LLV actions; Mukai and monodromy correspondences relate these actions to the K3 lattice.

Proof. The Looijenga–Lunts–Verbitsky theorem identifies the Lie algebra generated by all Lefschetz triples with the orthogonal algebra of the Mukai extension $H^2(M) \oplus U$. For S, the extension has signature $(4, 20)$. For $K3^{[n]}$ type with $n \geq 2$, the Beauville–Bogomolov lattice has signature $(3, 20)$, so adjoining the hyperbolic plane gives signature $(4, 21)$ [72, 91]. □

The LLV algebra is a classical cohomological symmetry. A Yangian deformation of the full orthogonal action consists of two further structures: a Drinfeld current presentation and a compatible spectral coproduct.

Chapter 128

Local ADE Yangians

An ADE configuration of (-2)-curves on S gives a root sublattice of $H^2(S,\mathbb{Z})$. Sheaves supported on the configuration produce the surface CoHA of the preceding part and hence a completed positive affine ADE Yangian. Nakajima quiver varieties supply compatible Yangian actions on their equivariant cohomology.

Thus K3 geometry contains local nonabelian Yangian sectors indexed by ADE enhancements of the Picard lattice. Picard monodromy transports the ADE root sublattices over the corresponding Noether–Lefschetz loci. A family of sheaf moduli and Hecke correspondences compatible with that monodromy transports the Yangian actions themselves.

Chapter 129

The K3 symmetry trichotomy

The three exact structures are

Structure	Carrier	Operation
Mukai Heisenberg	$\bigoplus_n H^*(S^{[n]})$	creation and annihilation
LLV $\mathfrak{so}(4,20)$	$H^*(S)$; separately $\mathfrak{so}(4,21)$ on $K3^{[n]}$ type	Lefschetz Lie action
Local ADE Yangian	curve-supported equivariant moduli	Hecke and current correspondences

A global K3 quantum algebra is a system that contains these three actions with explicit comparison homomorphisms. The two-Calabi–Yau BPS theorem supplies a global generalized Kac–Moody positive algebra. A full Yangian or double is specified by a compatible coproduct, spectral parameter, negative half, Cartan part, pairing, and completion.

Chapter 130

K3 correspondences on fixed coefficients

Fix rational cohomology and a collection $\mathscr{M}_S$ of smooth projective moduli spaces on a projective K3 surface S. Include the Hilbert schemes $S^{[n]}$ and close the collection under finite products. For $M, N \in \mathscr{M}_S$, define the graded kernel space

$$K(M,N) = H^*(M \times N, \mathbb{Q})[2\dim_{\mathbb{C}} M].$$

A kernel of degree r acts from $H^j(M)$ to $H^{j+r}(N)$ by

$$T_\alpha(v) = (p_N)_*(p_M^* v \cup \alpha).$$

The shift in $K(M,N)$ accounts for integration over M. All pushforwards in this construction are proper.

Proposition 130.1 (Kernel composition)**.** *The product*

$$\beta \circ \alpha = (p_{13})_*(p_{12}^*\alpha \cup p_{23}^*\beta)$$

defines an associative graded category with diagonal identities. The kernel action identifies $K(M,N)$ *with* $\mathrm{Hom}^*(H^*(M), H^*(N))$.

Proof. The Künneth isomorphism and Poincaré duality identify $H^*(M \times N)[2\dim M]$ with $H^*(M)^\vee \otimes H^*(N)$. Under this identification, T_α is the associated linear map. Integration over the middle factor contracts its dual pair, so $T_{\beta\circ\alpha} = T_\beta T_\alpha$. Faithfulness then proves associativity and the diagonal identity. External products give tensor products with their Koszul signs. □

A geometric symmetry is specified by selected kernels in this category. Nakajima incidence kernels connect different Hilbert schemes. Lefschetz multiplication and its adjoint act on each individual cohomology space. Using every cohomology kernel would instead give all linear maps; the selected geometric kernels carry the representation-theoretic content.

Chapter 131

The algebra generated by geometric actions

Put $V_S = \bigoplus_{M \in \mathscr{M}_S} H^*(M, \mathbb{Q})$. Choose a set of homogeneous operators T_λ on V_S, each sending a vector with finite support to a vector with finite support. The Nakajima operators on the Hilbert-scheme summand satisfy this condition: each changes the point number by a fixed integer. Individual Lefschetz operators, extended by zero on other summands, also satisfy it. Let F be the free unital graded algebra on symbols t_λ.

Proposition 131.1 (Geometric action algebra)**.** *The map*

$$\rho : F \longrightarrow \mathrm{End}_{\mathbb{Q}}(V_S), \qquad t_\lambda \longmapsto T_\lambda$$

defines a faithful action of $A_S = F/\ker\rho$. For every algebra B, maps $A_S \to B$ are exactly maps $F \to B$ that kill $\ker\rho$*. Products and commutators of the selected kernels are their convolution products and commutators on the relevant moduli spaces.*

Proof. Finite words in the selected operators preserve finite support, so ρ is defined by composition. The quotient acts faithfully by definition of its kernel. The factorization property is the universal property of an algebra quotient. The convolution assertion follows from Proposition 130.1. □

An equivariant ADE construction uses a separate coefficient field, for example $\mathrm{Frac}\, H^*_T(\mathrm{pt}, \mathbb{Q})$, and its prescribed localization and proper pushforwards. Joining it to A_S requires maps between the actual state spaces that intertwine those operators and their coefficients. A completion of A_S requires a specified topology and continuity of every generator. The indefinite Mukai norm alone supplies neither condition. The desired common K3 quantum algebra must construct these local-to-global actions and their mixed convolution relations.

Chapter 132

Tensor reconstruction and geometric kernels

Let $\mathscr{C}$ be an essentially small rigid linear monoidal category and let $\omega : \mathscr{C} \to \mathrm{Vect}_{\mathrm{fd}}(\mathbb{Q})$ be a strong monoidal functor. A choice of geometric kernels specifies the morphisms of $\mathscr{C}$; it is part of the input. Define

$$H = \left(\bigoplus_{M \in \mathscr{C}} \omega(M)^* \otimes \omega(M) \right) \Big/ \left\langle [\lambda \circ \omega(f) \otimes v]_M - [\lambda \otimes \omega(f)v]_N \right\rangle,$$

where $f : M \to N$, $v \in \omega(M)$, and $\lambda \in \omega(N)^*$.

Proposition 132.1 (Matrix-coefficient coalgebra)**.** *For a basis (e_i) with dual basis (e^i), the formulas*

$$\Delta[\lambda \otimes v]_M = \sum_i [\lambda \otimes e_i]_M \otimes [e^i \otimes v]_M, \qquad \epsilon[\lambda \otimes v]_M = \lambda(v)$$

are basis-independent and make H a coalgebra. Tensor product gives its multiplication, and duals give its antipode. Each $\omega(M)$ is a comodule by

$$v \longmapsto \sum_i e_i \otimes [e^i \otimes v]_M.$$

Every selected kernel is a comodule morphism.

Proof. The identity tensor $\sum_i e_i \otimes e^i$ is basis-independent. Inserting it twice proves coassociativity; evaluation proves both counit identities. The defining relations express naturality under f and are preserved by these formulas. The strong monoidal constraints identify the coefficient of $M \otimes N$ with the product of coefficients of M and N. Their pentagon and unit identities give the bialgebra axioms. The evaluation and coevaluation identities for duals give the antipode identities. The same relations prove that $\omega(f)$ intertwines the coactions. □

The linear dual of H represents linear natural endomorphisms of ω. Tensor-compatible natural automorphisms are the corresponding multiplicative, unit-preserving families, not all elements of this dual. An equivalence with a comodule category requires the additional exactness and reconstruction hypotheses stated in the Tannaka theorem below. The action algebra A_S acts by selected kernels; the coend records symmetries commuting with those kernels.

Book XXIII

The exact $K3 \times E$ protected completion

Chapter 133

The protected input

The K3 elliptic genus is $2\phi_{0,1}$. The Borcherds half determines the weight-five denominator D_5 and its Borcherds Lie superalgebra $\mathfrak{g}_\Delta$. Fix its positive root spaces with their parity and proper height. The positive algebra and its root Fock space are defined from this Lie superalgebra, independently of a compact Hall realization.

For each finite zero-potential quiver window $Q_{\Delta,H}$, form the Ginzburg algebra $\Gamma_{\Delta,H}$. Restriction to full subquivers gives the projections of Proposition 139.1. Write Γ_Δ for this specified inverse system. A non-degenerate continuous Calabi–Yau structure on a completed limit is additional data. An ordinary Hall category $\mathcal{B}_\Delta$ must carry its chosen quiver coefficients and extension correspondences. Its identification with a root presentation requires the corresponding Hall theorem on that precise datum.

Definition 133.1 (Root-datum realizations)**.** A protected root-datum realization records

$$\mathfrak{P}(K3 \times E) = (\Gamma_\Delta, \mathcal{B}_\Delta, \mathcal{H}_{\mathrm{ip}}, \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \widehat{U}_q(\mathfrak{g}_\Delta), \mathbb{A}_Z).$$

Here $\mathcal{H}_{\mathrm{ip}} = \bigoplus_{\alpha>0}(\mathfrak{g}_\Delta)_\alpha$ with its parity. The current-factorization, paired quantum-group, and Fredholm entries retain their own coefficient categories, completions, and defining hypotheses. Maps among these entries are specified realization maps, not consequences of using the same root labels.

The enveloping and Chevalley constructions are functorial under the primitive maps of Theorem 138.1. The root Fock trace is computed from the actual graded root spaces. A Fredholm realization adds a family of operators on a specified domain and the convergence and orientation data identifying its Pfaffian with the denominator. These are the analytic inputs to the square-root comparison below.

Definition 133.2 (Integral protected realization)**.** Let $\mathfrak{p}^+_X$ be the BPS primitive Lie algebra of a selected strong-oriented cohomological lift of the geometric E-equivariant Hall source. The integral protected realization is

$$\mathfrak{P}_{\mathrm{int}}(X, \Delta) = \left(\mathcal{H}^{E,\mathrm{mot}}_X, \mathfrak{P}(K3 \times E), \mathfrak{b}_{X\boxtimes\Delta}, r_X, r_\Delta, I^E_{X,\Pi}\right).$$

At the motivic level the first, second, and character components are defined. A strong orientation enhancement supplies $\mathfrak{p}^+_X$ and hence the amalgamated primitive component.

Proposition 133.3 (Primitive graph comparison)**.** *Suppose the two primitive Borels, their common Cartan action, and their completed amalgam are supplied as above. Their Cartan retractions induce retractions from the amalgam to each factor. For each specified primitive map φ preserving Cartan action, its graph quotient is canonically the geometric Borel.*

Proof. The identity on one Borel and the Cartan projection of the other agree on the common Cartan, so the amalgam gives the retraction. The quotient statement is Proposition 118.4. □

The motivic integration map and primitive comparison are different maps. Strong orientation gives the coefficient coherence of a cohomological Hall construction under its stated geometric hypotheses. Extracting a primitive Lie algebra additionally requires the appropriate BPS or coproduct construction.

Chapter 134

Cyclic open-string operations

Let A be a finite-dimensional, uncurved cyclic A_∞ algebra over $\mathbb{C}$, with a nondegenerate pairing of degree -3. Use the cohomological suspension $s : A \to A[1]$ of degree -1. The suspended operations $b_n : (A[1])^{\otimes n} \to A[1]$ have degree one. Our sign convention is the coderivation convention $b^2 = 0$ on the tensor coalgebra, with the Koszul rule for moving homogeneous inputs. Write ω for the resulting degree--1 symplectic form on $A[1]$. Fix the Hamiltonian convention $\iota_{X_F}\omega = dF$ and $[X_F, X_G] = X_{\{F,G\}}$.

For the universal degree-zero coordinate x on $A[1]$, put

$$S(x) = \sum_{n \geq 1} \frac{1}{n+1} \omega(x, b_n(x^{\otimes n})).$$

Cyclic invariance is read with the suspended Koszul signs. All functions are formal power series at the origin. An equivalent unsuspended convention writes this as $\frac{1}{2}\langle a, m_1 a\rangle + \sum_{n\geq 2} \frac{1}{n+1}\langle a, m_n(a^{\otimes n})\rangle$.

Proposition 134.1 (Cyclic master equation)**.** *The action satisfies* $\{S, S\} = 0$.

Proof. Cyclicity makes the $n+1$ derivatives of the arity-$n+1$ term equal, with the suspended Koszul signs. Thus $X_S(x) = \sum_{n\geq 1} b_n(x^{\otimes n})$. The coderivation identity implies $X_S^2 = 0$. The Hamiltonian identity gives $X_{\{S,S\}} = 2X_S^2 = 0$. Nondegeneracy of ω makes $\{S, S\}$ constant. Uncurvedness removes b_0, so S starts in order two and its bracket has no constant term. The constant is therefore zero. □

A finite-height Ginzburg realization requires such a cyclic model on the selected proper objects before this scalar action is applied. A Hall realization additionally requires a moduli problem with critical coefficient complexes. For an extension $A_0 \to F \to B_0$, put $L(A_0, B_0) = \det R\,\mathrm{Hom}(A_0, B_0)$. Three-Calabi–Yau duality gives

$$R\,\mathrm{Hom}(B_0, A_0) \simeq R\,\mathrm{Hom}(A_0, B_0)^\vee[-3].$$

Applying the determinant functor to the two derived Hom triangles gives

$$\det R\,\mathrm{Hom}(F, F) \cong \det R\,\mathrm{Hom}(A_0, A_0) \otimes \det R\,\mathrm{Hom}(B_0, B_0) \otimes L(A_0, B_0)^{\otimes 2}.$$

Indeed, the four subquotients are the two diagonal Hom complexes and the two cross complexes. Dualization and the odd shift cancel on their determinant lines, so the two cross determinants coincide. Orientation lines require square-root isomorphisms along this formula, compatible on triple flags and with any quotient descent. The cyclic master equation does not select these isomorphisms.

Chapter 135

An algebraic open–closed comparison

A finite-dimensional ordinary Lie algebra $\mathfrak{l}$ gives a precise coefficient model for an enveloping-algebra centre. Let $U = U(\mathfrak{l})$ over $\mathbb{C}$. The adjoint module U^{ad} has action $x \cdot u = xu - ux$. Its Chevalley cochains are

$$C^n(\mathfrak{l}, U^{\mathrm{ad}}) = \mathrm{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{l}, U),$$

with differential

$$\begin{aligned}(df)(x_1, \dots, x_{n+1}) = &\sum_i (-1)^{i+1} [x_i, f(x_1, \widehat{x_i}, \dots, x_{n+1})] \\ &+ \sum_{i<j} (-1)^{i+j} f([x_i, x_j], x_1, \widehat{x_i}, \widehat{x_j}, \dots, x_{n+1}).\end{aligned}$$

The second line places the bracket first and retains the order of the remaining arguments.

Theorem 135.1 (Hochschild–Chevalley comparison)**.** *Restriction followed by antisymmetrization defines a quasi-isomorphism*

$$C^{\bullet}_{\mathrm{HH}}(U, U) \longrightarrow C^{\bullet}(\mathfrak{l}, U^{\mathrm{ad}}), \qquad \phi \longmapsto \left[(x_1, \dots, x_n) \mapsto \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \phi(x_{\sigma(1)}, \dots, x_{\sigma(n)}) \right].$$

This is a comparison of ordinary cochain complexes.

Proof. The Chevalley bimodule resolution has terms $K_n = U \otimes \Lambda^n \mathfrak{l} \otimes U$. Its differential has the left-minus-right multiplication terms and the bracket terms with the signs dual to the displayed differential. Filter K_n by total PBW degree, counting each exterior generator as degree one. The multiplication terms preserve this degree and bracket terms lower it. The associated graded is the Koszul resolution of the diagonal in $\mathrm{Sym}(\mathfrak{l}) \otimes \mathrm{Sym}(\mathfrak{l})$, with regular sequence $x_i \otimes 1 - 1 \otimes x_i$. That Koszul complex is exact away from degree zero. The filtration is bounded below and exhaustive, and each element has finite filtration degree, so lifting a leading term proves exactness of $K_\bullet$ by induction. Its augmentation has image U. Thus $K_\bullet$ is a free bimodule resolution.

Antisymmetrization gives a map from $K_\bullet$ to the ordinary bar resolution, lifting the identity of U. Adjacent products in its bar differential pair to commutators; $xy - yx = [x, y]$ gives exactly the bracket terms of $K_\bullet$. The outer products give its left-minus-right terms. A comparison between projective resolutions lifting the identity is a homotopy equivalence. Applying $\mathrm{Hom}_{U \otimes U^{\mathrm{op}}}(-, U)$ gives the stated formula and quasi-isomorphism. □

This is the classical Cartan–Eilenberg comparison. For a finite Lie superalgebra, the same construction uses the super-exterior powers and permutation Koszul signs. Odd generators produce symmetric exterior directions, so the resolution can be unbounded; its algebraic direct sums and cochain products remain part of the convention. The PBW and super-Koszul proof applies with those conventions.

A physical mixed holomorphic–topological realization requires a map from its independently defined bulk observables to this cochain object. The theorem specifies the adjoint coefficients and the actual algebraic comparison. An E_2 refinement, a current-factorization realization, and compatibility with root-height transitions require maps preserving their additional operations. A quotient of Lie algebras alone does not provide a covariant map of Hochschild centres.

Chapter 136

The exact scalar and first-order observables

Set

$$\mathrm{Den}_\Delta(Z) = D_5(2Z), \qquad \jmath(Z) = Z/2, \qquad \jmath^*\,\mathrm{Den}_\Delta = D_5.$$

The one-particle root spaces give

$$\mathrm{sch\,Fock}(\mathfrak{g}_{\Delta_5+}) = \prod_{\alpha>0}(1-e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_{\Delta_5\alpha}}.$$

The oriented root operator gives

$$e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

The primitive reduced curve-counting theorem gives

$$\mathcal{Z}^{\mathrm{red}}_{K3\times E} = -(\jmath^*\,\mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}.$$

Theorem 136.1 (Protected square-root theorem)**.** *Assume the stated root-space Fredholm comparison with its orientation, Weyl shift, and convergence domain. Within that realization, the denominator section* Den_Δ *is simultaneously:*

(i) *the Weyl denominator of* $\mathfrak{g}_{\Delta_5}$*;*

(ii) *the Euler character of the bar–Chevalley complex;*

(iii) *the oriented Fredholm Pfaffian of* $\mathbb{A}_Z$*;*

(iv) *the PBW determinant of the parity-refined root superspace* $\mathcal{H}_{\mathrm{rp}}$*;*

(v) *the automorphic denominator whose half-coordinate pullback squares to the reduced scalar denominator.*

All five identifications preserve root degree and Weyl transport.

Proof. The first two are the denominator and bar–Chevalley theorems. The third is the Fredholm theorem. PBW for the Borcherds Lie superalgebra identifies the root superspace determinant, giving the fourth. The half-coordinate identity $\jmath^*\,\mathrm{Den}_\Delta = D_5$, the theta normalization, and the reduced curve-counting theorem give the fifth. □

Chapter 137

Physical meaning

The root-datum realizations retain several distinct constructions. The finite Ginzburg algebras and their projections form an inverse system. A completed three-Calabi–Yau sector requires a continuous nondegenerate Calabi–Yau structure and compatible cyclic models on the selected proper objects. The quiver Hall category retains its chosen coefficients and extension correspondences. A Hall theorem on that datum must identify its root presentation. The Borcherds root superspace defines the root-graded Fock space and its supercharacter. Its identification with physical one-particle states requires a parity-preserving graded comparison intertwining the primitive actions.

A realization on the elliptic curve must construct the current object with its local and wrapped supports, collision operations, and comparison maps. The quantum Borcherds algebra, with its specified pairing and completion, acts as a line-operator symmetry only through a representation preserving fusion and exchange. The root operator gives the Fredholm product on its specified convergence domain. A physical determinant interpretation requires an operator comparison and an orientation-preserving identification of the determinant lines and sections. The Fock character gives the scalar square. An operator trace additionally requires its state space and trace domain.

Simple-root states satisfying the defining relations give a primitive Lie map. The explicit enveloping and Chevalley extensions preserve products and brackets. A compact string realization must additionally construct its brane functor, Hall orientation maps, paired quantum coproduct, and trace-line comparison.

Chapter 138

Primitive comparisons and their algebraic extensions

Let $\mathfrak{n}_\Delta$ be the positive Borcherds Lie superalgebra. Choose homogeneous parity-preserving images of its simple generators in a charge-graded Lie superalgebra $\mathfrak{p}$. Assume these images satisfy every defining relation.

Theorem 138.1 (Extension of a primitive comparison)**.** *The chosen images determine a unique Lie-superalgebra map $f : \mathfrak{n}_\Delta \to \mathfrak{p}$. It induces*

$$U(f)(x_1 \cdots x_r) = f(x_1) \cdots f(x_r), \qquad C_*(f) = \mathrm{Sym}^c(f[1]).$$

These are respectively an enveloping-algebra map and a Chevalley chain-coalgebra map. Filtration-preserving maps extend to the specified inverse-limit completions. If f is surjective and both algebras have finite charge components and proper positive height, equality of their parity-refined PBW characters implies that f and $U(f)$ are isomorphisms.

Proof. The defining presentation gives f. Its tensor-algebra extension carries $xy - (-1)^{|x||y|}yx - [x, y]$ to the corresponding relation. It therefore descends to $U(f)$. The shifted symmetric-coalgebra map commutes with the Chevalley differential because f preserves brackets. Compatible maps on filtration quotients give maps on their inverse limits. Surjectivity of f gives surjectivity of $U(f)$. Character equality makes each finite charge component an isomorphism, separately in both parities. The PBW embeddings then imply injectivity of f. □

At level zero the current Lie conformal algebra is $\mathbb{C}[\partial] \otimes \mathfrak{n}_\Delta$ with $[x_\lambda y] = [x, y]$ on constant generators. The map $\partial^j x \mapsto \partial^j f(x)$ preserves this bracket and sesquilinearity. A nonzero central level additionally requires preservation of the specified invariant pairing. Hall convolution, Calabi–Yau structures, quantum coproducts, and determinant lines require their own comparison data.

Chapter 139

Finite quivers and Calabi–Yau completions

For a finite quiver Q with zero potential, let $\Gamma(Q, 0)$ be its path-length-completed Ginzburg algebra. Path products are read from left to right: if $a : i \to j$ and $b : j \to k$, then $ab : i \to k$. For each arrow $a : i \to j$, add $a^* : j \to i$ of degree -1. At each vertex add a loop t_i of degree -2. Original arrows have degree zero, and

$$da = da^* = 0, \qquad dt_i = e_i\Big(\sum_a [a, a^*]\Big)e_i.$$

Proposition 139.1 (Restriction to a full subquiver)**.** *If Q_0 is a full subquiver, retaining its generators and killing all other vertices, arrows, reverse arrows, and loops defines a continuous dg-algebra projection*

$$\Gamma(Q, 0) \longrightarrow \Gamma(Q_0, 0).$$

The inclusion of retained graded generators need not commute with the differential.

Proof. The projection respects source and target idempotents and extends multiplicatively. At a retained vertex, every term involving a discarded arrow vanishes. The other terms are precisely dt_i in the smaller algebra. All arrow differentials vanish, so the projection is a chain map. It preserves the path-length filtration and is continuous. For the final assertion, start with one vertex i and no arrows, where $dt_i = 0$. Add a vertex j and an arrow $a : i \to j$. In the larger algebra $dt_i = aa^*$, a nonzero length-two path. Thus the graded inclusion is not a dg map. □

Ginzburg completion supplies a three-Calabi–Yau algebra for each finite quiver in its stated setting. The displayed projections specify an inverse system for compatible zero-potential windows. A Calabi–Yau structure on a completed limit requires its own nondegeneracy and continuity argument. Comparisons with Hall categories and quantized Borcherds algebras must also preserve orientations, root relations, and the paired Hopf data.

Book XXIV

Quantum groups, exchange, and representation theory

Chapter 140

From positive half to full quantum group

Suppose the compact or local positive algebra H^+ is paired continuously with a negative half H^-, carries a completed coproduct, and admits Cartan data and a separated triangular completion. The completed double

$$\mathcal{U} = \widehat{D}(H^+)$$

has triangular decomposition $H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+$. Its universal R-matrix acts on continuous tensor products of modules. The classical limit is a Lie bialgebra whose cobracket is the first-order term of the coproduct asymmetry.

For $\mathbb{C}^3$, H^+ is the positive affine Yangian. For local ADE surfaces, it is the completed positive affine ADE Yangian. For the Igusa algebra, the paired Borel construction of the preceding part supplies a height-complete formal quantum group; a compact Hall realization maps to it through primitive, coproduct, and pairing comparisons.

Chapter 141

Tannaka reconstruction

Let $\mathcal{L}$ be an essentially small rigid $\Bbbk$-linear abelian monoidal category with finite-dimensional morphism spaces, finite-length objects, and exact tensor product. Let

$$\omega : \mathcal{L} \to \mathrm{Vect}_{\mathrm{fd}}$$

be a faithful exact strong monoidal functor. Assume the coend

$$H = \int^{M \in \mathcal{L}} \omega(M)^* \otimes \omega(M)$$

is formed by the matrix-coefficient relations. Coevaluation gives its coproduct, tensor product gives multiplication, and rigidity supplies the antipode. A braiding gives a coquasitriangular form. A meromorphic version additionally specifies its coefficient domains, expansions and continuous completed tensors.

Theorem 141.1 (Rigid Tannaka reconstruction)**.** *Under the preceding hypotheses, the comparison functor is a monoidal equivalence*

$$\mathcal{L} \simeq \mathrm{Comod}_{\mathrm{fd}}(H).$$

Braiding transports to the universal coquasitriangular form on H.

Proof. The exact faithful coalgebra reconstruction theorem identifies $\mathcal{L}$ with finite-dimensional comodules of this coend. The explicit coaction, multiplication and antipode are those of Proposition 16.8. The monoidal comparison is proved in Theorem 16.9. Taking matrix coefficients of the hexagon identities gives the coquasitriangular identities. □

This reconstruction explains why a line-operator category, a fiber functor, and exchange together produce a quantum group. The ordered boundary algebra provides ordered composition. A compatible bialgebra coproduct supplies tensor products of its modules, and exchange supplies additional braiding data.

Chapter 142

KZ and rational exchange

For a simple Lie algebra $\mathfrak{g}$, the Knizhnik–Zamolodchikov connection is

$$\nabla = d - \frac{\hbar}{2\pi i} \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

Flatness follows from the infinitesimal braid relations. Drinfeld–Kohno identifies its monodromy tensor category with the corresponding Drinfeld–Jimbo category after a choice of associator.

Rational difference exchange instead gives the Yangian R-matrix

$$R(u) = 1 - \frac{\hbar P}{u}$$

in the fundamental representation. The Yang–Baxter equation is the confluence identity for three exchanges. The KZ and Yangian constructions share the geometry of configuration spaces while using different flat connections.

Chapter 143

The centre and the double

The Hochschild centre $Z_{\mathrm{ord}}(A)$ controls central operations on the ordered algebra in its specified coefficient category. To form a Drinfeld centre of modules, first specify a monoidal module category. For an ordered chiral bialgebra satisfying Definition 120.1, use

$$\mathcal{M}_A = \mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$$

with the monoidal product of Theorem 120.2. An object of $Z(\mathcal{M}_A)$ is a module M together with natural equivalences

$$c_{M,N} : M \widehat{\otimes}^{!} N \xrightarrow{\sim} N \widehat{\otimes}^{!} M \qquad (N \in \mathcal{M}_A)$$

satisfying the half-braiding hexagon and unit identities, with their coherences. This categorical centre is braided. A comparison with the Hochschild centre must specify a realization functor, the source and target of the comparison, and the operations it preserves. Dualizability alone does not supply such a comparison. Derived loop-space descriptions apply in the geometric categories and under the hypotheses of their respective centre theorems [10].

The centre and the double answer complementary questions. The centre classifies central actions in the chosen category. The double reconstructs a quantum group from paired positive and negative halves. A field-theory comparison requires explicit realizations of the relevant line and bulk operations and a map compatible with both constructions.

Book XXV

A compact six-dimensional protected source

Chapter 144

Holomorphic cotangent BF theory on $K3 \times E$

Let S be a K3 surface with holomorphic symplectic form σ_S, let E be an elliptic curve with holomorphic form $\mathrm{d}z$, and put

$$X = S \times E, \qquad \Omega_X = \sigma_S \wedge \mathrm{d}z.$$

For a finite proper-height quotient $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$ of the positive Igusa Borcherds algebra, take fields

$$A \in \Omega^{0,\bullet}(X, \mathfrak{n}_{\Delta_5,+}{}^{(H)})[1], \qquad B \in \Omega^{0,\bullet}(X, (\mathfrak{n}_{\Delta_5,+}{}^{(H)})^\vee)[1].$$

The invariant evaluation pairing and Ω_X define the BV action

$$S_H(A,B) = \int_X \Omega_X \wedge \langle B, \bar\partial A + \frac{1}{2}[A,A]\rangle. \tag{144.1}$$

This is the holomorphic cotangent theory of the dg Lie algebra of $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$-valued Dolbeault forms.

Theorem 144.1 (Classical compact-background source)**.** *The action* (144.1) *satisfies the classical master equation and defines a holomorphic factorization algebra on the compact Calabi–Yau threefold X.*

Proof. The shifted cotangent pairing is invariant. The classical master equation is the Jacobi identity of $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$ together with $\bar\partial^2 = 0$ and the derivation property of $\bar\partial$. Local observables of an elliptic cyclic L_∞ algebra form a factorization algebra by the standard BV construction. □

Chapter 145

Tree exactness and the quantum master equation

Every interaction vertex in (144.1) has one B-leg and two A-legs. The propagator pairs an A-leg with a B-leg.

Lemma 145.1 (Loop bound)**.** *A connected Feynman graph of the cotangent theory has loop number at most one.*

Proof. A graph with V interaction vertices has exactly V internal or external B-half-edges. Every internal edge consumes one of them, hence $E \leq V$. The loop number is $E - V + 1 \leq 1$. □

The positive algebra is graded by strictly positive root height. For homogeneous $x \in \mathfrak{n}_{\Delta_s,+}{}^{(H)}$, the operator ad_x raises height. Its coadjoint operator lowers dual height. A product of adjoint operators labelled by positive roots has total nonzero height shift and therefore has zero trace.

Proposition 145.2 (Vanishing of one-loop Lie weights)**.** *For every finite positive-height quotient, the Lie weight of each one-loop graph in the cotangent theory vanishes.*

Proof. A one-loop graph has $E = V$. Every B-leg is consumed by an internal edge, so its external legs are A-legs. Its Lie weight is a cyclic trace of adjoint operators, or the dual trace with the conventionally equivalent signs. Each external label has positive root height. The product therefore strictly raises height on the finite-dimensional adjoint representation and has zero trace. The same argument applies to the dual representation. □

This vanishing removes the one-loop Lie coefficients in a perturbative construction with the stated propagator and vertex types. A renormalized quantum factorization algebra still requires admissible analytic kernels, local extensions, the quantum master equation, and compatible height transitions on the chosen observable spaces. These are the remaining quantum realization data.

Chapter 146

Classical reduction along K3

Choose a Kähler metric on S and normalize a closed harmonic $(0,2)$-form η by $\int_S \sigma_S \wedge \eta = 1$. Let $K = \mathbb{C}[\eta]/(\eta^2)$ with $|\eta| = 2$ and zero differential. There is a unital dg-algebra map

$$i : K \longrightarrow \Omega^{0,*}(S), \qquad \eta \longmapsto \eta.$$

Its image is closed under multiplication because $(0,4)$-forms vanish. The Dolbeault cohomology of a K3 surface is one-dimensional in degrees zero and two and zero in degree one, so i is a quasi-isomorphism. It preserves the trace defined by integration against σ_S.

Let $\mathfrak{l}$ be a finite-dimensional Lie algebra, and write $\mathfrak{d} = \mathfrak{l} \ltimes \mathfrak{l}^*$ for its cotangent Lie algebra, with invariant evaluation pairing. For an open $U \subset E$, tensor with $\Omega^{0,*}(U)$, using the completed smooth tensor product in the S direction. The field comparison is

$$K \otimes \Omega^{0,*}(U) \otimes \mathfrak{d} \xrightarrow{i \otimes \mathrm{id}} \Omega^{0,*}(S \times U) \otimes \mathfrak{d}.$$

The bracket is multiplication of forms followed by the Lie bracket, with Koszul signs.

Proposition 146.1 (Classical field comparison)**.** *The displayed map is a trace-preserving dg Lie quasi-isomorphism, compatible with restriction in U. Its induced minimal operations in the K3 factor have no higher products.*

Proof. Multiplicativity of i proves bracket compatibility. Hodge projection p and $h = \bar\partial^* G$ on the compact surface satisfy $pi = \mathrm{id}$ and $\bar\partial h + h\bar\partial = \mathrm{id} - i\,p$. Tensoring these continuous operators with forms on U gives the same contraction, with the tensor-product signs. Thus $i \otimes \mathrm{id}$ is a quasi-isomorphism. Restriction acts only on the U factor and commutes with it. The normalized trace on K is the restriction of the Dolbeault trace. Choose the contraction with $hi = 0$. In every higher transferred tree, the first product of inputs from $i(K)$ remains in $i(K)$ and the next internal homotopy kills it. Thus every higher transferred product vanishes. □

The coefficient algebra K retains both the degree-zero and degree-two K3 modes. The proposition compares classical field complexes and their brackets. Constructing an equivalence of quantum observable factorization algebras requires compatible quantization, support conventions, and the induced maps on observable completions. The ordinary Hochschild comparison of Theorem 135.1 is a separate algebraic calculation, not that quantum field map.

Chapter 147

Protected states and the Igusa trace

Let

$$\mathcal{H}_{\mathrm{ip}} = \widehat{\bigoplus}_{\alpha>0} \mathfrak{g}_{\Delta,\alpha}$$

with its Borcherds parity. The multiparticle state space is the completed super-Fock space

$$\mathcal{H}_{\mathrm{Fock}} = \widehat{\mathrm{Sym}}(\mathcal{H}_{\mathrm{ip},\bar{0}}) \otimes \widehat{\Lambda}(\mathcal{H}_{\mathrm{ip},\bar{1}}).$$

For Z in the absolute-convergence chamber, let e^{-H_Z} act by $e^{-\alpha(Z)}$ on the root space of degree α.

Theorem 147.1 (Protected Hamiltonian trace)**.** *The supertrace is*

$$\mathrm{str}_{\mathcal{H}_{\mathrm{Fock}}}(e^{-H_Z}) = \prod_{\alpha>0} (1 - e^{-\alpha(Z)})^{-\operatorname{sdim} \mathfrak{g}_{\Delta,\alpha}}.$$

After multiplication by the inverse Weyl line, its reciprocal is $\mathrm{Den}_\Delta(Z) = D_5(2Z)$*. After the half-coordinate pullback and doubling, the disconnected protected scalar is* $-(\jmath^* \mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}$ *in the Oberdieck–Pixton normalization.*

Proof. An even oscillator contributes $(1 - e^{-\alpha})^{-1}$ and an odd oscillator contributes $1 - e^{-\alpha}$. Multiplication over the root basis gives the product. The denominator identity and the reduced curve-counting theorem supply the final two equalities. □

Chapter 148

The compact-source comparison

The compact threefold provides the classical cotangent field theory and the dg Lie comparison of Proposition 146.1. The root datum separately gives the Borcherds algebra, its enveloping algebra, and the root-graded Fock space. The latter has the trace computed above. Proposition 145.2 supplies the finite one-loop Lie cancellation, while Theorem 135.1 computes the ordinary algebraic centre model with adjoint coefficients.

Problem 148.1 (Geometric compact comparison)**.** Construct objects E_i representing the simple branes and maps on their derived morphism complexes that preserve units, composition, and higher composition homotopies. Realize these maps on extension stacks with the reduced critical coefficients and orientation isomorphisms on triple flags. Construct compatible tensor and unit constraints on the resulting boundary functor. Compare the chosen quantum bulk action with the independently defined boundary centre, including support, collisions, and completion. Finally construct a trace and a map of its value line to the automorphic line in the same charge normalization.

A primitive map satisfying the root relations gives the enveloping and Chevalley maps of Theorem 138.1. A strong monoidal equivalence transports categorical centres. Neither construction determines the brane functor or its cyclic trace. These are the additional maps required for compact geometric reconstruction.

Book XXVI

Physics of the comparison maps

Chapter 149

Holomorphic–topological boundary theories

A three-dimensional holomorphic–topological theory on $C \times \mathbb{R}$ has local observables that are holomorphic along C and locally constant along $\mathbb{R}$. A boundary produces an ordered chiral algebra. A second topological direction permits exchange and gives an E_2 enhancement or a meromorphic braided line category. The bulk acts through the derived centre.

The finite cotangent BF model gives the exact local calculation:

$$\mathrm{Bar}(U(\mathfrak{l})) \simeq C_*(\mathfrak{l}).$$

For $\mathfrak{l} = \mathfrak{n}_{\Delta_5,+}{}^{(H)}$, this calculation realizes every finite-height Igusa denominator sector.

Chapter 150

Six-dimensional hCS on $K3 \times E$

Choose a holomorphic volume form $\Omega_S \wedge dz_E$. The hCS theory has classical fields

$$\Omega^{0,*}(S \times E, \mathfrak{a})[1]$$

and action S_{hCS}. Its factorization observables are native to the threefold. Pushforward along p_E gives a factorization algebra on E whose coefficient complex contains the Dolbeault modes of S.

A protected reduction to a chiral algebra consists of a deformation retract of the K3 Dolbeault complex, compatibility of the transferred L_∞ operations with the trace, a boundary or defect selecting ordered fusion, and a quantization trivializing the local anomaly. These data produce the ordered chiral algebra to which the Hall/BPS comparison may be applied.

The Hodge equality $\chi(\mathcal{O}_{S\times E}) = 0$ controls one compact supertrace. The gauge and gravitational anomalies are classes in the local BV deformation complex and are evaluated by their own characteristic forms.

Chapter 151

Omega backgrounds

Let a torus T act holomorphically on the local geometry and preserve the selected supercharge. For a generator V of the action, the equivariant differential is

$$Q_\Omega = Q + \iota_V, \qquad Q_\Omega^2 = \mathcal{L}_V.$$

After localization, fixed loci produce deformation parameters and often a spectral R-matrix. Toric threefolds and ADE surface resolutions carry such actions. A deformation family equipped with monodromy-compatible moduli correspondences and wall-crossing equivalences transports the localized algebra from algebraic or equivariant K3 fibers to the selected generic fiber.

Chapter 152

Anomalies as typed classes

The principal anomaly classes are

construction	deformation complex
hCS quantization	local BV cochains
critical CoHA orientation	Picard/gerbe theory of the critical stack
Hall coproduct	correspondence and localization complex
projective conformal blocks	Atiyah algebra of the determinant line
automorphic multiplier	character group of the modular cover.

A comparison theorem is a morphism between the corresponding complexes together with equality of the transported classes. This formulation turns numerical coincidences into geometric identities precisely when an intertwining map exists.

Book XXVII

Automorphic characters and scalar shadows

Chapter 153

Borcherds products as characters

Let $\mathfrak{n}_+$ be a locally finite positive Lie superalgebra. The Chevalley Euler character is

$$\chi C_*(\mathfrak{n}_+) = \prod_{\alpha>0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_\alpha}.$$

A Borcherds product becomes the Weyl-shifted character of $C_*(\mathfrak{n}_+)$. This construction produces a character of an existing algebra. Hall geometry supplies a second algebra, and a primitive comparison theorem identifies the two.

Chapter 154

The universal Borcherds weight formula

For a weight-zero weak Jacobi form

$$\varphi(\tau, z) = \sum c(n, \ell) q^n r^\ell,$$

the Borcherds lift has weight

$$\mathrm{wt}(\mathrm{Borch}(\varphi)) = \frac{c(0,0)}{2}.$$

For the K3 half-index, $c(0,0) = 10$ and the weight is 5. This formula lives entirely on the automorphic layer. Hodge Euler characteristics, Heisenberg ranks, central charges, and BV anomalies are evaluated by their own traces.

Chapter 155

The Gritsenko–Cléry atlas

The eight diagonal-divisor forms are

$$\Delta_{5},\ \Delta_{2},\ \Delta_{1},\ \Delta_{1/2},\ \nabla_{3},\ \nabla_{2},\ \nabla_{3/2},\ Q_{1}$$

with weights

$$5,\ 2,\ 1,\ \frac{1}{2},\ 3,\ 2,\ \frac{3}{2},\ 1.$$

Each form is an automorphic product on its own paramodular or Hecke congruence group with its own character or multiplier [21]. A geometric realization is specified row by row by a charge lattice, Hall problem, trace, and primitive algebra comparison.

Chapter 156

Scalarization hierarchy

The constructions descend through four levels:

$$\text{factorization or Hall object} \longrightarrow \text{primitive Lie algebra} \longrightarrow \text{PBW character} \longrightarrow \text{scalar partition function}.$$

Each arrow retains the quotient of structure displayed at its target. The first object contains supports, extension maps, and brackets. The second retains the infinitesimal algebra. The third retains signed graded dimensions. The fourth adds a trace and normalization. Upward reconstruction is supplied by the corresponding geometric operations and comparison morphisms.

Book XXVIII

The unified theorem system

Chapter 157

The foundational theorem

Definition 157.1 (Complete Calabi–Yau quantum-group datum)**.** A complete comparison datum specifies independent categorical, field, boundary, Hall, Hopf, and trace objects together with:

(i) PTVV-representable moduli for the smooth proper Calabi–Yau category;

(ii) a renormalized BV quantization satisfying the quantum master equation and a local compact factorization generator b;

(iii) a boundary reduction or proper locally trivial map to a curve satisfying factorization pushforward, together with the supplied curve-level faces and aligned distributive law;

(iv) a strong-oriented Hall coefficient theory whose pull–push maps satisfy base change and whose positive algebra is separated and complete;

(v) paired negative and Cartan halves, a completed coproduct, a continuous nondegenerate Hopf pairing, and a separated triangular completion on which the canonical tensors converge;

(vi) compatible open–closed and Hall–boundary morphisms;

(vii) a parity-refined primitive comparison for every automorphic identification.

Theorem 157.2 (Calabi–Yau quantum-group assembly)**.** *Let a complete datum consist of:*

(i) *a Calabi–Yau category and a representable moduli-of-objects problem;*

(ii) *a quantized cyclic elliptic BV theory and a boundary or defect generator;*

(iii) *a Hall extension correspondence with coefficient functor, base-change theorem, strong orientation data, and separated charge completion;*

(iv) *paired positive and negative bialgebras, Cartan data, a nondegenerate continuous Hopf pairing, and a completed double;*

(v) *every curve-level distributive, PBW, and analytic comparison used below.*

Then the datum determines functorially:

(i) *framed* E_2 *Hochschild cochains and the shifted-symplectic moduli structure;*

(ii) *the three curve-level algebraic faces and their three bars;*

(iii) *the positive Hall/BPS algebra and its primitive Lie algebra;*

(iv) *the completed quantum double and braided representation category;*

(v) *every supplied comparison among categorical, field-theoretic, Hall, current, quantum, and automorphic realizations;*

(vi) *the Weyl-shifted PBW/Chevalley character of any parity-refined primitive Lie-superalgebra identification.*

Proof. Cyclic Deligne and shifted symplectic geometry give the first output [14, 15, 84]. Boundary factorization gives the curve faces, and Volume I gives their bars and centres. The Hall correspondence and its base-change theorem give the associative product; strong orientation supplies the coefficient coherence required in the 3-Calabi–Yau setting [63]. The paired bialgebra data give the double and universal R-matrix. The comparison maps commute by hypothesis. PBW and Chevalley antisymmetrization give the final character. Each construction is natural under a morphism preserving the displayed data. □

The local quiver, equivariant surface, K3 correspondence, and Igusa Borcherds constructions have the carriers stated above. A geometric $K3 \times E$ comparison requires the primitive map and determinant-line map of Problem 148.1.

Chapter 158

Landscape of established realizations

Geometry	Positive algebra	Full or centred object	Character
A_2 quiver	$U_v^+(\mathfrak{sl}_3)$	$U_v(\mathfrak{sl}_3)$	PBW root product
$\mathbb{C}^3$	$Y^+(\widehat{\mathfrak{gl}}_1)$	$Y(\widehat{\mathfrak{gl}}_1)$	plane-partition and Fock traces
$\widetilde{\mathbb{C}^2/\Gamma}$	$Y^+(\widehat{\mathfrak{g}}_\Gamma)$	paired completion when the negative half and coproduct are supplied	surface Fock traces
$K3$	$U(\mathfrak{g}^+_{\mathrm{BPS}})$	Heisenberg and LLV actions	K3 Hall characters
$K3 \times E$	Borcherds root envelope; geometric reduced character	$U_q(\mathfrak{g}_{\Delta_5})$ and the root-current model and classical BF theory	$-\chi_{10}{}^{-1}$

The A_2 and $\mathbb{C}^3$ rows provide complete standard quantum-group models. The ADE row provides an established completed positive half together with its geometric current actions; its paired full double is formed from the negative half, Cartan data, coproduct, continuous Hopf pairing, and triangular completion stated in the table. The K3 row provides global two-Calabi–Yau BPS and cohomological symmetries. The final row records independently constructed root, character, and classical field objects. Their geometric Hall realization requires the comparison in Problem 148.1.

Chapter 159

Constructed comparisons and their geometric extensions

The Hall flag argument constructs associativity once its coefficient and orientation maps are defined. The tensor-action formula constructs representation fusion from a coassociative counital coproduct. The matrix-coefficient formulas construct the coend of a supplied fibre functor. The primitive presentation gives enveloping and Chevalley maps. The classical cyclic and K3 field calculations retain their actual differentials, traces, and brackets.

The quantum field realization requires admissible quantization and observable completions. The compact Hall realization requires the brane and orientation maps of Problem 148.1. The Fredholm and automorphic comparisons require their analytic domains and value-line isomorphisms. These constructions preserve the full geometric reconstruction problem beyond its finite algebraic and classical calculations.

Chapter 160

Actions and comparison maps

For a specified boundary algebra A_b, a coherent bulk action is a map

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A_b).$$

The target is the derived endomorphism object of the diagonal bimodule in the chosen factorization coefficient category. A Hall construction instead starts with

$$\mathcal{M}_{\alpha} \times \mathcal{M}_{\beta} \xleftarrow{p} \mathcal{E}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Critical coefficients, orientations, and admissible pull–push maps give its product. Framed moduli supply a representation after their action correspondences and triple-flag identities have been constructed. A paired Hopf structure gives $D(H^{+}, H^{\circ}, H^{-})$. Its coproduct defines tensor products of continuous modules, and a convergent universal R-matrix gives exchange.

A strong monoidal equivalence between this module category and a boundary category transports categorical centres. It does not identify that categorical centre with the object $Z_{\mathrm{ord}}(A_b)$ without a realization theorem. A trace on a specified representation then gives an amplitude. Its comparison with an automorphic section requires a map of value lines with the same normalization.

Chapter 161

Comparison data for Calabi–Yau constructions

Comparisons among Calabi–Yau categories, holomorphic field theory, chiral algebras, Hall algebras, quantum groups, and automorphic products require the following structures:

(i) for a smooth proper d-Calabi–Yau category with its nondegenerate negative-cyclic class, the categorical outputs are framed E_2 Hochschild cochains and, when the moduli stack of objects is representable in the PTVV setting, $(2-d)$-shifted symplectic moduli; the shift d does not change the Hochschild operadic level to E_n;

(ii) a geometric BV theory and boundary category produce factorization observables and transverse ordered chiral algebras; associative chiral convolution is recorded as a separate supplied structure;

(iii) pushforward to a curve uses an actual map $p : X \to C$; formal cycle-specialization notation alone specifies no pushforward;

(iv) extension stacks, coefficient sheaves, orientations, and completions define the Hall algebra;

(v) coproduct, Cartan, Hopf pairing, and topology define the double;

(vi) K3 Heisenberg, LLV orthogonal symmetry, local ADE Yangians, and global BPS generalized Kac–Moody algebras occupy distinct strata joined by explicit comparison maps where constructed;

(vii) the graph of the $K3 \times E$ Gram map becomes multiplicative in the polarized central extension;

(viii) point-local and wrapped sectors form a hybrid factorization problem;

(ix) the Igusa form is the Weyl-shifted PBW/Chevalley character of $\mathfrak{g}_{\Delta_5}$, and compact Hall geometry maps to it through the primitive comparison;

(x) anomaly identities are transported through explicit maps between deformation complexes.

The categorical, geometric, Hall, and automorphic operations thus have separate carriers and comparison maps.

Book **XXIX**

Technical appendices to the preceding book

Chapter 162

Typed data table

Object	Minimal data	Primary operation
CY category	nondegenerate negative-cyclic trace	Hochschild and shifted-symplectic structures
Factorization algebra	local observables and Weiss descent	disjoint-open product
Transversely ordered chiral algebra	factorization coefficient with an internal E_1 structure for $\otimes^{!}$	ordered fusion
Associative chiral algebra	an E_1 structure for chiral convolution $\star^{\mathrm{ch}}$	nonlocal chiral composition
CoHA	extension correspondence and coefficient theory	pull–push product
Bialgebra	compatible product and coassociative counital coproduct	fusion of representations
Double	positive and negative halves, Cartan data, Hopf pairing	universal R-matrix
Borcherds product	Jacobi or vector-valued modular input	graded character

Chapter 163

The three K3 brackets

The Heisenberg bracket is the excess intersection of two Nakajima incidence kernels. The LLV bracket is the commutator of diagonal Lefschetz kernels and their adjoints. The local Yangian bracket is the convolution commutator of ADE Hecke kernels with spectral equivariance. In $\mathrm{Corr}_{K3}(S)$ all three are compositions of geometric kernels. Their mixed commutators are the corresponding triple-intersection classes. The action algebra and Tannaka coend retain these classes while preserving the three geometric constructions.

Chapter 164

Exact arithmetic checks

The companion verification calculation establishes:

(i) the low Fourier rows of $\phi_{0,1}$ through q^3 by exact theta-series division;

(ii) the vector-valued cocycle identity for the Gram polarization;

(iii) the rational Yang–Baxter equation on $(\mathbb{C}^2)^{\otimes 3}$;

(iv) the A_2 Hall flag counts at ranks zero and one;

(v) the type-II Gram matrix and reflection identities.

Each calculation uses integer or rational arithmetic and appears as a worked formula in the relevant chapter.

Coherent boundary actions and BV fields

Boundary operations determine a deformation problem once their interface and coherence maps are supplied. A local realization gives a shifted cotangent BV theory. Comparing it with a physical bulk requires a map of fields, actions, and coherent boundary operations.

Book XXX

The full mixed boundary datum

Chapter 165

Three products and one interface

Holomorphic collisions and transverse composition act on different coefficient objects. A common boundary theory must specify their interaction on an interface. This interaction is part of the input to the following constructions.

Let X be a smooth complex curve and let $\mathcal{D}_X = D\text{-}\operatorname{Mod}(\operatorname{Ran} X)$ in a fixed six-functor enhancement. Write $\otimes^!$ for the pointwise product and $\star^{\mathrm{ch}}$ for chiral convolution.

Definition 165.1 (Complete mixed boundary datum)**.** A complete mixed boundary datum on X is a tuple

$$\mathbb{A} = (L, A^{\mathrm{ch}}, A^{\perp}, K, \iota, \lambda)$$

with the following components.

(i) L is a complete chiral Lie algebra.

(ii) A^{ch} is a complete associative algebra for $\star^{\mathrm{ch}}$.

(iii) $A^{\perp}$ is a complete E_1 algebra in $(\operatorname{Fact}_D(X), \otimes^!)$.

(iv) K is an $(A^{\mathrm{ch}}, A^{\perp})$ interface bimodule.

(v) $\iota : L \to (A^{\mathrm{ch}})_{\mathsf{Lie}}$ is a chiral Lie morphism.

(vi) λ is the aligned mixed distributive law relating the two module actions on K.

Every structure map is continuous for the chosen weight filtrations. The tuple includes the module actions, the mixed coherence homotopies, and their identities in the chosen enhancement. When a reduced bar is used, the relevant augmentations are supplied as additional data. Existence of this tuple for a physical boundary theory is a separate construction.

The three native bars are

$$\operatorname{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(L), \qquad \operatorname{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}), \qquad \operatorname{Bar}^{\perp}(A^{\perp}).$$

The interface has a fourth bar. Its degree-r terms are alternating words in the two associative colours with one marked interface letter. Its differential contracts adjacent compatible letters.

Construction 165.2 (Interface bar)**.** Let $\overline{A}^{\mathrm{ch}}$ and $\overline{A}^{\perp}$ be the augmentation ideals. Set

$$\operatorname{Bar}^{\mathrm{mix}}(\mathbb{A}; K) = \bigoplus_{p,q \geq 0} (s\overline{A}^{\mathrm{ch}})^{\otimes p} \otimes sK \otimes (s\overline{A}^{\perp})^{\otimes q}.$$

The differential is the sum of the internal differential, the two bar differentials, the left action on K, the right action on K, and the mixed terms supplied by λ.

Theorem 165.3 (Four-colour differential)**.** *The interface differential squares to zero. Its codimension-two terms are exactly the associative relations in the two colours, the two module relations, and the mixed distributive relation.*

Proof. Filter by total word length. Every component of the square contracts two adjacent edges. Disjoint contractions cancel by the determinant-line sign. Nested contractions in one colour cancel by associativity. A contraction meeting the interface cancels by the corresponding module axiom. The remaining square has one edge of each colour and is the aligned distributive square. The datum λ identifies its two composites. □

Chapter 166

Compatible modules and the mixed centre

An operation on a boundary theory must act on boundary-changing morphisms as well as on objects. Natural transformations retain this compatibility, and derived natural transformations retain its higher homotopies.

A compatible module is a triple $(M^{\mathrm{ch}}, M^{\perp}, \kappa)$ with an A^{ch} action, an $A^{\perp}$ action, and an interface comparison over K. Let $\mathrm{Mod}^{\mathrm{mix}}_{\mathbb{A}}$ be their stable category.

Definition 166.1 (Mixed derived centre)**.** The mixed derived centre is

$$Z_{\mathrm{mix}}(\mathbb{A}) = \mathrm{End}\Big(\mathrm{id}_{\mathrm{Mod}^{\mathrm{mix}}_{\mathbb{A}}}\Big).$$

It is computed by the endomorphisms of the diagonal compatible bimodule.

Theorem 166.2 (Universal interior action)**.** *$Z_{\mathrm{mix}}(\mathbb{A})$ carries an internal E_2 algebra structure. Every action already defined as a coherent natural endomorphism of the identity determines a point of this centre. The space of its factorizations through the representing endomorphism object is contractible. Morita-equivalent complete mixed data have equivalent centres.*

Proof. Endomorphisms of the identity functor form the higher centre of a module category. Composition and insertion of natural transformations give the brace operations and hence the E_2 structure. A supplied coherent natural action defines an endomorphism of the identity. A physical bulk insertion determines such an action only after its action on modules, morphisms, and higher compositions has been constructed. The factorization is the universal property of the endomorphism object. Morita equivalence identifies the module categories and their identity functors. □

Chapter 167

Relative induction from a chiral algebra

Let V be a chiral algebra with conformal vacuum. Its algebraic collar $\mathrm{Coll}(V)$ is the free transverse E_1 extension generated by one boundary copy of V, modulo the chiral relations already present in V. Weight completion gives $\widehat{\mathrm{Coll}}(V)$.

Theorem 167.1 (Universal collar)**.** *For every complete transversely ordered chiral algebra B and every chiral morphism $V \to B$, there is a unique continuous E_1 morphism*

$$\widehat{\mathrm{Coll}}(V) \longrightarrow B$$

extending the chiral map. The construction is left adjoint to restriction from transverse ordered algebras to chiral coefficients.

Proof. The algebraic collar is the relative operadic induction from the chiral operad to the mixed operad. Relative induction is left adjoint to restriction. Completion preserves the adjunction on positive complete objects because every map is determined weight by weight. □

When relative induction supplies a regular interface and a continuous aligned distributive law, these data define a mixed envelope with $L = V_{\mathsf{Lie}}$, $A^{\mathrm{ch}} = U^{\mathrm{ch}}(L)$, and $A^{\perp} = \widehat{\mathrm{Coll}}(V)$. Constructing that interface law in the selected geometric coefficient category is part of the envelope construction.

Book **XXXI**

The deformation algebra and its BV envelope

Chapter 168

The centre-extended deformation algebra

Varying a boundary product changes its compatibility with the interface and the central action. The deformation complex must therefore include all these operations. A local field realization is further data needed to turn their deformation equations into a BV action.

Fix a cofibrant coloured operad $\mathcal{P}_{\mathrm{mix}}$ presenting the structures of a complete mixed datum and its central action. Let $\mathcal{P}_{\mathrm{mix}}^{!}$ be its bar cooperad. The convolution complex

$$\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}} = \mathrm{Hom}_{\mathbb{S}}\left(\mathcal{P}_{\mathrm{mix}}^{!}, \mathrm{End}_{\mathbb{A}}\right)$$

contains the chiral operation, both associative products, the interface, and the centre action.

Theorem 168.1 (Complete deformation L_∞ algebra)**.** *The convolution complex has a canonical complete L_∞ structure. Its Maurer–Cartan points are complete mixed boundary structures together with a coherent central action. Twisting by the point defined by $\mathbb{A}$ gives the tangent algebra of deformations of the full datum.*

Proof. The infinitesimal decomposition maps of the bar cooperad insert collections of operations into one another. Composition in the endomorphism operad turns these decompositions into the convolution brackets. The cooperad identities imply the L_∞ identities. A Maurer–Cartan element is precisely an operad morphism from a cofibrant resolution of $\mathcal{P}_{\mathrm{mix}}$ to $\mathrm{End}_{\mathbb{A}}$. Twisting linearizes this mapping space at the selected structure. □

168.1 Field equations and Bianchi identities

Let α be a degree-one field in the localized deformation algebra. Its curvature is

$$\mathcal{F}(\alpha) = \sum_{n \geq 1} \frac{1}{n!} \ell_n(\alpha^{\otimes n}).$$

The linearization at α is

$$d_\alpha x = \sum_{n \geq 0} \frac{1}{n!} \ell_{n+1}(\alpha^{\otimes n}, x).$$

Theorem 168.2 (Universal Bianchi identity)**.** *For every α,*

$$d_\alpha \mathcal{F}(\alpha) = 0.$$

At a Maurer–Cartan point, $d_\alpha^2 = 0$.

Proof. Both identities are the exponential form of the L_∞ Jacobi relations. Expand the coderivation square on the cofree cocommutative coalgebra and evaluate it on e^{α} and $e^{\alpha} x$. □

Chapter 169

Shifted cotangent transgression

Let $\mathfrak{l}$ be a complete elliptic L_∞ algebra on a three-manifold M. Its shifted cotangent theory has fields

$$\mathcal{E} = T^*[-1]\mathfrak{l}[1] = \mathfrak{l}[1] \oplus \mathfrak{l}^\vee[-2].$$

The canonical action is

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle.$$

Theorem 169.1 (Universal BV envelope)**.** *The action satisfies the classical master equation. Its Euler–Lagrange equations are*

$$\mathcal{F}(\alpha) = 0, \qquad d_\alpha^\vee \beta = 0.$$

The tangent complex at a solution is the cotangent complex of the Maurer–Cartan moduli problem.

Proof. The Hamiltonian vector field of S is the cotangent lift of the homological vector field defining the L_∞ structure. The square of that vector field is zero, hence $\{S, S\} = 0$. Variation in β gives the Maurer–Cartan equation. Variation in α gives the dual linearized equation. □

Definition 169.2 (Local realization and chiral BV envelope)**.** A local realization of the complete mixed datum consists of an elliptic local L_∞ algebra $\mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}$ on $M = X^{\mathrm{an}} \times \mathbb{R}$, a specified continuous filtered L_∞ quasi-isomorphism

$$\Gamma(M, \mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}) \longrightarrow \mathrm{Def}_{\mathbb{A}}^{\mathrm{full}},$$

and the local pairing needed for cotangent transgression. Here Γ denotes the chosen complex of global sections with its declared topology. The comparison includes compatibility with the specified boundary operations. The dual fields use the density-valued continuous dual, with compact support or boundary conditions that make the integrated pairing and integration by parts defined.

For a supplied local realization, define

$$\mathbf{BVEnv}_3^{\mathrm{full}}(\mathbb{A}) = T^*[-1]\,\mathrm{MC}(\mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}).$$

Its action is the cotangent action displayed above. Constructing a local realization from an abstract mixed datum remains an additional existence problem.

Chapter 170

Intrinsic algebraic quantization

Let $\mathfrak{X}$ be a finite perfect derived formal moduli problem with a homological orientation μ: the corresponding Berezinian is invariant under the homological vector field. Functions on its shifted cotangent are completed polyvectors:

$$O(T^*[-1]\mathfrak{X}) = \widehat{\mathrm{PV}}(\mathfrak{X}) = \prod_{p \geq 0} \Gamma(\mathfrak{X}, \wedge^p T_{\mathfrak{X}})[p].$$

The homological orientation defines the divergence operator

$$\Delta_\mu : \widehat{\mathrm{PV}}(\mathfrak{X}) \longrightarrow \widehat{\mathrm{PV}}(\mathfrak{X})[1].$$

Theorem 170.1 (Oriented BV complex)**.** *The operator Δ_μ has order two and satisfies*

$$\Delta_\mu^2 = 0, \qquad [d_{\mathfrak{X}}, \Delta_\mu] = 0,$$

$$\Delta_\mu(fg) = \Delta_\mu(f)g + (-1)^{|f|} f \Delta_\mu(g) + (-1)^{|f|}\{f, g\}.$$

Consequently

$$\mathbf{QHdg}(\mathfrak{X}, \mu) = \left(\widehat{\mathrm{PV}}(\mathfrak{X})[\![\hbar]\!], d_{\mathfrak{X}} + \hbar\Delta_\mu\right)$$

is a square-zero quantum BV complex.

Proof. In oriented Darboux coordinates, Δ_μ is the sum of paired second derivatives. Their graded commutation gives $\Delta_\mu^2 = 0$. Invariance of the Berezinian under the homological vector field gives $[d_{\mathfrak{X}}, \Delta_\mu] = 0$. Expanding the operator on a product gives the bracket as the cross term. □

The orientation identifies polyvectors with differential forms by contraction. Under this identification,

$$\iota_\mu \Delta_\mu = d_{\mathrm{dR}}\, \iota_\mu.$$

Hence the BV complex is the Hodge-completed derived de Rham complex with differential

$$d_{\mathfrak{X}} + \hbar d_{\mathrm{dR}}.$$

The Rees parameter records Hodge degree.

Definition 170.2 (Intrinsic quantum complex of a mixed datum)**.** For a complete mixed datum $\mathbb{A}$ equipped with compatible homological orientations on its finite perfect charts, define $\mathbf{QHdg}_3^{\mathrm{full}}(\mathbb{A})$ by applying the preceding BV complex to finite perfect charts of

$$\mathrm{MC}(\mathbf{l}_{\mathbb{A}}^{\mathrm{full}})$$

and taking the separated inverse limit over the chart and weight filtrations.

Theorem 170.3 (Quantum master equation)**.** *A degree-zero action $S = S_0 + \hbar S_1 + \cdots$ defines a quantum state $e^{S/\hbar}\mu$ precisely when*

$$dS + \frac{1}{2}\{S, S\} + \hbar\Delta_\mu S = 0.$$

The equation is equivalent to

$$(d + \hbar\Delta_\mu)e^{S/\hbar} = 0.$$

Proof. Apply the second-order product formula to the exponential and collect powers of $\hbar$. □

Chapter 171

Perturbative factorization realization

The algebraic quantum differential does not specify a propagator or a space of singular kernels. Fix a local realization, elliptic gauge-fixing data, support and boundary conditions, and a renormalization prescription. A heat-kernel propagator is defined away from the diagonal. Extending its graph integrals to collisions requires counterterms and compatibility with the boundary conditions.

Definition 171.1 (Factorization realization data)**.** A factorization realization consists of renormalized local observable complexes, continuous disjoint-insertion maps, and coherent homotopies for their compositions. It includes extensions of graph weights to the required collision strata, their Stokes identities, and a quantum master equation in the permitted local functional complex. Weiss descent is a further required property of these actual complexes. A boundary realization also includes a comparison with the supplied mixed datum, preserving its two products, Lie operation, interface, and mixed law.

Under these hypotheses the disjoint-insertion maps and composition homotopies define the stated factorization object. The boundary comparison identifies its boundary operations with those of $\mathbb{A}$. The construction problem is to produce these data from the physical fields and action. A graphwise identity supplies the corresponding composition equation, whereas descent and boundary identification require their own maps and proofs.

Book XXXII

Boundary duality, exchange, and reduction

Chapter 172

The Koszul–Morita square

A change from modules to comodules must transport the interface as well as each algebra. The mixed bar supplies the comparison object. Exchange then asks for motion in an additional topological direction, while BRST reduction asks whether the differential respects every transported operation.

Let P be a dualizable boundary state for $A^{\perp}$. Its boundary dual is

$$A^{!}_{P} = \mathrm{RHom}_{A^{\perp}}(P, P).$$

The ordinary bar construction gives explicit twisted tensor functors. Their extension to the mixed datum must also transport the interface and its geometric operations.

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ be an augmented associative algebra in cohomological degree zero. Assume $A_0 = k$, augmentation projects onto A_0, and multiplication adds weights. Set $A = \prod_{w \geq 0} A_w$ and $C = \prod_{w \geq 0} B(A_{\mathrm{alg}})_w$, with degreewise products and the weight topology. The bar suspension lowers cohomological degree by one, with $b_2[a|b] = [ab]$. The degree-one twisting map $\tau : C \to A$ sends $[a]$ to a and vanishes on the empty word and longer words.

A module or comodule has the form $\prod_{w \geq w_0} M_w$, with differential preserving weight, for some integer w_0. Actions and coactions preserve weight. Comodule coactions are counital and coassociative, and use the tensor product

$$(C \widehat{\otimes} M)_w = \bigoplus_{u+v=w} B(A_{\mathrm{alg}})_u \otimes M_v.$$

The same convention defines $A \widehat{\otimes} M$. These sums are finite in each weight. Repeated reduced coactions vanish in weight w after more than $w - w_0$ steps. Invert quasi-isomorphisms in each weight to obtain $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$.

Proposition 172.1 (Ordinary twisted tensor equivalence)**.** *The functors*

$$F(M) = C \widehat{\otimes}_{\tau} M, \qquad G(N) = A \widehat{\otimes}_{\tau} N$$

induce inverse equivalences $D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C)$. Writing $\Delta c = \sum c_{(1)} \otimes c_{(2)}$ and $\rho n = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$d_F(c \otimes m) = d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m,$$
$$d_G(a \otimes n) = a \otimes d_N n - \sum a\tau(n_{(-1)}) \otimes n_{(0)}.$$

The counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a \epsilon_C(c) m, \qquad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The twisting equation makes both differentials square to zero. The action, coaction, and counit identities show that the displayed maps are chain maps and satisfy the triangle identities. For $GF(M)$, the extra degeneracy

of the two-sided bar inserts $1[\bar{a}_0|a_1|\cdots|a_r]m$ into $a_0[a_1|\cdots|a_r]m$, where $\bar{a}_0 = a_0 - \epsilon_A(a_0)1$. With simplicial boundary $\sum_i (-1)^i d_i$, it contracts the augmented bar. Multiplying length-r words by $(-1)^r$ gives the displayed differential convention. This contraction preserves weight, so the counit is a weightwise quasi-isomorphism. For $FG(N)$, filter each weight by the weight of the final N-factor. The reduced coaction lowers that weight. The associated graded unit is the contracted right-sided bar $k \otimes N_v \to B(k, A_{\mathrm{alg}}, A_{\mathrm{alg}}) \otimes N_v$. The filtration is finite in each total weight, so the unit is also a quasi-isomorphism. The same finite filtrations show that F, G preserve weightwise quasi-isomorphisms, since tensoring complexes over k preserves them. Thus the unit and counit give the asserted inverse equivalences. □

For a right module R and a left module L with the same lower-weight bounds, the two-sided bar has underlying weight-completed complex

$$\prod_n \bigoplus_{u+v_1+\cdots+v_r+w=n} R_u \otimes s\bar{A}_{v_1} \otimes \cdots \otimes s\bar{A}_{v_r} \otimes L_w, \qquad v_i > 0,$$

where the sum also runs over $r \geq 0$ and $\bar{A} = \ker(A_{\mathrm{alg}} \to k)$. Internal differentials and the two endpoint actions and adjacent products form its bar differential. Its realization computes $R \otimes^{\mathbb{L}}_A L$ in these weight-graded categories. Comparing it with a derived cotensor product requires a two-sided twisting comparison on the chosen comodule category and completion. Weight completion here is not identified with derived augmentation-adic completion.

Proposition 172.2 (Centre transport under a mixed equivalence)**.** *Suppose an equivalence between the compatible mixed module categories is supplied, with coherent compatibility with all four bar objects and the interface operations. It identifies the derived endomorphism objects of their identity functors.*

Proof. Conjugate a natural endomorphism by the equivalence and a quasi-inverse. The unit and counit homotopies give inverse maps on derived endomorphism objects and preserve composition and its coherent operations. □

Constructing such a mixed equivalence requires the restriction, collision, interface, and distributive-law comparisons for the actual geometric coefficients. Its unit and counit must respect those maps and the chosen inverse limits. A diagonal bimodule comparison must produce an actual compatible bicomodule before tensor products can be identified with geometric cotensor products. Transport of mixed Hochschild chains and traces further requires their cyclic comparison and the trace domains used in their definition. These are additional mixed Koszul–Morita constructions. The ordinary weight-graded equivalence supplies its stated one-colour functors, without identifying the other mixed carriers or their physical traces.

Chapter 173

Planar exchange and Tannaka reconstruction

A second topological direction gives paths exchanging parallel defects. Their homotopy classes form braid groups. Let $\mathscr{L}$ be an essentially small rigid braided linear category with a k-linear strong monoidal functor $\omega : \mathscr{L} \to \mathrm{Vect}_{\mathrm{fd}}(k)$. Its matrix-coefficient coend is

$$O(\mathbb{G}) = \int^{M \in \mathscr{L}} \omega(M)^* \otimes \omega(M).$$

Theorem 173.1 (Quantum symmetry from exchange)**.** *The coend is a coquasitriangular Hopf algebra. Its full linear dual acts naturally on every fiber. The coefficient form of the braiding lies in the completed dual tensor defined in Chapter 16. Naturality gives an RTT relation on a selected fiber. A presentation requires a further isomorphism from the algebra defined by those relations to the coend.*

Proof. Coevaluation gives $\Delta c_M(\lambda, v) = \sum_i c_M(\lambda, e_i) \otimes c_M(e^i, v)$, and evaluation gives the counit. Tensor product gives multiplication. Dual objects give the antipode. These formulas and their identities are proved in Proposition 16.8.

The braiding defines the coefficient form (16.14), and the hexagon identities imply its compatibility equations. Naturality gives the RTT relation. Further morphisms in the category can impose additional relations. □

A spectral version must specify its meromorphic coefficient domains, expansion maps and continuous tensors. Reconstruction of the whole category additionally uses the exactness hypotheses of Theorem 16.9.

Chapter 174

BRST reduction

Let Q be a square-zero derivation of a mixed datum. Extend Q factorwise to every bar word. Since Q is a derivation for each structure map, it anticommutes with every bar differential.

Theorem 174.1 (BRST–bar bicomplex)**.** *The total differential*

$$D = Q + b_{\mathrm{ch}} + b_{\perp} + b_{\mathrm{mix}}$$

satisfies $D^{2} = 0$. *A bounded-below complete filtration gives a convergent spectral sequence*

$$E_{1} = H_{Q}\big(\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)\big) \Longrightarrow H_{D}\big(\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)\big).$$

When Q-cohomology is concentrated and tensor-exact, the target is the mixed bar of the reduced datum.

Proof. The square vanishes by the derivation identity and the four-colour bar theorem. The filtration by BRST degree gives the spectral sequence. Concentration and tensor exactness identify the E_{1} differential with the bar differential of the reduced operations. □

Book XXXIII

Conformal covariance and Einstein completion

Chapter 175

Projective connections and the Virasoro tangent complex

A conformal boundary algebra transforms under changes of coordinates. Projective connections record its central term. Three-dimensional gravity supplies independently defined connection and coframe fields whose boundary transformation law can be compared with this algebraic structure.

Let $\mathcal{P}_c(X)$ be the formal moduli problem of projective connections with central parameter c. At a projective connection p, its tangent complex is locally

$$\Omega^{0,\bullet}(X, T_X) \xrightarrow{\mathcal{D}_{p,c}} \Omega^{0,\bullet}(X, K_X^{\otimes 2}),$$

where

$$\mathcal{D}_{p,c}(\epsilon) = \epsilon\, \partial p + 2(\partial \epsilon)\, p + \frac{c}{12} \partial^3 \epsilon.$$

Theorem 175.1 (Projective covariance)**.** *A conformal map* $\mathrm{Vir}_c \to A$ *defines a formal family of mixed structures over* $\mathcal{P}_c(X)$*. Its tangent algebra is the semidirect product of the Virasoro projective complex with the deformation algebra relative to a fixed projective connection.*

Proof. The conformal vector acts on fields by infinitesimal coordinate change. The central term is the affine cocycle in the transformation law of a projective connection. The action therefore gives a map from the projective tangent algebra to derivations of the mixed datum. The tangent algebra of the associated family is the semidirect product. □

Chapter 176

Three-dimensional gravity as a flat Cartan theory

Let M be an oriented three-manifold, let $\Lambda = -\ell^{-2}$, and let $\mathfrak{g}_\Lambda$ have basis J_a, P_a with

$$[J_a, J_b] = \epsilon_{ab}{}^{c} J_c, \qquad [J_a, P_b] = \epsilon_{ab}{}^{c} P_c, \qquad [P_a, P_b] = \ell^{-2} \epsilon_{ab}{}^{c} J_c.$$

The invariant pairing is $\langle J_a, P_b \rangle = \eta_{ab}$ and vanishes on the two diagonal summands. A Cartan connection is

$$\mathcal{A} = \omega^a J_a + e^a P_a.$$

Its curvature is

$$F_{\mathcal{A}} = \left(R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c \right) J_a + T^a P_a.$$

Theorem 176.1 (Cartan–Einstein equivalence)**.** *On the locus where e is invertible, $F_{\mathcal{A}} = 0$ is equivalent to*

$$T^a = 0, \qquad G_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \qquad g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu.$$

The Chern–Simons action for $\mathcal{A}$ equals the Einstein–Hilbert action with cosmological constant, up to its boundary term and normalization.

Proof. The P component of the flatness equation is the torsion equation. Invertibility of e gives the Levi–Civita spin connection. The J component states that the curvature two-form has constant sectional curvature $-\ell^{-2}$. In three dimensions the Riemann tensor is determined by the Ricci tensor, so this is the Einstein equation. Expanding the Chern–Simons three-form with the off-diagonal invariant pairing gives $e \wedge R + \ell^{-2} e^3$ plus an exact form [1, 92]. □

Chapter 177

Four equivalent linear complexes

Fix a constant-curvature solution (e, ω). There are four deformation complexes.

(i) The flat adjoint complex $\Omega^\bullet(M, \mathfrak{g}_\Lambda)$ with differential $d_{\mathcal{A}}$.

(ii) The Einstein–Cartan complex on $(\delta e, \delta\omega)$ with linearized torsion and curvature.

(iii) The metric Einstein complex on $h_{\mu\nu}$ with linearized Einstein operator.

(iv) The Calabi complex resolving local Killing fields by the Killing, linearized curvature, and Bianchi operators.

Theorem 177.1 (Chain equivalence of gravitational presentations)**.** *An invertible torsion-free dreibein gives chain maps among the four complexes. These maps are quasi-isomorphisms on every geodesically convex open set. Their cohomology describes infinitesimal isometries, physical linearized fields, and Bianchi identities in the four languages.*

Proof. The dreibein identifies internal translations with tangent vectors and internal Lorentz rotations with skew endomorphisms. Splitting $d_{\mathcal{A}}$ in this basis gives the linearized torsion and curvature operators. Eliminating $\delta\omega$ by the linearized torsion equation gives the metric Einstein operator. The symbol sequence is the Calabi symbol sequence and is exact on constant-curvature space. The Poincaré lemma for the flat adjoint connection and exactness of the symbol homotopies give local quasi-isomorphisms. □

Chapter 178

Brown–Henneaux and the projective boundary

Write the AdS connection as

$$A^{\pm} = \omega \pm \ell^{-1} e.$$

The Einstein action is the difference of two $\mathfrak{sl}_2$ Chern–Simons actions with level

$$k = \frac{\ell}{4G}.$$

Theorem 178.1 (Brown–Henneaux central charge)**.** *The classical asymptotic Poisson algebra of each chiral Chern–Simons sector has Virasoro central charge*

$$c = 6k = \frac{3\ell}{2G}.$$

The projective connection carried by the boundary stress tensor is the projective variable of the preceding conformal family.

Proof. Hamiltonian reduction of the boundary $\mathfrak{sl}_2$ current Poisson algebra gives the classical Virasoro bracket. Its central term is $6k$. Substitution of the gravitational level gives the second equality. The coadjoint transformation of the boundary stress tensor is the projective-connection transformation law [17]. □

Chapter 179

Einstein completion of a conformal pair

Let A_L and A_R be conformal mixed boundary data with equal Brown–Henneaux central charge. Their projective families map to the same $\mathcal{P}_c(X)$. Define their relative field algebra by the homotopy fibre product over the projective tangent algebra.

Definition 179.1 (Candidate Einstein BV completion)**.** Suppose the relative algebra and a continuous L_∞ action of the Cartan algebra have been supplied, together with their local realization. The associated cotangent candidate is

$$\mathbf{EinBV}_3(A_L, A_R) = T^*[-1]\,\mathrm{MC}\big(\mathfrak{g}_{\mathrm{grav}} \ltimes_{\mathrm{Ham}} \mathfrak{m}_{A_L,A_R}\big).$$

Here $\mathfrak{g}_{\mathrm{grav}}$ is the Cartan deformation algebra and $\mathfrak{m}_{A_L,A_R}$ is the relative mixed deformation algebra after projective matching.

Theorem 179.2 (Cotangent action for a supplied coupled algebra)**.** *Suppose projective matching and a continuous L_∞ action have constructed the displayed complete semidirect product. Suppose also that its local realization and integrated dual pairing are defined. Its shifted cotangent has action*

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle$$

and satisfies the classical master equation. Its Bianchi identity is the L_∞ identity of the coupled algebra.

Proof. The supplied semidirect product has a square-zero homological vector field. Its cotangent lift is Hamiltonian for the displayed action and remains square-zero. This proves the classical master equation and the Bianchi identity by the preceding cotangent construction. □

The gravitational summand of this cotangent action is the cotangent theory of Cartan deformations. It introduces independent dual fields. To identify it with Einstein BV theory requires a comparison preserving the BV pairing and the action, including boundary terms, on the invertible-coframe sector. The flatness equations alone do not provide that comparison. A physical coupled completion must also construct the matter action and its equivariant moment map. The intended Einstein reconstruction therefore retains these field and action comparisons as existence problems.

Chapter 180

Higher-spin completion

Replace $\mathfrak{sl}_2$ by $\mathfrak{sl}_N$ and choose its principal $\mathfrak{sl}_2$ embedding. The adjoint representation decomposes as

$$\mathfrak{sl}_N \cong \bigoplus_{s=2}^{N} V_{2s-2}, \qquad \sum_{s=2}^{N} (2s-1) = N^2 - 1.$$

Theorem 180.1 (Principal higher-spin spectrum)**.** *For the specified principal $\mathfrak{sl}_N$ Chern–Simons theory linearized about its gravitational background, the decomposition gives one frame-like field of each spin $2, 3, \dots, N$. With principal Drinfeld–Sokolov boundary conditions, reduction of the boundary affine Poisson algebra gives currents of these same spins.*

Proof. The principal decomposition (16.15) contains V_{2s-2} once for each $s = 2, \dots, N$. It is the internal representation of the spin-s frame-like field. Principal Drinfeld–Sokolov reduction of this theory produces one generating current in each conformal weight $2, \dots, N$ [19, 52]. □

For an independently supplied pair of W_N boundary theories, a bulk realization still requires a map from this reduction to those theories, preserving their Poisson operations and coupling. The spin multiplicities alone do not construct that map.

Book XXXIV

Exact gravitational complexes and quantum tests

Chapter 181

The four complexes in coordinates

The Cartan flatness equations and the metric Einstein equations describe the same classical solutions on the invertible-coframe locus. Their infinitesimal complexes make the comparison explicit. Quantum determinants additionally depend on gauge fixing, boundary conditions, and determinant-line data.

181.1 Flat adjoint complex

At a flat Cartan connection $\mathcal{A}$, the first complex is

$$\Omega^0(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^1(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^2(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^3(M,\mathfrak{g}_\Lambda).$$

Degree zero is the gauge parameter, degree one is the fluctuation, degree two is the linearized curvature, and degree three is the Bianchi identity.

181.2 Einstein–Cartan complex

Write a gauge parameter as (ξ^a,λ^a) and a fluctuation as $(u^a,v^a)=(\delta e^a,\delta\omega^a)$. The first differential is

$$\begin{aligned} u^a &= D_\omega \xi^a + \epsilon^a{}_{bc} e^b \lambda^c, \\ v^a &= D_\omega \lambda^a + \ell^{-2}\epsilon^a{}_{bc} e^b \xi^c. \end{aligned}$$

The second differential is

$$\begin{aligned} \delta T^a &= D_\omega u^a + \epsilon^a{}_{bc} v^b \wedge e^c, \\ \delta F^a &= D_\omega v^a + \ell^{-2}\epsilon^a{}_{bc} e^b \wedge u^c. \end{aligned}$$

The third differential is the linearized pair of Cartan Bianchi identities.

Proposition 181.1 (Chain identity)**.** *The composite of the first two Einstein–Cartan differentials vanishes on a constant-curvature torsion-free background.*

Proof. Apply D_ω^2 to ξ and λ. Replace the background curvature by $-\ell^{-2}e\wedge e$ and use $D_\omega e=0$. The curvature terms cancel the explicit ℓ^{-2} terms. The remaining expressions are Jacobi identities for $\epsilon_{ab}{}^c$. □

181.3 Metric complex

The dreibein identifies the translation parameter with a vector field. Modulo Lorentz rotations,

$$h_{\mu\nu} = 2\eta_{ab} e^a_{(\mu} u^b_{\nu)}.$$

The metric differential begins with the Killing operator

$$K(\zeta)_{\mu\nu} = \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu.$$

The linearized cosmological Einstein operator is

$$(E_\Lambda h)_{\mu\nu} = \frac{1}{2}\big(- \nabla^2 h_{\mu\nu} - \nabla_\mu\nabla_\nu h + \nabla_\mu\nabla^\rho h_{\rho\nu} + \nabla_\nu\nabla^\rho h_{\rho\mu} \\ - g_{\mu\nu}(\nabla^\rho\nabla^\sigma h_{\rho\sigma} - \nabla^2 h)\big) - \Lambda(h_{\mu\nu} - g_{\mu\nu}h).$$

It satisfies $E_\Lambda K = 0$ and $\nabla^\mu (E_\Lambda h)_{\mu\nu} = 0$.

Theorem 181.2 (Explicit metric reduction)**.** *Solving the linearized torsion equation for $v = \delta\omega$ and substituting into the linearized curvature equation gives $E_\Lambda h = 0$. The kernel of the map from (u, v) to h is the infinitesimal Lorentz action.*

Proof. The torsion equation is algebraic in the antisymmetric part of the covariant derivative of u and determines v. The symmetric part of u is $h/2$. Substitution into $\delta F = 0$ yields the standard variation of the Ricci tensor and the displayed Einstein operator. An antisymmetric $u_{\mu\nu}$ is exactly a Lorentz rotation of the dreibein. □

181.4 Calabi complex

On a constant-curvature manifold, the Calabi complex begins

$$\Gamma(TM) \xrightarrow{K} \Gamma(S^2T^*M) \xrightarrow{R'} \Gamma(\mathcal{R}) \xrightarrow{B'} \Gamma(\mathcal{B}),$$

where R' is the linearized Riemann tensor and B' is the linearized algebraic and differential Bianchi operator. In three dimensions the Riemann tensor is equivalent to the Einstein tensor, so the metric and Calabi complexes are locally quasi-isomorphic.

Chapter 182

Boundary BFV theory and the Virasoro moment map

The variation of Chern–Simons theory on a manifold with boundary gives the boundary symplectic form

$$\Omega_\partial = \frac{k}{4\pi}\int_{\partial M} \langle \delta A \wedge \delta A\rangle.$$

A chiral polarization selects $A_{\bar z}$ as coordinate. The residual moment map is the flatness constraint.

Theorem 182.1 (Boundary current algebra)**.** *Canonical quantization of the boundary symplectic form gives the affine current algebra at level k. Drinfeld–Sokolov reduction of the principal $\mathfrak{sl}_2$ constraint gives the Virasoro moment map*

$$T(z) = \frac{1}{2(k+h^\vee)} :J^a J_a:(z) + \textit{improvement},$$

with the classical Brown–Henneaux limit $c = 6k$.

Proof. The inverse of Ω_∂ is the current Poisson bracket. Quantization gives the affine central extension. The BRST complex of the principal constraint eliminates the gauge directions and retains the improved quadratic current. Its classical central term is $6k$ in the gravitational normalization. □

Chapter 183

Matter stress tensor and the Einstein moment map

Let $\mathfrak{m}$ be the relative field algebra of a conformal pair. Projective covariance gives a Hamiltonian action of vector fields and therefore a moment map

$$\mu_{\mathrm{matt}} : \mathfrak{m} \longrightarrow \Omega^{0,\bullet}(X, K_X^{\otimes 2})[-1].$$

Its classical component is the stress tensor.

Theorem 183.1 (Conservation from the master equation)**.** *The coupled BV action*

$$S_{\mathrm{tot}} = S_{\mathrm{grav}} + S_{\mathrm{matt}} + \langle h, \mu_{\mathrm{matt}} \rangle$$

satisfies the classical master equation precisely when the matter moment map is equivariant. The component of that equation linear in the diffeomorphism ghost is

$$\nabla^{\mu} T_{\mu\nu} = 0.$$

Proof. The master bracket of the coupling with itself is the defect of equivariance. The bracket with the gravitational action is the infinitesimal diffeomorphism of the moment map. Their cancellation is the Hamiltonian equivariance identity. Pairing the identity with the diffeomorphism ghost and integrating by parts gives stress-tensor conservation. □

Chapter 184

One-loop gravity and analytic torsion

Expand Chern–Simons theory around an acyclic flat connection $\mathcal{A}$. Gauge fixing gives the Laplacians of the flat adjoint de Rham complex.

Theorem 184.1 (One-loop determinant)**.** *The absolute value of the one-loop partition function is the square root of the Ray–Singer torsion of the flat adjoint bundle:*

$$|Z_{\text{1-loop}}(\mathcal{A})| = \tau_{\mathrm{RS}}(M, \operatorname{ad}\mathcal{A})^{1/2}.$$

The phase is the eta-invariant contribution fixed by the framing convention.

Proof. Bosonic and ghost Gaussian integrals give alternating determinants of the Laplacians in the flat de Rham complex. Their product is the analytic torsion. The odd signature operator supplies the determinant phase. This is the standard stationary-phase calculation for Chern–Simons theory [87, 92]. □

The same determinant complex is obtained from the Einstein–Cartan, metric, and Calabi presentations through their chain equivalences. One-loop equality is therefore a consequence of quasi-isomorphism together with compatible determinant-line orientations.

Chapter 185

BTZ, Cardy, and the area law

For a unitary modular boundary theory with left and right central charges c_L, c_R, the asymptotic entropy is

$$S_{\mathrm{Cardy}} = 2\pi\sqrt{\frac{c_L}{6}\left(L_0 - \frac{c_L}{24}\right)} + 2\pi\sqrt{\frac{c_R}{6}\left(\bar{L}_0 - \frac{c_R}{24}\right)}.$$

For the BTZ black hole,

$$L_0 - \frac{c}{24} = \frac{(r_+ + r_-)^2}{16G\ell}, \qquad \bar{L}_0 - \frac{c}{24} = \frac{(r_+ - r_-)^2}{16G\ell}, \qquad c = \frac{3\ell}{2G}.$$

Theorem 185.1 (Cardy–Bekenstein equality)**.** *Substitution gives*

$$S_{\mathrm{Cardy}} = \frac{2\pi r_+}{4G}.$$

This is the Bekenstein–Hawking entropy of the BTZ horizon.

Proof. Each square root becomes $\pi(r_+ \pm r_-)/(4G)$. Their sum is $2\pi r_+/(4G)$. The modular asymptotic is Cardy's theorem, and the geometric charges are the Chern–Simons holonomies of the BTZ solution [6, 20]. □

Chapter 186

Higher-spin holonomy and boundary charges

In $\mathfrak{sl}_N$ Chern–Simons gravity, regularity constrains the conjugacy class of the thermal holonomy. Such a constraint is imposed together with the boundary conditions. The functions

$$\operatorname{Tr}(\mathrm{Hol}^{s}), \qquad 2 \le s \le N,$$

are functions of the holonomy, whereas the local Drinfeld–Sokolov currents are functions of the connection. For a constant connection A, holonomy is $\exp(LA)$.

Proposition 186.1 (Recovering constant connection coefficients)**.** *For a regular semisimple constant companion connection, its holonomy conjugacy class determines its characteristic coefficients after a logarithm branch is fixed on a domain where the exponential is injective. Without that restriction the coefficients need not be determined.*

Proof. The branch recovers the eigenvalues of A from those of $\exp(LA)$, and hence its characteristic polynomial. For $A = \left(\begin{smallmatrix} 0 & 1 \\ q & 0 \end{smallmatrix}\right)$, the trace is $2\cosh(L\sqrt{q})$. At $L = 2\pi$, every $q = -n^2$, $n \ge 1$, gives holonomy equal to the identity. Proposition 16.11 proves both assertions. □

Hamiltonian reduction supplies the local W_N Poisson brackets. Identifying its charges with global holonomy functions requires the actual Hamiltonian comparison, including the boundary conditions. The characteristic polynomial of the connection does not give it.

Book XXXV

Anomalies, Hall sources, and automorphic completion

Chapter 187

Four native anomaly classes

A local anomaly, a determinant-line curvature, and an arity-zero bar term belong to different complexes. A cancellation in one complex becomes a geometric cancellation only through a comparison preserving the relevant class and its primitive. Hall and automorphic constructions require their coefficient and orientation maps for the same reason.

The construction carries four curvature classes.

(i) A local BV anomaly lies in the degree-one local deformation complex.

(ii) A projective anomaly is the curvature of the determinant line over parameter space.

(iii) A framing anomaly is the transgression of a characteristic class to a three-manifold.

(iv) A curved Koszul term is the arity-zero component of a curved coalgebra.

Theorem 187.1 (Curvature-killing extension)**.** *Let κ be a closed central degree-two element of a dg Lie algebra. Adjoining a central degree-one generator u with $du = \kappa$ makes κ exact. If κ is a curved Maurer–Cartan term, twisting by $-u$ makes that term zero.*

Proof. The differential squares to zero because $d\kappa = 0$. Its derivation identity on a bracket with u follows from $[\kappa, x] = 0$. The twisted curvature is $\kappa + d(-u) + \frac{1}{2}[-u, -u] = 0$. □

This extension can change the deformation problem. A geometric realization requires an independently constructed invertible theory with the opposite anomaly. A comparison must identify both classes in the same complex and normalization. Under those hypotheses their anomaly classes add to zero. The central-generator construction alone supplies neither that theory nor the comparison map.

Chapter 188

Hall and Calabi–Yau sources

Let C be a Calabi–Yau category with a Hall moduli problem. An oriented extension correspondence gives a Hall algebra H^+. A factorization pushforward to the boundary curve gives a chiral-convolution algebra. Relative induction supplies the remaining colours.

Proposition 188.1 (Transport under a supplied mixed Morita equivalence)**.** *Suppose a Hall construction supplies a complete mixed boundary datum and a Morita equivalence of its compatible module category with that of $\mathbb{A}$. The equivalence identifies their derived centres. A comparison of local deformation algebras, BV pairings, and actions is additional data for a BV comparison. Quantum line and amplitude comparisons additionally require the monoidal and cyclic structures used to define them.*

Proof. Conjugating natural transformations by the equivalence and its inverse identifies the endomorphisms of the two identity functors. The unit and counit homotopies provide the inverse comparison and preserve composition. This proves the centre comparison. It does not construct the additional local, symplectic, or cyclic maps. □

An algebra map alone does not induce the required forward centre map. For example, the homomorphism $\mathbb{C}[x] \to M_2(\mathbb{C})$ with $x \mapsto \operatorname{diag}(0,1)$ sends the central element x to a noncentral matrix. Its commutator with the matrix unit E_{12} is $-E_{12}$. Thus even the degree-zero condition fails for a centre map extending the given algebra map. Hall multiplication maps must be supplemented by an actual coherent module comparison before centre transport can be applied.

For $X = K3 \times E$, suppose a geometric Hall source supplies the complete charge-graded BPS Lie superalgebra and Cartan action required in Volume IV. Its semidirect product with the common Cartan defines $\mathfrak{b}_X$. Volume IV then constructs the completed Lie amalgam with the Igusa Borcherds Borel $\mathfrak{b}_\Delta$. Applying the completed enveloping functor gives the cocommutative Hopf algebra

$$H_{X\boxtimes\Delta} = \widehat{U}(\mathfrak{b}_X)\widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)}\widehat{U}(\mathfrak{b}_\Delta).$$

The two Hopf retractions preserve the geometric and automorphic sectors separately. A q-deformed mixed Hopf algebra is supplied by the quantum amalgam theorem when a geometric Hall coproduct and pairing are present.

188.1 Mixed realization of the amalgam

The Hopf retractions above are maps of their stated algebraic objects. A mixed realization must construct chiral and transverse operations, an interface, and a compatible distributive law. To realize the retractions, it must then supply maps preserving those operations and the chosen completions.

A BV retraction requires further maps between the local deformation algebras, together with compatible pairings and actions. The counterexample to forward centre functoriality prevents deducing these maps from the Hopf retractions alone. The intended geometric–automorphic theory asks for this mixed realization and for a primitive comparison quotient on its actual boundary operators. Its interaction

ideal is the kernel of that constructed quotient, rather than an ideal determined by equality of scalar characters.

Chapter 189

Igusa root states and the mixed-boundary comparison

The Gritsenko–Nikulin presentation defines the root Lie superalgebra $\mathfrak{g}_{\Delta_5}$ and its positive subalgebra $\mathfrak{n}_+$. Let $d_\alpha = \operatorname{sdim}(\mathfrak{g}_{\Delta_5})_\alpha$, and fix the positive cone, its proper height, and Weyl vector ρ. Put

$$P = \prod_{\alpha>0}(1-e^{-\alpha})^{d_\alpha}, \qquad \mathrm{Den}_\Delta = e^{-\rho}P = D_5(2Z), \qquad \jmath(Z) = Z/2.$$

The products are first interpreted in the positive-height completion.

Theorem 189.1 (Root character and reduced scalar)**.** *The root Fock space* $\mathrm{Sym}(\mathfrak{n}_+)$*, formed with super symmetry, has supercharacter* P^{-1}*. Consequently*

$$-\jmath^*\big(e^{2\rho}\operatorname{sch}(\mathrm{Sym}(\mathfrak{n}_+)^{\otimes 2})\big) = -(\jmath^*\,\mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}.$$

For a nonsingular projective K3 surface S and an elliptic curve E, this scalar also has an enumerative interpretation. Let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant in class (β_h, d), where β_h is nonzero and primitive with square $2h-2$. Use the incidence insertion $\beta_h^\vee \otimes [\mathrm{point}]$, with $\langle \beta_h, \beta_h^\vee\rangle = 1$. Then the series

$$\mathcal{Z}(u,q,s) = \sum_{g,h,d\ge 0} N_{g,h,d}\,u^{2g-2}q^{h-1}s^{d-1}$$

is the Laurent expansion of $-\chi_{10}(r,q,s)^{-1}$ under $r = e^{iu}$, by [81, Theorem 1].

Proof. An even root vector contributes $(1-e^{-\alpha})^{-1}$ to the supercharacter of its symmetric algebra. An odd root vector contributes $1-e^{-\alpha}$, because its symmetric algebra in super vector spaces is an exterior algebra. Multiplying the two contributions gives P^{-1}. Proper height makes every coefficient calculation finite. The Weyl factor gives $e^{2\rho}P^{-2} = \mathrm{Den}_\Delta^{-2}$. Pullback by $\jmath$ gives $D_5(Z)^{-2} = \chi_{10}(Z)^{-1}$. Oberdieck–Pixton, Theorem 1, identifies the separately defined reduced numerical generating series with its negative [81]. This last theorem is an enumerative input; the character calculation supplies the root side. □

A primitive realization starts with parity-preserving root classes in a charge-graded Lie superalgebra $\mathfrak{p}$ satisfying the defining relations. Require the classes to preserve charge. The free Lie presentation gives an even charge-preserving map $f : \mathfrak{n}_+ \to \mathfrak{p}$.

Applying f to tensor words constructs $U(f)$. With trivial coefficients, use Chevalley chains $C_*^{\mathrm{CE}}(\mathfrak{a}) = \mathrm{Sym}(\mathfrak{a}[1])$, where the homological suspension raises degree by one and the differential contracts pairs by the bracket. Applying the suspension of f to every factor gives a chain map because f preserves brackets.

For the level-zero Lie conformal current, define $\mathrm{Cur}(\mathfrak{a}) = \mathbb{C}[\partial]\otimes\mathfrak{a}$, with ∂ even. Its generator bracket is $[x_\lambda y] = [x,y]$. Extend it by $[\partial x_\lambda y] = -\lambda[x_\lambda y]$ and $[x_\lambda \partial y] = (\partial+\lambda)[x_\lambda y]$. The formula $\partial^j x \mapsto \partial^j f(x)$ defines $\mathrm{Cur}(f)$. Charge preservation makes these maps compatible with the specified proper-height quotients and their inverse limits.

A complete mixed boundary realization requires further geometric maps. The compact source consists of sheaves and extension flags on a common elliptic torsor. Its critical realization must supply reduced coefficient complexes, orientation cross-line maps, unit and flag coherences, and the admissible pull–push operations. A chiral comparison must preserve actual support and moving-insertion operations. A quantum-double comparison must preserve coproducts, pairings and the domain of exchange operators. The finite Calabi–Yau quiver completions require their specified dg maps; ordinary full-subquiver inclusions do not in general give those maps.

The mixed datum to which a BV envelope is applied must therefore include these operations and compatible maps between them. Identifying the primitive algebra of that envelope requires an actual map of its deformation complex to the root complex. A determinant comparison requires a skew perfect family, a square-root line and section, and an isomorphism to the automorphic line preserving transport and normalization. An operator-trace interpretation further requires its state space and trace domain. Constructing this datum and these comparisons is the Igusa mixed-boundary problem. The root character identity specifies the scalar that their final evaluation must recover.

Chapter 190

The carrier lattice

The complete theory uses the carrier set

$$\mathfrak{C} = \{\mathrm{diag}, \mathrm{ch}, \bot, \mathrm{int}, \mathrm{ex}, \mathrm{cyc}, \mathrm{mod}, \mathrm{BV}, \mathrm{Ein}\}.$$

For a subset $S \subset \mathfrak{C}$, let $\mathrm{Str}_S(\mathbb{A})$ be the homotopy limit of all mapping spaces attached to incidence cells supported in S.

Theorem 190.1 (Carrier descent)**.** *When S and T form a relation-excisive pair, the square*

$$\begin{array}{ccc} \mathrm{Str}_{S\cup T}(\mathbb{A}) & \longrightarrow & \mathrm{Str}_S(\mathbb{A}) \\ \downarrow & & \downarrow \\ \mathrm{Str}_T(\mathbb{A}) & \longrightarrow & \mathrm{Str}_{S\cap T}(\mathbb{A}) \end{array}$$

is a homotopy pullback. Thus a global structure is assembled from its carrier restrictions together with their common interface.

Proof. Relation-excision identifies the incidence category of $S \cup T$ with the pushout of the incidence categories for S and T over their intersection. Homotopy limits send this pushout to the displayed pullback. □

Definition 190.2 (Relation-complete carrier cover)**.** Let $U \subseteq \mathfrak{C}$ and let $\mathcal{U} = \{S_i\}_{i\in I}$ be a finite family with $U = \bigcup_i S_i$. The family is *relation-complete* when the canonical functor from the Cech homotopy colimit

$$\operatorname*{hocolim}_{\varnothing \neq J \subseteq I} \mathcal{I}_{\bigcap_{j\in J} S_j} \longrightarrow \mathcal{I}_U$$

is an equivalence. Equivalently, every incidence cell and every relation among its faces is contained in one member of the cover, with all overlaps recorded by the intersections.

Theorem 190.3 (Power-set coherence)**.** *The assignment*

$$S \longmapsto \mathrm{Str}_S(\mathbb{A})$$

is a contravariant functor on the full power set of $\mathfrak{C}$. For every relation-complete carrier cover $\mathcal{U} = \{S_i\}_{i\in I}$ of U, the restriction map is an equivalence

$$\mathrm{Str}_U(\mathbb{A}) \xrightarrow{\sim} \operatorname*{holim}_{\varnothing \neq J \subseteq I} \mathrm{Str}_{\bigcap_{j\in J} S_j}(\mathbb{A}).$$

Thus every subtheory, every intersection of subtheories, and the complete theory are values of one descent object.

Proof. An inclusion $S \subseteq T$ induces the full inclusion $\mathcal{I}_S \subseteq \mathcal{I}_T$ and therefore the restriction map on homotopy limits. Composition of inclusions gives functoriality. A relation-complete cover presents $\mathcal{I}_U$ as the displayed homotopy colimit of its intersection diagram. Mapping that colimit into the diagram of structure spaces converts it into the stated homotopy limit. The resulting equivalence is natural under refinement of the carrier cover. □

Definition 190.4 (Exact-support deformation complex)**.** For $S \subseteq \mathfrak{C}$, let $\mathrm{Def}_{\leq S}$ be the subcomplex generated by incidence cells supported in S, and set

$$\mathrm{Def}_{=S} = \mathrm{cofib}\Big(\operatorname*{hocolim}_{T \subsetneq S} \mathrm{Def}_{\leq T} \longrightarrow \mathrm{Def}_{\leq S}\Big).$$

Thus $\mathrm{Def}_{=S}$ retains the deformation cells whose minimal carrier support is exactly S.

Theorem 190.5 (Carrier-support spectral sequence)**.** *Let* $\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}}$ *be the cellular convolution complex of the complete incidence diagram. Filter it by the cardinality of minimal carrier support. The finite filtration gives a convergent spectral sequence*

$$E_1^{p,q} = \bigoplus_{\substack{S \subseteq \mathfrak{C} \\ |S| = p}} H^{p+q}(\mathrm{Def}_{=S}) \Longrightarrow H^{p+q}\Big(\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}}\Big).$$

A class on the S-summand is a deformation or curvature class whose first complete geometric carrier is S. The first differential is the signed incidence map between exact-support quotients.

Proof. The support-cardinality filtration is finite because $\mathfrak{C}$ is finite. Its pth associated-graded quotient is the direct sum of the exact-support complexes $\mathrm{Def}_{=S}$ with $|S| = p$. The spectral sequence of a finite filtered complex gives the displayed first page and converges strongly to the cohomology of the total convolution complex. The cellular determinant orientation gives the incidence signs. □

Chapter 191

Sixteen structural laws

The theory is governed by the following laws.

(i) A geometric carrier determines the type of its operation.

(ii) The diagonal stores the complete principal-part filtration.

(iii) The interval stores ordered fusion.

(iv) The plane stores exchange.

(v) The circle stores trace.

(vi) The stable curve stores modular sewing.

(vii) The transverse, associative-chiral, and chiral-Lie bars are distinct.

(viii) Chiral PBW compares the last two on enveloping objects.

(ix) The interface bar records the mixed distributive law.

(x) The derived centre is the universal interior action.

(xi) The deformation algebra is the tangent algebra of the centre-extended datum.

(xii) Maurer–Cartan curvature is the field equation.

(xiii) The L_∞ identity is the Bianchi identity.

(xiv) Shifted cotangent is the universal classical BV envelope.

(xv) Hodge-completed derived de Rham theory is the intrinsic algebraic quantum complex of the oriented formal moduli problem.

(xvi) A gravitational realization requires projective matching, a coupled local field algebra, and a comparison preserving the physical BV action and pairing.

Chapter 192

Coherent actions and gravitational reconstruction

A supplied mixed boundary datum defines compatible modules and their derived natural endomorphisms. A specified coloured operad organizes deformations of that datum together with its central action. These are constructions relative to the full operations and their coherence maps. The centre alone does not recover the datum from which the deformation algebra is formed.

A supplied local realization of the deformation algebra admits shifted cotangent transgression. With compatible invariant orientations, the finite perfect charts also carry the algebraic BV differential. The physical bulk is defined independently by its fields, action, gauge symmetry, boundary conditions, and quantization. Reconstruction requires a coherent action map from its observables to an independently defined geometric centre, followed by an equivalence theorem on the stated sector.

For gravity, the invertible coframe gives the classical Cartan–Einstein correspondence. The cotangent construction supplies a formal comparison theory with extra dual fields. Its identification with gravitational BV fields requires an action-preserving and pairing-preserving map. Quantum boundary currents, a continuum measure, global holonomy sectors, allowed fillings, and compatible states remain necessary for the quantum and global comparison.

Hall and Calabi–Yau geometry can supply boundary operations when their correspondences, coefficients, orientations, and comparison maps are constructed. Exchange additionally uses a monoidal category and a fibre functor. Cyclic pairings and sewing determine amplitudes only on their stated domains. These comparisons are the mathematical steps by which local boundary operations could reconstruct physical and geometric theories.

Book XXXVI

Technical appendices to the preceding book

Chapter 193

The Chern–Simons normalization

The gravitational charge calculation uses the action normalization. The mixed bar calculation uses the determinant orientation of its edges. The following formulas specify these conventions.

With invariant pairing $\langle J_a, P_b \rangle = \eta_{ab}$, the Chern–Simons action is

$$S_{\mathrm{CS}}[\mathcal{A}] = \frac{k}{4\pi} \int_M \langle \mathcal{A} \wedge \mathrm{d}\mathcal{A} + \frac{2}{3} \mathcal{A} \wedge \mathcal{A} \wedge \mathcal{A} \rangle .$$

For $A^{\pm} = \omega \pm \ell^{-1} e$ one finds

$$S_{\mathrm{CS}}[A^{+}] - S_{\mathrm{CS}}[A^{-}] = \frac{k}{2\pi\ell} \int_M \epsilon_{abc} e^a \wedge \left(R^{bc} + \frac{1}{3\ell^2} e^b \wedge e^c \right) + \int_{\partial M} (\cdots) .$$

The equality with the Einstein–Hilbert normalization gives $k = \ell/(4G)$.

Chapter 194

The principal embedding count

The exponents of $\mathfrak{sl}_N$ are $1, 2, \dots, N-1$. Under the principal $\mathfrak{sl}_2$ embedding,

$$\mathfrak{sl}_N = \bigoplus_{m=1}^{N-1} V_{2m}.$$

The dimensions satisfy

$$\sum_{m=1}^{N-1} (2m+1) = (N-1)(N+1) = N^2 - 1.$$

Writing $s = m+1$ gives one field at every spin $s = 2, \dots, N$.

Chapter 195

Low-arity mixed signs

Let $x|k|y$ be a suspended interface word. The two length-lowering contractions are

$$b_L[x|k|y] = (-1)^{|x|}(xk)|y, \qquad b_R[x|k|y] = (-1)^{|x|+|k|}x|(ky).$$

Applying the remaining contraction gives the two composites $(xk)y$ and $x(ky)$ with opposite determinant-line signs. Their sum vanishes by the interface associativity relation. A transverse and a chiral contraction acquire the product orientation

$$\det(E_{\mathrm{ch}} \oplus E_{\perp}) = \det(E_{\mathrm{ch}}) \otimes \det(E_{\perp}),$$

so reversing their order contributes the Koszul sign and yields anticommutation.

Bibliography

[1] Achucarro Achucarro and Paul K. Townsend, *A chern–simons action for three-dimensional anti-de sitter supergravity theories*, Physics Letters B **180** (1986), no. 1–2, 89–92.

[2] Luigi Alfonsi, Hyungrok Kim, and Charles Young, *Raviolo vertex algebras, cochains and conformal blocks*, 2024.

[3] Mats Andersson and Elizabeth Wulcan, *Direct images of semi-meromorphic currents*, Annales de l'Institut Fourier **68** (2018), no. 2, 875–900.

[4] Jonathan D. Axtell and Kyu-Hwan Lee, *Quantum generalized kac–moody algebras via hall algebras of complexes*, 2019.

[5] David Ayala, John Francis, and Hiro Lee Tanaka, *Factorization homology of stratified spaces*, Selecta Mathematica **23** (2017), 293–362, available at 1409.0848.

[6] Máximo Bañados, Claudio Teitelboim, and Jorge Zanelli, *The black hole in three-dimensional space-time*, Physical Review Letters **69** (1992), no. 13, 1849–1851.

[7] Ruggero Bandiera, *Cumulants, Koszul brackets and homological perturbation theory for commutative BV_∞ and IBL_∞ algebras*, 2020.

[8] Aliaa Barakat, Alberto De Sole, and Victor G. Kac, *Poisson vertex algebras in the theory of hamiltonian equations*, 2009.

[9] Alexander Beilinson and Vladimir Drinfeld, *Chiral algebras*, American Mathematical Society Colloquium Publications, vol. 51, American Mathematical Society, 2004.

[10] David Ben-Zvi, John Francis, and David Nadler, *Integral transforms and drinfeld centers in derived algebraic geometry*, Journal of the American Mathematical Society **23** (2010), no. 4, 909–966, available at 0805.0157.

[11] Yuri Berest, Giovanni Felder, Sasha Patotski, Ajay C. Ramadoss, and Thomas Willwacher, *Representation homology, lie algebra cohomology and the derived harish–chandra homomorphism*, 2015.

[12] Richard E. Borcherds, *Automorphic forms on $o_{s+2,2}(\mathbb{r})$ and infinite products*, Inventiones Mathematicae **120** (1995), 161–213.

[13] ———, *Automorphic forms with singularities on grassmannians*, Inventiones Mathematicae **132** (1998), 491–562, available at alg-geom/9609022.

[14] Christopher Brav and Tobias Dyckerhoff, *Relative calabi–yau structures ii: shifted lagrangians in the moduli of objects*, Selecta Mathematica (N.S.) **27** (2021), no. 4, 63, available at 1812.11913.

[15] Christopher Brav and Nick Rozenblyum, *The cyclic deligne conjecture and calabi–yau structures*, 2023.

[16] Alexander Braverman, Michael Finkelberg, and Hiraku Nakajima, *Towards a mathematical definition of coulomb branches of 3-dimensional $n = 4$ gauge theories, ii*, Advances in Theoretical and Mathematical Physics **22** (2018), no. 5, 1071–1147, available at 1601.03586.

[17] J. David Brown and Marc Henneaux, *Central charges in the canonical realization of asymptotic symmetries: An example from three-dimensional gravity*, Communications in Mathematical Physics **104** (1986), no. 2, 207–226.

[18] Damien Calaque, Vasiliy Dolgushev, and Gilles Halbout, *Formality theorems for hochschild chains in the lie algebroid setting*, Journal für die reine und angewandte Mathematik **612** (2007), 81–127, available at math/0504372.

[19] Andrea Campoleoni, Stefan Fredenhagen, Stefan Pfenninger, and Stefan Theisen, *Asymptotic symmetries of three-dimensional gravity coupled to higher-spin fields*, Journal of High Energy Physics **2010** (2010), no. 11, 007, available at 1008.4744.

[20] John L. Cardy, *Operator content of two-dimensional conformally invariant theories*, Nuclear Physics B **270** (1986), 186–204.

[21] Fabien Cléry and Valery Gritsenko, *The siegel modular forms of genus 2 with the simplest divisor*, Proceedings of the London Mathematical Society **102** (2011), no. 6, 1024–1052, available at 0812.3962.

[22] Kevin Costello, *Topological conformal field theories and calabi–yau categories*, Advances in Mathematics **210** (2007), 165–214.

[23] Kevin Costello and Owen Gwilliam, *Factorization algebras in quantum field theory, volume 1*, New Mathematical Monographs, vol. 31, Cambridge University Press, 2017.

[24] ———, *Factorization algebras in quantum field theory, volume 2*, New Mathematical Monographs, vol. 41, Cambridge University Press, 2021.

[25] Kevin Costello and Si Li, *Anomaly cancellation in the topological string*, 2019.

[26] Ben Davison, *Affine bps algebras, w algebras, and the cohomological hall algebra of $\mathbb{A}^2$*, Homotopical methods in geometry and physics, 2026, pp. 163–199.

[27] Ben Davison, Lucien Hennecart, and Sebastian Schlegel Mejia, *Bps algebras and generalised kac–moody algebras from 2-calabi–yau categories*, 2023.

[28] Ben Davison and Sven Meinhardt, *Cohomological donaldson–thomas theory of a quiver with potential and quantum enveloping algebras*, Inventiones Mathematicae **221** (2020), 777–871, available at 1601.02479.

[29] Alberto De Sole, Matteo Gardini, and Victor G. Kac, *On the structure of quantum vertex algebras*, 2019.

[30] Duiliu-Emanuel Diaconescu, Mauro Porta, Francesco Sala, Olivier Schiffmann, and Eric Vasserot, *Cohomological hall algebras of one-dimensional sheaves on surfaces and yangians*, 2026.

[31] Vasiliy Dolgushev, *Covariant and equivariant formality theorems*, Advances in Mathematics **191** (2005), no. 1, 147–177, available at math/0307212.

[32] Vladimir Drinfeld, *Hopf algebras and the quantum yang–baxter equation*, Soviet Mathematics Doklady **32** (1985), 254–258.

[33] ———, *Quantum groups*, Proceedings of the international congress of mathematicians, berkeley 1986, 1987, pp. 798–820.

[34] Martin Eichler and Don Zagier, *The theory of jacobi forms*, Progress in Mathematics, vol. 55, Birkhäuser, 1985.

[35] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik, *Tensor categories*, American Mathematical Society, 2015.

[36] Pavel Etingof and Victor Ginzburg, *Symplectic reflection algebras, calogero–moser space, and deformed harish–chandra homomorphism*, 2001.

[37] Pavel Etingof and Olivier Schiffmann, *Lectures on quantum groups*, International Press, 1998.

[38] John Francis and Dennis Gaitsgory, *Chiral koszul duality*, Selecta Mathematica **18** (2012), no. 1, 27–87, available at 1103.5803.

[39] Wee Liang Gan and Victor Ginzburg, *Almost-commuting variety, D-modules, and cherednik algebras*, 2005.

[40] Ezra Getzler and Mikhail Kapranov, *Modular operads*, Compositio Mathematica **110** (1998), 65–126.

[41] Victor Ginzburg, *Calabi–yau algebras*, arXiv preprint (2006), available at math/0612139.

[42] Suresh Govindarajan, *Borcherds–kac–moody lie superalgebras from counting twisted CHL dyons*, Journal of High Energy Physics **2011** (2011), no. 5, 089, available at 1006.3472.

[43] Suresh Govindarajan and K. Gopala Krishna, *Generalized kac–moody algebras from CHL dyons*, Journal of High Energy Physics **2009** (2009), no. 4, 032, available at 0807.4451.

[44] Suresh Govindarajan and Sutapa Samanta, *Borcherds–kac–moody lie superalgebras from counting twisted CHL dyons--ii*, Nuclear Physics B **948** (2019), 114770, available at 1905.06083.

[45] James A. Green, *Hall algebras, hereditary algebras and quantum groups*, Inventiones Mathematicae **120** (1995), 361–377.

[46] Valery Gritsenko and Viacheslav Nikulin, *Siegel automorphic form corrections of some lorentzian kac–moody lie algebras*, American Journal of Mathematics **119** (1997), no. 1, 181–224, available at alg-geom/9504006.

[47] ———, *Automorphic forms and lorentzian kac–moody algebras. part i*, International J. Math. **9** (1998), no. 2, 153–199, available at alg-geom/9610022.

[48] ———, *Automorphic forms and lorentzian kac–moody algebras. part ii*, International J. Math. **9** (1998), no. 2, 201–275, available at alg-geom/9611028.

[49] Alexander Grothendieck, *Techniques de construction et théorèmes d'existence en géométrie algébrique IV: les schémas de Hilbert*, Séminaire bourbaki, 1961, pp. 249–276.

[50] Owen Gwilliam and Theo Johnson-Freyd, *How to derive Feynman diagrams for finite-dimensional integrals directly from the BV formalism*, 2019.

[51] Owen Gwilliam, Eugene Rabinovich, and Brian R. Williams, *Factorization algebras and abelian cs/wzw-type correspondences*, 2021.

[52] Marc Henneaux and Soo-Jong Rey, *Nonlinear w_∞ as asymptotic symmetry of three-dimensional higher spin anti-de sitter gravity*, Journal of High Energy Physics **2010** (2010), no. 12, 007, available at 1008.4579.

[53] Gijs Heuts, *Koszul duality and a conjecture of francis–gaitsgory*, 2024.

[54] Jin Hong, *Center and universal r-matrix for quantized borcherds superalgebras*, Journal of Mathematical Physics **40** (1999), no. 6, 3123–3145, available at math/9810115.

[55] Jun-ichi Igusa, *On siegel modular forms of genus two*, American Journal of Mathematics **84** (1962), no. 1, 175–200.

[56] ———, *On siegel modular forms of genus two (ii)*, American Journal of Mathematics **86** (1964), no. 2, 392–412.

[57] Dominic Joyce and Markus Upmeier, *Orientation data for moduli spaces of coherent sheaves over calabi–yau 3-folds*, Advances in Mathematics **381** (2021), 107627, available at 2001.00113.

[58] Seok-Jin Kang and Olivier Schiffmann, *Canonical bases for quantum generalized kac–moody algebras*, Advances in Mathematics **200** (2006), no. 2, 455–478, available at math/0311089.

[59] Bernhard Keller, *Introduction to A_∞ algebras and modules*, Homology, Homotopy and Applications **3** (2001), 1–35, available at math/9910179.

[60] ———, *On differential graded categories*, 2006.

[61] ———, *Deformed calabi–yau completions*, Journal für die reine und angewandte Mathematik **654** (2011), 125–180, available at 0908.3499.

[62] Ahsan Z. Khan and Keyou Zeng, *Poisson vertex algebras and three dimensional gauge theory*, 2025.

[63] Tasuki Kinjo, Hyeonjun Park, and Pavel Safronov, *Cohomological hall algebras for 3-calabi–yau categories*, 2024.

[64] Maxim Kontsevich, *Deformation quantization of poisson manifolds*, Letters in Mathematical Physics **66** (2003), 157–216, available at q-alg/9709040.

[65] Maxim Kontsevich and Yan Soibelman, *Deformations of algebras over operads and the deligne conjecture*, 2000.

[66] ———, *Cohomological hall algebra, exponential hodge structures and motivic donaldson–thomas invariants*, 2011.

[67] Sabine Lechner, *A comparison of locally analytic group cohomology and lie algebra cohomology for p-adic lie groups*, 2012.

[68] Haisheng Li, *Nonlocal vertex algebras generated by formal vertex operators*, 2005.

[69] ———, *Quantum vertex algebras and their ϕ-coordinated quasi modules*, 2009.

[70] Jean-Louis Loday and Daniel Quillen, *Cyclic homology and the lie algebra homology of matrices*, Commentarii Mathematici Helvetici **59** (1984), 565–591.

[71] Jean-Louis Loday and Bruno Vallette, *Algebraic operads*, Grundlehren der mathematischen Wissenschaften, vol. 346, Springer, 2012.

[72] Eduard Looijenga and Valery Lunts, *A lie algebra attached to a projective variety*, Inventiones Mathematicae **129** (1997), 361–412.

[73] Ming Lu, *Quantum borcherds–bozec algebras via semi-derived ringel–hall algebras*, 2021.

[74] Jacob Lurie, *Higher topos theory*, Annals of Mathematics Studies, vol. 170, Princeton University Press, 2009.

[75] ———, *Higher algebra*, 2017. Available from the author's webpage.

[76] A. Matsuo and K. Nagatomo, *A note on free bosonic vertex algebra and its conformal vectors*, 1997.

[77] James E. McClure and Jeffrey H. Smith, *A solution of deligne's hochschild cohomology conjecture*, Contemporary Mathematics **293** (2002), 153–193, available at math/9910126.

[78] Andrei Mironov, Alexei Morozov, and Sergey Natanzon, *Complete set of cut-and-join operators in hurwitz–kontsevich theory*, Theoretical and Mathematical Physics **166** (2011), 1–22, available at 0904.4227.

[79] Alexander Molev, *Yangians and classical lie algebras*, Mathematical Surveys and Monographs, vol. 143, American Mathematical Society, 2007.

[80] Hiraku Nakajima, *Heisenberg algebra and hilbert schemes of points on projective surfaces*, Annals of Mathematics **145** (1997), no. 2, 379–388.

[81] Georg Oberdieck and Aaron Pixton, *Holomorphic anomaly equations and the igusa cusp form conjecture*, Inventiones Mathematicae **213** (2018), no. 2, 507–587, available at 1706.10100.

[82] Georg Oberdieck and Junliang Shen, *Reduced donaldson–thomas invariants and the ring of dual numbers*, Proceedings of the London Mathematical Society **118** (2019), no. 1, 191–220, available at 1612.03102.

[83] Andrei Okounkov and Rahul Pandharipande, *Gromov–witten theory, hurwitz theory, and completed cycles* (2002), available at math/0204305.

[84] Tony Pantev, Bertrand Toën, Michel Vaquié, and Gabriele Vezzosi, *Shifted symplectic structures*, Publications Mathématiques de l'IHÉS **117** (2013), 271–328, available at 1111.3209.

[85] M. Pevzner and Ch. Torossian, *Isomorphisme de duflo et la cohomologie tangentielle*, 2003.

[86] Stewart Priddy, *Koszul resolutions*, Transactions of the American Mathematical Society **152** (1970), 39–60.

[87] Daniel B. Ray and Isadore M. Singer, *r-torsion and the laplacian on riemannian manifolds*, Advances in Mathematics **7** (1971), no. 2, 145–210.

[88] Claus Michael Ringel, *Hall algebras and quantum groups*, Inventiones Mathematicae **101** (1990), 583–591.

[89] Olivier Schiffmann and Eric Vasserot, *Cohomological hall algebras of quivers and yangians*, 2023.

[90] Alexander Tsymbaliuk, *The affine yangian of* $\mathfrak{gl}_1$ *revisited*, Advances in Mathematics **304** (2017), 583–645, available at 1404.5240.

[91] Mikhail Verbitsky, *Cohomology of compact hyper-kähler manifolds and its applications*, Geometric and Functional Analysis **6** (1996), 601–611.

[92] Edward Witten, $(2+1)$*-dimensional gravity as an exactly soluble system*, Nuclear Physics B **311** (1988), no. 1, 46–78.